

ALL and AML

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Prognostic impact of chromosomal abnormalities and copy number alterations among adults with B-cell precursor acute lymphoblastic leukaemia treated on UKALL14A. Moorman^{1,*}, E. Butler¹, E. Barretta¹, A. Kirkwood², C. Schwab¹, T. Creasey¹, D. Leongamornlet³, E. Papaemmanuil⁴, P. Patrick², L. Clifton-Hadley², B. Patel⁵, T. Menne⁶, A. McMillan⁷, C. Harrison¹, C. Rowntree⁸, D. Marks⁹, A. Fielding¹⁰¹Translational and Clinical Research Institute, Newcastle University, Newcastle upon Tyne, ²Cancer Research UK & UCL Cancer Trials Centre, University College London, London, ³Sanger Institute, Cambridge, UK, ⁴Memorial Sloan Kettering Cancer Center, New York, USA, ⁵Department of Haematology, Queen Mary University of London, London, ⁶Department of Haematology, Newcastle upon Tyne Hospitals NHS Foundation Trust, Newcastle upon Tyne, ⁷Department of Haematology, Nottingham University Hospital NHS Trust, Nottingham, ⁸Department of Haematology, Cardiff And Vale University Health Board, Cardiff, ⁹Department of Haematology, University Hospitals Bristol NHS Foundation Trust, Bristol, ¹⁰Department of Haematology, UCL Cancer Institute, London, UK

Chromosomal abnormalities are established prognostic markers in adult ALL. Analysis of the UKALLXII trial revealed that patients (15–65 years) with *BCR-ABL1*, *KMT2A-AFF1*, low hypodiploidy (HoL) or a complex karyotype (CK) had an inferior outcome in multivariable analysis. In UKALL14, patients (25–65 years) with these genetic abnormalities were stratified to the high-risk arm to receive allogeneic stem cell transplant (alloSCT), where possible, or an unrelated transplant. Patients with *BCR-ABL1* received imatinib.

The objectives of this study were to determine the frequency and outcome of patients with high-risk chromosomal abnormalities and screen for novel gene rearrangements and copy number alterations. After a median follow-up time of 3.4 years, the 3 years outcome for B-cell precursor ALL patients was as follows: overall survival (OS) 52%, event-free survival (EFS), 45% and relapse rate (RR) 34%.

Among 653 patients, 319 (49%) harboured high-risk abnormalities: *BCR-ABL1* (31%), *KMT2A-AFF1* (8%), HoL (9%) and CK

(5%). The OS of patients with CK and HoL was 24% and 19% which was driven by high RR: 60% & 56%, respectively, despite alloSCT in first remission. Outcome was not obviously improved compared to UKALLXII (28%/22%); however, the patients were older. Patients with *KMT2A-AFF1* fusion had OS and RR similar to the small number of patients ($n = 9$) with other *KMT2A* fusions: 44% and 49% vs. 42% & 56%. The outcome of *BCR-ABL1* patients was not different from the remaining BCP-ALL patients (hazard ratios 0.92, $P = 0.5$ and 0.92, $P = 0.6$).

Additional chromosomal abnormalities were detected as follows: ABL-class fusions (*ABL1*, *ABL2*, *PDGFRB*, *CSF1R*) (1.3%), JAK-STAT abnormalities (*P2RY8-CRLF2*, *IGH-CRLF2*, *JAK2* fusions) (1.3%) and *ZNF384* fusions (3%). Patients with JAK-STAT abnormalities had high rates of MRD post phase 2 induction (87%), high RR (62%) and lower OS (35%) despite 82% being stratified as high risk. Half of the patients with an ABL-class fusion have had an event. In contrast, among the 12 patients with a *ZNF384* fusion only two have relapsed and none have died.

Secondary copy number alterations affecting key genes were determined by MLPA in 437 samples. 270 (62%) samples harboured one or more deletion at the following frequencies: *IKZF1* (39%), *CDKN2A/B* (37%), *PAX5* (22%), *BTG1* (11%), *ETV6* (8%), *EBF1* (3%), *RBI* (7%) and *PAR1* (1%). None of the individual deletions nor the number of deletions were associated with survival, either overall or when stratified by *BCR-ABL1* status. In particular, *IKZF1* deletions had no impact on RR: hazard ratios 1.05, $P = 0.8$ (all patients); 0.92, $P = 0.8$ (*BCR-ABL1*); 1.25, $P = 0.6$ (B-other). Neither of the recently validated paediatric CNA profiles (UKALL and IKZF1plus) showed a significant association with outcome in the whole cohort or any relevant subgroup.

Collectively these data indicate that chromosomal abnormalities remain stronger prognostic markers in adult ALL than the more recently identified secondary deletions. On the basis of these results, we propose an amendment to the genetic risk classification for adult ALL wherein groups with distinct outcomes comprise: (1) very high risk: CK, HoL or JAK-STAT abnormalities; (2) high risk: all *KMT2A* fusions; (3) Tyrosine kinase sensitive: *BCR-ABL1* and ABL-class fusions; (4) low-risk *ZNF384* fusions; (5) standard risk: all other patients. The integration of these primary genetic risk factors with other risk factor such as age, white cell count and MRD into an overall risk score is a key goal of our current work.

Disclosure of Interest: None Declared.

Transplantation

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The FLAMSA-Bu conditioning regimen does not improve outcome in patients allografted for high-risk acute myeloid leukaemia and myelodysplasia irrespective of pretransplant MRD: results of the UK NCRI FIGARO trial

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Abstract Content: Disease relapse represents the major cause of treatment failure in patients allografted for high-risk acute myeloid leukaemia (AML) and myelodysplasia (MDS). The advent of reduced intensity conditioning (RIC) regimens has substantially increased the number of older adults eligible for allo-SCT, but the optimal RIC regimen in high-risk AML remains unknown. Registry data have demonstrated improved outcomes using a sequential transplant regimen utilizing cytosine arabinoside (araC)/amsacrine (AMSA) cytorreduction followed by a fludarabine (Flu)/busulfan (Bu) based RIC regimen (FLAMSA-Bu). However, although the FLAMSA-Bu regimen is now widely used in adults with high-risk AML and MDS its benefit has not been evaluated in a randomized trial. Furthermore while retrospective analyses have identified that pretransplant measurable residual disease (MRD) is an important predictor of transplant outcome this has never been examined prospectively. We report the results of the first randomized trial evaluating the FLAMSA-Bu regimen compared with standard RIC regimens which also represents the first prospective evaluation of the impact of pretransplant MRD levels on transplant outcome. In total, 244 patients (median age 59 years) with high-risk AML ($n = 164$) or high-risk myelodysplasia ($n = 80$) were randomized 1:1 to a control arm determined by investigator's choice of either Flu/Bu₂/ATG or Flu/Bu₂/A *versus* an experimental arm of FLAMSA-Bu (Flu, araC 2 g/m² × 4 days, AMSA 100 mg/m² × 4 days, Bu -total dose 11.2 mg/kg). Patients over the age of 60 received an adjusted FLAMSA-Bu regimen utilising a reduced dose of araC (1 mg/m² × 4 days) and a total Bu dose of 6.4 mg/kg. Patients were transplanted using either an HLA identical sibling ($n = 49$) or matched (10/10 or 9/10) unrelated donor ($n = 195$). In total 155 patients with AML were in CR1 or CR2 at the time of transplant and nine had primary refractory disease. MRD was monitored by flow cytometry. Pretransplant MRD levels were measured in 201 patients (MRD positive = 80 (40%), MRD negative = 94 (47%), inadequate sample = 27 (13%)). Baseline characteristics including CR1/CR2 status, adverse cytogenetics and MRD levels were similar between regimens. Median follow-up was

35 months. Transplant outcomes were comparable between patients allografted in the control and FLAMSA-Bu arms. 2 years overall survival (OS) and cumulative incidence of relapse (CIR) were 61% and 30%, respectively, in the control arm *versus* 62% and 26% for the FLAMSA-Bu arm. Transplant-related mortality at 100 days was 3.0% in patients allografted using the control regimen *versus* 14% in patients allografted using the FLAMSA-Bu regimen. In the study cohort pretransplant MRD positivity was associated with both an increased CIR compared to patients testing MRD negative (2 years CIR 42% vs. 19%, $P = 0.009$) and decreased OS (2 years OS 52% vs. 71%, $P = 0.048$). The FLAMSA-Bu regimen failed to improve outcome in either MRD-positive or MRD-negative patients. This study represents the first demonstration in a prospective analysis that the presence of pretransplant immunophenotypic MRD is associated with reduced OS consequent upon an increased risk of disease relapse. Taken together these results support exploration of alternative strategies, such as pre or post-transplant pharmacological intervention, as the most promising strategy to reduce the risk of disease relapse post allograft.

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Post-transplant cyclophosphamide versus methotrexate-based regimen as graft-versus-host disease prophylaxis after allogeneic haematopoietic stem cell transplantation in patients with acute leukaemia

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Abstract Content: ALlogeneic haematopoietic stem cell transplantation (HSCT) is a potent curative treatment for haematological malignancies. Its curative potential is mainly based on immune-mediated graft-versus-leukaemia effects caused by donor T cells. However, donor T cells are also the cause of graft-versus-host disease (GVHD) which is a life-threatening complication of HSCT. Calcineurin inhibitor (CNI) and methotrexate (MTX) have been used as the standard GVHD prophylaxis in HLA-matched HSCT. Promising clinical trial data using high-dose post-transplant cyclophosphamide (PT-Cy) with or without additional immunosuppressive agents (IS) have shown that it is an effective, well-tolerated alternative and has crossed the HLA barriers of Haplo SCT.

To compare clinical outcomes of allogeneic HSCT after post-transplant cyclophosphamide *versus* methotrexate-based regimens, we enrolled 80 patients with acute leukaemia (53 males, 27 females) with age ranges from 18 to 60 years underwent fully matched allogeneic HSCT and haploidentical HSCT using reduced intensity chemotherapy at Maadi Armed Forces Medical Compound from 2014 to 2018 and median follow-up was 50 months after informed consent. They were randomized into two groups according to GVHD prophylaxis regimen. The first group (40 patients) received MTX in the following doses; day +2 (15 mg), day +4 (10 mg), day +6 (10 mg) with ciclosporin (5 mg/kg) from day -3 till day +90 and mycophenolate (MMF) (2-3 g/day) from day +1 till day +30. The

second group received PT-Cy (45 mg/kg/day) on days +3 and +4 with ciclosporin (5 mg/kg) from day +5 till day +30 & MMF (2–3 g/day) from day +5 till +30.

At median 1 year Post Allo, PT-Cy was associated with statistically significant lowering of chronic GVHD (cGVHD) incidence compared to MTX group (20%, 55%) respectively ($P = 0.002$). PT-Cy was associated with reduced risk of extensive cGVHD 5% compared to 30% in MTX group ($P = 0.003$). Incidence of acute GVHD (aGVHD) at day +100 was (40%, 32.5%) in group 1, 2 respectively ($P = 0.48$). About 30% of patients in group 1 suffered severe aGVHD, whereas only 12.5% of group 2 experienced it ($P = 0.056$). Patients received MTX had higher incidence of renal and liver toxicity when compared with PT-Cy group ($P = 0.018$, $P = 0.013$) respectively. PT-Cy group was associated with higher incidence of haemorrhagic cystitis (HC) when compared with MTX ($P = 0.006$). In MTX group, renal toxicity caused mortality of 1 patient (2.5%), Hepatotoxicity caused mortality of three patients (7.6%) and HC caused mortality of one patient (2.5%). In PT-Cy group, renal and

hepatic toxicity caused no mortality while HC caused mortality of three patients (7.5%). 87.5% of patients in group 1 and 90 % of patients in group 2 were engrafted with full donor chimerism on day +30 ($P = 0.72$). The risk of graft failure has no statistically significant difference between the MTX group (12.5%) and the PT-Cy group (10%). The cumulative incidence of relapse at 2 years after transplantation was 17.5 % (7 patients) for group 1 and 30 % (12 patients) for group 2 ($P > 0.05$), Cumulative 4 years progression-free survival (PFS) and overall survival (OS) in PT-Cy group were (45.7%, 46.9%), respectively, whereas (43.4%, 43.7%) in MTX group which is statistically non-significant ($P > 0.05$).

Our results demonstrate that PT-Cy with addition of IS drugs is a safe alternative which has statistically significant difference in reducing the incidence of cGVHD compared with MTX based regimen without affecting engraftment, relapse, PFS and OS.

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ALL and AML

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First analysis of the UKALL14 phase 3 randomised trial to determine if the addition of rituximab to standard induction chemotherapy improves EFS in adults with precursor B-ALL (CRUK/09/006)

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Abstract Content: UKALL14(NCT01085617) randomised 655 patients aged 25–65 years with B-precursor ALL, irrespective of Ph chromosome status or CD20 expression to determine if the addition of four doses of rituximab to standard induction chemotherapy (SOC+R) improved event-free survival (EFS). The primary analysis population comprises 577 patients recruited after an 04/2012 amendment to change SOC therapy. The trial was powered with an 84% chance to detect a 12% improvement in EFS [hazard ratio (HR) 0.71]. Secondary end-points included complete remission (CR), OS, non-relapse mortality (NRM), minimal residual disease (MRD) after induction and relationship of response to CD20 expression. SOC was daunorubicin 30 mg/m², vincristine 1.4 mg/m², dexamethasone 4 days blocks of 10 mg/m² on d1, 8, 15 and 22. Peg-asparaginase 1000 IU/m² was added on d4 and 18 (d18 only if >40 years) for Ph-ALL and continuous daily imatinib 600 mg for Ph+ ALL. Rituximab 375 mg/m² was given on days 3, 10, 17, 24. Two patients in the SOC+R arm did not start trial treatment. Analysis is by intention-to-treat. There were 288 patients in the SOC arm and 289 in the SOC+R arm, 273 (95.5%) of whom received all four doses of rituximab. The arms were well-balanced. Of note, 63.9% (SOC) and 62.6% (SOC+R) were aged >40 years at randomisation and 86 patients in each arm (29.9% and 29.8% respectively) had Ph+ ALL. CR rate, 92.7% SOC and 94.8% SOC+R, did not differ between the arms, nor did MRD. 121 (42.2%) of patients were MRD negative in the SOC arm vs. 120 (41.8%) in the SOC+R arm. Severe/adverse events and NRM did not differ between arms. At a median follow-up of 50.5 months (7 days to 83.6 months), 3 years EFS for SOC is 41.9% (95% CI: 35.8–48.0) vs. SOC+R 48.7% (42.4–54.8), HR 0.88 (0.71–1.11), *P* = 0.28. Preplanned subgroup analyses by cytogenetic/other risk groups (age, presenting WBC, cell surface CD20 expression) did not reveal any significant interactions. % blasts expressing CD20 was an independent poor prognostic factor; Youden cut-off suggested CD20 expression of 11.7% as the optimal cut-off. EFS HR for 10–20% CD20 was 1.74 (0.98–3.10) and >20% was 2.20 (1.27–3.81), compared to <10% CD20. Outcomes were analysed by post-induction treatment; myeloablative allogeneic stem cell transplant (MAalloSCT) for patients with high-risk ALL aged 40 and under, reduced intensity-conditioned RICalloSCT for patients over 40 years

and consolidation/maintenance for SR-ALL/no suitable donor. There was a large and statistically significant benefit to SOC+R among those who received MAalloSCT, SOC *N* = 53, 3 years EFS 50.7% (35.9–63.7) vs. SOC+R, *N* = 40, 3-year EFS 72.2% (54.9–84.2), HR 0.47 (0.23–0.95), *P* = 0.03 which was not evident in the RICalloSCT or maintenance groups and related to a reduction in relapse risk. Among those who received MAalloSCT, there was a preponderance of high-risk genetics in the SOC+R arm: 29 (64.4%) SOC vs. 28 (77.8%) SOC+R, suggesting that the benefit of SOC+R combined with MAalloSCT outweighed some of the adverse genetic risk. By contrast to the GRAALL-2005/R study, in which 16–18 doses of rituximab was given during all treatment phases, we did not find an overall statistically significant benefit for four infusions of rituximab during induction. The non-significant trend to a better outcome with SOC+R underscores the necessity for including 16–18 doses of rituximab to obtain the full benefit. However, our trial shows no treatment interaction of SOC+R with Ph+ status or CD20⁺ expression level.

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Gilteritinib prolongs survival in patients with FLT3-mutated relapsed/refractory acute myeloid leukaemia who have common AML co-mutations or a high FLT3-internal tandem duplication allelic ratio

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The *Fms*-like tyrosine kinase 3 (FLT3) inhibitor gilteritinib showed superior response and overall survival compared with salvage chemotherapy in patients with FLT3-mutated relapsed/refractory acute myeloid leukaemia (AML) in the phase 3 ADMIRAL study. We analysed the impact of baseline co-mutations and FLT3-internal tandem duplication (ITD) allelic ratio on response and overall survival. A total of 37 recurrently mutated genes in AML (Archer Core Myeloid Panel) were analysed by next-generation sequencing; the

Table.

Patients	CR/CRh (%)			Median overall survival (months)		
	Gilteritinib	Salvage chemotherapy	Gilteritinib	Salvage chemotherapy	Hazard ratio	P-value
ITT population (<i>n</i> = 371)	34.0	15.3	9.3	5.6	0.637	0.0007
Co-mutation-positive cohorts						
<i>NPM1</i> (<i>n</i> = 173)	32.2	12.1	8.3	5.1	0.419	<0.0001
<i>DNMT3A</i> (<i>n</i> = 115)	37.3	12.5	9.1	5.5	0.504	0.0031
<i>DNMT3A/NPM1</i> (<i>n</i> = 86)	40.0	9.7	10.8	5.0	0.252	<0.0001
<i>WT1</i> (<i>n</i> = 65)	35.6	5.0	9.1	3.4	0.309	0.0001

CR, complete remission; CRh, complete remission with partial haematological recovery; ITT, intention-to-treat.

cut-off for co-mutation positivity was ≥ 0.027 . Baseline *FLT3*-ITD allelic ratio (*FLT3*-ITD to *FLT3* wild-type DNA) was measured by the LeukoStrat[®] CDx *FLT3* Mutation Assay. The median *FLT3*-ITD allelic ratio value of 0.77 was used to define high (≥ 0.77) vs. low (< 0.77) *FLT3*-ITD allelic ratio. Analysis of 361 patients with *FLT3*-mutated relapsed/refractory AML identified four major co-mutation cohorts, each with $\geq 10\%$ of patients: *nucleophosmin 1* (*NPM1*, *n* = 173; 47.9%), *DNA methyl transferase 3 alpha* (*DNMT3A*, *n* = 115; 31.9%), *DNMT3A/NPM1* (*n* = 86; 23.8%) and *Wilms tumour 1* (*WT1*, *n* = 65; 18.0%). In addition, seven patients (1.9%) had all three co-mutations (i.e. *NPM1*, *DNMT3A* and *WT1*). The gilteritinib arm had superior response rates and overall survival across all four major co-mutation cohorts, with the greatest survival benefit in patients with *DNMT3A/NPM1* co-mutation positivity (Table). In *FLT3*-ITD allelic ratio analyses (*n* = 335), gilteritinib conferred longer overall survival than salvage chemotherapy in patients with a high or low *FLT3*-ITD allelic ratio (gilteritinib: high *FLT3*-ITD allelic ratio, 7.1 months vs. low *FLT3*-ITD allelic ratio, 10.6 months; salvage chemotherapy: high *FLT3*-ITD allelic ratio, 4.3 months vs. low *FLT3*-ITD allelic ratio, 6.9 months). In both arms, overall survival was longer in the low *FLT3*-ITD allelic ratio cohort than the high *FLT3*-ITD allelic ratio cohort, but the difference in the gilteritinib arm was not statistically significant (gilteritinib: hazard ratio = 1.341, *P* = 0.0712; salvage chemotherapy: hazard ratio = 2.01, *P* = 0.0021). The ADMIRAL trial shows that the clinical benefit of gilteritinib in *FLT3*-mutated relapsed/refractory AML is maintained regardless of *NPM1*, *DNMT3A*, *DNMT3A/NPM1* or *WT1* co-mutation positivity or high *FLT3*-ITD allelic ratio.

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General Haematology

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CIP2A: its validation as a predictive biomarker of subsequent progression to blast crisis in CML

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Disease progression to blast crisis remains a significant problem in CML, with a median overall survival of around 10 months. For patients presenting in chronic phase, several prognostic scoring systems based on age and simple haematological parameters can identify those at higher risk of progression, as can also the presence of certain 'major route' cytogenetic abnormalities. However, even in the presence of these, over 75% of patients remain progression-free at 8+ years, and these are therefore of limited practical use in the clinic.

Disease progression is a poorly understood multifactorial process likely initiated by both BCR-ABL dependent and independent mechanisms. Protein phosphatase 2A (PP2A) is a major cellular phosphatase and tumour suppressor, whose activity can be inhibited by Cancerous Inhibitor of PP2A (CIP2A). CIP2A protein overexpression can cause cellular transformation and increased proliferation. High CIP2A levels have been reported to be an adverse prognostic indicator in 21 different cancers including in chronic and acute myeloid leukaemia (CML and AML). We have previously shown in 72 selected local samples that patients expressing high CIP2A levels at diagnosis of chronic phase (CP) CML have a higher probability of progression (Lucas et al Blood 2011; 117: 6660-8; Leukaemia 2015; 29: 1514-23). However, this requires validation in a larger unselected multicentre cohort.

Here we measured CIP2A by flow cytometry in mononuclear blood cells from 172 unselected consecutive samples from the 814 patients enrolled into SPIRIT2, a UK-wide 1:1 randomised comparison of first line imatinib 400 mg daily *versus* dasatinib 100 mg daily in CP CML. CIP2A protein levels were independent of Sokal, Hasford and ELTs scores and major route cytogenetic abnormalities. Eighteen (9 imatinib; 9 dasatinib) of the 173 patients studied underwent disease progression, of whom 13 had high CIP2A levels, and patients with a high CIP2A protein level at diagnosis had an inferior freedom from progression and progression-free survival ($P = 0.02$ and $P = 0.03$). Interestingly, the 35 dasatinib recipients with high diagnostic CIP2A levels had a significantly worse freedom from progression than the 47 with a low CIP2A level ($P = 0.003$), though this difference was not seen in imatinib recipients (48 high and 42 low CIP2A levels respectively). High CIP2A patients also had an inferior time to MR2 and MR4.5 compared to low CIP2A patients ($P = 0.001$ and $P = 0.02$).

The present findings in an unselected large cohort broadly confirm our earlier reports on selected local samples, that a high diagnostic CIP2A level may predict disease progression. They also extend our earlier data by showing that this effect holds for dasatinib recipients, who were insufficient for comment previously. The absence of this effect in imatinib recipients is likely due to the high number of patients in the SPIRIT2 trial who switched treatment due to

perceived poor response, and the fivefold higher rate of stem cell transplantation in imatinib recipients overall (O'Brien et al; ASH 2018). CIP2A is a simple diagnostic biomarker that may help assess the prospective risk of disease progression.

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Efficacy of bosutinib in imatinib-resistant *versus* dasatinib/nilotinib-resistant chronic phase chronic myeloid leukaemia: results from the phase 4 BYOND study

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Abstract Content: Introduction: Bosutinib is approved for patients with Philadelphia chromosome (Ph) + chronic myeloid leukaemia (CML) resistant/intolerant to prior therapy and newly diagnosed patients in chronic phase (CP).

Methods: BYOND is an ongoing phase 4 study (NCT02228382) in patients with CML resistant/intolerant to prior tyrosine kinase inhibitors (TKIs). Patients received bosutinib 500 mg once daily (starting dose). Here, we report the efficacy of bosutinib in patients with Ph+ CP CML resistant to imatinib (but not nilotinib or dasatinib) *versus* patients resistant to ≥ 1 second-generation TKI (dasatinib and/or nilotinib), as well as in patients intolerant to all prior TKIs. Data are reported at ≥ 1 year after last enrolled patient; 85% of patients had a minimum follow-up of 2 years.

Results: Of 156 patients with Ph+ CP CML who received bosutinib, 52 had resistance only to imatinib, 31 had resistance to dasatinib and/or nilotinib and 73 were intolerant to all prior TKIs. Corresponding median treatment duration (range) was 24.1 (0.2–42.2), 8.9 (0.9–41.6) and 25.3 (0.4–41.9) months, and median dose intensity (range) was 360 (125–500), 431 (195–561) and 292 (80–500) mg/day. In all, 69.2%, 41.9% and 53.4% of imatinib-resistant, dasatinib/nilotinib-resistant and TKI-intolerant patients, respectively, were still receiving treatment as of the data cut-off date. The main reason for permanent treatment discontinuation was adverse events: 10 (19.2%), 8 (25.8%) and 21 (28.8%) imatinib-resistant, dasatinib/nilotinib-resistant and TKI-intolerant patients respectively. Corresponding discontinuations due to insufficient response occurred in two (3.8%), five (16.1%) and one (1.4%) patients. No patient

experienced on-treatment transformation to advanced phase CML or discontinued due to disease progression. In patients without baseline complete cytogenetic response (CCyR), CCyR rates by 1 year were 65.0%, 50.0% and 72.2% in imatinib-resistant, dasatinib/nilotinib-resistant and TKI-intolerant patients respectively. In patients without the respective baseline molecular response (MR), cumulative major molecular response rates by 1 year were 55.6%, 28.6% and 80.6% in imatinib-resistant, dasatinib/nilotinib-resistant and TKI-intolerant patients respectively. Correspondingly, MR⁴ rates were 36.1%, 12.5% and 59.6% and MR^{4.5} rates were 26.7%, 11.1% and 33.9%. In imatinib-resistant, dasatinib/nilotinib-resistant and TKI-intolerant patients, respectively, Kaplan–Meier estimated overall survival rates were 96.1%, 100% and 98.6% at 1 year, and 96.1%, 92.6% and 97.2% at 2 years with 3, 3 and 4 deaths occurring on study.

Conclusions: Response rates were similar between imatinib-resistant and TKI-intolerant patients. Despite the fact that dasatinib/nilotinib-resistant patients were more heavily pretreated and had a shorter treatment duration, responses were also seen in patients with resistance to second-generation TKIs. These results further support bosutinib use for patients with Ph+ CP CML and resistance/intolerance to prior TKIs.

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Decision tree application (DTA) to support accurate diagnosis of myeloid malignancies

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NICE guidance recommends multidisciplinary meetings (MDMs) as best practice in the diagnostic work-up of suspected haemato-oncology patients. Accurate diagnosis in a busy MDM can be difficult as WHO diagnostic criteria are complex, multifactorial and are subject to periodic updates. Poor documentation may lead to inefficiencies of clinical service, incorrect treatment strategies and poor quality of data for local and national collection. We have previously shown that digital solutions can improve the accuracy of MDS-subtyping in a single-centre retrospective study.

We have now developed a comprehensive decision tree application (DTA) that recapitulates the 2016 WHO diagnostic criteria for adult MDS, AML, MPN and MDS/MPN disease categories. This

multicentre study set out to validate the DTA by independent user testing. It shows our DTA is accurate and can improve precision in the diagnosis of commonly occurring myeloid malignancies.

The DTA was designed using the open source platform ESYN and is publicly available as a web app. Users enter clinical data (a minimum of 14 data points are required) which is fed into the decision tree and a suggested WHO diagnosis is displayed.

A clinician at each centre received training on using the DTA. Cases with a diagnostic bone marrow, documented local MDM discussion and a WHO diagnosis were randomly selected for analysis. Clinical data were logged using the DTA and the suggested diagnosis recorded. In cases of discrepancy between the DTA and the MDM outcome, a second independent clinical opinion was sought to identify whether i) the DTA had suggested an incorrect diagnosis or ii) the MDM diagnosis needed revision.

Sixty-two cases of myeloid malignancy were logged in the DTA from five UK centres, covering 24 of the possible 35 diagnoses in the DTA. Fifty-five cases (89%) were concordant between the DTA and the MDM diagnosis. On independent review of the seven discordant cases, three cases were recommended to revise the MDM diagnosis to the DTA-generated diagnosis. Four cases were due to DTA error. Therefore, overall the DTA was accurate in 58/62 cases (94%) (Table 1).

Minor adjustments were made to improve the DTA accuracy. For example three of the DTA errors were due to one missing logic step for cases of MDS-MPN-U. The clinical data from the 62 cases were re-run through the updated DTA and all 62 cases gave an output that matched the amended MDM diagnosis. To further validate the updated DTA, three additional cases with a known diagnosis of

Table.

WHO Chapter Disease Heading	Disease subtypes available in DTA ;(subtypes with no cases tested in the validation are marked in <i>italics</i>)	Number of correct cases diagnosed by DTA (correct disease and subtyping)
AML with recurrent mutations	t(8;21), t(16)/inv(16), t(15;17), NPM1, biallelic CEBPA, t(9;11), t(1;22), t(3)/inv(3), RUNX1	11/11
AML NOS	AML NOS	1/2
AML with MRC	AML with MRC	8/8
Therapy-related myeloid neoplasm	Therapy-related AML, therapy-related MDS / MPN	6/6
MDS	MDS-MLD, MDS-SLD, isolated del 5q, MDS-SLD-RS, MDS-EB1, MDS-EB2, <i>MDS-MLD-RS</i> , <i>MDS-U</i>	11/11
MPN	PV, ET, CML, primary MF, CNL, MPN-U, <i>prefibrotic</i> MF, <i>CEL NOS</i> , <i>CEL with recurrent mutation</i>	16/16
MDS/MPN	CMML-1, CMML-2, MDS-MPN-U, <i>atypical CML</i> , <i>MDS-MPN-RARS-T</i> , <i>CMML-0</i>	5/8
Total		58/62

MDS-MPN-U, that would have been mis-diagnosed by the original version, were tested – the correct diagnosis was returned in all three.

An interesting observation from one case was that greater clarity was required from the trephine report to be able to run the DTA. When the biopsy was reviewed and the report updated, this led to a recommended revision of the MDM diagnosis. Further testing is now needed to ensure that the rarer diseases covered by the DTA, that were not tested, are also accurate, before trialling the DTA in real time as a diagnostic aid in MDMs.

Increasingly complex data are generated in the diagnostic workup of haematological malignancies. This DTA provides an example of how technology can be used to support clinicians' assessment of large and complex data, improve diagnostic accuracy and MDM recording of outcomes to improve clinical care. The amended DTA is available for use at <http://esyn.org/DTA>

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BSH2020-OR-009

Splenectomy in immune thrombocytopenia: do changing treatment patterns for ITP affect outcome? Data from the UK ITP registry

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THE advent of medical alternatives to splenectomy, particularly thrombopoietin receptor agonists, has not only seen reduction in its use for ITP, but also use later in the treatment pathway. Splenectomy is major surgery associated with significant complications. We aimed to evaluate splenectomy use in the UK using data from the UK adult ITP registry, a national database of primary ITP cases set up in 2007. Cases entered to December 2018 were included. Demographics, response/relapse rates and complications were studied. The following response criteria were used: complete response (CR platelets $>100 \times 10^9/l$), partial response (PR platelets >30 but <100 , or doubling in baseline), overall response (OR = PR+CR). Duration of response (DOR) = time from splenectomy to next treatment or to latest platelet count if remained in remission.

There were 3824 registered patients with ITP diagnosis date between 1953 and 2018. About 351 patients (9.18%) (37.9% M, 61.5% F) had undergone splenectomy with median follow-up 8.2 years (IQR 4.1, 15.9) and total follow-up of 2511 years. The median age at splenectomy was 40 years (IQR 26–55). About 86.3% patients (296) were <65 years and 13.7% (47)³ 65 years. Seventy-nine patients were excluded from further analysis due to incomplete data including patients splenectomised <1989 .

Splenectomy use has decreased over time from 34.3% (1989–1998), 16.2%, 1999–2008 and 3.87% 2009–2018. Median time from ITP diagnosis to splenectomy was 1.5 years (IQR 0.44, 4.67) increasing from 1.17 years (IQR 0.48, 3.44) in 1989–1998, to 1.67 years (IQR 0.58, 4.73) in 2009–2018. This was associated with an increase in median number of treatments prior to splenectomy from 2 (IQR 1, 2.75) in 1989–1998 to 3 (IQR 2, 4) in 2009–2018. The most common treatments presplenectomy were prednisolone (92%), IVIG (50%), azathioprine (27%), rituximab (21%), dexamethasone (12%), mycophenolate (10%) and romiplostim (10%). Splenectomy was performed later in the treatment course over time (median 3rd line (IQR 2, 4) in 1989–1998 and 4th line (IQR 3, 5) in 2009–2018). The CR/PR rates were 64.4/24.8 at 1 month, 54.4/19.0 at 6 m, 46.2/13.6 at 12 m and 34.5/13.5 at 24 m.

About 53.7% patients who had an initial response postsplenectomy still had a response at latest follow-up (median 3.9 years (IQR 2, 7.5)). The response rates were lower in 65 years compared to <65 years (Table 1).

The overall median DOR was 3 years (IQR 1, 7) and fell from 10.3 years (1989–1998) to 3.8 years (1999–2008) and 1.7 years (2009–2018). Median DOR was 3.3 years in <65 years (IQR 1, 7.3) and 0.83 years in 65 years+ (IQR 0.42, 3.83). In patients who relapsed postsplenectomy, median time to next treatment was 2.2 years (IQR 0.5, 7.0).

Median DOR postsplenectomy was 2.7 years (IQR 0.8, 5.9) at 2nd line, decreasing to 2.2 years (IQR 0.5, 7.2) at 4th line and 1.5 years (IQR 0.8, 17.4) at 6th line.

Postsplenectomy, 44 patients (12.9%) reported infections and 52 patients (15.2%) reported thromboses (total 74 thromboses (48.6% arterial; 44.6% venous). Twelve patients (3.5%) died. One death was attributed to splenectomy.

In conclusion, splenectomy rates for ITP in the UK have fallen since 1989, associated with increased use of medical therapies. In refractory ITP where multiple medical therapies have failed, splenectomy has lower response rates. With the advent of medical therapies with large amounts on efficacy and safety data, the role of splenectomy is unclear and needs further analysis of markers of response.

Table 1 Shows the response rate to splenectomy fell over time. OR at 24 months was 48%.

	<65 years	65 years+	<65 years	65 years+	<65 years	65 years+
Response	1 m	1 m	6 m	6 m	12 m	12 m
CR	64.0	54.5	48.7	34.5	41.5	32.3
OR	87.3	81.7	66.1	44.8	54.5	38.8

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Fostamatinib as second-line therapy and in earlier stage disease in patients with immune thrombocytopenia

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Immune thrombocytopenia (ITP) is a rare autoimmune disease in which platelets targeted by circulating autoantibodies are marked for phagocytosis by Fcγ receptor-bearing macrophages. Fcγ receptor-dependent phagocytosis depends on intracellular spleen tyrosine kinase (SYK) activity. Fostamatinib is an oral SYK inhibitor evaluated for ITP in two phase 3 randomized placebo-controlled studies and one open-label extension study in adults. Fostamatinib received marketing authorization from the European Medicines Agency in January 2020 and from the US FDA in April 2018. The 2019 International Consensus Report on management of ITP included fostamatinib as a subsequent treatment with robust evidence following initial treatment with steroids or immunoglobulins. We performed a post hoc analysis of the phase 3 clinical study data to evaluate patients for whom fostamatinib was second-line therapy as well as in patients with earlier stage disease (<2 years duration).

In the phase 3 studies, 145 patients were treated with fostamatinib at a starting dose of 100 mg BID PO, which was increased to 150 mg BID if drug was tolerated to increase platelet count to a level sufficient to maintain haemostasis. These patients had persistent or chronic ITP, had failed ≥1 prior ITP treatment, and had ≥3 platelet counts <30 000/μl at screening. Fostamatinib was second-line therapy in patients who had previously received only steroids (± immunoglobulins) for the treatment of ITP. Patients were also categorized by duration of disease: 3 months to <1 year (persistent) ITP, 1 to <2 years ITP and ≥2 years ITP. A platelet response was defined as ≥1 platelet count of ≥50 000/μl at any visit in the absence of rescue therapy.

Overall, fostamatinib was second-line therapy for 32 of 145 patients. The baseline characteristics of the 32 second-line therapy patients included a median age of 50 years (range 20–88), a median ITP duration of 2.6 years and a median platelet count of 21 500/μl. A platelet response of ≥1 platelet count ≥50 000/μl was observed in 25/32 (78%) patients, which was higher than the 47% response rate in the 113 patients who were given fostamatinib as third or later line therapy. For the 32 second-line-therapy patients, common adverse events (AEs) were hypertension, diarrhoea, upper respiratory tract infection, nausea, vomiting, elevated liver enzymes, petechiae and headache. The majority of AEs were mild to moderate in severity and managed by dose adjustments and/or secondary medication. Serious AEs occurred in eight of 32 (25%) second-line therapy patients and 31 of 113 (27%) third or later line therapy patients. AEs were similar in type and frequency to AEs observed in the overall study population as well as in the overall safety database of >3500 patients across multiple diseases.

Out of the 145 patients, 29 had early-stage disease. A platelet response of ≥1 platelet count ≥50 000/μl was achieved by nine out of 10 (90%) patients with persistent ITP (<1 year), 11 out of 19 (57%) patients with 1 to <2 years of ITP and 58 out of 116 (50%)

patients with 2–53 years of ITP. A platelet response of 54% (78/145) was observed in the total phase 3 population.

These data demonstrate that fostamatinib may provide a better platelet response when used as second-line therapy for ITP compared with later lines of therapy. Fostamatinib may provide a higher response rate in early stage ITP, particularly in persistent patients.

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A phase 2 open-label study of danicopan (ACH-0144471) in patients with paroxysmal nocturnal haemoglobinuria who have an inadequate response to eculizumab monotherapy

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IN PAROXYSMAL nocturnal haemoglobinuria (PNH), a somatic mutation in the *PIGA* gene of one or more haematopoietic stem cells generates a clone of red blood cells (RBCs) that lack complement regulatory proteins CD55 and CD59. This leads to uncontrolled complement activation on affected RBCs and membrane attack complex (MAC) mediated intravascular haemolysis (IVH).

Standard of care of C5 inhibition blocks MAC formation thus MAC-mediated IVH. However, a published analysis showed approximately 70% of patients receiving a C5 inhibitor remain anaemic and >1/3 are transfusion-dependent presumably due to continuous extravascular haemolysis (EVH).

Factor D (FD), a serine protease, catalyses complement factor B cleavage, allowing formation of alternative pathway (AP) C3 convertase. By inhibiting FD, danicopan blocks AP C3 convertase formation and thereafter AP activation and amplification of all complement pathways. Consequently, C3 cleavage, C3 fragment deposition,

terminal pathway activation and MAC formation are all inhibited by danicopan, leading to control of both IVH and EVH.

To demonstrate danicopan as a potential treatment for PNH patients with an inadequate response to a C5 inhibitor, a phase 2, dose-finding, proof of concept trial was conducted in PNH patients who remained transfusion dependent (defined as ≥ 1 RBC transfusion within 12 weeks prior to screening) with eculizumab treatment. Danicopan was added to the eculizumab regimen at three times a day orally with a starting dose of 100–150 mg and dose escalation at specified periods based on clinical response. The final results from the study are reported herein.

Twelve patients received at least one dose of danicopan. One patient discontinued after 2 doses, due to a serious adverse event of worsening of an underlying condition (pulmonary hypertension/oedema), considered unlikely related to danicopan and is excluded from this analysis. The key laboratory results for the remaining 11 patients are summarized in Table 1. The primary end-point of haemoglobin

Table.

	results	
	Baseline N = 11	Week 24 N = 11
Laboratory parameters		
Hgb (g/dl), Mean (SD)	7.9 (1.42)	10.3 (1.66)
LDH (xULN), Mean (SD)	1.06 (0.321)	1.04 (0.181)
Reticulocytes ($10^9/\mu\text{l}$), Mean (SD)	219 (78.1)	135 (66.3)
Total bilirubin (mg/dl), Mean (SD)	2.17 (1.118)	1.35 (0.798)
Direct bilirubin (mg/dl), Mean (SD)	0.51 (0.220)	0.37 (0.207)
PNH RBC clone size (%), Mean (SD)	54 (24.7)	84 (22.1)*
C3 fragment deposition on PNH RBCs (%), Median (Range)	20 (6.3–80)	6 (0.10–57)
Quality of life		
FACIT-Fatigue†	34 (14.1)	45 (8.2)
Treatment regimen		
Oral danicopan TID	100 mg (n = 9) 150 mg (n = 2) N/A	100 mg (n = 2) 150 mg (n = 6) 200 mg (n = 3)
Eculizumab IV Q2WK	900 mg (n = 7) 1200 mg (n = 3) 1500 mg (n = 1)	900 mg (n = 8) 1200 mg (n = 2)‡ 1500 mg (n = 1)
Transfusions	24 weeks prior to screening N = 10§	24 weeks during trial N = 10§
Number of transfusions	34	1
Units of blood	58	2

Hgb, haemoglobin; IV, intravenous; LDH, lactic acid dehydrogenase; PNH, paroxysmal nocturnal haemoglobinuria; RBCs, red blood cells; TID, three times a day; ULN, upper limit of normal.

*N = 7; for four patients, samples were out of stability range.

†Scores based on the Functional Assessment of Chronic Illness Therapy Fatigue (FACIT) Fatigue Scale V4. Score range 0–52. A score of less than 30 indicates severe fatigue.

‡A US patient had eculizumab dose decreased per insurance company from 1200 to 900 mg prior to week 16 due to Hgb improvement.

§N = 10; Patient A excluded from transfusion results due to religious objection to receiving transfusions; baseline Hgb = 5 g/dl.

increase was achieved with a mean value of 2.4 g/dl at Week 24 relative to baseline. In parallel, an increase in FACIT Fatigue scores was observed with a mean value of 11 points at Week 24 relative to the baseline. In addition, total bilirubin was normalized while reticulocytes and direct bilirubin returned towards the normal range; PNH erythrocytes opsonized with C3 fragment decreased while PNH erythrocyte clone size approached PNH granulocyte clone size during the study. Lastly, transfusion needs also dramatically reduced, with one patient receiving 1 transfusion (2 units) during the trial, as compared to 34 transfusions (58 units) in the 6 months prior to screening.

Danicopan was generally well tolerated; 96% of treatment emergent adverse events (TEAEs) were mild to moderate in severity. No discontinuations due to TEAEs. Two patients had Grade 3 TEAEs which resolved.

Proof of concept is established with danicopan, an oral, FD inhibitor in the treatment of PNH in combination with eculizumab. Meaningful improvements in haemoglobin, transfusion needs, FACIT-Fatigue and parameters of interest were achieved. These

improvements demonstrate further benefit can be achieved by blocking AP at FD with danicopan, in combination with C5 inhibition. This benefit is likely due to the prevention of C3 fragment-mediated EVH, in addition to control of IVH. Danicopan targets an unmet need in PNH and will be further evaluated in a pivotal trial.

Disclosure of Interest: A. Kulasekararaj Conflict with: Alexion, Celgene, Novartis, Ra Pharma, Regeneron, Conflict with: Achillion, Celgene, Ra Pharma, A. Risitano Conflict with: Achillion, Alexion, Apellis, Biocryst, Novartis, Roche, Samsung, Amyndas, Conflict with: Alexion, Novartis, Alnylam, Ra Pharma, Conflict with: Alexion, Novartis, Pfizer, Apellis, J. Maciejewski Conflict with: Novartis, Conflict with: Alexion, Novartis, Celgene, R. Notaro Conflict with: Alexion, P. Browett Conflict with: Merck, Janssen, Roche, AbbVie, Conflict with: Roche, Beigene, Amgen, Conflict with: Internal Medicine Journal, J. W. Lee Conflict with: Achillion, Alexion, Apellis, Conflict with: Alexion, Conflict with: Alexion, M. Huang Conflict with: Achillion, M. Geffner Conflict with: Achillion, R. Brodsky Conflict with: Achillion, Alexion.

Education

BSH2020-OR-012

Transforming end of life care for patients with lymphoma – understanding training needs within the haematology department of a major acute teaching hospital

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WE have one chance to get end of life care (EOLC) right for our patients and those close to them, however, aspects of EOLC remain a challenge in haematology. The Transforming End of Life Care (TEOLC) team at University College London Hospitals NHS Foundation Trust (UCLH) train and support healthcare staff to deliver effective EOLC through a quality improvement strategy. The TEOLC framework consists of 4 key principles: recognising the dying patient (and initiating the Excellent Care in the Last Days of Life (ECILDOL) care plan when appropriate), compassionate communication (including Do Not Attempt Cardiopulmonary Resuscitation (DNACPR) discussions), care planning (with use of Coordinate My Care (CMC) records) and collaborative working. The haematology team at UCLH treat patients with lymphoma, including those undergoing transplants and chimeric antigen receptor T-cell (CAR-T) therapy. When the disease cannot be controlled, patients can have a complex dying phase. In order for the TEOLC to support the team to improve EOLC, the agreed first step was to complete a needs analysis. The aims were to understand: (i) Baseline haematology staff confidence related to EOLC. (ii) Current delivery of EOLC for patients with lymphoma and how this aligns to the TEOLC principles. A novel self-assessment staff confidence questionnaire, based on the TEOLC principles, was completed by staff working on the haematology unit in December 2019. The questionnaire utilised a 4-point Likert-scale ranging from 'not at all confident' to 'very confident'. In addition, an audit of the last 10 deaths of patients with lymphoma (who died between July and October 2019) was conducted against the TEOLC principles. Thirty-one haematology staff members completed the questionnaire (nurses $n = 25$; doctors $n = 4$; allied health professionals $n = 2$). Most (26/31) participants felt 'confident'/'very confident' in identifying when a patient was in their last days of life. Almost half (14/31) of the participants felt 'not so confident'/'not at all confident' in being able to sensitively ask patients or their family about what matters to them the most. Ten out of 29 participants felt 'not so confident'/'not at all confident' in initiating and using the ECILDOL care plan. A large majority (25/28) of the participants felt 'not so confident'/'not at all confident' in suggesting or completing a CMC record. The audit showed that four out of 10 patients died in ICU, and the remainder died under the care of haematology. Days between recognition of likelihood of dying and death varied from 0 to 8 days. Half the patients (5/10) were recognised as dying within a day or less of their death. Days between the completion of a DNACPR order and death varied from 0 to 23 days with most being completed within a day or less of death. Only two out of 10 patients had an ECILDOL care plan. To conclude, the questionnaire and audit have helped establish key educational needs. Focused training is required to support staff to have compassionate conversations, to confidently manage those who are in the last days

of life and to engage with care planning. Most patients died less than a day after recognition, despite staff rating themselves confident at identifying those who are in the last days of life, which indicates an area where perceived ability is not being translated into practice. Going forward, the TEOLC team are utilising this data to inform a bespoke training schedule to support the haematology team in their clinical activities.

Disclosure of Interest: None Declared.

BSH2020-OR-013

Simulating haematological emergencies: a pilot course for medical trainees

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THE immediate management of many haematological emergencies commonly relies on junior doctors. However, core medical trainees (CMTs; now internal medicine trainees) rotating into haematology jobs are often unfamiliar with complex treatment algorithms and therefore face a steep learning curve. Previous audits have cited poor knowledge of protocols from junior doctors as a major cause of low adherence rates to the one-hour antibiotic target for neutropenic sepsis¹. Simulation-based education has been shown to be an effective method of improving confidence in the management of major haemorrhage² and has been shown to improve preparedness starting a new clinical role³.

We designed and delivered a half-day simulation course within a high-fidelity simulation facility for nine CMTs starting their haematology rotation at the University College London Hospital, covering three main emergencies: neutropenic sepsis (NS), acute sickle cell chest crisis (SCC) and acute transfusion reaction (ATR). Faculty included simulation experts, consultants and advanced nurse practitioners. Quantitative and qualitative data were collected on participants' confidence and knowledge of protocols before and after the course for comparison.

A learning needs analysis of prior haematology CMTs ($n = 13$) highlighted that 77% felt unprepared for managing haematological emergencies on-call. On a 5-point Likert scale assessing confidence (1 = not confident, 5 = very confident) managing specific emergencies; NS, SCC and ATR obtained mean scores of 2.54, 1.77 and 1.85 respectively. Following the simulation course ($n = 9$), 100% improved their score on a 10-part multiple-choice questionnaire on the management of haematological emergency protocols administered before and after the course, and 100% improved their confidence in managing the three scenarios by at least 1 point on a 5-point confidence score. All participants strongly agreed the course would be of practical use to their day-to-day role on the haematology wards. Both participants and facilitators provided feedback that they felt the multidisciplinary involvement made the scenarios more realistic and that it was valuable to put 'key knowledge into practice'. Confidence and knowledge scores ($n = 3$) 6 weeks postcourse demonstrated persistent improvement for all three scenarios.

In conclusion, our simulation course improved junior doctor confidence in the management of haematological emergencies and knowledge of local protocols. We suggest that experiential learning through simulation is a time-efficient, effective and enjoyable method

Education

of training junior doctors transitioning into a new or unfamiliar specialty. Further studies are required to corroborate these findings with larger numbers of participants.

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Nursing

BSH2020-OR-014

Quality of life in survivors of adult haematological malignancy

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Introduction: The treatment for haematological malignancies is lengthy, complex and intense, leading to potentially debilitating symptoms and reduced health-related quality of life. Survivors can suffer long-term effects of both treatment and disease. The presentation is part of a mixed methods study and aims to deepen our understanding of the quality of lives of survivors of adult haematological malignancy. The study will contribute to the under-researched topic related to adult survivors of a haematological malignancy.

Objectives: The qualitative phase aims to explore in depth and improve understanding of the quantitative findings related to the quality of life among adults who have survived a haematological malignancy, identifying and addressing any unmet supportive care needs.

Methods: The participants ($n = 11$) were adults who had completed treatment for a haematological malignancy and were between 1 and 5 years post-treatment. A qualitative research process of inquiry was used to explore the quality of lives of survivors of haematological malignancy and identify unmet supportive care needs. In-depth face-to-face semi-structured interviews were conducted, based on the content and structure of the interviews envisaged to be grounded in the findings of the quantitative phase of the study.

Results: The key unmet supportive care needs identified in this cohort was the fear of recurrence, loss of continued connection with healthcare providers and uncertainty about the future.

Conclusion: Survivors of haematological malignancies have unmet supportive care needs. Enhancing their physical, psychological well-being and addressing supportive care needs optimises their QoL. Knowledge and understanding of these factors and addressing these needs may provide an insight into implementation measures to enhance the same.

Disclosure of Interest: None Declared.

BSH2020-OR-015

The haemato-oncology patient experience of the process of palliative care in the last year of life: a constructivist grounded theory study

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Haemato-oncology patients comprise 8.5% of all adult cancers within the UK. Despite advances in treatment, the majority will eventually enter a palliative care phase. Evidence suggests that palliative care arrives too late or not at all for haemato-oncology patients, resulting in suboptimal care and many patients dying in hospital. Various reasons have been cited that affect the integration of palliative care services. However, to date, there have been no studies about the patient's perspective of the process of palliative care in the last year of life.

Using Constructivist grounded theory, 21 semi-structured patient interviews were conducted with a sample from two different NHS trusts. Constant comparison analysis was used to collect and analyse data. The substantive theory of 'Suspended and silent status passage: The incurable haemato-oncology illness trajectory' was generated from the two core categories 'facing death' and 'talking about death'. This substantive theory describes the length of time, which is suspended, that haemato-oncology patients have to face death where, overtime, there is less opportunity to discuss dying and future care needs as they are silenced. This is a new conceptualisation regarding the last year of life.

The findings have serious implications for practice as they reconceptualise the incurable pathway as a 'status passage', with the inevitable outcome being death rather than the possibility of cure. As personalised care is central to the UK's health and social care integration agenda, it is a timely piece of work, illustrating that individuals diagnosed with an incurable haemato-oncology disease should be offered a holistic needs assessment which incorporates elements of palliative care and long-term remission. The findings also illustrate that palliative care service provision should be based upon critical episodes of care in combination with longer term follow-up, allowing more flexibility and a responsive process of palliative care.

Disclosure of Interest: None Declared.

BSH2020-OR-016

The role of a multiple myeloma patient support programme in patient management – interim data, December 2019

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THE HETEROGENEOUS nature of multiple myeloma and the various associated complications result in a patient treatment plan that requires input from a diverse multidisciplinary healthcare team, including clinical nurse specialists (CNSs). Patient access to CNSs has been shown to improve their outcomes and clinical experiences. However, many CNSs cover multiple disciplines throughout haematology services, which can reduce the time they have available for educating patients on their disease and explaining their individualised, often complicated, treatment plans.

An ongoing patient support programme (PSP) was started in September 2017. It was designed and implemented by Takeda UK (through the third-party provider APODI) to improve patient experiences and outcomes by providing those who are receiving ixazomib as a combination schedule with high-quality, personalised support from dedicated CNSs. To date, nine NHS Trusts are utilising PSP specialist nurses (SNs) and 131 patients have enrolled on the programme as of 31 December 2019 (Table).

All PSP SNs are experienced, field-based, haematology nurses working under the direction of the NHS Trust, and the existing haematology service. The PSP SNs maintain regular contact with patients via clinical consultations, phone calls and/or home visits. In addition to patient management, PSP SNs respond to questions during and after treatment, send reminder texts for patients to take their medication, respond to inbound phone calls (61 by year-end 2019),

effectively address, manage and report adverse events (AEs) and facilitate patient education using patient support materials. In the ongoing PSP, the overall patient-reported experience measure (PREM) satisfaction score was 95%. Of the 131 patients enrolled, 81 were active, 37 had been inactive (on hold) and 13 had died. Among the 37 inactive patients, five are currently on hold, 14 have resumed treatment after an 'on-hold episode' and 18 have left the programme after an on-hold episode. Overall, a total of 1,782 patient interactions occurred: most were during treatment Cycles 1–3 (34.2%), Cycles 7–12 (18.7%) or Cycles 13+ (22.9%), and interactions were via clinical consultations (35.1%), phone calls (52.9%) or home visits (11.7%). Analysis of interim subgroup data revealed 'very good' or 'excellent' responses (score 4 or 5) in 74.4%, 84.1% and 84.1% of patients, respectively, to the questions: 'How beneficial have you found support from the PSP SN?', 'How well do you feel the PSP SN listened to you and addressed your concerns?' and 'How well do you feel you understand your treatment with ixazomib?'. A total of 638 AEs were reported, primarily during Cycles 1–3 (27.1%) and Cycles 13+ (20.5%).

To date, the PSP data demonstrate a positive impact on the overall treatment and management of patients who are prescribed ixazomib as a combination schedule. These results suggest that the support offered from PSP SNs is providing suitable patient satisfaction, concordance and adherence, and is integrating well with the host service to ensure continuity of multidisciplinary healthcare within the NHS framework.

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Table. Interim data from the ixazomib PSP (as of 31 December 2019)

NHS Trusts enrolling patients, <i>n</i>	9
Ixazomib-treated patients enrolled, <i>n</i>	131
No. of active patients	81
No. of inactive (on-hold) patients	37
No. of patients currently on hold	5
No. of patients resuming treatment after an on-hold episode	14
No. of patients leaving the programme after an on-hold episode	18
Patient interactions by treatment cycle, <i>n</i>	1782
Cycles 1–3, <i>n</i> (%)	610 (34.2)
Cycles 4–12, <i>n</i> (%)	141 (7.9)
Cycles 4–6, <i>n</i> (%)	276 (15.5)
Cycles 7–12, <i>n</i> (%)	334 (18.7)
Cycles 13+, <i>n</i> (%)	408 (22.9)
Method of patient interaction, <i>n</i> (%)	
Clinic consultation	625 (35.1)
Phone call	943 (52.9)
Home visit	209 (11.7)
Patient experience feedback – patients with a 'very good' or 'excellent' response (score 4 or 5), <i>n</i> (%)	
'How beneficial have you found support from the PSP SN?'	32/43 (74.4)
'How well do you feel the PSP SN listened to you and addressed your concerns?'	37/44 (84.1)
'How well do you feel you understand your treatment with ixazomib?'	37/44 (84.1)
No. of adverse events	638

PSP, patient support programme; SN, specialist nurse.

Education

BSH2020-OR-017

Development of a regional palliative care guideline on the use of anticoagulation in palliative and end of life care

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THE evidence for the use of different anticoagulants in the setting of cancer-associated thrombosis (CAT) is a rapidly evolving field due to a number of CAT-specific clinical trials over recent years. This has led to a change in how we manage this cohort of patients where trials other than for the use of warfarin or low-molecular-weight heparin have previously been lacking. The impact has already been appreciated by many clinical specialties, including palliative care, as there are now more options available to treat patients with CAT. In order for learning from clinical trials to translate to improved patient care, it is critical that the evidence is appropriately and effectively disseminated to healthcare professionals (HCPs) across different specialties.

Pallaborative North West is a palliative care network in the North West of England whose aim is to use evidence-based medicine to improve the quality of care for people with life-limiting illness, using clinical audits, education and training and production of NICE accredited guidelines. A regional guideline for 'The Use of Anticoagulation in Palliative Care' was published in 2009. A guideline development group (GDG) was convened to update this guideline in light of recent evidence.

The GDG consisted of 10 members, including palliative care HCPs, a pharmacist, haematologist, patient and public representative and invited expert. Over 12 months, the GDG met bi-monthly to

coordinate activities leading to the completion of a final updated guideline. This included performing a comprehensive literature review of current evidence, as well as conducting supra-regional audits on anticoagulant use in palliative care. The guideline is currently undergoing internal review and is due for submission for external review before publication as a NICE accredited guideline.

Our literature search identified 833 papers which, following initial review and exclusions, was reduced to 100 for full text review. These were individually assessed and graded by at least two members of the GDG, and resulted in 32 clinical papers for inclusion in the guideline writing process.

The supra-regional audits were twofold: one focusing on HCPs' knowledge and confidence on the use of anticoagulation, and the other a case note review of current practice of anticoagulant prescribing. The audits were carried out using online surveys sent out to all specialist palliative care teams across the Network. There was a good uptake of the questionnaires, with responses from 10 specialist palliative care service group areas, made up of hospital, hospice and community settings.

Both the audit and literature search results were presented at the Pallaborative North West regional meeting, allowing comparison between current practice and the up to date evidence, and facilitating discussion about the proposed guideline standards and recommendations, to ensure practicability.

The development of this updated guideline was a substantial piece of work in an area where evidence is evolving rapidly. By using a NICE accredited process, a comprehensive review of current evidence and clinical practice was achieved. This resulted in a guideline designed to support palliative care professionals in making evidence-based decisions on the use of anticoagulation in palliative and end of life care – an area which is becoming increasingly complex due to the recent rapid evolution of evidence in this field, and the intricacies of clinical decision making.

Disclosure of Interest: None Declared.

Myeloma

BSH2020-OR-018

Four-year follow-up of the phase 3 POLLUX study of daratumumab plus lenalidomide and dexamethasone versus lenalidomide and dexamethasone alone in relapsed or refractory multiple myeloma

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Daratumumab (DARA) is approved in combination with standard-of-care for newly diagnosed multiple myeloma (NDMM) and relapsed or refractory MM (RRMM). In phase 3 studies, DARA-based regimens reduced the risk of progression or death and increased rates of $\geq$ complete response (CR) and minimal residual disease negativity in patients with NDMM and RRMM. In the POLLUX study (median follow-up 44.3 months), DARA plus lenalidomide and dexamethasone (D-Rd) reduced the risk of disease progression or death and significantly increased the overall response rate (ORR) versus Rd in patients with RRMM. Here, we present POLLUX efficacy and safety analyses after median follow-up of >4 years.

Patients with ≥ 1 prior line of therapy (LOT) were randomized to Rd $\pm$ DARA. Patients with high cytogenetic risk had t(4;14), t(14;16), and del17p abnormalities. Progression-free survival (PFS) on subsequent LOT (PFS2) was defined as time from randomization to progression after next LOT or death.

A total of 569 patients were randomized (D-Rd, $n = 286$; Rd, $n = 283$). At median follow-up 54.8 months, D-Rd prolonged PFS vs. Rd [median 45.0 vs. 17.5 months; hazard ratio (HR), 0.44; 95% confidence interval (CI), 0.35–0.54; $P < 0.0001$], and prolonged PFS vs. Rd regardless of prior LOT or cytogenetic risk status. D-Rd was

associated with a higher ORR vs. Rd (93% vs 80%) and higher $\geq$ very good partial response and $\geq$ CR. D-Rd prolonged time to next LOT vs. Rd (median: not reached vs. 22.8 months; HR, 0.38; 95% CI, 0.30–0.48; $P < 0.0001$) and prolonged PFS2 vs. Rd (median 53.3 vs. 31.6 months; HR, 0.53; 95% CI, 0.42–0.66; $P < 0.0001$). A total of 125 deaths were observed with D-Rd vs. 144 with Rd; overall survival (OS) follow-up is ongoing. Median treatment duration was 34.3 months with D-Rd vs. 16.0 months with Rd. The most common ($\geq 10\%$) grade 3/4 treatment-emergent adverse events with D-Rd vs. Rd were neutropenia (57% vs. 42%), anaemia (19% vs. 22%), thrombocytopenia (15% vs. 16%), pneumonia (16% vs. 10%) and diarrhoea (10% vs. 4%). Additional efficacy and safety data will be presented.

After a median follow-up of >4 years, D-Rd continued to show a significant PFS benefit and higher ORRs and deep response rates versus Rd in patients with RRMM. Significant PFS benefit was seen with D-Rd regardless of prior LOT or cytogenetic risk status; greatest benefit was seen in patients treated earlier with D-Rd. Significant PFS2 improvement suggests a potential survival benefit, but OS data are still immature. No new safety concerns were identified.

Study Support: Janssen.

Disclosure of Interest: G. Cook Conflict with: Janssen, Takeda, Sanofi, Karyopharm, Celgene, Conflict with: Celgene, Janssen-Cilag, Takeda, Conflict with: Amgen, Bristol-Myers Squibb, GlycoMimetics, Seattle Genetics, Sanofi, J. L. Kaufman Conflict with: AbbVie, Amgen, Bristol-Myers Squibb, Celgene, Incyte, TG Therapeutics, Takeda, Conflict with: Winship Cancer Institute of Emory University, Conflict with: Janssen, Karyopharm, Pharmacyclics, S. Z. Usmani Conflict with: Amgen, Bristol-Myers Squibb, Celgene, Janssen, Merck, SkylineDX, Takeda, Conflict with: Amgen Array Bio-pharma, Bristol-Myers Squibb, Celgene, Janssen, Merck, Pharmacyclics, Sanofi, Takeda, Conflict with: Amgen, Celgene, Janssen, Sanofi, Takeda, J. San-Miguel Conflict with: Amgen, Bristol-Myers Squibb, Celgene, Janssen, MSD, Novartis, Roche, Sanofi, Takeda, N. J. Bahlis Conflict with: AbbVie, Amgen, Celgene, Janssen, Takeda, D. White Conflict with: Amgen, Celgene, Janssen, Sanofi, Takeda, L. Benboubker: None Declared, M. Leiba: None Declared, P. J. Ho Conflict with: Celgene, Janssen, La Jolla, Novartis, K. Kim: None Declared, N. Takezako: None Declared, P. Moreau Conflict with: AbbVie, Amgen, Celgene, Janssen, Takeda, M. Krevvata Conflict with: Janssen, H. Pei Conflict with: Janssen, J. Ukropec Conflict with: Janssen, T. Renaud Conflict with: Janssen, S. Trivedi Conflict with: Janssen, R. Kobos Conflict with: Janssen, M. A. Dimopoulos Conflict with: Amgen, BMS, Celgene, Janssen, Takeda, Conflict with: Amgen, Genesis Pharma, Janssen, Sanofi Oncology, Takeda, Conflict with: Celgene.

BSH2020-OR-019

Tailoring treatment for patients with newly diagnosed high-risk myeloma – feasibility results of the UKMRA OPTIMUM (MUKnine) trial

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Background: Patients with high-risk (HiR) multiple myeloma (MM) or primary plasma cell leukaemia (PCL) are in urgent need of improved diagnostic and therapeutic strategies. In a recent analysis of over 1000 transplant-eligible newly diagnosed NCRI Myeloma XI trial patients, those with double-hit tumours (≥ 2 high-risk markers t(4;14), t(14;16), t(14;20), gain(1q) or del(17p)) and ISS2/3 had a median PFS of only 20 months (95% CI: 18–24 months) from initial randomisation. Outcome is equally adverse for gene expression profiling (GEP) SKY92-positive MM. We designed the UK OPTIMUM (MUK9) trial to address this unmet need by testing feasibility of central integrated risk profiling and efficacy of intensive immuno-biological therapy for HiR MM.

Trial design: Patients with suspected or newly diagnosed MM and fit for intensive therapy were consented to the OPTIMUMscreen (MUK9A) protocol for central molecular CD138⁺ bone marrow profiling of translocations, copy number aberrations (qRT-PCR; MLPA P425, MRC Holland) and SKY92 high-risk GEP signature (MMprofiler; SkylineDx). Patients found to have non-HiR MM received SOC (VTD, HD-MEL+ASCT), whereas patients with HiR MM by double-hit and/or SKY92 or PCL were offered participation in OPTIMUMtreat (MUK9B). Treatment in OPTIMUMtreat consisted of daratumumab, cyclophosphamide, bortezomib, lenalidomide, dexamethasone (Dara-CVRd) induction (6 cycles), bortezomib-augmented single HD-MEL+ASCT, Dara-VRd consolidation 1 (6 cycles), Dara-VR consolidation 2 (12 cycles) and Dara-R maintenance (until PD). Dose adjustments were permitted in order to maximize tolerability of long-term therapy. Primary end-point for OPTIMUMscreen is feasibility of central molecular testing. Primary end-point for OPTIMUMtreat is treatment efficacy, comparing against a predefined near-concurrent ‘digital comparator’ of matched HiR Myeloma XI patients using a Bayesian design.

Results: From September 2017 to July 2019, 472 patients with suspected MM or PCL were recruited across 39 NHS hospitals to OPTIMUMscreen. Of these, 379 patients were found to have symptomatic MM and 10 had PCL. The remainder were diagnosed with MGUS ($n = 36$), no MM ($n = 16$) or withdrew consent ($n = 31$). Central bone marrow CD138⁺ genetic and GEP risk profiles were successfully generated for 93% of patients. Low bone marrow aspirate quality was the commonest reason for technical failure of test. Median risk status turnaround time was 18 days (IQR 13–22), well below protocol defined maximum limit of 8 weeks. HiR status by genetic double-hit and/or SKY92 was detected in 38% of tumours, mirroring

retrospective results in Myeloma XI. The t(11;14) translocation was present in 50% of PCL patients, consistent with previous retrospective PCL datasets. A total of 108 patients with confirmed HiR MM or PCL were enrolled in OPTIMUMtreat, 19 were deemed ineligible by investigators and 11 withdrew consent. The majority of patients have now completed Dara-CVRd induction successfully and are continuing on trial. A total of 251 non-HiR patients are receiving SOC (VTD/ASCT) and are followed up in OPTIMUMscreen.

Discussion: We demonstrate for the first time feasibility of central genetic+GEP risk assessment in this UK multicentre trial for newly diagnosed MM. Recruitment completed 9 months ahead of projections, demonstrating the urgent need for novel approaches for HiR patients. Results also confirm the use of innovative trial concepts like the ‘digital comparator’ Bayesian design used in OPTIMUM for populations in need.

Disclosure of Interest: M. Jenner Conflict with: Abbvie, Amgen, Celgene, Novartis, Janssen, Sanofi, Takeda, A. Hall Conflict with: Celgene, Janssen, Karyopharm, A. Sherborne: None Declared, K. Walker Conflict with: Celgene, Janssen, Karyopharm, K. Sharp: None Declared, K. Bowles Conflict with: Abbvie, Janssen, M. Garg Conflict with: Sanofi, Janssen, Takeda, Amgen, Novartis, Conflict with: Janssen, Conflict with: Takeda, Janssen, S. Ellis: None Declared, G. Jackson Conflict with: Celgene, GSK, Amgen, Roche, Janssen, Sanofi, L. Flanagan: None Declared, M. Drayson Conflict with: Abingdon Health, Conflict with: Abingdon Health, R. DeTute: None Declared, R. Owen Conflict with: Celgene, Janssen, Conflict with: Celgene, G. Pratt Conflict with: Binding Site, Amgen, Takeda, Janssen, Gilead, G. Cook Conflict with: Janssen, Takeda, Sanofi, Karyopharm, Celgene, BMS, Amgen, Conflict with: Celgene, Janssen, Takeda, S. Brown Conflict with: Celgene, Janssen, Karyopharm, M. Kaiser Conflict with: Abbvie, BMS, Celgene, Takeda, Janssen, GSK Amgen, Abbvie, Karyopharm, Conflict with: Janssen, Celgene.

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Phase 3, open-label, non-inferiority study of subcutaneous *versus* intravenous daratumumab in patients with relapsed or refractory multiple myeloma: body weight subgroup analysis of Columba

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Daratumumab (DARA) is approved for intravenous (IV) administration as a single agent or in combination with standard-of-care regimens for treatment of multiple myeloma (MM). The phase 3 COLUMBA study demonstrated that a subcutaneous (SC) DARA formulation was non-inferior to DARA IV for overall response rate (ORR) and C_{trough} in patients with relapsed or refractory MM (RRMM). Here, we present a subgroup analysis by body weight.

DARA SC (1800 mg flat dose DARA + recombinant human hyaluronidase PH20 [ENHANZE® drug delivery technology, Halozyme, Inc]) and DARA IV (16 mg/kg) were given in 28-day cycles. Coprimary end-points were ORR and maximum DARA C_{trough} (predose concentration on Cycle 3 Day 1). Body weight subgroups were ≤ 65 , >65 –85 and >85 kg.

A total of 522 patients were randomized (DARA SC $n = 263$; DARA IV $n = 259$). Median baseline body weight was 73 kg. DARA SC achieved adequate exposure for all body weight subgroups. There was considerable overlap in maximum C_{trough} for DARA SC and IV across subgroups. However, mean DARA SC *versus* IV concentrations were 60% higher for patients ≤ 65 kg and 12% lower for patients >85 kg. Nevertheless, the range of C_{trough} across body weights for DARA SC was within the previously observed range (36–1764 $\mu\text{g/ml}$) for DARA IV. ORR was 41.1% vs. 37.1% for DARA SC vs. IV (intent-to-treat population) and was 43.6% vs. 38.0%, 37.3% vs. 39.0% and 43.9% vs. 32.8% for DARA SC vs. IV in the ≤ 65 , >65 –85 and >85 kg groups, respectively, suggesting that the slightly lower exposure at higher body weights was not clinically relevant. Incidence of grade 3–5 treatment emergent adverse events (TEAEs), serious TEAEs, and infusion-related reactions was similar across

body weight subgroups for both DARA SC and IV. Incidence of any grade TEAEs increased with decreasing body weight for DARA SC but was similar across body weight subgroups for DARA IV. Incidence of grade 3/4 thrombocytopenia and grade 3/4 neutropenia decreased with increasing body weight for DARA SC. Incidence of grade 3/4 hypertension increased with increasing body weight for DARA IV. In subjects ≤ 65 kg, incidence of maximum grade 3/4/5 TEAEs was similar, and incidence of serious TEAEs was lower with DARA SC vs. IV (23.7% vs. 30.4%).

ORR in all body weight subgroups was consistent with ORR in the overall population; DARA SC achieved adequate exposure and was well tolerated. The higher DARA SC concentration in patients ≤ 65 kg had no clinically relevant effect on safety, suggesting no weight-based dose individualization is needed.

Study Support: Janssen.

Offlabel Disclosure: This presentation/paper includes information/discussion of a subcutaneous formulation of daratumumab, which is currently under investigation in several clinical trials, but has not yet been approved. The intravenous formulation of daratumumab is approved as monotherapy and in combination with standard-of-care regimens for the treatment of MM.

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Pomalidomide, bortezomib and dexamethasone after one prior line of therapy in bortezomib-pretreated multiple myeloma: a subanalysis of the OPTIMISMM phase 3 trial

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In the Phase 3 OPTIMISMM trial (NCT01734928), pomalidomide (POM), bortezomib (BORT) and dexamethasone (DEX; Pvd) significantly improved progression-free survival (PFS) *versus* BORT and DEX (Vd) in relapsed refractory multiple myeloma patients (100% lenalidomide [LEN]-pretreated; 70% LEN-refractory; 1–3 prior lines of therapy). Because LEN and BORT are often used in the frontline treatment of multiple myeloma, we analysed the efficacy and safety of second-line Pvd *versus* Vd after frontline LEN ± BORT.

Eligible patients received (1:1) Pvd or Vd in 21-day cycles: POM 4 mg/day, days 1–14 (Pvd arm only), BORT 1.3 mg/m², days 1, 4, 8 and 11 cycles 1–8 and on days 1 and 8 cycles 9+, and DEX 20 mg/day (10 mg/day if > 75 years), days of and after BORT. The primary end-point was PFS. BORT-refractory patients (defined as refractory to once-weekly ≤1.3 mg/m² BORT) were included. Data cut-off was Oct 26, 2017.

Of the 559 patients enrolled, 226 (111 Pvd; 115 Vd) had 1 prior line of therapy; 59.3% had prior BORT (67 Pvd; 67 Vd) and 40.7% did not have prior BORT (44 Pvd; 48 Vd). In patients with prior BORT, median age was 63 years in the Pvd arm *versus* 64 years in the Vd arm, 16.4% vs. 10.4% were BORT-refractory, and 53.7% vs. 46.3% were LEN-refractory respectively. In patients without prior BORT, median age was 74 vs. 70 years and 63.6% vs. 70.8% were LEN-refractory in the Pvd *versus* Vd arm respectively. Pvd significantly improved PFS (median, 17.8 vs. 12.0 months; HR 0.47 [95% CI, 0.26–0.82]; *P* = 0.0068) vs. Vd in patients with prior BORT. Of note, median PFS values did not change after excluding BORT-refractory patients. In patients without prior BORT, median PFS was

20.7 months with Pvd vs. 9.5 months with Vd [HR 0.62 (95% CI, 0.35–1.11); *P* = 0.1055]. Pvd significantly improved ORR vs. Vd: 89.6% vs. 49.3%, *P* < 0.001 in patients with prior BORT and 90.9% vs. 62.5%, *P* = 0.002 in patients without prior BORT. Rates of ≥ VGPR were 62.7% vs. 16.4% and 59.1% vs. 31.3% respectively.

The safety population was comprised of 131 patients with prior BORT (67 Pvd; 64 Vd) and 90 patients without prior BORT (44 Pvd; 46 Vd). Neutropenia (44.8% vs. 10.9% prior BORT, 22.7% vs. 8.7% no prior BORT) and thrombocytopenia (23.9% vs. 26.6% prior BORT, 13.6% vs. 13.0% no prior BORT) were the most common haematologic grade 3/4 treatment-emergent AEs with Pvd vs. Vd respectively. Grade 3/4 infections (Pvd vs. Vd) occurred in 26.9% vs. 15.6% (pneumonia 7.5% vs. 4.7%) of patients with prior BORT and 31.8% vs. 15.2% (pneumonia 11.4% vs. 6.5%) of patients without prior BORT. The incidence of grade 3/4 peripheral sensory neuropathy (Pvd vs. Vd) was 10.4% vs. 0% in patients with prior BORT and 6.8% vs. 8.7% in patients without prior BORT.

After 1 prior line of therapy, Pvd significantly reduced the risk of progression or death by 53% in patients previously treated with BORT *versus* Vd. Regardless of prior BORT treatment, Pvd also improved ORR and resulted in deeper responses *versus* Vd. In summary, Pvd is an effective second-line treatment in patients previously treated with BORT and LEN. The safety of Pvd was consistent with the known profiles of each agent.

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Ixazomib-dexamethasone versus physician's choice in patients with relapsed/refractory systemic AL amyloidosis: primary data from the phase 3 TOURMALINE-AL1 trial

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Systemic AL amyloidosis is a clonal plasma cell disorder in which amyloid fibrils are deposited in tissues and organs, leading to multi-organ dysfunction. The most frequently involved organs are the heart and kidneys. Achievement of haematological response and improved organ function result in better outcomes. There are no approved treatments for AL amyloidosis; active, tolerable, disease-specific treatments that elicit a deep haematological response and improve organ function are needed. The oral proteasome inhibitor (PI) ixazomib (I) is approved with lenalidomide (L) and dexamethasone (Dex) for the treatment of multiple myeloma patients who have received ≥ 1 prior therapy. In the phase 3 TOURMALINE-AL1 trial, relapsed/refractory AL amyloidosis patients with major organ (cardiac/renal) involvement who required treatment after 1–2 prior therapies (and were not refractory to prior PI therapy) were randomised to receive I (4.0 mg on days 1, 8 and 15) plus Dex (20 mg on days 1, 8, 15 and 22) or physician's choice (Dex alone or plus melphalan [M], cyclophosphamide [C], thalidomide [T] or L). Randomisation was stratified by cardiac risk stage, relapsed versus refractory disease, and prior PI exposure. Treatment was given in 28-day cycles until disease progression or unacceptable toxicity (or best response plus 2 cycles or a maximum of 18 months' therapy/600 mg total dose for MDex). The primary end-points were overall haematological response rate and 2-year vital organ deterioration or death rate; key secondary end-points were overall survival and complete response (CR) rate. Overall, 168 patients were randomised: 85 to IDex and 83 to physician's choice

(47 LDex, 24 MDex, 10 CDex, 2 TDex). In the IDex versus physician's choice arm, median age was 65 (range, 38–84) vs. 66 (33–82) years; 56% vs. 63% of patients had cardiac and 66% vs. 58% had renal involvement (33% vs. 23% had both) at diagnosis. Haematological responses were seen in 53% (IDex) vs. 51% (physician's choice) of patients (odds ratio 1.10, $P = 0.762$). Several secondary end-points were also assessed. Higher CR rates were seen with IDex (26%) vs. physician's choice (18%). Time-to-event data favoured patients treated with IDex over physician's choice, including time to vital organ deterioration or death (34.8 vs. 26.1 months; Table). The vital organ response rate was 36% with IDex vs. 11% with physician's choice (cardiac / renal response rates: 18%/28% vs. 5%/7%). At data cut-off, median treatment duration was 11.7 (IDex) vs. 5.0 months (physician's choice); 21% vs. 6% of patients remained on treatment. For IDex versus physician's choice, grade ≥ 3 adverse event (AE) rates were 62% vs. 56%, 34% vs. 41% of patients had drug-related grade ≥ 3 AEs, 26% vs. 20% of patients had AEs resulting in discontinuation, and there were 6% vs. 5% on-study deaths. Common grade ≥ 3 AEs were fatigue (9% vs. 9%), anaemia (2% vs. 11%), cardiac failure and dyspnoea (each 6% vs. 4%), and peripheral oedema and pneumonia (each 5% vs. 5%). In summary, IDex resulted in improved CR and vital organ response rates and, although the first of two primary end-points was not met, all clinically relevant time-to-event end-point data favoured IDex over physician's choice. IDex was generally well tolerated and associated with a doubling of the treatment duration compared with physician's choice. TOURMALINE-AL1 is the first phase 3 trial in relapsed/refractory AL amyloidosis, demonstrating significant outcome improvements for IDex over physician's choice.

Table.

Median, months (unless stated)	IDex (n = 85)	Physician's choice (n = 83)	Hazard ratio; P-value* (unless stated)
Time to vital organ deterioration or death [†]	34.8	26.1	0.525; $P = 0.012$
2-year vital organ deterioration or death rate [‡] , %	40	45	Odds ratio, 0.83; $P = 0.608$
Overall survival	Not estimable	40.8	0.837; $P = 0.478$
Time to subsequent therapy	26.5	12.5	0.615; $P = 0.027$
Composite (haematological and vital organ) progression-free survival [†]	11.2	7.4	0.673; $P = 0.043$
Haematological progression-free survival [†]	20.1	16.7	0.790; $P = 0.283$
Vital organ progression-free survival [†]	18.0	11.0	0.615; $P = 0.019$
Time to treatment failure [†]	10.1	5.2	0.604; $P = 0.005$
Duration of haematological response	46.5	20.2	0.546; $P = 0.115$

*A hazard ratio or odds ratio of <1 indicates an effect favouring IDex.

†Per physician assessment.

‡Per adjudication committee. Disclosure of Interest: A. D. Wechalekar Conflict with: Amgen, Conflict with: Janssen-Cilag, Takeda, GSK, Celgene, A. Dispenzieri Conflict with: Akcea, Intellia, Janssen, Conflict with: Pfizer, Takeda, Celgene, Alnylam, E. Kastritis Conflict with: Amgen, Genesis Pharma, Janssen, Takeda, Pfizer, Prothena, Research Support from Amgen, Janssen, Conflict with: Amgen, Genesis Pharma, Janssen, Takeda, Pfizer, Prothena, S. O. Schönland Conflict with: Janssen, Prothena, Conflict with: Janssen, Prothena, Takeda, Medac, K. Kim: None Declared, V. Santhorawala Conflict with: Caelum, Proclara, Conflict with: Prothena, Celgene, Takeda, Janssen, H. J. Landau Conflict with: Karyopharm, Conflict with: Takeda, Amgen, Conflict with: Janssen, Celgene, Takeda, Caelum, Prothena, Pfizer, F. Kwok: None Declared, K. Suzuki Conflict with:

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Lymphoma/CLL

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ELEVATE-TN: a phase 3, multicentre, open-label study of acalabrutinib (Ab) combined with obinutuzumab (O) or Ab alone versus O plus chlorambucil (Clb) in patients (pts) with treatment-naïve chronic lymphocytic leukaemia (TN-CLL)

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Background: Ab is a highly selective, covalent irreversible Bruton tyrosine kinase inhibitor with minimal activity against other kinases. Ab alone or in combination with O has demonstrated durable responses in TN-CLL. Here, we present the interim results of the Phase 3 ELEVATE-TN trial (NCT02475681).

Methods: Eligible patients had TN-CLL requiring treatment per iwCLL criteria and were aged ≥ 65 years or < 65 years with co-existing conditions (CIRS score > 6 , creatinine clearance < 70 ml/min). Patients (pts) were randomised 1:1 to receive oral Ab (100 mg twice daily continuously) alone or combined with intravenous O (1000 mg on Days 1, 2 [split 100/900], 8, and 15 of Cycle 2, and Day 1 of subsequent 28-day cycles for a total of 6 cycles), or O plus oral Clb (0.5 mg/kg on Days 1 and 15 of each 28-day cycle for 6 cycles). Pts were stratified by del(17p) status, ECOG status (≤ 1 vs. 2) and geographical region. The primary end-point was independent

review committee (IRC)-assessed progression-free survival (PFS) with Ab+O versus O+Clb. Key secondary end-points included IRC-assessed PFS with Ab versus O+Clb, IRC-assessed overall response rate (ORR), overall survival (OS), and safety. Minimal residual disease (MRD) in peripheral blood or bone marrow was assessed in pts with investigator-assessed complete response (CR)/CR with incomplete marrow recovery. Crossover to Ab monotherapy was allowed in O+Clb arm with IRC-confirmed progression.

Results: A total of 535 pts were randomised to Ab+O ($n = 179$), Ab ($n = 179$) or O+Clb ($n = 177$) arms. The median age was 70 years (range 41–91 years); 69% had high- and 12% had very high-risk CLL IPI scores.

At a median follow-up of 28 months, Ab+O significantly prolonged PFS versus O+Clb (median not reached [NR] vs. 22.6 months; HR 0.10, 95% CI 0.06–0.18, $P < 0.0001$). Ab also prolonged PFS (median NR) versus O+Clb (HR 0.20, 95% CI 0.13–0.31, $P < 0.0001$). Estimated 30-month PFS rates with Ab+O, Ab, and O+Clb were 90%, 82% and 34% respectively. PFS improvement with Ab+O or Ab versus O+Clb was consistent across subgroups including del(17p) (HR 0.13, 95% CI 0.04–0.46; HR 0.20, 95% CI 0.06–0.64). Median OS was not reached in any arms while the estimated 30-month OS rates were 95%, 94% and 90% for Ab+O, Ab and O+Clb arms respectively. Five pts (3%) in the Ab+O arm, 11 pts (6%) in the Ab arm and 55 pts (31%) in the O+Clb arm had received subsequent therapy; 45 pts (25%) in the O+Clb arm crossed over to the Ab monotherapy arm.

IRC-assessed ORR was higher with Ab+O (94%; 95% CI 89.3–96.5%) vs. O+Clb (79%; 95% CI 71.9–83.9%; $P < 0.0001$); the ORR with Ab monotherapy was 85%. CR rates were higher with Ab+O (13%) vs. O+Clb (5%); there was 1 CR in the acalabrutinib monotherapy arm. MRD data will be presented. The median treatment duration was 27.7 months for Ab+O (range 2.3–40.3) and Ab (range 0.3–40.2) and 5.6 months (range 0.9–7.4) for O+Clb. AEs were similar between the Ab-containing arms. Infusion reactions were less frequent with Ab+O (13%) than with O+Clb (40%). AEs led to treatment discontinuation in 20 pts (11%) on Ab+O, 16 pts (9%) on Ab and 25 pts (14%) on O+Clb. AEs of interest (Ab+O or Ab versus O+Clb) were atrial fibrillation (any grade: 3% or 4% vs. 1%), bleeding (any grade/grade ≥ 3 : 43%/2% or 39%/2% vs. 12%/0%) and hypertension (Grade ≥ 3 : 3% or 2% vs. 3%).

Conclusions: Ab+O and Ab monotherapy significantly improved PFS versus O+Clb, with tolerable safety in pts with TN-CLL. Despite cross over for disease progression in the O+Clb arm, a trend towards improved OS was observed in both Ab arms although longer follow-up is needed.

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Table.

	Acalabrutinib + obinutuzumab (<i>n</i> = 178)		Acalabrutinib (<i>n</i> = 179)		Obinutuzumab + chlorambucil (<i>n</i> = 169)	
	Any Grade	Grade ≥ 3	Any grade	Grade ≥ 3	Any grade	Grade ≥ 3
Any, <i>n</i> (%)	171 (96)	125 (70)	170 (95)	89 (50)	167 (99)	118 (70)
Serious, <i>n</i> (%)	69 (39)	58 (33)	57 (32)	53 (30)	37 (22)	33 (20)
Common AEs, <i>n</i> (%)						
Headache	71 (40)	2 (1)	66 (37)	2 (1)	20 (12)	0
Diarrhoea	69 (39)	8 (4)	62 (35)	1 (1)	36 (21)	3 (2)
Neutropenia	56 (31)	53 (30)	19 (11)	17 (9)	76 (45)	70 (41)
Nausea	36 (20)	0	40 (22)	0	53 (31)	0
Infusion-related reaction	24 (13)	4 (2)	0	0	67 (40)	9 (5)
Thrombocytopenia	23 (13)	15 (8)	13 (7)	5 (3)	24 (14)	20 (12)
Anaemia	21 (12)	10 (6)	25 (14)	12 (7)	20 (12)	12 (7)
Pneumonia	19 (11)	10 (6)	13 (7)	4 (2)	5 (3)	3 (2)
Tumour lysis syndrome ^a	3 (2)	2 (1)	0	0	15 (9)	13 (8)
Febrile neutropenia	3 (2)	3 (2)	2 (1)	2 (1)	9 (5)	9 (5)

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KTE-X19, an anti-CD19 chimeric antigen receptor T-cell therapy, in patients with relapsed/refractory mantle cell lymphoma: results from phase 2 of ZUMA-2

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Outcomes with salvage regimens in patients with mantle cell lymphoma (MCL) who progress after Bruton tyrosine kinase inhibitor (BTKi) therapy are poor, with a median overall survival (OS) of 5-8 months (Martin, et al. *Blood*. 2016) and few patients proceeding to allogeneic stem cell transplant. Here, we present interim efficacy and safety results from ZUMA-2, the first Phase 2, registrational, multicentre study evaluating KTE-X19 autologous anti-CD19 chimeric antigen receptor (CAR) T-cell therapy in patients with relapsed/refractory (R/R) MCL.

Eligible patients (≥ 18 years) with R/R MCL, an ECOG of 0-1, and ≤ 5 prior therapies (including chemotherapy, an anti-CD20 antibody, and a BTKi) underwent leukapheresis and conditioning chemotherapy followed by KTE-X19 infusion at 2×10^6 cells/kg. Bridging chemotherapy was permitted. The primary end-point was objective response rate (ORR [complete response (CR) + partial response]), assessed by an Independent Review Committee per Lugano Classification (Cheson, et al. *J Clin Oncol*. 2014). Interim efficacy end-points were investigator-assessed using the revised International Working Group Response Criteria for Malignant Lymphoma (Cheson, et al. *J Clin Oncol*. 2007). Key secondary end-points included duration of response (DOR), progression-free survival (PFS), OS, frequency of adverse events (AEs), and blood levels of CAR T cells. Sixty patients received KTE-X19; here, we present results in patients with ≥ 1 year of follow-up. Updated results for all 60 patients will be reported in the presentation.

As of 30 May 2018, 28 patients received KTE-X19 with ≥ 1 year of follow-up (median, 13.2 months). The median age was 65 years; 43% of patients had an ECOG score of 1; 21% had blastoid morphology; 82% had stage IV disease; 50% had intermediate/high-risk MIPI; 86% received a median of four prior therapies; and 57% were refractory to last prior therapy. Eight patients received bridging therapy; all had disease present postbridging. Investigator-assessed ORR was 86% (95%

CI, 67-96%), with a CR rate of 57% (95% CI, 37-76%); 75% of responders remained in response, and 64% of treated patients had ongoing responses. The 12-month estimates of DOR, PFS, and OS were 83% (95% CI, 60-93%), 71% (95% CI, 50-84%) and 86% (95% CI, 66-94%) respectively; medians were not reached. The most common Grade ≥ 3 AEs were anaemia (54%), platelet count decrease (39%) and neutropenia (36%). Grade 3/4 cytokine release syndrome (CRS) assessed by Lee et al (*Blood*. 2014) and Grade 3/4 neurological events (NEs) were reported in 18% and 46% of patients, respectively (no Grade 5 CRS or NEs). All CRS events and most NEs (15/17 patients) were reversible. Median time to onset and resolution of CRS was 2 days (range, 1-7) and 13 days (range, 4-60) respectively. Median time to onset of NE was 6 days (range, 1-15), and median time to resolution was 20 days (range, 9-99). There was 1 Grade 5 AE of organizing pneumonia, considered related to conditioning chemotherapy. Median CAR T cell levels as measured by peak and area under the curve were 99 cells/ μ l (range, 0.4-2589) and 1542 cells/ μ l (range, 5.5-27 239) respectively. Peak CAR T cell expansion was observed between Days 8 and 15 and levels declined over time.

With ≥ 1 year of follow-up, KTE-X19 demonstrated significant and durable clinical benefit and a manageable safety profile in patients with R/R MCL for whom there are no curative treatment options.

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Five years on: a real-world observational study of patients receiving ibrutinib for relapsed/refractory MCL from the UK and Ireland

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Ibrutinib, a Brutons tyrosine kinase (BTK) inhibitor has transformed the management of relapsed and refractory (r/r) mantle cell

lymphoma (MCL) in recent years. Pooled survival data from trials show median progression free and overall survival (PFS & OS) of 12.5 and 26.7 months respectively with good tolerability. Long-term data to compare real-world with trial outcomes are lacking until now. We present the first ever 5 years, real-world observational study of survival and tolerability in patients with r/r MCL from 37 centres across the UK and Ireland receiving ibrutinib in the Named Patient Programme (NPP) from November 2014 to December 2015.

Patients with r/r MCL who met NPP criteria were identified from the Janssen-Cilag NPP database. Treating physicians were asked to complete standardised baseline and follow up questionnaires at 6 months and annual intervals until November 2019. Sixty-five patients were followed for 5 years from November 2014 until November 2019. On commencing ibrutinib, the median age was 67 years; 76% of patients were male; WHO performance status was: 0(21%); 1(45%); 2(24%); >2(10%). Forty-one per cent had a raised LDH and 18% had evidence of blastoid disease. The median number of prior treatment lines was 2, 70% had 2 or more prior lines and 26% >3 lines. About 65% had previously received an autologous stem-cell transplant (ASCT). The median time from end of previous therapy to initiation of ibrutinib was 6 months.

With 5 years of follow-up, the median time on ibrutinib was 10 months and 80% of patients had discontinued: progressive disease (69%); infection (8%); allogeneic SCT (8%); fatigue (4%); rash (2%); cardiac failure/atrial fibrillation (2%); other (8%). No fatal adverse events were reported. Dose reductions were applied in 11%. The median PFS and OS from point of commencing ibrutinib were 12 and 18.5 months respectively. The number of prior lines of therapy had a statistically significant effect on survival (OS for 1, 2 and >2 prior lines was 46, 24 and 9.5 months (P 0.005); as did prior ASCT vs. no ASCT [54 vs. 11 months (P 0.002)]. More durable responses were observed in patients with longer periods off treatment since prior therapy. Of the 52 patients who discontinued ibrutinib: 56% received supportive care (median OS 6 months); 8% – low-intensity chemo/radiotherapy (median OS 24 months); 12% – rituximab + bendamustine + cytarabine (median OS 24 months); 14% other novel agent (lenalidomide or venetoclax) (median OS 16 months); 10% allogeneic stem cell transplantation (median OS not reached).

To our knowledge, this is the first ever 5-year study of ibrutinib use for R/R MCL outside of a clinical trial setting. It demonstrates tolerability, duration of response and survival rates comparable with those observed in clinical trials, and 20% of patients remained on therapy at 5 years. The number of prior lines and duration of response to prior therapy are important factors in predicting survival. Responses to subsequent immuno-chemotherapy were encouraging, with durable responses observed in a small number who underwent consolidation with allogeneic stem-cell transplantation. **Disclosure of Interest:** D. Tucker Conflict with: Honorarium to attend educational events from Amgen and Takeda; advisory board member for Novartis, N. Morley Conflict with: ABBVIE – Conference Support and Speaker fees; Janssen – Speaker fees, Conflict with: Takeda – conference support, E. Vandenburghe: None Declared, J. Lambert: None Declared, E. S. Green: None Declared, C. Hall: None Declared, M. Macheta: None Declared, J. Wright: None Declared, P. MacLean: None Declared, S. Bowcock: None Declared, S. Booth: None Declared, S. Fernandes: None Declared, S. Iyengar: None Declared, N. Shah: None Declared, R. McCulloch: None Declared, L. Parisi Conflict with: Janssen-Cilag, S. Rule Conflict with: Advisory Board Member and Honorarium for Research from Janssen Cilag.

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A 10-year analysis of outcomes for advanced-stage Hodgkin lymphoma comparing extended follow-up of the multinational RATHL trial with real-world patients treated with ABVD or escalated BEACOPP in the East of England

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ABVD and escalated-BEACOPP (eBEACOPP) remain international standards of care for the treatment of advanced-stage Hodgkin lymphoma (HL). Although no prospective trials directly compare these regimens, it is widely accepted that eBEACOPP provides more effective disease control, at the expense of increased toxicity.

In 2017, at the 59th American Society of Hematology Annual Meeting, we presented a 10-year retrospective multicentre analysis of patients, aged 16–59, treated for advanced-stage HL in the Anglia Cancer Network (so-called ‘real-world’), and compared outcomes with age and international prognostic score (IPS)-matched patients in the multinational Response Adjusted Therapy for Hodgkin Lymphoma (RATHL) trial, in collaboration with the trial investigators.

We found comparable outcomes for first-line patients treated in both the real-world and trial setting, despite interim PET/CT-guided dose-escalation being restricted to trial patients, and demonstrated a first-remission progression-free survival (PFS) advantage for real-world patients treated upfront with eBEACOPP compared with ABVD, in both the real-world and RATHL trial. Subgroup analysis showed that this PFS advantage was preserved in real-world and RATHL IPS 3+ patients, and that real-world IPS 3+ eBEACOPP patients had an overall survival (OS) advantage compared with real-world IPS 3+ ABVD patients, but not compared with IPS 3+ RATHL patients.

We present an updated comparative analysis of real-world and RATHL patients, incorporating a further 2-year follow-up data for real-world eBEACOPP and RATHL patients. From 2004–2014, 221 real-world patients (aged 16–59) were treated for advanced-stage HL; 177 received upfront ABVD, and 44 eBEACOPP; with median follow-up of 87 and 88 months respectively. Whole cohort 5 years PFS was 82.5% and OS 93.9%. In RATHL, 1088 patients (aged 18–59) received upfront ABVD, with 5 years PFS 81.4% and OS 95.2%, and median follow-up 69.5 months.

Real-world ABVD patients retain highly similar 5 years PFS and OS rates compared with RATHL; PFS 79.2% vs. 81.4%, HR: 1.05 (0.73–1.50), $P = 0.79$; OS 92.9% vs. 95.2, $P = 0.71$. Despite there being proportionately more real-world IPS 3+ patients treated with eBEACOPP than ABVD ($P < 0.0005$), 5 years PFS remains superior for eBEACOPP (95.5% vs. 79.2%; HR 0.20 (0.05–0.84); $P = 0.027$), with similar OS (97.7% vs. 92.9%; HR 0.53 [0.12–2.33], $P = 0.40$).

Subgroup analysis of IPS 3+ patients, showed that real-world eBEACOPP patients maintain a 5 years PFS and OS advantage

compared with real-world ABVD patients (PFS 97.0% vs. 71.9%; HR 0.09 (0.01–0.70), $P = 0.021$; OS 100% vs. 84.5%, $P = 0.045$), and a 5 years PFS advantage compared with RATHL patients (97.0% vs. 76.3%; HR 0.10 [0.01–0.74], $P = 0.05$), but no OS advantage ($P = 0.16$). Despite similar relapse/refractory rates, our previous analysis found that real-world IPS 3+ ABVD patients had inferior OS compared with IPS 3+ RATHL patients, which might reflect an ‘era effect’ in salvage therapy options, as most real-world deaths from progressive disease occurred before brentuximab vedotin was available in the UK. However, with further follow-up of RATHL, this OS difference has been lost ($P = 0.17$).

Our updated analysis demonstrates a durable first-remission PFS advantage for advanced-stage HL patients treated upfront with eBEACOPP compared with ABVD, in both the real-world and trial setting, and suggests that higher-risk patients may benefit disproportionately from more intensive first-line therapy with eBEACOPP.

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Real-world data of high-grade lymphoma patients treated with CD19 CAR-T in the UK

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Background: England was the first European country to commission a national service for the delivery of CD19 CAR-T in high-grade lymphoma. Since December 2018, relapsed/refractory (r/r) high-grade lymphoma patients who failed two or more lines of treatment have had access to CD19 CAR-T through the NHSE Cancer Drug Fund (CDF). Treatment is delivered at seven commissioned centres in England (QEHBM Birmingham, University Hospital Bristol, KCH London, UCLH, The Christie Manchester, Manchester MRI, Freeman Hospital Newcastle). In addition, CAR-T centres in Glasgow and Cardiff have recently opened, alongside a *second wave* of centres in England (Cambridge, Leeds, RMH London), allowing more patients access to this specialised treatment in their local area. Treatment is approved through a weekly National CAR-T Clinical Panel (NCCP). Here, we report results of the first 125 lymphoma patients approved by the panel between December 2018 and July 2019.

Methods: All patients with r/r high-grade lymphoma referred to the NCCP between December 2018 and July 2019 and deemed eligible for treatment with CD19 CAR-T were analysed. Eligibility was assessed according to CDF criteria and confirmed at the centre's CAR-T multidisciplinary team meeting. The final decision on patient eligibility was made by consensus through the NCCP independent panel. The choice of CAR-T product (axicabtagene ciloleucel (axi-cel) or tisagenlecleucel (tisagen)) was at the discretion of the treating centre.

Results: A total of 133 cases were submitted to the NCCP of which 125 were approved for treatment with CD19 CAR-T. Seventy-nine patients were selected for axi-cel and 46 for tisagen. In total, 116 patients underwent leukapheresis and 91 proceeded to infusion. In the intention to treat (ITT) population, patients' median age was 62 years (range 18–75), 26% were 65 years or older. 89 (71%) patients had *de novo* diffuse large B-cell lymphoma, 23 (18%) transformed follicular lymphoma, six (5%) transformed marginal zone lymphoma and seven (6%) primary mediastinal B-cell lymphoma. About 76% of cases were advanced stage, 33% had bulky disease and 58% extranodal involvement. About 53/125 (42%) of patients had received three or more prior lines of treatment, 20 patients had previous autologous transplant, five previous allograft. 72% of patients had stable or progressive disease as best response to the last line of treatment.

About 83% patients received bridging therapy between the time of NCCP approval and CAR-T infusion (57% chemotherapy, 16% steroids, 10% radiotherapy). Of the 91 patients infused, 11% experienced grade 3+ cytokine release syndrome, 13% grade 3+ neurotoxicity and 34% of patients required intensive care support. Among 80 evaluable patients, the overall response rate at 3 months was 35% (20% complete responses). About 62.5% of patients showed early progression by 3 months. Two patients died from treatment-related toxicities. With a median follow-up of 4.8 months, the median event-free survival was 3.1 months, the median overall survival (OS) has not been reached. Median OS of the ITT cohort was 9.1 months. Updated efficacy analyses will be presented at the meeting.

Conclusion: NHSE has successfully implemented a national structure for providing broad and rapid access to licensed CD19 CAR-T products. This unique set-up allows capture of prospective real-world data of CAR-T treated patients in the UK, which will help the understanding of the clinical benefit and health economic implications of this complex treatment in daily practice.

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Sequential MATRIX-RICE therapy followed by autologous stem cell transplant in patients with diffuse large B-cell lymphoma and secondary central nervous system involvement: the International Extranodal Lymphoma Study Group-42 Phase II (MARIETTA) Trial

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Introduction: Secondary CNS dissemination (SCNSL) is a rare but lethal event in patients (pts) with diffuse large B-cell lymphoma (DLBCL). It can occur both at presentation, in association with systemic disease, or at relapse, during or after primary therapy. Following the experience from primary CNS lymphoma, pts with SCNSL are currently treated with high-dose methotrexate-based chemo and autologous transplant (ASCT). However, this strategy is associated with poor control of extra-CNS disease, and only 1 of 3 of pts proceed to ASCT. Thus, we designed a multicentre phase II trial addressing safety and efficacy of an intensified chemioimmunotherapy consolidated by ASCT in HIV-neg pts with SCNSL (MARIETTA; NCT02329080).

Methods: Inclusion criteria were as follows: histologically confirmed DLBCL; CNS involvement at presentation (concomitant to systemic disease) or relapse (isolated or concomitant to systemic lymphoma); age 18–70 years; ECOG-PS ≤3; no prior treatment with high-dose methotrexate. Registered pts received three courses of MATRIX

followed by three courses of RICE combined with intrathecal chemo and consolidated by BCNU-thiotepa/ASCT. The primary end-point was 1-year PFS. The Fleming design was used; to detect a difference in 1-year PFS from 50% (P0) to 65% (P1), 69 pts were required (one-sided test, type I error 5%, power 80%), with a dropout of 10%, 76 pts were needed.

Results: Between 3/2015 and 8/2018, 79 pts were enrolled at 24 centres in four countries; 75 pts (median age 58, range 23–70; 43% >60 years) were assessable. Patient characteristics; Table 1.

Out of 450 planned chemo courses, 320 (71%) were delivered. Seventy-eight SAEs were recorded in 42 pts, mostly due to FN and infections (64) or bleeding (5); 74 (95%) SAEs were followed by recovery. Dose reductions were indicated in 33 (10%) courses. Most common g4 toxicities were thrombocytopenia in 118 (37%), neutropenia in 113 (35%) courses.

Forty-nine pts (65%; 95% CI 54–76%) achieved a response after MATRIX-RICE induction, and 36 responders received ASCT; 13 responders did not receive ASCT due to insufficient mobilization ($n = 4$), PD due to treatment delay (5), frailty (2), neurological decline (1) and consent withdrawal (1). Forty-five pts (60%; 95% CI 50–70%) had responsive disease after the whole treatment.

At 1 year from registration, 41 pts were progression free (efficacy threshold ≥ 41). At a median follow-up of 25 (12–47) months, 31 pts are progression free, with a 2-year PFS of $42 \pm 6\%$ for the whole series and $75 \pm 7\%$ for the 36 transplanted pts (Fig A, B). Sites of relapse/progression were CNS in 10 pts, extra-CNS organs in 9 and both in 18. Overall, 33 pts are alive, with a 2-year OS of $42 \pm 6\%$ for the whole series and $82 \pm 7\%$ for transplanted pts. Causes of death were lymphoma (35) and toxicity (4); three pts died without evidence of disease due to neurological decline, PTE and sudden death.

Pts with CNS disease at presentation had the best outcome (Fig C), whereas CSF/meningeal disease (Fig D) and age >60 years were independently associated with poor outcome.

Conclusions: MATRIX-RICE followed by ASCT achieved the primary end-point in this very-poor-prognosis population, without major safety concerns. Survival figures of transplanted pts seems a little better than reported in prior trials, whereas pts with MATRIX-refractory disease had no benefit from crossing to RICE. The best survival figures were recorded in chemo-naïve pts treated at presentation and in pts without CSF/meningeal disease.

Table. Patient characteristics

	N (%)
Disease site at registration	
CNS involvement at presentation	32 (43%)
Isolated CNS relapse	15 (20%)
Concomitant CNS/systemic relapse	28 (37%)
CNS sites of disease	
Brain parenchyma	44 (50%)
CSF/meninges	8 (11%)
Both: parenchyma/CSF	19 (25%)
Intraocular lymphoma	18 (24%)
Extra-CNS sites at registration	
Lymphadenopathy	42 (56%)
Extranodal disease (other than CNS)	36 (48%)
Prior treatment	43 (57%)
	(R-CHOP 93%)

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Red Cell Disorders

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Resolution of sickle cell disease manifestations in patients treated with LentiGlobin gene therapy: updated results from the phase 1/2 HGB-206 Group C study

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LentiGlobin for sickle cell disease (SCD) gene therapy contains autologous CD34⁺ haematopoietic stem cells (HSCs) transduced with the BB305 lentiviral vector (LVV), encoding a human β -globin gene with the anti-sickling T87Q substitution (β^{A-T87Q}). The safety and efficacy of LentiGlobin for SCD is being evaluated in the ongoing phase 1/2 HGB-206 study (NCT02140554). The initial seven patients (Group A) were treated with LentiGlobin made from bone marrow harvested HSCs. The protocol was modified to improve HbA^{T87Q} production by preharvest red blood cell (RBC) transfusions, higher total busulfan exposure, and use of a refined LentiGlobin manufacturing process (Group B, $n = 2$). Here, we discuss data on Group C patients for whom HSCs collection by aphaeresis following plerixafor mobilisation was instituted.

Adults with SCD and complications that included recurrent severe vaso-occlusive crises (VOCs) and acute chest syndrome (ACS) were screened. Eligible patients received 240 μ g/kg plerixafor 4–6 h before aphaeresis. Collected CD34⁺ cells were transduced with BB305 LVV to make LentiGlobin drug product (DP). Patients then underwent myeloablative busulfan conditioning and were subsequently infused with LentiGlobin. Adverse events (AEs), laboratory parameters and SCD clinical manifestations were monitored. Data are shown as median (min–max).

As of 7 March 2019, 13 Group C patients were treated and followed for 9.0 (1.0–15.2) months postinfusion. All but one patient, who was not yet evaluable for engraftment as of the data cut date, achieved neutrophil and platelet engraftment at 19 (15–24) days and 28 (19–136) days respectively. All patients stopped red blood cell (RBC) transfusions within $\sim$ 3 months postinfusion. In patients with $\geq$ 6 months of follow-up ($n = 8$), median HbS contribution was $\leq$ 50% of total Hb. The total unsupported Hb in these eight patients at last visit was 11.5 (10.2–15.0) g/dl, with HbA^{T87Q} levels of 5.3 (4.5–8.8) g/dl. On average, $\geq$ 70% of RBCs from patients treated with LentiGlobin contain β^{A-T87Q} by month 9 post-treatment. Six of the eight patients with $\geq$ 6 months follow-up had a history of VOCs or

ACS; their annualised VOC+ACS rate decreased from 5.3 (3–14) pre-treatment to 0 (0–2) post-treatment. One Grade 2 VOC was observed at 35 months postinfusion, but no ACS or serious VOCs occurred post-treatment. At last visit, lactate dehydrogenase, reticulocyte count and total bilirubin levels were 225.0 (130.0–337.0) U/l, 150.0 (42.1–283.0) $\times 10^9/l$ and 22.2 (3.42–39.3) μ mol/l, respectively, and were lower than at baseline. The most common non-haematological Grade $\geq$ 3 AEs postinfusion were febrile neutropenia ($n = 10$) and stomatitis ($n = 7$). Serious AEs were reported in six patients postinfusion, most common being nausea and vomiting. To date, there have been no DP-related AEs or graft failure, vector-mediated replication-competent lentivirus or clonal dominance. Longer follow-up and additional patient data will be presented.

The safety profile of LentiGlobin for SCD remains consistent with that of single-agent busulfan conditioning and underlying disease. Patients in Group C experienced high-level, sustained, nearly pancellular expression of anti-sickling HbA^{T87Q}, with median HbS reduced to $\sim$ 50% of total Hb and median total Hb levels of 11.5 g/dl at 6 months. The cessation of clinical complications (no ACS or serious VOCs) and decrease in haemolysis suggest a strong therapeutic effect after LentiGlobin gene therapy in patients with SCD.

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Intravenous morphine achieved a more sustained pain response during acute pain crisis in adult sickle cell disease patients compared to intramuscular pethidineM. Opoku- Agyakwa^{1,*}, H. Lawson², C. Brown³, E. Olayemi⁴¹Family Medicine/Polyclinic, Korle Bu Teaching hospital,²Community Health (Family Medicine Unit), University of Ghana, School of Public Health, ³Medical laboratory Sciences, School of Allied Health Sciences, ⁴Haematology, University of Ghana Medical School, Accra, Ghana

Sickle cell disease (SCD) is a common chronic debilitating illness in sub-Saharan Africa. Acute episodes of pain, the most common reason for admission are often difficult to treat.

Opioids are widely accepted as the treatment of choice for acute vaso-occlusive crisis (VOC). However, there are variations in the type of opioid used, route and mode of administration.

In Ghana, the choice of opioid (usually pethidine and morphine) depends largely on the attending physician. Some physicians are wary of prescribing pethidine due to its more pronounced effect on the central nervous and respiratory systems; thus, when a physician's choice is unavailable, pain control is often delayed or inadequate.

The aim of this study was to compare parenteral pethidine and morphine in the elimination or reduction of the acute pain of VOC to tolerable levels in adult SCD patients.

In an open-label randomized study we compared intravenous (IV) morphine 5 mg 4 hourly to intramuscular (IM) pethidine 75 mg 4 hourly. We recruited 82 SCD participants from Korle Bu Polyclinic in Accra. All consenting adult SCD patients (≥18 years) who presented to the facility in VOC, with a minimum score of 7 on the McCaffery Pain Scale during the study period were recruited.

Pain was assessed using the McCaffery Pain Scale, a urine analysis was done for proteinuria (patients with proteinuria levels ≥100 mg/dl were excluded because of significant renal impairment). A data abstraction form was used to obtain demographics, haemoglobin phenotype, past medical and drug history. Participants were randomized into either the pethidine or morphine arm of the study; with 41 participants in each arm. Adequate pain control was defined as pain scores of 0–3 maintained for 24 h. Sustained pain response was defined as the time taken for a decrease in the pain score of ≥2 points from the initial score without any subsequent increase in pain score. The study was approved by Korle Bu Teaching Hospital Institutional Review Board (KBTH-IRB/004/2015), and registered with the Pan African Clinical Trial Registry database (PACTR201812917283565). $P < 0.05$ was considered significant.

There were 42 males and 40 females. Median age was 25 years (range 18–47 years); 67(81.7%) participants had HbSS phenotype, whereas 15(18.3%) were HbSC. In all, 32.5% (13/39) participants in the pethidine arm and 52.4% (22/41) in the morphine arm had a sustained response within 6 h, $P = 0.027$. In the pethidine arm, 61.5% (24/39) participants and 61% (25/41) of participants in the morphine arm had adequate pain control within 72 h of initiating therapy, $P = 0.296$. Most patients, 96.3% (79/82) had either no (74.4%, $n = 61$) or manageable side effects (22.0%, $n = 18$) to opioids. The

commonest side effects were nausea and vomiting ($n = 6$), generalized pruritus ($n = 7$) and headaches ($n = 5$). More patients in the pethidine arm (29.3%, $n = 12$) experienced side effects compared to those in the morphine arm (22.0%, $n = 9$), $P = 0.448$. None of the participants experienced features of CNS toxicity or respiratory suppression.

In conclusion, significantly more SCD patients on IV morphine achieved sustained pain response during VOC compared to those on IM pethidine ($P = 0.027$). There was no difference in both the tolerability and the side effect profiles of pethidine and morphine among the participants. None of the study participants experienced CNS toxicity or respiratory suppression. Therefore, physicians should confidently use morphine in cases of VOC.

Disclosure of Interest: None Declared.

BSH2020-OR-031

UK results from the International Sickle Cell World Assessment Survey (SWAY): patients with sickle cell disease (SCD) experience work impairment and decreased productivityB. Inusa^{1,*}, J. James², O. Rajkovic-Hooley³, T. Bailey³, G. Medalla⁴, B. Taylor⁴¹Paediatric Haematology, Evelina Children's Hospital, Guy's and St Thomas' Hospital, ²Sickle Cell Society, London,³Adelphi Real World, Bollington, ⁴Novartis Pharmaceuticals, London, UK

Evidence suggests individuals with sickle cell disease (SCD) encounter work impairment and associated problems. However, there is a need for greater quantification on the impact of SCD on work and finances. Here, we describe patients' and health care professionals (HCPs) perceptions of work impairment and decreased productivity associated with SCD in the UK.

SWAY was a cross-sectional survey of 2145 SCD patients and 365 health care professionals (HCPs) across 16 countries. Here we present data from UK patients, who were recruited via treating HCPs and the patient charity – The Sickle Cell Society. Patients aged 6 years and above were eligible with data collected using self-completed or proxy surveys. Agreement and satisfaction questions were asked based on a 7-point Likert scale with high agreement and satisfaction scores = 5–7.

Two hundred and ninety-nine patients from the UK completed the survey; 56% were female, mean (SD) age was 29.2 (12.4) years. Thirty HCPs from the UK completed the survey; 23% were SCD centre based, with a mean (SD) current SCD patient caseload of 76 (138) and a mean of 7 (9.3) newly diagnosed patients in the past year.

At the time of completing the survey, 52% of patients aged 12 and above were in full-/part-time employment, 24% were students, 13% were on disability/long-term sick leave. Patients reported a high impact on their ability to work as a consequence of SCD; 74% had seriously considered reducing hours at work with 61% having done so, 61% considered terminating their job, 55% required support from work to carry out their role, 48% changed from full- to part-time employment, 31% had been let go or made to stop working,

Table 7. Comparison of analgesia with duration for sustained response ($n = 82$)

Analgesia	Within 6 h (%)	Greater than 6 h (%)	χ^2	P -value	Total	Median (h)	P -value
Pethidine	13 (32.5%)	27 (67.5%)	3.31	0.069	40 (100.0%)	9.6	0.027
Morphine	22 (52.4%)	20 (27.6%)			42 (100.0%)	17.4	
Total	35 (42.7%)	47 (57.3%)			82 (100%)		

*Mann–Whitney U test.

30% voluntarily stopped working, 24% declined a promotion and 17% retired early.

Of the patients in full- or part-time employment ($n = 144$), 14% missed work within the past 7 days (absenteeism) with an average of 5.1 h missed. A high degree of impairment while working (presenteeism) was reported (42%) in addition to a high overall work impairment (47%). Patients reported a high impact on their career because of their disease – 72% report SCD has limited them to certain careers, 58% report reduced attendance at work as a result of their SCD, 54% report SCD has prevented them from progressing further in their career, 51% report SCD has prevented them from finding a suitable job. HCPs reported a high impact on patients' ability to enter employment (77%) and ability to maintain a job (80%).

SWAY demonstrates the substantial impact of SCD on patients' ability to work, highlighted by the fact that over half require support

at work. Patients report often missing work or having reduced work hours and HCPs agree that SCD has a high impact on patients' ability to enter employment and maintain a job. Further research is required to better understand the workplace-related factors influencing these work impairments. However, highlighting these important considerations for patients and their employers can help increase in understanding of how the employee is managing the work environment and what adjustments might need to be considered (e.g. being able to decide what tasks can be done or avoided, offering support, offering flexible working) in order to improve patients' work and productivity.

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Transfusion

BSH2020-OR-032

Cold stored platelets reverse thrombocytopenia following cardiac surgery

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Bleeding is common after cardiac surgery and is associated with increased morbidity and healthcare costs. Thrombocytopenia is a common contributor to bleeding and may arise from exposure to anti-platelet drugs or the mechanical effects of cardiopulmonary bypass (CPB). Platelet (PLT) transfusion is used widely after cardiac surgery to correct thrombocytopenia. However, the current standard, room temperature stored PLTs (RTP) have several potential disadvantages including; short shelf life (max. 7 days), risk of transfusion transmitted infection and deterioration in PLT function during storage. Cold stored PLTs (CSP) are a potential alternative product with longer shelf life, less infection risk and better-preserved haemostatic function. Using *ex vivo* mixing studies, we investigated whether CSP were better able to reverse the thrombocytopenia associated with cardiac surgery than RTP.

Blood samples were collected into 3.2% sodium citrate from 20 cardiac surgery patients (mean age of 62.6 years, mean time of CPB was 97.8 min, seven patients (35%) had taken recent anti-PLT medication). The patient samples were incubated for 10 min with the mitochondrial dye, DiOC6. Donor PLTs were prepared from clinical specification, pooled, buffy coat PLT units, split into two bags and stored at either 4°C (CSP), or 22°C (RTP) for up to 7 days. The donor PLTs were incubated with the dye CellTrace Violet before being mixed with the patient blood samples at 10% or 20% by volume to simulate PLT transfusion. The mixed samples were analysed using the TEG 5000 analyser (Kaolin TEG) and using a flow chamber (Ibidi μ -Slide VI 0.1) coated with collagen 50 μ g/ml, across which the samples were simultaneously infused at 1000/s to simulate arterial shear for 5 min. The flow chamber slides were fixed and analysed by confocal scanning microscopy to acquire Z stack images at 40 $\times$ magnification.

Patient samples were analysed alongside healthy controls ($n = 20$). Kaolin TEG parameters were similar between patient and control. In the flow chamber experiment, the maximum clot height and volume of patient sample were reduced compared to controls ($P < 0.0001$ and <0.0001), consistent with a thrombocytopenia arising from cardiac surgery.

After mixing the patient samples with either 10% CSP or 20% CSP, the TEG R times were shorter than in samples mixed with corresponding volumes of RTP ($P = 0.015$ for 10% mix and $P < 0.0001$ for 20% mix), indicating more rapid initiation of clot formation. There were no differences in the TEG MA or alpha angle between the samples mixed with CSP or RTP. In the flow chamber experiments, the total area of clot from mixed PLTs was greater in the patient samples mixed with either 10% CSP or 20% CSP compared with samples mixed with the corresponding volumes of RTP ($P = 0.001$ for 10% mix and <0.0001 for 20% mix) as were the maximum clot heights ($P = 0.012$ for 10% mix and 0.003 for 20% mix). The total volume of clot was greater in the patient samples mixed with either 10% CSP or 20% CSP compared with samples mixed with the corresponding volumes of RTP ($P = 0.008$ for 10% mix and <0.0001 for 20% mix).

These findings suggest that CSP, but not RTP can partially reverse the thrombocytopenia associated with cardiac surgery *ex vivo*, at clinically relevant mixing volumes. Reversal of thrombocytopenia by mixing CSP was greatest in the arterial shear model, which may indicate superior *in vivo* efficacy that requires confirmation in clinical trials.

Disclosure of Interest: None Declared.

BSH2020-OR-033

Results for the REAL pilot randomised trial of red cell transfusion in acute myeloid leukaemia: is there sufficient evidence of equipoise to support a definitive study?

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Patients with acute myeloid leukaemia (AML) are high users of red cell transfusions. There is uncertainty regarding the benefits and safety both of lower thresholds (e.g. risk of bleeding, reduced exposure to blood) and higher thresholds (e.g. complications of transfusions and costs, functional and quality of life). We aimed to assess adherence and tolerability between two haemoglobin (Hb) thresholds in AML.

We conducted an unblinded feasibility cross-over randomised controlled trial comparing restrictive *versus* liberal red cell transfusion strategies in patients with AML. Inclusion criteria: adults with AML and anticipated to receive at least two cycles of intensive chemotherapy. Exclusion criteria: history of myelodysplasia or acute promyelocytic leukaemia. Participants were randomised to receive red cell transfusion at a Hb threshold of 70 g/l during the first cycle and at 90 g/l during the second cycle (arm A) or vice versa (arm B), thus patients served as their own controls. EQ-5D-5L and EORTC QLQ-C30 questionnaires were administered at the beginning, middle and end of each treatment cycle. Follow-up was until the start of the third cycle of chemotherapy, or day +42 following the second cycle. Outcomes were adherence to protocol, achievement of a difference in Hb values between the two thresholds and clinical (bleeding, thrombosis, infections, number of transfusions, quality of life (QoL), transfusion reactions and death).

Of 84 eligible patients, 43 were randomised; 21 to arm A and 22 to arm B.

The percentage of pretransfusion Hb concentrations below the threshold of the assigned treatment group was 91% overall. Compliance was 77.2% for the restrictive cycles (exact 95% CI: 70.7–82.8%) and 99.3% in the liberal cycles (97.6–99.9%). Mean pretransfusion Hbs were 68.7 and 83.4 g/l in the restrictive and liberal cycles

respectively. After adjusting for subject as a random effect and any period effect due to the nature of the trial design, the difference in the pretransfusion Hb was 15.1 g/l (95% CI: 13.9–16.2). Both predefined primary outcome measures (adherence at least 70% and Hb difference at least 15 g/l) were met, and the study was deemed feasible.

The median number of red cell units transfused was 6 (IQR 4–9) in the restrictive cycles and 9 (IQR 7–11) in the liberal cycles. All participants in the trial received at least one red cell transfusion. More platelet transfusions were given during the liberal cycles; 298 units in the restrictive cycles *versus* 369 units given in the liberal cycles, and seven transfusions (IQR 6–10) per participant *versus* eight transfusions (IQR 4–14) per participant.

Completion rates for QoL questionnaires was 93.0% at the start of the study and 55.2% at the end. Two participants experienced a thrombotic event during the liberal cycle. Grade 3 or 4 bleeding was reported in one participant during the restrictive cycle and two patients during liberal cycles.

We have demonstrated feasibility of a randomised cross-over design for evaluating transfusion thresholds in patients with AML; predetermined adherence rates for pretransfusion Hb and Hb differences between the groups were met. Adherence to quality of life monitoring was good, and there were no concerns regarding safety outcomes. A larger study is required to provide evidence of improved efficacy of one transfusion threshold over the other, and lessons from this study indicate the importance of strategies to improve adherence to protocol and minimise deviations from protocol.

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BSH2020-OR-034

The use of e-learning in blood transfusion training: a review of the current situation

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Safe and appropriate blood transfusion practice forms part of the foundation curriculum for junior doctors. Yet, Serious Hazards of Transfusion (SHOT) has consistently reported very high rates of preventable transfusion errors (87.3% in 2018) – a value that could be minimised with optimal training and education. Currently, e-learning is increasingly used as the primary tool for delivery of blood transfusion training for all qualified clinical staff, both in the UK and internationally. However, there is sparse evidence demonstrating its effectiveness. This study aims to summarise the current state of e-learning in blood transfusion training.

Two independent investigators conducted a literature search of all published articles relating to online transfusion training in two major databases (EMBASE and MEDLINE). Articles relating to blood transfusion training using e-learning for all clinical staff were included. Abstract only articles were also included due to the sparsity of full-text articles. Data extraction was performed based on recommendations by UK National Health Service (NHS) Centre for Reviews and Dissemination. Included studies were analysed with qualitative synthesis methods.

Our initial search yielded a total of 249 articles. Following the application of exclusion criteria according to PRISMA guidelines, six articles were included in our final analysis. Four of the studies reported that e-learning was ineffective and is often viewed as a ‘tick box exercise’ or ‘clicking exercise’ when used in the context of trust mandated training. The remaining two studies reported that e-learning was beneficial and showed positive impacts on clinical practice. Both reported a reduction (>20%) in inappropriate blood product prescribing after the introduction of e-learning. Interestingly, both the studies that viewed e-learning positively were conducted outside the UK, in Denmark and Australia, whereas all the studies that viewed e-learning negatively were conducted in the UK.

The success of e-learning for the training of blood transfusion outside of the UK can likely be attributed to continued efforts in developing a successful e-learning platform. Successful platforms such as BloodSafe eLearning Australia (BEA), with more than 310 000 users, could serve as a model for other countries to follow. Furthermore, while transfusion training in Australia is standardised, the methods of transfusion training in the UK are widely varied both in content and methods of delivery, making it difficult to implement quality assurance.

The findings of this review demonstrates that the current standard of e-learning in blood transfusion training in the UK is inferior compared to its international counterparts. We recognise that given the high routine workload of clinical staff, e-learning is an attractive option as it is both cost and time effective. If e-learning is to be continued as the main method of blood transfusion training, a model similar to BEA, which is evidence-based and standardised nation-wide, should be implemented. Nevertheless, given the serious nature of transfusion errors, we suggest the current landscape of transfusion training be reviewed and evidence-based methods be used to identify the most effective way to deliver transfusion training to clinical staff. We hope that this review serves as a springboard for further research in what we believe is an important but overlooked aspect of clinical practice.

Disclosure of Interest: None Declared.

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Post-transplant outcomes of children with primary haemophagocytic lymphohistiocytosis given emapalumab to control disease

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Purpose: Primary haemophagocytic lymphohistiocytosis (pHLH) is characterized by hyper-inflammation and is driven by high production of interferon gamma (IFN γ). Haematopoietic stem cell transplantation (HSCT) is the only potentially curative option for pHLH and mortality remains high. Emapalumab is a human monoclonal antibody which neutralizes IFN γ and is the first FDA-approved treatment for paediatric and adult patients with pHLH with refractory or recurrent disease, or intolerance of conventional HLH therapy. In the pivotal trial (NCT01818492), overall response rate was 65% for all patients treated with emapalumab ($N = 34$). Outcomes of patients with pHLH who received emapalumab and proceeded to HSCT in the pivotal or extension trial (NCT02069899) were evaluated.

Methods: In the pivotal trial, patients were either treatment-naïve ($n = 7$) or had failed prior conventional HLH therapy ($n = 27$). Emapalumab (1 mg/kg, every 3 days, with the potential to increase the dose to 10 mg/kg) was administered with dexamethasone (initially 10 mg/m²/day, then weaning) until the start of HSCT conditioning. Approximately one-half of patients received an alemtuzumab-,

fludarabine- and melphalan-based preparative regimen, with the remainder receiving busulfan/treosulfan-based regimens. Most patients received a bone marrow graft from a matched, unrelated donor. The data cut-off applied was 20 July 2017. **RESULTS:** Overall, 64.7% (22/34) of patients proceeded to HSCT. Median time to transplant was 100 days (95% CI: 73, 155 days). About 86.4% (19/22) of HSCT recipients engrafted, and 90.9% (20/22) survived following HSCT. Three patients who failed to engraft received haploidentical transplants; two were rescued with a second allograft. Two patients died after HSCT due to septic shock or respiratory failure, two patients had HLH relapse after graft failure, and seven patients developed graft-versus-host disease (GVHD).

Conclusion: Patients who were treated with emapalumab and proceeded to HSCT experienced excellent post-HSCT overall survival, with a high rate of engraftment and an acceptable frequency of GVHD.

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BSH2020-OR-036

The long-term quality of life impact of vincristine neurotoxicity in paediatric acute lymphoblastic leukaemia survivors – a retrospective cohort study

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Despite the dramatic improvement in outcomes for paediatric acute lymphoblastic leukaemia (ALL) over the last 50 years, achieving cure rates in excess of 90%, treatment toxicity poses a significant challenge and can have a detrimental effect on the quality of life of survivors. It is widely acknowledged that nearly all children who receive vincristine chemotherapy experience vincristine-induced peripheral neuropathy (VIPN) (Ramchandren S et al, J Peripher Nerv Syst, 2009), however, the long-term effects that this may have on the peripheral nervous system are unclear. Assessing the burden of VIPN

Table.

	0	1	2	3	4
Tingling or Numbness	None	Fingers/toes	Extension to ankles/wrist	Extension to elbows/knees	Above elbows/knees or affecting function
Temperature sense	Normal	Reduced in fingers/toes only	Reduced up to wrist/ankle	Reduced up to elbow/knee	Reduced above elbow/knees
Strength	Normal	Mild weakness but can walk independently	Moderate weakness (can walk with assistance)	Severe weakness (unable to walk)	Paralysis
Constipation	None	Requiring change in diet	Requiring laxatives, such as lactulose/movicol	Requiring enemas or manual evacuation	Life-threatening complication of constipation
Hoarse voice	Normal voice/cry	Mild/ intermittent hoarseness	Persistent hoarseness	Able to whisper only	Breathing difficulties needing intubation or tracheostomy

in the paediatric population is further complicated by the fact that manifestations such as numbness, tingling and neuropathic pain can be difficult concepts for young children to describe, and several tools have been developed to try and objectively quantify these (Lavoie Smith EM et al, Cancer Nurs, 2013). There is little published data assessing the impact long-term neurological toxicity can have on a child's functioning and quality of life.

We conducted a retrospective cohort study at Birmingham Children's Hospital, a large UK paediatric tertiary treatment centre, to subjectively assess the long-term consequences of vincristine toxicity on patients who had completed treatment for ALL over the preceding 10 years, and the impact this can have on a child's functioning. Parents were asked to score the severity of neurological side effects during treatment and any persisting effects, to identify any long-term consequences using a modified version of the TNS-PV tool (Total Neuropathy Score – Paediatric Vincristine) (see Table).

Twenty-seven completed questionnaires were returned for evaluation (response rate 39%). Fourteen males and 13 females were included in analysis, with an age range of 15 months to 13 years at diagnosis. The median time since treatment completion of respondents was 4 years and 1 month (range 9 months to 9 years). About 58% patients experienced some degree of paraesthesiae during treatment, with 42% still experiencing long-lasting effects, and 10% patients perceiving that this impacts on their current functioning. All patients (100%) experienced weakness during treatment with residual weakness experienced by 58% respondents. About 16% patients were still having issues with constipation attributed to their vincristine treatment, with 21% suffering lasting voice changes, and 26% having enduring abnormalities in temperature perception. Median time to resolution of neurological symptoms was 12 months to 3 years in those individuals in whom symptoms had resolved.

About 84% respondents perceived that vincristine neurotoxicity had a moderate to severe impact on their child's quality of life during ALL treatment, with 36% saying that most activities were limited, and 26% required dose adjustment as a result of their vincristine toxicity. About 58% feel that vincristine neurotoxicity continues to impact their child's quality of life following completion of treatment, with 26% experiencing ongoing activity limitation as a consequence.

Our study clearly highlights the marked impact that vincristine-induced neurotoxicity can have on paediatric ALL survivors and the need for careful assessment of acute toxicity in order to make the necessary dose adjustments and minimise long-term consequences. As survival rates continue to improve, further studies are required to identify those at high risk of neurological complications and evaluate novel preventative and therapeutic strategies.

Disclosure of Interest: None Declared.

BSH2020-OR-037

Clinical outcomes following autologous haematopoietic stem cell transplantation with LentiGlobin gene therapy in the phase 3 Northstar-2 and Northstar-3 studies for transfusion-dependent β -thalassaemia

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In a phase 1/2 study of LentiGlobin gene therapy (autologous CD34⁺ cells encoding β^{A-T87Q} -globin gene for transfusion-dependent β -thalassaemia [TDT]), eight out of 10 patients with TDT and non- β^0/β^0 genotypes and three out of eight patients with β^0/β^0 genotypes achieved transfusion independence (TI). For patients achieving TI, reductions from baseline to month 48 were observed in median liver iron concentration (6.3 [$n = 11$] to 2.0 [$n = 7$] mg/g dw), median

serum ferritin levels (4829 [$n = 10$] to 937 [$n = 7$] pmol/l) and median transferrin saturation (96.5 [$n = 10$] to 49.0% [$n = 4$]). With up to 5-years' follow-up, integration was polyclonal and no oncogenesis was recorded. Herein, we present interim results of two phase 3 studies, Northstar-2 (NCT02906202; non- β^0/β^0 genotypes) and Northstar-3 (NCT03207009; β^0/β^0 , $\beta^0/\beta^{+IVS-1-110}$ or $\beta^{+IVS-1-110}/\beta^{+IVS-1-110}$ genotypes). CD34⁺ haematopoietic stem cells were collected via mobilisation/apheresis and transduced with BB305 lentiviral vector using a refined manufacturing process compared to the phase 1/2 study. Patients were infused with transduced cells following pharmacokinetic-adjusted, single-agent busulfan myeloablation (target area under the plasma concentration-time curve [AUC]: 3800–4500 [daily], 950–1125 [every 6 h] $\mu\text{mol/l}\cdot\text{min}$). Statistics are presented as median (min–max). As of 12 June and 30 September 2019, 34 patients (24 were ≥ 12 years of age) were treated in Northstar-2 and -3 with a follow-up of 11.6 (0.9–26.3) and 8.8 (2.5–20.0) months respectively. Treatment characteristics are shown in Table 1. Postinfusion non-haematological grade ≥ 3 adverse events (AEs) in ≥ 3 patients in either study were stomatitis ($n = 17$), febrile neutropenia ($n = 14$), pyrexia ($n = 3$), epistaxis ($n = 3$) and liver veno-occlusive disease (VOD; $n = 3$). VOD prophylaxis was used in 82% (28/34) of patients and all episodes of VOD were resolved with defibrotide. Drug product-related AEs were abdominal pain ($n = 3$), thrombocytopenia ($n = 3$), leukopenia ($n = 1$), neutropenia ($n = 1$) and pain in extremity ($n = 1$). All patients are alive and integration site analyses showed a polyclonal vector integration profile. In Northstar-2, 18 out of 20 patients with >5 months follow-up have not received a transfusion in >3.5 months. The primary end-point of TI (weighted average haemoglobin [Hb] of ≥ 9 g/dl without red blood cell transfusions for ≥ 12 months) was achieved by 9 out of 10 evaluable patients for an ongoing duration of 15.2 (12.1–21.3) months. Weighted average Hb during TI was 12.2 (11.4–12.8) g/dl. HbA^{T87Q} levels were 8.7, 9.3, 9.4 and 8.8 g/dl at month 6 ($n = 17$), 12 ($n = 11$), 18 ($n = 8$) and 24 ($n = 3$) respectively. Myeloid:erythroid ratios in patients who achieved TI were 1:1.6–1.9:1 at month 12 ($n = 9$) and 1:1.1 and 1:1.3 at month 24 ($n = 2$) compared to 1:2.1–1.7:3 at baseline, indicating improved erythropoiesis. In Northstar-3, 9/11 patients followed for >6 months have stopped transfusions for ≥ 3 months. At months 6 and 12, total unsupported Hb was 10.2 (8.5–13.2) g/dl ($n = 10$) and 13.8 (10.3–14.0) g/dl ($n = 3$), whereas HbA^{T87Q} was 8.3 (0–12.0) g/dl ($n = 11$) and 11.1 (8.8–12.6) g/dl ($n = 3$) respectively. TI was achieved by 2/2 evaluable patients. In conclusion, following treatment with LentiGlobin in Northstar-2 and -3, 18 out of 20 patients with non- β^0/β^0 genotypes and nine out of 11 patients with a β^0/β^0 genotype or an IVS-I-110 mutation and ≥ 6 months follow-up have stopped transfusions. In Northstar-2, 90% of evaluable patients achieved TI. The safety profile was consistent with single-agent busulfan myeloablation.

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BSH2020-OR-038

Analysis of second line treatments for children with persistent or chronic primary immune thrombocytopenia (ITP) in the UK

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We present the changing patterns of second-line treatments for persistent or chronic primary immune thrombocytopenia (ITP) in children from 2006 to 2019 in the UK. The source of information was the UK paediatric ITP registry. To be eligible for second line therapy patients must have had a platelet count $<30 \times 10^9/\text{l}$ at 6 months after diagnosis. Second-line therapies considered were thrombopoietin agonists (TPO-RA), rituximab and splenectomy. From 2006, 212 patients out of 1915 children on the registry at this time, would be eligible for second-line therapy with 23% of the eligible patients currently receiving a second-line therapy with a platelet count $<30 \times 10^9/\text{l}$ at 6 months.

The use of TPO-RA has risen in proportion to 47.6% of all therapies from January 2015 to January 2019 (Fig 1) compared to 23.8% rituximab and 9.5% splenectomy. This compares to 23.1% receiving TPO-RA, 34.6% receiving rituximab and 15.4% splenectomy between 2006 and January 2011. In our population, the patients most likely to receive second-line therapy were female, and of non-Caucasian ethnicity. They were also more likely to be older with an average age of 9.2 vs. 6.9 years, and a longer duration of disease of 54 vs. 36 weeks total follow-up.

The patients receiving second-line therapies are the more severe; compared to those who receive either no intervention (watch/monitor) or those receiving just rescue steroids or immunoglobulin. Based on the Bolton–Maggs severity score, patients on second-line therapies had a higher bleed severity score of 2.00 vs. 1.74 ($P = 0.02$). A total proportion of bleeds at Grade 2–3 of 71% vs. 67%, at Grade 3 of

Table. Treatment characteristics (median [min–max])

	Estimated average daily busulfan AUC ($\mu\text{mol/l}\cdot\text{min}$)	Neutrophil Engraftment (days)	Platelet Engraftment (days)	Hospitalisation duration (days)	Vector copy number (vector copies/diploid genome)	Cell dose ($10^6/\text{kg}$ CD34 ⁺ cells)
Northstar-2 ($N = 21$)	4428 (3709–8947)	23 (13–32)	46 (20–94)	44 (30–92)	3.3 (1.9–5.6)	7.9 (5.0–19.9)
Northstar-3 ($N = 13$)	4488 (3824–9087)	26 (14–38)	41 (21–64)	38 (29–68)	2.3 (1.2–4.3)	9.6 (5.9–15.8)

20% vs. 8.6%, finally at Grade 4 of 6% vs. 0%. Furthermore, a bleed frequency average 7.7 vs. 5.9 ($P = 0.017$), platelet count average of 13.6 vs. 15.7 ($P = 0.12$), and total follow-up period of 53 vs. 36 months.

New international guidelines will recommend an increase in second-line therapies for children with persistent or chronic ITP. The current data demonstrate the change in treatment trend since the start of the registry and provides a baseline analysis prior to the impact of these new recommendations.

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Updated results from the single-arm, open-label, long-term efficacy and safety study of subcutaneous (SC) romiplostim in children with immune thrombocytopenia (ITP)

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Background: Romiplostim is a thrombopoietin (TPO) receptor agonist approved for children and adults with ITP

Objective: Presenting results from a 3-year open label study evaluating safety and efficacy of rRomiplostim in children with ITP

Methods: Eligible children from 17 countries with ITP for ≥ 6 months and screening platelet count $\leq 30 \times 10^9/l$ (or uncontrolled bleeding) received SC romiplostim (1 $\mu g/kg$ titrated to 10 $\mu g/kg$ to maintain platelet counts of $50\text{--}200 \times 10^9/l$). In Europe, bone marrow was evaluated at baseline and after 1 or 2 years. The primary end-point was % time with a platelet response (platelet count $\geq 50 \times 10^9/l$, no rescue therapy in preceding 4 weeks) in months 0–6.

Results: A total of 203 patients (pts) received ≥ 1 dose; the median (interquartile range [IQR]) age was 10 (6–13) and median (IQR) platelet count 14 ($7\text{--}23.5 \times 10^9/l$). The median (IQR) duration of treatment was 145 (39–156) weeks, median (IQR) % of time with a platelet response in months 0–6 was 50% (17–83%), with 88% (179/203) of pts having a platelet response at least once. Median and lower quartile platelet counts were both consistently $>50 \times 10^9/l$ from week 12 and 48 respectively. Eleven pts maintained platelet counts $\geq 50 \times 10^9/l$ for ≥ 24 weeks.

Of 75 European pts with evaluable baseline bone marrow biopsies [modified Bauermeister scores: grade 0 (no reticulin, $n = 16$), grade 1 (fine fibres, $n = 54$) or grade 2 (fine fibre network, $n = 5$)], 27 had evaluable on-study biopsies after 1 year and 36 after 2 years. Of these, five pts developed increased reticulin at year 1 and 17 at year 2. One patient had an increase in modified Bauermeister score from baseline of ≥ 2 grades (increase from grade 0 to 2), four pts had an increase in 1 grade, two pts had a decrease in 2 grades, and three pts had a decrease in 1 grade in year 1. In year 2, 15 pts had an increase in 1 grade and three pts a decrease in 1 grade. No pts developed collagen and no bone marrow abnormalities were detected.

Conclusion: Over the course of the study with >30 months of treatment, 88% of children had a platelet response, median platelet counts were $\geq 20 \times 10^9/l$ above baseline 79% of the time and $>50 \times 10^9/l$ from week 12 on a median dose of 6.9 $\mu g/kg$. Bone marrow findings showed that children, like adults, did not develop clinically important fibrosis.

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Thrombosis and Haemostasis

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A novel assay of neutrophil extracellular traps (NETs) formation independently predicts DIC and identifies the rationale for anti-IL-8 therapies

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Neutrophils are the first line of defence against bacterial infection and formation of neutrophil extracellular traps (NETs) is an important protective mechanism. However, NETs can also cause harm by promoting intravascular coagulation and multiorgan failure (MOF) in animal models. Although increasingly considered as important therapeutic targets, there is currently no robust and specific measure of NETs formation to inform clinical care and enable precision medicine in patients on the intensive care unit (ICU). The aim of this study was to establish a novel assay for measuring NETs and assess its clinical significance.

A prospective cohort of 341 consecutive adult ICU patients was recruited, following written informed consent. The NETs-forming capacity of ICU admission blood samples was semi-quantified by directly incubating patient plasma with isolated healthy neutrophils *ex vivo*. The association of NETs-forming capacity with sequential organ failure assessment (SOFA) scores, disseminated intravascular coagulation (DIC) and 28-day mortality were analysed and compared with available NETs assays. Cytokine analysis together with inhibitor studies was performed to determine the driving factors of NETs in patients. To determine the pathological relevance of NETs, complementary *in vivo* studies were performed in mice models of sepsis (caecal ligation and puncture (CLP) or intraperitoneal injection of *Escherichia coli*), without or with anti-NETs therapy.

We observed that NETs were directly induced by heterologous healthy neutrophils incubated with plasma taken from ICU patients, but not from healthy donors (unless incubated with 100 nM PMA). Using the novel assay, we could stratify ICU patients into four groups, those with absent (22.0%), mild (49.9%), moderate (14.4%) and strong (13.8%) NETs formation respectively. Strong NETs formation was predominantly found in sepsis ($P < 0.0001$) and was associated with higher SOFA scores on admission and throughout the study duration (72 h postadmission). Adjusted by APACHE II, multivariate regression showed that measuring the degree of NETs formation in ICU admission could independently predict DIC and mortality whereas other NETs assays, for example cell-free DNA, myeloperoxidase and myeloperoxidase-DNA complexes, could not. Interleukin (IL)-8 levels were found to be strongly associated with NETs formation and inhibiting IL-8 significantly attenuated NETosis. Using mice models of sepsis, we specifically observed NETs positive staining (cit-H3) in the lung tissue, which could be monitored by measuring the degree of NETs formation by mice plasma. NETs

formation was associated fibrin deposition within the lung tissue and increases in lung injury scores, along with circulating markers of liver (BUN [CLP: $P = 0.005$, *E. coli*: $P < 0.001$]), kidney (ALT [CLP: $P = 0.01$, *E. coli*: $P = 0.002$]) and cardiac injury (cTnI [(CLP: $P < 0.001$, *E. coli*: $P < 0.001$])). By targeting IL-8 we were able to significantly inhibit NETs formation, fibrin deposition, organ injury and also improved survival times in septic mice ($P = 0.004$).

Our new NETs assay directly measures the NETs-forming capacity in patient plasma. This could guide clinical management and enable identification of NETs-inducing factors in individual patients for targeted treatment and personalised ICU medicine. We identify IL-8 as a major driving factor in sepsis, with anti-IL-8 therapy in septic mice significantly reducing NETs-induced organ damage and mortality.

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Abstract Withdrawn.

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Gain of function variant of prothrombin (p.R541W) causes dominant heritable thrombosis and is associated with antithrombin resistance

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Relatively common causes of hereditary venous thromboembolism (VTE) are the loss-of-function variants in the natural anti-coagulants antithrombin, protein C and S, along with gain-of-function variants in procoagulant factors V (factor V Leiden) and II (prothrombin G20210A). Miyawaki et al. (*NEJM* 2012) reported a gain-of-function prothrombin variant (p.R596L) associated with resistance to antithrombin and susceptibility to VTE.

Here, we report a family with early onset of VTE. The index was under investigation because of recurrent, unprovoked VTE from the age of 15 years. His sibling had a first unprovoked VTE at the age of 18 years. Family members with and without thrombosis were consented to the NIHR Bioresource Rare Diseases project and whole genome sequencing was performed.

Variants in *F2*, *ILF3*, *BACE1*, *PCSK6* and *IGF1R* co-segregated with the clinical phenotype. Of these, the very rare c.1621C>T variant in *F2* causing a p.R541W substitution has the highest pathogenic plausibility. Interestingly, this variant has simultaneously been reported in a Dutch large pedigree with thrombosis. The arginine R541 is located within the sodium binding region of prothrombin, an area of critical importance for stabilising the thrombin-antithrombin (TAT) complex and promoting efficient thrombin inhibition. We hypothesised that this variant could be associated with antithrombin resistance.

To characterise the molecular mechanisms inducing the prothrombotic phenotype of p.R541W, plasma from family members, with and without the p.R541W variant and not on anticoagulation, was characterised using thrombin generation and prothrombin activity:antigen (Ac:Ag) ratios. Recombinant, wild-type and prothrombin p.R541W were produced and analysed for antithrombin resistance and TAT complex formation. Quantitative and qualitative measurement of prothrombin in plasma samples showed similar levels between the wild-type family control and p.R541W heterozygote as well as unchanged Ac:Ag ratios. This indicates that intrinsic procoagulant catalytic activity of the variant is unmodified. However, thrombin generation assay revealed a significantly increased Endogenous Thrombin Potential with the p.R541W variant as well as increased time-to-tail indicating a reduced thrombin inhibition. The resistance of recombinant p.R541W to inhibition by antithrombin was compared with wild-type prothrombin. The p.R541W variant had statistically significant reduced antithrombin-mediated inhibition of activity associated with reduced TAT complex formation.

In conclusion, in this family with inherited VTE, we have identified a rare gain-of-function prothrombin variant. The p.R541W substitution resulting from this variant disrupts thrombin-antithrombin interaction and causes thrombin to display pathological antithrombin resistance properties.

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When is immune thrombocytopenia not immune thrombocytopenia? Using high-throughput sequencing to distinguish patients with inherited thrombocytopenia and diseases of immune dysregulation

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Immune thrombocytopenia (ITP) is a disease of exclusion, there is no diagnostic test, and it is difficult to treat with many patients trying several unsuccessful treatments before achieving remission. Inherited thrombocytopenia can present similarly and therefore a proportion of patients are misdiagnosed with ITP but carry a genetic

causal variant. Patients with ITP may also have a causal variant associated with rare diseases of immune dysregulation or auto-inflammation as defined by the International Union of Immunological Societies Classification of Immunodeficiencies. It is important to differentiate these patients to tailor treatments to the specific cause.

To further explore potential genetic causes of ITP, we used whole genome sequencing (WGS) or targeted sequencing in 94 patients (20 children and 74 adults) with a diagnosis of chronic ITP: forty-one patients underwent WGS through BRIDGE NIHR, 41 additional patients had targeted sequencing using the thrombogenesis (TG) gene panel and 12 had targeted sequencing for the primary immune deficiency (GRID) panel. For the WGS, all rare (MAF < 0.01) and highly conserved (CADD > 10) non-synonymous missense variants in the target genes (from the TG and GRID panel) were included. Variants confirmed in the literature as benign or identified in a single allele for a recessive disease were excluded. In total, 82 patients underwent variant testing for the TG genes and 53 patients for the GRID genes. Variants identified through these platforms are discussed in the relevant MDT; a specific haemostasis/thrombosis or primary immunodeficiency specialist meeting where both clinicians and geneticists come to a joint consensus on pathogenicity.

Overall, seven patients (7%) had pathogenic or likely pathogenic variants confirmed at MDT; five in genes from the TG panel (ANKRD26, ETV6, GP1BB (2), TUBB1) and two in genes from the PID panel (NLRP12 and TNFRSF13B). One further variant in ANKRD26 has not yet been confirmed by the MDT but is in the same promoter region as reported pathogenic variants. Three of the six patients with pathogenic variants in TG genes had thrombocytopenia which did not require treatment; three were managed as ITP although they did not respond to immunosuppression but had lasting responses to thrombopoietin receptor agonists. Half of the patients with pathogenic variants in the TG genes had a positive family history. Two patients with pathogenic variants in GP1BB were diagnosed in childhood, suggesting this gene tends to cause the phenotype earlier in life. While one of the adults with ANKRD26 had a phenotype typical for this variant (mild chronic thrombocytopenia), the second presented later in life with features typical of ITP, although only responded to a thrombopoietin receptor agonist.

We identified a further 20 patients with variants of uncertain significance (VUS); eight of which have been confirmed in specialist MDTs. Two patients had two rare variants in GRID genes; NOD2 and NLRP12 in one patient, and CASP10 and PLCG2 in the other. The CASP10 variant is more likely to be pathogenic because it is

Table. Number of patients with variants identified in each gene panel

	Inherited thrombocytopenia (TG panel) <i>n</i> = 82		Immune dysregulation and auto-inflammation (GRID panel) <i>n</i> = 53	
	Number of patients	Specific genes	Number of patients	Specific genes
Pathogenic or likely pathogenic (MDT decision)	5 (6%)	ANKRD26 ETV6GP1BB (2)TUBB1	2 (4%)	NLRP12TNFRSF13B
Pathogenic (not yet reported by MDT)	1 (1%)	ANKRD26		
Variant of unknown significance (MDT decision)	3 (4%)	RUNX1ITGA2BTUBB1	5 (9%)	MEFVNOD2CASP10PLCG2 (2)
Variant of unknown significance (not yet reported by MDT)	6 (7%)	RUNX1TUBB1DIA PH1MECOM (2)SLFN14	8 (15%)	PRF1 (2)PLCG2 (2)CASP10TNFRSF1ASAMHD1STXBP2

*Number of patients with a variant in this gene.

associated with autoimmune lymphoproliferative syndrome and the patient presented with lymphadenopathy.

In conclusion, sequencing of patients with ITP identified a rare pathogenic variant in 9% of patients, and a VUS in a further 21%. This highlights the importance of sequencing patients with ITP to distinguish rare pathogenic variants which can guide clinical management as well as screening for haematological malignancy.

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Molecular dynamics analysis provides mechanistic insights into ACTN1-related thrombocytopenia

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ACTN1-related thrombocytopenia (A-RT) is a non-syndromic form of heritable macrothrombocytopenia caused by heterozygous variants in ACTN1, which encodes the dimeric cytoskeletal protein α -actinin 1. Variants associated with A-RT are distributed throughout the functional domains of the protein (actin-binding domain (ABD), central rod domain (RD) and calmodulin-like domain (CaM)). However, there is no clear relationship between genotype and the severity of A-RT phenotype, and a common mechanism through which ACTN1 variants cause thrombocytopenia has not been defined. ABD variants could potentially result in macrothrombocytopenia by directly disrupting α -actinin 1–actin interactions which are necessary for the cytoskeletal remodelling that underlies normal platelet production. However, this mechanism is unlikely for variants in the RD or CaM, because these domains do not directly interact with actin. Other possible molecular mechanisms include disruption of α -actinin 1 dimerization, calcium binding or alteration of the overall protein structure. We used structural modelling and molecular dynamics (MD) to investigate ACTN1 variants outside the ABD that are implicated in A-RT.

A human α -actinin 1 structural homology model was generated from the published crystal structure of chicken α -actinin 1. Variant models were made by introducing variants associated with A-RT (R320Q (RD) and R738W (CaM)) in heterodimeric and homodimeric states. The GROMACS-4.6.7 suite of software was used to set up and perform the MD simulations in the presence and absence of calcium. Simulations were run over 50 ns and structures saved every 0.1 ns for analysis. The root-mean-square deviation (RMSD) of the backbone atoms of the protein was measured and expressed in Ångström as deviation over time and mean deviation by residue. Principal component analysis (PCA) was performed using the GROMACS tools. Trajectories were combined at the protein backbone level. Covariance for the calcium-bound and calcium removed cases were separately calculated and global matrices compiled to establish the most significant motions (principal components). Global principal components (PC) 1 and 2 were then calculated from the global covariance matrices. PC1 and PC2 were analysed for the WT, variant homodimer and variant heterodimer models, allowing direct comparisons because the same PC were being analysed.

Simulations demonstrated that the mean RMSD was similar in all three functional domains of the calcium-bound compared to the calcium-unbound models, indicating that the presence of calcium had little impact on α -actinin 1 stability. PCA demonstrated that PC1 and PC2 for the WT dimer do not overlap with those of the homodimeric or heterodimeric R320Q and R738W substituted proteins

regardless of the presence of calcium. PCA demonstrated that the two different substitutions were very similar to one another and easily distinguished from WT.

Although a preliminary finding based on analysis of only two A-RT substitutions, this suggests that both variants result in altered mechanical properties of the substituted α -actinin 1 in homodimer and heterodimer forms. This provides a credible mechanistic explanation for macrothrombocytopenia since the α -actinin 1 dimer forms a mechanical cross-link between critical structural elements in platelets, including the $\alpha_{IIb}\beta_3$ integrin, subintegrin apparatus and the actin cytoskeleton.

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Factor VIII use in the treatment of breakthrough bleeds in persons with haemophilia A without inhibitors on emicizumab prophylaxis: the phase III HAVEN 3 study experience

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HAVEN 3 investigated the use of emicizumab as prophylaxis in adult and adolescent (≥ 12 years old) persons with haemophilia A (PwHA) without factor VIII (FVIII) inhibitors (NCT02847637). The primary analysis of HAVEN 3 included an intrapatient comparison of 48 participants who received FVIII prophylaxis in a non-interventional study (NIS) prior to enrolment in HAVEN 3. In this intrapatient comparison, emicizumab prophylaxis resulted in an annualised bleed rate that was 68% lower than the rate with previous FVIII prophylaxis (1.5 vs. 4.8, $P < 0.001$). No dosing guidance was provided regarding the use of on-demand FVIII in HAVEN 3, and investigators prescribed FVIII at their own discretion. Herein, we characterise the dose and frequency of replacement FVIII used for the treatment of breakthrough bleeds (BTB) in these 48 participants.

The primary comparisons are of on-demand FVIII use for BTB, whereas participants took FVIII prophylaxis during the NIS versus emicizumab prophylaxis during HAVEN 3. Any use of on demand FVIII other than to manage BTB (e.g. prophylaxis; preventative dosing prior to activity; treatment for surgery) was not included in our analyses. Annualised on-demand FVIII use was calculated by dividing by the number of days in the efficacy period and multiplying the resulting daily consumption by 365.25 days. The number of infusions and cumulative doses of on-demand FVIII use are described at the participant and individual bleed levels, and are presented descriptively for both the NIS and HAVEN 3 exposure periods. No formal statistical inferences have been conducted. All analyses were based on a 4th October 2018 data cut-off.

The analysis cohort consists of 48 participants treated with FVIII prophylaxis during the NIS and then treated with emicizumab prophylaxis during HAVEN 3. Annualised infusion rates of on-demand FVIII per participant and cumulative doses of on-demand FVIII (in

international units [IU] per kilogram) per participant were higher during the FVIII prophylaxis period than during the emicizumab exposure period (mean 15.3 vs. 7.2; median 3.6 vs. 0.6 annual infusions per participant and mean 602.4 vs. 209.0 IU/kg; median 75.5 vs. 19.1 IU/kg respectively). At the individual bleed level, FVIII infusions and total cumulative dose suggested that participants were administered a similar amount of medication to treat bleeds during both the NIS and HAVEN 3 study periods: median number of infusions per bleed were 1.0 (interquartile range [IQR] = 1.0) vs. 2.0 (IQR = 3.0) and median cumulative doses were 43.5 (IQR = 35.1) vs. 50.0 (IQR = 72.7) IU/kg respectively (Table 1). Further characterisation of treated bleeds showed similar rates of joint and muscle bleeds between the NIS and HAVEN 3 exposure periods (57% vs. 61%; 17% vs. 14% respectively).

This analysis revealed a lower annualised infusion rate and a correspondingly lower annualised cumulative dose of FVIII for treatment of BTB during emicizumab prophylaxis compared with FVIII prophylaxis. At the individual bleed level, the amount of on-demand FVIII used per bleeding episode was comparable between NIS and HAVEN 3 exposure periods. Thus, based on this single analysis, participants received less on-demand FVIII during emicizumab prophylaxis compared with FVIII prophylaxis, as a result of overall reduction of bleed frequency, whereas the treatment of individual bleeds was similar regardless of the prophylaxis therapy administered.

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Abstracts for “The Crucible” session: What lessons can Haematology learn from others?

Lessons from Oncology and Palliative Care – It’s About Time

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Increasing awareness led to closer integration between clinical haematology and specialist palliative care (SPC) services. However, with <50% of haematology patients being referred to SPC¹, they remain less likely to receive palliative care or hospice services than those with other cancers. This is despite haemato-oncological patients having a substantial illness burden, poor quality of life (QoL), and aggressive care at the end of life with greater likelihood of dying in hospital.

Many barriers to timely SPC referrals have been identified, including unpredictable disease trajectories, inaccessibility to often fragmented services and ill-conceived physician perceptions. However, haematology can learn one thing from oncologists and palliative care physicians – that SPC is not synonymous with end-of-life care. Rather, SPC describes a philosophy of care, working with the other patients’ doctors to provide an extra layer of support². Aside from managing a variety of symptoms including: pain, mucositis, dyspnoea, fatigue, nausea, constipation, and diarrhoea, palliative care specialists also help patients engage in emotional work to facilitate coping, accepting, and planning.

The emphasis is on the well-being of patients and families at any point along their disease trajectory, regardless of their illness state. Mounting evidence suggests that earlier SPC input results in better outcomes. Three seminal, well-designed, randomised controlled trials have clearly established the feasibility and benefits of integrating early palliative care concurrently with standard oncology care³. Also, various other solid tumour studies documented: reduced symptom burden³, better QoL⁴, improved depression scores^{4,5}, higher mean scores for global health with better emotional and social functioning⁶, and even significantly longer survival^{4,5}. Early SPC referral decreased receipt of aggressive interventions at end-of-life and significantly lengthened duration of hospice care.

Haematology should take a page from Oncology and appreciate the scope for major research in this area. Hopefully, this will lead to earlier SPC intervention for our patients.

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What can haematology learn from linguistics? Back to basics

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On the day that you started on your training path to where you are today, you unwittingly started on a journey to learn the art of a new language – the language of medicine. From there you have now flourished into someone who uses this complicated language fluently and expertly. This skill took years, perhaps decades, to truly master. Medical jargon allows for efficient and clear discussions to be had between medical professionals and is critically important in this respect.

Now imagine that you are a person who has suddenly become very unwell and you are being immersed into this language with absolutely no prior training or interest in the subject. In fact, you wish you didn’t have to hear any of this. You feel really ill – the worst you’ve ever felt in your life – and just want to go to sleep and let this all wash over your head. You’re being hit with a barrage of scary-sounding words that you don’t know. They said “acute” – you think that means painful... or does it mean small? You don’t even have the energy to ask.

Linguistics theorises that jargon has two main effects, the first being to increase efficiency of communication. However, the second is to exclude outsiders who do not have prior understanding of the topic being discussed. This unfortunate side-effect is causing a rift between haematologists and patients which can only be bridged by conscious attempts to simplify communication based on our audience. As haematological treatment options become more plentiful and advanced, the language we use becomes more complicated by the day. I propose that haematology learns this simple lesson of the unwanted effects of jargon to allow patients to join the conversation with us.

What lessons can haematology learn from others? Valid consent and shared decision-making in clinical practice

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“It is trite law that in general a doctor is not entitled to treat a patient without the consent of someone who is authorised to give that consent”¹. What constitutes valid, informed consent? In 2015, an obstetric case (Montgomery v Lanarkshire Health Board) redefined the law on consent and rejected previous case law and by

inference medical paternalism². Should haematologists take heed from this lesson learnt on the labour ward?

Within haematology, certain conditions carry such a high untreated morbidity and mortality that treatment is clinically imperative. However, equipoise exists in many circumstances, for example in the context of low-grade disorders, palliative treatment, or where research evidence does not provide clarity to guide therapeutic choice. Haematology is a field of rapidly evolving treatments, adding complexity to decision-making.

Fundamentally, informed consent requires knowledge of the indications and material risks of proposed interventions. The National Chemotherapy Board provides guidance³ relating to systemic anti-cancer treatment and has recently collaborated with Cancer Research UK to publish regimen-specific consent forms providing comprehensive patient information.

However, a thorough, signed consent form is simply one component. Patients vary in their desire for information and involvement in choices regarding their care, but evidence suggests one third of haemato-oncology patients are dissatisfied with their role in decision-making^{4,5}. "Shared" or "supported" decision-making is increasingly recognised as crucial in promoting patient autonomy and enabling valid consent. Shared decision-making involves the collaborative consideration of evidence-based information alongside patients' own priorities and values. It has been promoted by both the Royal Colleges of Obstetricians and Gynaecologists, and of Surgeons, in response to the Montgomery ruling^{6,7} and is highlighted in the current draft revision of GMC consent guidance. Should we follow in our surgical colleagues' footsteps to focus training and resources to ensure that our patients have the best opportunity to be involved in decision-making?

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Physician know thyself: learning to value our lived experience

Anna Dillon

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In 2016, my daughter died. Her life and her passing have changed my life and my life as a doctor.

Our profession stigmatises vulnerability: we are taught to keep personal experiences outside of the workplace. However, it has been my experience as a patient, a carer and a bereaved relative that has taught me the most. I learnt how it felt when 21st century medicine couldn't help: what life was like on the other side of the consultation room.

I have learnt to listen to my colleagues and the importance of their journeys outside of the workplace. Many of us are experiencing significant personal experiences outside of work: by valuing and supporting these experiences we can become better haematologists and reduce burnout within our teams.

Most importantly, I have learnt to truly listen to my patients. Feeling unheard is one of the commonest reasons for litigation and complaint because it is one of the most fundamental aspects of the doctor-patient relationship.

In this era of personalised medicine, perhaps we need to learn the most important personalisation of all: what is important to the patient in front of us.

Too often, as doctors we listen to reply; by valuing our lived experience we can learn to listen to understand.

This year we are celebrating 60 years of BSH and looking to the future of our speciality. Many of us have been inspired by amazing haematologists: what made these people great mentors was their ability to listen to their trainees and patients.

I look forward to a changing climate in haematology where our personal experiences are valued for their role in helping us to listen to ourselves, each other and our patients.

Why separate the inseparable? Integrating Mind and Body Care in haematology

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'The mind is such a fragile thing. How can someone begin to heal physically if they are mentally broken?'

The haematology landscape has undergone revolutionary transformations; however, we often neglect the mental and social-health needs of our patients and carers throughout their journey from diagnosis to recovery and beyond.

The scale of the problem is significant. Anxiety and depression are 3-4 times more likely where physical health issues co-exist. 30% of patients with a long-term condition, including many with haematological conditions, will have a mental health problem¹ and they have poorer outcomes when they have a blood cancer². Evidence supporting psychological services within haemato-oncology is summarised in NICE guidance³ and more recent research⁴. Further, specialist psychiatric care is necessary to manage complex treatment effects and drug interactions, alongside other issues⁵.

Many haematological services already attempt integrated 'Mind and Body' care, with psychology or counselling services available for some groups.

Is this sufficient? Evidence suggests more is needed to care for mental and physical needs equally and simultaneously, developing integrated Mind and Body service 'whole person' models.

Abstracts for “The Crucible” session: What lessons can Haematology learn from others?

This necessary shift would see routine mental health and wellbeing screening for *all* patients on *every* haematology pathway, together with clinically appropriate, personalised responses.

An enhanced Mind and Body programme has potential to transform patient experience, enabling optimised psychological, social, and clinical care concurrently.

A compelling case supports transformation of physical, mental, and social care delivery across haematology pathways, but requires skills derived from a team of specialist psychiatrists, psychological therapists, and social workers. This multi-professional collaboration embedded within routine haematological care offers a unified focus to benefit patient experience, outcomes, and wellbeing.

Haematology has much to learn from others and embedding a Mind and Body team within haematology will create a culture shift, departing from conventional compartmentalised care to a truly holistic model.

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ALL and AML

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Whole genome profiling of adult B-other acute lymphoblastic leukaemia on the UKALL14 trial

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B-other ALL represents a working definition for patients with B cell precursor (BCP) ALL without a known primary chromosomal abnormality. In this study we use whole genome sequencing (WGS) to characterize adult B-other cases (age ≥ 25 years) from the UKALL14 trial (NCT01085617).

Of 652 patients aged 25–65 years enrolled onto UKALL14, 333 (51%) had B-other ALL. Sufficient material was available to screen 156/333 B-other cases for recurrent Ph-like fusion events (*CLRF2*, *JAK2*, *ABL1*, *ABL2* & *PDGFRB*) using FISH and MLPA (kit P335). This identified 28 (18%) “Ph-like” fusion events (21 *CRLF2*, 5 *ABL*-class fusions and 2 *JAK2*). 57/128 B-other cases, had available samples for tumor normal paired WGS (read depth 60× and 30× respectively). Bioinformatic analysis was performed to determine small somatic mutations (SSMs); single nucleotide variants (SNVs) and insertion/deletions (INDELs) as well as copy number aberrations (CNA) and structural variants (SV). We also undertook de novo motif analysis to identify RAG mediated deletions.

We present data for 51/57 cases (6 cases failed sample QC), median age of diagnosis 42, range 25–65. Within this cohort we identified 1649 SVs, 165 530 SNVs and 8508 indels, with each case having a median representation of 28 (10–137) SVs, 1663 (912–77 183) SNVs and 103 (11–2749) indels. RAG heptamer sites were found at 64% (462/724) of breakpoint resolved deletions.

Fusion gene analysis identified 19/51 (37%) cases with recurrent rearrangements this includes *TCF3-PBX1* and *P2RY8-CRLF2* cases which escaped screening. Of the remaining 17; 5 were *IGH-DUX4*, 4 *ZNF384r*, 3 *MEF2Dr*, 2 *PAX5alt* (Gu et al. 2019) and 2 Ph-Like candidates (*MYO18A-FGFR1*, *IGH-EPOR*). A further 3 cases had previously identified *PAX5alt* concurrent coding drivers (p.R38H/C & p.R140L).

We found no evidence of driver subgroups clustering with age, although 3/4 *ZNF384r* cases were 26 years at diagnosis just above the trial entry threshold. Mutation burdens between driver subgroups were not significant; *MEF2Dr* cases had high translocation burden (25, cohort median 2) and low RAG mediated deletion burden (1, cohort median 7.5); *PAX5alt* had a higher INDEL burden compared to all other subgroups (183, cohort median 94). As expected we found recurrent CNA in *CDKN2A/B* (59%; 30/51), *PAX5* (27%; 14/51), *IKZF1* (27%; 14/51), *ETV6* (9%; 5/51) and *BTG1* (4%; 2/51) across the cohort.

Within the unknown driver subgroup ($n = 29$); 5 cases had either *IGH-CEBPA/B* or *ZEB H1038R* these are likely to be the G12 subtype identified by Li et al. 2018; 5 “complex” cases including an MMR hypermutator & translocation “chain”; a single “masked” hypodiploid previously described by Carroll et al. 2009 and a single case had a high RAG deletion burden (45 events); 10 “Like” candidates – 9 with Ph-like & a single *ETV6* dominant negative requiring RNA-seq confirmation; of the remaining 7 cases, 3 had MDS mutations (2x *IDH2* R140Q & *ASXL1* LOH).

Whole genome sequencing enables characterization of disease defining alterations in B-other ALL that are of clinical relevance in 50% of cases. 15% (8/51) of cases with a normal karyotype had large CN events by WGS, suggesting a bias in karyotype detection particularly in lower purity samples. Distinct classes of genomic instability are identified in B-other ALL this include a RAG mutation phenotype, high SV burden and MMR hypermutator. This approach sets the premise to design an extended genotype-clinical correlative study on B-ALL cases enrolled in UKALL14 and 60+ trials.

Disclosure of Interest: None Declared

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Regulation of acute myeloid leukaemia maintenance by Wilms tumour 1 in an isoform specific manner

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The gene encoding Wilms Tumour 1 (*WT1*) is recurrently upregulated in Acute Myeloid Leukaemia (AML). *WT1* transcript levels are associated with primary refractory disease and with relapse (Lambert et al., Oncotarget, 2014). We have recently described the gene regulatory networks in purified blasts from AML patients with various driver mutations or translocations (Assi et al., Nature Genetics, 2019); *WT1* was an AML specific regulatory node in all networks examined. Furthermore, in knockdown experiments as part of a shRNA depletion screen in vitro and in murine xenotransplantation experiments, we showed that *WT1* was essential for leukaemic maintenance in t(8;21) AML (Martinez-Soria et al., Cancer Cell, 2018). In our current work, we have examined the role of *WT1* in primary AML patient blasts and in cell lines by performing a comprehensive phenotypic and multi-omic characterisation.

WT1 produces 8 primary isoforms, by employing one of two transcription start sites and alternatively splicing exon 5 and exon 9, all of which are expressed in haematopoietic cells. To understand how the different isoforms of *WT1* are involved in leukaemic maintenance, we cloned each of them into a doxycycline-inducible lentiviral vector and transduced them into a t(8;21) AML cell line, Kasumi-1, which well recapitulates primary t(8;21) AML.

We next expressed each of the 8 *WT1* isoforms to levels seen in AML patient samples. Expression of isoforms lacking 3 amino acids in the zinc finger DNA binding domain (*WT1* -KTS) significantly reduced cell growth and colony formation, increased apoptosis, caused a G1 cell cycle arrest and caused myeloid differentiation of leukemic blasts. However, expression of isoforms with the 3 amino acids (*WT1* + KTS) in the DNA-binding domain had the opposite

effects, thus playing a major role in the maintenance of a leukemic phenotype. Consistent with this idea, *WT1* + KTS isoforms were expressed at higher levels than *WT1* -KTS isoforms in purified blasts from AML patients.

We then performed the first ever *WT1* chromatin immunoprecipitation (ChIP-seq) in AML. The *WT1* -KTS and *WT1* + KTS isoforms display different binding patterns with *WT1* -KTS binding to many more (9451 sites) and different sites than *WT1* + KTS (1355 sites). We integrated this data with gene expression as assayed by RNA-seq and with open chromatin sites as assessed by DNase I-seq, in each cell population. We found differential binding to cis-regulatory elements and isoform specific altered expression of genes involved in proliferative, cell cycle, apoptosis and differentiation pathways, explaining many of the phenotypic findings.

We validated these findings in primary FLT3-ITD AML patient blasts and again found that expression of *WT1* -KTS reduced colony formation compared to the empty vector control or with expression of *WT1* + KTS. However, CD34⁺ cells isolated from the bone marrow of healthy patients were unaffected by expression of either *WT1* isoforms. Therefore, as already indicated by our gene regulatory network analysis, *WT1* is only of importance in an AML specific context.

Several vaccine studies and more recently a T cell receptor study targeted against *WT1* have been undertaken but they have shown mixed efficacy in AML (Chapuis et al, Nature Medicine, 2019). Based on our results that show antagonistic effects of different isoforms of *WT1*, we suggest that a more effective therapeutic strategy would be to selectively target *WT1* + KTS isoforms only.

Disclosure of Interest: None Declared

BSH2020-PO-003

BCAT1 increases sensitivity to Cytarabine and confers CXXC motif derived resistance to pro-oxidant treatment in Acute Myeloid Leukaemia cells

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Recently increased branched-chain amino transferase (BCAT1) expression has been associated with poor prognosis in AML making it an attractive potential target for the development of novel therapies. BCAT1 features a CXXC motif characterised by redox active cysteine thiols, a feature common to antioxidant enzymes which utilise thiol groups as reducing agents. We previously demonstrated BCAT1 can metabolise H₂O₂ *in vitro* and overexpression in U937 cells results in a decrease in cellular reactive oxygen species (ROS). Given excessive ROS can induce apoptosis we hypothesised BCAT1 CXXC antioxidant properties may protect against pro-oxidant treatment. Here we demonstrate for the first time BCAT1 decreases ROS and increases survival when challenged with pro-oxidant Rotenone.

To evaluate the BCAT1 CXXC motif we utilised site directed mutagenesis to modify the C-terminal Cysteine of the CXXC motif creating a Cysteine → Serine mutant (BCAT1-CXXS) as previously described. This mutation eliminated the C-terminal thiol significantly reducing the ability to metabolise H₂O₂ *in vitro* compared to BCAT1-WT. Following lentiviral transduction stable expression in U937 cells was confirmed by western-blot and qPCR; BCAT1-WT (5.57 ± 1.31 fold increase) and BCAT1-CXXS (5.93 ± 2.6 fold increase).

Transgenic U937 cells were challenged with Rotenone and cell density was measured after 72 h. Dose response analysis revealed the LD₅₀ was significantly higher for BCAT1-WT (40.9 ± 1.14 nmol/l) compared to either BCAT1-CXXS (20.8 ± 1.14 nmol/l) and EV

control (20.7 ± 1.18 nmol/l) (*n* = 3, *P* < 0.001). This finding suggests that the BCAT1 CXXC provides protection against ROS mediated cell death. To support this we measured intracellular ROS following Rotenone treatment using dichlorofluorescein diacetate. Treatment with 60 nmol/l Rotenone resulted in a significantly decreased ROS signal in BCAT1-WT (491 ± 39 nmol/l) compared to either BCAT1-CXXS (707 ± 29 nmol/l) and EV control (659 ± 22 nmol/l) (*n* = 3, *P* < 0.05).

Previously BCAT1 has been shown to promote cellular proliferation in a variety of cancer cell lines. First line AML therapy relies on Cytarabine (AraC) a nucleoside analogue that interrupts cell division however to date there has been no examination how BCAT1 expression affects treatment with AraC. We theorised the proliferative advantage of BCAT1 overexpressing cells may render them vulnerable to AraC. Firstly we confirmed BCAT1 provides a proliferative advantage in a U937 model, we seeded a T75 flask with 2 × 10⁵ cells/ml and monitored cell density at 24 h intervals. BCAT1 cells reached a higher peak cell density after 96 h (1.57 × 10⁶ ± 8.53 × 10⁴ cells/ml) compared to EV control (1.1 × 10⁶ ± 7.71 × 10⁴ cells/ml) a relative increase of 42%. (*n* = 9, *P* < 0.01). Moreover at mid-exponential phase (48 h) BCAT-WT cells displayed a consistently higher percentage of cells in G2M phase (23.2 ± 1.4%) compared to EV control (16.4 ± 1.4%) (*n* = 4, *P* < 0.05). Next cell lines were incubated with AraC and cell density was measured after 72 h. LD₅₀ values were significantly lower for BCAT1-WT (0.442 ± 1.34 μmol/l) compared to EV control (2.36 ± 1.18 μmol/l) (*n* = 3 *P* < 0.0001), confirming increased AraC sensitivity in BCAT1-WT over-expressing cells.

In summary here we have demonstrated for the first time that BCAT1 expression protects against pro-oxidant treatment whilst increasing vulnerability standard therapeutic treatment. This data opens up a new therapeutic avenue for clinicians seeking novel strategies in the stratified treatment of BCAT1^{high} and BCAT1^{low} expressing patients.

Disclosure of Interest: None Declared

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Exploration of the variant composition in the genome of pediatric and adult recurrent acute myeloid leukemia

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Acute myeloid leukemia (AML) is the most common form of acute leukemia, characterized by a high frequency of relapse and a low long-term survival rate. Patients with recurrent AML show poorer overall survival, accelerated disease progression and they often do not respond to conventional treatment. One reason for the overall

poor outcome is high inter- and intratumor heterogeneity, where it is thought that different sub-clones potentially have different functional properties, jointly responsible for relapse and drug resistance. Further, the mutational landscape and the tumorigenesis of adult *versus* pediatric AML is marked by age-specific characteristics. To overcome this complexity, a personalized treatment approach is required to increase event free survival and expand treatment options for relapse and refractory pediatric and adult AML patients.

Here we report genome-wide next-generation sequencing analyses of patient-matched longitudinal specimens from 26 pediatric and 49 adult AML patients from the Nordic countries. These comprised samples collected at diagnosis ($n = 54$), relapse ($n = 80$), as well as primary resistant specimens ($n = 6$), alongside with remission specimens ($n = 18$) and normal bone marrow stromal cells ($n = 45$ cases) as a source of germline DNA.

Our findings reveal recurrent relapse specific mutations in genes not previously reported in de novo AML (*CSF1R* [2.7% of our relapse cases]) or previously identified mutated at a very low frequency (e.g. *ARID1A* [6.1%*], *H3F3A* [6.1%*], *MGA* [10.2%*], *of our adult cases) in studies of AML cohorts solely consisting of specimens from initial diagnosis. Of specific note are the remarkably high frequencies of *WT1* mutations in relapse/primary resistant adult (22.4%) and pediatric (26.9%) AML compared to previous findings for non-acute promyelocytic leukemia (APL) AML cases. Further, we report novel specific differences in the mutational spectrum between adult *versus* pediatric relapsing AML, with recurrent internal tandem duplications (ITDs) in *UBTF* found solely in pediatric relapsing AML ($n = 3$; 11.5%), while *H3F3A* ($n = 3$; 6.1%) and *MGA* mutations only were identified in adult AML cases. We further describe two cases of very late relapse, both harboring a subset of identical alterations at relapse as found in their respective founder clone, while gaining for instance p.K27M mutations in *H3F3A* during the course of evolution. Finally, our results emphasize the importance of applying whole genome sequencing, over targeted gene panels and even over whole exome sequencing, to be able to identify the full repertoire of alterations (including rare small nucleotide variants, long ITDs and copy number alterations) when attempting to characterize AML at relapse.

This study is the first multi-genome sequencing study of both pediatric and adult relapsing AML, and it has provided critical new insights into the biologic basis of this disease. Knowledge gathered from this study will hopefully help to pave the way for improved and individualized treatment strategies of this high-risk cancer type and lead to prolonged patient survival.

Disclosure of Interest: None Declared

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Towards clinical whole genome sequencing (WGS) in acute leukaemias – learning from the 100 000 Genomes Project

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WGS promises substantial benefit in the clinical evaluation of AML and ALL owing to its potential to detect all types of actionable mutation. A major strength of the technology is a genome wide “agnostic” approach, meaning mutation detection is not limited to a small subset of specifically targeted abnormalities. Existing WGS analytical pipelines however, have primarily evolved as research tools and modification and development is required if the full potential of WGS in

a clinical setting is to be realised. In this study we use the Haematological Malignancy cohort of the 100 000 Genomes Project Cancer Programme ($n \Rightarrow 1000$ patients) to inform improvements to the WGS pipeline required for clinical use.

Comparisons between results obtained for 100 000 Genomes Project participants using the Genomics England Cancer Analytical Pipeline (GeCAP) and results obtained using orthologous local standard of care clinical testing demonstrated high sensitivity and specificity for detection of clinically-relevant somatic variants using WGS data (sensitivity >99%, precision 91.3% for single nucleotide variants and sensitivity 94.8%, precision 85.8% for indels). However, this comparison identified issues in the GeCAP with the detection of structural variants (SVs) and in sample quality control procedures that needed addressing. It was shown that tumour in normal (TIN) contamination of the matched germline/normal samples required for WGS analysis severely affected sensitivity of somatic SV calling, e.g. on occasions *PML-RARA* fusions were missing in AML patients with TIN contamination. As obtaining uncontaminated germline samples in acute leukaemia is challenging we are developing a statistical test for detecting TIN and where significant amounts are identified, clinically relevant SVs will be called with unmatched germline. Preliminary analysis demonstrated this analytical approach did not compromise sensitivity compared with analysis employing matched germline and specificity only decreased 10–20%. We further identified that *FLT3* Internal Tandem Duplications (ITDs) were not reliably detected by the Manta SV calling algorithm. Accordingly, locus specific detection of *FLT3* ITD using Pindel was introduced, allowing known *FLT3* ITDs to be recovered, except for one instance with low variant allele frequency (VAF) of 5%. Interrogation of undetected gene fusions identified using orthogonal testing uncovered that deletions >10 kb are filtered out as default by Manta, resulting in non-reporting of fusions between two genes residing on the same chromosome, e.g. *EBF1-PDGFRB*. This was rectified by including such breakpoints. Low tumour content significantly reduced the sensitivity of calling SVs evidenced by non-reporting of disease-defining translocations in this cohort. Therefore, the GeCAP v2 estimates tumour content from sequencing data and highlights samples with a derived tumour content of <30% and a high fraction of variants with a VAF < 6%, returning results with a warning for reduced sensitivity of SV calling.

These modifications to the 100 000 Genomes Project GeCAP should significantly increase the utility of WGS in a clinical setting. This will allow patients with AML & ALL to fully benefit from the potential diagnostic and therapeutic opportunities afforded by WGS as we move towards implementation in the NHS Genomic Medicine service.

Disclosure of Interest: None Declared

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Abstract Withdrawn.

BSH2020-PO-007

Targeting SET-PP2A complex as a new avenue of treatment for Mixed Lineage Leukemia (MLL) treatment

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Mixed lineage leukemia (MLL) is a distinct and rare form of leukemia which mainly affects children. This aggressive disease arises as a

result of rearrangements affecting the *MLL(KMT2A)* gene located on chromosome 11q23. Despite extensive research to develop a targeted therapy, MLL patients have a poor prognosis and efficacious treatments are needed.

Protein Phosphatase 2A (PP2A) is a crucial regulator of pro-survival pathways such as Akt, Erk, GSK3-Beta, which are found constitutively deregulated and hyperactivated in MLL. SET is an endogenous binding partner and inhibitor of PP2A. Phosphorylation of SET at Ser9 regulates its accumulation in the cytosol, where it can form an inhibitory complex with PP2A. Inhibition of PP2A upon SET binding promotes the activation of pro-survival oncogenic pathways. Conversely, inhibition of SET using the SET antagonist, FTY720, rescues the activity of PP2A reducing malignant proliferation in several forms of solid tumours and leukemia, highlighting SET as an interesting target for therapy.

We characterized for the first time SET molecular profile and regulation in MLL cell lines and primary samples with the aim to explore whether SET could be a novel therapeutic target.

We found that SET is over-expressed in MLL cell lines and primary samples compared to the healthy controls. Analysis of SET phosphorylation and localization confirmed that, in MLL cell lines, SET was phosphorylated on Serine residues and mostly localized within the cytoplasm. We also found that SET formed a protein-protein interaction complex with PP2A in MLL. In addition, SET knock down largely blocked colony formation in MLL cell lines, whereas it did not affect the ability of non MLL cell lines such as K562 and REH to form colonies in semi-solid medium. These findings indicate an important role for SET in the self-renewal of MLL cells. Likewise, the pharmacological inhibitor of SET, FTY720, induced a specific block in proliferation in MLL cell lines, whereas it did not affect non MLL cell lines. In MLL cell lines, FTY720 disrupted SET-PP2A complex, it reduced the levels of SET, phospho-PP2A and phospho ERK, without affecting other PP2A downstream targets, including Akt and GSK3Beta. These results might indicate that ERK pathway is the major pathway regulated by PP2A in MLL cell lines and that a complete deletion of SET would be required to shut down the rest of PP2A targets.

Taken together, our data indicate that inhibition of PP2A by SET is key for the tumorigenic potential of MLL leukemia cells and suggest a potential use of targeting PP2A as a novel therapeutic intervention in MLL.

Disclosure of Interest: None Declared

BSH2020-PO-008

Therapeutic potential of allogeneic gamma delta T cell therapy in relapsed/refractory acute myeloid leukaemia – preliminary data from a phase 1 study

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For Acute Myeloid Leukemia (AML) patients with chemotherapy-resistant disease or for those ineligible for allogeneic haematopoietic cell transplant, treatment options are limited and prognosis is poor. Novel approaches that treat disease while minimising safety risks are urgently needed. $\gamma\delta$ T cells have endogenous anti-leukaemia innate reactivity; however, autologous $\gamma\delta$ T cells isolated from AML patients can have impaired antitumorigenic function. We have developed TCB002 – *ex vivo* expanded, purified allogeneic V γ 9V δ 2 + CD3 + T

lymphocytes currently being studied in a Phase 1 trial in relapsed/refractory (R/R) AML.

Adult patients with R/R AML meeting inclusion criteria were recruited to the dose escalation study. TCB002 was generated from HLA matched (haploidentical) donor apheresis material. Prior to TCB002 infusion, patients underwent a 5-day standard lymphodepleting pre-conditioning treatment with fludarabine and cyclophosphamide. The co-primary endpoints are incidence of treatment-emergent adverse events (TEAEs) and dose limiting toxicity (DLT). Secondary endpoints include number of patients achieving complete remission (CR) and overall survival (OS). (Full trial details: NCT03790072)

Four patients with multiple R/R AML have received ≥ 1 infusion of TCB002 (1×10^6 – 1×10^7 cells/kg). No signs of GvHD or organ injury have been reported, with only two potential TEAEs (CRP rise, Grade 2; and Fever, Grade 1) over a mean observation of 74 days (range 14–112 days). No DLTs have been identified. Patient 1 experienced fatal respiratory failure due to bilateral pneumonia, not related to drug, at Day 14. Two patients in Cohort 1 (1×10^6 cells/kg) achieved significant decrease in % blasts in bone marrow (BM) by Day 14 and achieved morphologic leukemia-free status by Day 28. Both patients received additional infusions of TCB002 with no adverse reactions. Patient 2 succumbed to fatal GI haemorrhage (not related to treatment) at Day 112. Patient 3 discontinued at Day 100 owing to disease progression. Patient 4 received a higher dose (1×10^7 cells/kg) of TCB002 and achieved morphological complete remission with incomplete haematological recovery (CRi) at Day 14, and CR at Day 28. The patient received an additional infusion of TCB002 on Day 36 and remains in remission with excellent performance status reported at Day 70.

These preliminary results from a first-in-man dose escalation study highlight the encouraging safety profile and therapeutic potential TCB002 for patients with R/R AML. Achievement of CR and improvement of performance status post-treatment suggests a potential role for TCB002 as a bridge therapy. The trial remains open for enrolment. Dose escalation continues.

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Efficacy of FMS-like Tyrosine Kinase 3 (FLT3) inhibitors in patients with acute myeloid leukemia

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The FMS-like tyrosine kinase 3 (FLT3) gene is mutated in 30% of all patients with acute myeloid leukemia (AML). Mutation in the internal tandem duplication (FLT3-ITD) domain is most commonly encountered, often indicating high leukemic burden, higher relapse rates, and poor prognosis in patients. Recent clinical studies with the next-generation FLT3Is have shown greater potency and less off-

target effects, suggesting the potential benefit of incorporating FLT3I in combination regimens for treatment of AML. We conducted a systematic review and meta-analysis of randomized controlled trials (RCT) to explore the overall efficacy of FLT3I in patients with AML.

We conducted a systematic search of Medline, Embase, ClinicalTrials.gov databases, and meeting abstracts through October 31, 2019, to find out all the RCTs comparing a FLT3 inhibitor-based regimen with other agents in the treatment of patients with AML. Pooled estimates of hazard ratios (HR) with respective 95% confidence intervals (CI) for overall survival (OS) and event-free survival (EFS) were calculated using the generic inverse variance method. We used the random effects model. Heterogeneity of effect size was quantified using I^2 statistic and Cochran's Q. All statistical analyses were performed with *Review Manager* (RevMan Version 5.3, Copenhagen: The Nordic Cochrane Centre, The Cochrane Collaboration).

A total of five RCTs (two phase II and three phase III) randomizing 1919 patients (1088 in the FLT3I arms and 831 in control arms) were included in the final analysis. Randomization ratio was 2 to 1 in QuANTUM-R and ADMIRAL trials, and 1 to 1 in other studies. Sorafenib was used as FLT3I in two studies; midostaurin, quizartinib, and gilteritinib were used in one study each. FLT3I was used in newly diagnosed AML patients in three studies. Recently published QuANTUM-R and ADMIRAL trials tried FLT3I as salvage therapy for patients with relapsed/refractory AML. All participants were FLT3 mutated in RATIFY, QuANTUM-R, and ADMIRAL trials. Median age of patients ranged from 47.9 to 68 years. Including data from all studies, pooled HR for OS was not statistically significant for FLT3I compared to control [0.81, 95% CI: 0.65–1.02, $P = 0.07$]. In patients with newly diagnosed AML ($n = 1181$), pooled HR for OS showed trend towards statistical significance [0.85, 95% CI: 0.72–1.00, $P = 0.05$] on analysis of three studies. A significant pooled OS benefit was observed for FLT3I in the setting of salvage therapy for patients with relapsed/ refractory AML ($n = 738$) [pooled HR 0.69, 95% CI: 0.57–0.83, $P < 0.0001$]. Pooled HR for EFS did not reach statistical significance overall (four studies), and in patients with newly diagnosed AML [0.87, 95% CI: 0.68–1.11, $P = 0.26$, and 0.86, 95% CI: 0.61–1.22, $P = 0.40$, respectively]. No to substantial heterogeneity was noted among studies, depending on type of analysis.

FLT3Is demonstrated a significant improvement in OS in patients with relapsed/ refractory AML and a trend towards improvement in OS in newly diagnosed AML patients. However, pooled EFS across therapy settings were not found to be statistically significant. Future randomized studies exploring further novel agents, especially the next generation FLT3Is should help formulate optimal combination regimens to improve the outcomes in this high-risk subset of AML patients.

Disclosure of Interest: None Declared.

BSH2020-PO-010

Combination therapies targeting apoptosis pathways in paediatric AML

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Leukaemia is the most frequently (30%) occurring type of paediatric cancer. Of these, approximately 80% are acute lymphoblastic leukaemia (ALL) with acute myeloid leukaemia (AML) cases making up the remaining 20% alongside other leukaemias. Unfortunately, children with AML do not have promising prognosis with only 60% surviving 5 years or longer. It has been highlighted recently the need for age-specific therapies for AML patients, with paediatric AML cases

having a different mutational landscape compared with AML diagnosed in adult patients.

Drug Repurposing is a recognized strategy in drug discovery and development where an already approved drug is used for diseases other than originally indicated. We aim to identify novel combination therapies with the promise of providing alternative more effective and less toxic induction therapy options.

Our *in-silico* analysis highlighted 'cell death and survival' as an aberrant, potentially targetable pathway in paediatric AML patients. On this basis, 83 apoptotic inducing compounds were screened. A preliminary single agent screen was also performed to eliminate potentially toxic chemicals, then drugs were constructed into a pooled library with 10 drugs per well over 160 wells, with 45 possible pairs and 120 triples in each well. Seven cell lines were used during this study to represent the clonality of AML in paediatric patients (Kasumi-1, CMK, CMS, MV11-14, PL21, THP1, MOLM-13). Cytotoxicity was assessed up to 72 h using CellToxTM Green reagent. Fluorescence readings were normalized to a DMSO control. Z-Score was assigned to each well based on the mean and standard deviation of all the data. Combinations with a Z-Score <2 were eliminated and the remaining wells were taken forward for further analysis. A well was considered 'successful' if each drug individually demonstrated a Z-Score <2 , while the combination exhibited a Z-Score >2 .

Each of the ten compounds in one well (155) had minimal or no effect as single agents on cell viability however, a combination of two or more of the compounds resulted in a substantial increase in cell death, therefore the ten compounds were de-convoluted to identify a possible synergistic pair/triple combinations.

The screen identified two possible 'novel' drug pairing, with BCL2 inhibitor ABT-737, combined with either a CDK inhibitor Purvalanol A, or AKT/ PI3K inhibitor LY294002. (ABT-737- 100 nmol/l + Purvalanol A- 1 μ mol/l) (ABT-737- 100 nmol/l + LY294002- 2 μ mol/l). Three possible triple combinations were identified (LY2409881 + Akti-1/2 + Purvalanol A, SU9516 + Akti-1/2 + Purvalanol A, and ABT-737 + LY2409881 + Purvalanol A), which will be taken forward for examining their efficacy at varying concentrations and dosing schedules, across multiple paediatric AML cell lines for optimisation of maximum synergy.

Our combinations showed interesting therapeutic potential, which to our knowledge has not been reported by other groups investigating these drug pairings. We believe that our combination screening approach has potential for future use with a larger cohort of drugs including FDA approved compounds and patient material.

Disclosure of Interest: None Declared

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Combination therapies targeting DNA damage signalling as a therapeutic strategy in acute myeloid leukaemia

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Acute myeloid leukaemia (AML) is an aggressive disease with an unmet clinical need. Current treatment regimens rely on standard chemotherapy with potential stem cell transplants which are poorly tolerated by patients >60 years. Due to its heterogeneous nature, patients often relapse resulting in low 5-year survival. Genome instability and impaired DNA damage response represent key features of

leukaemic cells, indicative of the potential to target this signalling pathway as a therapeutic strategy in AML.

Previously we identified an essential role for the epigenetic regulator, histone lysine-specific demethylase 4A (KDM4A) in AML cell survival. We and others have found a potential role for KDM4A in parallel with Poly (ADP-ribose) polymerase (PARP) signalling in response to DNA damage and repair. Therefore, we hypothesise that a combination therapy via pharmacologically targeting KDM4A together with PARP inhibitors (PARPi) may represent a novel, effective treatment for AML. To address this, we developed a series of novel KDM4A inhibitors (KDM4Ai). As a monotherapy, KDM4Ai induced leukaemic cell differentiation and apoptosis, with an IC_{50} of $3.2 \pm 0.2 \mu\text{mol/l}$ in MLL-AF9 driven THP1 cells after 48 h treatment ($n = 3$), similar efficacy was observed in other human AML cell lines ($n = 3$) including FLT3-ITD AML, MV411 ($2.7 \mu\text{mol/l}$), and MOLM13 ($1.7 \pm 0.7 \mu\text{mol/l}$), t(8;21)(q22;q22) fusion harbouring Kasumi1 ($2.69 \pm 0.1 \mu\text{mol/l}$) and NPM1 mutant OCI-AML3 ($4.9 \pm 1.0 \mu\text{mol/l}$) alongside primary patient blasts ($3.8 \mu\text{mol/l}$), importantly identifying a therapeutic window of effect on CD34⁺ leukaemic stem cell enriched population in comparison with donor cells. Combination treatment of KDM4Ai with PARPi for 48 h in THP1 cells ($n = 3$) exhibited a Combination Index of ~ 0.69 indicative of a synergistic effect. The IC_{50} of KDM4Ai in combination with the apparent EC_{50} of the PARPi is reduced to $1.86 \mu\text{mol/l}$, with the most synergism identified with simultaneous administration of $3 \mu\text{mol/l}$ KDM4Ai and $5 \mu\text{mol/l}$ PARPi. This combination treatment exhibits a significant decrease in proliferation ($>15\%$ reduction) ($P = 0.0164$) and increased apoptosis ($>20\%$) ($P = < 0.001$) in MLLr-AML cell lines compared with KDM4Ai or PARPi alone.

Global transcriptomic profiling by RNA-seq provided a better understanding of the mechanism and molecular consequences responsible for KDM4Ai induced leukaemic killing in THP1 cells ($n = 3$). Pathway analysis showed an enrichment of IL4 & IL13 signalling (Enrichment score (ES) = -0.53 , $q = 0.017$), and neutrophil degranulation (ES = 0.53 , $q = 0.017$), accompanied by significant upregulation of DNA damage response pathways (ES = 0.6 , $q = 0.012$). These results were further investigated and confirmed in AML cell lines with an accumulation $>20\%$ of γH2AX and cleavage of PARP. Combination treated THP1 cells have been subjected to global RNA-seq to confirm the detailed molecular mechanism underlying the synergistic effect and require further analysis.

In summary, our data show that pharmacological inhibition of KDM4A using novel compounds effectively eliminates leukaemic cells with a synthetic lethality observed through combination with PARPi. Our data further supports the role of KDM4A in myeloid oncogenesis whilst identifying a parallel role with DNA damage and repair pathways exhibiting a synergistic targeting potential. This promotes the future clinical evaluation of KDM4A and other key components in DNA damage/response signalling pathways as potential therapeutic vulnerabilities in AML.

Disclosure of Interest: None Declared

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Use of next generation sequencing to identify AML with germline predisposition; an essential diagnostic test

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Acute myeloid leukaemia (AML) with biallelic mutation of *CEBPA* is a distinct subtype of AML with a favourable prognosis. In approximately 10% of cases, the mutation of one *CEBPA* allele is germline enabling a classification of "AML with germline *CEBPA* mutation". Patients with a germline mutation typically present at a young age and are prone to multiple relapses. This prospect brings stem cell transplant (SCT) into the management plan and thus the need for timely genetic screening of related, potential donors.

Here we report on a case of AML with germline *CEBPA* mutation detected at 1st relapse, how it reflects advances in molecular diagnostics and impacts on patient management.

The patient initially presented aged 5 with pancytopenia and circulating blasts. He had no past medical history and no known family history of haematological malignancy. The diagnosis of AML was confirmed by bone marrow aspirate (BMA) with a typical immunophenotype (CD34⁺, CD117⁺, CD13⁺, CD33⁺, CD117⁺, CD7⁺). A normal karyotype was observed (46, XY) with no mutation within *FLT3* or *NPM1*. He was recruited to the MRC AML15 trial and treated with a non-transplant protocol (2 cycles of FLAG-Ida induction and 3 cycles of high dose cytarabine consolidation). He achieved CR1 and completed treatment without significant complications. There was no evidence of long-term therapy related toxicity on routine follow up.

Twelve years later, aged 17, he presented with an upper respiratory tract infection. A full blood count demonstrated a neutropenia ($1.59 \times 10^9/l$) with circulating blasts ($<5\%$ ANC). Further investigations demonstrated AML. The immunophenotype was identical to that at diagnosis. The karyotype was also normal (46, XY). No mutation within *FLT3* or *NPM1* was observed. As the initial working diagnosis was therapy-related AML he was commenced on Vyxeos chemotherapy with a plan for allogeneic SCT at CR1.

To further characterise his disease, a next generation sequencing (NGS) panel was performed (OncoPrint Myeloid assay). Two pathogenic variants within *CEBPA* were identified: c.190dup [p.(Ile64Asnfs*44)] and c.931_932insGAC [p.(Thr310_Gln311insArg)]. The disease was thus classified as AML with biallelic *CEBPA* mutation. Of note, the variant allele frequency (VAF) of the c.190dup variant was 52% whereas the VAF of the c.931_932insGAC variant was 33% raising the possibility the c.190dup variant represented a germline mutation. NGS and conventional sequencing of the relevant *CEBPA* region in DNA from a subsequent remission sample and saliva sample demonstrated the c.190dup but not the c.931_932insGAC variant. These results indicate the presence of a germline *CEBPA* mutation with subsequent acquisition of a second somatic *CEBPA* mutation, consistent with the molecular pathogenesis of this disorder.

CEBPA variants are often complex insertion/deletions. These results demonstrate successful incorporation of an amplicon based NGS strategy for the identification of germline and somatic *CEBPA* variants in AML with a normal karyotype and the importance of germline assessment in patients with AML with biallelic *CEBPA* mutation.

In retrospect the clinical presentation is typical of AML with germline *CEBPA* mutation. Beyond implications for prognosis, family screening and family planning the presence of a germline mutation is particularly important to establish in the context of a planned

related donor SCT. New pathways must therefore be developed to allow timely access to Clinical Genetics.

Disclosure of Interest: None Declared

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Inhibition of myeloid cell leukaemia 1 (MCL1) overcomes bromodomain and extra-terminal (BET) inhibition resistance in acute myeloid leukaemia (AML) cells

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Acute Myeloid Leukaemia (AML) remains to be a therapeutic challenge mainly due to its heterogeneity. So far, treatment with chemotherapy agents has not been effective in curing it. As a result, more research is needed in predicting suitable drug combinations. Recent studies have shown that AML is associated with epigenetic changes, in particular the Bromodomain (BRD)-containing protein families. The first prototypic BET protein family inhibitor (JQ1) has been shown to have antiproliferative effects in lymphoma, myeloma cell lines and leukaemia, in particular AML. Several of these compounds are currently in clinical trials but none have received FDA approval yet, mainly due to poor tolerability and modest activity in patients with relapsed or refractory AML. The reasons for the inadequate therapeutic effects are unclear. Accumulating evidence demonstrates that combining a BET inhibitor with a Myeloid Cell Leukaemia (MCL-1) inhibitor has a greater apoptotic effect on AML cells. However, data is limited. Our previous work showed that treating AML cells with JQ1, increases the antiapoptotic protein, MCL-1, which could be contributing to JQ1 resistance. The aim of the present study is to investigate the effect of the combination treatment (BET and MCL-1 inhibition) on AML cells. Gene and protein levels of BCL-2, BCL-XL, CMYC and MCL-1 were determined in AML cell lines and primary samples. Cells were treated with a selective Brd4 inhibitor (JQ1) and/or a MCL-1 inhibitor (AZD 5991). Proliferation was assessed using WST-1 assay. Apoptosis was assessed by flowcytometry and caspase C assay. The experiments were performed in two AML cell lines (OCI-AML3 and THP-1) and four patient bone marrow samples. JQ1 treatment downregulated BCL2, BCL2L1 and CMYC and upregulated MCL1 gene levels in OCI-AML3 cells, which is in-line with our previous microarray data. AZD5991 induced up-regulation of all these genes while the addition of JQ1 to AZD5991 restored down-regulation of BCL2 and MYC but not BCL2L1. JQ1 (250 nmol/l) and AZD5991 (100 nmol/l) induced cell death levels of 34% and 37% respectively in 48 h. Combination treatment of OCI-AML3 cells with both JQ1 and AZD5991 produced synergistic cytotoxicity, determined by comparing actual and expected cell viability. JQ1 (250 nmol/l) and AZD5991 (100 nmol/l) resulted in actual viability of 16.5% compared with expected viability of 47.8%. JQ1 (250 nmol/l) was also tested with a lower concentration of AZD5991 (10 nmol/l), which demonstrated synergy (actual viability of 54.1% compared with expected viability of 75.3%). Using primary cells, two out of four samples showed apparent synergy (actual viability of 12.0% compared to expected viability of 20.5%). We find that JQ1 (250 nmol/l) and AZD5991 (100 nmol/l) induce both early and late apoptosis of OCI-AML3 cells with JQ1 having a larger early apoptotic effect than AZD5991. In OCI-AML3 cells, Annexin-V positivity was 16.9% for 250 nmol/l JQ1 and 17.9% for 100 nmol/l AZD5991 but 55.2% for the combination, which is in excess of the expected level of 31.8% and consistent with synergistic cytotoxicity. In conclusion, JQ1 and AZD 5991 were shown to work synergistically in OCI-AML3 cell line and three out of the four patient samples. This study,

although small, provides novel data on the effect of the combination of MCL-1 and BET inhibitors on the apoptosis of AML cells.

Disclosure of Interest: None Declared

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Analysis of treatment patterns and outcomes for patients with therapy-related acute myeloid leukaemia and acute myeloid leukaemia with myelodysplasia-related changes diagnosed in England between 2011–2016 using hospital episode statistics[®]

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Two subgroups of the World Health Organisation 2016 diagnostic criteria for acute myeloid leukaemia (AML) include therapy-related AML (t-AML) and AML with myelodysplasia related changes (AML-MRC). Patients with t-AML have received prior chemotherapy or radiotherapy. AML-MRC is defined as either: (i) history of myelodysplastic syndrome (MDS) or myelodysplastic/myeloproliferative neoplasm, (ii) MDS-related cytogenetic abnormality, or (iii) multilineage dysplasia in >50% of ≥ 2 cell lineages in the absence of *NPM1* or biallelic *CEBPA* mutations. Patients with t-AML or AML-MRC historically had poor outcomes with conventional chemotherapy, and the timeframe of this analysis (2011–2016) is prior to European Medicines Agency approval of any agents specifically for t-AML or AML-MRC (e.g. CPX-351). The aim of this retrospective, observational study was to describe the treatment patterns and analyse outcomes for patients with *de novo* AML versus t-AML or AML-MRC using data from the Health Episode Statistics (HES) database.

HES is a National Health Service (NHS) database containing details of all admissions, accident and emergency attendances, and outpatient appointments at NHS hospitals in England. Adult patients (≥ 18 years) who were diagnosed with AML between NHS years 2011 and 2016 were identified through ICD-10 diagnosis codes (C92*, excluding C921, C922, and C927). There were no ICD-10 codes specific to t-AML or AML-MRC, and these patients were identified through a history of a relevant cancer and specific OPCS-4 procedure codes for chemotherapy (X7*) or radiotherapy (X65*), or a history of MDS (ICD-10 D46*), chronic myelomonocytic leukaemia (ICD-10 C931), or diagnosis of AML with multilineage dysplasia (ICD-10 C928). Patients with AML who did not fall into these criteria were classified as *de novo* AML. Based on chemotherapy and transplant OPCS-4 codes, patients were allocated to the following AML treatment pathways: intensive chemotherapy (IC) $\pm$ haematopoietic cell transplant (HCT), azacitidine $\pm$ HCT, or low-dose cytarabine (LDAC). Patients who did not receive active systemic therapy (i.e. best supportive care alone) were excluded.

Overall, 9758 patients with AML were identified, comprising 7465 (77%) with *de novo* AML and 2293 (23%) patients with t-AML or AML-MRC. Patients with *de novo* AML were typically younger (median [interquartile range] age: 63 [50–72] vs. 69 [61–75] years) and had a lower proportion of males (56% vs. 62%) versus patients with t-AML or AML-MRC. Patients with *de novo* AML were more likely to receive IC $\pm$ HCT (68% vs. 43%) and less likely to receive palliative azacitidine alone (15% vs. 37%). The use of IC $\pm$ HCT was higher for patients with *de novo* AML versus t-AML and AML-MRC across all age deciles but most notably for patients aged 60–69 years (76% vs. 55%) and 70–79 years (36% vs. 20%). Patients with *de novo* AML had significantly longer overall survival versus t-AML or AML-MRC (median: 1.79 vs. 0.94 years; unadjusted hazard

ratio = 1.66 [95% confidence interval: 1.58–1.75]; nominal $P < 0.0001$).

In conclusion, this large, retrospective analysis using the HES database has confirmed the poor historical outcomes for patients diagnosed with t-AML and AML-MRC in England between 2011 and 2016. Compared with patients with *de novo* AML, those with t-AML or AML-MRC were less likely to receive IC $\pm$ HCT than azacitidine $\pm$ HCT or LDAC across all age groups, and overall survival was significantly shorter. Copyright © 2020. Re-used with the permission of NHS Digital. All rights reserved.

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BSH2020-PO-015

Synergistic inhibition of FLT3-ITD acute myeloid leukaemia cell growth in vitro using combination of FLT3 inhibitor quizartinib and standard of care chemotherapeutic agent cytarabine

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Mutations of the Fms-like tyrosine kinase 3 (FLT3) is the most common mutation in adult acute myeloid leukaemia (AML), with an internal tandem duplication (ITD) present in 25% of all AML cases correlating with high leukaemic burden and poor prognosis. Presence of FLT3-ITD leads to constitutively activated FLT3 kinase and activation of its downstream molecular pathways, including PI3K/AKT/mTOR, RAS/RAF/MEK, JAK/STAT5 and MAPK/ERK that can be inhibited using FLT3 small molecule inhibitors. However, the use of these multi-targeted inhibitors is limited by a lack of selectivity causing off-target effects. Quizartinib is a highly potent and selective type II inhibitor of FLT3 kinase that can induce a haematological improvement in comparison to standard chemotherapy, but its favourable response as monotherapy is limited due to the rapid development of resistance. To improve efficacy and find ways to reduce its toxicity, strategies including combination of FLT3 inhibitors with other agents e.g. chemotherapy, are currently under pre-clinical evaluation.

In this study, we aimed to investigate whether quizartinib efficacy could be enhanced if combined with chemotherapeutic agent cytarabine/AraC in human FLT3-ITD AML cell lines. In a panel of human AML cell lines including FLT3-ITD (MOLM-13 and MV4-11) and FLT3 wildtype (HL-60 and THP-1) IC50 values were determined in a resazurin-based assay, which is an oxidation-reduction indicator of cell viability. The IC50 value at 48 h for MOLM-13 was 0.72 ± 0.18 nmol/l and for MV4-11 0.58 ± 0.12 nmol/l, whereas HL60 and THP1 were insensitive to quizartinib, suggesting that FLT3-ITD is the key driver of survival in the responsive cell lines. Based on the IC50 concentrations obtained from resazurin-based assays, we sought to evaluate whether the drug induced apoptosis or cell cycle arrest. Using flow cytometry, we observed that 1 nmol/l

quizartinib treatment is moderately apoptotic, but induced G1 cell cycle arrest with an increase of G1 phase of $36 \pm 9\%$ in MOLM-13 and $55 \pm 9\%$ in MV4-11 cells over untreated control.

Furthermore, combination of quizartinib with cytarabine at respective IC50 concentrations displayed a Combination Index (CI) of less than 1 indicating synergy in MV4-11 and MOLM-13 cells at 48 h (calculated with CompuSyn). For MOLM-13 cells, 1.4 nmol/l quizartinib combined with 500 nmol/l cytarabine resulted in a CI value of 0.48. As for MV4-11 cells, CI value was 0.33 when 1.2 nmol/l quizartinib was combined with 20 nmol/l cytarabine. Thus, the combination treatment is more synergistic with MV4-11 than MOLM-13 cells. Dose reduction indices (DRI) revealed that to achieve 50% inhibition, cytarabine had to be used at 15.7 nmol/l for MV4-11 but 130 nmol/l with MOLM-13 in combination with 1.2 and 1 nmol/l quizartinib, respectively, suggesting that MOLM-13 cells require approximately 8-times more cytarabine to achieve the same level of inhibition in combination than MV4-11 cells. It is speculated that sequential administration of drug treatment could further improve synergy, as cytarabine inhibits synthesis of DNA when the cell is in the S phase, its mode of action could be impeded by quizartinib-induced cytostatic effects. Furthermore, by taking advantage of RNA-seq we expect to identify new targets in future experiments.

Keywords: AML, chemotherapy, FLT3, quizartinib
Disclosure of Interest: None Declared

BSH2020-PO-016

Evaluate correlation between NUDT15 variants and 6-mercaptopurine sensitivity in vietnamese pediatric acute lymphoblastic leukemia

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Background: Thiopurines (i.e. 6-mercaptopurine [MP]) plays a critical role in the treatment of pediatric acute lymphoblastic leukemia [ALL], but it probably results in myelosuppression. NUDT15 variants have been suggested to be strongly correlated with the extent of bone marrow suppression caused by 6-MP in Asian patients. This study aims to investigate the NUDT15 variant frequency and assess their impact on 6-MP's myelotoxicity in Vietnamese pediatric ALL patients.

Methods: We used Sanger sequencing genotyping to identify NUDT15 variants. MP dose intensity during maintenance therapy was a marker for drug-induced toxicities assessment and was defined as a 28-day average dose. The association between the variants and myelotoxicity was analyzed.

Results: A total of 70 pediatric patients treated according to FRALLE 2000 ALL protocol were enrolled in the study. Among them, NUDT15 variants were identified in 23 patients (32.9%). We found 4 novel coding variants: NUDT15 *2 (p.V18_V19ins and p.R139C), *3 (p.R139C), *6 (p.V18_V19ins) and NUDT15 p.R11Q. Particularly, p.R11Q variant was not previously reported. The NUDT15 CC, CT, and TT genotypes accounted for 67.1%, 30%, and 2.9%, respectively. During 12 months of 6-MP administration, patients with NUDT15 CT and TT were given a median MP dose intensity of $52.3 \text{ mg/m}^2/\text{day}$ vs. $68.7 \text{ mg/m}^2/\text{day}$ in patients with NUDT15 CC, $P < 0.0001$. Patients with TT genotypes were substantially sensitive to MP, with an average dose intensity of 33.3%, compared with 73.3% and 92% of those with CT and CC genotypes, respectively, $P < 0.0001$.

Conclusion: In Vietnam, the use of 6-MP in the treatment for childhood ALL warrants the investigation of NUDT15 variants. Our results suggest that 6-MP dose adjustment to NUDT15 variants is needed to avoid complications caused by excessive drug sensitivity.

Keywords: Pediatric acute lymphoblastic leukemia, NUDT15, 6-Mercaptopurine.

Disclosure of Interest: None Declared

BSH2020-PO-017

Whose choice is it anyway? A multi-national and multi-stakeholder perspective on shared treatment decision making in older adults with acute myeloid leukaemia

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Acute Myeloid Leukaemia (AML) is a rare haematologic cancer primarily affecting older people (median 68 years old at diagnosis). Heterogeneity in the presentation, functional status and comorbidities among patients presents unique challenges for treatment decisions. There is a lack of consensus regarding optimal treatment; options include intensive chemotherapy [IC], low intensity therapy, best supportive care, and hospice care. Given the diversity in available treatments and new therapies in development, and numerous factors influencing treatment decisions, it is important to understand how patients, their families, and physicians make such decisions, notably the trade-offs that each is willing to make. However, qualitative research among stakeholders is sparse. This study aimed to explore the treatment decision-making process from the perspective of physicians, patients, and their family members.

Ten physicians in the US ($n = 4$), UK ($n = 3$) and Canada ($n = 3$), 15 US AML patients (>65 years old, not taking IC) and 15 of their family members each took part in a one-on-one, 60-min semi-structured telephone interview. Additional interviews with patients and family members are scheduled for the UK and Canada and will be available at presentation. Interview questions started open-ended to elicit spontaneous content, followed by more focused

questions if needed. Each participant also rated a series of treatment features on a scale from 0 (not at all important) to 3 (very important) to determine their importance in AML treatment decision-making.

Data collected thus far revealed that the treatment decision-making process most commonly involved the patient and their physician, although some patients mentioned the involvement of their family. Most patients ($n = 9/15$) reported their physician had the most influence. However, patient and physicians' priorities differed, with more patients ($n = 9/15$) than clinicians ($n = 1/10$) rating avoidance of hospitalization as 'very important'. Furthermore, patients described a fear of treatment side effects being worse than their symptoms as the reason for not taking treatment, while physicians rarely considered this factor. These divergent priorities highlight the need to increase patient's role in treatment decision-making. However, the suddenness of diagnosis and associated emotional trauma left some patients unable to process information and make decisions. Some patients also perceived a lack of available treatment options.

Patients, their family members and physicians were concordant in their opinion that the patient's health (including age, comorbidities, and being "unwell") played a big part in treatment decisions (reported by $n = 10/10$ physicians, $n = 11/15$ patients and $n = 11/15$ family members). When asked about factors that would influence their decision to agree to treatment, patients prioritized good health-related quality of life, limited treatment side effects, and longer survival, followed by proximity to medical sites. Future analysis will explore how priorities vary (if at all) across countries and medical systems.

The findings underline the key role physicians play in the AML treatment decision-making process, as well as the potential gaps in patient understanding of available treatments. Given stakeholders' differing priorities, the dynamics of stakeholder interactions should be considered when supporting patients and delivering treatment information to ensure that treatment decisions are patient-centered.

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BSH2020-PO-017A

Genetic and genomic characterisation of older adults with acute lymphoblastic leukemia treated on the UKALL14 and UKALL60 + clinical trials

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Acute lymphoblastic leukemia (ALL) is characterised by recurrent chromosomal abnormalities and genomic microdeletions. Although the incidence of adult ALL increases with age, few studies have examined the genetic aberrations in adults aged ≥ 60 years.

The objectives of the study were to define the frequency of recurrent chromosomal abnormalities and characterise the copy number profile of ALL in older adults.

All patients from UKALL14 and UKALL60+ clinical trials aged ≥ 60 years at diagnosis of ALL were considered in the study. Cytogenetic and fluorescence in situ hybridisation (FISH) results at diagnosis were used to classify patients in genetic subgroups: BCR-ABL1, TCF3-PBX1, KMT2A rearranged, low hypodiploidy (HoTr) and high hyperdiploidy (HeH). B-cell precursor (BCP) patients lacking a primary chromosomal abnormality (B-other) were selected for extended FISH screening using dual colour break apart probes for ABL1,

ABL2, PDGFRB/CSF1R, CRLF2, JAK2, IGH@, ZNF384 and MEF2D. Separately, single nucleotide polymorphism (SNP) arrays were performed to identify copy number abnormalities (CNAs). Multiplex ligation-dependent probe amplification (MLPA) was used to confirm CNAs in EBF1, IKZF1, CDKN2A/B, JAK2, PAX5, ETV6, BTG1, RB1, and PAR1.

207 patients aged ≥ 60 years were identified from UKALL14 ($n = 91$) and UKALL60+ ($n = 116$). Median age was 64 years (range 60–83) and 50% were male. Cytogenetic data were available for 86% patients. Frequencies of individual genetic subgroups are shown in table. FISH screening of B-other cases identified rearrangements in CRLF2 (5%), ZNF384 (2%) and MEF2D ($<1\%$). IGH@ translocations were found in 9% (incl 6 IGH-CRLF2). No variant ABL1, ABL2, PDGFRB/CSF1R or JAK2 rearrangements were detected.

SNP arrays were performed on 83 patient samples using Illumina CytoSNP 850k ($n = 52$) or Affymetrix Cytoscan HD ($n = 31$) arrays. Recurrent whole chromosome and whole arm abnormalities seen in >3 cases are shown in the table.

Recurrent deletions were identified affecting IKZF1 (52%), CDKN2A/B (45%), PAX5 (39%), RB1 (22%), ETV6 (21%), EBF1 (19%), BTG1 (12%), CD200/BTLA (16%) and ATP10A (13%). The frequency of individual deletions varied by genetic subtypes. Recurrent novel and less-well described intragenic microdeletions were also seen in COL11A1 (13%), MEF2C (10%), MBNL1 (8%), PTEN (7%), NF1 (5%), LEMD3 (6%), KDM6A (5%). These deletions will need to be validated by a second technique.

In this cohort we confirm $>25\%$ patients have BCR-ABL1. HoTr is rare in younger patients but was the second most common abnormality and good risk genetic subgroups were rare. Additionally, we confirm the rarity of T-ALL in older adults ($<5\%$). Within B-other, CRLF2 rearrangements were the most common kinase activating abnormality (5%), with IGH-CRLF2 in 6/8 cases. ABL-class fusions were absent in this cohort, differing from published studies. IKZF1 and CDKN2A/B deletions were seen in 52%/45% patients. Importantly, these are frequently included in high risk copy number profiles, e.g. IKZF1plus. A number of novel intragenic microdeletions, including tumour suppressor genes, were identified but require validation. Collectively, these data further confirm the high risk nature of ALL in older adults. These patients appear genetically distinct to younger adults with primary genetic abnormality unidentified in a significant proportion. Hence the need for further genomic studies in older adults.

Disclosure of Interest: None Declared.

Education

BSH2020-PO-018

Findings from the UK NEQAS for leucocyte immunophenotyping haematological malignancy bone marrow aspirate assessment programme

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Assessment of bone marrow aspirates is an important part of the haemato-oncology diagnostic pathway. In 2018 UK NEQAS for Leucocyte Immunophenotyping (UK NEQAS LI) introduced the Haematological Malignancy Bone Marrow Aspirate Assessment programme. This programme is intended for both clinicians and laboratory staff working in the area of haemato-oncology and is designed to assess their ability to examine a bone marrow aspirate and report their findings.

The programme issues participants with a digital bone marrow aspirate image on which they are required to enumerate the percentage of blast cells, identify 5 pre-selected cells, provide commentary on the cellularity of the aspirate, haematopoiesis and presence/absence of dysplasia. Participants are then requested to state which further testing they feel is required to fully diagnose the case. Following submission of results, via an online data entry form, data is analysed by UK NEQAS LI and personalised reports are produced for each participant, highlighting how their results compare to the consensus results for the exercise.

For the enumeration of blast cells, analysis of the results submitted for the 8 exercises issued so far found that in 5 of 8 exercises all respondents were within 10% of the consensus value, with 2 participants outside of this limit for exercise 171 802 and 1 outside this limit for exercise 181 902. Greatest variation was seen in exercise 192 002, a B-cell prolymphocytic leukaemia, which gave two major groups of responses with blast percentages of 0–10% reported by 8/22 (36.4%) of participants and an identical number reporting a range of 81–90%.

For the identification of pre-selected cells phase of the exercise, 100% agreement was only seen in 10/40 of the cells (5 cells per exercise and 8 exercises issued). If a consensus level of >50% was used then 37/40 pre-selected cells were identified successfully. Greatest variation was seen in exercise 192 002, a case of B-PLL, where one of the 5 pre-selected cells was identified by participants as 10 different cell types.

When reviewing the results returned for the 7/8 exercises requesting the assessment of cellularity, haematopoiesis and dysplasia (not requested in the first exercise issued) it was found that for cellularity and haematopoiesis an agreement level of >50% was seen in every case, with 100% agreement in participant responses seen in three out of the seven cases. However, when reviewing dysplasia results there were no instances of 100% agreement seen and a level of agreement of >50% was not reached in one out of the seven exercises (exercise 181903).

When identifying which further testing would be useful, flow cytometric testing was deemed necessary by >80% of participants in all exercises. Additionally, the majority of participants also selected molecular testing as necessary, although the nature of the molecular testing varied between participants.

From exercises issued, variation has been seen in blast cell number counting, cellular identification and assessment of bone marrow aspirate cellularity, haematopoiesis and dysplasia. This highlights the importance of ensuring adequate training for any staff performing bone marrow examination in a clinical laboratory and underlines the need for an educational EQA programme to monitor performance and support clinicians and scientists in harmonizing working practices to improve patient care.

Disclosure of Interest: None Declared

BSH2020-PO-019

Optimising diagnosis and management of acute chest syndrome in sickle cell disease through a full cycle audit of adherence to the BSH guidelines

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Acute chest syndrome (ACS) is a sickle cell disease (SCD) complication with a high mortality and morbidity, especially if there is a delay in diagnosis or management. It accounts for 25% of premature deaths in SCD patients.

The aim of this project was to assess adherence to national ACS guidelines in a London teaching hospital trust.

Each audit cycle included 20 consecutive SCD and 10 consecutive ACS patient admissions. Data sources were paper and electronic patient notes and an electronic observations system. Adherence with the following criteria was determined: (i) diagnosis – 4-hourly observations with appropriate escalation, daily chest examination, (ii) investigation – chest x-ray, basic bloods, cultures, ABG, nasopharyngeal aspirate, (iii) treatment – oxygen, fluids, analgesia, incentive spirometry, chest physiotherapy, antibiotics, blood transfusion as indicated, review by HDU/ITU if deteriorating, bronchodilators if wheeze, and hydroxycarbamide after acute admission.

The first audit cycle was in December 2018. Over the next few months we initiated several educational initiatives: (i) teaching for junior doctors (ii) eye-catching “Think acute chest syndrome!” infographic posters in clinical areas, (iii) an update to local SCD guidelines with a simplified ACS management flowchart, and (iv) a conspicuous warning box added to patient protocols to indicate those at high risk of ACS with a reminder to closely monitor them. The second audit cycle was in November 2019.

On re-audit, we found a significant improvement in adherence to the diagnostic criteria. Compliance with 4-hourly observations improved from 30% to 60%, daily chest examinations from 22% to 65%, and escalation of low saturations from 90% to 100%. We maintained 100% compliance with basic investigations for ACS and showed improvements in both ABGs (40% to 70%) and blood cultures (from 60% to 70%). In terms of treatment there were significant improvements in adherence with correct antibiotics (80–100%), blood transfusions where indicated (80–100%) and HDU/ITU review for deteriorating patients (70–100%).

The proportion of patients managed in accordance with BSH guidelines improved between our first and second audit cycles. These improvements can be attributed to our interventions to raise awareness amongst clinicians about the need to monitor for ACS

development in SCD and its appropriate management. This is supported by the results of a pre- and post-teaching survey from junior doctors in which 100% of doctors reported improved confidence in both recognising and treating ACS. After the teaching, 100% of doctors were able to demonstrate awareness of national guidelines (from 30% before teaching) and 100% knew how to access these guidelines (from 15% before teaching).

We plan to build on these improvements by holding regular teaching sessions for junior doctors and the ED team to remind them to be aware of ACS and its appropriate management. We are also considering how to involve patients by ensuring they are aware of their risk of ACS and understand how they can highlight this to clinicians on admission. We will re-audit in 12 months.

Disclosure of Interest: None Declared

BSH2020-PO-020

Graphical description of performance characteristics of laboratories and assays using data from the UK NEQAS for haematinics

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Clinicians may be aware that different Diagnostic Manufacturers assays can differ, but they may not be aware of the scale of these differences nor the changes that can and do occur over time.

The UK NEQAS for Haematinics has been assessing the performance of participating laboratories for many years, but only in the last few years when the scheme operation came under the auspices of Birmingham Quality, part of University Hospitals Birmingham Trust, have participants had their data expressed in the easy to assimilate, graphical rich format that has been used in the Clinical Chemistry EQA domain for some time.

We present here data for Serum Folate which has seen some radical shifts in assay performance, some planned and some less so, which impact on both the numerical values obtained and how that value is interpreted by the laboratory. The scheme operates on the usual UK NEQAS Chemistry model whereby participants receive three specimens every month which they analyse and interpret. The data is entered into a web portal and at the end of the monthly Distribution cycle a personalised report is produced. Data is calculated at the Specimen level as well as an accumulated Rolling Time-Window level synthesising data on a six-monthly moving average.

Data from August 2019 not only highlights the between-method differences at the normal/low boundaries but also how the values are being interpreted. At a Folate overall consensus mean concentration of approximately 4.0 µg/l, we observed that less than 10% of Abbott users considered Specimen 263C to be normal while well over 40% of Roche users did. Another key feature is that even for users of the same method who obtained the same numerical values, different interpretations were made by participants.

The use of long term trend statistics allows us to show a Laboratory's individual history, colour coded by analytical performance; the so-called Bias B score *versus* time 'Seismograph Plot' is a game changer in letting participants see their performance within their group. The complementary method-specific Bias B *versus* Consistency of Bias C 'Penalty Box Plots' show a snapshot of relative method characteristics which allows, with little statistical uncertainty, the true performance of the assay systems. Concentration-dependent bias is also assessed.

The use of graphical representation of data puts into stark images the findings which we hope will generate more of a sense of priority. Anecdotal evidence from feedback from Participant Roadshows and

general Scheme feedback when discussing performance issues with Participants would suggest that participants like and appreciate the displays, though we have not undertaken a formal study. Likewise Diagnostic Manufacturers find the new formats helpful.

We continue to have strong statistical data which underpins the graphs, but the immediacy of the displays direct the Participant to focus on those areas which need addressing and for Diagnostic Manufacturers, data from a high frequency, multi-specimen EQA Scheme provides data that they cannot get from any other source.

Disclosure of Interest: None Declared

BSH2020-PO-021

A review of bleeding events associated with anticoagulants that have led to patient harm with a primary focus on transitions of care

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The potential for medication discrepancies to cause adverse drug events (ADEs) during transition from hospital to care home setting identified anticoagulants to be third most frequent drug to result in ADEs. 30-day readmissions were classified as one of the categories that can be potentially preventable and these admissions could have been avoided if care was altered in the previous hospital contact. Many studies have acknowledged that there is a strong association between anticoagulants adverse events and transition of care but there is no data available in literature to identify the rates of these adverse drug events that are linked to transition of care in United Kingdom. The objectives of this clinical audit were to identify availability of information regarding all oral anticoagulants on transition of care (ToC) documents and to characterize patients who presented within 30 days from their last contact (≤30 days cohort) at Barts Health NHS Trust (BH). Records of patients who presented with a major bleeding event at Royal London Hospital from 2013 to 2016 were traced retrospectively to identify information pertaining anticoagulants that were available on their ToC documents. Themes that were identified from literature review that were pertinent to ToC forms for anticoagulants were information on disease and management, follow-up care and laboratory test results. For patients in ≤30 days cohort, ToC document and case notes were further analysed to identify additional bleeding risk factors that were present. A total of 115 patients presented with major bleeding events to RLH and 29.6% ($n = 34$) of patients were admitted for a bleeding event within 30 days after a last contact with a health care professional at BH. More than 50% of all ToC documents analysed from BH had no information on indication for anticoagulation, target international normalised ratio, duration of anticoagulation treatment and follow-up plan. 58.8% ($n = 20$) patients in ≤30 days cohort did not have any parameters relevant to anticoagulant mentioned in their ToC document. The bleeding risk, measured using modified HASBLED scores, in the ≤30 days cohort was almost double (78.9%) compared to the >30 days cohort (42.5%). In conclusion, availability of information provided on ToC documents is poor and more apparent in ≤30 days cohort. ToC documents should include basic information on anticoagulants and other risk factors that potentiate risk of bleeding to ensure patient receives optimal care.

Disclosure of Interest: None Declared

BSH2020-PO-022

Bolus: education delivered stat podcastS. Thomas^{1,*}, S. Jordan¹, G. Cooper¹¹Haematology, UCLH NHS Trust, London, UK

Introduction: We believe that better informed clinicians make better clinical decisions. Healthcare technology and innovation moves as fast as the rising expectations of patients. Keeping clinical knowledge up-to-date is an ever-present challenge. Often, clinicians' day-to-day experience can only provide a narrow viewpoint of a haematology patient's journey; sometimes this lacks the wider context of new therapies and advancements in treatment modalities. Formal teaching opportunities can be infrequent, difficult to organise, complicated to facilitate and require time and financial commitments many organisations can't support.

Background: Floor-based teaching is a time-tested way of keeping staff updated but it has to be repeated many times to reach a significant number of staff. Technologies like eLearning are expensive and time consuming to design, deploy and maintain; what is needed is a flexible, efficient contemporary solution for continuing professional development. *Podcast* technology, whilst not new, has not been fully explored as a core learning solution in healthcare. Surveying haematology staff ($n = 120$) at University College Hospital in London, 80% of respondents (70, 58%) identified a suggested Podcast education delivery method as a desirable way to keep up-to-date with the latest developments in specialist haematology care.

Methods: A small team of senior clinical staff and practice educators assembled to develop the Podcast resource using inexpensive and easy to use equipment. Clinical experts in particular fields were recruited to deliver content in a conversational style. The format allows a host, to ask questions of the expert based partly on user feedback and partly on historical educational deficits. The resource became formally named, '*Bolus – Education Delivered Stat*' and editions were released via popular podcast applications online. Prospective evaluation surveys were given to the clinical workforce to provide intelligence on *Bolus's* reception with staff and evaluate the effectiveness of the delivery method.

Evaluation: The Podcast team monitors listening statistics, produced by the web hosting service. The response has been unexpectedly good with >15 000 total downloads, from >50 countries around the world in less than 6 months. User evaluation survey data reports improved knowledge and increased confidence across all employee groups. Respondents identified a number of key areas that were reported on including: accessibility, time management, resource utility and content relevance.

Conclusion: The Podcast format is a surprisingly viable, effective and, importantly, cost-effective education solution which has the potential to educate large groups of clinical staff across global geographical locations. The 30-min Podcasts can be listened to anywhere and on a variety of formats and devices.

Disclosure of Interest: None Declared

BSH2020-PO-023

On arterial blood gas: how can we reduce avoidable sampling?A. Pons^{1,*}, N. Jones¹, C. Gerrard¹¹Royal Papworth Hospital, Cambridge, UK

Arterial Blood Gas (ABG) is the most frequently ordered test in the Intensive Treatment Unit. (1–3). As part of the Choosing Wisely initiative (1) which aims to increase awareness of unnecessary medical tests, treatments, and procedures, some authors have described too frequent sampling routines (2,3). These can lead to iatrogenic anemia

and other undesirable events. In this work, we aimed to monitor the number of ABG performed at Royal Papworth Hospital and to understand the reasons behind sampling. Simultaneously, educational actions were undertaken. We wanted to assess the efficiency of such measures in terms of reduction in the number of ABG.

The number of ABG was monitored from October 2018 to October 2019. In September 2019, educational efforts were undertaken to raise consciousness about this issue. In October 2019, we designed and distributed a 24-h survey. The nursing team reported the time of the procedure, 11 checkable possible reasons, and the possibility to include if potential clinical actions were undertaken as a result. Data was collected during six sampling periods. We used a statistical analysis to interpret the resulting information.

In September 2019, 8171 ABGs were performed in the Critical Care Area (CCA) (mean = 30.6 per patient). We analyzed $N = 30$ patients in total. 136 ABG were performed, of which 65 (47.8%) were done between 8 AM and 1 PM, and 16.7%, and 3.7% were sampled at the beginning and at the end of the shift respectively. The three more frequent reason for ABG were (1) to check electrolyte levels (63.3%), (2) to check blood sugar levels (52.7%) and (3) to check lactate levels (28.6%). We noticed that these three reasons were often concurrent. Of note, in 39% of tests, no follow-up clinical intervention was performed. The checking of electrolyte and blood sugar levels often led to clinical action – 86% and 81%, respectively –.

8171 and 9480 ABGs were performed during September and October respectively in the Critical Care Area (CCA). This significant reduction in the number of samplings (*versus* 30000 ABGs in March) coincided with educational strategies, to increase awareness on the need to avoid unnecessary testing. However, 39% of the testing doesn't lead to medical action and a significant number of ABGs remain redundant and therefore avoidable. This data encourages the implementation of ongoing educational efforts directed at further reducing ABG and optimizing indication for sampling.

Acknowledgements: RPH nursing team for completion of the survey

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Disclosure of Interest: None Declared

BSH2020-PO-024

'Guidelines and chips': food for thought in supporting traineesA. Aggarwal^{1,*}, N. Al-Ali¹, R. Frewin¹¹Gloucestershire Hospitals NHS Foundation Trust, Cheltenham, UK

Specialist Trainees are under numerous pressures contributing to burnout. Causes include social isolation from repeated moves during training, significant clinical commitments, and the pressure of studying for professional exams.

Access to training is also impeded by significant Consultant shortages along with an increasing workload. This results in the prioritisation of clinical service provision over teaching commitments.

Within Gloucestershire, 2–3 Specialist Trainees have 6–12 months attachments within the 2 hospitals (Gloucester Royal Hospital and

Cheltenham General), which are 6 miles apart. There are also 3 Speciality Doctors working across both sites along with an Internal Medicine Trainee and a Foundation program doctor, whom are often not present during normal working hours due to other rota commitments. It is therefore difficult to foster a team environment and to deliver regular teaching with such disparate working patterns.

Therefore, a consultant with educational interests proposed a fortnightly evening teaching session within her own home, with attendance purely voluntary as it occurred out of normal hours. It is structured as a social event, with all trainees bringing a dish and the teaching is structured around dinner. The trainees take turns to present a guideline and a seminal paper, after which a relevant essay question for part IMRCPath is set.

We feel that there are many benefits to this style of teaching. Most importantly, it has provided a relaxed social environment aiding team bonding and increasing morale. The majority of the dinner talk is not on work related matters and allows the team members to be viewed as individuals.

Secondly, it has helped to develop critical appraisal and presentation skills across the group. The in depth discussion of guidelines on a regular basis, without time pressures or frequent work interruptions, helps trainees to be familiar with the concepts for use in practice. Various presentation styles have been employed catering to different learning styles and allowing for experimentation with different teaching methods in a comfortable and non-threatening environment, with immediate feedback. This enables the refinement of teaching methods over the course of the year.

These sessions are also invaluable as evidence of continuing professional development (CPD) for specialty doctors.

Finally, the essay practice for those preparing for exams reinforces the teaching done that week and gives them useful essay feedback.

Therefore, although this model does require a commitment of the team to a regular meeting out of normal hours, the benefits both socially and educationally are significant and of benefit to all.

Disclosure of Interest: None Declared

BSH2020-PO-025

CDx report program – assessment of the current reporting practice for haematology companion diagnostics (CDx) markers in the UK and Ireland

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In the era of personalised medicine, CDx (companion diagnostics) has become the basis for optimal patient care. Whilst technical aspects of a CDx assay performance are assessed regularly by participating in external quality assessment (EQA) programs, the reporting of these CDx markers is rarely scrutinised. The aim of this CDx report program was to get a detailed picture of haematology CDx marker reporting in the UK.

We invited 28 clinical labs in the UK and Ireland to participate in the programme. Ten clinical labs provided information for up to seven biomarkers used for treatment decisions in haematology in their laboratory. The 7 CDx markers covered in this CDx report program were: BCR-ABL1 in CML (diagnosis), BCR-ABL1 in CML (MRD), IDH1 and IDH2 in AML, FLT3-ITD in AML, FLT3-TKD in AML, IGHV in CLL, TP53 in CLL. The information requested included two anonymized reports: one with a

mutant result and one with a wild type/negative result. The anonymised reports were reviewed by an expert panel according to pre-agreed review criteria. In addition, labs responded to a short online survey which gathered information about test volumes, turnaround time (TAT), positivity rates, participation in an EQA program, accreditation status and test reimbursement. Overall, we received 49 survey datasets and 46 sets of anonymised reports. A review of the anonymised reports according to pre-defined criteria revealed inconsistencies in the way CDx results are represented and interpreted in clinical reports. Examples include the risk attributed to a result (specifically a FLT3-ITD) which indicates a targeted therapy for that marker, and reference (or lack of) to specific clinical guidelines relating to prognosis. This observation suggests that a programme for more standardised reporting is likely to be beneficial.

Not all markers covered in this program were performed by all participating labs, the number of datasets per CDx marker ranged from 4 to 10 (BCR-ABL1 (diagnosis): 10 datasets, BCR-ABL1 (MRD): 9, FLT3-ITD: 9, FLT3-TKD: 6, IDH1/2: 4, IGHV: 4, TP53: 6). The 49 survey datasets represented approximately 3000 tests per month with an average TAT across all markers of 9 days. In 46/49 (94%) of the datasets, labs participated in EQA programs. For 49/49 (100%) datasets labs were accredited according to ISO15189. For 21/49 (43%) datasets reimbursement reported as sufficient.

An identical CDx report program is currently being performed in 10 other countries including US, Germany and China. The anonymised and aggregated data generated provides the basis for other initiatives and may support guideline updates for the potential international harmonisation of CDx reporting in onco-haematology.

Disclosure of Interest: P. Evans Conflict with: Diaceutics, Astellas and Novartis, D. Wilkinson Conflict with: Diaceutics, J. Mason Conflict with: Diaceutics, Jazz Pharmaceuticals, Novartis and Asuragen, S. Delic Conflict with: Diaceutics, N. Atkey Conflict with: Diaceutics

BSH2020-PO-026

The Haematology Summit 2014–2019: evolution of a national medical education programme

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In 2013, a medical education steering committee (MESC) identified an unmet need for balanced, cross-malignancy medical education among the UK haematology-oncology community. To address this, Janssen established the Haematology Summit a year later, which has since become an annual event in the haematology-oncology medical education calendar. The meeting was developed in collaboration with the MESC to facilitate a balanced exchange of clinical experience and best practice, with the goal of improving standards of care and patient outcomes.

This analysis explored the methods by which an educational meeting of this nature can be adapted and expanded to ensure it continually addresses the needs of the multidisciplinary team (MDT). The Haematology Summit programme has evolved each year to reflect changes in the therapeutic landscape, and to incorporate both quantitative and qualitative delegate feedback. These adaptations involved addition of new topics and tumour types, enhanced interactivity via different session formats and new technology, an increase in faculty diversity and coordination with patient advisory

groups. Discipline-specific sessions, including the Pharmacy Academy and Nurse Networking Forum, have also been introduced to provide the MDT with focussed educational content. As a result, attendance at the Haematology Summit increased from 66 delegates in 2014 to 210 in 2019, with over half of all delegates returning to attend more than one meeting. Feedback has remained consistently high since 2014, with delegates scoring their likelihood of recommending the Haematology Summit to a colleague as a mean of 9.2/10 ($n = 570$). The quality and relevance of the scientific content received mean scores of 9.4 ($n = 508$) and 9.1 ($n = 513$), respectively, and objectivity and balance received a mean score of 8.9 ($n = 503$). Overall delegate satisfaction, as measured by the Net Promoter Score, increased from 54 in 2014 ($n = 57$) to 86 in 2019 ($n = 127$).

In conclusion, the Haematology Summit has demonstrated the value in holding a national medical education meeting that covers multiple therapy areas in a way that is accessible and relevant to the entire MDT. This is supported by consistently positive feedback and an increase in attendance over 6 years of the meeting. Continuous revision and evolution of educational meetings like the Haematology Summit, in response to changing therapeutic landscapes and delegate feedback, is appropriate to ensure that unmet educational needs are satisfied year-on-year. Moving forward, future meetings should continue to address the most pertinent topics in the field, and embrace new technologies and digital platforms to further engage with participants.

Disclosure of Interest: N. Pryn Conflict with: Janssen employee, G. Jackson Conflict with: Honoraria, J. Gribben Conflict with: Honoraria, G. Follows Conflict with: Honoraria, L. McDougall Conflict with: Employee

BSH2020-PO-027

An audit on blood picture request forms submitted to haematology laboratory in a tertiary care hospital in Sri Lanka

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Modern clinical practice has become more and more dependent on reliable laboratory results, thus diagnostic errors due to inaccurate laboratory results can lead to serious consequences including death. Recent studies have clearly demonstrated that 70% of laboratory errors occur in pre analytical phase which comprises all the procedures from the time a physician has made the decision to request for a certain laboratory test until the specimen is analyzed at the laboratory. Properly filled request form will aid in delivering an accurate laboratory report, hence better and improved patient care.

Thus we have carried out an audit in order to evaluate the blood picture request forms received to the haematology laboratory in Colombo South Teaching Hospital Colombo to determine the frequency of incompletely filled request forms and the information that has been overlooked.

The blood picture request forms received at haematology laboratory between 17th September to 23rd of October 2019 were retrospectively analyzed to measure the compliance of referring clinicians in adequate completion of request forms. Colombo South Teaching Hospital is an academic, tertiary hospital in the Western province of Sri Lanka. The haematology laboratory receives specimens from both in and outpatient sections of the hospital. The information on request forms were entered into a Microsoft excel sheet and analyzed. Each form was assessed for the presence and completeness of the information requested therein: demographic details, the type of

investigation required, a clinical history (defined as a clinical history and/or differential diagnosis), designation and signature of the requesting clinician, and request date. These data should be present on 100% of requests if completed correctly.

During this study period, 1000 request forms were received and the results obtained are as follows.

No clinician's designation was entered in 223 forms (22.3%) and 15 (1.5%) had no signature of the clinician. Gender details were not recorded in 102 (10.2%) and ward was not written in 18 forms. (1.8%). Eight forms (0.8%) did not have the Bed Head Ticket number (BHT).

Only one form was found with no name and gender recorded on it.

Clinical history was not given in 286 request forms (28.6%). Out of 714 forms where adequate clinical information was provided 32 forms (4.4%) had non standard abbreviations. Six forms had illegible handwriting (0.6%).

Nine hundred twenty three forms did not specify the specimen type collected (92.3%) and 3 forms did not have the date (0.3%). Thus out of 1000 request forms we have analyzed, 426 forms (42.6%) were incompletely filled.

Even though many studies have been carried out to emphasize the importance of pre analytical phase in laboratory testing only few have been done to find out the frequency and impact of incomplete laboratory request forms. Colombo South Teaching Hospital do not have an established ward based electronic requesting and the investigations are ordered via manually filled laboratory request forms. Incompletely filled forms may affect the diagnostic processes including results interpretation by the haematologist. Even though this audit is limited only to blood picture request forms received by a haematology department in a teaching hospital, it is likely that similar results would have been possible in other departments and hospital set ups where ward based requesting is not established properly like ours.

Disclosure of Interest: None Declared

BSH2020-PO-028

Education and training of healthcare scientists in haematology - fit for the future?

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Over the last 60 years Haematology & Transfusion Science laboratories have undergone major changes. However, the current service is a mishmash of poorly connected specialist and general laboratories. The historical evolution of the NHS laboratory service has resulted in a lack of concordance in training and career progression for Healthcare Scientists (HCS) nationally within Haematology & Transfusion Science. This results in the loss of HCS from the workforce and stagnation and under use of HCS with potential to become specialists of the future. HCS includes Biomedical Scientists (BMS) and Clinical Scientists (CS). Haematology & Transfusion Science historically employs BMS with CS confined to specialised laboratories. This matters as there is an established route to Consultant CS but scientific career progression as a BMS is difficult with able BMS often 'lost' to senior laboratory management roles. I hoped the 2010 education and training strategy for HCS, 'Modernising Scientific Careers' (MSC) would remove these artificial barriers by introducing a national curriculum and harmonising career pathways. However, the Scientist Training Programme (STP) (National School of Healthcare Science (NSHCS)), a highly competitive post-graduate programme that attracts the brightest remains under-utilised in Haematology. There are perceptions that there is no role for CS in

general haematology and confusion about how to use these knowledgeable CS. At a time when there is a workforce shortfall of medical Haematology & Transfusion Consultants there should be an expansion of CS undergoing Higher Specialist Training (HSST) with clear recognition of their role in the profession. HSST is a doctorate level programme and is jointly managed by the NSHCS and the Royal College of Pathologists (RCPath). HCS training is delivered by a 'confusion' of routes. BMS training (Institute of Biomedical Science (IBMS)) has basic (BSc) and advanced level (MSc) training with assessment based on a portfolio of evidence and viva which is subjective. CS training (NSHCS) is MSc level, and is similar to advanced BMS training. CS follow a national curriculum and have an objective structured final assessment (OSFA). An apprenticeship route is under development that leads to HCPC registration as a CS – assessment methods are uncertain. HCS wishing to progress to Consultant grade must be CS. These highly trained BMS must undertake 'Equivalence' to the national CS STP under the remit of the Academy of Healthcare Science (AHCS) or the IBMS who also offer an experiential route. This is a significant burden of work for the BMS who generally has to navigate the system alone with acceptance of equivalence status subjective. In conclusion, if we want a Haematology & Transfusion Science scientific workforce fit for the next 60 years we need to rethink our approach to the education and training of HCS. We cannot risk this valuable human resource because of artificial barriers and historical division. Education and training of HCS routes must be simplified. There should be agreed national curricula and objective assessment that all routes share. Existing professional bodies need to co-operate with joined up approaches to objective assessment and clear benchmarks for career progression. This will generate a larger pool of eligible staff for HSST which would ameliorate some of the medical workforce issues. The abstract contains the personal views of the author (SO'C).

Disclosure of Interest: None Declared

BSH2020-PO-029

Knowledge, perception and opinion of health workers on bone marrow transplantation in public sector hospital

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Bone marrow transplantation (BMT) cures Haematopoietic disorders, poorly Functioning Bone Marrow, Primary immunodeficiency syndromes and many Oncological diseases.

It's a sophisticated and highly skilled treatment. The infusion of Haematopoietic stem cells generally follows a Conditioning regimen consists of drugs to create marrow space, suppression of recipient's immune system and to prevent rejection by the host. It is a multidisciplinary procedure, which requires the services of many health care personnel such as medical doctors in various specialties, nurses, pharmacists, psychologists, technicians, dieticians, radiographers, social workers etc involved in the management of patients during the management.

But unfortunately the poor understanding of the health care providers and administration and the limited resources of the country are the hurdles for establishment of BMT units in Public sector Hospitals.

Bone marrow transplantation (BMT) cures Primary Haematopoietic disorders, Primary immunodeficiency syndromes and many

Oncological diseases. It's a sophisticated and highly skilled treatment. It is a multidisciplinary procedure, which requires the services of many health care personnel such as medical doctors in various specialties, nurses, pharmacists, psychologists, technicians, dieticians, radiographers, social workers etc involved in the management of patients during the management. But unfortunately the poor understanding of the health care providers, administration and limited resources of the country are the hurdles for establishment of BMT units in Public sector Hospitals.

This study was carried out to enhance the basic knowledge of Health care providers and determine their current level of knowledge, preparedness and perception about BMT.

This was a single center cross-sectional study among health care workers in Pediatric tertiary center, Children Hospital, Lahore, Pakistan. Ethical committee approval was taken and a questionnaire was administered to employees aged more than 18 years during 1 week duration. The questionnaire has 10 questions and each question scores 0–5. Score were not mentioned on questionnaire provided to Participants. We considered persons having ≥ 45 score well aware, 35–44 some aware, 25–34 poorly aware while < 25 were considered unaware. Each person's identity and marks were kept confidential.

A total of 100 participants including 40 doctors, 25 nurses, 8 pharmacists and pharmacy technicians, 10 laboratory scientists and laboratory technicians and 17 other health care workers such as dietitians, social workers and radiographers participated in the study. Our study shows 55% well aware, 23% some aware, 14% poorly aware and 8% were considered unaware about BMT. The highest level of awareness was among doctors. The mean knowledge score of all participants was 40 ± 2.06 .

The Health care providers have limited knowledge of the Bone Marrow Transplantation. Most of the Personnel consider it a Surgical procedure and to be done by the surgeons. Even few Doctors were unaware and had poor knowledge.

This study elevated the basic information and knowledge of health care personnel on BMT that helped us in the establishment of first ever Public sector Bone marrow Transplant Center in Pakistan. We recommended Continuous Medical Education in Pakistan for every health care provider about new innovations and Advancements in the health sector.

Table Knowledge, Perception and Opinion of Health Workers on Bone Marrow Transplantation in a Public Sector Hospital

1	What is meant by Bone Marrow Transplant?
2	Who does the Bone Marrow Transplant?
3	Is BMT Surgical or Medical Management?
4	What are indications where BMT is indicated?
5	How much each routine BMT cost in PKR in Pakistan?
6	Is BMT available in Pakistan?
7	Is BMT possible to be done in Public sector Hospital?
8	What is basic need of BMT?
9	Should we open BMT center in our hospital?
10	How many centers should be in Pakistan?

Disclosure of Interest: None Declared

BSH2020-PO-030

A different kind of OOPe, volunteering to improve haematology services in MalawiM. Player^{1,*}¹*Haematology, University Hospitals Of Leicester, Leicester, UK*

For my OOPe I spent a year volunteering at Queen Elizabeth Central Hospital (QECH), Blantyre, Malawi to assist with the development and provision of Haematological services

QECH is a 1000-bed tertiary referral hospital for Southern Malawi serving a population of at least 7 million people. Currently in Malawi there are only two Haematologists. One is based in Lilongwe and one at QECH in Blantyre, serving a population of over 18 million people between them.

The need for Haematology input is extensive with a diverse range of presentations including; anaemia, pancytopenia, PUO and hepatosplenomegaly. The burden of HIV is extremely high.

During my time in Malawi I focused on the area of diagnostics, as suitably trained morphologists are in extremely short supply and this appeared to be the area of greatest need. As well as reporting samples this also involved training and supervising laboratory staff.

A diverse range of Haematological diagnoses were encountered including large numbers of Haematological malignancies such as CML, lymphoproliferative disorders (CLL, Tropical Splenic Lymphoma, Hodgkin's, Burkitt's, DLBCL), ALL, AML and Myeloma.

Another major clinical need was the investigation of cytopenias. The causes of these are diverse; infectious (TB, HIV, salmonella), drugs, hypersplenism, liver cirrhosis, haematinic deficiency (iron/megaloblastic anaemia) and primary haematological disorders (myelodysplasia, aplastic anaemia and fibrotic marrows).

I was also exposed to Haematological diseases rarely encountered in the UK including: Tropical Splenomegaly Syndrome and Castleman's disease.

I also undertook General Medical ward rounds, led Haematology clinics, provided Liaison Haematology input to ward patients and provided teaching to Medical students and Doctors on a range of Haematology subjects.

Managing Haematology patients at the hospital was particularly challenging given the limited resources available;

Treating VTE is extremely difficult as Warfarin is the main available anticoagulant but access to safe treatment and testing is severely limited

- Blood Products for Transfusion are in very short supply
- Minimal Laboratory Diagnostics exist with no access to: haematinics, coagulation screens, HPLC or serum electrophoresis
- Only a small range of antibiotics are available
- Cross sectional imaging is rarely available
- The range of Chemotherapy drugs is limited

I feel the experience has significantly enhanced my skills as a Haematologist developing my;

- Blood film and Morphology competencies
- Laboratory Management understanding
- Clinical Acumen
- Ability to maximise available resources
- Understanding of basic pathophysiology, investigation and management of Haematological diseases
- Understanding of pathophysiology and management of infectious diseases/tropical medicine
- Teaching skills

Following my visit, I have continued my involvement with the hospital; this has included helping with production of guidelines and training documents, undertaking further education projects, arranging exchange visits between colleagues in the UK + Malawi and

seeking further resources for the hospital. I have also joined the BSH Global Haematology SIG.

I hope through sharing my experience this will hopefully encourage others within our society to consider pursuing similar activities to support the provision of Haematology in low income settings where there is a massive need.

Disclosure of Interest: None Declared

BSH2020-PO-031

Haembase: improving the quality of a free open access medical education resource targeted at haematologists in trainingT. Bull^{1,*}¹*Haematology, Cambridge University Hospitals NHS Foundation Trust, Cambridge, UK*

The term Free Open Access to Medical education (FOAM) was coined in 2012 by the team behind the emergency medicine project Life in the Fast Lane. Since then, haematologists have come to benefit from a number of excellent free digital resources, some of the author's favourite examples being TeamHaem, Buku Haematology & Practical-Haemostasis. A recent addition to the FOAM community is www.haembase.com, a website launched in June 2018 with a specific focus on haematology registrars preparing for their fellowship examinations.

The aim of this quality improvement project was to identify, and then implement, the necessary changes required to address the learning needs of Haembase's users.

An open invite survey was advertised on Haembase and the accompanying @TomboxaneA2 Twitter account in April 2019. The survey asked nine questions, covering the respondents' demographics, opinion on the value of Haembase, and their priorities for changes required to the site. Following the implementation of suggested improvements, a second open invite survey was launched in September 2019 to review the response to these changes.

The April survey received 50 responses. In this group, the typical visitor was a haematology registrar (66%), living in the UK (74%), and visiting Haembase in order to prepare for the Fellowship of the Royal College of Pathologists (FRCPath) Part One examination (44%). The survey also received responses from visitors living in Egypt, India, Oman, Pakistan, Saudi Arabia and Sri Lanka. 98% of respondents found what they were looking for, and would recommend the site to others. Respondents were divided on their preferred priority upgrades to Haembase but a request for additional Part 1 examination style practice questions was the most common choice (24%).

In response to the survey, the number of practice part 1 style questions was increased from 17 to 33 and new pages discussing ways to prepare for the FRCPath examinations were added to the site.

The repeat survey in September received 60 responses. The demographics of the typical visitor were the same as in April, and an additional question found that 73% of respondents were visiting Haembase more than once per week.

34 (57%) of the respondents had completed both surveys. Of these, 8 (24%) found much improvement, 19 (55%) some improvement, and 7 (21%) no change in the site since April.

The responses to both surveys confirm that there is a desire for quality revision materials and that Haembase is providing this worldwide in a free, open access digital format. The online world of medical education has the power to keep content up-to-date with the latest in clinical and laboratory practice without delay. However, this also poses the risk of getting lost in a sea of new knowledge.

Education

Table. If you completed the first survey in April this year, and have returned to the site since then, what do you think of the changes?

Response	No. of respondents	Percentage (%)
I have noticed much improvement	8	13
I have noticed some improvement	19	32
I had not noticed any change	7	12
It has got worse	0	0
I did not complete the April survey	26	43

Targeted use of visitor feedback helps to ensure that time put into maintaining a FOAM resource is well spent and applicable to its users' needs. Finally, in the spirit of ongoing quality improvement for the site, the author would welcome contact from any haematology colleagues interested in joining or contributing to Haembase.

Disclosure of Interest: None Declared

General Haematology

BSH2020-PO-032

Serum ferritin and transferrin saturation are not reliable predictors of mild to moderate iron overload detected by R2-MRI in patients with non-HFE mutated hyperferritinaemia

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Increasing numbers of patients with high serum ferritin are referred from primary care to haematology clinics. While iron overload can result from genetic haemochromatosis and secondary causes such as alcoholic liver disease or steatohepatitis, ferritin is also an acute phase reactant that can confound its utility as a marker of iron stores. In hyperferritinaemia due to homozygous HFE gene mutation, BSH guidelines recommend venesection targeting a ferritin <50 mcg/l, but the management of HFE-negative hyperferritinaemia is less clear. It is difficult to ascertain if a raised ferritin represents true iron overload in non-transfusion dependent patients, though a high transferrin saturation is often considered a marker of excess iron stores. BSH guidelines suggest patients with a normal transferrin saturation and serum ferritin <1000 µg/l may be discharged back to primary care with lifestyle advice. Our aim was to ascertain whether there is a correlation between serum ferritin, transferrin saturation and liver iron concentration (LIC) as measured by R2-MRI (Feriscan®) in non-HFE mutated hyperferritinaemic patients, and the degree of iron overload.

Thirty-five patients were identified at the Trust to have had a R2-MRI of the liver between 2010 and 2019 without genetic haemochromatosis, transfusional iron overload or significant haemoglobinopathy. We retrospectively reviewed R2-MRI reports, serum ferritin and transferrin saturation at the time of the scan, and any documentation of other factors influencing ferritin levels such as inflammation, liver disease, metabolic disorders, or excess alcohol.

51.4% (18/35) of hyperferritinaemic patients had evidence of iron overload by R2-MRI (>1.8 mg/g dry liver tissue). The remaining 48.6% (17/35) had no evidence of excess iron. In iron overloaded patients, the mean LIC was 4.41 mg/g dry liver tissue ranging from 1.9 to 19.7 mg/g dry liver weight. In non-iron overloaded patients, the mean LIC was 1.28 mg/g (Range: 0.5–1.7 mg/g). The mean serum ferritin in the group with excess LIC was 1190 mcg/l and mean transferrin saturation was 40.4% (Range: 20%–84%). There was no statistically significant difference in mean ferritin and transferrin saturation between patients with and without excess LIC. However, amongst the iron overloaded patients as determined by R2-MRI, there was a positive correlation between serum ferritin level and LIC ($R^2 = 0.3228$, $P = 0.0139$), but no correlation was found in the non-iron overloaded group.

These results show that in patients with mild to moderate iron overload, serum ferritin and transferrin saturation do not predict the presence or absence of iron overload. This suggests that either R2-MRI or T2* MRI should be used to determine whether there is true iron overload to guide further management. While potential end-organ damage in these patients may not be comparable to those with non-transfusion dependent thalassaemia, guidelines of the International Federation of Thalassaemia recommend chelation therapy if LIC is >5 mg/g. Only a few of our patients met this criterion but

long-term follow up may be appropriate to assess for increasing iron accumulation. Studying a larger cohort over a longer follow-up period is required to assess the true significance of mild to moderate iron overload in this group of patients.

Disclosure of Interest: None Declared

BSH2020-PO-033

Suggestions for the monitoring and follow up of patients with systemic mastocytosis based on a single centre tertiary experience

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There are no agreed guidelines for the monitoring and follow up of patients with systemic mastocytosis (SM). We undertook a review of our cohort of SM patients to determine trends in their follow up to help address this. Patients were identified from clinics between May 2018 – April 2019. Hospital and laboratory records were used to collect demographic details, blood results including molecular profiles, bone mineral density (BMD) and abdominal ultrasound (US) results. Serial mast cell tryptase (MCT) levels were collected with a minimum interval of 3 months between results. The trends and outcomes were documented and analysed to formulate suggestions on the monitoring and follow up these patients. Our SM cohort consisted of 22 patients (male = 9, female = 13) with a median age of 59 years (range 28–90 years). Their diagnoses were indolent SM ($n = 13$), SM with associated haematological neoplasm ($n = 3$), smouldering SM ($n = 2$) and aggressive SM ($n = 2$). An additional entity known as pre-diagnostic SM ($n = 2$) was also included where the WHO criteria for SM were not fully met. MCT levels at diagnosis were available for 21/22 patients with a median value of 50.0 ng/ml (range 6.3–550 ng/ml). The diagnostic MCT was unavailable for 1 patient as their diagnosis pre-dated our electronic records and was not documented elsewhere. 17/22 were found to harbour c-KIT mutations (D816V $n = 16$, D816Y $n = 1$). 2/22 were negative for c-KIT mutations and results were unavailable for 3/22 (diagnosed in other regions or pre-dating our current molecular laboratory records). Cutaneous involvement was seen in 16/22 patients. Follow-up data was available for 20/22 patients with a median follow up of 58.9 months (range 7.7–178.3 months). Across all patients and diagnoses there was an average increase in MCT levels of 1.7% with an average interval of 255 days between measurements. 17/20 patients had a baseline US with only 1 patient found to have radiological splenomegaly. Of those who had a normal US result at baseline, 9 patients went on to have serial scans (total number of scans per patient range 2–3). Of these, no patients went on to develop radiological splenomegaly. 18/20 patients had baseline BMD recorded. 6/18 patients were found to be osteopenic and 3/18 were osteoporotic. Of the 9 patients with normal BMD scans initially, 5 had serial scans (total number of scans per patient range 2–4). Of these, no patients developed osteopenia or osteoporosis. In our cohort 3/20 patients had procedures requiring general anaesthetic. They were given a standard pre-operative prophylaxis regime to minimise the risk of anaphylaxis without any adverse outcomes. 4/20 patients had a history of anaphylaxis. 5/20 patients had repeat prescriptions for Epi-pens. Of the 15/20 patients without regular Epi-pens, there were no

episodes of anaphylaxis during their follow up. Based on these findings we suggest SM patients have a baseline clinical assessment and examination for organomegaly (especially spleen size) with standard bloods including MCT levels. An abdominal US for spleen size and BMD should be performed. Thereafter, we propose a standard annual review (shortened to 6 monthly at clinician's discretion). Interval US should only be performed if clinically indicated. BMD is performed 3 yearly. We recommend an Epi-pen if significant mast cell mediator systems or history of allergy / anaphylaxis. A standard pre-procedural anaphylaxis prophylaxis protocol should be followed.

Disclosure of Interest: None Declared

BSH2020-PO-034

A study of Telomere length as a predictor for multi-system complications and outcomes in bone marrow failure disorders

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Telomereopathies predominantly cause bone marrow failure (BMF) but variably cause other systemic disorders, with variable penetrance. However telomere length (TL), as an independent marker of systemic complications in BMF, with or without mutations in telomere genes (TGC), is not well characterised. We systematically review a large cohort of 472 patients in a single centre with BMF using TL and TGC analysis as a discriminator, to study risk of manifesting with liver/lung and skin complications, cancer predisposition and likelihood of a family member presenting with BMF.

We screened 1060 consecutive patients at a single centre from the years 2010–2018, with BMF or unexplained cytopenias for telomere length (TL) analysis using a multiplex qPCR. 472 (44.5%) patients had TL less than the 25th centile, of whom 243 had <1st centile, 122 had <10th and 107 had TL between the 10–25th centile. These 472 patients underwent TGC mutation analysis on a customised panel of 12 TGC genes (*TERT*, *TERC*, *DKC1*, *TINF2*, *NHP2*, *NOPI0*, *RTEL1*, *CTC1*, *USB1*, *WRAP53*, *ACD* and *PARN*) using deeply parallel sequencing and were studied alongside their clinical parameters for multi-system complications and overall survival (OS).

The median age and gender across the 3 cohorts studied (TL 10–25th centile, <10th centile and <1st centile) were comparable, with a median age of 43 for the entire cohort. 127 (26.9%) of patients were detected to have a total of 142 mutation in the TGC.

A third of patients with BMF and TL < 1st centile had a TGC mutation, but the prevalence did not statistically differ with the other cohorts of TL < 10th centile and between 10–25th centile. Patients with TL < 1st centile can have mutations in 2 telomere genes, but this is less frequently seen when TL < 10th centile and not seen in the cohort with TL between 10–25th centile.

TL does not correlate with the manifestation and presence of liver, respiratory and skin problems of telomere disease. TL in BMF does not associate with increased predisposition to cancer, as their presence was not statistically significant across the 3 groups.

Disease manifestation with BMF, haematological indices, need for blood transfusions, presence of a PNH clone and karyotypic abnormalities, including complex and monosomal karyotype were not dissimilar across the 3 TL cohorts.

Not surprisingly, there were more family members with features of cytopenias and BMF, on screening of patients with a confirmed TGC mutation.

TL did not predict a difference in treatment strategies used across the 3 groups, although more patients with a TGC mutation received anabolic steroids.

The risk of clonal evolution to PNH or MDS/AML and OS was again similar across the 3 TL cohorts with an OS of 93% at 1078 days follow-up for the entire cohort.

We conclude TL can reliably be used as a screening tool to investigate patients for TGC mutation analysis in patients with BMF. Heterozygous state mutations in *TERT* are the commonest, and can be associated with mutations in other TGC genes, particularly *RTEL1* and *TERC*. This may cause a compounding factor in shortening the TL further. However, TL < 1st centile does not associate with more severe cytopenias or predict more multi-system manifestation or increased cancer pre-disposition. Indeed, in our cohort it failed to show an increased risk of clonal evolution or decreased OS. Telomere biology is a dynamic process and may require the interplay of other factors as a second hit for multi-systemic disease predisposition.

Disclosure of Interest: None Declared

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The impact of immune thrombocytopenia (ITP) on activities of daily living: findings reported by UK patients and physicians from the ITP World Impact Survey (I-WISH)

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ITP is an immune disease that affects up to 4000 patients in the UK, and is characterised by a reduced platelet count, clinically defined as a platelet count of less than $100 \times 10^9/l$. Treatments aim to elevate patients' platelet counts and maximise quality of life (QoL). The I-WISH study captured perceptions from patients and treating physicians on treatment patterns in ITP and their perceptions on treatment goals.

I-WISH is a cross-sectional survey of ITP patients and physicians across 13 countries. Patients have been recruited via both patient support groups (ITP Support Association, UK) and physicians. Participating patients and physicians completed a 30-min survey which captured demographics, diagnosis pathway and symptoms, impact of ITP, treatment patterns and patient-physician relationship and a steering committee of expert physicians and patient advocacy ITP specialists designed and endorsed the survey materials. Patients were asked to complete the ITP Life Quality Index (ILQI), which assessed the burden of their disease on various aspects of their daily life. Physicians were also asked to indicate their perception of how ITP affected patients' lives, including impact on physical activity, sex life and travel plans on a 7-point Likert scale (7 is a great deal).

31 physicians and 120 patients completed a survey in the UK. Mean (SD) patient age was 55 (14.17) years and of patients were female (68%). Patients reported having been diagnosed for 11.4 (10.25) years and 69% of patients described their current health state as ≥ 5 using a 7-point Likert scale (7 is excellent health). All physicians were specialists in either haematology or haematology-oncology and reported an ITP caseload of 47.5 (39.22).

Patients reported ITP to have a high impact on their ability to be physically active, with 73% indicating ITP at least 'sometimes' impacts their capacity to exercise. Physicians align with patients' perception of the impact of ITP on physical activity; 70% reported high impact on overall physical exercise, 60% impact on moderate

exercise and 84% impact on intense exercise. Patients also believe ITP to impact on sexual activity, with 46% indicating their condition affects their sex life at least 'sometimes'. 58% physicians believe ITP has a negative impact on sexual activity. Further, the majority of physicians (81%) perceive ITP to have a high impact on patients' travel plans due to concerns around bleeding and most believe ITP has a high impact on patients in relation to travel insurance and taking medication abroad (84% and 74% respectively). 84% patients also believe ITP to at least 'sometimes' impact their energy levels and 68% report ITP to at least 'sometimes' affect their ability to undertake everyday tasks. Physicians have a similar perception; 67% indicate ITP to have a high impact on daily activities.

Overall, data from this study demonstrate the wide-reaching impact of ITP on patients' lives, such as the ability to exercise, sexual activity and travelling. Patients and physicians agree that ITP has a substantial impact on patients' ability to perform activities of daily living, highlighting the need for greater disease control.

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Prolonged corticosteroid use in newly diagnosed primary immune thrombocytopenia

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Corticosteroids are the standard initial treatment for symptomatic or high bleeding risk immune thrombocytopenia (ITP). The high initial response rates must be balanced against a significant profile of adverse effects. With the advent of proven alternative therapeutic options for the treatment of ITP, international guidelines and consensus statements have consistently recommended treatment regimens that would reduce corticosteroid use, both with shorter total courses and more rapid weaning schedules. Current recommendations from Provan et al⁽¹⁾ suggest a maximum corticosteroid course of 56 days, with weaning from no later than 21 days for responders and 14 days for non responders. A maximum daily dose of 80 mg is advocated.

Furthermore, British Society for Haematology guidelines⁽²⁾ recommend an assessment of the risk of osteoporosis with a FRAX score for all patients aged greater than 40, as well as bisphosphonate treatment for all patients greater than 70 or younger patients at high risk of fragility fracture.

We studied the use of corticosteroids for all newly diagnosed primary ITP patients at a tertiary referral centre between July 2017 and June 2019. Patients with a documented contraindication to corticosteroid use, or with secondary, pregnancy related or mild ITP without a treatment indication were excluded. Of 44 newly diagnosed patients, 19 were treated with corticosteroids as their first line therapy. A response, defined as an improvement in the platelet count to more than $50 \times 10^9/L$, was observed in 89.5% of patients and 24% of responders relapsed on subsequent weaning. The recommended maximum steroid dose of 80 mg was exceeded in 10.5% of patients. Of those above 40 years of age, no patients had FRAX scores calculated and for patients above 70, only 25% were given a bisphosphonate.

For responders, the mean initial corticosteroid course was 116 days (range of 18–373 days). For these patients, corticosteroid weaning was commenced at a mean of 14.9 days into treatment

(range 3–60 days). A corticosteroid steroid sparing agent was used in 65% of patients after a mean interval of 20 days with mycophenolate mofetil the most common choice (63%).

For non-responders, the mean corticosteroid course was 61 days and weaning began at a mean of 9 days (range 28–93 days). Second line treatment was introduced at a mean interval of 21 days.

These data demonstrate that prolonged courses of corticosteroids continue to be given for primary ITP with insufficient assessment of steroid related side effects such as osteoporosis. Corticosteroid weaning is commenced within a good time frame for the majority of both responders and non-responders. The majority of responders receive a steroid sparing agent, whilst non-responders have second line treatment introduced promptly. Further efforts are required to minimise total corticosteroid doses and mitigate their side effects, both on a local and also likely on a national scale.

References: (1) Provan et al., Updated International Consensus Report on the investigation and management of primary immune thrombocytopenia, *Blood Advances* 2019;3:3780-3817.

(2) Hill et al., The prevention of glucocorticoid induced osteoporosis in patients with immune thrombocytopenia receiving steroids: a British Society of Haematology Good Practice Paper, *British Journal of Haematology* 2019;185:410-417.

Disclosure of Interest: None Declared

BSH2020-PO-037

Patients' and health care professionals' perspectives on treatment goals in sickle cell disease – a UK perspective: results from the International Sickle Cell World Assessment Survey (SWAY)

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Treatment of Sickle Cell Disease (SCD) focuses on the prevention and treatment of symptoms and complications. However, the treatment goals of patients and their treating healthcare professional (HCP) has yet to be explored. Here we describe these goals from both the patient and HCP perspective in the UK.

SWAY was a cross-sectional survey of 2145 SCD patients and 365 HCPs across 16 countries. Here we present data from UK patients, who were recruited via their treating HCPs and the patient charity – The Sickle Cell Society. Patients aged 6 years and up were eligible to participate with data collected using self-completed or proxy surveys. Patients and HCPs were asked: 'Other than a cure for SCD, what are your 3 most important treatment goals?', with 1 being the treatment goal considered the most important. Agreement and satisfaction questions were asked based on a 7-point Likert scale with high agreement and satisfaction scores = 5–7.

299 patients from the UK completed the survey; 56% were female, mean (SD) age was 29.2 (12.4) years. 30 HCPs from the UK completed the survey; 23% were SCD-centre based, with a mean (SD) current SCD patient caseload of 76 (138).

The most common treatment goal for both patients and HCPs was 'improving quality of life', 69% and 80% respectively, with 'improvement in overall symptoms' stated 34% and 37% accordingly. However, there were some differences with 57% of patients selecting 'preventing worsening of my SCD' compared to 10% of HCPs selecting the comparative statement; 40% of HCPs selecting 'reducing

chronic pain levels' compared to 12% of patients, and 37% of HCPs selecting 'reducing VOCs' compared to 21% of patient.

Overall, 53% of patients stated their agreement that their current treatment approach was helping them achieve their treatment goals. Most HCPs (73%) selected high agreement that patients understand treatment goals and 87% high agreement that they discuss/agree on treatment goals with patients.

46% of patients strongly agreed that their main HCP had similar management and treatment goals to them. 51% of patients were highly satisfied with their main HCPs' understanding and support of their treatment goals, and 50% were highly satisfied with their consideration of their needs when designing treatment goals.

These results are the first of their kind within the UK and indicate that patients and HCPs align on overall treatment goals in SCD with reference to 'improving quality of life' and 'improvement in overall symptoms'. However, differences that exist demonstrate further need for alignment in helping achieve optimal treatment strategies. The disparity between patients and HCPs in the goal 'preventing worsening of SCD' highlights the need for greater emphasis on listening to, and communicating better with, patients to understand their perception of the care they receive and achieve better alignment on collective goals of their disease management. Greater alignment on treatment goals will help patients and HCPs improve overall disease management and enhance patients' quality of life. These findings call for a review of the barriers to successful communication with patients in order to investigate the most effective methods to adopt. This may involve redesigning patient information resources and investigating the role of mobile health tools, e-learning or social media to reach the different sectors of clients.

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BSH2020-PO-038

Iron deficiency diagnosis in pregnancy using reticulocyte haemoglobin equivalent

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In the U.K., approximately 90% of anaemic pregnant women suffer from iron deficiency (ID). This can lead to maternal mortality, premature delivery and low birth weight. However, laboratory tests offered for antenatal ID assessment in current U.K. practice are limited to full blood count (FBC) and ferritin for those with a known haemoglobinopathy or considered at high risk of iron deficiency. Reticulocyte haemoglobin equivalent (RET-He), the amount of haemoglobin (Hb) in reticulocytes, reflects functional iron currently available for haemoglobin production and could be added to antenatal FBC samples with minimal additional work. Studies have shown the clinical utility of RET-He as a discriminator of ID in non-pregnant patients but have not been conducted in pregnancy.

One thousand nine hundred and sixty-six antenatal haemoglobinopathy screening samples from the first trimester of pregnancy were utilised to measure RET-He level along with full blood count. Hb < 110 g/L was used to define anaemia in the first trimester. Serum ferritin (SF) assay was performed on one thousand two hundred and eighteen serum samples received for virology testing. Serum ferritin < 30 micrograms/l was used to define iron deficiency in pregnancy. Statistical significance of RET-He value was calculated depending on iron and anaemia status, using SF and haemoglobin levels respectively. Receiver operating characteristics analysis (ROC) of RET-He was performed against (i) SF to determine

Table. Summary of results

Non-anaemic (Hb ≥ 110 g/l)	1) 29.3% were iron deficient 2) 7.4% were iron depleted 3) RET-He significantly lower in patients with iron depletion (<i>n</i> = 90) compared to those with SF ≥ 15 µg/l (<i>n</i> = 1097) [32.4 ± 3 pg vs. 34 ± 1.8 pg] <i>P</i> < 0.001 4) RET-He was significantly lower in patients with ID (<i>n</i> = 354) compared with the iron replete (<i>n</i> = 833) [33.3 ± 2.2 pg vs. 34.2 ± 1.8 pg] <i>P</i> < 0.001
Iron depleted (SF < 15 µg/l)	1) Anaemic patients (<i>n</i> = 11) have significantly lower RET-He compared to non-anaemic patients (<i>n</i> = 90) [26.1 ± 2.9 pg vs. 32.4 ± 3 pg] <i>P</i> < 0.001
Iron deficient patients (SF < 30 µg/l)	1) Anaemic patients (<i>n</i> = 15) have significantly lower RET-He compared to non-anaemic patients (<i>n</i> = 354) [28.5 ± 4.9 pg vs. 33.3 ± 2.2 pg] <i>P</i> < 0.001
Iron replete (SF > 30 µg/l)	1) Anaemic patients (<i>n</i> = 7) have significantly lower RET-He compared to non-anaemic patients (<i>n</i> = 833) [28.1 ± 6.8 pg vs. 34.2 ± 1.8 pg] <i>P</i> = 0.006
Anaemic (Hb < 110 g/l)	1) There was no significant difference in RET-He between iron deficient (<i>n</i> = 15) and iron replete patients (<i>n</i> = 7) [28.5 ± 4.9 pg vs. 28.1 ± 6.8 pg] <i>P</i> = 1.000

which was better at elucidating the cause of microcytic/normocytic anaemia and (ii) haemoglobin to determine which was a better indicator of iron status.

RET-He mean (±SD) of patients with normal haemoglobinopathy screening results (34.1 ± 1.7 pg) was similar to previous reports (33.7 ± 1.5 pg). RET-He values of non-anaemic iron depleted/deficient patients were higher than those with iron deficiency anaemia but lower than those with normal haemoglobin and iron stores (*P* < 0.001). Better areas under the curve were observed when RET-He was compared to 1) SF (0.817 vs. 0.667), and 2) haemoglobin (0.720 vs. 0.708).

This study suggests that RET-He can detect iron deficiency before anaemia develops in the first trimester of pregnancy. Given its potential to aid early diagnosis of iron deficiency and, thereby, facilitate prompt treatment with the intention of reducing obstetric complications, further clinical and economic study should be considered to determine if this test should be incorporated into current practice.

Disclosure of Interest: None Declared

BSH2020-PO-039

Prevalence of anaemia in late pregnancy - a measure of whether national guidelines have been successfully implemented

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Anaemia in pregnancy remains an important issue with clinical consequences for mother and baby. The most common cause is iron

deficiency, as requirements increase significantly during pregnancy and many women have insufficient stores to meet this demand. Current available UK data suggest the prevalence of anaemia in pregnancy to be 44% at booking or 28 weeks and disappointingly, 2/3 of women who are anaemic at booking are still anaemic at 28 weeks. National BSH guidelines exist on management of iron deficiency in pregnancy, but these figures suggest a failure of these guidelines or their implementation.

Following 4 years of concerted efforts to promote implementation of the 2012 BSH guidelines in our centre, we evaluated the prevalence of anaemia at booking, 28 weeks but also 34 weeks and pre-delivery. This enabled us to assess whether anaemia is being effectively managed and consider whether anaemia in late gestation was associated with increased risk of post-partum haemorrhage (PPH), length of stay and need for transfusion.

We undertook a retrospective review of all women giving birth at Oxford University Hospital Trust in 2018. This identified 7402 women. At booking, mean and median Hb was 129 g/l, with only 2.3% of women anaemic. At 28 weeks, mean and median Hb was 115 g/l, with 12.5% patients' anaemic; highlighting a group of women that were non-anaemic but iron deficient earlier in pregnancy and had not received intervention. At 34 weeks, 11% women were anaemic, suggesting anaemia was effectively managed in approximately 1/8 cases. However, the percentage of anaemic women did not increase between 28 and 34 weeks and just 6.4% women were anaemic pre-delivery. 5.4% of women received ferrinject antenatally.

There was an increased rate of severe PPH (Estimated blood loss (EBL) >2 L) in anaemic women compared to those not anaemic at 34 weeks and pre-delivery. The risk increased with anaemia severity. In women Hb < 105 g/l at 34 weeks, 3% had severe PPH and 33% women with Hb < 80 g/l had severe PPH. In those with Hb > 105 g/l, only 1% had severe PPH. A similar pattern is seen pre-delivery; 3% of women who were anaemic had severe PPH, whereas only 1.5% of those not anaemic had severe PPH. Where Hb < 80 g/l, 14.3% patients had severe PPH.

We also identified that EBL correlated with length of hospital stay, with mean stay 2.3 days in women with EBL < 1L, 3.8 days where EBL 1-2L and 4.8 days when EBL > 2L. Red cells were given to some women with PPH and more units were ordered for those in the anaemic group pre-delivery.

These results suggest better implementation and compliance with the National guidelines in our centre compared to other UK centres that have contributed to the literature, as they demonstrate lower rates of antenatal anaemia and a reduction in numbers of women affected by anaemia as pregnancy progresses beyond 28 weeks. They also highlight that where iron deficiency is not managed effectively early in pregnancy, women may then become anaemic in the third trimester, a time when Hb is not routinely checked nationally. Women found to be anaemic in the third trimester appear to be at increased risk of developing severe PPH, requiring red cell transfusion and prolonged hospital stay. This supports a key new recommendation in the recently updated BSH guidelines on this topic, that non-anaemic women at high risk of iron deficiency should be identified in early pregnancy and given empirical oral iron supplementation to prevent anaemia developing.

Disclosure of Interest: None Declared

BSH2020-PO-040

Final results from the apixaban length-of-stay pulmonary embolism study - Hospital Admissions (ALPHA-PE)

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Patients presenting with venous thromboembolism (VTE), including pulmonary emboli (PEs), in the UK have historically been treated in hospital with warfarin. More recently, direct oral anticoagulants (DOACs) have been introduced. Unlike warfarin, patients receiving DOACs need less frequent blood test monitoring, and also benefit from standardised dosing and quicker onset of action. Findings from phase 3 randomised controlled trials supported by recent real-world evidence from primary care showed that patients who receive DOACs are at reduced risk of major bleeding compared with patients who received warfarin.¹ A database study in England also found that after the introduction of apixaban (a DOAC), patients experienced shorter hospital length of stay (LOS) to treat acute PE.² This study of real-world UK clinical practice compared LOS in patients initiated on anticoagulation for acute PE before and after licensing of apixaban.

A retrospective, observational study was undertaken at five hospitals in the UK (Birmingham, North Tyneside, Sunderland, Glasgow, and Oxford). Data were extracted from medical records of adult patients aged 18 years or older who presented with acute PE and were initiated on anticoagulation (parenteral anticoagulation and warfarin, or apixaban) during the hospital admission, and in receipt of anticoagulation on discharge. A before and after study design was used, with two cohorts: Cohort 1 (before apixaban licensing) with eligible patients admitted between 01 January 2013 and 30 June 2013, and Cohort 2 (after apixaban licensing) with patients admitted between 01 January 2017 and 30 June 2017. Descriptive and Inferential analyses were conducted. Statistical inferences were obtained from linear mixed models.

A total of 538 patients (269 in Cohort 1 and 269 in Cohort 2) were available for analysis. At baseline, patients in Cohort 1 (before apixaban licensing) and Cohort 2 (after apixaban licensing) were comparable in age (mean age 66.9 years vs. 64.2 years), gender (51.3% male vs. 44.6%) and other demographic and clinical characteristics. In Cohort 1, 40.7% of patients were classified as low risk according to simplified PESI (sPESI) risk scores and 59.4% as high risk. By comparison, patients in Cohort 2 were equal divided between low and high risk classifications (50.0% in each group).

Most patients in Cohort 1 received oral anticoagulation with warfarin (79.6%), and with apixaban in Cohort 2 (72.1%). The mean (±SD) length of stay (LOS) for patients in Cohort 1 was 7.43 (±9.39) days, compared with 4.64 (±7.31) days in Cohort 2. This represents a 2.79-day reduction ($P < 0.005$) in mean LOS between the two cohorts. In patients who received warfarin, mean LOS was similar in Cohorts 1 ($n = 214$) and 2 ($n = 14$), at 7.31 days (SD 8.10) and 7.21 days (SD 5.96), respectively). Patients who received apixaban in Cohort 2 ($n = 194$) had mean LOS of 4.48 days (SD 7.80).

Use of apixaban to treat acute PE was associated with reduced length of stay compared with warfarin, with a reduction of approximately 3 days. Shorter hospital stays reduce the burden to patients and may also benefit local healthcare services and the NHS in potential resource use savings.

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Disclosure of Interest: R. Alikhan Conflict with: Study support from BMS/Pfizer, Conflict with: Speaker fees/advisory boards from Bayer, Boehringer Ingelheim, BMS, Daiichi Sankyo, Pfizer, C. Church Conflict with: Study support from BMS/Pfizer, S. Shapiro Conflict with: Study support from BMS/Pfizer, Conflict with: Speaker fees and/or meeting sponsorship from Bayer, Pfizer/BMS and Boehringer Ingelheim, S. Lister Conflict with: Employee of BMS; shares owned in BMS and Pfizer, N. Smith Conflict with: Study support from BMS/Pfizer, Conflict with: Lecture fees from BMS, S. Chatterjee Conflict with: Study support from BMS/Pfizer, C. Bomken Conflict with: Study support from BMS/Pfizer, S. Grundy Conflict with: Employee of BMS Ltd., J. Gordon Conflict with: Funded by BMS/Pfizer, D. Sugrue Conflict with: Funded by BMS/Pfizer, N. Hill Conflict with: Employee of BMS; shares owned in BMS, L. Howard Conflict with: Study support from BMS/Pfizer, Conflict with: Honoraria for lectures from Bayer; honoraria for scientific advisory committees from Bayer and BMS

BSH2020-PO-041

Mutant specific anti calreticulin antibody (CAL 2) immunohistochemistry as a screening test for calreticulin (CAL R) mutation testing

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Background: CALR mutation is seen in more than half of JAK 2 negative Essential thrombocythemia (ET) and Primary Myelofibrosis (PMF). About 50 different mutations are identified, all leading to development of common new protein C terminus, irrespective of the type of mutation. Hence, a logical approach of targeting this terminal epitope through the mutation specific antibodies can identify the CALR mutations.

Material Methods: Molecular detection of CALR mutation was done using allele specific PCR followed by fragment analysis. Corresponding bone marrow biopsy was subjected to IHC using mutation specific antibody DIA-CAL from Dianova. The results from molecular methods and IHC were compared.

Results: CALR mutation was tested in 48 cases of JAK 2 V617F negative PMF and ET. Mutated CAL R was noted in 20 cases accounting for 11.5% (9 of 78) cases of PMF and 17.7% (11 of 68) cases of ET. Different types of mutations noted were 52 bp deletion (56.2%), 45 bp del (12.5%), 5 bp insertion (31.3%). Corresponding BM biopsy were available in 45 cases (19 CALR mutated and 26 CALR wild type). A positive staining pattern was noted in all the CAL R mutation positive cases, while a negative pattern was noted in all the negative cases. Staining pattern was similar in different types of mutations. A 100% concordant rate was noted between molecular and IHC methods.

Conclusion: CAL2 IHC detects all types of CAL R mutation. This can act as a sensitive, specific, rapid and cost effective screening test for CALR mutation analysis.

Key Words: Calreticulin Mutation, CAL 2 IHC, Essential Thrombocythemia, Myelofibrosis

Disclosure of Interest: None Declared

BSH2020-PO-042

Abstract Withdrawn.

BSH2020-PO-043

Immunological effects of ponatinib therapy in patients with chronic myeloid leukaemia

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Introduction: Ponatinib is the most potent tyrosine kinase inhibitor (TKI) currently available and retains activity against all currently recognised ABL1 kinase domain mutations. Ponatinib displays moderate SRC kinase inhibitory activity, with an IC50 of 5.4 nmol/l (O'Hare, Cancer Cell, 2009), however its potential immunosuppressive effect has not been studied. The SRC family kinase LCK plays a critical role in signalling from the T cell receptor (TCR) with immediate downstream targets including ZAP70 and LAT. NK cells also possess an abundance of SRC family kinases which play a pivotal role in signalling from activating NK cell receptors. We analysed immune cell function in a cohort of patients treated with this third generation TKI.

Methods: We performed a two-phase functional analysis of the immunomodulatory effects of ponatinib in 4 patients and 3 healthy controls. We first performed phosphoflow cytometry in T effectors (Teff), T regulatory cells (Treg) and NK cells. We analysed the effect of ponatinib on key signalling molecules downstream from the TCR and NK cell activating receptors, including activated/phosphorylated ZAP70 and LAT, as well as the critical signalling molecule in T cells and NK cells, STAT5. 11-colour flow cytometry was performed after cells were activated with H2O2 for 15 min, due to its activity as a potent phosphatase inhibitor. We then performed 10 colour intracellular flow cytometry assessing the impact of ponatinib on T cell cytokine production including TNF α , IFN γ , IL-2, IL-4 and IL-10 after stimulation with OKT3.

Results: Patients on ponatinib had significantly reduced phosphorylation of LAT compared with controls, with a mean increase in median fluorescence intensity (MFI) of 1.9 vs. 19.1 in CD4⁺, 1.8 vs. 28.8 in CD8⁺ and 5.7 vs. 24 in CD56^{br}NK cells ($P = 0.005$, $P = 0.0001$, $P = 0.006$). Reduced phosphorylation of ZAP70 was also seen in all cell subsets analysed in patients on ponatinib but statistical significance was not reached. Patients on ponatinib also had significantly reduced phosphorylation of STAT5 with a mean increase in MFI of 4.3 vs. 32.2 in CD3⁺, 3.4 vs. 26.3 in CD4⁺, 5.1 vs. 41.6 in CD8⁺, 3.7 vs. 26.3 in Tregs and 9 vs. 43.2 in CD56^{br} cells ($P = 0.0001$, $P = 0.0001$, $P = 0.004$, $P = 0.004$). Patients on ponatinib had a reduced increase in expression of TNF α in CD4⁺ and CD8⁺ cells when compared with controls, with an absolute increase in expression of 16.3 vs. 33.3 and 10.2 vs. 42.8 respectively ($P = 0.039$, $P = 0.032$). Patients on ponatinib also had a reduced increase in expression of IFN γ when compared with controls, with an absolute increase in expression of 8.5 vs. 23.1 in CD4⁺ cells and 11.9 vs. 25.5 in CD8⁺ cells ($P = 0.014$, $P = 0.09$).

Conclusion: Ponatinib inhibits signalling within immune effector cells with an associated resultant decrease in expression of proinflammatory cytokines in T cells. The strongest inhibition of signalling was seen against pSTAT5 which plays a key role in T cell proliferation

through IL-2 signalling and also regulates NK cell function primarily via IL-15. This is, to our knowledge, the first study to report on the immunosuppressive qualities of ponatinib and has important implications with regards to potential combination therapy with immunomodulatory agents.

Disclosure of Interest: P. Harrington Conflict with: Incyte, Conflict with: Bristol Myers Squibb, D. Radia Conflict with: Blueprint, Novartis, M. Lim: None Declared, R. Dillon: None Declared, S. Kordasti: None Declared, K. Rezvani: None Declared, C. Harrison Conflict with: Novartis, Celgene, Gilead, Incyte, Shire, Conflict with: Novartis, H. de Lavallade Conflict with: Novartis, Pfizer, Bristol Myers Squibb, Incyte, Conflict with: Bristol Myers Squibb, Incyte

BSH2020-PO-044

When should haematologists see patients with asymptomatic lymphocytosis?

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Increasing numbers of patients are referred to haematology clinics for investigation of lymphocytosis picked up incidentally on routine blood tests. There are no clear guidelines on what level of lymphocytosis should be referred for haematology clinic review or immunophenotyping. Our department is developing a 'Population Haematology' strategy in conjunction with General Practitioners (GPs).

We struggle to give clear guidance to GPs on the appropriate thresholds for referral. To aid decision making and facilitate discharge from clinic, the practice at our centre is for patients seen in clinic with persistent lymphocytosis and no clear cause to undergo immunophenotyping.

Most patients have no clonal population, whilst patients with a clonal population are given a diagnostic label and often long-term follow-up. This can negatively impact quality of life [Levin et al, 2007]. The benefits of investigating and monitoring patients with a monoclonal B cell lymphocytosis (MBL) are unclear. The rate of progression from MBL to progressive CLL is estimated at only 1–2% per year [Rawstron et al, 2008]. Furthermore, immunophenotyping and follow up clinics carry a significant financial cost to the NHS.

We set out to investigate the long-term outcomes for patients with asymptomatic lymphocytosis. We performed a retrospective analysis of patients who underwent immunophenotyping at our centre over the course of 18 months (01/01/2013 – 30/6/2014) with 5 years or more of follow-up. All patients with request clinical details of 'lymphocytosis', 'raised WCC' or '?CLL' were included. Patients were excluded if there were no clinical records, they had a pre-existing diagnosis of a lymphoproliferative disorder (LPD) or had constitutional symptoms or cytopenias at the time of immunophenotyping.

Of 167 patients, 93 met inclusion criteria, of whom 20 (21.5%) had a clonal population. These cases were CLL (12 patients), CLL-like MBL (5), splenic marginal zone lymphoma (1) and B-LPD NOS (2). Of the 20 patients with a clonal population, 2 patients (10%) underwent treatment after developing symptoms during the follow-up period, (treatment at 4 and 24 months after diagnosis). The remainder continued under 6–12 monthly monitoring (9 patients), died from other causes (3) or were lost to follow-up (6). Both patients needing treatment presented with an initial lymphocyte count of $>20 \times 10^9/l$. All patients with non-clonal lymphocytosis (73) had initial lymphocyte counts of $<10 \times 10^9/l$.

In the absence of clear guidelines on lymphocytosis thresholds for referral, we sought peer opinion. We sent out a national survey to look at the approach of other haematology centres. The 23 responses

showed a marked variation in lymphocytosis referral thresholds (ranging from >3 to >20 , most common thresholds were 5 and $10 \times 10^9/l$). We also found variation in follow-up strategies and direct access to immunophenotyping for GPs.

Our results will influence the advice we give GPs about referral to haematology clinics, with levels of $10\text{--}20 \times 10^9/l$ perhaps being appropriate thresholds for detection of clinically relevant conditions. These also suggest similar thresholds for consideration of immunophenotyping when a patient is in haematology clinic, to aid discharge and monitoring. Patients with MBL are unlikely to benefit from an "early diagnosis", which may instead have a negative impact on their quality of life, as well as being a significant burden on haematology services.

Disclosure of Interest: None Declared

BSH2020-PO-045

UK adult immune thrombocytopenia (ITP) pregnancy registry: collaborative research to better understand outcomes: the first 18 months

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In August 2018, the UK Adult Immune Thrombocytopenia (ITP) pregnancy registry was added to the UK adult ITP registry to collect prospective data regarding the management and outcome of pregnancy in patients with ITP with the aim of furthering our understanding of this disorder in pregnancy. Inclusion criteria included: Pregnant women with primary ITP diagnosed prior to pregnancy or primary ITP diagnosed in pregnancy where: the platelet count falls to $<50 \times 10^9/l$ or treatment for ITP is required during pregnancy or up to 3 months post-partum. Data links to the main arm of the UK adult ITP registry. The data sets captured are listed in table 1 and align with the data sets from UKOSS published in 2017.

To date, 19 women have been recruited, 15 have delivered and we have complete data sets for the first 12 patients. The median age is 32 and ethnic mix ($n = 14$) is 57% White (English, Welsh, Scottish, Northern Irish, and British), 35.7% Asian (Indian, Bangladeshi or other) and 0.07% Black (any other Black, African, Caribbean background). 57% were diagnosed with ITP before pregnancy and 43% were diagnosed in pregnancy. The mean lowest platelet count during pregnancy was $55 \times 10^9/l$ (range $7\text{--}127 \times 10^9/l$). The mean platelet

count at delivery was $94.6 \times 10^9/l$ (range $55\text{--}138 \times 10^9/l$). The mean platelet count 3 months post-partum was $125 \times 10^9/l$ (range $58\text{--}235 \times 10^9/l$).

Of those with complete data, 6 out of 12 experienced some bleeding during their pregnancy. The median lowest platelet count of those that suffered bleeding was $52 \times 10^9/l$ (range $7\text{--}127 \times 10^9/l$). The median lowest count of those without bleeding was $59 \times 10^9/l$ (range $26\text{--}93 \times 10^9/l$). The main bleed sites were gynaecological bleeding ($n = 4$) and epistaxis/oral cavity ($n = 2$). Four of the 12 women required treatment during pregnancy: all received steroids and 1 also received IVIG. Four of the 12 patients had a caesarean section whilst the other 8 had vaginal delivery. There was only 1 reported post-partum haemorrhage in these cases.

This study will provide invaluable data on the management and outcomes with ITP in pregnancy to help better risk stratify women as they head into pregnancy. Pregnancy ITP is a rare overall occurrence and using the pre-existing infrastructure of the UK ITP Registry has allowed a linked dataset to further our knowledge in this area.

Table 1. UK adult immune thrombocytopenia pregnancy registry data sets

Medical and obstetric history	Participant and Clinician Details Problems in previous pregnancies Morbid events in previous pregnancies
Current pregnancy information	Comorbid conditions Anthropometric and lifestyle information Bleeding events
ITP treatments during pregnancy	Supportive therapies Medication
Delivery information	Details and type of delivery Anaesthesia method at delivery
Haematological and coagulation values	During pregnancy At delivery Post-partum
Maternal outcome	Peri-partum bleeding events RBC transfusion requirement Comorbid conditions during pregnancy or up to 3 months post-partum
Infant details	Morbidity and treatments Platelet counts Breastfeeding

Disclosure of Interest: None Declared

BSH2020-PO-046

Transition to generic imatinib: a single-centre experience of patient and financial outcomes

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Background: Imatinib was the first tyrosine kinase inhibitor (TKI) approved for clinical use in Chronic Myeloid Leukaemia (CML) and it transformed the outcomes in terms of progression free survival and overall survival in these patients. Until January 2017, imatinib was marketed and sold exclusively as the branded form Glivec® in the UK. Since the expiry of its patent protection, a number of generic forms of imatinib have become available for use in patients with

CML. At Norfolk and Norwich University Hospital (NNUH) all patients with CML taking Glivec® were switched to generic imatinib as soon as it became available. Through this lookback exercise, we sought to examine issues of tolerability of generic imatinib and confirm adequate disease-control following the switch.

Patients and Methods: We identified all patients with CML at NNUH being treated with imatinib in January 2017 ($n = 43$). We confirmed that all patients had their prescription changed to generic imatinib at this time.

Patient notes between January 2017 and August 2019 were reviewed retrospectively for documentation of new reported side effects. Laboratory results were reviewed to assess disease status, as defined by the BCR-ABL1/ABL (IS), before and after the switch.

43 patients were identified, 15 (35%) were female, 28 (65%) male. The median length of time taking imatinib prior to the switch was 6 years (range 1 month – 18 years).

Results: Immediately prior to changing to the generic formulation, disease responses were as follows: CHR only 8/43 (19%), CCyR 3/43 (7%), MMR 9/43 (21%), MR4 9/43 (21%), MR4.5 11/43 (26%), CMR 3/43 (7%).

Two patients were lost to follow up after moving out of area (no change in disease status when last seen). The most recent disease control on imatinib of the remaining evaluable patients following the switch was unchanged (23/41 patients, 56%) or continuing to improve (18/41, 44%).

During follow up, four patients had their therapy changed to second generation TKIs due to ongoing failure to achieve MMR (no change in disease control after switching to generic imatinib), and one died from an unrelated pre-existing illness (liver failure).

10/43 (23%) patients reported experiencing side effects on switching to generic imatinib. These included recurrence of cramp, transient skin rash, subconjunctival haemorrhage, and gastrointestinal upset. All were mild and most were transient. No patient required a switch back to Glivec®, although some patients reported improvement with a different brand of generic imatinib, especially with regards gastrointestinal upset.

Based on NHS indicative pricing, generic imatinib can be up to 97% cheaper than the same dose of Glivec®. The cost saving to the NHS of switching our patients to generic imatinib was approximately £79 000 per month.

Conclusions : Switching from branded to generic imatinib did not lead to any change in disease control for our cohort of patients and new formulations were generally well tolerated. These findings provide reassurance of the safety of switching to generic formulations and, along with the significant savings to the NHS, support the continued use of generic imatinib.

Disclosure of Interest: None Declared

BSH2020-PO-047

Refining the erythrocytosis diagnostic pathway: an audit of current practice compared to British Society for Haematology guidelines

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The British Society for Haematology (BSH) published new guidelines in 2019¹ which provide an updated pathway for investigation of erythrocytosis, i.e. a persistently raised haematocrit. The initial investigations should include full blood count, renal/liver function, carboxyhaemoglobin/erythropoietin level, imaging (chest x-ray,

abdominal ultrasound) and *JAK2* V617F mutation testing, alongside full history and examination. This should help to identify the majority of patients with *JAK2*-positive polycythaemia vera (PV) and those with any common secondary acquired causes of polycythaemia, including smoking, sleep apnoea, etc. If these initial investigations do not reveal any underlying cause, the next steps are stratified according to erythropoietin levels; if high, then imaging such as CT head and neck for cerebellar haemangioblastoma, meningioma or a parathyroid tumour should be considered. If normal or low then *JAK2* exon 12 mutational analysis and bone marrow biopsy are recommended. If no cause has been identified then testing for mutations in genes known to be associated with erythrocytosis such as high affinity haemoglobin or HIF1a, usually by a next generation sequencing (NGS) gene panel, could be requested when there is a clinical suspicion of a congenital cause.

We present an audit of current practice in investigation of GP referrals for erythrocytosis at a tertiary centre compared with the recently published BSH guidelines, focussing upon utility of NGS testing which is an expensive resource.

We identified 26 patients for whom NGS gene panel was requested in work up of erythrocytosis between April 2016 and January 2019 and undertook a retrospective review of patient records. All were negative for *JAK2* V617F mutation and all had low or normal erythropoietin levels. 21/26 patients (80.8%) had carboxyhaemoglobin levels checked, and in 9/21 (42.9%) were elevated. 23/26 cases (88.5%) had *JAK2* exon 12 mutation analysis requested and all were negative and 14/26 (53.8%) underwent bone marrow biopsy, all lacked evidence of PV. Concerning NGS, only 1/26 patient (3.8%) had a pathogenic mutation affecting *SH2B3* and 7/26 (26.9%) had variants of unknown significant (VUS) in genes known to be associated with erythrocytosis. Of the patients with elevated carboxyhaemoglobin, 3 (33.3%) had VUS.

The low pick-up rate for the NGS panel could be further improved by more diligent use and review of the initial investigations and avoiding pressures to condense the pathway usually driven in our centre by patients travelling for a tertiary opinion. Of extended note, future testing for VUS did not yield any new data.

References:

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Disclosure of Interest: None Declared

BSH2020-PO-048

Opioid crisis in Nigeria: feeling the pain of sickle cell disease patients mistaken as drug seekers

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Objectives: To examine the perception of healthcare providers concerning patients with sickle cell disease in pain crisis vis-a-vis opioid use among people who live with sickle cell disease.

Background: Patients with sickle cell disease (SCD) often seek care in the emergency department (ED) for pain associated with vaso-occlusive crises (VOC). Research has shown that negative provider attitudes serve as a barrier to care in this patient population.

In the wake of global opioid crisis possibly due to westernization, low middle income countries such as Nigeria is plagued with opioid dependence and addiction thereby increasing the suspicion of healthcare providers to sickle cell disease patients. These patients unfortunately are deemed opioid seeking and may continue to suffer from their unfortunate disease. In this study we aimed to assess the prevalence of opioid abuse in these patients to better inform providers on their care.

Methods: Cross sectional study was conducted in the sickle cell disease clinic at Lagos State University Teaching Hospital (LASUTH). The study population were (i) patients older than 18 years old who provided consent ($n = 350$ patients) and (ii) health care providers providing care for patients with SCD ($n = 61$ providers). The "ASSIST" survey instrument developed and validated by the WHO to screen for hazardous, harmful and dependent use of alcohol, tobacco and other psychoactive substances was used. The ASSIST was scored and interpreted as: Low -0-3; Patient is at low risk of health and other problems from current pattern of use. Moderate 4-26; Moderate risk of health and other problems from current pattern of use. High 27+; High risk of experiencing severe problems (health, social, financial, legal relationship) as a result of current pattern of use and are likely dependent. Perception of consenting health care workers was assessed by another validated structured questionnaire to determine their views on liberal versus restrictive use of opioids in SCD vis-a-vis risk of abuse and addiction.

Results: 78% of patients never used opioids while 15.1% had. In the past 3 months 87.4% had never used opioids, only 2.3% had used it

Table 1. Diagnostic pathway with audit results.

Initial assessment - history and examination 26 (100%)			
Full blood count26 (100%)	Renal/liver profile20 (76.9%)	Carboxyhaemoglobin21 (80.8%)	Erythropoietin26 (100%)
ImagingChest x-ray - 12 (46.1%)			
Ultrasound abdomen - 18 (69.2%)			
JAK2 V617F analysis26 (100%)			
JAK2 V617F negative26 (100%)			
Low/normal erythropoietin26 (100%)		High erythropoietin 0 (0%)	
JAK2 exon 12 analysis23 (88.5%)	Bone marrow biopsy14 (53.8%)	ImagingCT/MRI head - 2 (7.7%)	
Gene sequencingPathogenic mutation (<i>SH2B3</i>) - 1 (3.8%)VUS - 7 (26.9%)			

weekly and 1.7% daily 94.3% never had a strong desire or urge in the past 3 months, only 2% had a daily strong desire to use opioids. Also a high percentage of them 95.7% had no problem about taking control, cut down or stops using opioids. A total of 93.4% had never used any other drug injection for non medical use. Almost all participants (96.9%) confirmed use of opioids has not led to health, social, legal or financial problems. 27.9% of providers believed 20–40% of patients are addicted to opioid

Discussion: Contrary to providers' prejudice, our study reported only 2% of SCD patients were dependent on opioid compared with the perception of 54.1% of providers who believed that >20% of sickle cell disease patients were addicts. This is a misplaced fear of causing or exacerbating addictions among SCD patients. These patients are mislabeled as drug-seekers and inappropriately considered as addicts. As the opioid crises continues to ravish the country, SCD patients will bear the brunt of policies put in place to address this issue. While the issue of drug addiction should not be ignored, the appropriate treatment of SCD patients in Nigeria who are at disadvantage to getting high quality care should be appropriately addressed.

Disclosure of Interest: None Declared

BSH2020-PO-049

A clinical audit on compliance with British Society of Haematology guidelines for diagnosis and management of Philadelphia negative myeloproliferative neoplasms

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Myeloproliferative Neoplasms (MPN) are a group of disorders affecting multipotent progenitor cells and are classified on the presence or absence of the BCR-ABL fusion gene on the Philadelphia chromosome. BCR-ABL-negative MPN include Primary Myelofibrosis (PMF), Polycythemia rubra vera (PRV) and Essential Thrombocythemia (ET). British Society of Haematology (BSH) guidelines provide diagnostic and treatment recommendations for MPN. A risk-adapted strategy is recommended to minimize risk of thrombosis and improve survival. Variability in practice is observed in real world in management of MPN. The primary objective of this study was to assess compliance with British society of haematology guidelines for diagnosis and management of Philadelphia negative myeloproliferative disorders. The secondary objectives included the description of demographics and clinical characteristics of patients at time of diagnosis, type of management strategy used and outcome. Record of 145 patients with a diagnosis of MPN treated at New Cross Hospital and Cannock Chase Hospital (Royal Wolverhampton NHS trust) from August 2016 to August 2019 was audited. Median age at time of diagnosis was 67 (range 34–94) years. Male: female ratio was 1:1.5. Diagnosis included ET [$n = 94$ (65%)], PRV [$n = 40$ (28%)], PMF [$n = 6$ (4%)] and Pre-fibrotic MF [$n = 2$ (1%)]. Three patients (2%) were diagnosed as MPN without further sub classification. In terms of clinical presentation, 59(41%) patients were asymptomatic. The three most commonly reported symptoms were fatigue in 16 (12%), headache in 10(7%) and acquagenic pruritis in 8(5.5%). Presenting symptom was not documented in 5(3%) patients. Splenomegaly was present in 15% of all patients [ET (7%), PRV (15%), PMF (100%), Pre-fibrotic MF (50%)] at diagnosis. Bone marrow biopsy as part of diagnostic work-up was done in 68% of all patients [ET (75%), PRV(42.5%), PMF and Pre-fibrotic MF(100%)]. Molecular testing was done in 97% of patients. Molecular markers were positive for JAK-2 in 76%, MPL in 4% and CALR in 6%. Rest of patients

(14%) were triple negative. Discussion in MDT took place in 60% of all cases [ET(58%), PRV(60%), PMF(100%), Pre-fibrotic (100%)]. Risk stratification was not documented in 37% of all cases [ET (28%), PRV(47%), PMF(83%)]. Across the 145 patients, a total of 182 individual management strategies (defined as any clinical decision on patient management) were initiated. Management strategies included wait and watch [$n = 22$ (12%)], Hydroxycarbamide [$n = 111$ (61%)], Anagrelide [$n = 5$ (3%)], Ruxolitinib [$n = 8$ (4%)], Busulphan [$n = 5$ (3%)], radiophosphorus [$n = 1$ (1%)] and Venesection combined with hydroxycarbamide [$n = 30$ (16%)]. Disease transformation to MF and/or AML was seen in 4/94(4.3%) ET and 5/40 (12.5%) PRV patients. At the time of record review, 19/145 (13%) patients were dead. [ET(9.5%), PRV(12.5%), PMF(66.6%)]. Cause of death was disease related in 10% cases. In conclusion, compliance with BSH guidelines was satisfactory in terms of diagnostic workup and treatment stratification for cytoreduction in high risk patients. Compliance was found to be poor in performing bone marrow biopsy as part of workup, discussing cases in MDT and documenting risk stratification and prognostic scoring. The recommendations are to update trust guidelines as per BSH recommendations, incorporate risk stratification/ prognostic scoring in treatment pathways for all MPN and conduct educational session to standardize practice across the trust

Disclosure of Interest: None Declared

BSH2020-PO-050

Disease progression in patients with essential thrombocythaemia receiving anagrelide within a single institution

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Introduction: Anagrelide is currently used as a second-line treatment to lower platelet counts in essential thrombocythaemia (ET). Since the PT1 study¹, there has been increased interest in the potential risk of accelerated myelofibrotic transformation in ET patients treated with anagrelide. Current BSH guidelines advocate monitoring bone marrow trephines at 3 yearly intervals for those treated with anagrelide to monitor for signs of emerging fibrosis².

Aim: To identify clinical features and outcomes of ET patients who received anagrelide in our local centre.

Methods: A retrospective chart review was performed to identify all patients prescribed anagrelide between January 2012 to December 2018 from pharmacy records. Clinical data was collated from electronic patient records.

Results: During the 6-year period (2012 to 2018), our institution treated 37 patients for a minimum of 2 consecutive months on anagrelide. The average duration of treatment was 43 months (range 3 – 132 months). Of those prescribed anagrelide during this period, 57% discontinued for either disease transformation, lack of adequate response or side effects. For those who discontinued for transformed disease (AML or MF), the average treatment duration was 34.2 months. Of those that discontinued for non-disease transformation indications, 70% stopped due to cardiac toxicity. 16 (43%) patients remained on anagrelide at the end of the study period (average treatment duration >62 months).

27% of patients (10/37) transformed to myelofibrosis and 3% (1/37) developed AML. Of those that transformed to myelofibrosis, 80% had the JAK2V617F mutation. In comparison, a similar study at our institution reviewing the outcome of 27 patients taking hydroxycarbamide over a similar time window identified 7% of patients

progressing to myelofibrosis and 7% developing AML. Therefore the rate of myelofibrotic transformation was significantly higher in patients treated with anagrelide with a Chi squared value of 3.9 and *P* value of 0.047.

There was no difference in age of presentation between those that did and did not transform to myelofibrosis (57 years vs. 56.7 years). 11/37 patients (30%) had a pre-anagrelide trephine biopsy. Of 4 of these who later progressed to myelofibrosis, 50% had fibrosis grade 1 prior to anagrelide exposure. 28% (2/7) of patients receiving anagrelide who did not transform to myelofibrosis had evidence of fibrosis (grade 1) prior to treatment.

Conclusion: These findings highlight the higher incidence of fibrosis development in patients with ET treated with anagrelide. Screening for baseline fibrosis/evidence for pre-fibrotic myelofibrosis with a bone marrow biopsy prior to the commencement of anagrelide is important, with careful monitoring for signs of progression in patients on treatment.

References: 1. Harrison CN, Campbell PJ, Buck G, et al (2005). *New England Journal of Medicine*, 353(1):33–45

2. Harrison CN, Butt D, Campbell P et al (2010). Guideline for investigation and management of adults and children presenting with a thrombocytosis. *British Journal of Haematology*, 149, 352–375.

Table

	Anagrelide	Hydroxycarbamide
Myelofibrosis	10 (6.94) [1.35]	2 (5.06) [1.85]
No myelofibrosis	27 (30.06) [0.31]	25 (21.94) [0.43]

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Second primary malignancies in patients receiving treatment with ruxolitinib for myelofibrosis – a regional experience

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Patients with myeloproliferative neoplasms (MPN) are at an increased risk of second primary malignancies (SPM) compared with matched-controls and the general population. The Surveillance, Epidemiology and End Results (SEER) database (US cancer registries) reported that patients with primary myelofibrosis (PMF) had an incidence of SPM of 5.3% after a median duration of 1.8 years (range 0.04–9.1 years). These patients were at increased risk of respiratory malignancies and lymphoma when compared with the US general population.

Ruxolitinib (Jakavi, Novartis) is a protein kinase inhibitor that targets Janus-associated kinase (JAK) signalling and is approved for use in selected adults with primary and secondary myelofibrosis (PMF). There is conflicting published data for the occurrence of SPM in ruxolitinib treated MF patients. There appears to be a possible increased incidence of non-melanoma skin cancers (NMSC) in these patients but the risk of other SPM is less clear.

We undertook an audit of our regional experience of SPM in ruxolitinib treated MF patients. Local Trusts within the Cheshire & Merseyside Cancer Alliance were invited to participate. All patients with a diagnosis of PMF or secondary MF who had received ruxolitinib were included for analysis. Data were collected and analysed for patient demographics, disease characteristics, past history of malignancy, duration of ruxolitinib therapy and any occurrences of SPM whilst on therapy.

There were 29 patients (male = 19, female = 10) included in the audit covering a 6 years period (January 2013 to September 2019). Of these, 15 had PMF and 14 had secondary MF. The median age at diagnosis was 73 years (range 35–94 years). Four patients had a past history of malignancy prior to ruxolitinib exposure. These included: NMSC (*n* = 3), breast cancer (*n* = 1 – this patient also had concurrent NMSC) and bladder carcinoma (*n* = 1). The median age at commencement of ruxolitinib was 73 years (range 47–94 years). The median duration of treatment was 18 months (range 2–79 months). The total duration of ruxolitinib exposure across the entire cohort was 60 years. SPM were seen in 6 patients (21%). The median duration of treatment prior to SPM diagnosis was 19 months (range 12–63 months). The SPM diagnoses were as follows: NMSC (*n* = 2), oropharyngeal basaloid squamous cell carcinoma (SCC, *n* = 1), diffuse large B-cell lymphoma (DLBCL, *n* = 1), SCC of the lung (*n* = 1) and myelodysplastic syndrome with multi-lineage dysplasia (*n* = 1). The case of DLBCL did not have a preceding history of lymphoproliferative disorder. Four patients with SPM (67%) had no prior history of malignancy. One patient with previous NMSC developed multiple aggressive NMSCs after commencing ruxolitinib with the suspicion of a pulmonary metastasis. The case of the SCC of the lung originated from a previously stable lung nodule that had been under surveillance for 5 years. After 24 months of ruxolitinib the lung nodule progressed into lung carcinoma and the patient died 12 months later secondary to this.

Our regional audit confirms the increased incidence of SPM in ruxolitinib treated MF patients when compared to the SEER comparison population (21% versus 5.3%). This is comparable to other published data. On this basis, we counsel all patients being considered for ruxolitinib for MF of the risk of NMSC and the potential risk of other SPMs. A larger UK audit of real world incidence of SPM in ruxolitinib treated MF patient would be valuable.

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Stopping IM B12: a collaborative service development between primary and secondary care

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Vitamin B₁₂ is essential for formation of red blood cells, development and myelination of the CNS as well as for the maintenance of its normal function.

B12 is routinely checked as part of a macrocytic anaemia screen, dementia screen or in patients with concerning neurological symptoms. It is also often checked in non specific lethargy. B12 treatment advice is available in UK trusts and from the British Society of Haematology. Whilst its clear how to manage those with lifelong IM B12 requirements other patients may not require this, exposing patients unnecessarily to IM injections and healthcare appointments.

There are no specific guidance regarding stopping B12 and the likelihood of patients needing to restart B12. This service development project aims to address this.

Service development was undertaken in a 15 000 patient GP practice, those receiving IM B12 were identified through pharmacy records. Patients with confirmed Pernicious anaemia (PA), gastric band or required to stay on IM for other reasons were identified and excluded.

The remaining patients were reviewed, with baseline data assessed pre-IM B12. The service improvement was discussed in person or by telephone.

A guidance flow chart was developed. Blood samples were planned for 6, 12 and 24 months post last injection.

247 had B12 IM on their medication list, 11 were excluded. 79 remained on B12; Indications were Haematological (25), Gastroenterological (44) and GP discretion/other (10).

Intrinsic factor (IF) in whole cohort was checked in 63.9% (151/236), data missing for 5.9% (14). Of those who had IF checked 10 were positive and remained on IM B12

Cessation of B12 patients: had IM B12 stopped, 105 were female. Mean Hb on starting B12 was 127 g/l. The majority were normocytic (64%), 39 had high MCV, 9 had no data, 8 had a low MCV. 6 had the classical macrocytic anaemia

133/157 completed 2 years of monitoring or longer (84.7%). 9 (5.7%) DNA'd follow-up blood tests despite repeated invites. 4 left the practice during the project.

17 patients have died during the project, 6 completed >2 years off B12 before death.

20 patients restarted B12, mean time from stopping to restarting B12 of 22.5 months (range 17–36 months). Median % of B12 below normal range was 9% (range 1% >38%). 11 had an oral course of B12, 5 started IM and 4 had oral then IM due to repeated oral courses

There is limited guidance regarding stopping B12 with advice often stating a blood test 6 months post B12 cessation. With one IM B12 dose expected to last 2–10 months there is concern patients require longer monitoring.

This service improvement shows that stopping B12 in selected patients is safe, reducing exposure to unnecessary injections. It also provides guidance to a previously 'ad hoc' process. Patient education is important for ensuring monitoring and symptom awareness.

There are several issues with stopping B12. There is not the 'perfect' B12 assay, confirming a diagnosis of PA is now difficult. There are patients who have a 'false normal' B12, and those who have low B12 with no symptoms who do not require treatment. It is therefore essential for education of healthcare staff to be aware of this.

Stopping B12 in selected patients following a structured follow-up of blood tests is safe, reducing patient exposure to unnecessary medication and improved medicines management. It should be remembered the limitations of B12 testing, patients should seek GP advice if there are clinical concerns

Table: I have been unable to include the guidance flow chart this will be included if the abstract is accepted

Disclosure of Interest: None Declared

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Renal function over 10 years in patients with chronic myeloid leukaemia treated with tyrosine kinase inhibitors: a single centre experience

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Treatment of chronic myeloid leukaemia (CML) was revolutionised with the development of tyrosine kinase inhibitors (TKI). Over more recent years second and third generation TKIs have led to further options for patients who develop imatinib resistance or intolerance. Despite ongoing studies to stop TKI therapy in eligible patients, there is still a large cohort of patients who remain on TKIs long term with sustained major molecular response (MMR). Many side effects of TKIs are well documented but some of the potential long-term side effects are less well characterised. There is some evidence that patients on long term TKIs have a gradual decline in their renal

function as measured by estimated glomerular filtration rate (eGFR). The recognition of this side effect is all the more important because of increasing experience with patients successfully stopping TKIs and maintaining MMR.

To gain further insight and audit our own experience within our trust, we conducted a retrospective analysis of all patients treated with TKIs for CML at our trust for at least 2 years. We collected data on eGFR calculated using the Modification of Diet in Renal Disease Study (MDRD) equation. MDRD eGFR values were collected for patients from initiation of TKI for up to 10 years at 3 monthly intervals for the first 3 years of treatment, followed by 6 monthly intervals thereafter. Where data was not available for the specific month a weighted average of the two closest results was calculated.

We analysed 56 patients; the median age at TKI initiation was 48 (range 12–77). In terms of specific TKI use; 43 patients received imatinib, 8 patients received nilotinib, 3 patients received dasatinib and 2 patients received bosutinib. Patients were on single TKI treatment for a median of 72 months (Interquartile range 38–105 months). The mean baseline eGFR was 82.8 ml/min/1.73 m². eGFR declined gradually over time with a 5-year mean eGFR of 76.2 ml/min/1.73 m² ($n = 33$) and a 10-year mean eGFR of 60.6 ml/min/1.73 m² ($n = 9$). Across all patients, there was a significant decline from the first to the final eGFR recorded whilst on TKI treatment ($P < 0.001$ Wilcoxon signed rank test). The mean eGFR change per year was -2.8 ml/min/1.73 m² (95% confidence interval $+6.1$ ml/min/1.73 m² to -11.7 ml/min/1.73 m²). eGFR decline occurred across all TKIs but was more marked in patients on imatinib, with a mean eGFR change of -3.2 ml/min/1.73 m² per year for patients on imatinib ($n = 43$) versus -1.4 ml/min/1.73 m² per year for patients on other TKIs ($n = 13$).

Our results demonstrate evidence that eGFR declines over time in patients with CML on TKIs, particularly imatinib. Our real-world data confirms findings from other studies and highlights the importance of monitoring renal function in patients with CML on TKIs. This is part of our centre's routine practice and we consider referring all patients with declining renal function to our renal colleagues. The decline in renal function appears to occur over a long period of time and emphasises the importance of maintaining pharmacovigilance for long-term adverse effects of TKIs. Although generally the decline in renal function is mild, it may be another factor to consider when deciding whether to trial stopping TKI treatment in patients in MMR.

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Low haemoglobin A1c (HbA1c) result: lessons learnt in clinical practice

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Glycated Haemoglobin (HbA1c) is a well-established test in both primary and secondary care for monitoring diabetes control and has a significant impact on treatment decisions. Since 2011, it has been recommended by the World Health Organisation (WHO) for the diagnosis of diabetes mellitus. A lower HbA1c target is generally desirable in patients with known diabetes mellitus, but certain conditions or medications can falsely lower HbA1c, mainly by reducing the red cell life span. Examples include: haemolysis, haemoglobinopathies, acute blood loss, blood transfusion, hypertriglyceridaemia and chronic liver disease.

In Sheffield Teaching Hospitals, we assessed the number of HbA1c results below the reference range (19–47 mmol/mol) over a 12 months period (November 2017 to November 2018). The aim was to address the following questions:

- How many patients had a newly diagnosed haemolytic/red cell disorder as a result of the low HbA1c?
- How many patients had a known haemolytic/red cell disorder prior to testing and HbA1c was still used for diabetic screening or monitoring?
- Can a specific level of HbA1c be used as a predictor of haemolysis?
- Were there any other identifiable causes for the low HbA1c result?

During the specified period, HbA1c results below the reference range were recorded from the Apex laboratory system. This identified 50 low HbA1c results for 42 patients, ranging from 4–19 mmol/mol

- 6 patients had a new diagnosis related to their low HbA1c (4 patients had autoimmune haemolytic anaemia, 1 had an unspecified red cell membrane disorder and the remaining had anaemia of unknown cause).

Haemolysis was diagnosed at varying levels of less than 10 mmol/mol but also as high as 18 mmol/mol.

- Potential drug causes were identified in 3 patients: (1 patient was on Dapsone and 2 were on antiretroviral treatments for Human immunodeficiency virus- HIV)

Other causes: 6 patients had chronic liver disease on its own which was likely to be the cause

Lessons learnt:

- The need to consider haemolytic/red cell disorders in patients with unexplained low HbA1c (36% of patients).
- The degree of low HbA1c doesn't appear to predict presence of haemolysis based on our small numbers.
- HbA1c is still being requested inappropriately for patients with known haemolytic/red cell disorders as a screening or monitoring test for diabetes. An alternative test should be used for these patients such as blood sugar monitoring or Fructosamine.
- A knowledge gap is demonstrated and requires education among junior doctors, general practitioners and clinical nurse specialists in both secondary and primary care on the potential significance of a low HbA1c result.
- Laboratory prompts have been introduced routinely for low HbA1c results advising on the unreliability of the test for diabetes monitoring or screening in this situation and prompting consideration of haemolysis.
- This demonstrates the importance and value of Haematology liaison work and collaboration with other specialties such as the diabetes team/ general physicians and general practitioners.

Disclosure of Interest: None Declared

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Efficacy and safety of bosutinib by charlson comorbidity index in previously treated patients with chronic myeloid leukemia: results from the phase 4 BYOND study

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Bosutinib is approved for the treatment of Philadelphia chromosome-positive (Ph+) chronic myeloid leukemia (CML) resistant/intolerant to prior therapy and newly diagnosed Ph+ chronic phase (CP) CML. The phase 4 BYOND study is further examining efficacy and safety of bosutinib (starting dose 500 mg once daily) for the treatment of CML resistant/intolerant to prior tyrosine kinase inhibitors (TKIs). Here, the efficacy and safety of bosutinib by baseline comorbidities in previously treated patients with Ph+ CP CML were investigated. Comorbidities were analyzed using the Charlson Comorbidity Index (CCI), a validated measure of the influence of comorbid conditions and age on mortality. Scores were derived from baseline data and patients grouped by CCI score 2–3, 4–5, and ≥ 6. This analysis was based on ≥ 1 year of follow-up. 156 patients with Ph+ CP CML ($n = 42$ [26.9%], 48 [30.8%], and 66 [42.3%] for CCI 2–3, 4–5, and ≥ 6) received bosutinib; 83 ($n = 22$, 27, and 34 for CCI 2–3, 4–5, and ≥ 6) were resistant to ≥ 1 prior TKI and 73 ($n = 20$, 21, and 32 for CCI 2–3, 4–5, and ≥ 6) intolerant to all prior TKIs. At the data cutoff date, 61.9%, 68.8%, and 43.9% of patients with CCI 2–3, 4–5, and ≥ 6 were still receiving bosutinib. Median treatment duration was 28.9, 23.9, and 20.1 months and median dose intensity 366.7, 385.3, and 291.5 mg/day in the respective groups. Among all evaluable patients with a valid baseline assessment, 79.5%, 74.5%, and 65.1% with CCI 2–3, 4–5, and ≥ 6 achieved/maintained major molecular response (MMR); among those without a baseline response, 87.5%, 66.7%, and 85.7% of TKI-intolerant patients, and 64.3%, 50.0%, and 30.0% of TKI-resistant patients, achieved MMR. Grade 3/4 treatment-emergent adverse events (TEAEs) rates were 64.3%, 72.9%, and 80.3% for CCI 2–3, 4–5, and ≥ 6. Grade 3/4 TEAEs differed between groups; diarrhea and increased alanine aminotransferase (ALT) were more common for CCI 2–3 (21.4% and 23.8%, respectively) and 4–5 (18.8% and 16.7%) and pleural effusion more common for CCI ≥ 6 (12.1%). For CCI 2–3, 4–5, and ≥ 6, 19.0%, 20.8%, and 31.8% of patients permanently discontinued treatment due to AEs and 2.4%, 2.1%, and 9.1% due to insufficient response. Most common AEs leading to discontinuation in the respective groups were increased ALT and aspartate aminotransferase (7.1% each), increased ALT (8.3%), and pleural effusion (3.0%). 10 deaths occurred ($n = 0$, 1, and 9 for CCI 2–3, 4–5, and ≥ 6), 8 due to AEs unrelated to bosutinib, 1 CML-related, and 1 of unknown cause. No on-treatment transformations to advanced CML occurred. Results demonstrate efficacy of bosutinib

across CCI scores. Patients with CCI ≥ 6 showed a trend toward higher rates of TEAEs, discontinuations due to AEs, and deaths unrelated to bosutinib treatment. CCI stratification may enable identification of patients who are at a higher risk of developing TEAEs and require more careful monitoring.

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Patients' and physicians' perceptions of treatments and goal setting in immune thrombocytopenia (ITP): findings reported by UK patients and physicians from the ITP World Impact Survey (I-WISH)

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ITP is an immune disease that leads to a shortage of platelets, affects up to 4000 patients in the UK, and is defined as a platelet count of less than $100 \times 10^9/L$. Treatments aim to elevate patients' platelet counts and maximise quality of life (QoL). The I-WISH study captured perceptions from patients and treating physicians on treatment patterns in ITP and their perceptions on treatment goals.

I-WISH is a cross-sectional survey of ITP patients and physicians across 13 countries. Patients have been recruited via both patient support groups (ITP Support Association, UK) and physicians. Participating patients and physicians completed a 30-min survey which captured demographics, diagnosis pathway and symptoms, impact of ITP, treatment patterns and patient-physician relationship and a

steering committee of expert physicians and patient advocacy ITP specialists designed and endorsed the survey materials. Patients were asked about the treatments prescribed to manage their ITP. Patients and physicians were asked to select their top three treatment goals, and answered perceptual questions pertaining to the patient-physician relationship in terms of discussing and agreeing treatment goals, asked on a 7-point Likert scale where 7 is strongly agree. Agreement is classified as a rating of ≥ 5 .

31 physicians and 120 patients completed a survey in the UK. Mean (SD) patient age was 55 (14-17) years and most patients were female (68%). Patients reported a mean time since diagnosis of 11-4 (10-25) years and 69% of patients described their current health state as ≥ 5 using a 7-point Likert scale (7 is excellent health). All physicians were specialists in either haematology or haematology-oncology and reported an ITP caseload of 47-5 (39-22).

At the point of data collection, patients had been on their current treatment for a mean of 31-1 (45-98) months. The top three treatment goals selected by patients were 'healthy blood counts', 'increasing energy levels' and 'preventing worsening of ITP' (73%, 62% and 39% respectively). Physicians reported reduction in spontaneous bleeds, improving QoL and symptom improvement to be their leading treatment goals (74%, 61% and 55% respectively). Although patients and physicians indicate agreement that symptom goals are aligned (67% and 74% respectively), there is disconnect between patients and physicians as to whether they discuss treatment plans (49% and 90% respectively), and less than half (43%) patients believe they are kept informed about new treatment options (versus 87% physicians). Over a quarter (26%) of patients do not believe their current treatment plan is helping to reach their goals, and just under a third (31%) of patients do not believe their physician is aware of their treatment goals.

These data suggest that there is a disparity between patients and physicians in the UK with regard to what they prioritise when treating ITP, with patients believing a healthy blood count is most important while reducing spontaneous bleeds is the priority goal among physicians. There is a clear disconnect between patients and physicians in terms of how treatment goals are communicated, indicating a potential area for improvement in the patient-physician relationship to ensure patients are optimally treated.

Disclosure of Interest: N. Cooper Conflict with: Novartis, Amgen, Rigel, Dova, Griffiths, UCB, Principia, D. Provan Conflict with: Novartis, Amgen, Rigel, ONO, M. Morgan Conflict with: Novartis, UCB, T. Bailey: None Declared, B. Taylor: None Declared

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Comparison of 8G and 11G trephine needle size on sample quality, sufficiency and ease of procedure

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Introduction: Haematohistopathologists require a sample of sufficient volume and quality to be both sensitive in recognising the presence of pathology and to make a specific diagnosis. An 8G trephine needle has a 3-43 mm internal diameter, larger than the 11G at 2-39 mm, which translates to a 2x larger area. Not only can this provide substantially more material, it may allow for a more robust sample, less likely to suffer a crush artefact. However, the 8G needle can be intimidating, and there are concerns for patient tolerance and bleeding risk. There is no study comparing trephine gauge, so we

performed a prospective observational study comparing these two sizes.

Methods: Practitioners performing bone marrow biopsies at a large teaching hospital selected either 11G or 8G trephine needles for the procedure. They completed a questionnaire on the ease of performing the biopsy, whether several attempts were taken, if there was any prolonged bleeding, the ease of sample extraction and the approximate length. A haemato-histopathologist completed a questionnaire commenting on whether the sample was crushed, diagnostic, of sufficient quantity and the overall quality.

Results: Data was collected on 22 11G and 24 8G biopsy attempts, all using a 4" needle. The length of samples obtained were comparable (mean 1.95 vs. 2.13 cm, 11G vs. 8G). Ease of procedure appeared to be higher in 11G (86%) vs. 8G (75%). However, ease of sample extraction was lower in 11G (82%) vs. 8G (92%). A further attempt was required in 23% of 11G and 12.5% of 8G. Regular haemostasis was achieved less in 11G (86%) than in 8G (95%). Haemato-histopathologist assessment revealed similarly diagnostic samples from the 11G (90%) and 8G needles (92%), not including failed attempts. Crush artefact was seen in 15% of 11G vs. 25% of 8G. Overall quality was comparable between 11G (mean 4.3 of max score of 5) and 8G (mean 4.0). Sufficient tissue quality was lower in 11G (82%) vs. 8G (92%). Haemorrhage artefact was lower in 11G (5%) vs. 8G (21%). No results demonstrated significance in a chi-squared test.

Conclusion: Our study comparing 11G and 8G trephine needle suggests a similar overall sample quality between the needle sizes. With around double the area of sample, theoretically only half the length is needed to obtain the same volume of tissue. Despite this, operators were obtaining similar lengths with both biopsy needles. Some differences were seen, with the 11G needle holding advantages in the ease of taking the sample, and having less haemorrhage artefact. The 8G needle was easier to extract the sample from, had less repeated attempts and more often provided sufficient tissue quantity, but had no advantage diagnostically over the 11G. Haemostasis appeared to be paradoxically slightly better in the 8G biopsies, though this may be due to operator selection based on patient characteristics. Overall quality, as judged by the haemato-histopathologist, were similar. We believe this is the first comparison of bone marrow trephine needle sizes. There may be operator reasons for selecting the 11G needle, especially in frail or anticoagulated patients, however this study confirmed our suspicion that the 8G needle provided samples which were more often of sufficient quantity, required less repeat attempts, and additionally did not demonstrate worse haemostasis. This information may of use for departments wishing to select between these two needle sizes.

Disclosure of Interest: None Declared

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Ruxolitinib in myelofibrosis: a multicentre analysis of real-world patients in England, Northern Ireland and Wales

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Ruxolitinib (RUX), the oral Janus kinase (JAK)-1/2 inhibitor, is licensed to treat disease-related symptoms or splenomegaly in primary myelofibrosis (PMF), post-essential thrombocythaemia (PET)-MF, or post-polycythaemia vera (PPV)-MF. From 2012, MF patients with intermediate (int)-1/2 or high-risk disease, based on the International Prognostic Scoring System (IPSS), could access RUX via the Cancer Drugs Fund (CDF) in England. At that time, access to RUX in Northern Ireland (NI) and Wales was limited mainly to clinical trials. In 2016, RUX was commissioned by the NHS for int-2 and high-risk patients, but there is increasing evidence that lower-risk patients also benefit significantly from treatment.

We performed a multicentre retrospective analysis of real-world MF patients treated with RUX outside of clinical trials, at 13 hospitals in England, Wales and NI, from 2011–2020, to assess overall survival (OS), safety and efficacy.

We identified 178 MF patients, with median age at the start of therapy of 69 years (29–91) and 55% were male. There were 76 (43%) PMF, 51 (29%) PPV-MF, 37 (21%) PET-MF, and 14 (8%) post-myeloproliferative neoplasm-unclassified MF patients; consisting of 6 (3%) low-risk, 39 (22%) int-1, 57 (32%) int-2 and 76 (43%) high-risk patients. A driver mutation was found in 88% of patients (124/141); 80% JAK2+, 6% CALR+, 2% MPL+. RUX was used first-line in 54 (36%) patients, and 111 (62%) were treated via the CDF.

The median OS for all patients was 3.5 years (0.02–8.12) with median follow-up of 3.5 years (95% CI 3.4–3.8). The median OS was 1.7 years for high-risk patients, 8.1 years for int-2, and was not reached in low/int-1 risk patients. The IPSS remained prognostic with 3-year OS of 100%, 76%, 57%, and 32% for low, int-1, int-2 and high-risk patients (HR 2.4 [95% CI 1.7–3.2]; $P < 0.0005$). The median duration on therapy was 2.7 years (0.0–9.0) with estimated 1, 3 and 5-year discontinuation rates of 32%, 55%, and 64%, respectively.

Data collection of objective clinical response measures, including spleen size, weight and symptom-burden, is underway. Of the data collected to date, 68% (69/101) experienced a reduction in spleen size. As previously observed, weight gain (71%; 68/96 patients) remains predictive of survival ($P = 0.0003$). The most common grade 3/4 AEs were thrombocytopenia (25%) and anaemia (38%), with 77 patients (43%) requiring dose adjustments/interruption. Infection was also common, with 62 patients receiving antimicrobial therapy. Seventeen patients (9.6%) transformed to acute myeloid leukaemia (AML).

At the time of analysis, 98 (55%) patients had stopped therapy; including 29 with progressive disease, 29 who died, 22 with AEs, 9 who had stem cell transplantation, and 2 by patient choice. In total, 83 (47%) patients have died; including 21 from disease progression/high-grade transformation, 18 from infection, 5 from bleeding, 2 from second cancers, 2 from post-allogeneic stem cell transplant complications, and 1 from myocardial infarction. Eighty (45%) patients remain on therapy.

Our data reflect contemporary clinical trials that show the efficacy of RUX in improving MF-related splenomegaly and symptoms in both int-1/2 and high-risk patients. As expected, haematological AEs were common but readily managed. Longer follow-up is needed to assess the impact on survival and further studies into the use of RUX in lower-risk patients are required.

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Abstract – steroid induced hyperglycemia and diabetes on the haematology ward: are we getting it right? Re-audit following initiation of JBDS-IP guidelines and teaching

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High doses of pulsed steroids are commonly integrated in chemotherapeutic regimens used for treating haematological malignancies. This can worsen glucose control in people with pre-existing diabetes (steroid induced hyperglycaemia (SIH)), or induce diabetes (steroid induced diabetes (SID)). Risk factors include high doses and/or prolonged duration of steroid treatment, obesity (BMI > 30), increasing age, personal history of previous SIH/SID, and family history of diabetes. The prevalence of SIH/SID in the outpatient setting has been reported to be as high as 40%, but the prevalence of SIH/SID in haematology inpatients and the effects of different steroid regimens used in various haematological malignancies on glycaemic control in inpatients are not largely known.

The Joint British Diabetes Societies for Inpatient care (JBDS-IP) guidelines recommend in people without pre-existing diabetes on steroid therapy to monitor capillary blood glucose once daily prior lunch or evening meal, and test people with diabetes four times daily.

We evaluated the management of SIH/SID and implemented a protocol for detection and management of SIH/SID on a haematology ward.

Data from a single-centre haemato-oncology inpatient service receiving therapy as per the NICE guidelines for their respective haemato-oncologic condition was collected using a standard proforma over a 2 months period. An initial group of 18 patients were screened in 2018 (Group 1) to determine the prevalence and management practice. Re-audit of a further group of 18 patients in 2019 (Group 2) to evaluate change in practice was conducted. Teaching sessions were delivered and an algorithm for screening and management of SIH/SID based on JBDS-IP guidelines was introduced post Group 1 analysis.

Patients included had high grade lymphoma (10/18 versus 6/18), plasma cell dyscrasia (4/18 versus 7/18), acute lymphoblastic leukaemia (2/18 versus 1/18), acute myeloid leukaemia (1/18 versus 0/18), chronic lymphocytic leukaemia (1/18 versus 3/18) and myelofibrosis (0/18 versus 1/18). The average hydrocortisone equivalent dose given over a 24-h period was 155 mg (range 100–625 mg) in Group 1 versus 462 mg in Group 2 (range 100–1332 mg) ($P = 0.49$) with an

average duration of steroid treatment of 14 days versus 8.6 days in Groups 1 and 2 respectively.

Only 28% of patients had appropriate monitoring of capillary blood glucose (CBG) in Group 1. Following teaching and implementation of the protocol, appropriate monitoring of CBG improved to 94% in Group 2. With improved monitoring of CBG, 78% of patients were found to have SIH/SID in Group 2, compared to 44% in Group 1. Escalation of diabetes treatment plan including gliclazide or insulin initiation based on the protocol was also proactively and promptly started in 6 patients in Group 2 before diabetes team review, compared to none in Group 1.

Awareness of SIH/SID was low among Haematology in the inpatient setting; in particular SIH. Our intervention led to significant improvement in screening, detection and prompt treatment of SIH/SID. We achieved the recommended national standard of monitoring for SIH/SID, in 94% being appropriately screened (JBDS-IP audit standard 90%). Further work is needed to improve post discharge care planning. Similar practice review is currently being planned in the outpatient setting.

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The emotional and psychological impact of blood cancer in patients in England

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Blood Cancer UK (formerly Bloodwise) is the UK's leading blood cancer charity, funding clinical trials and biological research into all blood cancers, and providing information, support, and advocacy to those affected by the disease.

The emotional and psychological impact of cancer is well documented, and its impact increasingly understood and recognised. However, like in a number of other aspects, the emotional and psychological impact of blood cancer is different from other cancer types. For example, approximately 30% of individuals with blood cancer symptoms must visit their GP three or more times before being referred to a hospital for diagnosis (comparing unfavourably with other cancer types), and several blood cancers are chronic conditions often with periods of active monitoring (also known as 'watch and wait') meaning individuals affected never live 'beyond' cancer. Accordingly, the emotional and psychological impact of blood cancer was determined to be a priority for Blood Cancer UK with a particular patient need identified for patients on active monitoring (Patient Needs report, 2015).

The annual NHSE Cancer Patient Experience Survey (CPES) provides insight into the care experienced by cancer patients across England treated as day cases or inpatients to monitor cancer care progress; drive quality improvements; and inform the work of care providers and stakeholder groups. CPES respondents are given the opportunity to be approached for an additional follow-up survey.

To better understand the unique emotional and psychological impact of blood cancer; the extent to which these needs are recognised and addressed by the NHS in England; and whether the impact and care needs and experiences of different patient groups vary, Blood Cancer UK performed a pan-blood cancer follow-up survey of CPES respondents. The survey – performed in collaboration with the charity Anthony Nolan and the specialist health and social care survey organisation Quality Health – contained questions covering diagnosis and symptoms, the emotional and psychological impact of blood cancer, professional and non-professional support offered /

received, and information needs. A number of free text sections were also included to further explore themes.

In total, 1430 patients completed the questionnaire, a response rate of 73%. In order to understand the unique emotional and psychological impact of active monitoring (a patient group excluded in the CPES follow-up cohort), an identical online survey was performed in parallel by a self-identified audience (204). Both surveys were performed in such a manner to additionally allow for questionnaire responses to be interrogated by blood cancer type, gender, age, NHS vs. private funding, urban vs. rural, socioeconomic status, ethnicity, and stem cell transplant status.

Analysis of survey results (fieldwork undertaken between August and October 2019) is ongoing but preliminary data shows a tentative unmet patient need. While 80% of female and 70% of male patients felt blood cancer impacted upon their mental health (with those on active monitoring expressing a particular need for support), approximately half of patients (48%) were not offered any mental health support by the NHS. A full analysis and description of identified discrepancies between patient groups shall be discussed in due course.

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Managing patients with systemic mastocytosis with associated haematological neoplasm: UK single centre experience

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Advanced systemic mastocytosis (AdvSM) is a rare haematological neoplasm sub-classified into either aggressive SM, SM with an associated haematological neoplasm (SM-AHN) or mast cell leukaemia. Approximately 60–70% of patients with AdvSM are categorised as having SM-AHN; a heterogeneous disorder with regard to both clinical phenotype and prognosis and associated with significant therapeutic challenges. Median overall survival in SM-AHN is 24 months with SM-MPN patients having longer survivals compared to those with either SM-CMML, SM-MDS or SM-acute leukaemia. Recent genomic characterisation revealed that *SRSF1*, *ASXL1* and *RUNX1* ('SAR') mutations confer a worse prognosis in AdvSM. Clinicians should be aware that patients may present with indolent SM and subsequently develop an AHN, hence acuity to this potential is necessitated.

We characterised a total of 31 patients with SM-AHN. Myeloid AHNs were the most frequent: 10 CMML, 11 MPN, 2 AML, 2MDS and 3 MDS/MPNu. Plasma cell dyscrasias (PCD) and lymphoproliferative diseases comprised 2 cases and 1 patient had concurrent diagnoses of PCD, CMML and SM. A total of 13/31 SM-AHN patients had an extended myeloid gene panel performed. 'SAR' and *TET2* mutations were identified in 6 CMML patients (2 patients had 1 ≥ SAR mutations and 4 had detectable *TET2* mutations). *JAK2* V617F mutation was present in 8 patients with SM-MPN; 2 of whom also had *TET2* mutations. 'SAR' mutations were identified in one patient with MPNu.

The variable natural history of AHN complicates therapeutic decision-making and timing of intervention. Establishing whether the mast cell or non-mast cell element is the predominant problem is necessitated to facilitate appropriately targeted therapy. Determining the mast cell and *c-KIT* allelic burdens, tryptase levels and degree of dysplasia or fibrosis in MDS and MF respectively, can be of value. Potential targeted therapies for the AHN component may be required (e.g. ruxolitinib in MPN or azacitidine in CMML/MDS)

whilst those with 'high SM components' benefit from KIT inhibition (avapritinib, ripretinib or midostaurin). Patients with MPN-AHNs and low risk of leukaemic transformation may be best managed as per 'non-SM' associated MPN and reduction of perceived thrombotic risk prioritised.

We present 3 illustrative clinical cases:

- 50 years old male diagnosed with indolent SM (D816V+), noted to develop a persistent thrombocytosis and reclassified as SM-MPN: *JAK2* V617F mutated Essential Thrombocythaemia, complicated by cerebral sinus thrombosis. Received cytoreduction with Pegylated IFN, hydroxycarbamide and anti-coagulation.
- 60 years old female presenting with indolent SM (D816V+), needing symptomatic mediator treatment and noted to have a raised PCV. Diagnosed with *JAK2* V617F mutated polycythaemia and treated with venesections. Progressed to post-PV myelofibrosis with anaemia and splenomegaly which is responding to ruxolitinib.
- 77 years old male presenting with fatigue, weight loss, cytopenias and a raised serum tryptase (593 ug/l). Diagnosed with AHN: SM+MDS/MPNu (D816V+, *JAK2* V617F, *SRSF2*, *ASXL1* and *RUNX1* mutations). Underwent treatment with midostaurin to good effect.

Conclusions: Patients with ISM can develop AHN or *vice versa*. Patients diagnosed with ISM need longterm follow up with comprehensive haematological and clinical reviews. Management of patients with SM-AHN requires personalised therapeutic stratification incorporating both clinicopathological and mutational features.

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Prevalence of latent iron deficiency in early pregnancy in a tertiary care hospital in Sri Lanka

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Maternal iron deficiency can affect morbidity and mortality of pregnant women (PW). It increases the risk of iron deficiency in newborn leading to impaired psychomotor or mental developments. Thus early detection of iron deficiency is vital for an optimum pregnancy. Latent iron deficiency (LID) is when serum ferritin level (SFL) is <30 µg/l, while haemoglobin (Hb) is at its normal level. If LID is not diagnosed early in pregnancy, it could lead to iron deficiency anaemia. Screening for LID by Hb alone is insufficient. SFL is a better test, yet for high cost is not a routine procedure.

This study aimed at:

- estimating prevalence of LID among PW,
- assessing LID in relation to parity, age, education level and household income,

- finding correlations between LID and red cell indices, RDW and red cell morphology; in order to suggest cost effective screening procedure for LID.

A cross sectional study was conducted at obstetrics and gynaecology units of North Colombo Teaching Hospital, Sri Lanka. All PW booked for antenatal care <20 weeks of gestation having normal Hb level (≥ 11 g/dl in <12 weeks and >10.5 g/dl in 13–20 weeks of gestation) were the potential study subjects. Among them, those with haematological diseases, active infections, liver diseases, chronic connective tissue disorders, malignancies and haemoglobinopathies were excluded. Written informed consents were obtained from the selected subjects. Data were gathered through interviews and antenatal records. Blood film report was from three independent observers.

Study consisted of 355 PW with 192 in LID group and 163 in normal group, indicating prevalence of LID at 54%. Majority (56%) were young, between 21–30 years. In LID, 56.2% were in 13–20 weeks of POA while in normal group 53.4% were in <12 weeks of POA. Primigravida pregnancies were noted among 43.2% in LID group while it varied in the normal group. Majority (72%) of subjects was well educated to GCE O/L or above, yet 89.6% of them were unemployed. Nearly half were of low income families with monthly earnings <LKR 50 000 (278 USD). Statistical significance ($P = 0.006$) for the gap between pregnancies being <2 years was observed but not for PW's age, parity, POA, income and education.

Blood film morphology depicted statistically significant ($P = 0.001$, 0.007) presence of hypochromic microcytic red cells and pencil cells. Considering FBC parameters, in spite of no significant difference between Hb, RBC, MCV, MCH and MCHC; the RDW was markedly significant ($P = 0.001$). Among PW in LID group, 25% had ferritin level in iron deficiency range (<15 μ g/l). Though 43% of PW in the cohort were on iron supplements, there was no statistical difference for haematinics between two groups. ROC was constructed for red cell indices to identify predictive values. Significant AUC and P values ($P = 0.0001$) were noted only for RDW.

Study results reveal high prevalence of LID in early pregnancy and one in four of women having ferritin in iron deficiency region. They are at high risk of developing iron deficiency anaemia. Presence of raised RDW, hypochromic microcytic red cells, pencil cells and <2 years gap between pregnancies were indicators of LID. To identify PW with LID, blood film, Hb and RDW could be recommended as basic tests. Among such identified, Ferritin test could be recommended as a follow-up and as an affirmative test. The suggested step-wise testing is a cost effective procedure.

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Physician associate (PA) led myeloproliferative neoplasm (MPN) telephone clinics: is this the way forward to a revolutionised service?

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Myeloproliferative Neoplasms (MPN) are chronic haematological disorders affecting 3890 people per year. The 5-year survival rate is 93.5%, illustrating a high number burden of patients for long term follow up in the outpatient clinic. This provides the challenge of maintaining the quality and sustainability of an ever growing service.

The majority of chronic MPN are stable patients with straightforward assessment and management. In early 2012 meetings were initiated and funded by Shire Pharmaceuticals with NHS Commissioners and Haematology Nurse Specialists to develop a tool-kit that will aid the development of MPN Nurse-Led Clinics. This was then adopted

by some Haematology departments across the country to reduce the number of stable MPN patients filling the clinics.

Telephone clinics emerged as a new way of managing chronic conditions both in primary and secondary care. On the other hand, there is growing interest in physician associates (PAs), what they do, and how they fit with the established roles and systems.

At Sheffield Teaching Hospitals we identified the following problems in our MPN service:

- Very large patient numbers and overbooked clinics
- Longer waiting times to see new patients
- Doctors and nurses staffing issues
- Patient dissatisfaction and complaints due to long waiting times
- Delays in outpatient pharmacy due to the high number of prescriptions
- Crowded waiting rooms and difficult parking

Questions raised?

- Do all patients need to be seen face to face?
- Do we have patients who are appropriate for telephone clinics?
- Who is the appropriate healthcare professional to do the clinic?
- Can the patients have their blood tests prior to appointments?
- Can patients pick their prescription on a separate visit?

The idea then came to set up a telephone clinic led by our PA to overcome some of the issues highlighted. Criteria for patients selected for the telephone clinic:

- Diagnosis of Essential thrombocythaemia or Polycythaemia Rubra vera who are on hydroxycarbamide or venesections, stable on three monthly follow up.
- Patients with low MPN 10 scores

A service evaluation questionnaire was given to 49 patients fitting the criteria above. Results include

- 81.6% of patients agreed to go on to telephone clinic
- 16% of patients were unhappy to be switched to telephone clinics due to reasons such as: "too many visits", short term memory loss, language barrier, not always at home
- 92% of patients had their blood tests done at the hospital with an average waiting time for blood tests of 17 min (range of 5–45 min)
- 36 % of patients rated the service as good (mainly due to lengthy waiting times)

Telephone clinics have been running since November 2019. Patients have their blood tests done at their preferred location either hospital or GP 1 week prior to the appointment and pick up the medications post appointment.

Feedback on the telephone clinic experience so far:

- 100% happy with PA review and got enough time over the telephone
- 100% confident in the quality of care provided by the PA
- 100% feel this is an improvement to the service

This illustrates a transferrable model of excellent quality, efficiency and cost effective care for patients with a chronic stable condition, led by a physician associate. It also demonstrated the change in MPN management over time and perhaps sheds the light into using MPN10 scores to select the appropriateness of patients being selected for telephone clinic follow up.

Disclosure of Interest: None Declared

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Capillary glucose monitoring in haematology inpatients on glucocorticoidsP. Withanage¹, V. Ruparel^{1,*}, M. M. Khan¹¹Haematology, NHS Grampian, Aberdeen, UK

Glucocorticoids are an important part of the treatment of a wide spectrum of haematological conditions such as Immune Thrombocytopenic Purpura (ITP), Autoimmune Haemolytic Anaemia (AIHA), Multiple Myeloma (MM) and lymphoma. However, there are several associated side effects including steroid induced hyperglycaemia and diabetes.

The aim of this audit was to evaluate whether our approach to hyperglycaemia monitoring in patients commenced on glucocorticoid therapy is consistent with standards set out by the Joint British Diabetes Society (JBDS). We prospectively analysed capillary blood glucose (CBG) monitoring in patients initiated on steroids for management of their haematological condition over a 2-month period. The data was collected by reviewing blood glucose monitoring charts and inpatient records. We assessed whether patients had adequate hyperglycaemia monitoring based on achievement of the following 3 criteria:

- Initiation of CBG monitoring once daily before or after lunch or evening meal
- Continued once daily blood glucose monitoring, if CBG is less than 12 mmol/l.
- If CBG greater than 12 mmol/l, escalate CBG monitoring to 4 times a day(QDS)

We also evaluated whether adequate glucose control was maintained in both of the following patient groups:

- Patients with steroid induced hyperglycaemia (steroid driven rise in glucose in existing diabetic patients)
- Patients with steroid induced diabetes (steroid driven rise in glucose in non-diabetic patients)

The JBDS guidelines indicate that adequate hyperglycaemia monitoring should occur in at least 90% of patients. Whilst maintenance of adequate glucose control should be achieved in 75% of cases.

During the study period, a total of 65 patients were commenced on glucocorticoid therapy as part of management of their haematological condition. Fifty-four (83%) patients on steroids were commenced on CBG monitoring of which only 16 (30%) were screened before or after lunch or tea time. All other patients had random CBG readings taken. Eleven (69%) continued once daily CBG monitoring if CBG was less than 12 mmol/l. Fourteen (22%) patients were found to have CBG readings above 12 mmol/l and 6 (43%) of these patients were escalated to QDS monitoring. One patient was an existing diabetic and they maintained adequate glucose control of steroid induced hyperglycaemia after starting Gliclazide as recommended by the JBDS guidelines. There were 14 (22%) patients with no prior diagnosis of diabetes who developed steroid induced diabetes. Of these, only 5 (36%) had adequate glucose control after starting Gliclazide.

In summary, this audit demonstrates that whilst 83% of patients are having some form of CBG monitoring after commencing steroid therapy, only a small proportion are undertaking this in line with current recommendations. The results also show that a significant proportion of patients with no previous history of diabetes went on to develop steroid induced diabetes, highlighting the importance of monitoring CBGs. It is possible that if the recommended CBG monitoring criteria were adhered to more stringently, there may have been more cases of steroid induced diabetes detected. Furthermore, the audit showed adequate glucose control was only achieved in approximately one third of these patients. This audit demonstrates the need

for an increased awareness of current standards in CBG monitoring for patients receiving steroids for haematological indications.

Disclosure of Interest: None Declared

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A review of the use of rasburicase for tumour lysis syndrome at Nottingham University Hospitals NHS TrustS. A. Kazmi^{1,*}, F. Wadelin¹¹Nottingham University Hospitals NHS Trust, Nottingham, UK

British Committee for Standards in Haematology (BCSH) guidelines recommend the use of rasburicase in patients with a haematological malignancy who are at risk of, or develop, established tumour lysis syndrome (TLS). Patients are stratified as having low, intermediate or high risk of TLS. Guidelines recommend that high risk patients receive increased hydration and a prophylactic single fixed 3 mg dose of rasburicase. Low and intermediate risk patients should be managed with hydration and allopurinol.

BCSH guidelines recommend rasburicase dose of 0.2 mg/kg/day ('full dose') once daily for up to 5 days in adult patients with established TLS. They also recommend no addition of potassium to hydration and no use of urinary alkalisation or peritoneal dialysis in the treatment of patients with TLS.

We did a retrospective audit of rasburicase use over 9 months from April to October 2019. The objective of our audit was to assess the adherence of the Haematology department in Nottingham University Hospitals Trust (NUH) to BCSH and local Trust guidelines on TLS. All electronic patient records were thoroughly searched to collect the data.

Twelve patients who received rasburicase for a total of 16 episodes were identified. Diagnoses included: Burkitt's lymphoma (3), high grade or transformed lymphoma (5), Acute Myeloid Leukaemia i.e. AML (2), Myelodysplastic Syndrome i.e. MDS (1) and Acute Lymphoblastic Leukaemia i.e. ALL (1). The median age was 65, (age range 9 to 82). 10 were males and 2 females. 3 patients received >1 episode of rasburicase, with 1 receiving 3 episodes and 2 others receiving 2 episodes each. Patients were assessed for risk of TLS in 15 of 16 episodes (94%). In 12 episodes, (75%) the patient was identified as being at high risk of TLS. A 3 mg prophylactic dose was used in only 1 of the 12 high-risk episodes (6% of total). Hydration was administered as per the guideline in 6 of 16 episodes (38%), not as per guideline in 2 (12%), and not administered in 8 episodes (50%). Of those who received appropriate hydration, fluid without potassium was given in 5 episodes (31%) and with potassium was given in 1 (6%). Established laboratory TLS was identified in 11 of the 16 episodes (69%). Full dose Rasburicase (0.2 mg/kg/day) was used in all 16 of the episodes and was continued until resolution in 10 episodes (63%) not continued in 3. Of the remainder, 2 did not have laboratory TLS and 1 patient died after electing for palliative approach. Urinary alkalisation or peritoneal dialysis was not used for any of the episodes.

Our findings show that rasburicase use is compliant with our local Trust guidelines which recommend full dose rasburicase for high risk patients as opposed to a prophylactic dose. Despite a higher dose of rasburicase being administered to these patients compared to the BCSH guideline, established laboratory TLS occurred in the majority, though no patient required dialysis. Noncompliance with the BCSH guidelines regarding hydration may explain the incidence of established laboratory TLS. Additional departmental teaching has focussed on the importance of adequate hydration in patients at high risk of TLS, and a repeat audit is planned.

Disclosure of Interest: None Declared

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Unique differences in clinical presentation between JAK2 splanchnic vein thrombosis and WHO defined JAK2 myeloproliferative neoplasmS. George¹, J. MacLeod^{1,*}, S. Francis¹¹Sheffield Teaching Hospitals NHS Trust, Royal Hallamshire Hospital, Sheffield, UK

Splanchnic vein thrombosis (SVT) encompasses portal vein thrombosis, Mesenteric vein thrombosis, splenic vein thrombosis and Budd chiari. Myeloproliferative neoplasms (MPN) are an important systemic cause of SVT. Large studies have shown JAK2 SVT has distinct clinical characteristics compared to WHO defined MPN. The prevalence of JAK2V617F mutation varies between 28–41% in SVT.

Patients with JAK2 SVT were identified from MDT records from 2006 to 2019, and were all diagnosed and treated in Sheffield. WHO defined MPN patients were selected from patients with MPN 10 scores at baseline from MDT records in Sheffield. 38 patients with MPN 10 scores and WHO defined MPN were assessed at diagnosis. The MPN10 score and age was recorded. The average age was 65.4 years (45–84 years). This group comprised 19 patients with JAK2 polycythaemia rubra vera (PRV) and 19 patients with JAK2 essential thrombocythemia (ET). In the PRV group there were 17 males and 2 females. The MPN 10 score ranged between 1–53. In the ET group there were 9 males and 10 females. The MPN score ranged between 4–28. There were 17 patients with JAK2 splanchnic vein thrombosis, of whom 10 had a haematocrit <0.45 and platelets <450. These 10 patients were classified as Myeloproliferative neoplasm – unclassifiable (MPN-U). The 7 patients with JAK2 SVT, 2 patients had an elevated haematocrit consistent with PRV and 4 patients had thrombocytosis consistent with ET. The average age was 53.8 years (43–67 years). There were 14 patients with portal vein thrombosis and 3 patients with mesenteric vein thrombosis. The JAK2 splanchnic vein thrombosis group comprised 10 females. The MPN 10 for the JAK2 splanchnic vein group ranged between 13–45.

There was a significant difference between the mean age of the JAK2 MPN group compared to the JAK2 splanchnic thrombosis ($P < 0.05$). The average MPN 10 score for the JAK2 MPN group was 13, while for the JAK2 splanchnic vein thrombosis group the MPN 10 score was 18. There was a significant difference in the MPN 10 scores between the groups ($P < 0.05$). Patients with JAK2 splanchnic vein thrombosis had higher individual MPN 10 scores for early satiety and abdominal discomfort.

The 38 patients with JAK2 MPN were treated with Hydroxycarbamide and aspirin ($n = 30$), venesections and aspirin ($n = 4$) and aspirin only ($n = 4$), as per BCSH guidelines. Of the 17 patients with JAK2 splanchnic vein thrombosis, 7 patients were treated with hydroxycarbamide due to either an elevated haematocrit or platelet count greater than 450, while the other 10 patients had normal FBC results and did not receive cytoreductive treatment. All patients with JAK2 splanchnic vein thrombosis were treated with warfarin with a target INR range of 2–3.

There are unique differences between patients with JAK2 splanchnic vein thrombosis and WHO defined JAK2 myeloproliferative neoplasms. Patients with JAK2 SVT are younger age, and have higher MPN 10 scores compared to WHO defined JAK2 MPN patients. JAK2 SVT patients have higher individual MPN 10 scores for abdominal discomfort and early satiety. 10 patients with JAK2 SVT had no evidence of proliferative blood features and were not treated with cytoreductive treatment. All JAK2 SVT patients received life long warfarin, while only 2 patients with WHO defined MPN were on lifelong warfarin. The optimal treatment for JAK2 SVT is unclear,

but needs to incorporate prevention of recurrent thrombosis, treatment of proliferative counts and support of liver dysfunction.

Disclosure of Interest: None Declared

BSH2020-PO-067

Abstract Withdrawn.

BSH2020-PO-068

The diagnostic yield of bone marrow biopsy in patients with liver failureT. Ko Ko^{1,*}, H. Wood¹, S. Patel², T. Pirani², S. Kassam¹¹Haematology, ²Critical care, Liver Intensive Care and ECMO, King's College Hospital, London, UK

Haematological diagnoses can be associated with liver failure. Cytopenias are also common in patients presenting with liver failure and following liver transplantation. Distinguishing a primary haematological/immunological cause from secondary causes can be difficult and a bone marrow biopsy (BMB) remains a frequently performed investigation.

To evaluate the diagnostic utility of a BMB and its influence on patients management in the UK's largest liver unit

We retrospectively collected clinical, biochemical and pathological data on consecutive referrals to the haematology department for BMB in patients with liver failure or a liver transplant between October 2012 and December 2018. Both paper and electronic patient records were used.

We defined cytopenias as Hb <100 g/l, platelets < 100 × 10⁹/l and neutrophils < 1.8 × 10⁹/l present in more than 2 cell lines.

108 BMBs were performed in 103 patients. Five patients had two BMBs. The median age was 47.7 years (range 18–76 years) with 57 (55%) male patients. Forty-three (42%) patients had received a liver transplant.

The indications for BMB were possible haemangiolymphocytosis (HLH) in 28 (27%), cytopenias in 47 (46%), post-transplant lymphoproliferative disorder (PTLD) in 14 (14%), myeloproliferative neoplasm (MPN) in 10 (10%) patients with Budd-Chiari syndrome, myeloma in 2 (2%), lymphoma in 1 (1%) and storage disorder in 1 (1%).

The final diagnosis after BMB included HLH in 8 (8%) patients, hepatosplenic T-cell lymphoma in 2 (2%), T-cell large granular leukaemia (T-LGL) in 1 (1%), aplastic anaemia in 4 (4%), telomeropathy in 2 (2%), probable MDS in 4 (4%), immune thrombocytopenia in 1 (1%), and MPN in 4 (4%). All 4 with MPN had evidence of the JAK2 V617F mutation. The BMB did not inform the final diagnosis in 77 (75%) patients.

In those who were confirmed to have HLH, the trigger was likely viral with CMV and/or EBV viraemia in 6, EBV and HSV viraemia in 1 and HHV8 viraemia in 1. Four patients were treated actively with various combinations of ganciclovir, rituximab, IVIG, steroids and etoposide, however, 6/8 patients died.

In 4 patients diagnosed with aplastic anaemia, 1 patient was treated with ciclosporin followed by allogeneic stem cell transplant (SCT), 1 patient is awaiting a SCT and 2 patients are under active surveillance only. In the 4 patients with probable MDS, 1 was treated with supportive (G-CSF and erythropoietin), 2 are under observation and 1 has died. The 2 patients with telomeropathy, 1 with immune thrombocytopenia and 1 with T-LGL are all under active surveillance. Both patients with hepatosplenic T-cell lymphoma have died; 1 prior to chemotherapy and 1 patient died with from relapsed disease.

Although we did not diagnose PTLD from the BMB, 8 (8%) patients subsequently received this diagnosis from another biopsy site. Four of the 8 patients have died. In the 4 patients diagnosed with an MPN, all 4 patients (4%) were treated with hydroxycarbamide. The 5 patients who had repeat BMBs were referred for possible HLH of which the diagnosis was subsequently made in 4. Overall, 42 (41%) patients have died.

Although the majority of BMBs (75%) did not reveal a diagnosis, a haematological/immunological diagnosis was made in 25% of referrals. Therefore, a BMB remains an important investigation in patients with liver failure and cytopenias with a wide range of potential diagnoses that have implications for therapy.

Disclosure of Interest: None Declared

BSH2020-PO-069

A review of a case series of post renal-transplantation erythrocytosis, a study of aetiological factors and outcomes

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Post renal-transplant erythrocytosis (PTE) is a rare complication, where the implicating aetiological factors, management strategies and outcomes are poorly defined. We performed an observational case control study to define these in a large cohort of 346 post renal transplantation (RT) patients with follow-up at a single centre over a period of 4 years, for the development of erythrocytosis, their management and outcomes.

Elevated haematocrit (Hct) > 51% after RT, 8 weeks apart was used as the defining criteria. Standard criteria were used to define normalization of Hct, grafted kidney and renal functions and immunosuppressive treatment (IST). We reviewed the cause of end-stage renal disease (ESRD), comorbidities present, pre-transplant reticulocyte count, type of renal transplant – live or cadaveric kidney – and if the patient's native kidney was removed.

11 patients (3.1%) were identified with PTE at a median of 372 days (range 64 – 1088) post RT. All were male and the median age of the cohort at PTE was 56 years. All had end stage renal failure, with 8 having received pre-transplant haemodialysis. 8 had cadaveric and 3 had live related donors. All were matched for blood groups. Antecedent history of smoking was present in 4 patients, and none continued after RT. All patients had a normal reticulocyte count pre-RT. 3 patients had received exogenous erythropoietin (EPO) pre-RT.

4 patients had polycystic kidneys, 9 with hypertension and 4 with co-existing diabetes, 1 with lupus/drug-induced and 1 without any identifiable risk factors. 3/11 received post PTE angiotensin-inhibitor (ACE-I) treatment. Native kidneys were preserved in all the patients. All patients received and continued IST.

All patients had normal white cell and platelet counts at the presentation of PTE. Average Hct at presentation was 54.7. All screened JAK2 and Exon 12 mutation negative. Only one patient had an elevated Erythropoietin level. Screening tests for hypoxemia and renal artery stenosis were negative.

3 patients had bone marrow tests, that showed erythroid dominance, but nil of significance otherwise. Early renal dysfunction post RT was seen in 3/11 patients (within 30 days). None required renal supportive treatment.

6 patients were symptomatic (headaches and dizziness). None suffered any VTE event. 2 patients developed worsening renal functions but none required renal supportive treatment. 9 (82%) patients commenced regular venesections, 6 were commenced on Aspirin and

ACE-I. None of the patients had resolution of PTE, but were alive at a median follow-up of 222 days.

There is significant male preponderance to the development of PTE, and is secondary (low or normal EPO levels). Early renal dysfunction post RT as a contributing factor and ACE-I as prophylaxis against development of PTE need to be further explored in systematic prospective studies.

Disclosure of Interest: None Declared

BSH2020-PO-070

Fertility preservation in male oncology patients – emergency sperm and spermatogonial stem cell retrieval

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Background: Patients receiving chemotherapy or bone marrow transplant in oncological or haematological context are at high risk of losing their fertility hence the recommendation for sperm cryopreservation from semen. However, patients presenting with either azoospermia or inability to provide semen, face the risk to have to sacrifice their possibility to father their biological children. To those patients our andrology unit aims at offering timely preservation of their fertility by surgical sperm/stem cell retrieval. **Methods:** Patients were enrolled prospectively and data analysed retrospectively.

Results: We screen on annual basis about 500 patients necessitating chemotherapy or bone marrow transplant, of which 10.5% present azoospermia. From January 2018 to September 2019, 40 patients (median age of 18.4 years old and average testosterone at 7.1 nmol/l) underwent surgical sperm/stem cell retrieval which could be offered within 1 week from semen-analysis and preceding 1 day from chemotherapy start date (medians expressed). Primary diagnoses were sarcoma in 11, leukaemia in 1, lymphoma in 10, testicular tumour in 5, multiple myeloma in 1, prostate cancer in 1, while 3 patients had non-oncological haematological diseases. 14 patients underwent TeSE, 5 had oncoTeSE, 5 had electroejaculation and TeSE, 1 had TeSA and 15 had spermatogonial stem cell retrieval. The success rate for surgical sperm retrieval was 78.9% to which could be added the 13 patients who had spermatogonial stem cell retrieval. Average Johnsen score was 5.08.

Conclusion: Emergency sperm/stem cell retrieval in oncological context is a valid treatment option with high success rate for patients unable to cryopreservation semen.

Disclosure of Interest: None Declared

BSH2020-PO-071

The efficacy and predictive value of primary care 2-week wait referrals for suspected haematological cancer: quality evaluation of GP referrals and their outcomes

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Background: The 2-week wait (2WW) pathway allows general practitioners to urgently refer patients presenting with suspected cancer symptoms, expediting investigation, diagnosis and treatment. It is believed that the 2WW referral pathway for suspected haematological

malignancy is often overused, with many forms deemed inappropriate or lacking in clinical information. With the average haematology outpatient appointment costing the NHS £160, responsible use of such resources should be optimised.

There is currently sparse data regarding the quality of haematological cancer referrals and their outcomes. A high-quality referral is appropriate at a multi-dimensional level: in necessity, timeliness, destination and process. The aims of this study are to review each of these domains and the correlation between diagnostic criteria and referral outcome (predictive value).

Methods: Retrospective case review of GP referrals under the suspected haematological cancer pathway between January–April 2019 was undertaken using hospital Clinical Data Archive and Somerset databases. Completeness of clinical reason, documented blood results, timeliness of initial specialist appointment and subsequent secondary care investigations were noted.

Appropriateness of referral was assessed using adherence to NICE guidance (NG12) and whether the referral form was marked inappropriate by the lead consultant following clinic. Final diagnosis was determined through hospital clinic letters, investigation reports and MDT cancer proformas.

Results: 104 referrals were reviewed. 97% of patients were seen in secondary care within 14 days. 13.5% (14) of referred patients were diagnosed with a haematological malignancy. These were lymphoma (8), leukaemia (3) and multiple myeloma (3). 5.8% were diagnosed with MGUS. Of those with a non-malignant diagnosis, 38.5% underwent further investigation: 3.8% were invasive procedures such as bone marrow aspirate and trephine.

The most common indication for 2WW referral was lymphadenopathy (32%) with a predictive value of 18%. 53.8% of referrals were deemed inappropriate with 29% based on symptoms alone (no prior investigation or blood test). Only 67% of referring clinicians used the correct paperwork to refer despite wide distribution of the forms to the relevant practices and prior education. Those that used the correct form were less likely to refer inappropriately.

Referral forms were incomplete in 56% of cases. In 39% of cases, the fact that a referral has been made for suspected haematological cancer was not adequately discussed with the patient.

Conclusion: A large proportion of 2WW referrals were incompletely filled out by the referring clinician and an alarming number did not document whether the referral had been discussed with the patient. The low positive predictive value (13.5%) in this study suggest that most referrals are made inappropriately or that the criteria, on which the referrals are based, lack specificity. To improve this, updated forms and educational worksheets will be distributed to local GP practices to offer stricter inclusion and exclusion referral criteria. Despite poor detection rates found in this study, it is in keeping with the national average for all cancer subtypes (11%).

Disclosure of Interest: None Declared

BSH2020-PO-072

Symptom burden in immune thrombocytopenia (ITP): findings reported by UK patients and physicians from the ITP World Impact Survey (I-WISH)

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ITP is an immune disease that leads to a shortage of platelets and affects up to 4000 patients in the UK. Patients with platelet levels

below $50 \times 10^9/l$ have an elevated risk of bleeding from trauma and a count below $20 \times 10^9/l$ can lead to spontaneous bleeding. While these serious symptoms of ITP are well reported, there are limited data on the other symptoms experienced by ITP patients in the UK and how this is reported by treating physicians. The I-WISH study aims to understand the frequency and severity of symptoms reported by ITP patients and treaters.

I-WISH is a cross-sectional survey of ITP patients and physicians across 13 countries. Patients have been recruited via both patient support groups (ITP Support Association, UK) and physicians. Patients and physicians completed a 30-min survey which captured demographics, diagnosis pathway, symptoms, impact of ITP and patient-physician relationship. A steering committee of expert physicians and patient advocacy ITP specialists designed and endorsed the surveys. Patients were asked to select symptoms from a pre-coded list experienced at diagnosis and currently. Patients were asked to rate the severity of symptoms they were currently experiencing. Physicians selected the top five symptoms most commonly reported by patients overall and at diagnosis and were required to rate all symptoms on their impact on quality of life (QoL).

31 physicians and 120 patients completed a survey in the UK. Mean (SD) patient age was 55 (14.17) years and most patients were female (68%). Patients reported a mean time since diagnosis of 11.4 (10.25) years and 69% of patients described their current health state as ≥ 5 using a 7-point Likert scale (7 is excellent health). All physicians were specialists in either haematology or haematology-oncology with a mean ITP caseload of 47.5 (39.22).

At diagnosis, 66% patients reported experiencing fatigue, whereas 55% physicians did not include fatigue as a top five symptom experienced at diagnosis. 61% patients experiencing fatigue at data capture reported a severity of ≥ 5 using a 7-point Likert scale (7 is worst imaginable). Physicians reported a similar impact of fatigue, with 65% indicating an impact of ≥ 5 using a 7-point Likert scale (7 is a great deal). At diagnosis, 18% patients experienced depression, although this was considerably less frequently reported by physicians, with 3% believing this to be a top five symptom patients present with. Despite this, 62% physicians reported depression to have a high impact on QoL. This aligns with 50% of patients experiencing depression indicating a high severity. 18% patients reported increased moderate-to-severe headaches/migraines at diagnosis in comparison with 3% physicians ranking it as a top five symptom in patients. 32% patients and 48% physicians reported a high severity/impact of headaches/migraines. Dizziness was reported by 13% patients at diagnosis, while less than 1% physicians perceived this to be a commonly experienced symptom presented by patients. However, patients (33%) and physicians (38%) both do report a high impact of this symptom.

Overall, these data indicate that ITP patients in the UK present with a number of symptoms which contribute to a significant disease burden. While it is clear that physicians recognise the severity of symptoms such as fatigue and depression associated with ITP, it is also evident that physicians may underestimate the proportion of patients who present with these symptoms.

Disclosure of Interest: N. Cooper Conflict with: advisory boards and educational meetings from Novartis, Amgen, Rigel, Dova, Grifols, UCB, Principia., D. Provan Conflict with: Honoraria from Novartis and Amgen Consulting fees from Novartis Amgen Rigel ONO Research support from Amgen Novartis and Rigel, M. Morgan Conflict with: Novartis – I-WISH Steering Committee UCB – ITP Patients Council, T. Bailey: None Declared, B. Taylor: None Declared

BSH2020-PO-073

The clinical characteristics, diagnosis and management of patients with mastocytosis in a tertiary centre

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Mastocytosis is a heterogeneous group of disorders in which neoplastic mast cells accumulate in various organs. It ranges from being relatively indolent with predominantly cutaneous involvement to an aggressive multi-systemic disease. The World Health Organisation (WHO) has defined categories of Mastocytosis to reflect the varying degrees of organ involvement and severity. Cutaneous Mastocytosis (CM) is sub classified into three categories; Urticaria Pigmentosa (UP), Diffuse CM and Solitary Mastocytoma of the skin. Systemic Mastocytosis (SM) is classified into indolent cases and advanced SM. Advanced SM includes patients with SM and an associated haematological neoplasm (SM-AHNMD), aggressive SM, and mast cell leukaemia. Both diagnosis and management require a multi-disciplinary team approach. Due to its rarity, there has been little systemic study to date.

A service evaluation was carried out in order to establish the characteristics, diagnosis and management of patients with Mastocytosis across South Yorkshire and North Derbyshire. Retrospective data for 31 patients was collected and analysed. Of those patients, 28 had SM (8 of whom were deceased) and 3 had CM alone. Of the SM group, 13 cases had a prior diagnosis of CM. The average time from CM diagnosis to SM diagnosis was 5 years. The most common variant of SM was indolent, followed by SM-AHNMD. Two patients were documented as having the WHO aggressive subtype. There were no cases of mast cell leukaemia. The most common subtype of CM was Urticaria Pigmentosa (14 cases) followed by 2 cases of diffuse CM. The average age at diagnosis for those with SM was 59 years compared with 17 years for those with CM. There was no gender bias in the SM group.

As per WHO guidelines, major and minor criteria must be met to reach a diagnosis of SM. In our patient population, the diagnosis was reached based on one major and one minor criterion in 85% of cases and on three minor in 15%. The tissue used to diagnose SM was bone marrow (BM) in 90% of cases. In one case the diagnosis was made on duodenal tissue as the patient was unable to tolerate a BM trephine. The average tryptase level at diagnosis in the SM patients was 85 ng/dL. The following immunohistochemistry (IHC) was used in diagnosis; CD25 (6 cases), CD2 (13 cases) and CD117 (16 cases). There was no immunochemistry performed in 7 cases. Of the 17 cases known to have had a c-kit mutation analysis performed, 86% were tested on BM and 18% on a blood sample.

Further investigation in SM includes identification of organomegally in order to assess disease severity. Within the patient cohort there were 8 cases of splenomegaly and 2 cases of hepatosplenomegaly. On-going management of these patients requires a holistic approach and monitoring of bone density is important. Of the 28 patients with SM, 57% had a DEXA scan with 2 cases showing osteoporosis. 11 patients were receiving bisphosphonate treatment and/or vitamin D supplementation.

SM has been shown to be a disease of middle age with CM manifesting in young adults. Almost half of the SM cases had a prior diagnosis of CM with a relatively short interval between each diagnosis. In-depth examination of BM was required to reach the diagnosis of SM using IHC and molecular testing in the majority of cases. The most common group of SM patients had indolent disease which requires monitoring of symptoms and regular assessment of bone density.

Disclosure of Interest: None Declared

BSH2020-PO-074

An evaluation of double or triple positivity in JAK2 positive essential thrombocythaemia and primary myelofibrosis

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The *JAK2* c.1849G>T; p.(Val617Phe) mutation, commonly referred to as *JAK2* V617F, is identified in approximately 50% of cases of Essential Thrombocythaemia (ET) and Primary Myelofibrosis (PMF). Other somatic mutations commonly identified in ET and PMF include those in the *MPL* gene which are found in approximately 4–7% of patients, and the *CALR* gene which are detected in approximately 20–35% of cases. Different clinical phenotypes have been described when mutations in these genes occur in isolation; however, there is little published literature on the prevalence of dual or triple positivity for driver mutations in *JAK2*, *CALR* and *MPL* in these conditions.

The aim of this study was to identify the proportion of patients with *JAK2* V617F positive ET or PMF who have additional somatic mutations in the *CALR* and *MPL* genes and to determine whether additional genetic testing is necessary in patients found to have the *JAK2* V617F mutation.

This was a retrospective analysis of all *JAK2* V617F positive cases identified at our centre. The cases were subdivided into the following disease groups: Polycythaemia Vera (PV), ET or PMF. ET and PMF cases which were also tested for common *CALR* and *MPL* mutations between August 2014 and February 2019 were identified. The proportion of dual or triple mutations was determined for both ET and PMF.

DNA extracted from blood samples was analysed for the following mutations: *JAK2* c.1849G>T; p.(Val617Phe) [V617F]; *MPL* c.1544G>T; p.(Trp515Leu) [W515L]; insertions or deletions in *CALR* exon 9 based on GenBank sequences NM_004972-3, NM_005373-2 and NM_004343-3, respectively. *JAK2* and *MPL* mutations were detected using fluorescent allele specific PCR and capillary electrophoresis. *CALR* insertions and deletions were detected using fluorescence based PCR fragment length analysis.

A total of 116 cases of *JAK2* positive ET were detected, of these, 97 patients were also tested for *CALR* and *MPL* mutations. There were only 2 (2.1%) cases which were *JAK2* and *CALR* positive. Dual positivity for *JAK2* and *MPL* accounted for only 3 (3.1%) of cases. No cases were triple positive.

16 patients were diagnosed with *JAK2* positive PMF, of which, 12 were tested for presence of other mutations. There were no cases of dual or triple positive PMF.

In this cohort there was a very low detection rate for dual positivity for driver mutations in the *JAK2*, *CALR* and *MPL* genes and there were no triple positive cases. Interestingly, all the cases of dual positivity were in ET patients with none of the PMF cohort having an additional somatic mutation in one of these genes. Given the very low prevalence of dual/triple positivity and lack of data as to whether there is a clinical impact of having a mutation in more than one of these genes, there is an argument to not test for *MPL* or *CALR* in *JAK2* positive ET or PMF patients. The analysis was limited to the most common mutations and did not exclude the presence of less common mutations in *JAK2* exon 12 or *MPL* exon 10; however, the vast majority of *JAK2* positive patients have the *JAK2* c.1849G>T; p.(Val617Phe) mutation and about 80% of *MPL* mutations are reported to be *MPL* c.1544G>T; p.(Trp515Leu).

Disclosure of Interest: None Declared

BSH2020-PO-075

An audit of tumour lysis syndrome prophylaxis in patients with haematological malignanciesA. Cairns^{1,*}, M. M. Khan¹¹Haematology, Aberdeen Royal Infirmary, Foresterhill Health Campus, Aberdeen, UK

Tumour lysis syndrome (TLS) is a well recognised complication of chemotherapy which can lead to biochemical abnormalities as well as renal dysfunction. The British Committee for Standards in Haematology (BCSH) published guidelines in 2015 providing recommendations on risk stratification of patients and what prophylaxis they should receive depending on whether they were classified as low, intermediate or high risk.

The aim of this audit was to review patients in a level 3 haematology centre providing autologous haematopoietic stem cell transplantation centre were:

- Appropriately assessed for tumour lysis risk
- Received prophylactic treatment appropriate to their level of risk
- Went on to develop tumour lysis syndrome

Data was collected prospectively over a 3 months period from February to April 2019. Any patient admitted to the ward who was highlighted as potentially at risk of tumour lysis was included in the audit. For each case, the notes were reviewed to ascertain whether the patient had been formally assessed for TLS risk. The prescription records were reviewed to see if the patient received TLS prophylaxis, and if so, what treatment they received. Patient's biochemical parameters were monitored to see if TLS developed following delivery of chemotherapy.

Over the 3 months study period 20 patients were identified by the clinical team on admission as being at risk of tumour lysis syndrome. For each case, the risk category was determined by review of the diagnosis and laboratory parameters including Full Blood Count (FBC) and biochemistry. 8 patients were high risk, 9 intermediate risk and 3 low risk using the BCSH risk categories. There was however no documentation of the TLS risk stratification for any patient recorded in the case notes. Out of the 20 patients 13 (65%) received TLS prophylaxis appropriate for their risk status. Seven (35%) patients received treatment not required for their level of risk. One low risk patient received fluids and Allopurinol whilst 6 intermediate risk cases received given fluids and Rasburicase. Only 1 patient went on to develop tumour lysis syndrome despite prophylaxis. They were a high risk status and received fluids and Rasburicase.

In summary the majority of patients at our centre are being treated appropriate to their risk status. However, approximately one third of patients received TLS prophylaxis regimens not deemed appropriate for their level of risk based on the current BCSH guideline. Whilst there may be good clinical reasons for the TLS prophylaxis regimen used in these cases, this was not documented in the patient's case notes.

This audit therefore highlights the need to appropriately document the TLS risk profile for a patient due to receive chemotherapy. The proposed prophylaxis regimen should also be recorded, giving clear reason(s) if it is deemed necessary to deviate from the guideline recommendations. One way to approach this is to include TLS prophylaxis as part of the multidisciplinary team (MDT) outcome proforma.

Disclosure of Interest: None Declared

BSH2020-PO-076

Abstract Withdrawn.

BSH2020-PO-077

Retrospective analysis of full blood counts and iron studies on patients diagnosed with bowel cancerT. Ball^{1,*}, H. Lusta¹, W. Thomas¹¹Haematology, University Hospital Plymouth NHS Trust, Plymouth, UK

There is a common stereotype that patients with bowel cancer tend to have hypochromic microcytic anaemia on presentation. Making associations between more general lab results and disease can be useful in modern healthcare, however it is imperative that these ideas aid diagnosis.

A retrospective study was conducted on patients diagnosed with colon cancer using endoscopy at Derriford Hospital over a 6-month period beginning January 2019 ($n = 88$). Where possible, full blood counts (FBC) and haematinics were evaluated in these patients at diagnosis and 6 and 12 months prior to this. All patients had FBCs within 12 months of diagnosis, in which 56% (49/88) had anaemia (haemoglobin < 120 g/l in women, < 130 g/l in men) within this time frame. Of those patients 55% (27/49) had microcytic anaemia (MCV < 80 fl), and in 69% (34/39) the anaemia was hypochromic (< 28 pg).

Anaemia was not invariable during the 3 timeframes that we reviewed the blood counts:

-On or around admission for the endoscopy or surgery: Anaemia was found in 16 of 28 (57%) patients tested. 7 microcytic (44%), 9 normocytic (56%), hypochromic 10/16 (36%).

-0 to 6 months before endoscopy or surgery: Anaemia was found in 35 of 69 (51%) patients tested. 18 microcytic (51%), 16 normocytic (46%), 1 macrocytic, 25 hypochromic (75%).

-6 to 12 months before endoscopy or surgery: Anaemia was found in 14 of 33 (42%) patients tested. 5 microcytic (36%), 9 normocytic, 8 hypochromic (57%).

Haematinics were not conducted on all patients, with only 42 patients having serum ferritin measured, with 26 (62%) of these patients having serum ferritin < 30 mcg/l. 21 patients had transferrin saturation tested and of these 18 (86%) had transferrin saturations below 20%. Interestingly, only 53% (18/34) of those patients with hypochromic anaemia had a serum ferritin < 30 mcg/l.

This data shows the importance of not ruling in/out bowel malignancy from haematological results. It is clear that this patient group does not conform to the classic stereotype and thus should be challenged. Further investigation in more patients of the same population would provide a clearer picture with regards to confirming these results. Haematinics were not regularly tested in this patient group and the importance of investigating the causes of a patient's anaemia should be reiterated.

Disclosure of Interest: None Declared

BSH2020-PO-078

Neutropenic sepsis: can we do better?M. Caruana Dingli^{1,*}, S. Cauchi¹, M. Borg¹, R. Abela¹, D. J. Camilleri¹¹Mater Dei Hospital Malta, Msida, Malta

Febrile Neutropenia refers to a life threatening haematological emergency, defined as a fever of $\geq 38.5^{\circ}\text{C}$ on one occasion, 38°C maintained over an hour, or symptoms/signs of shock in a patient with a neutrophil count of $<0.5 \times 10^9$ cells/l.

In these circumstances, our local current guidelines, based on the last audit in 2014, suggest the empiric use of piperacillin/tazobactam +/- gentamicin depending on the Multinational Association of Support in Cancer Care (MASCC) score and carbapenemase producing enterobacteriaceae status.

In this study we assessed whether these antibiotic guidelines were still appropriate.

We recorded 144 episodes of positive blood cultures from septic neutropenic patients over a span of 2 years in the haematology ward in Malta. We further documented the demographic data and antibiotic sensitivities for the bacteria grown.

Following this analysis, results indicated that 72% of the cultured organisms were gram negative bacteria and 28% were gram positive bacteria. *Klebsiella pneumoniae* (23%) was the commonest organism followed by *Escherichia coli* (16%), *Staphylococcus epidermidis* (12%) and *Pseudomonas aeruginosa* (10%).

With regards to the underlying haematological condition being treated, an equal number of leukaemias and lymphomas (45% each) were noted, with Non-Hodgkin's being the overall most common condition (39%) followed by acute myeloid leukaemia (31%).

Analysis of antibiotic sensitivities revealed that gentamicin was the commonest sensitive antibiotic (59%), followed by fosfomycin (52.8%), amikacin (51.4%), imipenem (51.4%) and meropenem (50.7%). Interestingly, piperacillin/tazobactam was only sensitive in 35.4% of cases.

Antibiotic sensitivities were also assessed for the four most commonly cultured organisms. In these cases we noted that piperacillin/tazobactam sensitivity was 70.8% for *Escherichia coli*, 36.3% for *Klebsiella pneumoniae*, and 73.3% for *Pseudomonas Aeruginosa*. Comparatively, sensitivity to amikacin was at 95.8%, 66.7%, and 93.3% respectively, and sensitivity to gentamicin 95.8%, 57.6% and 80% respectively. Meanwhile, tigecycline, vancomycin and linezolid were the most effective antibiotics for the treatment of *Staphylococcus epidermidis*, as it was sensitive in all cases recorded in this study. When comparing aminoglycoside sensitivity in cases where the organisms were resistant to piperacillin/tazobactam, amikacin held a slight advantage in the treatment of *Klebsiella pneumoniae*, covering 55% of cases as opposed to the 35% of cases covered by gentamicin.

Consequently, alterations to the current neutropenic sepsis guidelines are needed, with gentamicin being used first line with piperacillin-tazobactam if the MASCC score is ≥ 21 and amikacin being used instead of gentamicin if the score is < 21 to better empirically cover gram negative organisms, in particular *Klebsiella pneumoniae*.

In addition to this, where gram positive cocci are seen on gram stain or gram positive infections are highly likely, then the addition of vancomycin would be beneficial so as to cover *Staphylococcus epidermidis*.

This study highlights the importance of re-auditing and re-evaluating local practices and causative organisms with their sensitivities in collaboration with local infection control teams, so as to maintain a leading edge in the management of neutropenic sepsis.

Disclosure of Interest: None Declared

BSH2020-PO-079

Impact of stopping long-term immunoglobulin therapy in haematological secondary antibody deficiencyS. Goddard¹, D. Hughes¹, L. Diwakar¹, J. Graham^{2,*}¹Immunology, ²Haematology, University Hospitals North Midlands, Stoke-on-Trent, UK

The Commissioning Criteria for the use of therapeutic immunoglobulin (Ig) in immunology, haematology, neurology and infectious diseases in England were updated by NHS England in January 2019; with the colour coding scheme, previously devised for demand management but often utilised as a commissioning tool, being replaced by categorisation of Ig use in to routinely commissioned or not commissioned categories based on the strength of evidence.

Following these changes, we trialled stopping Ig infusion for all secondary antibody deficiency patients who did not meet the updated criteria; hypogammaglobinaemia (IgG < 4 g/L excluding paraprotein) AND recurrent or severe bacterial infection despite continuous oral antibiotic therapy for 6 months. Further details can be found on mdsas.com.

Twenty-nine patients were identified on long-term Ig infusion at UHNM, of which 17 (59%) were attributable to haematological disease/treatment; chronic lymphocytic leukaemia (CLL 6), non-Hodgkins lymphoma (NHL 6), marginal zone lymphoma (MZL 1), Waldenstrom macroglobulinemia (WM 1), monoclonal gammopathy (MGUS 1), post-transplant lymphoproliferative disease (PTLD 1), and benign B cell lymphocytosis (1). Average age was 69.5, with 53% male. Mean duration of Ig replacement was 5.3 years (range 1–13), with estimated cost £23,000/patient/year for product and day case costs with 25 g IVIg 3 weekly. Average immunoglobulin G (IgG) level prior to commencement was 3.34 g/l (0.3–6.1 g/l). Six patients (35%) had had a documented trial of prophylactic antibiotics prior to starting Ig. None of the patients met all the new commissioning criteria.

Ig infusions were stopped in spring 2019 and all patients were given a prophylactic antibiotic (azithromycin, doxycycline or septrin), an alternative standby course of antibiotics and instructions about how to manage any infections. Additional clinic appointments were provided.

As of January 2020, all haematological patients remain off Ig infusion, with an average duration of 9.12 months (range 6–12). Mean baseline IgG level is 5.05 g/l, with 11 of 16 patients (69%) having an IgG level > 4 g/l (i.e. no longer eligible for replacement). Three patients (17.6%) were admitted with suspected infection; confirmed influenza A and remains off Ig, no confirmed evidence of infection and remains off Ig and died due to lung cancer. No patient has experienced a worsening rate of infection, with some patients having fewer infections off Ig.

The main challenges to taking long-term patients off Ig was reassuring patients that a prophylactic antibiotic was safe and effective, providing clear instructions to patients about antibiotic management, collecting data to demonstrate infection rate and providing extra clinic slots to monitor patients closely and provide appropriate support. Our experience demonstrates that trials of stopping Ig treatment can be done safely and confidently in haematological patients by clinicians and patients; ensuring unnecessary treatment can be identified, NHS costs reduced, and precious Ig saved.

Disclosure of Interest: None Declared

BSH2020-PO-079A

UK Patient perceptions on the symptomatic and emotional burden of vaso-occlusive crises resulting from Sickle cell disease**B. Inusa^{1,*}, J. James², O. Rajkovic-Hooley³, T. Bailey³, G. Medalla⁴, B. Taylor⁴**¹Paediatric Haematology, Evelina Children's Hospital, Guy's and St Thomas' Hospital, ²Sickle Cell Society, London, ³Adelphi Real World, Bollington, ⁴Novartis Pharmaceuticals, London, UK

Sickle cell disease (SCD) is characterised by vaso-occlusive crises (VOCs), which impact patients' quality of life (QoL) and increase the risk of organ damage and early death. There are limited published data describing the outcome of VOCs despite the chronic damage and fatality risk following a single VOC. Understanding the burden of SCD on patients can inform optimal disease management and improve patients' QoL. The International Sickle Cell World Assessment Survey (SWAY) was a cross-sectional survey of 2,145 SCD patients and 365 healthcare professionals (HCPs) across 16 countries. Here we present data from UK patients, who were recruited via their treating HCPs and the patient charity – The Sickle Cell Society. Patients aged from 6 years were eligible to participate, with data collected using self-completed or proxy surveys. The survey captured demographics, symptoms, VOC burden and disease impact. VOCs were defined as 'severe pain crises' and 'severe' as patient defined. Analyses were performed to evaluate impact of VOC burden on emotional wellbeing. VOC burden was grouped into 0–1, 2–4, 5–10, and 11+ VOCs experienced in last 12 months. Impact statements on emotional burden and activities of daily living were asked using a 7-point Likert scale with high agreement and satisfaction scores = 5–7. 299 UK patients completed the survey; 56% were female, mean (SD) age was 29.2 (12.4) years. 14% of patient surveys were proxy-completed. The mean age at diagnosis was 3.4 (6.6), the majority of patients knew their genotype (91%), most having SS Disease (74%). The majority (94%) of patients reported experiencing at least one VOC in the past 12 months (mean 5.4, SD 5.1), with 18%

experiencing 0–1, 41% 2–4, 24% 5–10 and 16% reporting 11+. Of all VOCs experienced, 32% led to hospitalisation, 42% were managed at home, 13% required ER (not overnight) and 13% required medical assistance (not ER/overnight). When patients manage VOCs at home ($n = 167$) they do it mainly by drinking fluids (92%), resting/sleeping (86%), and taking opioid pain killers (76%). The main motivations for managing VOCs at home were 'poor experience at the ER or hospital' (56%) and thinking 'medical professionals do not understand their disease' (40%). Over two thirds of patients (69%) reported a high impact on emotional wellbeing. Patients report a higher emotional impact with increasing VOC burden; 0–1 (52%), 2–4 (66%), 5–10 (77%), 11+ (86%) VOCs per year. Patients report high impact in terms of frustration with having to put up with symptoms (78%) and worry that their SCD will get worse (71%). Patients report SCD has a high impact on relationships with family/spouse (48%), daily activities (e.g. housework) (44%) and sexual desire/activity (44%). 29% of all patients received professional emotional support (e.g. psychiatrist, psychologist, counselling), and 72% of all patients reported a desire to receive additional support. This analysis indicates that SCD patients in the UK experience a high VOC symptom burden, a large proportion of which patients choose to manage at home due to fear of poor experience with healthcare professionals. The high emotional impact of SCD in over two thirds of patients reflects the overwhelming need for professional support and may contribute to frequent hospitalisation. Improving management of VOCs would enhance patients' QoL, with likely improvement in individual emotional wellbeing and reduce the economic burden of SCD and UK healthcare system costs.

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Table. Antibiotic Sensitivities of the Most Commonly Cultivated Organisms

	<i>E. coli</i>	<i>K. pneumoniae</i>	<i>P. aeruginosa</i>	<i>S. epidermidis</i>
Piperacillin-Tazobactam	70.8%	36.3%	73.3%	/
Meropenem	100%	69.6%	66.7%	/
Imipenem	100%	69.6%	66.7%	/
Ertapenem	100%	66.7%	/	/
Gentamicin	95.8%	57.6%	80%	23.5%
Amikacin	95.8%	66.7%	93.3%	/
Fosfomycin	95.8%	75.8%	/	94.1%
Cefotaxime	75%	51.5%	/	/
Ceftazidime	79%	39.4%	60%	/
Cefepime	79%	48.5%	53.3%	/
Ciprofloxacin	45%	51.5%	53.3%	23.5%
Nitrofurantoin	100%	9%	/	94.1%
Trimethoprim	33.3%	36.4%	6.7%	58.8%
Colistin	8.3%	45.5%	100%	/
Rifampicin	/	/	/	94.1%
Tigecycline	/	36.3%	/	100%
Linezolid	/	/	/	100%
Teicoplanin	/	/	/	58.8%
Vancomycin	/	/	/	100%

Lymphoma/CLL

BSH2020-PO-080

Abstract Withdrawn.

BSH2020-PO-081

Updated results of a phase Ib/II randomised study: polatuzumab vedotin (Pola) plus bendamustine (B) and rituximab (R) in relapsed/refractory diffuse large B-cell lymphoma (R/R DLBCL)

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Pola is an antibody-drug conjugate targeting CD79b on B-cell non-Hodgkin lymphoma. In the primary analysis of GO29365 (NCT02257567) Pola-BR significantly improved efficacy *versus* BR in patients (pts) with transplant ineligible R/R DLBCL. Here, we report updated safety and efficacy results for DLBCL cohorts with an additional year of follow-up.

Safety was evaluated in Phase (Ph) Ib and Ph II randomised Pola-BR arms ($N = 45$) *versus* randomised BR ($N = 39$) arm; including longer-term safety outcomes for peripheral neuropathy (PN; a typical Pola-associated adverse event [AE]) and second malignancies. Efficacy, including investigator (INV)-assessed PFS, OS, and DOR for pts with persistent responses (defined as having two consecutive responses of CR or PR per INV), was reported for randomised arms only.

As of 15 March 2019, median follow-up in Ph Ib ($N = 6$) and Ph II arms ($N = 80$; Pola-BR [$N = 40$]; BR [$N = 40$]; 39 treated per arm) was 46 and 30 months (mo), respectively. Baseline characteristics were similar in randomised arms (Pola-BR vs. BR): 75% vs. 85% of pts were refractory to their last prior treatment; 28% vs. 30% of pts had 1 prior line of therapy, 28% vs. 23% had 2, 45% vs. 48% had ≥ 3 ; median prior lines of therapy were 2 in both arms. In the Pola-BR arm, 40% (18/45) of pts reported PN events; all were Grade (Gr) ≤ 2 . Median time to PN resolution was ~ 8 days (range 0–69); 56% (10/18) of pts had complete resolution of PN at data cut-off.

Of pts without resolved PN, further follow-up for PN resolution was not possible in 4 pts (early discontinuation due to death from disease progression or AE), 3 had unresolved Gr 1 PN, 1 had improving PN (Gr 1). Second malignancies occurred in 4% (2/45) of Pola-BR-treated pts and 5% (2/39) of BR-treated pts.

Updated median INV-assessed PFS (95% confidence interval [CI]) in randomised arms (Pola-BR vs. BR) was 7.5 (4.9, 17.0) vs. 2.0 (1.5, 3.7) mo (hazard ratio [HR] 0.33; 95% CI 0.20, 0.56); median OS (95% CI) was 12.4 (9.0, 32.0) vs. 4.7 (3.7, 8.3) mo (HR 0.41; 95% CI 0.24, 0.71), respectively. Median time to first response was 2 mo (range 1.8–5.3). For all responding pts in Pola-BR ($N = 28$) vs. BR ($N = 13$) arms, median DOR (95% CI) was 12.7 (5.8, 27.9) vs. 4.1 (2.6, 12.7) mo (HR 0.42; 95% CI 0.19, 0.91). For persistent responders in Pola-BR ($N = 19$) vs. BR ($N = 7$) arms, median DOR (95% CI) was 27.9 (10.3, NE) vs. 12.7 (7.7, NE) mo (HR 0.44; 95% CI 0.14, 1.32). In the Pola-BR arm, 47% (9/19) of pts with persistent responses remained event-free at data cut-off (8 pts in CR and 1 pt in PR [downgraded from CR based on modified Lugano criteria due to missing bone marrow biopsy]). DORs of 8/9 pts ranged from 22+ to 34+ months (1 pt consolidated with allogeneic transplant), 1 pt withdrew early from the study after 14.5 mo of response. In the BR arm, 2 pts maintained responses without events; both had consolidative therapy (1 radiation, 1 allogeneic transplant).

In this updated analysis, no new safety signals were identified. PN events were low grade, manageable and mostly reversible. Risk of second malignancies was not increased with Pola-BR vs. BR. Median time-to-event endpoints have not changed significantly since the primary analysis. However, longer follow-up demonstrates notable durability of response in a proportion of Pola-BR-treated pts. Pola-BR represents a promising new treatment for pts with transplant-ineligible R/R DLBCL.

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Safety and efficacy results from TRANSCEND NHL 001, a multicenter phase 1 study of lisocabtagene maraleucel (liso-cel) in relapsed/refractory large B-cell lymphoma

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Liso-cel is an investigational, CD19-directed, defined composition, 4-1BB chimeric antigen receptor (CAR) T cell product administered at target doses of CD8+ and CD4+ CAR+ T cells. We present results from the large B-cell lymphoma (LBCL) cohort of TRANSCEND NHL 001 (NCT02631044).

Patients aged ≥ 18 years had relapsed/refractory diffuse LBCL not otherwise specified (including transformed from any indolent lymphoma), high-grade B-cell lymphoma with *MYC* and *BCL2* and/or *BCL6* rearrangements, primary mediastinal B-cell lymphoma, or follicular lymphoma grade 3B after ≥ 2 lines of therapy, and Eastern Cooperative Oncology Group Performance Status (ECOG PS) of 0–2. Patients with mild/moderate organ dysfunction and secondary central nervous system (CNS) lymphoma were eligible. Bridging therapy was allowed with reconfirmed positron emission tomography (PET)-positive disease before lymphodepletion with fludarabine/cyclophosphamide. Liso-cel was given at 1 of 3 dose levels (DLs): 50×10^6 (DL1), 100×10^6 (DL2), or 150×10^6 (DL3) viable CAR+ T cells. DL2 was chosen for dose confirmation. Primary endpoints were treatment-emergent adverse events (TEAEs) and objective response rate (ORR). Secondary endpoints were complete response (CR) rate, duration of response (DOR), progression-free survival (PFS), and overall survival (OS). Cytokine release syndrome

(CRS) was graded per 2014 Lee criteria. ORR was assessed by independent review per Lugano criteria (2014).

Of 344 patients leukapheresed, 269 received liso-cel (DL1, $n = 51$; DL2, $n = 177$; DL3, $n = 41$). No product could be manufactured for 2 patients, and 25 patients received nonconforming product. In the optimized manufacturing process, median time from leukapheresis to liso-cel availability was 24 days. Median age was 63 years (range, 18–86 years; ≥ 65 years, 42%; ≥ 75 years, 10%). Overall, 26% of patients had ≥ 4 lines of prior systemic therapy (median, 3; range, 1–8), 67% were chemo-refractory, 44% had never achieved CR, and 59% had bridging therapy. Twenty-five patients received liso-cel as outpatients. Outcomes were similar across DLs, so data were pooled. 79% of patients had grade ≥ 3 TEAEs, mostly cytopenias (neutropenia, 60%; anemia, 38%; thrombocytopenia, 27%). 47% of patients had CRS and/or neurological events (NEs), with late onset (median, 5 and 9 days, respectively). Incidence of grade ≥ 3 CRS (2%) and NEs (10%) was low. Four grade 5 liso-cel-related TEAEs occurred (diffuse alveolar damage, pulmonary hemorrhage, multiple organ dysfunction syndrome, and cardiomyopathy). Safety was similar among patient subgroups. All primary and secondary efficacy endpoints were met. Of 256 efficacy-evaluable patients, the ORR was 73% (95% confidence interval [CI], 67–78) and the CR rate was 53% (95% CI, 47–59). Responses were similar across patient subgroups (Table). Median DOR was not reached (NR; 95% CI, 8.6–NR) with 12.0 months of median follow-up; median DOR for patients in CR was NR (95% CI, NR–NR). Median PFS was 6.8 months (95% CI, 3.3–14.1). Median OS was 21.1 months (95% CI, 13.3–NR).

Liso-cel showed durable clinical activity with a favorable safety profile across relapsed/refractory LBCL histologic subgroups and in patients with poor prognosis, including chemotherapy refractory, older age, comorbidities, and high tumor burden. Incidence of severe CRS and NEs was low, with late onset, allowing for outpatient administration.

Disclosure of Interest: J. Abramson Conflict with: Gilead Sciences, Seattle Genetics, Celgene, Novartis, Amgen, Juno Therapeutics, Karopharm, Verastem, Bayer, AbbVie, Janssen, Merck, Kite Pharma, Genentech, EMD Serono, MorphoSys, Conflict with: Seattle Genetics, Celgene, Genentech, Juno Therapeutics, AI Therapeutics, M. L. Palomba Conflict with: Merck & Co Inc, advisory board from Pharmacyclics, Conflict with: Honoraria: Merck & Co Inc, advisory board from Pharmacyclics, Novartis; Honoraria to immediate family member: Flagship Ventures, Evelo, Jazz Pharmaceuticals, Therakos, Amgen, Seres Therapeutics; Royalties to immediate family member: Juno Therapeutics, Seres Therapeutics; Stock to immediate family member: Seres Therapeutics, L. Gordon Conflict with: Advisory Boards: Kite/Gilead and Bayer, M. Lunning Conflict with: AbbVie, Bayer, Celgene, DAVA, Gilead Sciences Inc, Janssen, Juno Therapeutics, Kite, Novartis, OnLive, Portola, Seattle Genetics, Spectrum, TG Therapeutics, VANUUM, Verastem, Conflict with: ADC Therapeutics, Celgene, Curis, Janssen Scientific Affairs LLC, Juno Therapeutics, MiRagen, TG Therapeutics, M. Wang Conflict with: Janssen, AstraZeneca, Acerta Pharma, Kite Pharma, Pharmacyclics, Loxo Oncology, Celgene, Pulse Biosciences, MoreHealth, Guidepoint Global, Targeted Oncology, Nobel Insights, Conflict with: Janssen, AstraZeneca, Acerta Pharma, Kite Pharma, Pharmacyclics, Loxo Oncology, Celgene, BeiGene, Velosbio, Avira, Verastem, Conflict with: Honoraria: Janssen, AstraZeneca, Acerta Pharma, Kite Pharma, Pharmacyclics, Loxo Oncology, Celgene, Pulse Biosciences, MoreHealth, Guidepoint Global, Targeted Oncology, Nobel Insights; Royalties: MoreHealth; Advisory Boards: Janssen, AstraZeneca, Acerta Pharma, Kite Pharma, Pharmacyclics, Loxo Oncology, Celgene, Pulse Biosciences, MoreHealth, Guidepoint Global, Targeted Oncology, Nobel Insights, J.

Table. Adverse Events and Efficacy Outcomes

TEAEs of special interest	All liso-cel-treated patients (<i>N</i> = 269)		Response	ORR, <i>n</i> (%)	CRR, <i>n</i> (%)
	Any Grade	Grade $\geq$ 3			
CRS, <i>n</i> (%)	113 (42)	6 (2)	LBCL cohort, all patients	186 (73)	136 (53)
Time to onset, median (range), days	5 (1–14)	6.5 (3–12)	DLBCL NOS (<i>n</i> = 131)	89 (68)	64 (49)
NEs, <i>n</i> (%)	80 (30)	27 (10)	Transformed from FL (<i>n</i> = 57)	48 (84)	36 (63)
Time to onset, median (range), days	9 (1–66)	9 (2–44)	Transformed from other indolent lymphoma (<i>n</i> = 18)	11 (61)	7 (39)
Tocilizumab alone*, <i>n</i> (%)	18 (7)		PMBCL (<i>n</i> = 14)	11 (79)	7 (50)
Corticosteroids alone*, <i>n</i> (%)	22 (8)		HGBCL (<i>n</i> = 33)	25 (76)	20 (61)
Both tocilizumab and corticosteroids*, <i>n</i> (%)	34 (13)		FL grade 3b (<i>n</i> = 3)	2 (67)	2 (67)
Efficacy	Efficacy-Evaluable Patients (<i>N</i> = 256)		Age $\geq$ 65 years (<i>n</i> = 108)	82 (76)	65 (60)
Median PFS (95% CI), all patients, months	6.8 (3.3–14.1)		LDH $\geq$ 500 U/L (<i>n</i> = 57)	36 (63)	23 (40)
Median PFS, patients with CR, months	NR (NR–NR)		SPD $\geq$ 50 cm ² (<i>n</i> = 70)	43 (61)	23 (33)
Median PFS ratio [†]	0.681		Chemotherapy refractory (<i>n</i> = 171)	120 (70)	90 (53)
Median OS, all patients, months	21.1 (13.3–NR)		Received prior HSCT (<i>n</i> = 87)	68 (78)	55 (63)
Median OS, patients with CR, months	NR (NR–NR)		Received bridging therapy (<i>n</i> = 150)	101 (67)	68 (45)

CI, confidence interval; CR, complete response; CRR, complete response rate; CRS, cytokine release syndrome; DLBCL, diffuse large B-cell lymphoma; FL, follicular lymphoma; HGBCL, high-grade B-cell lymphoma; HSCT, hematopoietic stem cell transplantation; LBCL, large B-cell lymphoma; LDH, lactate dehydrogenase; NE, neurological event; NOS, not otherwise specified; NR, not reached; ORR, objective response rate; OS, overall survival; PFS, progression-free survival; PMBCL, primary mediastinal B-cell lymphoma; SPD, sum of the product of perpendicular diameters; TEAEs, treatment-emergent adverse events.

*Treatment for CRS and NE.

[†]PFS ratio was the ratio of PFS on the most recent line of therapy before liso-cel to the PFS after liso-cel.

Arnasson Conflict with: Advisory Boards: Regeneron and Celgene/Juno., A. Mehta Conflict with: Pharmacyclics, Celgene, Spectrum, and Imbrium Therapeutics, Conflict with: Roche-Genentech, Incyte, Kite/Gilead, Takeda, Rhizen, ADC Therapeutics, Forty Seven Inc., Juno/Celgene, Affimed, Seattle Genetics, Astex, TG Therapeutics, and miRagen, Conflict with: Speakers Bureau: Kyowa Kirin, Celgene, Kite/Gilead, Spectrum, Seattle Genetics, and AstraZeneca, E. Purev Conflict with: Salary: Genentech, D. Maloney Conflict with: Juno, a Celgene/BMS company, Celgene, a BMS company, and Kite Pharma, a Gilead company, Conflict with: Honoraria: Juno, a Celgene/BMS company, Celgene, a BMS company, and Kite Pharma, a Gilead company, BioLineRX, Gilead, Novartis, Pharmacyclics, A2 Biotherapeutics; Patents: Juno, a Celgene/BMS company; Royalties: Juno, a Celgene/BMS company; Stock: A2 Biotherapeutics, C. Andreadis Conflict with: Celgene, Novartis, Gilead, Jazz, Conflict with: Novartis, Celgene, Amgen, Merck, Conflict with: Stock: Roche; Salary: Genentech (spouse), A. Sehgal Conflict with: Juno/Celgene, Kite/Gilead, Merck, S. Solomon: None Declared, N. Ghosh Conflict with: Celgene, TG Therapeutics, Seattle Genetics, Janssen, and Gilead, Pharmacyclics, Conflict with: Celgene, TG Therapeutics, Pharmacyclics, Genentech, Conflict with: Speakers Bureau: Celgene, Seattle Genetics, Janssen, and Gilead, Pharmacyclics, AbbVie, AstraZeneca; Research Bureau: Forty Seven Inc, T. Albertson Conflict with: Salary and ownership interest from BMS, J. Garcia Conflict with: Salary and ownership interest from BMS, A. Kostic Conflict with: Salary

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UK CLL forum 5 years update on 315 relapsed refractory CLL patients treated with ibrutinib in 66 UK and Ireland centres

U. CLL Forum^{1,*}

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UK and Ireland clinicians gained non-trial access to ibrutinib for relapsed CLL through a named patient scheme (NPS) for 7 months in 2014, and the UK CLL Forum has been following a cohort of NPS patients treated with ibrutinib at 66 UK and Ireland hospitals. We now update this dataset with up to 5 years follow-up. Of the original 315 patients, 213 have stopped ibrutinib and 159 patients have died, with all living patients having a survival update in 2019. With 55.6 months median follow-up, median DFS was 33.5 months

(95% CI: 28.4–29.3) and median OS was 55.9 months (95% CI: 42.5–N/A). Four years after starting ibrutinib, 37.0% of patients were still taking ibrutinib and 52.4% were alive.

Both Discontinuation Free Survival (DFS) and Overall Survival (OS) were inferior for older (age > median) and PS 2+ patients. Patients with 17p- have inferior DFS and OS compared to patients with other defined cytogenetic aberrations (13q-, 11q-, t(12)) and more patients with 17p- compared with non-17p- stopped ibrutinib for progressive CLL (PD) (26.4% vs. 18.8%; $P = 0.01$) and Richter's Syndrome (12.1% vs. 8.3%; $P = 0.05$). First relapse patients had more than a 3 fold lower rate of ibrutinib discontinuation due to CLL progression (7.1% versus 24.5%; two-tailed Fisher's exact test; $P = 0.001$) compared with patients receiving ibrutinib at later lines of therapy, although DFS and OS for these groups were similar. With multivariable analysis of pre-treatment factors, older age, PS2+, 17p- and > 1 prior line of therapy retained significance for earlier ibrutinib discontinuation and age, PS2+ and 17p- associated with inferior OS.

To analyze any potential impact of reduced ibrutinib dosing during the first year of therapy, we originally classified all patients alive at 12 months into 4 groups, depending on first-year ibrutinib treatment. *Group A*: No dose reductions or breaks >14 days; *Group B*: Dose reductions, but no breaks >14 days; *Group C*: Breaks >14 days but back on ibrutinib by 1 year; *Group D*: stopped ibrutinib permanently but still alive at 1 year. With extended follow-up, groups A and B retain similar DFS and OS initially, although beyond 2 years the curves separate (DFS HR: 1.62 (1.05–2.50), OS HR: 1.52 (0.86–2.70)). Interestingly, the dose-reduced group do not have a higher incidence of disease progression, (PD A:24.8%, PD B:23.8%), but have a higher chance of discontinuation for all non-PD causes (A:23.6%, B:42.9%). Patients from groups C and D had significantly impaired OS compared with group A, with 4 year OS 39.4% and 16.7% respectively compared with 76.4% for group A ($P < 0.001$).

Considering all patients who discontinued ibrutinib, stopping for AEs decreased over time (year 1: 57%, 2: 53%, 3: 30%, 4: 21%). The risk of stopping ibrutinib due to PD-CLL increases with time (year 1: 11%, 2: 18.6%, 3: 42.5%, 4: 57.6%). Of the PD-CLL patients, treatment with venetoclax had the best OS ($n = 43$; 1 year OS: 77.5%) compared with the 7 treated with rituximab / idelalisib (1 year OS: 42.9%) and 14 treated with other palliative strategies (1 year OS: 10%).

With approaching 5 years of follow-up, around one third of real-world UK / Ireland patients treated with ibrutinib for relapsed CLL remain on ibrutinib. Our data suggest that younger, better performance status patients who have had fewer prior lines of therapy and no evidence of 17p deletion seem most likely to remain on ibrutinib long-term. Treatment of PD-CLL with venetoclax post ibrutinib shows encouraging results.

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Earlier steroid use with axicabtagene ciloleucel in patients with relapsed/refractory large B cell lymphoma

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Axicabtagene ciloleucel (axi-cel) is an autologous anti-CD19 chimeric antigen receptor (CAR) T cell therapy approved for the treatment of patients with relapsed/refractory large B cell lymphoma (R/R LBCL) with ≥ 2 prior systemic therapies. In Cohorts 1 + 2 (C1 + 2) of ZUMA-1, the multicenter, single-arm, registrational Phase 1/2 study of axi-cel in patients with refractory LBCL, Grade ≥ 3 cytokine release syndrome (CRS) and neurologic events (NEs) occurred in 11% and 32% of patients, respectively (Locke FL et al. *Lancet Oncol.* 2019). With 39.1 months' median follow-up, the median overall survival (OS) in C1 + 2 was 25.8 months, and the 3-year OS rate was 47%. A safety expansion cohort (Cohort 4 [C4]) evaluated the impact of earlier steroid use on adverse events (AEs). Here we present the primary analysis of ZUMA-1 C4 with a greater number of patients and longer follow-up.

Eligible patients with R/R LBCL were leukapheresed, could receive bridging chemotherapy (not allowed in C1 + 2), and received conditioning chemotherapy before axi-cel infusion at 2×10^6 cells/kg. C4 patients received early steroid intervention at Grade 1 NE and Grade 1 CRS after 3 days of supportive care. Primary endpoints were incidence and severity of CRS and NEs. Objective response rate (ORR) and CAR T cell levels were compared by tumor burden.

As of May 6, 2019, 41 patients in C4 received axi-cel, with a median follow-up of 8.7 months. Only 1 patient had not reached ≥ 6 months of follow-up due to a delay in dosing after the primary analysis trigger. Patients who received bridging therapy (68%) all had evidence of disease after bridging. The median age was 61 years. Disease types included DLBCL (63%), TFL (24%), PMBCL

(5%), HGBCL (7%). Nearly half of patients (49%) had an ECOG of 1; 70% had disease stage III/IV; 68% were refractory to $\geq$ second-line therapy; 63% had ≥ 3 prior lines of therapy; and 20% had relapsed after autologous stem cell transplant. C4 patients generally had lower median tumor burden by sum of product diameters (SPDs; C4, 2100 mm²; C1 + 2, 3723 mm²).

A greater proportion of C4 patients received steroids and tocilizumab versus C1 + 2 (73% and 76% vs. 27% and 43%). Fewer patients in C4 experienced Grade ≥ 3 CRS (2%) and NEs (17%) than in C1 + 2. The ORR in C4 was 73%, with a complete response rate of 51%. Among patients with ≥ 6 months of follow-up, response was ongoing in 54% of C4 patients versus 44% of C1 + 2 patients. Although most patients in C4 had lower SPD than in C1 + 2, responses were comparable when evaluated by tumor burden. Median duration of response was 8.9 months, consistent with that observed for C1 + 2 (8.1 months; Locke FL, AACR 2017). Median progression-free survival was 11.7 months; median OS was not reached. Peak CAR T cell levels were 42 cells/ μ L blood in C1 + 2 versus 59 cells/ μ L in C4. C1 + 2 had a median CAR area under the curve of 462 cells/ μ L $\times$ days vs. 512 cells/ μ L $\times$ days in C4. CAR T cell expansion by tumor burden was comparable between cohorts. Levels of NE-associated biomarkers including ferritin (pre- and post-treatment) and IL-2 (post-treatment) appeared lower in C4 than in C1 + 2.

Earlier steroid use may reduce the rates of CAR T cell treatment-related CRS and NEs without affecting efficacy. Conclusions are limited by the nonrandomized study design and differences in population sizes and baseline characteristics between cohorts. Optimizing AE management is important to improve the benefit-risk profile of CAR T cell therapy.

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Escalated BEACOPDac as first line therapy in 100 Hodgkin lymphoma patients reduces red cell transfusion requirements and may shorten time to menstrual period recovery whilst maintaining efficacy compared to escalated BEACOPP

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Introduction: It is increasingly common practice to modify escalated BEACOPP (eBPP) by removing oral procarbazine and replacing it with intravenous dacarbazine (250 mg/m² D2-3) to reduce haematopoietic stem cell and gonadal toxicity. However, published data of 'escalated BEACOPDac (eBPDac)' regimen are very limited.

Methods: We collected retrospective data from 15 centres in the UK, Ireland and France, that offer eBPDac therapy for first line advanced stage Hodgkin Lymphoma, and compared outcomes with matched patients treated with eBPP at 4 UK centres. Most patients were treated as per HD15 or HD18 protocol. The 24 patients treated in Paris followed the AHL2011 protocol with two courses of eBPDac given upfront and if iPET2 negative were deescalated to 4 cycles of ABVD.

Results: From 2009, 152 patients were managed first line with either eBPP ($n = 52$) or eBPDac ($n = 100$) with median follow-up 41.4 months for eBPP and 14.2 months for eBPDac patients. Patients were well matched with no significant differences in age

Lymphoma/CLL

(median: 28), sex, stage (stage 3/4: 82%) and international prognostic score (IPS3+:66%).

More patients treated with eBPDac received only 4 cycles of treatment (44% vs. 12%; $P < 0.001$) reflecting recent publication of HD18 trial data¹. In total, 73% patients achieved iPET2 Deauville score 2 or 3 and 96% patients achieved PET negative remission by end of treatment. Of eBPDac patients, 74% achieved iPET Deauville 2 or 3 which was statistically similar to the eBPP cohort (69%; $P = 0.538$) and matched the 76% iPET D2/3 reported in HD18¹. Of 152 patients, 150 are alive and 147 continue in first remission. Two eBPP patients have relapsed at 13 and 41 months and the latter died of refractory disease. One eBPDac patient had primary refractory disease, another relapsed 7 months after treatment, and one 56-year-old

eBPDac patient with high IPS died with bowel perforation during cycle 1.

Toxicity was compared over the first 4 cycles. There was no difference in day 8 ALT between the two regimens although the mean day 8 neutrophil count was lower in eBPDac than eBPP patients (1.93 vs. 2.35; $P = 0.083$; G-CSF given day 9). There was a trend to fewer non-elective days of in-patient care for eBPDac compared with eBPP (mean: 2.91 vs. 6.00; $P = 0.0689$), and eBPDac patients received fewer red cell transfusions during cycles 1 to 4 compared with eBPP patients (Mean 1.93 units vs. 4.33 units; $P < 0.001$). Women aged <35 , who completed ≥ 4 cycles of eBPDac/eBPP with >6 months post chemotherapy follow-up had a similar rate of return of menstrual cycles (eBPP: 20/21; eBPDac: 14/16), although

Table.

Baseline Characteristics	Escalated BEACOPP <i>N</i> = 52	Escalated BEACOPDac <i>N</i> = 100	<i>p</i> -value
Median Age (range)	22 (16–54)	28 (16–62)	U = 3015,
Male sex (%)	25 (48%)	61 (61%)	$P = 0.107$
Stage 2B / 2X / 2XB			Fisher, $P = 0.168$
3	11 (21%)	18 (18%)	Fisher, $P = 0.908$
4	7 (13%)	16 (16%)	
IPS 0–2	34 (65%)	66 (66%)	
3–4	13 (25%)	34 (34%)	
5–7	29 (55%)	48 (48%)	
IPS ≥ 3	10 (19%)	18 (18%)	Fisher, $P = 0.274$
39 (75%)	66 (66%)		
Treatment Outcomes			
Total cycle number 2	0	22 (22%)	
4	6 (12%)	44 (44%)	Fisher, (4 vs. 6) $P = 4.9E-7$
5	5 (10%)	3 (3%)	
6	40 (77%)	31 (31%)	
7	1 (2%)	0	
iPET Deauville score 1–2	10 (24%)	25 (26%)	
3	19 (45%)	48 (49%)	
4	13 (31%)	25 (26%)	
≤ 3	29 (69%)	73 (74%)	Fisher, $P = 0.538$
No iPET	10	2	
Mean day 8 ALT (cycles 1–4) [SD]	37.4 (± 28.6)	44.1 (± 26.0)	$t(98) = 1.33$ $P = 0.185$,
Mean day 8 neutrophils (cycles 1–4) [SD]	2.35 (± 1.37)	1.93 (± 1.15)	$t(95) = 1.75$ $P = 0.083$
Mean days non-elective admission (cycles 1–4) [SD]	6.00 (± 7.61)	2.91 (± 4.04)	U = 1560, $P = 0.0689$
Mean number of red cell units transfused (cycles 1–4) [SD]	4.33 (± 4.27)	1.93 (± 2.76)	U = 1197, $P = 0.000206$
Median follow-up from diagnosis in months (range)	41.4 (3.6–109.5)	14.2 (0.92–32.6)	U = 553, $P = 1.90E-15$
Mean number of months for return of menstrual period	8.65 (± 5.68)	3.86 (± 1.83)	$t(24) = 3.52$, $P = 0.0017$
post-chemotherapy [SD]	5.86 (± 0.36)	4.63 (± 0.96)	U = 60, $P = 0.000119$
Mean number of cycles completed by women <35 years of age [SD]			

eBPDac patients appeared to restart menstruation earlier post chemotherapy completion (mean: 3.85 vs. 8.65 months, $P = 0.0018$). However, this could also reflect the higher mean chemotherapy cycle number completed by the eBPP women (5.86 vs. 4.67; $P < 0.001$). The use of monthly Goserelin to suppress ovulation varied between centres.

Conclusions: Accepting the limitations of a retrospective study, we suggest that substituting dacarbazine for procarbazine is unlikely to compromise the efficacy of eBPP in frontline therapy of patients with advanced Hodgkin Lymphoma and may have some toxicity benefits. As it is highly unlikely that this single drug substitution will ever be tested in a prospective trial, publishing real-world data from eBPDac patients is important.

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Brentuximab vedotin plus bendamustine in relapsed and refractory classical Hodgkin lymphoma in children, adolescents and young adults: a single centre experience

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The majority of children and young people with classical Hodgkin lymphoma (cHL) are cured with first line treatment. Treatment failure rates with paediatric regimens are approximately 10% in early stage, and 15 to 20% in advanced stage disease.¹ Cure may be achieved in the relapsed/refractory (R/R) setting but the optimal salvage treatment has not been defined in children and young people. Conventional first-line salvage chemotherapy yields complete response (CR) rates ranging from 20 to 60%.² Brentuximab vedotin

(Bv) is a novel agent increasingly used in R/R cHL, with CR rates of 33% when used as monotherapy in a phase I/II paediatric trial.³ A number of studies have reported that the combination of Bv plus bendamustine (Bv+B) achieves superior outcomes to either agent alone.^{2,4} Achievement of complete metabolic remission (CMR) prior to consolidation with high dose chemotherapy (HDCT) and autologous stem cell transplantation (SCT) is highly predictive of long term progression free survival in R/R cHL.⁵

We present a series of 19 consecutive children and young people with R/R cHL, treated with Bv+B as second line or later relapse therapy at our institution. Patients (age range 9–27 years) were treated from May 2015 to June 2019. 14 patients (79%) had primary refractory disease. Median number of prior treatment lines was two, including ASCT in four patients. Patients received bendamustine (90 mg/m²) on days 1 and 2, and Bv (1.8 mg/kg) on day 2 of a 21 days cycle, with responses assessed after two cycles. Median number of cycles received was three (range two to six). Response was assessed by FDG-PET after two cycles and reported according to the revised response criteria for malignant lymphoma⁶. CMR was defined as Deauville score of 1, 2 or 3.

Of 19 patients, 14 (74%) achieved a CMR, one achieved a partial response and one achieved stable disease. Three patients (16%) did not complete two cycles of treatment due to infusion related reactions (IRRs) and so were not included in the efficacy analysis. CMR amongst the 16 efficacy-evaluable patients was 88%. All 16 evaluable patients received consolidation therapy, with ASCT in nine patients and with allogeneic transplant in the remainder.

Overall, IRRs affected 13 patients (68%). These were of severity grade 1 or 2 in ten patients, grade 3 in one patient and grade 4 in two patients. IRRs primarily occurred on day 1 of the second cycle, during or after the bendamustine infusion. The nature, timing and consequence of IRRs was similar to that reported by LaCasce in the phase I/II trial⁴. The protocol for that trial was amended to require pre-medication with 100 mg methylprednisone (or equivalent) and anti-histamine (50 mg diphenhydramine or equivalent) on days 1 and 2 of all cycles. The protocol at our institution has subsequently also been amended, incorporating 100 mg methylprednisolone and IV chlorphenamine administered prior to bendamustine on days 1 and 2 of all cycles. Other grade 3/4 adverse events included cytopaenias (six patients; 32%) and infection (two patients; 10%).

This series indicates that Bv+B is a safe and highly active salvage regimen for children and young people, achieving CMR in 74% of the entire cohort treated (and 88% in efficacy-evaluable patients). The CMR rate in this retrospective series is superior to reported outcomes of either drug as single agent. This combination of Bv+B can facilitate a bridge to HDCT/ASCT in a state of CMR in the majority of cases.

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The addition of Molecular High Grade classification to CNS International Prognostication Index and Cell of Origin may improve prediction of CNS relapse in Diffuse Large B Cell Lymphoma

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Accurate predication of CNS relapse in patients with DLBCL is essential to appropriately target preventative measures to reduce the risk of CNS relapse.

Methods to predict CNS relapse have used clinico-pathological data in a CNS international Prognostication Index (CNS IPI). Addition of cell of origin (COO) classification to CNS IPI where ABC and unclassified subtypes, where labelled as higher risk improved risk stratification (CNS-IPI-C model).

The REMoDL-B study identified a distinct subset of 'molecular high grade' (MHG) patients using gene expression profiling with a particularly poor prognosis. We hypothesised that the addition of MHG to COO+CNS-IPI model would further improve stratification of CNS relapse risk.

The overall risk of CNS relapse in REMoDL-B was 0.93% (10/1076). 20.1% (216/1076) patients were classed as high CNS IPI whose risk of CNS relapse was 2.78% (6/216). 41.2% (443/1076) patients were classified as adverse COO (ABC or Unclassified) whose risk of CNS relapse was 0.90% (4/443). 7.9% (85/1076) patients had a combination of adverse COO and CNS IPI whose risk of CNS relapse was 2.35% (2/85). 10.13% (109/1076) patients had a combination of adverse COO or MHG and CNS IPI whose risk of CNS relapse was 3.67% (4/109).

This data suggests that incorporating MHG into the CNS-IPI + COO model may improve stratification of DLBCL patients for CNS relapse risk.

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Adoption of rapid obinutuzumab infusions: real world experience

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Obinutuzumab is a type II monoclonal antibody directed against the CD20 antigen. Its use in combination with cytotoxic agents has entered mainstream haematological practice following recent positive trial data in the treatment of chronic lymphocytic leukaemia (CLL) and follicular lymphoma (FL). In common with other monoclonal antibodies, infusion-related reactions have been highlighted as a frequent adverse event during trials. The obinutuzumab monograph recommends first administration is undertaken with slow rate incrementation and close observation, with a minimum infusion duration of more than 250 min. In the absence of significant reactions during the first dose, all subsequent infusions are advised to commence at a rate of 100 mg/hour, increased every 30 min by 100 mg/hour increments. This requires a minimum of 175 min of patient and chemotherapy chair time and mandates 3 rate changes by nursing staff, placing a significant burden on the physical capacity and nursing time of stretched chemotherapy units.

Authors of the GATS and GATHER studies have recently reported the safe administration of obinutuzumab over a 90-min infusion from the second cycle of treatment onwards in selected patient groups. The Norfolk and Norwich University Hospital (NNUH) haematology unit adopted this approach with a protocol allowing rapid 90-min fixed-rate infusions from cycle 2 onwards for patients with follicular lymphoma and patients with CLL with a lymphocyte count $<5 \times 10^9/l$ at the start of cycle 2. Patients who experienced grade 3 or 4 infusion reactions during cycle 1 are excluded. Demographic data, medical notes and nursing notes of patients who have been treated using this protocol have been retrospectively reviewed and analysed.

We report the outcomes for 15 patients at NNUH (median age 74 years; 6 patients with FL and 9 patients with CLL) who have received a total of 52 rapid obinutuzumab infusions between October 2018 and January 2020. 7 patients had experienced prior grade 1 or 2 reactions during cycle 1 obinutuzumab infusions. 11 patients (73%) commenced rapid infusions from cycle 2. Patients completed a median of 3 rapid infusions (range 1–5). No patients experienced infusion related reactions during rapid 90-min infusions.

The experience at NNUH supports the administration of obinutuzumab via rapid fixed-rate infusion after the initial cycle, demonstrating that it can be safely administered in this way to selected patient groups. This intervention can virtually halve infusion durations, and has saved over 73 h of patient, nursing and chemotherapy chair time at NNUH to date, as well as making the infusion easier for nurses to administer by removing the need for 3 rate changes. This data provides further evidence of the safety of rapid obinutuzumab infusions from the second cycle onwards in routine clinical practice, which has the potential to optimise chemotherapy unit efficiency alongside enhancing patient experience.

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Brentuximab vedotin plus chemotherapy for stage III/IV classical Hodgkin lymphoma: 4-year update of the ECHELON-1 study

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In the primary analysis of the international, phase 3 ECHELON-1 trial, treatment with brentuximab vedotin plus doxorubicin,

vinblastine and dacarbazine (A+AVD) resulted in a statistically significant improvement in the primary endpoint of modified PFS per independent central review compared with doxorubicin, bleomycin, vinblastine and dacarbazine (ABVD) in patients with newly-diagnosed Stage III/IV classical Hodgkin lymphoma (cHL) (NCT01712490; Connors JM, et al. *N Engl J Med* 2018;378:331–44). A subsequent analysis of standard progression-free survival (PFS) at 3-years' median follow-up showed that the benefit of A+AVD over ABVD on PFS was maintained (3-year PFS per investigator: 83.1% vs. 76.0%, respectively; hazard ratio [HR] 0.704 [95% CI: 0.550, 0.901]; *P* = 0.005). Here, we present efficacy and safety data from a longer follow-up at a median 48.4 months. Newly diagnosed patients with Stage III or IV cHL were randomized 1:1 to receive A+AVD (*n* = 664) or ABVD (*n* = 670) intravenously on days 1 and 15 of each 28-day cycle for up to 6 cycles. The present follow-up analysis of PFS is exploratory and per investigator assessment. Patients with ongoing peripheral neuropathy (PN) at end of treatment were followed for resolution or improvement (defined as improved by ≥ 1 grade from worst grade as of the latest assessment) during the post-treatment follow-up period. At this extended follow-up, treatment with A+AVD versus ABVD was associated with a 31% reduction in the risk of progression or death per investigator (HR 0.691, *P* = 0.003; Table). Four-year PFS rates were 81.7% with A+AVD versus 75.1% with ABVD. PFS numerically favoured patients treated with A+AVD versus ABVD across subgroups, including Stage III and Stage IV disease, age, extranodal sites, and International Prognostic Scores (IPS), with overlapping confidence intervals. Exploratory subgroup analyses of PFS in PET2(+) and PET2(-) patients showed a treatment effect in favour of A+AVD that was independent of PET2 status (Table). In the primary analysis, 67% of patients in the A+AVD arm and 43% of patients in the ABVD arm experienced PN. At the cut-off date for this analysis, 83% (365/442) of patients who had developed PN in the A+AVD arm had either complete resolution (68%, 300/442) or improvement (15%, 65/442) of their symptoms compared with 84% (240/286) with either complete resolution (76%, 217/286) or improvement (8%, 23/286) in the ABVD arm. Among patients with ongoing PN at their last follow-up (21% of A+AVD patients, 10% of ABVD patients), most cases were Grade 1/2 events (125/142 [88%] in the A+AVD arm and 65/69 [94%] in the ABVD arm). Overall survival data are not yet mature; per protocol, the final analysis will be performed after 112 deaths have occurred. In summary, with a median follow-up of 48.4 months, A+AVD continues to provide a robust and durable benefit over ABVD for

Table.

Patient population	4-year PFS rate, % (95% CI)		Difference at 4 years, %	HR (95% CI)*	P-value†
	A+AVD (<i>n</i> = 664)	ABVD (<i>n</i> = 670)			
All patients (ITT)	81.7 (78.3, 84.6)	75.1 (71.4, 78.4)	6.6	0.691 (0.542, 0.881)	0.003
PET2(-) patients	84.5 (81.1, 87.3) <i>n</i> = 588	78.9 (75.2, 82.2) <i>n</i> = 578	5.6	0.680 (0.515, 0.899)	0.006
PET2(+) patients	59.8 (43.9, 72.4) <i>n</i> = 47	44.5 (30.8, 57.4) <i>n</i> = 58	15.3	0.664 (0.371, 1.189)	0.164

ITT, intent to treat; PET2, PET scan after cycle 2.

*HRs (A+AVD/ABVD) and 95% CIs are based on a Cox proportional hazard regression model, which is stratified for the ITT population and unstratified for subgroup analyses.

†P-values are calculated using a log-rank test, which is stratified for the ITT population and unstratified for subgroup analyses.

patients with previously untreated Stage III or IV cHL. The PFS benefit in favour of A+AVD is evident regardless of PET2 status, disease stage, age and IPS. PN continues to resolve or improve in both arms, with most patients achieving complete resolution of their symptoms. Together, these data further support the clinical advantages of A+AVD over ABVD as frontline treatment for patients with Stage III/IV cHL.

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Structural and functional variability of the tumour B-cell receptor indicates a role for environmental influences on the behaviour of mantle cell lymphoma

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The B-cell receptor (BCR) surface immunoglobulin (sIg) is a functional requirement for mature normal and tumour B cells (Lam K, 1997). Mantle cell lymphoma (MCL) is a mature CD5 + B-cell tumour defined by overexpression of cyclins. In chronic lymphocytic leukaemia (CLL), IGHV status and sIgM levels have provided key information on the variability of CLL behaviour (D'Avola A, 2016). MCL express variably high sIgM levels and IGHV status seems to associate with different clinical outcome (Navarro, 2012). Sensitivity to BCR-kinase inhibitors suggests that BCR structure and function are also important for MCL cell behaviour.

Our aim was to investigate the hypothesis that the variability of sIgM levels and function may be consequent to environmental influences acting on the BCR.

Peripheral blood mononuclear cells from 36 MCL patients were investigated. BCR structure was analysed by IGHV-D-J transcript sequencing whilst the functional status of the sIg was assessed by immunoblotting of the tumour sIgM μ chain constant region. MCL sIgM/D expression was determined by immunophenotyping (0, 24–72 h in culture). sIgM increasing >20% over baseline was used as an indicator of prior down modulation by antigen *in vivo*. Signalling capacity was measured by calcium mobilization and phosphoflow (pSYK Y525/526) following anti-IgM stimulation. Data were compared to a CLL cohort from the UK Mature Lymphoid Malignancies Observational Study (NIHR/UKCRN ID: 31076).

Analysis of the tumour IGHV ($n = 34$) revealed a cohort with equal mutated (M, homology to germline <98%, 17/34) and unmutated (U) IGHV cases. sIgM levels were highly variable (MFI range 47–2808, median 942) while sIgD levels were consistently low (range 1–253, median 29). Median sIgM, but not sIgD, level was significantly higher in MCL than CLL ($P < 0.01$). In non-aggressive MCL (excluding blastoid MCL, $n = 30$), sIgM expression was significantly higher in U-MCL (median 1219 vs. 542, $n = 0.03$).

Signalling capacity, measured by calcium mobilization, was high but variable (range 11–100%, median 89%). Variability positively correlated with sIgM levels ($r = 0.86$, $P < 0.01$). Although MCL signalling was significantly higher than CLL ($P < 0.01$) a sub-group of MCL ($n = 8$) had low signalling capacity (CLL-like).

7/8 (88%) low signalers (M-IGHV ($n = 4$), U-IGHV4-39 ($n = 1$), 3–21 ($n = 2$)) recovered sIgM expression during 'antigen-free' *in vitro* culture, in contrast to only 1/9 (11%) high signalers. Basal

pSYK was significantly higher in the low signalling CLL-like group ($P = 0.047$). Analysis of the tumour sIg μ chain identified an immature glycosylation pattern, indicating prior activation ($P < 0.01$) (Krysov, 2010).

In this selected cohort, MCL sIgM expression is variable, correlates with signalling capacity and associates with IGHV status. Although sIgM levels and signalling capacity are generally higher in MCL compared with CLL, in this cohort we identify a subset with low sIgM and signalling. This subset had BCR-related features similar to CLL and that associate with an indolent course (M-IGHV/U-IGHV3-21; Walsh, 2003). The pattern of sIgM recovery, high basal pSYK and immature glycosylation pattern are highly suggestive of the effect by antigen *and point to* ongoing selective environmental influences on the BCR *in vivo*. This process may be prominent in indolent MCL where genomic aberrations are less frequent. This evidence claims analogies with CLL, where variable antigen drive plays a biological and clinical role.

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An international multicentre study of consecutively diagnosed patients with ALCL: outcomes following first-line therapy in routine clinical practice

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Introduction: Standard first-line (1L) therapy for systemic ALCL (sALCL) is CHOP-based chemotherapy, with or without consolidative autologous stem cell transplant (ASCT) (Schmitz, Blood, 2010). Approximately 40–65% of patients experience relapsed/refractory disease, for which brentuximab vedotin (BV) as monotherapy has established efficacy (Pro, Blood, 2017)

BV is currently FDA-approved in combination with CHP (CHP+A) as first-line treatment for CD30 + PTCL, based E2 results (34% reduction in risk of death), an effect only demonstrable in ALCL (70% of E2 cohort) (Horwitz, Lancet, 2019). We investigated outcomes of unselected ALCL patients treated in routine practice prior to widespread approval of BV in 1L.

Methods: Consecutively diagnosed patients with sALCL from 6 UK and Australian centres ($n = 166$) were studied (Dec 2004–Dec 2018).

Patients ≥ 16 years were included irrespective of Alk status. Treatment allocation was clinician choice and included best supportive care (BSC). Post-mortem diagnosis and 10 patients treated on E2 were excluded. Principal outcomes were PFS and OS following 1L treatment. Additional outcome measures were frequency of ASCT and use of BV for r/r ALCL. Data were retrospectively collected following GDPR guidelines and declaration of Helsinki.

Results: Baseline patient characteristics are shown in table 1. The most frequent 1L treatment regimen was CHOP in 104 pts (67%), 26 (18%) received intensified regimens (CHOEP, CHOP/IVE/MTX, CODOX-M, ALCL99), 4 received other regimens (2.6%) and 14 had best supportive care (9%). Treatment-related mortality was 5.6% (8/

Table.

	ALCL ($n = 156$)
Median Age (range)	57.5 (16–93)
Baseline ECOG PS, n(%)	60 (45)
0	39 (27)
1	19 (13)
2	21 (14)
3	9 (6)
4	
Male, n(%)	96 (62)
ALK positive, n(%)	53 (35)
Stage by CT, n(%)*	20 (16)
I	28 (22)
II	26 (21)
III	51 (41)
IV	31
Missing, n	
BM involvement, n(%)	13 (12)
Missing, n	46
EN sites, n(%)	68 (44)
0	46 (30)
1	24 (16)
2	15 (10)
>2	3
Missing, n	
Elevated LDH, n(%)	73 (54)
Missing	20
IPI score	37 (35)
0–1	19 (18)
2	28 (26)
3	20 (19)
4	3 (3)
5	49
Missing, n	
PIT score	59 (62)
0–1	23 (24)
2	11 (12)
3	2 (2)
4	61
Missing, n	

142). 12 (8%) patients underwent ASCT in first response, of whom 10 had received intensified-CHOP regimens.

Of 141 evaluable pts, ORR was 71% and CR rate was 52%. Median follow up of all patients was 40 months. For the CHOP-treated cohort ($n = 104$), 3-year PFS and OS were 47% and 57%. Outcomes for ALK-negative patients treated with CHOP ($n = 78$) were inferior to ALK-positive; 3-year PFS, OS of 35% and 44% vs. 66% and 82% respectively ($P = 0.003$ for both, figure 1).

BV was given to 24 pts (15%), mostly for 2L therapy (16/24). Median doses received were 5 (range 1–14). ORR to BV was 11/24 (45%) and CR rate was 7/24 (30%). Median follow up after BV was 18 months, at which point PFS and OS rates were 33% and 35% respectively (Figure 1). Bridging to ASCT or alloSCT occurred in 3 and 4 pts respectively, 6 remain alive at data cutoff.

41 patients were staged with CT and PET, with concordant staging in 63% (9/41 PET-upstaged). At data cutoff, 20/72 patients progressed after PET-defined CR (28%), whereas 13/48 patients did not progress after PET-defined PR (21%).

Conclusions: We describe a large unselected cohort of sALCL treated in routine clinical practice, Alk pos/neg distribution matched E2 data. Survival outcomes of our CHOP-treated cohort are similar to those in the E2 control arm, notwithstanding 28% of our patients with ECOG ≥ 3 (E2 exclusion criterion) and also comparable to data from the international T-cell lymphoma project (Shustov, Haematol Oncol, 2019). A minority of pts effectively underwent ASCT consolidation. Outcomes of r/r ALCL treated with single-agent BV were inferior in our cohort as compared to the pivotal Phase 2 data (Pro, Blood, 2017). Our data represent an important benchmark as an unselected sALCL population treated in routine practice; future studies of CHP+A in the real-world setting will be crucial to assess the wider impact of BV in front-line sALCL treatment.

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Ibrutinib at first relapse for mantle cell lymphoma: a United Kingdom real world analysis of outcomes in 185 Patients

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Background: Ibrutinib has transformed the clinical approach to relapsed mantle cell lymphoma (MCL) and those receiving it at first relapse appear to gain greatest benefit (Rule et al, 2019). However, the general applicability of clinical trial findings is not established and concerns regarding tolerability persist, especially in non-trial populations enriched for frailer patients. In the United Kingdom ibrutinib is now standard of care at first relapse and this retrospective cohort study has analysed the clinical effectiveness of this approach in a non-clinical trial, real world population.

Method: 35 NHS Trusts across the UK contributed data from consecutive patients treated with ibrutinib for MCL at first relapse. Patients received standard dose 560 mg OD, unless documented, and commenced treatment between January 2015, when first available on the CDF, and June 2019. Study primary outcome was PFS.

Results: 185 patients were included in the study. Median age at start of ibrutinib was 73 years (range 33 to 97), 68.6% were male. Frontline treatment was cytarabine based in 28.5% (26.3% HSCT consolidation) and 8.6% received low intensity frontline therapy. Median PFS with frontline therapy was 20.5 months (95% CI 14.6 to 26.5) and 53.2% progressed within 2 years. On ibrutinib commencement 56.2% were MIPI high risk, 23.8% were ECOG ≥ 2 and 3.8% had CNS involvement.

ORR to ibrutinib was 69.9% with 28.1% achieving CR/CRu. Median PFS with ibrutinib was 18.9 months (95% CI 12.7 to 25.0) and median OS 23.9 months (95% CI 13.3 to 34.5), with median follow up 24 months. On univariate analysis ECOG score ≥ 2 (HR 2.14, $P = 0.001$) and time to progression < 24 months with frontline therapy (HR 1.93, $P = 0.001$) were strongest predictors of PFS, whereas age ≥ 75 years was not predictive (HR 1.33, $P = 0.141$).

10 patients (5.4%) received attenuated dose from start due to frailty and 22 patients (11.9%) underwent dose reductions due to drug toxicity. Of 118 patients to discontinue ibrutinib 73 (61.9%) were due to progressive disease (PD), 16 for consolidation HSCT (13.6%), 8 due to drug toxicity (6.8%), 6 due to co-morbidities (5.1%), and 15 due to deaths while on therapy (12.7%): 7 deaths of unknown cause, 3 deaths relating to co-morbidities, 4 deaths from sepsis and 1 traumatic subdural haematoma.

For patients stopping due to PD ($n = 73$) median survival from date of ibrutinib discontinuation was 1.2 months. However, in the 22 of 63 patients (35%) who received further systemic therapy (14 received R-BAC or R-B) median OS from date of ibrutinib discontinuation was 9.3 months.

Conclusion: This study population was enriched for several recognised adverse prognostic markers, including older age and poor performance status, which is likely to explain differences in PFS relative to clinical trial data. Despite these features response rates were similar and it is reassuring that the proportion of patients stopping ibrutinib due to toxicity was similar to trial data and bleeding events requiring alterations to therapy were rare.

Although ibrutinib represents a significant breakthrough with clear benefit to outcome irrespective of patient age, the results also highlight that response in many cases was limited. 40% of patients progressed within 1 year of starting ibrutinib and outcome for those progressing through ibrutinib was often very poor. Improved treatment strategies in the post-ibrutinib setting remains a priority.

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BSH2020-PO-093

Finding the five percent: predicting adult T-cell leukaemia/lymphoma (ATLL) in human T-cell leukaemia virus-1 (HTLV-1) carriers using a novel flow cytometric method

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Background: Adult T-cell leukaemia/lymphoma (ATLL) is an aggressive T cell malignancy with a dismal prognosis. Directly associated with the retrovirus Human T-cell leukaemia Virus-1 (HTLV-1), the lifetime incidence in HTLV-1 carriers is 5%. High proviral load (PVL; > 4 copies of HTLV-1 per 100 cells) is a known predictor of transformation but only 20% of high PVL patients will transform to ATLL. Targeted strategies are needed to identify those at the highest risk to allow early intervention/prevention. Oligoclonality within HTLV-1 infected cells has been quantified previously but methods have either lacked sensitivity for low-frequency clones or have been highly sensitive but time-consuming, costly and unsuitable for large-scale clinical application.

Patients and Methods: Peripheral blood mononuclear cell (PBMC) samples from 133 HTLV-1 carriers with no history of ATLL (94 with PVL $\geq 4\%$), attending the National Centre for Human Retrovirology, Imperial College Healthcare NHS trust (www.htlv.eu) were analysed. High PVL carriers were followed for 367 person-years (median 16 months). Clonal populations were quantified using a flow cytometry method gating on T-cell receptor $\nu\beta$ subunit expression and oligoclonality was scored between zero to one (OCI-flow); with one representing a monoclonal population and zero representing equal (polyclonal) distribution of all populations. CD7 and Ki-67 expression in clonal populations was quantified, as CD7 expression is typically low and Ki-67 high in ATLL. PBMC samples from known cases of ATLL ($n = 30$) and HTLV-1 carriers with no progression to ATLL ($n = 38$) were used for training purposes.

Results: Based on training samples, the OCI-flow threshold was set at > 0.770 ('high OCI-flow'). Fifteen of ninety four (16%) of high PVL HTLV-1 carriers were above the threshold. Subjects in the high OCI-flow group were significantly more likely to have a family history of ATLL and have a higher PVL and lymphocyte count than subjects with OCI-flow ≤ 0.770 ('low OCI-flow'). Clonal populations were identified in 13/15 (87%) of carrier with high OCI-flow. CD7 and Ki-67 expression were not significantly different in clonal populations compared to other HTLV-1-infected cells. Three carriers with a high OCI-flow transformed to ATLL (2 acute subtype, 1 lymphoma) at 19, 41 and 69 months post-sampling. No carriers in the low OCI-flow group transformed to ATLL ($P = 0.003$). The transformation rate in carriers with high OCI-flow is 90 per 1000 person-years (vs. 8 per 1000-person-years in all high PVL carriers).

Discussion: A novel method for identifying HTLV-1 carriers at the highest risk of ATLL has been developed and validated. The OCI-flow score stratifies carriers into distinct groups, which is supported by demographic and haematological differences and three transformations observed in the high OCI-flow group. The ten-fold increase in transformation rate in patients with high OCI-flow strengthens the clinical utility of this model. Carriers at highest risk can thus be identified using this method even before typical immunophenotypic changes have occurred, at least 18 months prior to disease

presentation. Longer follow-up of the group, including genomic analysis, is ongoing.

OCI-flow analysis is therefore recommended in all HTLV-1 carriers with PVL $\geq 4\%$. Those with high OCI-flow should undergo increased surveillance and be considered for intervention studies to prevent ATLL development.

Disclosure of Interest: None Declared

BSH2020-PO-094

Investigating the clinical utility of a next generation sequencing B-cell clonality assay. case study: identification ighv4-34 rearrangement and assess its prognostic value in hairy cell leukaemia

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Distinguishing between reactive and malignant B-cell proliferations remains very challenging in some situations and often standard of care morphological and immunophenotyping testing is insufficient to make this distinction. Demonstration of clonal immunoglobulin gene rearrangements can assist with this process and developments in molecular techniques have meant these are easier to identify. PCR clonality assays in molecular diagnostic testing are considered to be the gold standard technique. However, owing to speed and increasing affordability, next generation sequencing (NGS) technologies have broadened the potential of clonality assays. These massively parallel sequencing technologies allow both detection of the presence of clonal immunoglobulin rearrangements and also the identification of the specific immunoglobulin heavy chain variable sequence and its frequency. This can be useful diagnostically and for measurable residual disease (MRD).

This project had two aims: to demonstrate that an NGS-based clonality assay would be a satisfactory replacement for a conventional clonality assay utilising capillary electrophoretic (CE) fragment analysis even for formalin fixed-paraffin embedded (FFPE) tissues likely to yield low concentrations of fragmented DNA, and to evaluate the clinical utility of the assay to identify the presence of IGHV4-34 hairy cell leukaemia variant (HCL-v), a biomarker associated with a poorer prognosis than non-IGHV4-34 classic HCL (HCL-c).

Analysis of immunoglobulin heavy chain (IGH) and light kappa chain (IGK) gene rearrangements was performed using the LymphoTrack assay. 15 patient samples were tested using assays for all IGH frameworks (FR1/2/3) and IGK using the MiSeq platform. Genomic DNA (gDNA) quality was assessed using a TapeStation to evaluate DNA suitability for downstream analysis. The results were analysed using LymphoTrack software. Clonality was defined as a major rearranged sequence $\geq 2.5\%$ for IGH and/or $\geq 5\%$ for IGK of the total reads and when more than double the percentage of reads for the 3rd most frequently rearranged sequence, as per manufacturer guidelines. The assay results were compared with those obtained using CE.

Successful assessment of clonality was obtained for 66% of samples for FR1/2/3 and 73% for IGK by NGS compared to 80% and 66% respectively when using CE.

A higher failure rate compared to the one that has been reported in other validation studies using the same assay, was likely due to poor quality DNA less suited for NGS. However, these sample types

are more representative of those provided in a clinical practice. Several technical issues are to be considered, related to highly fragmented FFPE-derived DNA and difficulties analysing results due to inadequate coverage, particularly problematic in the variable regions where alignment is intrinsically harder.

When assessing the clinical utility of detecting the IGHV4-34 variant in HCL-v it was demonstrated that the assay could differentiate HCL-v from HCL-c. 60% of *BRAF* wild-type patients were found to carry the IGHV4-34 gene region usage, supporting a diagnosis of HCL-v ($P = 0.001$). Furthermore, comparison of progression-free survival survival between patients with VH4-34 HCL compared to non-VH4-34 HCL demonstrates median survivals of 2 and 9.5 years respectively from diagnosis. The study demonstrated that the NGS-based assay is a feasible method for B-cell clonality assessment of FFPE specimens.

Disclosure of Interest: None Declared

BSH2020-PO-095

Cost-effectiveness of a 24-month fixed duration of venetoclax in combination with rituximab in relapse/refractory chronic lymphocytic leukaemia

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Introduction: Venetoclax is a first-in class targeted BCL2 inhibitor which is an inducer of apoptosis in Chronic Lymphocytic Leukaemia (CLL) cells. When combined with rituximab (R) in the Phase 3 trial (MURANO), Venetoclax plus Rituximab (VEN+R) achieved durable responses for patients who were Relapsed / Refractory (R/R) to prior CLL therapies. This study aims to build a valid and transparent economic model to show the full economic value of VEN+R for R/R CLL patients.

Method: A partitioned survival analyses (PartSA) model was developed from a UK perspective using three health states: progression free, progressed, and death. The model had a cycle length of 28 days and a lifetime time horizon. VEN+R was compared to bendamustine plus rituximab (BR), fludarabine plus cyclophosphamide plus rituximab (FCR), ibrutinib (IBR), idelalisib + R (IR), and ibrutinib + BR (IBR +BR). Data to inform the model inputs were identified from, the MURANO clinical trial (NCT02005471), systematic literature review, UK Specific national reference costs and from clinical experts. The model was populated with survival data extrapolated from a more recent datacut of the MURANO trial, where the limited OS data was leveraged with the more mature PFS data using a joint-estimation method. The relative efficacy of VEN+R versus other comparator treatments not included in the trial was estimated using Matching Adjusted Indirect Comparison (MAIC). Costs included treatment and administration, routine care and monitoring, adverse events, costs associated with disease progression and end of life care. Uncertainty was tested through deterministic, probabilistic and scenario analyses.

Results: Over the 30-year time horizon, R/R CLL patients receiving VEN+R were estimated to incur costs of £154 974 (Del(17p)/TP53: £152 605, Non-Del(17p)/TP53: £155 536) and accrual of 6.355 Quality-adjusted-life-years (QALYs) (Del(17p)/TP53: 5.837, Non-Del

(17p)/TP53: 6-548). The incremental benefits measured in QALYs for VEN + R were: 3-910 vs. IR, 3-486 vs. FCR, 2-462 vs. BR, 2-007 vs. IBR, and 1-196 vs. IBR+BR. Compared to treat to progression IBR containing regimens, VEN+R was more efficacious with lower total costs and therefore IBR and IBR+BR were dominated. IR, FCR, and BR were less costly and less effective, leading to incremental-cost-effectiveness-ratios (ICERs) of £16 865, £36 603, and £44 778 respectively when compared to VEN+R.

The cost-effectiveness model in this study shows an overview of the current treatment landscape of the overall R/R CLL population. The slope of the line which compares the cheapest non-dominant comparators with VEN+R also represents the ICER. All the dominated treatment options are above to the left of the frontier (line). In summary the results show that VEN+R either dominates relevant comparators or is cost-effective at a £45 000 willingness to pay (WTP) threshold. The mean probabilistic results have strong agreement with the deterministic results in that the ICER vs. FCR is similar, and dominance is maintained vs. ibrutinib.

Conclusion: For R/R CLL patients, VEN+R is a cost-effective treatment and should be considered as a standard treatment option. In addition this study introduces an expert validated novel joint survival modelling approach which leverages the mature PFS data to inform the OS extrapolations.

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Serial troponin measurement during anthracycline-containing chemotherapy may predict early cardiotoxicity that is amenable to primary prevention with ACE inhibitors

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Despite well-established cardiotoxicity profiles, anthracyclines are widely used in haematological malignancies. The European Society of Cardiology recommends a thorough assessment of baseline cardiac risk factors and using serial cardiac biomarkers to identify those at risk. Recent studies show that echocardiography and clinical features of heart failure are late findings of cardiotoxicity and fail to capture early changes. They also emphasise that patients started on cardioprotective medication (eg. ACE inhibitor or beta blocker) on the basis of rising cardiac biomarkers have better long-term outcomes.

This prospective study focused on patients receiving ABVD or RCHOP chemotherapy for Hodgkin's or Diffuse Large B cell Lymphoma from November 2016 to December 2018. Patients troponin levels were measured at baseline and prior to each cycle of chemotherapy; echocardiography at baseline, at the end of treatment, and 1 year post chemotherapy. For the purposes of this study, a significant troponin rise was defined as a 50% rise in troponin from baseline. If this occurred, patients were started on 1-25 mg ramipril daily as cardioprotection. Patients' cardiac risk factors were analysed for correlation with troponin rise. Echocardiogram reports were used to detect any decline in left ventricular function.

From a total of 65 patients who completed at least 2 cycles of ABVD or RCHOP; 54 (83%) exhibited significant troponin rise and of these 61 started ramipril. Data showed a progressive increase in troponin with each cycle of chemotherapy, whilst echocardiography failed to reflect any significant changes. Only 2 patients had a drop in their ejection fraction to below 50%. Of note, both these patients had cardiac risk factors which may have played an additive role for early decompensation. 19 (79%) patients without any cardiac risk factors and who were below the age of 65 still had a significant troponin rise. Pre-existing cardiac risk factors did not appear to play a significant role in the degree of troponin elevation. 2 patients stopped ramipril due to side effects and 1 patient choice.

In conclusion, the majority of patients in our study exhibited significant troponin rises regardless of baseline cardiac risk factors. Conversely, echocardiography did not show an overall trend of cardiac compromise despite the significant troponin rises. This finding is in keeping with early myocyte damage detected by biomarkers before progression to overt cardiac dysfunction. Although limited by a small population size, our study supports starting an ACE inhibitor in patients receiving anthracycline regimens as a primary prevention strategy. Data collection is ongoing to determine the impact of early intervention on long term cardiac outcomes.

Disclosure of Interest: None Declared

BSH2020-PO-097

Successful employment of spatial imaging mass cytometry in follicular lymphoma confirms a central role of follicular helper T-cells in disease pathogenesisE. Phillips^{1,2,*}, N. Petrov³, F. Spada⁴, M. Terranova-Barberio⁴, W. Townsend⁵, J. Salisbury⁶, S. Devereux^{7,8}, A. Pepper⁹, P. Patten^{7,8}¹Division of Cancer Sciences, University of Manchester, Manchester, ²Comprehensive Cancer Centre, School of Cancer and Pharmaceutical Sciences, King's College London, ³NIHR BRC Flow Core Facility, Guys Hospital, King's College London, ⁴NIHR BRC Flow Core Facility, Guys Hospital, King's College London, ⁵Haematology, University College London Hospitals, ⁶Histopathology, ⁷Haematology, King's College Hospital, ⁸School of Cancer and Pharmaceutical Sciences, Comprehensive Cancer Centre, King's College London, London, ⁹Brighton and Sussex Medical School, Brighton, UK

Follicular lymphoma (FL) is a germinal centre (GC) B-cell neoplasm that is critically dependent on its non-malignant tumour microenvironment (TME) for survival. Different T-cell subsets can either enhance or inhibit FL growth. Follicular helper T-cells (Tfh), which are necessary for normal GC B-cell development, support FL survival, whilst other T-cell subsets mediate anti-tumour immunity. However, their roles in FL pathogenesis are poorly defined, partly due to the inability of conventional histology studies to assess complex cell interactions *in situ*. Imaging mass cytometry (IMC) is a novel technique that can assess expression of >40 antigens in a single tissue section simultaneously. Our aims were to explore the feasibility and utility of IMC for highly-multiplexed, *in situ* analysis of the FL TME.

Archival formalin-fixed, paraffin-embedded (FFPE) tissue was obtained from 12 FL patients, selected from extremes of clinical outcome, with progression of disease within 24 months defining a poor-risk group. Tissue sections were imaged by Hyperion mass cytometer (Fluidigm).

Firstly, using a 20-antigen panel on single tissue sections, we were able to simultaneously identify lymphoma cells, T-cell subsets and other non-malignant cells, along with markers of proliferation and exhaustion. One section was excluded due to poor staining quality. We subsequently developed an automated analysis pipeline, using established image processing and machine learning tools (CellProfiler and Ilastik) to identify, quantify and assess spatial interactions between >15 cell types.

Next, we demonstrated that IMC can recapitulate findings produced by other imaging techniques. We have previously shown that Tfh numbers correlate with FL B-cell proliferation using confocal immunofluorescence microscopy (Townsend, Haematologica 2019). IMC similarly confirmed a correlation between the number of Tfh (CD3+CD4+PD1+ICOS+) and proliferating FL B-cells (Ki67+CD20+; $N = 11$, $r = 0.62$; $P = 0.03$). Moreover, neighbourhood analysis confirmed a close spatial correlation. The mean distance from Tfh to the nearest Ki67+ B-cell was 22.0 μm , compared with 34.8 μm with non-Tfh CD4+ T-cells and 36.7 μm with CD8+ T-cells ($P < 0.001$ for both, $n = 11$). Together, these data suggest that Tfh influence FL B-cell proliferation.

Finally, we assessed the relationship of T-cell subsets with prognosis. Tfh were more prevalent in poor-risk patients ($n = 5$) compared with good-risk cases ($n = 6$; $P = 0.059$). Conversely, we found a relative paucity of CD8+ T-cells ($P = 0.006$) and non-Tfh CD4+ T-cells ($P = 0.058$) in poor-risk cases compared with good-risk cases. However, other T-cell subsets, defined by FoxP3, T-bet and Tim3 expression, did not differentiate good- and poor-risk cases. Similar results

were obtained when these analyses were restricted to follicular areas, delineated by CD21+ follicular dendritic cell networks.

This work demonstrates that IMC on FFPE biopsies can be effectively employed to deep phenotype and analyse the TME in FL. Our data provides spatial characterisation of the T-cell component in FL, confirming a central role for Tfh in driving FL growth. While both CD8+ T-cells and Tfh cells may correlate with prognosis, further studies using samples from larger numbers of uniformly-treated patients are required. Elucidation of cellular networks within the TME and their prognostic relevance will enhance our understanding of disease pathogenesis and facilitate rational design of immune therapies.

Disclosure of Interest: None Declared

BSH2020-PO-098

Minimal residual disease and the prediction of survival outcomes in chronic lymphocytic leukaemia: a pooled analysis of observational and clinical trial dataE. Ngonga Kemadjou¹, R. Johnston^{2,*}, A. Martin³¹JB Medical Limited, Sudbury, ²Medical Affairs Oncology, AbbVie, Maidenhead, ³Crystallise Limited, Basildon, UK

This study aimed to evaluate the association of MRD with survival outcomes (OS, PFS) and the predictive relevance of MRD in patients with CLL. We searched PubMed, Embase, Heero.com, Cochrane Library and other grey literature sources, screened 3065 studies, included 31 in a review and identified 15 with adequate data for meta-analysis. The association of MRD with outcomes was assessed by line of therapy (first-line, subsequent-line). Random effect models were used to obtain pooled hazard ratios (HR) with 95% confidence intervals (CIs) and the predictive interval (PI). Where HRs for outcomes were not reported in the original studies, HRs were estimated using either individual patient data (IPD) from Kaplan-Meier (KM) curves or survival data at specific time-points were computed. Scenario analysis was performed including studies that had been previously excluded on the grounds that the proportional hazard (PH) assumption did not hold. For OS, relevant data was provided by 3 out of 5 studies in first-line therapy and by 4 out of 7 studies in subsequent-line therapy. Meta-analysis of all 7 studies (706 patients with a negative MRD test [MRD-], 562 with a positive MRD test [MRD+]) showed that a MRD-negative status was significantly associated with increased OS as compared with MRD+ (HR 0.40, 95% CI 0.24–0.66). Scenario analysis supported the base case findings for OS in all 4 studies ($n = 583$ MRD-, $n = 415$ MRD+) with a HR of 0.53 (0.31–0.90) and in 3 first-line studies only ($n = 528$ MRD-, $n = 337$ MRD+) with a HR of 0.43 (0.29–0.63). All HRs for OS were in the same direction favouring those patients who achieved MRD-negativity. However, 95% PI included null, indicating variation in effect estimates in different settings and what can be expected in future studies. For PFS, relevant data was provided by 6 out of 8 studies in first-line therapy and in 8 out of 12 studies in subsequent-line therapy. Regardless of line of treatment, meta-analysis of 14 studies ($n = 1021$ MRD-, $n = 1289$ MRD+) showed that MRD negativity was a strong predictor for PFS: HR 0.29 (0.21–0.38), with a narrow PI of 0.11–0.77. Pooled data from 6 first-line studies ($n = 644$ MRD-, $n = 818$ MRD+) also showed a strong association between MRD negativity and PFS: HR 0.22 (0.19–0.26), with a narrow PI of 0.16–0.31. Scenario analysis (3 studies, $n = 644$ MRD-, $n = 818$ MRD+) supported the base case findings for PFS: HR 0.22 (0.18–0.27), with a narrow PI of 0.14–0.35. Meta-analysis of all 8 subsequent-line studies ($n = 377$ MRD-, $n = 471$ MRD+) gave a significant HR for PFS of 0.34 (0.20–0.59), with a 95% PI including null. This picture was

consistent in a scenario analysis (5 studies, $n = 249$ MRD-, $n = 236$ MRD+), estimating a HR of 0.29 (0.1–0.79). Exclusion of the 2 studies that compared patients at extreme ends of the MRD status (i.e. undetectable MRD vs. high MRD) gave a HR of 0.48 (0.42–0.54), with a PI of 0.37–0.61. All studies included in the meta-analyses consistently confirmed the significant association of MRD status with OS and PFS, regardless of therapies used. This indicates that predictive value of MRD status is independent of study design, MRD assessment and type of treatment used. A narrow PI indicates the benefit of MRD testing in the prediction of PFS; however, lack of high-quality estimates for OS resulted in some uncertainty. These results support the role of MRD status as a surrogate endpoint for outcomes in patients with CLL.

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BSH2020-PO-099

Incidence of BTK mutations in chronic lymphocytic leukaemia treated with BTK inhibitors in a single UK institution

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Bruton tyrosine kinase inhibitors (BTKi) have dramatically changed the outcome in chronic lymphocytic leukaemia (CLL) but relapses on therapy still occur. BTK and PLCG2 mutations have been identified in progressive disease and are considered to be responsible for BTKi resistance. However, data on their incidence, development, clinical impact on disease and outcome are limited.

We analyzed retrospectively 115 patients with CLL, treated with BTKi, as single agent ($n = 95$) or combined ($n = 20$) at St James's University Hospital, to establish BTKi discontinuation and efficacy. Subsequently, we performed sequencing for BTK and PLCG2 mutations in progressive disease, looking for variants on specific encoding regions of BTK (exon 15) and PLCG2 (exons 19–20), with a minimum reporting threshold for single nucleotide variants and indels of five percent minor allele fraction and 100× read depth. Then, we analyzed clinical characteristics in different patients' groups.

The population had high-risk features; 56.5% unmutated IGHV and 29% 17p deletion. BTKi was the first line in 27% of patients. Ibrutinib was the most used BTKi (71%), followed by acalabrutinib (17%), zanubrutinib (9.5%) and tirabrutinib (2.5%). Cumulative incidence of discontinuation at 36 months was 43.6% (95% CI, 33.8–53.8), due to relapse, transformation and other causes. Discontinuations for transformation and other reasons occurred mainly during the first 24 months, while relapses after 24 months. Incidence of progression/transformation at 36 months was 18.3% (95% CI, 10.6–29.7). With a median follow up of 30.8 months, 36-month overall survival (OS) was 72.8% (95% CI, 62.8–80.9) and progression-free survival (PFS) was 69.6% (95% CI, 58.3–78.9). We retrospectively identified twelve available samples on twenty-one relapses for high throughput sequencing, prepared samples for extraction of genetic material and tested for mutations on specific exons of BTK and PLCG2. Cys481 BTK mutations were observed in five cases, all involving nucleotide 1442G>C and/or 1441T>A. All cases were significant mutations, with variant allele fraction (VAF) always >0.2,

with the exception of one mutation (VAF 0.05) in a patient who had two BTK anomalies on different nucleotides. We did not identify any PLCG2 mutations. It was difficult to compare clinical features and outcome of different patients' groups due to the small sample size, however, we observed a different exposure time to BTKi, which was longer (72.5 months, 56.8–79.2) with BTK mutations than without (35.8 months, 16.5–52.0, $p = 0.0052$). All patients with mutated BTK were alive; next therapy with good response was started in two cases, although one relapsed after 14 months of venetoclax. A further twenty-one patients, without relapse, were tested for mutations at different times (before, during or after discontinuation of BTKi for other causes), without any detected mutations.

Although BTKi improved long-term responses and outcome in CLL, disease progression and discontinuation still occur. BTK mutations seem to play a role in relapse and their occurrence might depend, at least in part, on BTKi exposure time. However, only a proportion of patients developed these anomalies, suggesting that other unknown mechanisms may contribute to drug resistance. For this reason, different therapeutic strategies, with new drugs and combinations, and possible intermittent treatment with BTKi should be studied to avoid or bypass BTK and PLCG2 mutations.

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BSH2020-PO-100

Lymph node core biopsy: outcomes from 554 consecutive biopsies and prospective patient experience survey demonstrate a safe, rapid, reliable diagnostic method for lymphoma

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Lymph node excision has historically been preferred to core biopsy for diagnosing lymphoma, given the greater quantity of histological material obtained and consequent reported higher diagnostic yield. However, core biopsy is associated with lower morbidity and is more cost effective. Advances in radiological and histological techniques in extracting and processing material warrants a re-look at the efficacy of core biopsy as a viable primary diagnostic method, including the patient experience and complication rate.

Over a 3-year period (January 2016 – December 2018), all patients who underwent core biopsy for investigation of possible lymphoma or relapse at our centre, were identified from a local database. Argon medical BioPinceTM needles were used for the majority of procedures. Over 6 months, patients undergoing core biopsy were contacted and gave consent to participate in a telephone survey, prior to receiving their final diagnosis.

554 consecutive patients were included. 225 (40.1%) patients had a prior lymphoma diagnosis whilst 329 (59.4%) were new presentations. 78 (14.1%) biopsies were guided by prior PET imaging of which 69/78 (88.5%) were to investigate possible relapsed disease. Three or more cores were taken in 420 (75.8%) cases. The median time from request to biopsy was 2 days (range 0–40 days) whilst median time from biopsy to histology report was 7 days (range 1–24 days). Final diagnoses were: 195 (35.2%) reactive/non-malignant, 346 (62.5%) lymphoma [88 (15.9%) Diffuse large B cell lymphoma (DLBCL), 76 (13.7%) Follicular lymphoma, 56 (10.1%) Hodgkin lymphoma, 37 (6.7%) T-cell lymphoma, 28 (5.1%) Chronic lymphocytic leukaemia, 61 (11.0%) other B cell lymphoma], 12 (2.2%) other malignancy and 1 (0.18%) inadequate for interpretation. 510/544 (93.8%) biopsies were diagnostic inclusive of 74/78 (94.9%) PET-directed biopsies and 35/37 (94.6%) biopsies confirming T-cell

lymphoma. There was no significant diagnostic difference between new or relapsed lymphoma ($P = 0.445$), PET-directed ($P = 0.507$), for a diagnosis of T-cell lymphoma ($P = 0.468$) or nodal vs. extra nodal ($P = 0.693$). There was a clinical need for further tissue in 38 cases, who went on to have a second biopsy. For the patient experience study, 40 patients were contacted, of whom 36 (90%) agreed to take part. 5 (13.9%) reported minor bleeding or bruising. No other complications were reported. Discomfort was graded on a scale of 1–10, with a median of 1 (range 1–6), and a median duration of discomfort of 30 min. There was no significant difference in discomfort based on needle gauge ($P = 0.329$) or diagnosis (benign vs. malignant) ($P = 0.2997$). The mean waiting time was 24.3 (range 0–180) min and patients were kept for observation for no longer than 20 (range 0–20) min. Patients graded their overall impression of the service from 1 to 5 (1 = poor, 5 = excellent). 29 (80.6%) graded the service as 'excellent', with the remaining 7 patients grading it as 'very good'.

Core biopsy performed by experienced interventional radiologists and analysed by expert haemato-pathologists is a reliable and convenient method for diagnosing lymphoma and confirming relapse including T-cell lymphoma. Sufficient material can be obtained by this method, with most cases having three or more cores taken, without the need for further biopsy in the majority. Core biopsy is safe and well tolerated by patients, and can be performed safely, effectively and quickly in the outpatient setting.

Disclosure of Interest: None Declared

BSH2020-PO-101

Exploiting VDJ recombination of T cells to dissect the anti-cancer immune interactions in T cell lymphoma

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Introduction: T cell lymphoma presents a unusual immune-cancer interaction, as the key cell of our anti-cancer adaptive immune response, the T helper cell, is targeting the same cell type within the microenvironment. This presents two challenges. Investigating the pathophysiology is hampered by the similarity between these cell types, which are both CD3+ CD4+ and can have similar morphological appearance. This can lead to confounding results when investigating disease progression. Tumour infiltrating lymphocytes (TIL) have an important prognostic role in cancer, but they are anatomically and phenotypically close, and so are hard to preserve in anti-cancer treatment. We investigated the tissue microenvironment of cutaneous T cell lymphoma, exploiting the physiological genetic self-editing that a T cell undergoes early on, to discriminate between tumour and TIL.

Methods: 49 skin samples were obtained from patients with mycosis fungoides, dissociated and analysed using TCR-specific antibodies to determine clonality. The TCR-specific antibody was used to discriminate between tumour and TIL, and a panel of immune checkpoint markers, cytokine production and transcription markers were analysed using flow cytometry. Four samples were analysed using single cell RNA-seq with combined immunoprofiling, retrieving whole exome gene expression and the TCR sequence for each cell.

Results: Immune checkpoint receptor expression was consistently elevated in the TIL population, whereas the tumour cell population

showed marked heterogeneity of surface markers, clustering into three subgroups. IL-17 production dominated the microenvironment, which occurred in both tumour and TIL, whereas IL-4 production was elevated only in the tumour population. Single-cell RNA-seq demonstrated TCR clonality in the tumour cell population in three samples, and differential gene expression analysis revealed several recurrently upregulated genes in tumour.

Conclusion: The balance between tumour cells and TIL can vary markedly between patients with cutaneous T cell lymphoma, and this can confound analyses which do not discriminate between these opposed cell populations. High dimensional single-cell techniques are necessary to define the two populations, and they can be effective in illuminating differences in features. Further use of these techniques can help us to design treatments which can leverage the TIL population against the tumour cells, either by targeting over-expressed markers, or enhancing tumour-specific TIL.

Disclosure of Interest: None Declared

BSH2020-PO-102

An immunocompetent, spontaneous mouse model of DLBCL for studying the response to R-CHOP and identifying new treatment targets

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First line therapy for diffuse large B cell lymphoma (DLBCL) has not improved for 20 years, despite numerous randomized trials. Often the new agent combinations in these trials have greater toxicity, with little or no improvement in efficacy, compared with the standard R-CHOP (rituximab, cyclophosphamide, doxorubicin, vincristine, prednisolone). The new agents chosen for such trials had demonstrated single-agent efficacy in relapsed DLBCL, but had rarely been properly characterised in combination with R-CHOP in pre-clinical (mouse) models. Most pre-clinical models, when used, are xenografts of a human cell line into immune-compromised mice. Rational development of new R-CHOP combinations requires a more sophisticated model, which captures both the heterogeneity of newly diagnosed DLBCL and the role of the immune system in mediating the response to therapy. We have developed a spontaneous immunocompetent mouse model of DLBCL, of the activated B cell (ABC) subtype, and we demonstrate the feasibility of CHOP combined with a murine depleting anti-CD20 antibody as a model of R-CHOP. This will be a powerful tool for identifying and testing new treatment combinations.

ABC-type DLBCL is characterised by activation of the NFkB transcription factor family, and data from our lab indicate that unopposed NFkB activation leads to plasmablastic tumours (unpublished). In DLBCL, plasmablastic differentiation is blocked by various mutations such as loss of PRDM1 (PR domain zinc finger protein 1) or overexpression of BCL6. We therefore generated mice combining overexpression of MYC, IKKb (inhibitor of kappaB kinase beta) and deletion of *Prdm1* specifically in B cells, using Cre recombinase under the control of the CD19 locus. These mice develop a high-grade large B cell lymphoma after 6–8 months. These lymphomas recapitulate many of the features of human ABC lymphoma, including widespread lymphadenopathy with frequent extranodal manifestations (11/14 mice), large B cell morphology, expression of IRF4 but not BCL6, and high proliferative index by Ki67 staining. The inclusion of marker proteins GFP and human CD2, in the IKKb and MYC alleles, respectively, allows recombined cells to be identified by flow cytometry.

Treatment was started when mice show signs of lymphoma, such as abdominal swelling, reduced movement or weight loss. Cyclophosphamide, vincristine, doxorubicin and anti-CD20 (clone 5D2) were given by intravenous injection on day 1, and prednisolone by oral gavage on days 1–5. Serial assessments by ultrasound tomography on day 1 and day 8 showed a mean reduction in spleen volume of 33%. Histology demonstrated a reduction in GFP+ tissue, compared with saline-treated mice, and, by flow cytometry the proportion of GFP+ hCD2+ cells in 5D2-CHOP treated animals was 16% vs 45% in saline-treated animals. To confirm that the antibody and CHOP elements are independently effective, further cohorts were treated with CHOP or 5D2 alone, and, in these groups, reduction in spleen size was also seen. We hypothesise that the surviving tumour cells are enriched for biological pathways conferring resistance to 5D2, CHOP or the combination, and are undertaking RNA sequencing of sorted populations to identify these pathways. The effect of targeting these pathways, pharmacologically or genetically, will be assessed in combination with R-CHOP in this *in vivo* model to identify the most promising candidates for clinical trials.

Disclosure of Interest: B. Maybury Conflict with: CRUK Clinical Fellowship, J. Fitzgibbon Conflict with: CRUK, D. Calado Conflict with: Leuka, Bloodwise, CRUK

BSH2020-PO-103

Response to brentuximab vedotin plus CHP according to CD30 expression in the ECHELON-2 trial

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Brentuximab vedotin is an antibody–drug conjugate that targets CD30. The phase 3 ECHELON-2 trial demonstrated significantly longer progression-free ($P = 0.0110$) and overall survival ($P = 0.0244$) with brentuximab vedotin plus cyclophosphamide, doxorubicin and prednisone (A+CHP) versus cyclophosphamide, doxorubicin, vincristine and prednisone (CHOP) in the frontline treatment of patients with CD30+ peripheral T-cell lymphoma (PTCL) (Horwitz S, et al *Lancet* 2019;393:229–40). Eligible histologies for inclusion in ECHELON-2 included ALK+ systemic anaplastic large cell lymphoma (sALCL) (International Prognostic Index ≥ 2), ALK– sALCL, PTCL not otherwise specified (PTCL-NOS),

angioimmunoblastic T-cell lymphoma (AITL), adult T-cell leukaemia/lymphoma, enteropathy-associated T-cell lymphoma and hepato-splenic T-cell lymphoma. The complete remission (CR) rate (68% vs 56%; $P = 0.0066$) and overall response rate (ORR) (83% vs 72%; $P = 0.0032$) were also significantly higher with A+CHP versus CHOP. Patients with CD30+ ($\geq 10\%$ by local review) PTCL were included in the trial; expression of CD30 is universal in systemic anaplastic large-cell lymphoma (sALCL), but variable among non-sALCL subtypes of PTCL. As ORR is a direct measure of antitumour activity, we examined response to A+CHP in ECHELON-2 according to CD30 expression in non-sALCL histologies. The relationship between CD30 expression (assessed by Ber-H2 antibody immunohistochemistry) above and below the median value (median CD30: 18% in PTCL-NOS, 25% in AITL) and CR rate, ORR and duration of CR (DOCR) were analysed in patients with AITL or PTCL-NOS who had been treated with A+CHP. Most AITL patients (26/29, 90%) had CD30 expression between 10% and 30%, whilst those with PTCL-NOS had CD30 expression levels ranging from 10% to 100%. The analyses showed that CD30 levels were neither predictive of response (Table) nor associated with DOCR in patients with AITL ($P = 0.30$) or PTCL-NOS ($P = 0.90$) (log-rank test). Hence, CD30 expression above versus below the median (or at 10%) does not appear to predict response to A+CHP in patients with non-sALCL subtypes of PTCL, as responses are seen across CD30 levels. This finding may be due to intra- and inter-tumoural heterogeneity of CD30 expression, limitations of the immunohistochemical technique, the nature of CD30 on the cell surface, multiple mechanisms of action of brentuximab vedotin, or low patient numbers. Further evaluation of the expression–response relationship in PTCL patients with CD30 levels of $<10\%$ is warranted.

Table

Histology	CD30 expression	Patients, n	Complete remission, n (%)	Partial remission, n (%)
AITL	CD30 > median	14	8 (57)	1 (7)
	CD30 $\leq$ median	15	8 (53)	3 (20)
	CD30 = 10%	8	5 (63)	0
PTCL-NOS	CD30 > median	14	8 (57)	2 (14)
	CD30 $\leq$ median	14	10 (71)	2 (14)
	CD30 = 10%	6	4 (67)	2 (33)

Disclosure of Interest: T. Illidge Conflict with: Takeda and Industry Takeda, S. Horwitz Conflict with: Astex Pharmaceuticals; Celgene Corporation; Kura Oncology, Inc.; Kyowa Hakko Kirin Co., Ltd.; Portola Pharmaceuticals, Inc.; Seattle Genetics, Inc.; Takeda Pharmaceuticals North America, Inc.; Verastem, Conflict with: ADC Therapeutics; AILERON; Celgene Corporation; Daiichi-Sankyo Co.; Forty Seven, Inc.; GlaxoSmithKline; Merck & Co., Inc.; miragen; Portola Pharmaceuticals, Inc.; Seattle Genetics, Inc.; Takeda/ Millennium; TRILLIUM; VERASTEM, S. Iyer Conflict with: Daiichi Sankyo, Legend, Crispr, Conflict with: Rhizen, Seattle Genetics, Merck, Affimed, Corvus, Daiichi Sankyo, Legend, Crispr, N. Bartlett Conflict with: Research support to institution Seattle Genetics/Takeda, W. S. Kim Conflict with: Research Funding from Roche, Kyowa-Kirin, Novartis, J&J, Mundipharma, Celltrion, Donga, D. Belada Conflict with: Takeda, T. Feldman Conflict with: Seattle Genetics/BMS/Celgene, Conflict with: (Not to her but to) Hackensack University Medical Center/Celgene/Seattle Genetics/AbbVie/Pharmacyclis/Janssen/KITE/BMS, Conflict with: Takeda/Celgene/Seattle Genetics/AbbVie/Pharmacyclis/Janssen/KITE/BMS, Á. Illés Conflict with: Janssen, Celgene, Novartis, Pfizer, Takeda, Roche, Conflict with: Takeda, Seattle Genetics, E. Jacobsen Conflict with: Consulting or Advisory Role -

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BSH2020-PO-104

IGHV analysis utilising the Ion GeneStudio™ S5 System and the Oncomine™ BCR IGH-LR assay – early access experience

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In 1999 two studies found that patients with Chronic Lymphocytic Leukaemia (CLL) could be separated into two categories, those with mutated IGHV genes (<98% identity with the germline gene, M-CLL) having a more favourable prognosis than those with unmutated IGHV genes (U-CLL). IGHV gene mutational status has continued to be one of the most robust prognostic markers in CLL; furthermore, it has a strong predictive value for response to treatment with U-CLL displaying shorter progression-free survival after chemo-immunotherapy with the fludarabine, cyclophosphamide and rituximab (FCR) regimen compared to M-CLL.

The standard method for IGHV analysis has been to Sanger sequence (SS) the IGHV rearrangement of CLL B cells. Recently, Next Generation Sequencing (NGS) assays have been developed and here we describe our early access experience of the Oncomine™ BCR IGH-LR assay (ThermoFisher). The assay is RNA-based and can be used to assess repertoire diversity, clonal expansion, B cell isotype and subtype and mutational status, employing multiplex FR1 and isotype specific primers.

We assessed 48 previously SS samples; 27 from the same time point and 21 with a median difference in time point of 36 months. 6 had been easy to SS, 8 had been difficult and 34 had borderline mutation identity (97–98.99%), this was to assess possible issues with mutational status assignment due to FR1 primers excluding the 5' sequence of the variable gene.

NGS detected the same rearrangements as previously detected by SS in 47/48 cases and for 1 previously unproductive (UP) only result, NGS identified the productive (P) sequence. 10 NGS results, including 4 of the difficult cases, did show some variation; 6/10 had the same major clone and additional minor clones, with 1 of these minor clones being P by NGS but UP by SS (1 nucleotide insertion), 2 showed clonal variation over time, 1 had the same sequence except a 3 nucleotide deletion present in the SS which was absent by NGS and 1 had a predominant rearrangement by NGS that was not detected by SS. Mutational status was concordant in 13/14 cases that

had previously been sequenced using a FR1 primer, with 1 case previously borderline now U-CLL due to the absence of 1 mutation with NGS. 10/32 cases previously sequenced using leader primers (entire variable region covered), did cross a cut-off but, 2 difficult cases where identity previously could not be determined, now gave a result. NGS isotype analysis identified 9 CLL cases showing evidence of some class-switching and 4 of these were predominantly switched.

Overall, our experience of the early access Oncomine™ BCR IGH-LR assay has been positive; for all productive rearrangements the same sequence was detected by NGS, the observed differences in mutational status due to priming in FR1, should not be an issue going forward as a leader primed assay is currently in development, but the 2 cases with variation in deletion-insertions will need further investigation. The main advantage of the assay is the detection of all rearrangements separately, which obviates the issues encountered with direct Sanger sequencing when there is more than one rearrangement or polyclonal B cells present. The assay will also enable the detection of a clone at very low frequency e.g. following treatment or at clonal evolution. In addition, isotype identification may be of interest in translational research and potentially provide information about the dynamics of the clone.

Disclosure of Interest: Z. Davis Conflict with: Early access reagents provided by ThermoFisher

BSH2020-PO-105

Outcomes for patients with diffuse large B cell lymphoma in the West of Scotland in the pre-CAR-T era

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The majority of patients with Diffuse large B-cell lymphoma (DLBCL) are cured by chemo-immunotherapy (CI) regimens. However, those who are primary refractory to CI or suffer disease relapse have poor outcomes and remain a patient group with significant unmet clinical need. Chimeric-Antigen Receptor-T (CAR-T) cell therapies hold significant promise for patients with relapsed or refractory DLBCL. We sought to assess the outcomes of patients diagnosed with DLBCL in the West of Scotland (WoS) from 01/01/2013 to 31/12/2014 (excluding Primary Mediastinal B Cell Lymphoma), to evaluate how many patients in this region would be eligible for licensed CAR-T therapy on an annual basis, and to serve as a comparator data set for assessing outcomes of future CAR-T recipients.

359 patients were diagnosed with DLBCL in this 2 years period with median age 70 years (range 37–92 years). First line chemotherapy was R-CHOP in 216 patients (60.2%), R-miniCHOP in 59 patients (median age 80 years, range 60–92 years), R-CODOX-M-IVAC in 6 patients, with other regimens in 38 patients. 40 patients were not fit for systemic treatment.

The median overall survival (OS) for the whole cohort was 72 months, with 5 years overall survival (OS) 56% (95% CI 50.6–61.4%) and 5 years progression-free survival (PFS) 56.3% (95% CI 50.6–60.2%). The 5 years OS of the 216 patients treated with RCHOP was 69% (95% CI 65.3–71.7%). In terms of response to first line CI, 61% of patients achieved a complete response (CR), 9% a partial response (PR) and 8.4% were refractory. 11.4% of patients were not treated and response data was not evaluable in 10% of patients.

48/220 (22%) patients who achieved a CR subsequently relapsed. The median time to relapse was 18 months (range 3–55 months). The most commonly utilised regimens at relapse were R-DHAP ($n = 11$), R-GDP ($n = 6$), R-ESHAP ($n = 5$) and palliative

radiotherapy ($n = 20$). 24 patients were refractory to second line treatment. 9 achieved a CR and 11 a PR. 25% achieving a CR or PR to salvage therapy suffered a second relapse. All patients treated with third line treatment died of progressive disease.

Notably, 30 patients in this series were identified as having primary refractory disease. 6 of these 30 patients received intensive second line treatment with none achieving a CR. 3 achieved a PR, one of whom received autologous stem cell transplant (ASCT) and remains alive and in CR at the time of analysis. Thus, the median OS of patients with primary refractory disease was only 7.3 months, with 29/30 patients ultimately dying due to DLBCL.

11 patients in this series had ASCT. 5 patients who were in a CR pre-ASCT remain alive and in CR at the time of data analysis in 2019. 2 of the 6 who were in PR ultimately suffered progressive disease and died.

This analysis has demonstrated that, in line with published literature, outcomes for patients with relapsed/refractory DLBCL in WoS are extremely poor. Of the 359 patients diagnosed in this 2 years period, 330 had sufficient data to evaluate suitability for CAR-T therapy. Using current eligibility criteria, 16 patients were retrospectively identified as being fit for curative treatment with adequate performance status and had relapsed or refractory disease after two lines of therapy. Extrapolating data from the WoS (population ~2.5 million), this would predict that in Scotland approximately 18 patients per year would be eligible for CAR-T therapy for relapsed/refractory DLBCL.

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BSH2020-PO-106

Phase Ib/II trial of polatuzumab vedotin plus obinutuzumab and lenalidomide in patients with relapsed/refractory follicular lymphoma: primary analysis of the full efficacy population

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In a Phase Ib/II trial of patients (pts) with relapsed/refractory follicular lymphoma (R/R FL), polatuzumab vedotin (Pola) + obinutuzumab (G) showed activity and tolerability (Phillips et al. Blood 2016). A Phase II study of the doublet combination of G + lenalidomide (Len) showed activity and acceptable safety in pts with R/R FL (Morschhauser et al. Lancet 2019). Here, we present the full primary analysis of efficacy and safety of Pola-G-Len in pts with R/R FL from the Phase Ib/II study, GO29834 (NCT02600897).

GO29834 is an open-label, multicentre study of pts with R/R FL (Grade <3b) who had received ≥ 1 prior anti-CD20-containing chemo-immunotherapy regimen. The recommended Phase II dose (RP2D) for Pola+Len was defined in a 3+3 dose-escalation phase. In

the Phase II expansion cohort, pts received induction treatment with six 28-day cycles of: G 1000 mg IV (Cycle [C]1: Day [D]1, D8, D15; C2-6: D1); Pola 1.4 mg/kg IV (D1), Len 20 mg PO (D1-21). Responders received maintenance treatment for 24 months: G 1000 mg (D1 every 2 months); Len 10 mg (D1-21, Months 1-12). The primary endpoint was Independent Review Committee (IRC)-assessed complete response (CR) at end of induction (EOI), based on positron emission tomography-computed tomography (PET-CT) scans (modified Lugano 2014 criteria). Progression-free survival (PFS) was assessed by the investigator.

As of 12 August 2019, 56 pts were enrolled and had entered induction (Phase Ib/II cohorts); median follow-up was 16.6 and 15.1 months in safety- and efficacy-evaluable populations, respectively. Baseline characteristics were: median age, 62 years; male, 59%; Ann Arbor Stage III-IV, 88%; Follicular Lymphoma International Prognostic Index high-risk (≥ 3), 55%; bulky disease (≥ 7 cm), 16%; ≥ 2 prior lines of therapy, 77%; refractory to last line of prior therapy/any anti-CD20 treatment, 59%/71%, respectively. All pts had ≥ 1 adverse event (AE), 32 (57%) had a serious AE, 47 (84%) had a Grade 3-4 AE. The most common Grade 3-4 AEs were neutropenia ($n = 31$, 55%), thrombocytopenia ($n = 15$, 27%), infections ($n = 11$, 20%), and anaemia ($n = 8$, 14%). AEs led to dose reduction or interruption of any drug in 19 (34%) and 43 (77%) of pts, respectively; the majority were modifications of Len. AEs led to the discontinuation of any study drug in 17 (30%) pts. One Grade 5 AE was reported (septic shock); this was not considered study treatment-related as the pt was receiving a new anti-lymphoma treatment following disease progression (PD). In the primary efficacy population ($n = 46$), the IRC-assessed objective response rate was 76%, the CR rate was 63% (Table). Subgroup analysis showed that 60% (15/25) of pts who were refractory to their last treatment achieved a CR. Median PFS was not reached.

Our study of the novel triplet combination, Pola-G-Len demonstrates a safety profile consistent with the known profiles of the individual drugs. CR rates at EOI were high in this heavily pre-treated and refractory population, which compares favourably with currently available R/R FL therapies. These findings support further investigation of Pola-G-Len in a larger pt population. Follow-up is ongoing to determine the median PFS.

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Table. Responses at end of induction ($N = 46$)

	Modified Lugano 2014 ¹		Lugano 2014	
	INV	IRC	INV	IRC
Best overall response, n (%)				
Objective response	38 (83)	35 (76)	38 (83)	35 (76)
CR	28 (61) ²	29 (63) ²	34 (74)	33 (72)
PR	10 (22)	6 (13)	4 (9)	2 (4)
SD	3 (7)	4 (9)	3 (7)	4 (9)
PD	3 (7)	1 (2)	3 (7)	1 (2)
Missing/unevaluable	2 (4)	6 (13) ³	2 (4)	6 (13) ³

CR, complete response; CT, computed tomography; EOI, end of induction; INV, investigator assessment; IRC, independent review committee assessment; PD, progressive disease; PET, positron emission tomography; PR, partial response; pts, patients; SD, stable disease.

¹Modified Lugano requires a negative bone marrow biopsy to confirm PET-CR, and PET-PR must also meet CT-PR criteria.

2CR downgraded to PR due to missing bone marrow biopsy in six pts by INV and three pts by IRC.

3Three patients had early PD, EOI not reached, and scans not submitted to IRC, and were therefore classified as missing. Disclosure of Interest: C. S. Diefenbach Conflict with: Genentech/Roche, Celgene, A. McMillan Conflict with: Genentech/Roche, Conflict with: Pfizer, Conflict with: Roche, BMS, Celgene, Novartis, Pfizer, Gilead, Sandoz, MSD., B. S. Kahl Conflict with: Genentech/Roche, F. Miall Conflict

with: Takeda - advisory board honorarium, Roche, L. Banerjee: None Declared, J. Briones Conflict with: Roche, Conflict with: Roche, R. Cordoba Conflict with: Janssen, Servier, Kyowa-Kirin, Gilead, Conflict with: Roche, Gilead, P. Abrisqueta Conflict with: Janssen, Abbvie, Roche, Celgene, Conflict with: Janssen, Abbvie, Roche, Celgene, J. Hirata Conflict with: Roche, Y. Chang Conflict with: Roche, L. Musick Conflict with: Roche

Myeloma

BSH2020-PO-107

Real world, multi-centre experience of daratumumab monotherapy and response to subsequent treatments in relapsed/refractory myeloma

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Introduction: Daratumumab (DARA) is an anti-CD38 monoclonal antibody that is approved for treatment of relapsed/refractory multiple myeloma (RRMM). It has been available as a monotherapy at 4th line via the Cancer Drugs Fund (CDF) in England since January 2018. We review here the real-world effectiveness of DARA monotherapy at 3 English centres and response to subsequent therapies.

Methods: Patients were identified at University College London Hospital, Leeds Teaching Hospitals and Huddersfield Royal Infirmary who had commenced 1 cycle of DARA monotherapy (received ≥ 1 dose) according to the licensed regimen (16 mg/kg IV weekly (8 weeks (w)), fortnightly (16w), monthly thereafter). Patients were managed as part of a clinical trial, as private patients or via the CDF. Data were collected from April 2016 to October 2019. High risk cytogenetics (CG) were defined as t(4;14), t(14;16), t(14;20) and del 17p. Response was assessed according to IMWG criteria and progression free survival (PFS) was determined using Kaplan-Meier analysis. **Results:** Ninety-four patients were identified. The median age at DARA was 67 years (40–91). Disease isotypes were: IgG, 54 (57%); IgA, 19 (20%); light chain, 18 (19%); and non-secretory, 3 (3%). Seventy-one patients (75.5%) had received an autologous stem cell transplant. Patients were refractory as follows: 26 (28%) to bortezomib; 13 (14%) to thalidomide; 57 (61%) to lenalidomide; and 67 (71%) to immunomodulatory drugs (IMiD) and proteasome inhibitors (PI). Clinical relapse was evident in 56 (60%) patients: 38 (40%) had only serological relapse. Median time from diagnosis to DARA was 62 months (m) (11–190). The majority of patients (82 (87%)) received DARA at 4th line. Eighty-four (89%) patients had assessable disease as per IMWG criteria, of whom 66 (70%) had CG information available from diagnosis.

The median number of cycles delivered was 5 (1–36). After a median follow-up of 13.6 m (0.2–42.6), 30 (31%) patients remained on DARA. Thirty (32%) patients died, of whom 29 (97%) died of progressive MM. Overall response rate ($\geq$ PR) was 38%: PR in 18 (21%); $\geq$ VGPR in 14 (17%). Stable disease was achieved by 12 patients (14%), MR by 19 (23%) and 21 (25%) had primary progression on DARA. Median PFS on DARA was 4.8 m. Deeper response conferred improved median PFS (VGPR/PR vs MR/SD/PD, 14.7 m vs 3.1 m ($p < 0.0001$, 95% CI 0.11–0.41)). Patients with standard risk CG ($n = 44$) had a longer median PFS than patients with

high risk CG ($n = 22$) (5.0 m vs 2.5 m, $P = 0.01$, 95% CI 0.27–0.91). Patients with extra medullary disease ($n = 14$) had poorer median PFS than those without (1.6 m vs 7.4 m, $P = 0.0002$, 95% CI 0.12–0.38).

The time to next treatment from DARA was median 5 m (0.6–20). Forty-one patients (44%) commenced at least 1 further line of treatment after DARA (T1); 13 (13.8%) patients received 2 (T2); and 2 (2.1%) patients received 3 (T3) subsequent lines. At T1 CG information was available for 27 (66%) of whom 9 (22%) had high risk CG. At T1 30 patients received a pomalidomide-based regime, 6 enrolled in a clinical trial, and 5 patients received other palliative regimen. Encouragingly median PFS at T1 was 5.3 m in patients with assessable disease ($n = 38$).

Conclusion: DARA monotherapy shows clinical benefit in a heavily pre-treated population who are largely refractory to PI and IMiD. Response rates align with those reported in clinical trials. Most patients who subsequently relapse can receive further effective therapies.

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Myeloma: not just bones at risk of fracture – assessing the increasing demand on services

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Myeloma management has seen significant changes in the last 5 years. The International Myeloma Working Group (IMWG) updated the criteria for treatment initiation in 2014 to include Myeloma Defining Events (MDE). In addition, NICE has approved several new therapies over this time period. We assessed the impact of these changes on the myeloma clinic workload in a District General Hospital.

We retrospectively analysed data on the myeloma clinic from 2014 (before application of IMWG changes) and 2019. Data was collected from electronic patient notes at these 2 time points. We excluded patients with MGUS and solitary plasmacytoma. Data was analysed for age, diagnosis date, symptomatic vs asymptomatic disease, indication for first line treatment, number of treatment lines received, number of clinic attendances and if patients were transplanted and at what age.

Data from 405 patients and 2733 clinic visits were reviewed. Despite fewer new diagnoses made in 2019 (48 compared to 57 in 2014), the total number of patients seen increased by 32% to 302 in 2019 compared to 228 in 2014. This reflects the increased number of

patients in follow up: 237 (84%) in 2019 compared to 171 (75%) in 2014. Consequently, there was a marked increase of 57% in total clinic visits (1663 in 2019 vs 1070 in 2014). Patients on treatment were seen 1104 times compared to 696 in 2014, a 59% increase. In 2019, median age of patients seen was 65 yrs (32–86) compared with 55 yrs (47–90) in 2014. More newly diagnosed patients were referred for transplant in 2019: 12(25%) vs 9(16%) in 2019 and 2014 respectively. Median age of transplanted patients was 68 yrs (52–72) in 2019 and 59 yrs (49–70) in 2014. There was a 16% increase in the number of patients on treatment in 2019 compared to 2014 (136 vs 99 respectively). Of these patients 32% were initiated for MDEs, reflecting an additional 48 patients undergoing treatment. The number of patients on 3rd line treatment or greater increased markedly in 2019 to 57 compared to 24 in 2014, an increase of 95%. 32 patients (11%) treated in 2019 received Daratumumab either as 2nd line combination therapy or 4th line monotherapy reflecting an additional day unit burden.

In the 5 years period studied, there was a 32% increase in the number of myeloma patients leading to a 57% increase in the number of clinic visits. This increase is likely to reflect patients receiving additional lines of treatment and living longer, as demonstrated by nearly double the number of patients on 3rd line treatment or above. In 2019, 32% of patients were commenced on treatment due to MDEs and fewer patients were started due to conventional CRAB criteria, 88 compared to 99 in 2014. One possible explanation for this shift could be that we are treating more people earlier in their disease course although additional data would be needed to validate this. Overall, there has been a significant increase in patient numbers receiving myeloma care, with a significantly increased demand on outpatient and day unit services, pharmacy demands and of course drug budgets. We have to anticipate that these numbers will only increase further, putting strains on already strained services. There has to be a national strategy to aid appropriate provision of care going forwards.

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Real world outcomes in multiple myeloma in patients aged 75 and over

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Introduction: Outcomes for older patients with myeloma are inferior to younger patients. Frailty, comorbidity and poor performance status adversely effect outcomes in this patient group. These patients account for a large proportion of myeloma diagnoses but have been underrepresented in clinical trials. We report the outcomes of patients over the age of 75 treated for myeloma according to NICE treatment pathways diagnosed over a 5 years period within Northumbria Healthcare NHS Trust.

Method: We retrospectively identified all cases of myeloma diagnosed within the Trust between 01/01/2013 and 31/12/2017. Electronic records were reviewed to identify demographic information, performance status and comorbidity at diagnosis, therapy received, survival and cause of death. Data was collected up to 18/04/2019.

Results: There were 148 new diagnoses of myeloma in this period. 53 (36%) were over the age of 75 (range 75–94). Median overall survival and disease specific survival from diagnosis were 39 and 60 months. By the end of the study period 45 patients had received treatment. Overall and disease specific survival from first treatment were 38 and 55 months. Of the untreated patients 4 had died and 4

continued observation. The percentage of patients who had received 1st, 2, 3 or 4 lines of treatment was 85%, 40%, 26% and 2%. In first line 60% of treatment was thalidomide (30 CTDA, 12 MPT) and 27% bortezomib based (5 VMP, 4 VD, 3 VCD). In second line therapy 62% was bortezomib based (9 VCD, 4 VD). The remainder consisting of 2 thalidomide based, 2 carfilzomib, 1 lenalidomide and 1 C-weekly. All 14 patients treated at 3rd line received lenalidomide and dexamethasone with 1 receiving ixazomib, lenalidomide and dexamethasone. 1 patient received 4th line pomalidomide and dexamethasone. All treatment was consistent with NICE approvals at the time of treatment. There was no significant difference in OS between patients aged 75–79 and over 80 years old (median OS 65 vs 34 months, $P = 0.185$). Poor ECOG performance status, 0–1 vs 2+ (median OS 56 vs 27 months, $P = 0.011$) and comorbidity, HCT-CI 0–2 vs 3+ (median OS 60 vs 25 months, $P = 0.011$) were significantly associated with worse outcome. 27 (51%) patients died during the study period. 20 deaths (38%) were myeloma related. 7 deaths (13%) were from causes unrelated to myeloma. These consisted of 3 s malignancies (AML, cholangiocarcinoma, colorectal carcinoma), and 1 each of heart failure, stroke, frailty and death from unknown causes in a patient in remission.

Conclusion: The survival data for our cohort is comparable with the ONS data for myeloma survival in England for patients diagnosed between 2011–15 in this age group (31% 5 years survival vs 32% from first treatment or 41% from diagnosis in our cohort). This suggests that our data is representative of real world UK outcomes for patients treated with funded therapies on NICE pathways. Poor performance status and comorbidity identified a high risk group of patients with a poor outcome. Whilst more complex tools are available to assess frailty and comorbidity these are not yet routinely used in clinical trials or general haematology practice. We demonstrate that routinely collected clinical data can identify high risk patients. This group has an unmet need for effective well tolerated therapy and supportive care to improve outcomes.

Disclosure of Interest: None Declared

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The UK experience of daratumumab monotherapy in relapsed systemic AL amyloidosis

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Daratumumab is a CD38-targeting monoclonal antibody used in the treatment of patients with myeloma. Small retrospective studies suggest significant activity in patients with relapsed systemic AL amyloidosis. We report here the experience of patients treated with daratumumab monotherapy (2016–2019), identified from the database at the UK National Amyloidosis Centre.

Haematological and disease responses were defined as per amyloidosis consensus guidelines. The Kaplan-Meier method was used to estimate median progression-free survival (PFS) and overall survival (OS) with correlations made using log-rank methods.

53 patients were identified. Fifty patients were evaluable for haematological response at 3 months (2 uninterpretable light chains, 1 died prior to response assessment). Of evaluable patients, median

age was 68 (42–85) years; 64.2% male; Disease stage at diagnosis: Mayo 1, 11 (20.8%), Mayo 2, 19 (35.8%), Mayo 3A, 18 (34.0) and Mayo 3B, 5 (9.4%); Organs involved were: cardiac 39 (73.6%), renal 30 (56.6%), liver 14 (26.4%), soft tissue 15 (28.3%), peripheral nerve 6 (11.3%), autonomic nerve 5 (9.4%) and gastrointestinal 4 (7.5%). Patients had a median of 3 (1–4) prior lines of therapy inclusive of bortezomib in 49 (92.5%) patients and lenalidomide in 44 (83.0) patients. Autologous stem cell transplantation had been performed in 13 (24.5%) patients.

Haematological responses at 3 months were: complete response (CR) – 19 (38%), very good partial response (VGPR) – 14 (28%), partial response (PR) – 9 (18%) and no response – 8 (16%). Median time to response was 1 (1–6) months and best response was 1 (1–6) months. Of assessable patients, cardiac, renal and hepatic responses were seen in 43.8%, 25.0% and 0% of patients whilst progression occurred in 25.0%, 12.5% and 37.5% respectively. Patients were followed up for a median of 9 (2–35) months from start of daratumumab therapy. During the period of follow up, 34 (64.2%) patients continue on daratumumab, 11 (20.8%) patients died, 4 (7.5%) patients stopped treatment and 4 (7.5%) patients moved to next line therapy. Median PFS was 19.9 months (95% CI 8.2–31.8 months) whilst median OS was not reached. Patients achieving a CR had a significantly longer median OS (not reached) compared to those in a lesser haematological response (median 22.7 months [95% CI 17.0–28.4 months]) ($P = 0.036$). Furthermore, patients achieving a rapid response (≤ 1 month) had a significantly longer median PFS (not reached) than those responding at a later time point (9 months [95% CI 5.8–12.2 months]) ($P = 0.013$).

Serious adverse events were: 6 (11.3%) patients required admission to hospital (2 fluid overload, 2 falls [1 secondary to postural hypotension, 1 unexplained]), 1 non-cardiac chest pain and 1 anaemia requiring blood transfusion). The commonest grade I–II AEs were infusion reaction 7/53 (13.2%), thrombocytopenia 6/53 (11.3%), fatigue 6/53 (11.3%), infection 5/53 (9.4%), anaemia 5/53 (9.4%) and fluid overload 4/53 (7.5%).

Daratumumab monotherapy is an effective therapy in patients with multiply-relapsed systemic AL amyloidosis and, contrary to the experience in relapsed multiple myeloma, responses are rapid, seen in 84% of patients and long lasting, especially in patients who respond within 1 month. Given the greater side effects of chemotherapy in patients with AL, daratumumab monotherapy should be considered, if available, in patients who have not received daratumumab as part of a combination earlier in therapy.

Disclosure of Interest: None Declared

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Analysis of frontline treatment of non-transplant elderly myeloma patients in Wales: comparison of bortezomib versus lenalidomide based treatment

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There are no head to head studies comparing bortezomib (V) based fixed duration therapy with continuous lenalidomide and dexamethasone (Rd), available in Wales since July 2016. Real world data can justify a new approach. The FIRST trial (Benbouker et al NEJM

2014¹) showed a progression free survival (PFS) of 25.5 months for continuous Rd. Real-world data comparing the FIRST data to real world has been presented (Sharpley, ASH abstract 3404, 2017²). However, in this publication the control (Rd) arm data was taken from MM-20 trial subjects rather than real world patients and the FDT cohort included a variety of inferior regimens (CD/CTD/MP/MPT) as well as V based treatment. PFS in this analysis showed an impressive advantage for Rd of 21.9 months vs 9.1 for other regimens².

We have compared these two treatment approaches based on 3 years of real world data in Wales, to analyse whether practice now favours first line Rd above V based regimens and to assess the efficacy of the two treatments in terms of PFS and clinical characteristics at baseline. Data was analysed for all frontline non-transplant eligible (NTE) patients across three health boards in Wales starting on 01/03/2016 and data cut off was taken at 01/03/2019. From a total of 121 patients, 112 received V based regimens and only 7 received Rd. There were 68 males and 53 females, median age (at diagnosis) was 75.9 and 76.0 for males and females respectively. ISS staging was available for 112 (92.5%) of patients of which 60 were ISS 3 (53.6%), 39 ISS 2 (34.8%) and 13 were ISS 1 (11.6%). Numbers of ISS 3 patients were similar to Sharpley² data but higher than in the FIRST¹ trial (40%). Median PFS for patients receiving first line V based regimens was 281 days/9.2 months compared to 273 days/8.9 months for Rd.

Genetic data was available for 64 (52.8%), high risk defined as the presence of one or more of deletion del(17/17p), t(4;14), t(14;16), t(14;20). Of this cohort, 40 were standard risk (median PFS 333.5 days/10.9 months) and 24 were high risk (median PFS 315 days/10.3 months). Of the 40 standard risk patients, 38 received first line V (median PFS = 333.5 days/10.9 months) and 2 received first line Rd (median PFS = 214 days/7.0 months). Of the 24 high-risk patients, 23 patients received first line V median PFS = 357 days/11.7 months), and 1 received first line Rd (median PFS = 273 days/8.9 months). ISS staging was available on 112 (92.5%) of 121 patients, of the 112, 13 were ISS 1 (median PFS = 430 days/14.1 months), 39 were ISS 2 (median PFS = 294 days/9.6 months) and 60 were ISS 3 (median PFS = 275 days/9.0 months).

This data shows that uptake of frontline Rd has been slow, most patients continue to receive V based regimens. Median PFS for V based treatment was 9.2 months vs 8.9 months for Rd and this small improvement for was maintained across cytogenetic groups. This data suggests a reluctance amongst myeloma treating physicians to divert from established V based regimens in the elderly, perhaps due to the potential increased myelosuppression and toxicities that can be seen in the elderly with IMiDs and also a desire to reserve this treatment for later lines of therapy. Elderly patients may also be less able to tolerate continuous treatment and want a treatment free interval due to reduced life expectancy. This data also suggests that the PFS advantage from the FIRST trial and previous real-world analysis² may not be reflected in real world UK patients.

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VIM: the results of a phase 1/2 trial of bortezomib, idarubicin and melphalan in relapsed or refractory multiple myelomaK. Boyd^{1,2,*}, F. Davies³, G. Morgan³, A. Bonner¹, L. Conneely¹, M. Kabir⁴, G. Machell¹, C. Pawlyn², R. Saso¹, L. Smith¹, E. Strong¹, M. Kaiser^{1,2}¹Department of Haemato-oncology, Royal Marsden Hospital, ²The Institute of Cancer Research, London, UK, ³Perlmutter Cancer Centre, NYU Langone Health, New York, USA, ⁴Statistics Department, Royal Marsden Hospital, London, UK

Preclinical data suggests that bortezomib can inhibit genotoxic stress response pathways and thereby restore sensitivity to DNA-damaging chemotherapeutic agents. This was the rationale for this study combining bortezomib with two chemotherapy agents. The trial recruited patients from 2006 to 2013 but has never been analysed or reported and provides insights into the role of standard chemotherapy agents in relapsed myeloma, for which there is little published evidence.

VIM was a single centre phase 1/2 study of bortezomib, idarubicin and melphalan in relapsed or refractory myeloma. Phase 1 was a 3+3 design with escalating doses of idarubicin and melphalan in combination with 1.3 mg/m² bortezomib administered on D1,4,8,11 on a 4 weeks cycle. Phase 2 was dose expansion at the maximum tolerated dose (MTD) in a further 20 patients to explore efficacy. Treatment was for 3 cycles. Following VIM, patients were allowed to directly proceed to other treatments including autologous transplant (ASCT). The primary objective was safety and tolerability. Secondary endpoints were response, Progression Free and Overall Survival (PFS, OS). Key inclusion criteria were relapsed or refractory myeloma following ≥1 previous therapies, platelet count ≥70 x10⁹/L, white cell count ≥3.0 x10⁹/L and creatinine clearance of ≥40 mls/min. Dose limiting toxicity was defined as any grade 4 non-haematological toxicity, any grade 3 non-haematological toxicity not resolving to grade 2 within 2 weeks of end of cycle and any grade 4 haematological toxicity except neutropenia not resolving to grade 3 within 2 weeks of end of cycle.

The study recruited 3, 3, 7 patients on dose levels 1, 2 and 3 respectively during the dose escalation phase. The MTD was defined as level 2 which was: Idarubicin 5 mg/m² PO D4, 5; melphalan 15 mg/m² IV D4; bortezomib 1.3 mg/m² D1, 4, 8, 11 on a 4 weeks cycle. An additional 20 patients were recruited to the dose expansion phase. Median patient age was 62. Median prior lines of therapy was 4 (range 2–8). Adverse Events at the MTD level are listed in table 1. The main toxicity was haematological, with 74% of patients having grade ≥3 neutropenia, 78% grade ≥3 thrombocytopenia. 26% of patients experienced grade ≥3 infective complications. There were 3 deaths on treatment; all had progressive or refractory disease at time of death, two with concurrent respiratory infection but neither were neutropenic or directly treatment related. In the intention to treat (ITT) cohort, the overall response rate (ORR) was 79%, with 46% achieving ≥very good partial response (VGPR) of which 14% were near complete response (nCR, defined as no paraprotein by serum protein electrophoresis and <5% plasma cells on trephine but without immunofixation to confirm CR). In the MTD cohort the ORR was 85%, with 50% ≥VGPR and 20% nCR. PFS reflects the heterogeneity of treatment after VIM, including ASCT. Median OS was 19.0 months in the ITT cohort and 26.1 months in the MTD cohort.

Treatment options at relapse have multiplied since this trial was designed. However, VIM is a generic, relatively inexpensive, steroid free regimen with defined toxicities and significant efficacy. The main side effects, as expected, were haematological. Responses were higher than expected for median 5th line therapy suggesting possible synergy

in this combination and reminding us that standard chemotherapies can have a role in relapsed myeloma if other options are unavailable and blood counts are reasonable.

Table.

Event type	Grades 1–2		Grades ≥3		Total	
	No.	%	No.	%	No.	%
Cardiovascular	4	17.4	1	4.3	5	21.7
Constipation	5	21.7	0	0.0	5	21.7
Constitutional	11	47.8	1	4.3	12	52.2
Identified infection	13	56.5	6	26.1	19	82.6
Nausea	10	43.5	1	4.3	11	47.8
Neutropenia	0	0.0	17	73.9	17	73.9
Other	14	60.9	6	26.1	20	87.0
Other GI tox	10	43.5	3	13.0	13	56.5
Other haem tox	2	8.7	11	47.8	13	56.5
Other neuropathy	4	17.4	0	0.0	4	17.4
Pain (not neuropathic)	11	47.8	4	17.4	15	65.2
Pyrexia/non identified infection	2	8.7	4	17.4	6	26.1
Sensory neuropathy	6	26.1	1	4.3	7	30.4
Thrombocytopenia	0	0.0	18	78.3	18	78.3

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Risk of pneumonia and tolerability in patients with multiple myeloma treated with daratumumab: a systematic review and combined meta-analysis of six phase III randomised controlled trialsT. W. Htut^{1,*}, K. Z. Thein², M. A. Win³, E. M. Phyu³, A. Lawrie¹, J. Tighe¹, G. Preston¹¹Department of Haematology, Aberdeen Royal Infirmary, Foresterhill Health Campus, Aberdeen, UK, ²Investigational Cancer Therapeutics, The University of Texas MD Anderson Cancer Center, Houston, TX, ³Internal Medicine, LewisGale Medical Center, Salem, VA, USA

Multiple myeloma is a hematologic cancer of plasma cell, accounting for approximately 13% of all hematologic malignancies. Recent studies have shown survival benefits with daratumumab, either monotherapy or in combination with proteasome inhibitors, immunomodulatory agents and/ or other antimyeloma therapies.

Daratumumab is a human, IgG1κ monoclonal antibody that targets CD38 on myeloma cells and also has direct antitumor effect and an immunomodulatory component. Nonetheless, there are some safety concerns. Hence, we performed a combined meta-analysis of phase III randomised controlled trials (RCTs) to evaluate the risk of treatment discontinuation (TD) and death due to treatment-related adverse events (TRAE) and pneumonia in patients with newly diagnosed multiple myeloma (NDDM) and relapsed or refractory multiple myeloma (RRMM) treated with daratumumab.

We systematically conducted a comprehensive literature search using MEDLINE, EMBASE databases and meeting abstracts from inception till 10th December 2019. Phase III RCTs utilising daratumumab in patients with multiple myeloma were incorporated in the analysis. Mantel-Haenszel (MH) method was used to calculate the estimated pooled risk ratio (RR), with 95% confidence interval (CI). Heterogeneity was assessed with I^2 and Cochran's Q- statistic. A 'p value' of less than 0.05 was considered significant and $I^2 > 50\%$ was considered substantially heterogeneous.

Six phase III RCTs (POLLUX, CASTOR, CANDOR, ALCYONE, CASSIOPEIA and MAIA studies) with a total of 4,008 patients with NDDM ($n = 2503$) and RRMM ($n = 1505$) were included in our meta-analysis. Studies compared daratumumab (D) + bortezomib (V) + melphalan (M) + prednisolone (P) vs VMP, D + lenalidomide (R) + dexamethasone (d) vs Rd, Dvd + thalidomide (T) vs VTd, DVD vs Vd and Carfilzomib (K) + d + D vs Kd. Daratumumab was utilised in relapsed and refractory multiple myeloma in the POLLUX, CASTOR and CANDOR studies, and as first-line treatment for patients with multiple myeloma in the ALCYONE, CASSIOPEIA, and MAIA studies. The randomisation ratio was 2:1 in the CANDOR study and 1:1 in other studies. The I^2 statistic showed heterogeneity among RCTs and random-effect model was applied. TD due to TRAE was 209 (10.0%) in study group vs 231 (12.0 %) in control group and RR was near significant at 0.77 (95% CI: 0.59–1.00; $P = 0.05$). The pooled RR for TD due to TRAE was calculated for each subset; NDDM (RR 0.61, 95% CI: 0.39–0.95; $P = 0.03$) and RRMM (RR 0.93, 95% CI: 0.73 – 1.19; $P = 0.59$). Treatment-related deaths were reported in 69 (3.3%) in daratumumab arm vs 77 (4.0%) in control arm (RR 0.88, 95% CI: 0.60 – 1.30; $P = 0.52$). Any-grade pneumonia was occurred in 303 (14.5%) versus 164 (8.5%) in control group with RR of 1.67 (95% CI: 1.27 – 2.19; $P = 0.0002$). The rate of high-grade pneumonia was 4.1% higher in daratumumab combination regimen compared to control arm and the pooled RR was statistically significant at 1.63 (95% CI: 1.230–2.17; $P = 0.0007$).

In conclusion, TD due to TRAE was significant at RR of 0.61 in patients with NDDM receiving daratumumab, favouring daratumumab combination regimens and there was no statistical difference in treatment-related deaths due to TRAE. Despite increasing survival benefits and attaining better tolerability, the addition of daratumumab to standard multiple myeloma regimen contributed to higher incidence of all grades of pneumonia. Antimicrobial prophylaxis should be considered in the high risk population.

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Autologous stem cell transplant for multiple myeloma patients above 65 years of age

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Treatment options for multiple myeloma (MM) are evolving quickly, including novel triplet/quadruplet therapies combining good tolerability and efficacy profiles. Despite this, autologous stem cell transplant (ASCT) is the first line treatment to consolidate the effects of chemotherapy for fit patients. A limited number of studies show comparable effects of ASCT in selected patients over 65 to those younger. More evidence is needed to understand the toxicity and long-term outcomes in older patients following ASCT.

This study aims to compare the toxicity and long-term efficacy of ASCT in patients up to 65 years of age vs above 65 years.

Patients who underwent ASCT for the treatment of MM at Manchester Royal Infirmary between January 2010 and December 2017 were included in this retrospective study. Patients were excluded from the study if they had previously received an ASCT for the treatment of MM. Age-adjusted dosing of melphalan was not used; all patients received 200 mg/m² apart from those with renal impairment (clearance creatinine <30 ml/min or Creatinine >200 μmol/L) where 140 mg/m² was used.

A total of 302 patients were stratified into 2 groups based on their age at the time of transplant: ≤65 years (209 patients) >65 years (93 patients). Ages ranged from 30–75 years. Each age group had similar comorbidity (average HCT-CI score 2) and a similar proportion of patients stratified into the higher risk ISS 3 group (26% ≤65 years vs 23% >65). Both groups were evenly matched in renal function (CKD 3 or above in 24% in ≤65 years vs 28% >65 years).

Patients in all groups received on average 6 cycles of chemotherapy. Good responses to initial chemotherapy were seen in both groups, with 32% achieving CR and 46% achieving VGPR in the ≤65 age group and 22% CR and 63% VGPR in the >65-year group. The most common chemotherapy regimens included velcade (44% ≤65-years vs 37% over 65) or lenalidomide (26% ≤65-years vs 23% over 65), with cyclophosphamide being the most common additional drug (75% ≤65-years, 77% over 65).

Median time to neutrophil engraftment ($>1.0 \times 10^9/l$) was equal in both groups (13 days), as was time to haemoglobin $>100 g/l$ (16 days). Time to platelets $>50 \times 10^9/l$ was 20 days in ≤65-year olds and 19 days in >65 years. Transfer to ITU was seen in 13 patients (6%) in the younger group and 4 patients (4%) in the older group, and both groups had a median of 4 days with CRP over 50. Median length of hospital stay was identical for both groups (19 days), but readmission rates were markedly higher for older patients (5% of ≤65-year olds, 24% for over 65s).

Three months after ASCT the response rates improved, with 47% of ≤65-year olds and 36% of over 65s achieving complete remission. VGPR was seen in 42% of ≤65s and 49% of over 65s. This response was maintained at 12 months with 58% under 65 and 38% over 65 in CR, and VGPR was seen in 28% ≤65-year olds and 43% over 65. The median OS for the group with age ≤65 years was 113 months vs 77 months for the group >65 years ($P = 0.0339$).

This study suggests that ASCT is safe and effective in selected patients above 65 years. The time to engraftment and the toxicity are superimposable apart from the rate of readmission which is more common in elderly patients. The long-term survival is excellent in both groups, although better of patients ≤65 years, possibly related to the availability of subsequent treatment options.

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Real world experience of tandem autologous stem cell transplant for high risk multiple myeloma at a single UK centre

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Optimal management of newly diagnosed patients with high risk multiple myeloma (MM) is challenging and these patients have poorer outcomes, despite novel agents. Emerging data support tandem autologous stem cell transplant (ASCT) in patients with high risk cytogenetics or R-ISS II-III with improved progression free survival (PFS) and overall survival (OS). Trial data suggest a high proportion of those assigned to tandem ASCT proceed with this treatment.

We undertook a retrospective analysis of high risk patients who underwent tandem or single ASCT between November 2017–November 2019. High risk was defined as: adverse risk cytogenetics: (t(4;14), t(14;16), t(14;20), del 17p)), plasma cell leukaemia, extramedullary disease, or suboptimal response to induction therapy (≤partial response). Patient characteristics and clinical outcomes were recorded. PFS was defined as time from D0 ASCT-1 to progression or death.

44 patients were identified, of which 33 were considered for tandem. Ultimately 17 patients (39%) underwent tandem and 27 (61%) single ASCT. Patient characteristics are included in Table 1. Reasons for not proceeding to tandem included toxicity from 1st ASCT ($n = 7$, 44%), performance status ($n = 6$, 35%), patient choice/psychological factors ($n = 4$, 25%), cytopenias ($n = 1$, 6%), inadequate harvest ($n = 1$, 6%), inability to store cells due to infection ($n = 1$, 6%), and disease progression ($n = 1$, 6%).

All patients received a proteasome inhibitor (PI) or immunomodulatory agent (IMiD) for induction, with PI+IMiD more common in the tandem vs single ASCT arm (88% vs 63%). Patients received a median of 2 lines (1–3) pre-ASCT in the tandem arm and 1 line (1–3) in the single ASCT arm. 11 tandem patients (65%) and 13 single ASCT patients (48%) required salvage pre-ASCT (Table 1). At time of proceeding to ASCT, overall response rate (ORR) was 94% for tandem patients (vs 81% for single ASCT), with 47% achieving ≥ very good partial response (VGPR) (vs 44% for single ASCT). Median time to engraftment was unaffected by ASCT (12 days for both ASCT-1 and ASCT-2 in tandem arm), and 12 days for the single ASCT arm. In the tandem arm, Melphalan dose was more often reduced in ASCT-2 compared to ASCT-1 (41% vs 12%).

ORR at 3 months was 88% for tandem (vs 81% for single ASCT), 76% achieving ≥ VGPR (vs 50% for single ASCT). 5 patients (29%) in the tandem group had improvement in response between ASCT-1 and ASCT-2.

Median follow up was 17 months for tandem and 13 months for single ASCT patients. 1 year PFS was 69.1% (95% CI: 35.9% to 87.5%) for tandem and 68.8% (95% CI: 46.9% to 83.2%) for the single ASCT arm. 1 year OS was 86.9% (95% CI: 56.6% to 96.6%) for tandem and 87.1% (95% CI: 64.8% to 95.7%) for single ASCT patients. 1 patient (6%) in the tandem arm and 7 patients (26%) in the single ASCT arm relapsed early (<12 months post ASCT). There were 2 peri-transplant deaths in the tandem arm attributable to infection and acute lung injury, and 1 in the single ASCT arm from stroke. PFS and OS data remain immature due to short follow-up.

Our data highlight the potential benefits and challenges of delivering tandem ASCT in the real world setting. ORR was favourable in the tandem arm, with deepened responses between 1st and 2nd ASCT in 29% of patients. However, 48% of patients considered for tandem did not proceed to 2nd ASCT. Better understanding of the risks, benefits and feasibility of tandem ASCT will help define its place in the real world management of this high risk group.

Table 1. Patient characteristics

	Tandem ASCT	Single ASCT
Number of patients	17	27
Sex	Male: 10 (59%) Female: 7 (41%)	Male: 11 (41%) Female: 16 (59%)
Median age	52 years (29–69)	60 years (31–74)
ISS Stage	4 (24%)	5 (19%)
Stage I	6 (35%)	12 (44%)
Stage II	6 (35%)	5 (19%)
Stage III	1 (6%)	5 (19%)
Unknown		
FISH	9 (53%)	18 (67%)
High risk	7 (41%)	8 (30%)
Standard risk	1 (6%)	1 (4%)
Unknown		
Extramedullary disease	2 (12%)	2 (7%)
Plasma cell leukaemia	3 (18%)	0 (0%)
Lines of therapy pre-ASCT:	6 (35%)	14 (52%)
One	6 (35%)	10 (37%)
Two	5 (29%)	3 (11%)
Three		
1st line Induction therapy:	2 (12%)	10 (37%)
PI alone	0 (0%)	0 (0%)
IMiD alone	15 (88%)	17 (63%)
PI+IMiD		
Salvage regimen 1	7 (44%)	9 (56%)
PACE-based	2 (13%)	0 (0%)
ESHAP-based	6 (44%)	7 (44%)
Other		
Melphalan dose	ASCT-1: 200 mg/m ² : $n = 15$ (88%) 140 mg/m ² : $n = 2$ (12%)	200 mg/m ² : $n = 19$ (70%) 140 mg/m ² : $n = 8$ (30%)
	ASCT-2: 200 mg/m ² : $n = 10$ (59%) 140 mg/m ² : $n = 7$ (41%)	
Median time from diagnosis to ASCT-1	9 months (7–25)	10 months (7–28)
Maintenance/consolidation	3 (18%)	4 (15%)
Response at 3 months post ASCT-1	4 (24%)	7 (26%)
CR	7 (41%)	6 (35%)
VGPR	6 (35%)	8 (30%)
PR	0 (0%)	4 (15%)
PD	0 (0%)	1 (4%)
Death	0 (0%)	1 (4%)
Unknown		

Table. (Continued)

	Tandem ASCT	Single ASCT
Response at 3 months post ASCT-2	6 (35%)	
CR	7 (41%)	
VGPR	2 (12%)	
PR	0 (0%)	
PD	2 (12%)	
Death		

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Pharmacovigilance analysis of peripheral neuropathy with newer generation proteasome inhibitors: Using the FDA Adverse Event Reporting System (FAERS)

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Peripheral neuropathy (PN), a common adverse effect of oncologic therapies, can significantly impact the quality of life in cancer patients. Proteasome inhibitors (PIs) are increasingly utilized in the treatment of multiple myeloma (MM). Carfilzomib (CFZ, approved in 2012), Ixazomib (IZB, approved in 2015) and Bortezomib (BTZ, approved in 2003) are three PIs currently approved for MM in the United States. PN is a predominant and well-studied adverse effect of BTZ and occurs in a significant number of patients. The newer-generation PIs, CFZ, and IZB, are believed to have less off-target side-effects such as PN. However, there is a relative paucity of data in the literature highlighting the risk of developing PN in these recently approved agents. Therefore, the aim of this study is to investigate the degree of association of PN occurring in CFZ and IZB.

This is a pharmacovigilance study utilizing the Food and Drug Administration's Adverse Events Reporting System (FAERS) in which adverse events are voluntarily reported by health care professionals or consumers. Specifically, from 2016-2019, FAERS was queried for "Peripheral neuropathy", "Sensory peripheral neuropathy", "Motor peripheral neuropathy" and "Sensorimotor peripheral neuropathy" with the drugs "Bortezomib", "Carfilzomib", and "Ixazomib".

Reported PN attributed to the newer-generation PIs, CFZ, and IZB, were preliminarily compared to the total number of cases of PN in FAERS. Signal disproportionality analysis was performed by calculating the reporting odds ratio (ROR) with a 95% confidence interval. ROR was reported to be significant if the 95% CI did not cross 1.0.

The total ROR (95% CI) for PN from 1/2016-9/2019 with all three PIs combined was 19.2 (18.2–20.2). Of the total cases of PN reported, 53% were males and 47% were females with a mean age of 65.5 (± 12.4) years. Individually, the ROR for CFZ, IZB, and BTZ compared to total FAERS was 5.6 (95% CI 4.77–6.59), 13.2 (95% CI: 11.9–14.6) and 31.4 (95% CI: 29.5–33.4), respectively.

Consistent with results illustrated in pivotal trials, our results suggest that newer-generation PIs have an estimated lower association of PN in comparison to BTZ. Nonetheless, patients receiving either CFZ or IZB remain at a considerable risk of developing PN and should be advised of this possibility given its potential implications on quality of life. However, the risk of PN should not dissuade clinicians from recommending these agents given the beneficial outcomes on survival for patients receiving these agents.

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Panobinostat in combination with bortezomib and dexamethasone for heavily pre-treated myeloma: a UK real-world multi-centre cohort

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Panobinostat (Pan), the first in-class histone deacetylase inhibitor is approved in combination with bortezomib (Bor) and dexamethasone (Dex) for relapsed/refractory multiple myeloma (RRMM) after ≥ 2 prior lines of therapy, including Bor and an IMiD, based on data from PANORAMA-1 trial. However, the trial excluded patients who are refractory to prior proteasome inhibitors (PIs). In clinical practice, PanBorDex is often introduced as 5th or later line of therapy

Table 1.

	Carfilzomib	Ixazomib	Bortezomib	Total PIs
Number of PN reported	151	365	1176	1,692
Total number of adverse events	8,217	8,526	11936	28,679
Mean age($\pm$ SD)	67.1 (± 9.3)	68.3 (± 8.9)	63.6 (± 14)	65.5 (± 12.4)
Males/Females	70/123 (57%)	182/343 (53%)	482/912 (53%)	734/1,378 (53%)
	53/123 (43%)	161/343 (47%)	430/912 (47%)	644/1,378 (47%)
ROR (vs FAERS)(95% CI)	5.6(4.77–6.59)	13.2(11.9–14.6)	31.4 (29.5–33.4)	19.2 (18.2–20.2)
ROR (vs other PIs)(95% CI)	0.24(0.21–0.29)	0.65(0.58–0.73)	3.2(2.9–3.6)	-

using a range of pre-planned attenuated dosing schedules to maintain therapy and improve tolerability. We conducted a UK multicentre retrospective study to evaluate efficacy and toxicity outcomes of PanBorDex in a heavily pre-treated, myeloma cohort treated in routine care.

Sixty two patients were eligible for inclusion, of whom 58.1% were male, 46% were ISS 3 staging and 32.3% had a high cytogenetic risk (HR), defined as one or more of the following: $t(4;14)$, $t(14;16)$, $del(17p)$, $t(14;20)$, 1q gain. Median age was 72 years (IQR 62–76 years). Median time from diagnosis was 5.6 years.

Median number of prior therapies was 4 (range 1–7); 98.4% of patients were PI-exposed, all patients received a prior IMiD. 46.8% of patients were PI-refractory, 82.3% were IMiD-refractory, and 43.5% were double refractory.

Patients received a median of 4 cycles (IQR 2–8). Eighteen patients (29%) completed ≥ 8 cycles. Twenty patients (32.3%) initiated PanBorDex at full dosing, and the remainder at a reduced dose or frequency or both of Pan and/or Bor. The overall response rate in the total cohort was 45%; VGPR, PR and PD rates were 14.5%, 30.6%, and 30.6%, respectively.

At a median follow up of 35.9 months, median PFS for the total cohort was 5.1 months (95% CI 2.0–8.1 months). Median OS was 9.5 months (95% CI 5.5–13.4 months). Median PFS according to depth of response was 8.2 months and 17.7 months for PR and $\geq$ VGPR, respectively. PFS was significantly shorter in double refractory compared to non-PI and non-IMiD-refractory patients (2.1 vs 11.4 months, $P = 0.001$). Patients who were either PI-refractory or IMiD-refractory only had a median PFS of 7.2 months. HR cytogenetics subgroup demonstrated a very short median PFS of 2.2 months.

Most commonly observed grade 3/4 adverse events (AEs) were thrombocytopenia (45.1%) and diarrhoea (17.7%). Other any grade AEs included infections (32%), fatigue (21%) and peripheral neuropathy (21%). 25 patients had a baseline ECGs then repeat ECGs during therapy; 2 patients had QTcF prolongations (>60 msec) from baseline, and both had pre-existing heart failure.

AEs led to discontinuation of either Pan or Bor or both in 11.3%, 1.6% and 17.7% of patients respectively. Dose reductions (at least one) of Bor or Pan or both were introduced in 8%, 16.1% and 22.6% of patients respectively. The mean relative dose intensity for Bor and Pan were 67.5% and 64.2%, respectively.

This is the largest reported real-world PanBorDex cohort. It is characterised by more heavily pre-treated and refractory myeloma patients compared to participants of PANORAMA-1 trial. Median PFS is consistent with 5.4 months reported in PANORAMA-2 study investigating this combination in Bor-refractory patients. These results are consistent with outcomes observed with pomalidomide and dexamethasone, and with daratumumab monotherapy when used in similar heavily pre-treated cohort. Lower rates of grade 3/4 AEs in this cohort can be explained by dose attenuation compared to the trial cohort.

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Impact of change of mobilisation method on adequacy of stem cell collection in multiple myeloma patients – single centre experience

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Background: Despite the emergence of many new treatments, high dose Melphalan and autologous stem cell transplantation (ASCT) remains the standard of care for younger patients with Multiple Myeloma (MM). Effective ASCT is dependent on an adequate collection of haematopoietic stem cells (HSC) from the patient, usually by peripheral stem cell harvest. Critical to this process is effective mobilisation of HSC from the bone marrow into peripheral blood. In our centre, a decision was made to move from a preference for mobilisation with cyclophosphamide and G-CSF, to G-CSF alone in most cases. This study evaluates the impact of this change on overall rates of stem cell collection adequacy.

Methods: In broad terms, the two main mobilisation strategies are the use growth factors alone (Granulocyte-colony stimulating factors, (G-CSF), or of growth factors in combination with chemotherapy (Chemo-mobilisation). In addition, chemokine receptor antagonists (Plerixafor) can be used to improve the efficacy of mobilisation by both methods. The methods are known to have differing advantages and disadvantages. Chemo-mobilisation may increase adequacy compared to G-CSF alone, but is known to increase toxicity and requires chemotherapy administration. In these times of limited inpatient bed resources, in 2017, decision was made to increase the number of patients receiving chemo-free mobilisation in order to reduce the need for patient stays in our haematology department for chemo-administration and reduce chemotherapy induced toxicity in our patients. The impact of this policy change on adequacy of stem cell harvest was then evaluated for the subsequent 2 years. An adequate collection is considered to be a harvest of a minimum of 2×10^6 CD34+ cells/kg for a single ASCT and a least 4×10^6 CD34+ cells/kg for two ASCTs. Mobilization failure is usually defined as the inability to procure 2×10^6 CD34+ cells/kg despite 3 apheresis sessions

Table.

Results of collection	2017 G-CSF (n = 5)	2017 G-CYCLO (n = 43)	2018 G-CSF (n = 35)	2018 G-CYCLO (n = 14)	2019 G-CSF (n = 41)	2019 G-CYCLO (n = 3)
FAILED TO ACHIEVE TARGET FOR 1 ASCT	0	0	2 (6%)	1 (8%)	1(3%)	1(33%)
TARGET CD34 ACHIEVED FOR 1 ASCT	0	5 (12%)	9 (26%)	3 (21%)	12(29%)	0
TARGET CD34 ACHIEVED FOR 2 ASCT	5 (100%)	38 (88%)	24 (68%)	10 (71%)	28(68%)	2(67%)

Results: During the 2017–2019 period a total of 141 Myeloma patients underwent HSC collection (2017, $n = 48$, 2018, $n = 49$, 2019 $n = 44$).

The change in centre mobilisation policy led to marked differences in the rates of chemo-free vs chemo-mobilisation (2017 = 10.4% vs 89.6%; 2018 = 71.4% vs 28.6%; 2019–93.2% vs 6.8%).

Rates of adequacy of collection were not significantly different when comparing the two mobilisation regimes, chemo-free mobilisation vs chemo-mobilisation (2017–100% vs 88%, 2018–94% vs 92%, 2019–97% vs 67%).

Conclusions: In conclusion, in our single centre experience, a switch to a preference for chemo-free mobilisation can be undertaken without impact on adequacy of stem cell collection. This could lead to savings in the need for day unit capacity and reduce inconvenience for patients. Further evaluation of the impact on inpatient admissions with sepsis following chemo-mobilisation versus chemo-free should be undertaken.

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Feasibility and safety of 90-min rapid rate daratumumab in myeloma: a prospective multicentre observational study

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Daratumumab is the first in class anti-CD38 monoclonal antibody, used either as monotherapy or in combination with lenalidomide and dexamethasone (DRd) or bortezomib and dexamethasone (DvD) to treat relapsed/refractory multiple myeloma (RRMM).

Daratumumab is generally well tolerated, but infusion-related reactions (IRRs) were reported in about half of clinical trial patients. The majority of IRRs occur during the first infusion. As such, the need to incrementally increase the infusion rate resulting in long administration times of 6.5, 4.5 and 3.5 h for the first, second and subsequent infusions, respectively. Incidence of IRRs is much lower after the 1st infusion; safety of an accelerated 90-min infusion from the third infusion has previously been reported in a small single centre study.

We conducted a prospective multi-centre observational study to evaluate safety of rapid rate (RR) 90-min daratumumab infusion. Patients must have received at least the first two infusions at the manufacturer's recommended standard rate (SR) and tolerated their last infusion at the SR without any $\geq$ grade 1 IRRs. Previous IRRs with SR infusion were not an exclusion criterion. IRRs were graded according to CTCAE v4.03.

Sixty nine patients were eligible for inclusion. Median age was 69 years (range 43–91 years), M/F 42/27. Pre-existing cardiac and respiratory co-morbidities were present in 30 (43.5%) and 9 (13%) patients, respectively.

Forty patients (58%) received RR daratumumab monotherapy, 27 (39.1%) as part of DvD and 2 patients as part of DRd. Patients received a median of 4 SR infusions (range 2–51) before starting RR. 23 patients (33.3%) had experienced a previous IRR with SR infusion; which were all of G1/2 and occurred with the first infusion.

During the evaluation period, a total of 383 RR infusions were observed. Patients received a median of 6 RR infusions (range 1–15). 35 patients on daratumumab monotherapy received pre-medication with dexamethasone at 12–20 mg on day 1 and post-infusion dexamethasone at 4 mg on days 2 and 3. Three patients received prednisolone 75 mg pre-infusion and 25 mg for 2 days post-infusion. All 29 DvD or DRd patients received pre- and post-daratumumab steroids, which were also part of the triplet combination. 53 patients (76.8%) received pre-medication with montelukast 10 mg. All patients received pre-medication with paracetamol and chlorphenamine.

Of all patients receiving RR infusions, 4 patients (5.7%) experienced a grade 1–2 IRR. One patient experienced drop in oxygen saturation with the 1st RR infusion. The second patient tolerated the 1st and 2nd RR infusions well, but experienced a temperature spike with the 3rd RR infusion. The third patient developed a mild rash with the 1st RR infusion. The fourth patient tolerated the 1st RR infusion well but experienced dyspnoea, vomiting and headache with the 2nd infusion. Standard of care measures for the management of IRRs including slowing the infusion rate were implemented. All patients completed the infusion and none required admission. Subsequent RR infusions were well tolerated with no re-occurrence of IRRs.

This is the largest prospective dataset reporting the safety of rapid rate daratumumab. Our data shows that a 90-min infusion is well tolerated, with very low incidence and grades of IRRs. We demonstrated that daratumumab can be safely administered at this rate from the third dose onwards either as monotherapy or in combination regimes.

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A single centre review of bortezomib home-administration in myeloma patients: portsmouth experience

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The number of myeloma patients is increasing. In our unit we had 75 newly diagnosed patients during 2018/19 compared to 51 in 2014/15. Many receive Bortezomib, which forms the backbone of 4 NICE approved regimens. Bortezomib is given once or twice-weekly as a subcutaneous injection. This places considerable burden on Haematology Day Unit (HODU) resources and adversely affects patient experience with frequent hospital visits.

In 2017 our centre instigated a Homecare protocol to teach patients to self-administer Bortezomib with training by designated ward or HODU nurses or Clinical Nurse Specialists (CNS). Aims were to improve patient quality of life by reducing hospital attendances and increase HODU capacity. Allocated slots for Bortezomib are 30 min on HODU.

Since its introduction in 2017, the majority of patients have been treated on the Homecare protocol (24 in 2018, 47 in 2019). To assess the effectiveness of this programme, we conducted a retrospective review of patients receiving Bortezomib between January and October 2019 to establish the:

- 1) Proportion able to commence the Homecare protocol
- 2) Proportion successful in completing treatment on the Homecare protocol
- 3) Amount of additional CNS time required to support the Homecare protocol
- 4) Number of associated adverse incidents

Data was obtained from pharmacy records and CNS log of patient visits and phone-calls.

Of 61 patients treated with Bortezomib, 52 were included in this study (exclusions: 2 inpatient treatment only, 3 transferred elsewhere, 3 early cessation (2 for toxicity, 1 for rapid progression). Average age was 74 years (range 51 to 90 years). 29/52 received Bortezomib as first line treatment, 19/52 second line and 4/52 subsequent lines. Average treatment duration was 5.3 cycles (range 2–9).

1) 45 patients (86.5%) proceeded with the Homecare protocol. 4 were unable to self-administer. The remaining 4 were given Bortezomib as part of a DVD protocol on HODU (3 later switched to Homecare). 25 patients self-administered, 17 had a relative administer (3 unknown).

2) 43 patients (82.6%) were commenced on the Homecare protocol. 2 patients reverted to HODU/CNS administration due to difficulty managing.

3) 38 patients were CNS trained (with 2 assessed as unsuitable for Homecare). The remaining patients were trained by ward/HODU nurses or did not require training (8 were able to self-administer from training). Of CNS trained patients, the average initial consultation was 43 min (32% <30 min, 42% 31–45 min, 21% 46–60 min, 5% >60 min). Mean time for additional CNS input was 55 min (range 10–190 min). 7 patients required re-training. 13 had additional consultations for other issues: 2 required CNS administration initially, 2 to request HODU administration and 9 for extra support. Queries requiring extra support included:

- Injection site queries
- Further reassurance during administration
- Syringe query
- Administrative query
- Spillages

4) 6 patients sustained adverse events; 3 reported spillages (with no harm caused). 3 had injection site incidents (2 – rash, 1 – pain at injection site)

The majority of patients can safely proceed with the Homecare protocol, significantly increasing HODU capacity. Sufficient CNS resources are required for training and support. Patient experience is likely to be improved by reducing hospital attendances. We plan to explore this further with a patient experience survey.

Disclosure of Interest: None Declared

BSH2020-PO-121

End of life management of patients with myeloma at University Hospitals of Birmingham

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Patients with haematological cancers have low rates of hospice enrolment and higher rates of aggressive medical care at the end of life (EoL) in comparison to patients with solid organ malignancies, as identified by Thi *et al* (2011) and O'Connor *et al* (2014). They also identified the rising use of Hospice services but concomitant rises in 'late' enrolment (≤ 3 days before death). In a study by Odejide *et al* (2018), it was recognised that symptom burden increased as disease progressed and care should be focussed on quality of EoL care.

In this audit we aimed to assess how aggressive EoL management of patients with myeloma at University Hospitals Birmingham (UHB) was and what EoL process was in place.

Our study was a retrospective analysis of patients with myeloma who died between April 2018 and March 2019. We collected data from electronic, MDT and data management records. Indicators of aggressive EoL care used were those established by Earle *et al* (2004); Chemotherapy use within 14 days of death, ≥ 2 ED visits within 30 days of death, ≥ 2 hospitalisations within 30 days of death, hospital stay > 14 days and within 30 days of death, at least 1 ICU admission within 30 days of death and death in a hospital. Patients with myeloma or plasmacytoma were included irrespective of age or date of diagnosis. We excluded patients with amyloidosis (2) or incomplete data (2).

There were 30 deaths within our cohort, 21 males and 9 females (mean age = 75.2 years). 93% of patients had at least 1 indicator of aggressive EoL care, most frequent was death in a hospital ($n = 25$) and least frequent was ICU admission ($n = 2$). 25% of patients died within 3 years of diagnosis and duration of illness was between 0 – 4 years. 18 patients died from myeloma and treatment related complications (myeloma $n = 10$, bronchopneumonia $n = 5$, neutropenic sepsis $n = 1$, malignant encephalopathy $n = 1$, multi-organ failure $n = 1$, pulmonary embolism $n = 1$, plasma cell leukaemia $n = 1$). 6 patients died from non-myeloma related causes (myocardial infarction $n = 3$, intracranial haemorrhage $n = 1$, pulmonary oedema $n = 1$, cancer of unknown primary $n = 1$). The cause of death for 4 patients was unknown.

25 patients died in hospital, 3 died at home and 2 in a hospice. 12 patients were on treatment within 14 days of death and 21 patients were on treatment within 30 days of death. For patients who had ≥ 3 lines of treatment, more aggressive EoL care was used, with some patients having 2–3 aggressors on average. No correlation was found between time of diagnosis to death vs aggressiveness of EoL care. 18 patients had a RESPECT form (EoL decision making and documentation of resuscitation status tool), 4 of which were completed on the day of death. For 12 patients, there was no documentation of RESPECT forms.

Recommendations from this audit included timely discussions with patients regarding EoL and the processes involved, potentially from the time of progression beyond the 3rd line of treatment. Discussion of EoL care depended on clinicians recognising EoL accurately. An exploration of clinician reticence to discuss and initiate EoL planning would be important to explore. At UHB, a clinical nurse specialist (CNS) was introduced in July 2019. Re-auditing patients to establish whether this has changed practice would be interesting. Preferred place of death was not included in the remit of this audit and would be a suitable addition for a further re-audit.

Disclosure of Interest: None Declared

BSH2020-PO-122

Real-world outcomes of daratumumab monotherapy in relapsed/refractory myeloma

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Daratumumab is the first anti-CD38 monoclonal antibody approved for use in multiple myeloma (MM). The initial license was for use as monotherapy in patients with relapsed refractory multiple myeloma (RRMM) who have previously received a proteasome inhibitor (PI) and an immunomodulatory agent (IMiD) and who are refractory to their last therapy. The initial approval was based on two single-arm phase 1–2 studies including relatively limited number of patients. We conducted a retrospective multi-centre study to present additional efficacy and safety data on daratumumab monotherapy when used in routine care.

Fifty-five patients, treated with daratumumab monotherapy between January 2016 and April 2019, were eligible for inclusion. The median age was 72 years (IQR; 64–77), median time from diagnosis was 4.9 years (IQR; 2.8–7.2). 65% were male, 34.5% were ISS 3 and 20% had known high-risk cytogenetics, defined as one or more of the following; t(4;14), t(14;16), del(17p), t(14;20), or 1q-gain+1p(del). Myeloma isotype was IgG (43.6%), IgA (25.5%), and light chain only (25.5%), 6 patients had AL amyloid.

Median number of prior therapies was 3 (range 1–6). Fifty-three patients (96%) received ≥ 3 prior therapies; 98.2% and 96.3% of patients had prior PI and IMiD exposure, respectively; 54.5% were PI-refractory, 76.4% were lenalidomide-refractory and 47.2% were double refractory.

Patients received a median of 14 infusions (IQR 9–17). Median duration of treatment was 5.7 months (95% CI 3.8–7.5). Overall response rate for the total cohort was 50%; 11% and 38.8% achieved a VGPR and PR respectively. Responses also include 2 patients who had negative immunofixation suggesting complete serological response to therapy.

At a median follow up of 9.2 months, median PFS for the total cohort was 5.1 months (95% CI 3.4–6.7). Patients who achieved $\geq$ PR has significantly longer PFS compared to those with $\leq$ MR (9.8 months vs 2.7 months respectively, $p < 0.001$). PFS was significantly shorter in patients with high-risk compared to standard-risk cytogenetics (2.3 vs 6.7 months, $P = 0.001$). Double refractory patients had a median PFS of 2.7 months (95% CI 1.9–3.4). 24 patients received a subsequent therapy; median time to next treatment was 7.7 months. Median OS for the total cohort was 15.9 months (95% CI 12–19.7).

Ten patients (18%) experienced an infusion related reactions (IRR) with the first infusion. The majority of patients had a grade 1–2 IRR, only one patient had a grade 3 IRR. 20 patients (36%) experienced an adverse event (AE) leading to hospital admission. Most commonly, admissions were due to infective episodes: pneumonia or URTI/LRTI (15 patients), sepsis (2 patients), E.coli bacteraemia/infective endocarditis (1 patient), and CMV reactivation (1 patient).

This is the largest reported real-world cohort of RRMM patients receiving daratumumab monotherapy. Response rates and PFS in this cohort compare favourably to the 31% ORR and 4 months median PFS reported in a pooled analysis GEN501 and SIRIUS phase 1–2 trials, although the latter had more heavily pre-treated and double

refractory patients. Patients with high-risk disease and those who were double refractory fared poorly and remain challenging to treat. Daratumumab monotherapy offers a well-tolerated, efficacious treatment option for certain patient groups in RRMM, but its appreciated that monotherapy use will decline as the antibody is used in combinations in less refractory groups at earlier relapses.

Disclosure of Interest: N. Maouche Conflict with: honorarium, consultancy fees, speaker fees, conference attendance, grant from Takeda UK; speaker fees from Celgene and Novartis; conference fees from Janssen and Abbvie., H. Leary: None Declared, F. Collings Conflict with: Speaker fees: Takeda, Celgene, B. Tseu: None Declared, F. Djebbari: None Declared, G. Vallance Conflict with: Jazz Pharmaceuticals, K. Ramasamy Conflict with: Janssen, Conflict with: Janssen - Research grant, speaker fees, honoraria, J. Kothari: None Declared

BSH2020-PO-123

Prevention and management of steroid-induced hyperglycaemia in outpatient myeloma chemotherapy - a tertiary centre audit and quality improvement project

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Long term high-dose corticosteroids (steroids) are commonly used in myeloma chemotherapy regimes delivered in an outpatient setting. Steroids cause hyperglycaemia and can result in steroid-induced diabetes mellitus (SIDM). Myeloma is predominantly a disease of older adults and old age is a risk factor for SIDM. Unfortunately blood sugar monitoring is often overlooked in haematological care, despite frequent follow-up. A recent mortality in our Trust secondary to undiagnosed SIDM highlighted the need for an in-depth review of local monitoring and management practice.

We audited myeloma patients starting their first cycle of first or subsequent lines of chemotherapy over 8 months at the Bristol Haematology and Oncology Centre. All patients received high-dose pulsed steroids (mean doses: 148 mg dexamethasone or 360 mg prednisolone per month). The incidence of pre-existing diabetes and SIDM risk factors in these patients was measured. We reviewed chemotherapy pre-assessment clinic (PAC) notes, letters and consent forms to record how frequently patients were formally consented for SIDM, patients were involved in a documented discussion on steroid side effects, capillary blood glucose (CBG) and HbA1c were measured according to guidelines, SIDM was diagnosed and managed according to guidelines and primary care services were informed of SIDM risk and monitoring.

All audit targets were based on national (Joint British Diabetes Societies, NICE) and local Trust guidelines for steroid-induced hyperglycaemia. 100% of patients were expected to meet audit targets.

7.6% of patients had pre-existing diabetes or prediabetes. 100% possessed at least 1 risk factor for SIDM and 49.9% possessed 2 or more.

PAC notes analysis showed low levels of compliance with guidelines and poor clinician awareness of SIDM. No patients had CBGs tested at each PAC or had appropriate CBG monitoring arranged. HbA1c was measured prior to steroid therapy in less than 40% of patients and rechecked appropriately in none. Only 31% and 15% had diabetic state and SIDM risk factors identified and documented in notes respectively. Only 35% of patients were formally consented for SIDM and 38% had a documented discussion on steroid side effects. There were no formal diagnoses of SIDM. Liaison between

haematology services and primary care took place in solely 4% of patients.

In response to our findings, we collaborated with the local myeloma and diabetes teams to implement the following changes: education for outpatient clinical staff on SIDM risk factors, monitoring and management; a steroid-specific side effect section on the PAC proforma; serum HbA1c and glucose added to myeloma clinic blood test profile; plan for CBG to be checked and recorded at each PAC by nursing staff; and clinicians to highlight risk of SIDM to primary care physicians on clinic letters. We are currently working to disperse CBG monitors to all patients for community monitoring. We aim to re-audit the above standards after 6 months of the changes being implemented.

Monitoring of blood sugars in haematological patients receiving high-dose steroids can be easily overlooked due to time constraints and limited experience in a non-haematological complication. Unrecognised hyperglycaemia is however a potentially life-threatening complication in the high-risk myeloma population. Simple measures with clear communication between care providers can significantly reduce under-diagnosis and improve management of SIDM.

Disclosure of Interest: None Declared

BSH2020-PO-124

INSIGHT MM study of routine clinical practice in multiple myeloma: analysis of duration of therapy and treatment sequencing in UK patients

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United Hospitals Bath NHS Foundation Trust, Bath, ⁸Nottingham University Hospitals NHS Trust, Nottingham, ⁹Takeda UK Limited, High Wycombe, UK, ¹⁰Millennium Pharmaceuticals, Inc., A Wholly Owned Subsidiary Of Takeda Pharmaceutical Company Limited, Cambridge, MA, USA

Treatment pathways for patients with newly diagnosed multiple myeloma (NDMM) and relapsed/refractory multiple myeloma (RRMM) vary worldwide and currently there is no global standard of care. Clinical trial data inevitably inform treatment approval and access; however, real-world data are becoming increasingly important to fully define treatment effectiveness in the clinical setting.

INSIGHT MM (NCT02761187) is the largest global, prospective, non-interventional, observational study in multiple myeloma to date, following 4,321 patients. Adults with NDMM or RRMM (1–3 prior therapies) from 15 countries in Europe, the USA, Latin America and Asia were enrolled and continue to be followed prospectively for ≥5 years. Data were collected at baseline and continue to be collected every 3 months. The UK enrolled 474 patients (11%) into this global study. Here, we present a UK-specific INSIGHT MM analysis for 373 patients and examine treatment selection, duration of therapy and reasons for discontinuation.

At data cut-off; (median follow up, 12.3 months), 126 patients with NDMM and 247 with RRMM were enrolled, including 161/48 out of the total patient population who had reached second-/third-line therapy respectively (patients could receive more than one line). Additional patient and disease characteristics are shown in the Table. The most frequently recorded specified regimens per line of treatment were: first-line treatment with bortezomib–thalidomide–dexamethasone (VTD), bortezomib–cyclophosphamide–dexamethasone (VCD) or bortezomib–melphalan–prednisone (VMP); second-line treatment with CTD or VCD; and third-line treatment with lenalidomide–dexamethasone (Rd) or ixazomib–Rd (IRd). Analysis of the reported myeloma therapies showed that 248 permanent discontinuations of prescribed therapy occurred among 107 patient reports across first, second and third lines of therapy (multiple discontinuations could be recorded for individual patients in more than one line). Reasons for discontinuation varied with line of treatment. Overall, end of planned therapy was the most common reason for

Table 1. Patient and disease characteristics

Characteristic	NDMM* n = 126	RRMM* n = 247	Total n = 373
Male/female, %	53/47	67/33	62/38
Median age, years (range)	67 (38–89)	68 (39–87)	67 (38–89)
Median time from initial diagnosis to study entry, months (range)	2 (0–81)	53 (7–378) [n = 241]	38 (0–378) [n = 367]
Creatinine clearance (<30 mL/min), %	11	5	7
Elevated (>11 mg/dL) calcium, %	8	7	7
Bone lesions present at diagnosis, %	52	32	39
ISS disease stage at initial diagnosis, %			
I II III	292521 [†]	121814 [‡]	182116 [§]
History of peripheral neuropathy, %	14	44	34
Type of treatment facility, %			
Academic/University	71	82	78
Community	29	18	22

ISS, International Staging System; NDMM, newly diagnosed multiple myeloma; RRMM, relapsed/refractory multiple myeloma.

*Data shown are as at study entry unless otherwise stated

[†]25% unavailable/missing data;

[‡]56% unavailable/missing data;

[§]45% unavailable/missing data.

treatment discontinuation and was most frequently cited at second line compared with other lines of treatment. Discontinuations due to remission decreased with successive treatment lines, whereas instances of discontinuation due to relapse increased. Adverse events (related to therapy) were cited as a reason for discontinuation at a similar frequency across all treatment lines.

Of those patients who received a proteasome inhibitor-based regimen as first-, second- or third-line therapy, 23%, 24% and 6%, respectively, had another proteasome inhibitor-based regimen as their subsequent therapy, whereas 34%, 55% and 21%, respectively, followed with a regimen containing an immunomodulatory drug.

In this UK-specific analysis from the global study, there is clear variability in treatment selection and sequencing, and reasons for treatment discontinuation change as patients move through lines of therapy. INSIGHT MM provides a valuable opportunity to understand patterns of multiple myeloma care in day-to-day clinical practice.

Disclosure of Interest: G. Cook Conflict with: Janssen, Takeda, Amgen, Celgene, Karyopharm, Sanofi, Conflict with: Takeda, Janssen, Celgene, Conflict with: Speaker Bureau: Janssen, Takeda, Amgen, Celgene, Sanofi, J. Ashcroft Conflict with: Amgen, Celgene, Janssen, Conflict with: Treasurer, BSH; UK Chair, Myeloma Forum, K. Boyd Conflict with: Janssen, Takeda, Celgene, Novartis, Conflict with: Speaker fees: Janssen, Amgen, Takeda, Celgene, Novartis. Educational Meeting Attendance Support: Janssen, Celgene, Takeda, M. Garg Conflict with: Advisory Boards: Sanofi, Takeda, Novartis, Janssen., Conflict with: Janssen, Conflict with: Travel Support: Takeda, Janssen. Honoraria: Janssen, Takeda, R. Hall: None Declared, B. Kishore Conflict with: Travel And Conference Grants: Celgene, S. Moore Conflict with: Advisory Boards: Takeda, Celgene, Chugai, Janssen, Conflict with: Speaker At Educational Meetings: Riche, Janssen, D. Smith Conflict with: Celgene, Takeda, Amgen, AbbVie, Conflict with: Amgen, Conflict with: MTA: Takeda. Conference Attendance Expenses: Takeda, Amgen., N. Crofts Conflict with: Takeda: Employment, K. Ren Conflict with: Takeda: Employment, G. Pratt Conflict with: Advisory Boards/Honoraria: Amgen, Binding Site, Celgene, Gilead, Janssen, Takeda

BSH2020-PO-125

Efficacy of daratumumab combination regimen in patients with multiple myeloma: a combined analysis of six phase III randomised controlled trials

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Multiple myeloma (MM) is a hematological malignancy of abnormal clonal plasma cell proliferation which accounts for 1% of all cancers. Over the last few years, introduction of novel agents has become a

therapeutic landmark in the management of newly diagnosed MM (NDMM) and relapsed or refractory MM (RRMM), in both transplant – eligible and transplant – ineligible patients. Studies have shown that daratumumab, a human IgG kappa monoclonal antibody targeting CD38 on myeloma cells, monotherapy or in combination with proteasome inhibitors, immunomodulatory agents and/ or other antimyeloma therapies increased survival in the treatment of MM. We undertook an updated meta-analysis of phase III randomised controlled trials (RCT) to determine the efficacy of daratumumab combination regimen in patients with NDMM and RRMM.

We systematically conducted a comprehensive literature search using MEDLINE, EMBASE databases and meeting abstracts from inception through 10th December 2019. Phase III RCTs utilising daratumumab in patients with NDMM and RRMM were incorporated in the analysis. A generic inverse variance method was used to calculate the estimated pooled hazard ratio (HR) for progression-free survival (PFS) with 95% confidence interval (CI). Mantel-Haenszel (MH) method was used to calculate the estimated pooled risk ratio (RR) for overall response rate (ORR), and complete response (CR) and stringent complete response (sCR). Heterogeneity was assessed with I² and Cochran's Q -statistic. A 'p value' of less than 0.05 was considered significant and I² >50% is considered substantially heterogeneous.

Six phase III RCTs (POLLUX, CASTOR, CANDOR, ALCYONE, CASSIOPEIA and MAIA studies) with a total of 4,025 patients with NDMM (*n* = 2528) and RRMM (*n* = 1497) were included. Studies compared daratumumab (D) + bortezomib (V) + melphalan (M) + prednisolone (P) vs VMP, D + lenalidomide (R) + dexamethasone (d) vs Rd, Dvd + thalidomide (T) vs Vtd, DVD vs Vd and Carfilzomib (K) + d + D vs Kd. Daratumumab was utilised in relapsed and refractory multiple myeloma in the POLLUX, CASTOR and CANDOR studies, and as first-line treatment for patients with multiple myeloma in the ALCYONE, CASSIOPEIA, and MAIA studies. The randomisation ratio was 2:1 in the CANDOR study and 1:1 in other studies. The I² statistic showed some heterogeneity among RCTs and random effects model was applied. The pooled HR for overall PFS was statistically significant at 0.47 (95% CI: 0.38–0.57; *P* <0.00001). The pooled HR for PFS was calculated for each subset; NDMM (HR 0.52, 95% CI: 0.44–0.61; *P* <0.00001) and RRMM (HR 0.43, 95% CI: 0.29–0.63; *P* <0.0001). The PFS benefit was observed in both standard risk cytogenetic and high risk cytogenetic cohorts; standard risk cytogenetic cohort (HR 0.38, 95% CI: 0.30–0.48; *P* <0.00001) and high risk cytogenetic cohort (HR 0.63, 95% CI: 0.48–0.82; *P* = 0.0005). ORR was 90.16 % in daratumumab arm vs 78.92% in control arm and the pooled RR was significant at 1.17 (95% CI: 1.08–1.27; *P* = 0.0002). The rate of CR and sCR was 18.38 % higher in daratumumab combination regimen compared to control group. The pooled RR was statistically significant at 2.03 (95% CI: 1.66–2.47; *P* <0.00001).

Our meta-analysis showed that daratumumab combination regimens significantly improved PFS, ORR and CR and sCR compared to control arm in patients with NDMM and RRMM. The improvement in PFS was observed in both high and standard risk cytogenetic cohorts.

Disclosure of Interest: None Declared

Nursing

BSH2020-PO-126

A narrative inquiry into patients' experience of receiving blood transfusions in a day unit setting whilst in a palliative stage of a haematological malignancy

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Most patients in a palliative stage of a haematological malignancy will become severely anaemic and be given a blood transfusion every 2–3 weeks, generally, in an acute hospital day unit setting. A Cochrane review suggests blood transfusions (BT) increase mortality in patients with an advanced cancer and highlight ethical issues around cost, scarcity and futility. Furthermore, two recent systematic reviews suggest limited evidence to support BT administration in patients with advanced cancer. Limited qualitative research has been undertaken that addresses palliative haematology patients' perspective of BT. None were found which addressed haematology patients' perspective of receiving BT in a hospital day unit. The overriding purpose of this Professional Doctorate in Nursing study was to gain a deep insight into the everyday life experiences of this patient population, who were receiving BT in a day unit setting.

As a senior nurse committed to providing compassion and excellence in nursing care and reducing patients' suffering, research objectives set were: to explore participants expectations and challenges; to explore the healthcare professional role on participants experience; to explore the effect of place; and explore change in participants experience with disease progression.

A narrative inquiry approach, underpinned by a pragmatic philosophical foundation, was employed. Twenty-two longitudinal, unstructured interviews with a purposive sample of eight participants allowed exploration of rich narratives of everyday-life blood transfusion experiences. During data collection and narrative analysis of transcripts, experiences were explored through three dimensions: sociality, place and temporality. Analysis produced eight narrative accounts. Meaning, implicit within and across these accounts, was revealed through resonant plotlines. Plotlines of special interest were those which illuminated feelings, hardships and concerns. Five overarching narrative storylines were developed from plotlines. Storyline findings were supported by Found Poetry.

Storyline One: Everyday blood transfusion life hangs in the bag; Two: Everyday blood transfusion life hangs in the hands of others; Three: Everyday blood transfusion life involves waiting and anxiety; Four: More individualised care associated with the hospice day unit; Five: An everyday SAD life during the late to terminal palliative phase of a haematological malignancy, where Somnolence overwhelms daily life; Adversities overpower everyday life and participants demonstrate Desperation to hang onto life.

Discussion and implications for clinical practice were addressed through a palliative lens focussed on the six C's of nursing: compassion, care, communication, confidence, competence and courage. Essentially, to provide compassionate blood transfusion care healthcare professionals should: communicate realistic expectations when explaining benefits of BT; consider early involvement of the palliative care team to aid with difficulties around goal setting; reconsider the setting where BT are administered; significantly reduce waiting in clinical areas; raise awareness of hardships associated with chelating

agents; and develop professional competence, courage and commitment to achieve individualised patient-centred care.

Disclosure of Interest: None Declared

BSH2020-PO-127

Improving haematology clinics: Chesterfield Royal Hospital experience

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Significant issues with the Haematology clinics had been identified by staff and patient feedback over several years up to 2018. Specifically, patients frequently experienced unreasonable waiting times for ODBT (on the day blood tests), doctor- or nurse-led clinic ('clinic wait') and the Haem-Oncology Pharmacist (where required). Inadequate clinic capacity had led to overbooking and all staff groups working in excess of their sessions. A major piece of service improvement (SI) project was therefore initiated in February 2018 and the improvements to date are presented here.

At the launch of the project a SI lead was appointed by the Trust to kick off baseline assessment and brainstorming meetings; colleagues from all points in the patient pathway were included (reception, phlebotomy, clinical and administration). From October 2018 onward, all SI work became team lead.

Process mapping identified that our waiting times within clinic were chaotic. Analyses confirmed this was primarily due to lack of a booking facility for ODBT, with patients arriving anywhere from 6 h prior to 30 min after their actual clinical appointment (instead of the preferred timeframe of 1 h prior). The median wait for ODBT of 45 min in turn led to significant wasted time with medical and nursing staff spending at least 30 min per staff member per clinic checking for blood results. Laboratory turnaround time was found to be within expectations. Following a number of trials using the Plan Do Study Act approach, a Medway ODBT booking system was devised and fully implemented by the end of March 2019. Re-audit in July 2019 indicated that the median wait for ODBT was 0 min (range -35 to +37 min).

Demand-capacity analyses resulted in an expansion of numbers for both medical and specialist nursing staff. A programme of training and competency assessments introduced in 2018 allowed expansion of our specialist nurse clinics (approximately 250% increase in patient numbers) and a slight reduction in medical staff clinics. All staff now work within their job plans. A further stream of work focused on reducing variations, which included appropriate and timely discharge, reduction in blood test requests, rational selection of patients for Pharmacist consultation. We established a process to deal with clinic bank holiday capacity issues and implemented partial booking for patients not on treatment. Overbookings have virtually resolved.

Re-audit in July 2019 showed that the median clinic wait had improved from 65 min (ranging from -65 to 175 min) to 47 min (ranging from -30 to 130 mins). A concurrent patient survey showed 83% were 'very happy' or 'quite happy' with their clinic wait. On the same re-audit, waiting time for Pharmacist consultation was a

median of 15 min (range 0 to 32 min). The recent local results from the cancer patient survey (April to June 2019) confirmed this with 76% of patients saying the overall waiting time in clinic was 'very short' or 'about right'.

Our reflections on this SI work are:

- There is a need for Trust level engagement where capacity is a major contributor to poor clinic experience for patients and staff
- Detailed process mapping enables identification of key targets for change
- SI must be driven by the staff involved, as they are familiar with the issues and already have many of the solutions

Disclosure of Interest: None Declared

BSH2020-PO-128

Abstract Withdrawn.

BSH2020-PO-129

Abstract Withdrawn.

BSH2020-PO-130

Optimising services by offering care closer to home: the introduction of telephone consultations for patients on treatment for liver Graft vs Host Disease

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The University Hospital of Wales in Cardiff is a level 4 treatment centre performing haematopoietic stem cell transplantation, for patients living in Mid, West and South Wales. On average, at the University Hospital of Wales we perform 160 stem cell transplants each year to treat predominantly haematological malignancies.

Graft vs Host Disease (GvHD) is the most common complication post-transplantation and due to prolonged treatment with immunosuppressive therapies requires frequent assessment of the response to immunosuppression, in order to guide weaning.

Previously patients with liver GvHD who had no other complications, were travelling to Cardiff from as far as Aberystwyth to be reviewed and guided on dose adjustments, this proved challenging and tiring for patients. The decisions surrounding treatment was based on blood results. As an Advanced Nurse Practitioner taking over reviewing these patients, I thought we should consider getting bloods taken locally and conducting telephone consultations.

The aim of this was to improve patient experience and reduce their travel costs, reduce usage/cost of hospital transport. In addition to reducing through flow in day unit which is at capacity and reduce risk of acquiring respiratory viruses while in hospital.

Potentially eligible patients were identified and approached. Contact made with GP's/local hospitals to take bloods and forms given. Patients booked into virtual clinic each week for telephone consultation with an Advanced Nurse Practitioner.

Evaluated via patient feedback which has stated patients feel able to return to a more 'normal' lifestyle.

There have been no adverse events noted due to decreased visits. Engagement with local services has improved, if any additional

problems noted on consultation, they have been able to be reviewed and treated locally.

Keywords: care closer to home, nurse-led service, telephone consultation

Disclosure of Interest: None Declared

BSH2020-PO-131

Rolling out adult and paediatric, patient centred Emicizumab education, counselling and administration training: Experience from a Haemophilia commissioned network covering a large catchment area

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Since NHS England commissioned the use of emicizumab in patients with severe haemophilia A, with and without inhibitors; there has been growing interest from individuals to switch to this novel, more convenient, subcutaneous treatment. The Southern Haemophilia Network (SHN) with Basingstoke comprehensive care centre (CCC) at its hub, has a large catchment area spanning 3 counties. Therefore we designed a patient centred emicizumab education and counselling provision that could be readily accessed.

As a network it was decided that eligible patients would have emicizumab offered at local clinic appointments, if interest was shown they would be invited back for an hour appointment with the CCC haemophilia consultant and a member(s) of the nursing team. As there was a significant cohort of interested adult Dorset patients, we set up a satellite group education and counselling session at Royal Bournemouth hospital. This was led by the CCC and local haemophilia consultant with support of haemophilia nurse specialists. All patients verbally consented to participate in the group education and counselling session. Satellite paediatric group education sessions will be made available once a small cohort is identified. Home visits by the CCC consultant and nurse specialist were also carried out in order to provide equal access of emicizumab education and counselling to our elderly patients with mobility problems.

Our centre devised an 'emicizumab information pack' which was distributed to interested parties. The first two injections had to be administered at the CCC however, subsequent doses could be given locally with nurse specialist supervision if still required alongside Haemtrack education as necessary. Patients then returned to their local centre for week five bloods and then seen 3 and 6 monthly by the CCC consultant either at a satellite clinic or CCC.

So far our CCC has started 6 adults (Basingstoke 4, Dorset 1, Salisbury 1) and 1 child (Salisbury) on emicizumab. Over the next month, we are scheduled to have a further 7 adults (Basingstoke 3, Dorset 3, Portsmouth 1) and 3 children (Basingstoke 2, Dorset 1) on emicizumab. With more patients switching to emicizumab within the SHN, the team is aware of the need to be flexible and mobile to meet the needs of our patients.

Disclosure of Interest: H. Cordial Conflict with: Roche/Chugai, L. Oyesiku Conflict with: Roche/Chugai, CSL Behring, NovoNordisk, C. Scott: None Declared, J. Yeaman: None Declared, U. Reddy: None Declared, T. Everington: None Declared, J. Mainwaring: None Declared, S. Mangles Conflict with: Roche, Takeda, NovoNordisk

BSH2020-PO-132

Implementing a venetoclax nursing care plan for a safe administration in a haematology settingR. Bowen^{1,*}, A. Etherington¹, C. Bloodworth¹, N. Elmusharaf¹¹Haematology, Cardiff and Vale university health board, Cardiff, UK

Background: Venetoclax is a BCL2 inhibitor used in the treatment of chronic lymphocytic leukaemia (CLL). In February 2019, NICE approved the use of Venetoclax with Rituximab for previously treated CLL patients who received at least one line of treatment. Venetoclax is administered in a dose escalation regime for 5 weeks and the main side effect is tumour lysis syndrome. Specific blood tests are required at specific time points to monitor for the possibility of an evolving tumour lysis syndrome.

Objectives: To provide the nursing staff, throughout haematology both inpatient and outpatient clinical areas at University Hospital of Wales, with a clear and precise nursing care plan to be used over the 5 weeks escalation period to minimise confusion and ensure blood investigations are done at the appropriate time points.

Assessment: A nursing care plan has been created with exact investigations required, which include exact time points for the nurses to take bloods, administer Venetoclax, administer fluids and supportive medications if needed. We highlighted the importance of waiting for ALL blood results to come back and seek doctors' advice.

Recommendations: Our aim is to enable the administration Venetoclax safely and reduce the risk of tumour lysis. This is the first draft of the Nursing care plan, which will be evaluated continuously and feedback from nursing staff will be vital on how to improve the care plan. The care plan has generated interest from other health boards in South Wales who are keen to use it in their departments, as a result this will be shared, and we aim to include it in the South Wales Cancer Network treatment plans.

Disclosure of Interest: None Declared

BSH2020-PO-133

Becoming a nurse principal investigatorE. Williams^{1,*}, Z. Hilton², S. Elias²¹Haematology, ²University Hospital of Wales Cardiff, Cardiff, UK

Research is often misunderstood and there can be a number of barriers to its implementation within clinical practice Seidlecki (2016), Woodward (2007). The aim was to find a suitable study within the Haematology portfolio that could be carried out by a nurse acting as Principal Investigator. Currently within the Health Board there are three nurse Investigators, two being at a Nurse Consultant level. Due to Emma Williams' many years of working in both Cancer and Haematology it was felt that she could drive this project forward within three different departments with the support of her administrative team within the Clinical Research Group.

The overall aim was to give underrepresented cancer patients a voice and to enable them to become empowered and proactive within a research trial. It was also an important aim to demonstrate that not just the senior members of the nursing team can participate within research projects.

During the project we collaborated with a research naïve Gynae Clinical Nurse Specialist –Sarah Elias and gave her the relevant training and reassurance that she could adapt this into her role.

The study was delivered over three clinical areas within the Health Board, in Llandough Breast centre, Gynae Outpatients and

Haematology Outpatients in the Heath Hospital. The overall trial aimed to consent 3,000 patients nationally and reached this target in April of this year.

This work was inspired from a review of the literature pertaining to nursing research within acute hospital settings whilst conducting an MSC pathway. Some of the findings being that there is a huge lack of nursing research and an undertone within the literature that medical research can be favoured over nursing Seidlecki and Albert (2016). An extension of these findings was conducted during the leadership module of the Masters pathway and found that nurses are being deterred from research due to work place demands and that there needs to be leaders driving these changes forward. In turn we will create a positive, dynamic and innovative nursing research culture that will continue to improve health and social care services Sissling (2003). Such leaders can be found at all levels within the NHS not just within senior positions and we therefore need to champion such staff.

The Horizons study is run at the University of Southampton with Chief Investigator, Professor Claire Foster and her Macmillan Survivorship Research Group and is a longitudinal study design looking at patients journeys before during and after their treatment and uses a series of questionnaires to capture patient data throughout their cancer pathways. It looks at the pertinent issues that matter to these patients and how these are similar and may also differ to those of the more common cancer groups

The study has now closed and we have recruited to time and target and the collaborative team are keen to share their experiences with others who may have an interest in conducting research within their areas. Since this work we are going to partake in another nurse PI study, alongside our bone marrow transplant Clinical Nurse Specialist and hope to set up our own Principal Investigator Research and Development trading account to enable us to fund any relevant training or conferences to extend our research knowledge further.

Disclosure of Interest: None Declared

BSH2020-PO-134

An eight week pilot of a weekly haematology day unit information evening eventH. Cosh¹, R. Bowen¹, N. Coles¹, J. Sully¹, C. Bloodworth^{1,*}¹Haematology, University Hospital Of Wales, Cardiff, UK

One of our annual patient forums raised the issue of a gap in knowledge regarding local support services despite all patients receiving a new starter pack with leaflets and an information file for patients to browse through during treatments. Individual patients already have the opportunity to have a needs assessment and are sign posted to services from this, however people still felt that there were gaps in their knowledge regarding services they did not know about. To help bridge this gap, together with patients, we developed the idea of a weekly patient information evening, where local services were invited in to demonstrate their services, whilst having an opportunity to meet key workers and a chance to look round the day unit.

An eight week pilot of a once weekly information evening event was commenced. It was planned to run once a week between the hours of 6 pm- 7 pm when the unit was closing, using a different theme for each evening, such as hypnotherapy tasters, a chance to hear from local charities and professionals such as dieticians and physiotherapists. The evenings were staffed by two CNSs, one day unit chemotherapy nurse and two volunteers.

The trial was evaluated using anonymous evaluation forms for those who attended the evenings and half way through the trial a further questionnaire was sent out to people having treatment who

were invited but not attending, to ask their reasons of non-attendance.

For those attending, the evaluations on the day rated the evenings as overwhelming excellent. All (100%) attending considered the evenings as “extremely valuable”, they liked the timeliness, length of the evenings and use of volunteers/staff. However, actual attendance was disappointingly low, numbers ranged from 3 to 7 attendees, with an average of only four attendees. Of those whom returned the non-attendance questionnaires, the main reasons given for not attending were; timing of the evenings (dark winter evenings), a wish not to attend an additional appointment and a lack of transport.

The staff felt that the effort and energy required to put on the nights outweighed the benefit the few were getting out of them. As a result of the evaluation of the pilot we are now developing alternative ideas such as videoing the support service sessions, editing them to short segments and putting them on-line into a website accessible from home, or alternatively running the evenings once a month on a weekend. Overall, it was agreed that any additional steps to improve patient support were worth doing, however the challenge remains on how best to achieve this to enable equal access to all patients.

Disclosure of Interest: None Declared

BSH2020-PO-135

Nurse Led DVT service over the last 17 years: IVDU (intravenous drug users) comparison to general population

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The nurse led DVT clinic based in University hospitals Plymouth NHS Trust (large teaching hospital of 900 beds), was set up in 2002 with the backing and support of 2 consultants, to improve the journey of patients presenting to the hospital with symptoms and signs of Deep Vein Thrombosis (DVT). The service required significant modernisation, from referral to assessment, investigation, diagnosis

and treatment, using Patient Group Directives [PGD] to facilitate a fully nurse led Clinic. This has evolved over time with changes in treatment duration guidelines and the introduction of direct-acting oral anticoagulants (DOACs), with ongoing support from Haematology.

This is a retrospective study, reviewing the IVDU patients seen over 17 years. Several aspects of the DVT clinic are examined in this patient group, including the modified wells score, D-dimer levels and scan results. This particular group of patients can be quite unique in their presentation and treatment plans.

Total patients referred (not including DNA's, cancelled attendance or refused) over a 17 years period from 2002 to 2019 was 18384, of whom 7840 were male, 10543 female and 1 transgender. 3943 had DVT confirmed (21.4%); male 2214 (28.2%) and female 1726 (16.3%).

Of the IVDU population over the same period, total referred was 678 of whom 474 were male, 204 female. 368 had DVT confirmed (54.2%) overall pick-up rate, with male 261 (55%) and female 107 (52.4%).

D-Dimer was positive in 221 IVDU patients. 19 had a negative D-dimer; D-dimer was not done in 128 patients

Over the years the IVDU patients referred to the clinic have been getting older, a few even into their 60's however we are seeing less in their 20's

These data demonstrate that the IVDU patients, as a group presenting to UHP Trust, DVT clinic, have a specific phenotype. They generally have a higher D-Dimer level at presentation and a significantly higher pick-up rate of 54% compared to 21% over the 17 years. Finally, they are somewhat younger than the general population who develop DVT's, but their age is gradually increasing.

Table.

Wells score	1	2	3	4	5
Number	40	99	146	69	14

Disclosure of Interest: None Declared

Paediatrics

BSH2020-PO-136

Capillary blood sampling for anti-Xa monitoring in paediatric patients- a practice changing quality improvement project at Birmingham Children's Hospital

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Low molecular weight heparins (LMWHs) are the most frequently used anticoagulant of choice in the paediatric setting, yet their wide variation in dosing and response warrants frequent anti-Xa level testing for monitoring. Obtaining such Anti-Xa levels in young children poses a number of challenges, including difficulties with venous access and obtaining a free-flowing sample, as well as avoiding heparin contamination when samples are obtained from central lines. Capillary blood sampling is often used in children as an alternative to venepuncture, however, activation of the sample that occurs as a result of this process interferes with a number of coagulation assays, including the APTT, with the result that coagulation tests are generally only done on venous samples. Anti -Xa assays monitor the effect of heparin and should therefore not be affected by sample activation. Following on from previous work at Sheffield Children's Hospital, we conducted a study at Birmingham Children's Hospital to ascertain whether capillary chromogenic anti-Xa levels correlate with venous samples and can therefore be used as an alternative, less-invasive approach for monitoring LMWH.

All patients receiving therapeutic LMWH with a target anti-Xa level of 0.5–1.0 IU/ml during the study period were eligible for inclusion in the study. A pre-study target sample size of 20 was calculated to ensure our results were representative. 20 paired venous and capillary samples were collected on 10 different patients over an 8-month period from October 2018–June 2019. Capillary samples were collected from the side of heel in infants or the ball of the thumb in older children. Chromogenic anti-Xa levels were performed on a Sysmex CS2000i analyser.

Data was inputted onto an excel spreadsheet and each value was plotted onto a calibration curve.

Our results demonstrated that venous and capillary anti-Xa levels were comparable, in agreement with Sheffield's work, and confirmed our hypothesis that capillary anti-Xa levels are a suitable alternative for children receiving LMWH therapy in whom venous access is challenging.

As a result of our study, we have implemented a new policy at Birmingham Children's Hospital which enables children to have their anti-Xa levels monitored using capillary samples. This has led to significant improvements in obtaining anti-Xa levels at the correct time, and ultimately improved compliance with anti-Xa monitoring and maintaining therapeutic anti-Xa levels.

Disclosure of Interest: None Declared

BSH2020-PO-137

Significant levels of endogenous formaldehyde in humans – a potential driver of Fanconi anaemia?

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Damage to our DNA can result in premature aging and cancer. This is exemplified by the genetic disease Fanconi anaemia (FA), a congenital bone marrow failure and cancer syndrome in children and young adults. The genetic mutations in FA result in deficiency in the FA DNA repair pathway, leading to breakages in chromosomes and sensitivity to DNA-crosslinking chemotherapy agents such as cisplatin. However, the physiological cause of the DNA damage that drives bone marrow failure and an increased cancer risk in FA is not well understood. Recent advances using mouse models of FA have highlighted formaldehyde, a highly reactive by-product of metabolism, as the physiological genotoxin that drives the disease (Pontel et al., Mol Cell 2015; Burgos-Barragan et al., Nature 2017). Indeed, mice deficient in both the FA pathway and Adh5, the enzyme that catabolises formaldehyde, present with a much more severe bone marrow failure and acute leukaemia phenotype, recapitulating the key features of human FA disease. We wanted to address whether levels of formaldehyde produced in humans could be high enough to drive FA. To this end, we set out to measure formaldehyde concentration in human serum.

Between September and December 2019, we consented and recruited 51 volunteers attending the haemostasis clinic at the Haemophilia Centre, Cambridge University Hospitals NHS Foundation Trust for investigation and management of bleeding disorders. None had a diagnosis of or were receiving active treatment for bone marrow failure syndromes or haematological malignancies. The cohort represented a wide age group of 16 – 83 years (median 36 years), with 21 (41%) female vs 30 (59%) male and an average weight of 79.9 kg (range 37 kg – 124.4 kg). Donations of approximately 3 ml of serum were collected. As native formaldehyde is highly reactive, we treated the serum with a derivatisation agent (O-(2,3,4,5,6-Pentafluorobenzyl)hydroxylamine hydrochloride) that is chemically modified on reaction with formaldehyde. The derived product is then measured and quantified using gas chromatography and mass spectrometry.

Our results show that the average level of serum formaldehyde was 7.1 µM (range 3.9 – 11.2 µM; 95% confidence interval 6.4 – 7.6 µM). To assess whether this level of formaldehyde could affect cell viability, we measured the growth of a haematopoietic cell line treated with formaldehyde for 4 days. Whilst wildtype cells were unaffected by formaldehyde concentrations of less than 30 µM, cells deficient in the FA pathway treated with 10 µM formaldehyde showed a 20% growth reduction. Strikingly, cells deficient in both the Adh5 enzyme and FA pathway exhibited a 40% growth reduction following only 5 µM formaldehyde treatment. This suggests a significant proportion of volunteers produced sufficient endogenous

formaldehyde to pose a threat to cell growth in the absence of the FA DNA repair pathway and/or Adh5 enzyme. We wanted to test for clinical factors that could account for the variation in human serum formaldehyde levels. Thus far, we have observed no correlation with gender, weight, age, creatinine clearance, or serum liver enzymes.

In summary, we have been able to measure levels of formaldehyde in human serum and have found that it is present at concentrations that are able to cause *in-vitro* cellular toxicity if not protected by the FA DNA repair pathway. It is therefore possible that formaldehyde is the major genotoxin driving the development of human FA disease.

Disclosure of Interest: None Declared

BSH2020-PO-138

Intensive care admissions for children with acute lymphoblastic leukaemia: a 13 years single centre experience

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The outcomes for children with acute lymphoblastic leukaemia (ALL) are steadily improving, with cure rates of over 93%. However, treatment-related mortality is still significant; infection being the leading cause of morbidity and mortality in these immunocompromised patients. Children with ALL who suffer from disease or treatment complications may require escalation of care to the paediatric intensive care unit (PICU).

Using the PICANet national database, we reviewed the PICU admissions and chemotherapy records of patients diagnosed with ALL under the age of 16 years in Leeds Children's Hospital over a 13 years period (June 2004– May 2017), regardless of trial participation.

Over the 13 years study period, approximately 450 under 16s were treated for ALL in the study centre, and 67 (15%) of these had at least one PICU admission (84 admissions in total). Fifty-three patients had one PICU admission and fourteen had more than one; over half of second admissions occurring within a month of the previous PICU discharge. Infection was associated with 61 (73%) PICU admissions. The most common causes for admission were sepsis (22/84), respiratory infection (16/84), and respiratory failure (13/84). The mean length of stay for patients with infection was 8.9 days compared to 3.9 days in patients without an infection. Complete treatment information was available for fifty-eight patients. Of these, nine patients had relapsed disease (11 admissions), three were infants (10 admissions), two had Philadelphia positive ALL (3 admissions), and two had had a haematopoietic stem cell transplantation (3 admissions). In the remaining forty-two patients, there were 49 PICU admissions: 27/49 in induction, 4/49 in consolidation, 7/49 in delayed intensification, and 11/49 in maintenance. Of the total 84 admissions, 49 admissions required ventilation, 73% of which were due to infection. In 45 of the admissions, the patient required inotropic support; 80% of these admissions were secondary to infection. Twenty of the patients who died were ventilated, eleven of which were on inotropic support. There were twenty-two deaths within the first 30 days from discharge from PICU. Seven of those deaths were in patients with relapsed disease. Patients who died in PICU had shorter admissions with a mean

duration of 5 days. The mortality rate for ALL patients in PICU was 16 per 100 person-days at risk. Crude mortality rate ratio in patients at induction versus other phases was 3.5 $P = 0.05$. The mortality rate was higher in those who were ventilated, however, only inotrope requirement reached statistical significance: crude rate ratio 3.5 $P = 0.03$, increased to 4.6 $P = 0.02$ controlling for ventilation status.

Most admissions to PICU in this cohort of patients occurred during induction, however, a significant number of admissions happened in maintenance treatment with 3 deaths in 11 PICU admissions during this phase. Children with ALL who are admitted to PICU have poor outcomes; nearly 1 in 3 children died. Patients with relapsed ALL are at particularly high risk of mortality in PICU (6/9). Most admissions to PICU were secondary to infective causes.

Disclosure of Interest: None Declared

BSH2020-PO-139

Transcranial doppler screening programme in children and adolescents with sickle cell disease: experience in a District General Hospital in the UK

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Background and aims: Surveillance programmes through annual transcranial Doppler (TCD) to prevent stroke in children and adolescents with Sickle Cell Disease (SCD) are a standard of care. However, across the world, screening is inadequate, with rates between 22–44% of eligible children in the United States being screened; there are no European data on feasibility or adherence to the screening program.

We aimed to investigate the feasibility of TCD screening program according to the Stroke Prevention Trial in Sickle Cell Anaemia (STOP) criteria, and screening rates, in a district general hospital (DGH) in the UK.

Methods: Eligible children underwent local TCD screening as a tertiary paediatric support programme from Oxford University Hospital. A dedicated specialist nurse liaised with children and families through home and school visits, facilitating attendance. Patients' clinical details, TCD time-averaged mean of the maximum velocity (TAMMV) and screening outcomes were recorded at each visit.

Results: All 49 eligible SCD children attended the screening program (100% rate) between 2016 and 2019; they were classified as per Table 1 and treated accordingly.

Conclusions: TCD screening for SCD is feasible at a UK DGH, with risk stratification close to the STOP trial recommendations. Specialist nurses working in the community have a critical role in promoting compliance to the programme.

Disclosure of Interest: None Declared

Table.

	NORMALTAMMV <170 cm/s	CONDITIONALTAMMV 170-199 cm/s	HIGH RISK TAMMV >200 cm/s	INCONCLUSIVE
PATIENTS (%)	32(65)	11(23)	5(10)	1(2)
MEAN AGE (SD)	7.49(5.11)	6.18(3.22)	7(3.24)	2
MEAN TAMMV (SD)	113.48(16.60)	175.73(5.83)	203(4.24)	Not applicable

BSH2020-PO-140

Complex paediatric apheresis support for advanced therapies

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Advanced therapies with genetically modified autologous stem cells or T-cells are used increasingly in paediatrics, including gene therapies for severe congenital immunodeficiencies and thalassaemia major, and chimeric antigen receptor T-cell therapies (CAR-T; licensed for certain indications). This is an evolving field, mainly in the research arena. Apheresis mobilization/cell collection and processing methods can vary between research protocols and may impact on the ability to generate the final therapeutic product, its quality and the potential for errors. Despite the apheresis procedure being a significant intervention including line insertion and possible use of red cell prime, it is frequently not detailed in trial protocols. Developing more standardised procedures and quality monitoring would reduce complexity and could lead to more consistent outcomes.

Apheresis collections at a tertiary paediatric centre, Jan 2016-Dec 2019 inclusive, were reviewed, both peripheral blood stem cells (PBSCs: CD34+) and lymphocytes. Variables included patient diagnosis, age, venous access, type of trial/licensed therapeutic product, apheresis collection target, collection volume/cell count, purity of collection product, number of procedures required per collection cycle, collection outcomes, virology and quality monitoring.

There were a total of 96 apheresis collection cycles from 90 patients for advanced therapies: 33 PBSC collections (30 patients) requiring 56 procedures (target reached for 27/32); 63 lymphocyte collections (60 patients) requiring 71 procedures (target reached for 62/63). PBSC collections were for gene therapy for immunodeficiency (31; 8 trials) and thalassaemia (2; 1 trial). Lymphocyte collections were for CAR-T therapy (61: 45 for 3 trials – leukaemia/neuroblastoma; 16 for Novartis Kymriah licensed leukaemia therapy – one failing to generate a therapeutic product) and immunomodulatory therapies post stem cell transplant (2; 2 trials). There was variation in target cell count, reflecting evolving knowledge, (for PBSCs between 10 and 50x10⁹/kg; for lymphocyte procedures between not specified and CD3 1x10⁹, total nucleated cells 1–2 x 10⁹) and collection volume (2–4 blood volumes, usually 2–2.5); also in virology screening requirements. Some trials required specific modifications of collection techniques, such as minimising granulocyte contamination (for 4/6 lymphocyte protocols) or adjustment for thalassaemia erythroid precursors. Quality monitoring of collection efficiencies for lymphocytes is less well established than for PBSCs and was not employed. Quality assurance processes vary: the stem cell collection facility is regularly and rigorously regulated and accredited by the Human Tissue Authority and the Joint Accreditation Committee ISCT-Europe & EBMT (JACIE) but several trial sponsors insisted on their own additional quality inspections.

These results demonstrate successful collection of autologous cells for advanced therapies across a diverse research portfolio (15 trial/licensed), with multiple trial protocols and disease states. This reflects

close teamwork between an expert, flexible apheresis service and clinical/research teams, with attention to measures of success. Ongoing therapeutic diversification/development, including impact of collection variables on outcomes, will require international collaboration with manufacturers, sponsors and regulatory bodies to improve standardisation, efficiency and quality.

Disclosure of Interest: None Declared

BSH2020-PO-141

Does a specific factor VII mutation predict a risk of neonatal intracranial haemorrhage?

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Factor VII deficiency is a rare, autosomal recessive, coagulation disorder. The clinical phenotype is variable and multiple mutations to the *F7* gene have been identified. We present a series of four children from two families with severe factor VII deficiency. Although unrelated, the families originate from the same area of India. They share a common *F7* intron 5 c.572-12T>A mutation. Patient A is the first child of non-consanguineous parents. He presented at birth with a large intracranial haemorrhage and was found to have a severe FVII deficiency, with baseline levels <0.01 IU/ml. He was treated with plasma derived FVII concentrate, initially at treatment doses and then as daily prophylaxis. He developed an inhibitor at the age of 7 months for which he received immune tolerance induction therapy. His prophylaxis was switched to recombinant factor VII after developing muscle bleeds at the age of 3. He continues on daily prophylaxis and has had no significant bleeds since. His brother, patient B, was diagnosed with severe FVII deficiency at birth. Cranial US was undertaken on day 0 and day 4 and showed no bleeds, but at day 8 of life he was noticed to have abnormal movements, and CT scan revealed a large intracranial haemorrhage. Treatment was initiated with recombinant FVII concentrate until resolution of the bleed, and he remains well on daily prophylaxis. Patients C and D are sisters born to consanguineous parents. Both had significant intracranial haemorrhages in infant life which were treated with FVII concentrate. They were established on prophylaxis but have continued to demonstrate a severe bleeding phenotype, with particularly debilitating menorrhagia.

We present here four patients from two unrelated families with a common FVII mutation and a clinical phenotype characterised by severe bleeding, particularly intracranial haemorrhage. Further investigation of other families with the same mutation would be useful to determine the possible role of the mutated protein in contributing to the phenotype of the disease.

Disclosure of Interest: None Declared

BSH2020-PO-142

Health-related quality of life following initial diagnosis of childhood immune thrombocytopenia

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Health-related quality of life (HRQoL) is one of the outcomes that should be considered in children with immune thrombocytopenia (ITP). ITP is a generally benign illness, that is usually self-limiting with a good prognosis, and therefore it is important to consider the

effect of treatment and other disease related parameters on the HRQoL of these children.

The objective of this study was to assess how HRQoL changes in the 6 weeks and six months following a diagnosis of ITP, taking into account differences in platelet counts and treatment methods between patients. The data used for the purposes of this study were taken from the UK paediatric ITP registry. HRQoL in these patients was measured using the UK version of the Kids' ITP Tools.

139 patients were included in this study; these were patients who had KIT scores recorded at diagnosis and with at least one more KIT score recorded in the following presentation. 86 patients had a repeat platelet count at 6 weeks following diagnosis and these patients form the focus of this report.

Of the patients whom had a count available at 6 weeks post presentation 30 (35%) had a platelet count $<3010^9/L$, 56 (65%) had recovered to $\geq 3010^9/L$. 53 patients had no platelet count recorded at 6 weeks but were recorded at later time point(s).

In the 6 weeks following diagnosis 109 (80%) children were managed without therapy (watch and monitor), 14 (10%) with immunoglobulin and 13 (10%) with steroids. In addition two patients had combination therapy and the detail of therapy was not complete in one child.

There was a significant ($p<0.001$) improvement in HRQoL for the whole cohort at both 6 weeks and six months after diagnosis ($p<0.05$). All of the subgroups showed improvements in HRQoL after 6 weeks, however these were not statistically significant. The subgroup that had the greatest improvement in HRQoL 6 weeks after presentation was those who were treated with IVIG.

In conclusion, this study shows that there is a general improvement in HRQoL in the 6 weeks that follow a diagnosis of ITP and that improvement is more likely following platelet recovery to over 30 or following IVIg treatment.

Disclosure of Interest: None Declared

Red Cell Disorders

BSH2020-PO-143

Effects of luspatercept on iron overload and impact on responders to luspatercept: results from the BELIEVE Trial

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The phase 3 BELIEVE trial (NCT02604433) evaluated the efficacy and safety of luspatercept, a first-in-class erythroid maturation agent, in adults with beta-thalassaemia requiring regular red blood cell (RBC) transfusions. Here we report the effects of luspatercept on iron loading and the impact of baseline iron levels on response to luspatercept.

Eligible patients were adults with beta-thalassaemia or haemoglobin E/beta-thalassaemia requiring regular transfusions of 6–20 RBC units in the 24 weeks prior to randomisation (with no transfusion-free period more than 35 days). 336 patients were randomised 2:1 to receive luspatercept 1.0 mg/kg (titrated up to 1.25 mg/kg) or placebo subcutaneously every 3 weeks for at least 48 weeks.

Over the 24 weeks prior to Week 96, mean change from baseline in serum ferritin level and liver iron concentration was -412.16 mcg/L and -0.38 mg/g dry weight, respectively, for patients receiving luspatercept, versus +94.92 mcg/L and +1.86 mg/g dry weight for

patients receiving placebo. In the 130 (78.8%) luspatercept-treated and 38 (37.6%) placebo-treated responders (response defined as at least 33% reduction in RBC transfusion burden with a reduction of at least 2 RBC units during any 12 weeks in the first 48 weeks), mean serum ferritin change from baseline was -490.21 mcg/L and +161.65 mcg/L, respectively. Mean change (SD) in liver iron concentration was -1.33 (8.22) mg/g dry weight in luspatercept responders versus +5.64 (7.25) mg/g dry weight in luspatercept non-responders. Mean reduction from baseline in myocardial iron T2* MRI values with luspatercept remained within the normal myocardial iron range, and so had little clinical significance. There was no significant change from baseline in mean daily dose of iron chelation therapy between luspatercept-treated and placebo-treated patients.

Luspatercept treatment resulted in statistically significant reductions in serum ferritin, and there was a trend for decrease in liver iron concentration. These data suggest that luspatercept may contribute to improved control of body iron in patients with beta-thalassaemia who require regular RBC transfusions, particularly in those achieving at least a 33% reduction in RBC transfusion burden.

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C3 inhibition with pegcetacoplan in the treatment of subjects with paroxysmal nocturnal hemoglobinuria

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Paroxysmal nocturnal hemoglobinuria (PNH) is an acquired hematologic disease characterized by chronic complement-mediated hemolysis and life-threatening thrombosis. Intravascular hemolysis (IVH) is mediated by the membrane attack complex and extravascular hemolysis (EVH) is facilitated by C3 opsonization, a process upstream of C5. Although eculizumab (ECU), a C5 inhibitor and the first approved treatment for PNH, addresses IVH, 70% of individuals remain anemic and 36% require ≥ 1 transfusions per year due to C3-mediated EVH. Pegcetacoplan (APL-2), a C3 inhibitor, was evaluated for its potential to better control hemolysis.

This phase 1b, open-label, prospective, non-randomized, single- and multiple-ascending-dose study (NCT02264639) was designed to assess the safety, tolerability, pharmacokinetics (PK) and pharmacodynamics (PD) of pegcetacoplan as an add-on to standard of care (SOC) in subjects with PNH. Adults diagnosed with PNH were eligible if they had received ECU for ≥ 3 months, and either had hemoglobin (Hb) levels < 10 g/dL at screening or had received ≥ 1 transfusion within 1 year prior to screening. Four dose escalation cohorts were enrolled in this study; data presented are for six Cohort 4 subjects who received 270–440 mg/day subcutaneous pegcetacoplan for up to 2 years. Primary endpoints were the number and severity of treatment-emergent adverse events (TEAEs) and PK parameters. Additional PD endpoints included Hb, absolute reticulocyte count (ARC), lactate dehydrogenase (LDH) and total bilirubin. The Functional Assessment of Chronic Illness Therapy (FACIT)-fatigue score and number of transfusions were recorded.

All subjects had quantifiable concentrations of pegcetacoplan by Day 2 and steady state serum concentration was reached 6–8 weeks after treatment initiation. Mean (SD) Hb increased from 8.78 (1.41) g/dL at baseline to 11.50 (0.51) g/dL over 2 years of treatment, mean change from baseline (CFB) 2.63 (1.90) g/dL (95% CI: -0.39, 5.64; $P = 0.0695$). Mean baseline (SD) ARC reduced from 333.93 (121.21) $\times 10^9/L$ to 82.67 (31.02) $\times 10^9/L$, mean CFB -239.40 (48.97) $\times 10^9/L$ (95% CI: -361.04, -117.76; $P = 0.0137$). Mean (SD) LDH reduced from 280.2 (131.1) U/L to 249.0 (89.5) U/L, mean CFB 39.3 (98.5)

U/L (95% CI: -117.5, 196.0; $P = 0.4837$), and total bilirubin reduced from 1.92 (0.45) mg/dL to 0.50 (0.16) mg/dL, mean CFB -1.60 (0.27) mg/dL (95% CI: -2.03, -1.17; $P = 0.0013$). Mean (SD) FACIT-fatigue score increased from 32.50 (11.04) to 45.25 (4.11), mean CFB 9.00 (7.53) (95% CI: -2.98, -20.98; $P = 0.0966$). Once pegcetacoplan serum concentration reached steady state, no subjects received a transfusion. In total, 427 TEAEs were reported with 68 possibly related to study drug, 57 of these were injection site reactions. No TEAEs resulted in discontinuation of pegcetacoplan. Eight serious TEAEs occurred in 2 subjects. Two of 6 subjects in Cohort 4 discontinued for reasons unrelated to study drug. SOC adjustments were made in the remaining 4 subjects with ECU doses being initially reduced and then discontinued between Days 457–626, after which all subjects received pegcetacoplan monotherapy.

This is the first study to evaluate the efficacy, safety, and tolerability of pegcetacoplan for up to 2 years in individuals with PNH who experienced persistent anemia while on ECU. Pegcetacoplan improved hematological response by achieving broad hemolysis control, enabling subsequent ECU discontinuation. In this study, pegcetacoplan was generally well-tolerated.

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One-year efficacy of ravulizumab (ALXN1210) in adult patients with paroxysmal nocturnal hemoglobinuria naive to complement inhibitors

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Ravulizumab, a new C5 complement inhibitor, was approved for the treatment of paroxysmal nocturnal hemoglobinuria (PNH) in the US and in several other countries. In the largest to date phase 3 study in complement inhibitor-naïve PNH patients (pts) (NCT02946463), ravulizumab, administered every 8 weeks (q8w), was non-inferior to eculizumab, administered every 2 weeks (q2w) for all primary and secondary endpoints after 26 weeks (wks). To evaluate long-term efficacy and safety with up to 52 weeks of treatment, all pts had the option to continue ravulizumab in an extension phase.

In the extension study, we report the efficacy and safety of ravulizumab through 52 weeks of treatment and evaluate the efficacy profile of ravulizumab after switching from eculizumab in eculizumab-naïve PNH patients.

This randomized, active-controlled, open-label study was conducted in 123 centres in 25 countries. Pts were randomized to receive eculizumab ($n = 121$) or ravulizumab ($n = 125$) for 26 weeks, after which pts in the ravulizumab arm continued ravulizumab maintenance therapy (R-R arm), while pts in the eculizumab arm were then switched to ravulizumab (E-R arm). Results of the co-primary (lactate dehydrogenase normalization [LDH-N], transfusion avoidance) and secondary endpoints (breakthrough hemolysis [BTH] and LDH levels) are provided. Serum free C5 levels were obtained through 52 weeks.

During the first 26 weeks, 74% of pts in the R-R arm avoided transfusion, vs 77% avoiding transfusion in weeks 27–52. More than 90% ($n = 83$) of pts who avoided transfusion during the initial 26 weeks maintained this response through 52 weeks, and 38% ($n = 12$) who required transfusion in the 0–26 wk period avoided transfusion in weeks 27–52. In the E-R arm, 66% (26 weeks) vs 67% (52 weeks) avoided transfusion; 87% ($n = 69$) who avoided transfusion for 26 weeks maintained this response through 52 weeks.

Additionally, 28% ($n = 11$) who required transfusion during the 26 weeks while on eculizumab avoided it through 52 weeks after the switch to ravulizumab. In the R-R arm, LDH-N occurred in 48% of pts ($n = 60$) at 26 weeks and in 44% of pts ($n = 54$) at 52 weeks. Similarly, in the E-R arm, at 26 weeks, LDH-N was 42%, and was 40% at 52 weeks. At the end of 26 weeks, pts on ravulizumab had a 77% mean reduction in LDH from baseline and this was maintained through 52 weeks (77%). All pts ($n = 119$) in the ravulizumab group continued to maintain free C5 <0.5 mg/ml at all time points through 52 weeks. In pts initially randomized to eculizumab, the switch to ravulizumab showed improved free C5 control, and no patients had free C5 >0.5 mg/ml after the switch. Of the 125 pts initially on ravulizumab, 5 pts (4%) had BTH in the first 26 weeks vs 4 pts (3%) during weeks 27–52. In the E-R arm, 13 pts (11%) experienced BTH during the first 26 weeks vs 2 pts (2%) in weeks 27–52 after the switch to ravulizumab. During weeks 27–52, none of the BTH events in both arms were associated with free C5 ≥ 0.5 mg/ml, the threshold associated with complete inhibition of hemolysis. These results demonstrate maintenance of complete free C5 inhibition in pts receiving ravulizumab. The most common treatment-related adverse events decreased in frequency with increased treatment duration.

Ravulizumab demonstrated durable efficacy over 52 weeks of treatment and was well tolerated. Importantly, all pts who had sub-optimal free C5 inhibition receiving eculizumab achieved complete free C5 inhibition after the switch to ravulizumab which was associated with a decreased incidence of BTH.

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One-year efficacy and safety from a phase 3 trial of ravulizumab in adult patients with paroxysmal nocturnal hemoglobinuria receiving prior eculizumab treatment

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In ALXN1210-PNH-302 (NCT03056040) the long-acting C5 complement inhibitor ravulizumab (q8w) was noninferior to eculizumab (q2w) after 26 weeks for all primary and key secondary endpoints in eculizumab-experienced paroxysmal nocturnal hemoglobinuria (PNH) patients (pts). After 26 weeks, all pts had the option to receive weight-based dosing of ravulizumab in the extension. We report on the efficacy and safety of ravulizumab through wk 52.

In the extension PNH pts (≥ 18 yr) who were stable on eculizumab for ≥ 6 mo and with lactate dehydrogenase (LDH) of $\leq 1.5 \times \text{ULN}$ at screening were randomly assigned 1:1 to switch to ravulizumab or to continue eculizumab for 26 weeks. After 26 weeks, pts in the ravulizumab arm continued ravulizumab therapy (R-R arm), and pts in the eculizumab arm were switched to ravulizumab (E-R arm). For the 52-wk data, the primary efficacy endpoint was percent change in LDH from baseline (BL); key secondary endpoints included pts with breakthrough hemolysis (BTH), transfusion avoidance, improvement in FACIT-Fatigue total score, and stabilized hemoglobin (HGB-S) levels. Additional endpoints included change in serum free C5 levels from BL and safety evaluations.

Overall, 191 pts entered the extension (R-R arm: $n = 96$; E-R arm: $n = 95$). Pts in both groups showed a durable response for percentage change in LDH up to 52 weeks, similar to what was observed during the first 26 weeks. At 52 weeks, pts in the R-R arm had an 8.8% increase in LDH from BL, while pts in the E-R arm had 5.8% change in LDH from BL. Mean LDH levels in both groups were maintained at $1.0 \times \text{ULN}$ (< 246 U/L). During weeks 0–26, no pts in the R-R arm had BTH vs 3 during weeks 27–52. In the E-R arm, 5 pts had BTH during weeks 0–26 vs 1 in weeks 27–52. During weeks 27–52, no BTH events were associated with free C5 of ≥ 0.5 $\mu\text{g/ml}$ (threshold for complete C5 inhibition). During weeks 0–26, 88% of pts in the R-R arm avoided transfusion vs 87% in weeks 27–52; in the E-R arm, 83% (0–26 weeks) vs 83% (27–52 weeks) avoided transfusion. FACIT-Fatigue scores were maintained in both treatment groups through 52 weeks. The proportion of pts with HGB-S was 76% in each arm during weeks 0–26, and 81% in each arm during weeks 27–52. All pts in the R-R arm maintained free C5 of < 0.5 $\mu\text{g/ml}$ at all time points through 52 weeks ($n = 96$). All pts on ravulizumab maintained free C5 of < 0.5 mg/ml at all time points through 52 weeks ($n = 96$). No pts had free C5 of ≥ 0.5 $\mu\text{g/ml}$ after the switch to ravulizumab. In the extension, 79% in the R-R arm had a treatment-emergent adverse event (TEAE) vs 75% in the E-R arm. The most frequently reported TEAEs in the R-R arm were fatigue (14%) and upper respiratory tract infection (URTI; 9%); in the E-R arm these were fatigue (14%) and headache (11%). Eight pts (8%) in the R-R arm and 5 pts (5%) in the E-R arm experienced

serious AEs; none led to discontinuation or death. No new treatment-emergent antidrug antibody-positive responses were reported during weeks 27–52. There were no meningococcal infections, deaths, or discontinuations due to AEs.

Pts on stable eculizumab therapy who received ravulizumab over 52 weeks experienced durable efficacy; pts in the E-R arm had an efficacy response consistent with pts in the R-R arm. All pts who had suboptimal free C5 inhibition on eculizumab achieved complete free C5 inhibition after switching to ravulizumab; no BTH events were associated with free C5 levels of ≥ 0.5 $\mu\text{g/ml}$. Ravulizumab was well tolerated through wk 52 with no new safety concerns.

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Assessment of longer-term efficacy, safety, and haematological improvement in the phase 3, randomised, double-blind, placebo-controlled MEDALIST trial of luspatercept to treat anaemia in patients with lower-risk myelodysplastic syndromes with ring sideroblasts who require red blood cell transfusions

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Luspatercept is a first-in-class erythroid maturation agent that binds several TGF- β superfamily ligands to enhance late-stage erythropoiesis. We report longer-term outcomes of the MEDALIST trial (NCT02631070), a phase 3 study evaluating the efficacy and safety of luspatercept in patients with anaemia due to lower-risk myelodysplastic syndromes (LR-MDS) with ring sideroblasts (RS).

Eligible patients were aged at least 18 years; had Revised International Prognostic Scoring System (IPSS-R)-defined Very low-, Low-, or Intermediate-risk MDS with RS; were refractory, intolerant, or unlikely to respond to erythropoiesis-stimulating agents; and required regular red blood cell (RBC) transfusions (at least 2 units per 8 weeks in the 16 weeks before randomisation). The primary endpoint was RBC transfusion independence (RBC-TI) for at least 8 weeks (Weeks 1–24). A total of 229 patients were randomised 2:1 to luspatercept (1.0 mg/kg with titration to 1.75 mg/kg) or placebo subcutaneously every 3 weeks.

As of the new July 1, 2019 cutoff date, 41 of 153 (26.8%) patients remained on luspatercept treatment; none of the 76 patients in the placebo arm remained on treatment. RBC-TI for at least 8 weeks any time during the treatment period was achieved by 73 (47.7%) and 12 (15.8%) patients in the luspatercept and placebo arms, respectively. Clinical benefit (defined as achieving RBC-TI for at least 8 weeks and/or modified haematologic improvement (HI)-erythroid [mHI-E] per International Working Group [IWG] 2006 criteria) was achieved by 98 (64.1%) and 20 (26.3%) patients in the luspatercept and placebo arms, respectively.

Of patients with baseline neutropenia ($n = 25$; baseline neutrophils $< 1 \times 10^9/L$) or baseline thrombocytopenia ($n = 14$; baseline platelets $< 100 \times 10^9/L$), 3 of 15 (20.0%) and 1 of 10 (10.0%) achieved HI-neutrophil (HI-N; per IWG 2006 criteria), and 5 of 8 (62.5%) and 2 of 6 (33.3%) achieved HI-platelet (HI-P; per IWG 2006 criteria) during Weeks 1–48 in the luspatercept and placebo arms, respectively.

The treatment-emergent adverse events more commonly reported with luspatercept caused few discontinuations. Grade 3 or 4 neutropenia occurred in 13 patients (4.6% vs 7.9%, luspatercept vs placebo) and was not considered treatment-related; no grade 3 or 4 thrombocytopenia was reported.

In this longer-term analysis, 14 months after the original analysis, more luspatercept-treated patients achieved RBC-TI for at least 8 weeks and clinical benefit versus placebo. In the few patients with baseline neutropenia or thrombocytopenia receiving luspatercept, cytopenias did not worsen following treatment. Luspatercept is a potential new therapy for the treatment of anaemia in patients with LR-MDS with RS.

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Evaluating luspatercept responders in the phase 3, randomised, double-blind, placebo-controlled BELIEVE trial of luspatercept in adult beta-thalassaemia patients who require regular red blood cell transfusions

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Luspatercept is a first-in-class erythroid maturation agent that binds several TGF-beta superfamily ligands to diminish Smad2/3 signaling and enhance late-stage erythropoiesis. We aimed to evaluate the number of responses, duration of clinical benefit, and safety in luspatercept responders in the phase 3 BELIEVE study (NCT02604433).

Eligible patients were aged at least 18 years, had beta-thalassaemia or haemoglobin E/beta-thalassaemia, and required regular transfusions (6–20 red blood cell [RBC] units) in the 24 weeks prior to randomization with no transfusion-free period more than 35 days. Patients were randomised 2:1 to luspatercept 1.0 mg/kg (titrated up to 1.25 mg/kg) or placebo subcutaneously every 3 weeks for at least 48 weeks. Achievement of clinical benefit response (at least 33% reduction in RBC transfusion from baseline over any consecutive 24 weeks), number of separate response episodes (defined as non-overlapping periods in which at least a 33% reduction from baseline in RBC transfusion over any 24 weeks was achieved), duration of clinical benefit (defined as time from first response to discontinuation), and safety were assessed.

As of the January 7, 2019 cutoff date, 101/224 (45.1%) patients randomised to luspatercept had achieved clinical benefit (at least 33% reduction in RBC transfusion over any 24 weeks) versus 3/112 (2.7%) placebo-treated patients ($P < 0.0001$). Of the luspatercept responders, 74 (73.3%) had at least 2 separate responses. Eleven (10.9%) luspatercept responders experienced multiple episodes of response during any 24 weeks. Median duration of clinical benefit for luspatercept responders was 76.3 weeks; 5 responders achieved RBC transfusion independence for at least 24 weeks (median 94.6 weeks). The mean reduction in RBC units in luspatercept responders was 6.69 units over any 24 weeks and 8.36 units in patients with > 15 units/24 weeks, compared with baseline. Bone pain, arthralgia, and dizziness occurred more frequently in the luspatercept arm compared to the placebo arm (20.2% vs. 8.3%, 21.2% vs. 14.7%, and 12.1 vs. 4.6%, respectively). New onset of these adverse events (AEs) was largely transient and these AEs were not associated with treatment discontinuation.

Most luspatercept responders experienced multiple episodes of meaningful transfusion reduction and had durable clinical benefit over time. Luspatercept treatment in adult patients with beta-thalassaemia who require regular RBC transfusions resulted in a significant reduction in RBC transfusion burden, with an acceptable safety profile.

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Unilateral versus bilateral peripheral access using ultrasound machine guidance in patients undergoing red blood cell exchange transfusion: a comparative results

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Apheresis machine operators frequently face a significant challenge when attempting venous access prior to automated red blood cell exchange transfusion. According to the National Institute for Health and Care Excellence (NICE) Guidelines, the committee was advised that venous access can be difficult for patients having exchange procedure particularly in very young children[1]. Cannulation of deep peripheral veins or central veins in the neck region or groin may require ultrasound guidance. The American Society for Apheresis (ASFA) currently recommends using peripheral access, if there is a safe and effective option[2].

From July 2017, the apheresis unit at St Georges University Hospitals NHS Foundation Trust has undertaken red blood cell exchange transfusion using ultrasound guided peripheral one arm cannulation as an alternative to two arm cannulation. The risk of recirculation was the main concern in performing the red blood cell exchange using the unilateral peripheral access.

A retrospective data analysis from July 2017 to February 2018 was performed identifying the patient's peripheral access either inserted unilaterally or bilaterally with the use of ultrasound machine. The data included the pre- and post- results of haematocrit, haemoglobin S percentage (HbS%), the percentage reduction of the HbS% and the site of inlet and return access used during the procedure.

A Chi-Squared test as recommended by Campbell (2007) and Richardson (2011) was used to compare the averages of percentage HbS% reduction.

During the 7 months period, a total of 331 red blood cell exchange procedures were performed with 193 had unilateral access and 127 with bilateral access. Both unilateral and bilateral access were analysed with 11 procedures excluded as there was no post blood results or the sample has clotted. The average reduction of HbS% was 68-65% when bilateral access was utilised and 65-45% when procedures were undertaken with the unilateral approach ($P = 0.5529$)

No significant difference in HbS% reduction efficiency was identified when comparing the two different cannulation approaches. The findings indicate that a unilateral peripherally ultrasound guided inserted access provide a satisfactory results in reducing the HbS% after the red blood cell exchange procedure.

Proficiency in the use of ultrasound machine and the anatomical knowledge of vein sites are some of the keys to a successful

peripheral insertion to undertake unilateral access in red blood cell exchange transfusion. However, patients with poor upper arm muscle tone, absence of large cephalic, brachial or basilica veins, vein structure and arm contractures are some of the reasons that may limit to attain unilateral cannulation access for the procedure.

Staff training in the use of ultrasound is highly recommended.

[1] NICE Guidelines, "Spectra Optia for automated red blood cell exchange in patients with sickle cell disease", Medical Technologies guidance [MTG28], Published date: March 2016.

[2] ASFA, "Five things patients and providers should question", Choosing Wisely: An initiative of the ABIM Foundation, Released April 2018.

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Breakthrough hemolysis in adult patients with paroxysmal nocturnal hemoglobinuria treated with ravulizumab: results of a 52-week extension from two phase 3 studies

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In patients (pts) with paroxysmal nocturnal hemoglobinuria (PNH) receiving eculizumab, ~11%–27% experience breakthrough hemolysis (BTH), which may be associated with suboptimal C5 inhibition or complement-activating conditions (CACs). This study defined BTH based on literature review and expert consensus: ≥ 1 new or worsening sign or symptom of intravascular hemolysis (fatigue, hemoglobinuria, abdominal pain, dyspnea, anemia [hemoglobin < 10 g/dL], major adverse vascular event [including thrombosis], dysphagia, or erectile dysfunction) in the presence of elevated lactate dehydrogenase (LDH) ($\geq 2 \times$ ULN) after prior LDH reduction to $< 1.5 \times$ ULN on therapy.

Ravulizumab, administered every 8 weeks (q8w) was studied in ALXN1210-PNH-301 (301; NCT02946463) and ALXN1210-PNH-302 (302; NCT03056040), and was noninferior to eculizumab 900 mg, administered every 2 weeks (q2w), for all primary and key secondary

endpoints. Each study had a 26-wk randomized primary evaluation period, with an extension where pts could receive ravulizumab. Here, we assess causes and clinical parameters associated with BTH incidences up to 1 yr in both studies.

Eligible pts were ≥ 18 with PNH. Complement inhibitor-naïve (Study 301) or -experienced pts (Study 302) received weight-based dosing of ravulizumab q8w or the approved eculizumab dose (900 mg q2w) for 183 d; in the extension, eligible pts continued (R-R arm) or switched to ravulizumab (E-R arm). The key outcome was the incidence, cause, and clinical parameters associated with BTH during the extensions of 301 and 302 (weeks 27–52), which was categorized as related to suboptimal C5 inhibition (free C5 ≥ 0.5 $\mu\text{g/ml}$), CAC, or unrelated to either event.

Study 301: 243 pts entered the extension (R-R arm: $n = 124$; E-R arm: $n = 119$). In both groups, 99% of pts who had no BTH during weeks 0–26 experienced no BTH incidents during weeks 27–52. Four pts in the R-R arm experienced BTH during weeks 27–52; 3 of these had BTH during weeks 0–26. In the E-R arm, there were fewer BTH events during weeks 27–52 compared with weeks 0–26. Two pts had BTH after switching to ravulizumab in the E-R arm, of which 1 had experienced BTH during weeks 0–26. During weeks 27–52, 2 BTH events (1 per arm) were associated with infection; 4 BTH events in the R-R arm and 1 in the E-R arm were unrelated to free C5 elevation or reported infection. Study 302: 191 pts entered the extension (R-R arm: $n = 96$; E-R arm: $n = 95$). Of these, 97% in the R-R arm and 100% in the E-R arm had no BTH during weeks 0–26 and had no BTH event during weeks 27–52. Three pts in the R-R arm had BTH during weeks 27–52. In the E-R arm, there were fewer BTH events during weeks 27–52 versus weeks 0–26. One pt experienced BTH after switching to ravulizumab; this pt had BTH during weeks 0–26. During weeks 27–52, 2 BTH events in the R-R arm and 1 event in the E-R arm were associated with infection; 1 BTH event in the R-R arm was unrelated to free C5 elevation or reported infection. In both studies, no BTH incidents were associated with free C5 of ≥ 0.5 $\mu\text{g/ml}$ during weeks 27–52 while on ravulizumab.

In both studies, a similar portion of pts had BTH. During weeks 27–52 in both studies, the number of pts who experienced BTH events was stable over time in the R-R group; fewer pts experienced BTH events after switching in the E-R group. No BTH events were associated with elevations in free C5 ≥ 0.5 $\mu\text{g/ml}$ in pts taking ravulizumab. Similar numbers of pts initially receiving ravulizumab or eculizumab experienced infection-related BTH events.

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Patient preferences for the treatment of paroxysmal nocturnal hemoglobinuria: results of a patient survey of ravulizumab (ALXN1210) and eculizumab in the UK

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Eculizumab administered every 2 weeks (q2w) has been the standard of care for paroxysmal nocturnal hemoglobinuria (PNH) since its approval in 2007. However, q2w dosing frequency may negatively impact patients' quality of life. Ravulizumab, a novel C5 complement inhibitor, is administered every 8 weeks (q8w) and has demonstrated noninferiority to eculizumab in two phase 3 trials on all primary and key secondary efficacy measures. In global substudy ALXN1210-PNH-302s, a significantly higher proportion of patients (pts) preferred ravulizumab over eculizumab. In regions where two treatment options are available for PNH, understanding patient preferences for ravulizumab or eculizumab could help guide future treatment decision making.

In the phase 3 open-label study (ALXN1210-PNH-302; NCT03056040), adult PNH pts stable on eculizumab for ≥ 6 months received weight-based ravulizumab or eculizumab in accordance with labeling for 26 weeks (randomization period). Pts were then enrolled in an extension period, all receiving ravulizumab. This substudy enrolled PNH pts who participated in the extension period and had received a minimum of 2 maintenance doses of ravulizumab during the extension phase. In this substudy, patient treatment preference was evaluated at one time point using an 11-item PNH Patient Preference Questionnaire (PNH-PPQ[®]). Differences in preference between ravulizumab and eculizumab were examined with an exact binomial test (Question [Q] 1), frequency distributions (Q2–3), and paired *t* tests (Q4–11), with $P < 0.05$ considered statistically significant. Standardized effect sizes (*d*) were calculated for differences between evaluations of ravulizumab and eculizumab on Q4–11.

Overall, 98 pts were enrolled: 95 pts from 8 countries completed Q1 of the PNH-PPQ, including 35 pts from the UK. The mean age

of participants was 53 years; 54.3% were female. Mean time since diagnosis was 13.7 years and mean number of days between the last randomized study treatment and completion of PNH-PPQ was 278.1. Overall, 91.4% of pts ($n = 32$) preferred ravulizumab vs 8.6% ($n = 3$; $P < 0.001$) who had no preference ($n = 2$) or preferred eculizumab ($n = 1$). For specific aspects of treatment, ravulizumab was predominantly preferred on frequency of infusions (100%), ability to plan activities (97%), convenience of receiving treatment (97%), overall quality of life (88%), and effectiveness of medication until the next infusion (76%). 'Frequency of infusions' was selected by the most participants ($n = 15$; 43%) as the most important factor determining treatment preference, followed by 'being able to plan activities' ($n = 5$; 14%) and 'overall quality of life' ($n = 5$; 14%). Moderate to large effect sizes were observed for factors favoring ravulizumab over eculizumab, including the frequency of infusions disrupting everyday life ($d = -2.14$, $P < 0.001$), feeling fatigued after infusions ($d = -0.71$, $P < 0.001$), and being able to enjoy life while receiving treatment ($d = 0.95$, $P < 0.001$).

In this clinical substudy of eculizumab-experienced PNH pts, a majority of UK pts preferred ravulizumab compared with eculizumab because of reduced infusion frequency (q8w vs q2w), better ability to plan activities, a more convenient treatment option, improved overall quality of life, and effectiveness of medication until the next infusion.

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Comparison of lost productivity due to eculizumab and ravulizumab treatments for paroxysmal nocturnal hemoglobinuria in France, Germany, Italy, Russia, Spain, the UK, and the USA

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Little is known about the productivity loss experienced by patients and caregivers due to treatment of paroxysmal nocturnal hemoglobinuria (PNH) in countries of different economic status. Internationally, some health technology assessment bodies require economic evaluations of direct medical costs (excluding productivity costs such as absenteeism, presenteeism, time lost from normal activities, and societal burden of illness) to be submitted by pharmaceutical manufacturers. However, patient and caregiver testimonies indicate that productivity costs are significant for PNH, as substantial time is required from both patients and caregivers for intravenous treatment administration at infusion clinics. When patients rely on unpaid caregivers, a complete accounting of costs should include caregivers' time loss. We assessed productivity loss for PNH

Table. Productivity costs (USD) associated with treatment with eculizumab q2w and ravulizumab q8w^a

Country	Eculizumab, Only In-Clinic Infusions (q2w)		Ravulizumab, Only In-Clinic Infusions (q8w)	
	Productivity Cost Without Caregivers Mean (SD) [range]	Increase in Productivity Cost per 10% Increase in Caregivers ^a Mean (SD) [range]	Productivity Cost Without Caregivers Mean (SD) [range]	Increase in Productivity Cost per 10% Increase in Caregivers ^a Mean (SD) [range]
United States	4,306,790 (1,323,653) [627,915-8,884,342]	414,865 (84,523) [288,675-573,179]	1,535,099 (471,858) [223,840-3,167,103]	147,892 (30,131) [102,907-204,328]
Germany	815,264 (270,883) [120,449-1,656,174]	79,543 (11,406) [64,110-101,247]	290,627 (96,565) [42,938-590,396]	29,006 (6,564) [21,085-41,680]
United Kingdom	575,711 (209,774) [93,505-1,323,222]	57,759 (6,835) [50,443-71,406]	205,230 (74,781) [33,333-471,704]	20,590 (2,437) [17,982-25,455]
France	564,088 (216,997) [8,764-1,523,924]	58,692 (15,001) [31,773-89,395]	201,087 (77,355) [3,124-543,251]	20,923 (5,347) [11,327-31,868]
Italy	408,217 (167,383) [21,934-1,021,167]	41,859 (17,790) [11,888-64,712]	145,522 (59,669) [7,819-364,027]	14,922 (6,342) [4,238-23,069]
Spain	405,431 (174,532) [0-1,082,426]	42,648 (14,654) [17,840-64,142]	144,529 (62,217) [0-385,865]	15,203 (5,223) [6,360-22,866]
Russia	343,969 (285,427) [0-1,735,393]	31,240 (25,327) [-16,816 to 67,906]	122,619 (101,750) [0-618,635]	11,137 (9,029) [-5,995-24,207]

q2w, once every 2 weeks; q8w, once every 8 weeks; SD, standard deviation; USD, United States dollars.

^aRefers to the increase in lost productivity costs for every 10% increase in the number of patients accompanied by a caregiver from baseline.

patients and caregivers due to treatment infusion in France, Germany, Italy, Russia, Spain, the United Kingdom and the United States (US).

Using a cost-consequence approach and published inputs, productivity costs were estimated for patients with PNH receiving eculizumab every 2 weeks (q2w) or ravulizumab every 8 weeks (q8w) in infusion clinics in 7 countries for 2 years. Inputs included estimated prevalence of PNH (1 in 500,000; Schuller Y, et al. *Orphanet J Rare Dis.* 2015), the published mean annual income for each country (World Bank, 2019). Estimation of total treatment duration included travel time to clinic, infusion time for loading and maintenance doses, and recovery time (240 minutes for eculizumab; 330 minutes for ravulizumab), and was assumed to be the same across all countries. Lost wages were estimated using the human capital approach based on the hourly wage rate (derived from each country's 2018 gross national income per capita) and assumed a 40-hour work week. All costs were converted to 2018 USD. Time and costs of caregivers were not considered in baseline analyses. We undertook 11 simulations (each made up of 1000 iterations) where patient population size and lost wages were normally distributed random variables and the proportion of patients accompanied by caregivers increased by 10% intervals.

Mean annual income varied 6-fold between countries. The table shows total productivity loss due to eculizumab and ravulizumab treatment for PNH patients and the incremental cost increases for every 10% increase in the number of patients with accompanying caregivers in each country. For eculizumab, lost productivity ranged from \$344,000 (Russia) to \$4.3 million (US), without caregivers. In each country, lost productivity due to ravulizumab therapy was approximately three-quarters lower than lost productivity for eculizumab due to less frequent dosing, ranging from \$123,000 (Russia) to \$1.5 million (US), without caregivers. On isolating population size and lost wages, the latter had larger proportionate effects in lower-income countries (data not shown).

Omitting the productivity costs of unpaid caregiving in PNH underestimates the total societal burden of illness. Use of an q8w over an q2w treatment creates substantially smaller productivity losses. Savings were numerically greater in higher-income countries (US and Germany), where the opportunity cost of patient time is greater. In chronic diseases requiring lifelong therapy such as PNH, societal burden of illness and relative treatment effects should be considered when assessing treatment value.

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BSH2020-PO-153

Association between protein C, protein S and cerebral blood flow velocity in sickle cell disease patients with high risk of cerebrovascular accident

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The main trigger of thrombosis in sickle cell patients is hyper-coagulable state; which may due to deficiency of natural anticoagulant such as protein C and protein S among other abnormalities of the coagulation factors. This study aimed to determine association between protein C and S *vis-a-vis* cerebral blood flow velocity in sickle cell anaemia patients with high risk of cerebrovascular accident (CVA).

This was a case-control study involving HbSS patients who have were categorized based on risk of cerebrovascular accidents using transcranial doppler (TCD), and HbAA age-matched controls. All participants' samples were assayed for protein C, S, full blood count, prothrombin time (PT) and activated partial prothrombin time (APTT). The association between the assayed parameters and TCD values were established.

A total of 150 participants with Hb SS phenotype and 80 HbAA controls were recruited. There was a statistically significant increase ($P < 0.05$) in the time average mean velocity (TAMV) of the participants with sickle cell anaemia (184.29 ± 4.39 secs) in comparison to the control group (119.00 ± 1.42 secs). Similarly, decreased levels of proteins C and S observed in the participants with sickle cell anaemia (46.21 ± 5.56 %, 128.90 ± 3.86 %) respectively in comparison with the control group (65.76 ± 3.87 , 155.13 ± 3.50 %), were also statistically significant ($P < 0.05$). As the risk of CVA increased from standard/normal risk, conditional risk to high risk, the levels of protein C significantly ($P < 0.05$) decreased. However, the observed decrease in the levels of Protein S was not statistically significant.

The result of this study showed significant difference ($P < 0.05$) in the plasma levels of protein C and S in the HbSS group compared with the HbAA controls. The TAMV was also significantly increased in the HbSS group. There was also a significant negative regression coefficient between high TAMV and protein C levels ($r^2 = 0.073$, $P = 0.03$) but not with protein S ($r^2 = 0.019$, $P = 0.028$).

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BSH2020-PO-154

Formation of multivalent complexes in the presence of more than one conventional antibody to complement factor C5

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When considering development of therapeutic monoclonal antibodies, it is essential to account for the properties of the circulating

immune complexes formed with the target antigen. In particular, the valency of such complexes can affect clearance, effector function and uptake by phagocytic cells involved in antigen processing and inflammation. Therefore, significant care and consideration must be given to when a recipient is exposed to more than one immunoglobulin G (IgG) antibody which bind to a monovalent target at different sites, potentially leading to formation of multivalent immune complexes. For example, this could arise when a patient being treated with one anti-complement component 5 (C5) IgG is switched to a different anti-C5 IgG that binds to a different site on C5, wherein finite levels of the first antibody remain in circulation to maintain continuous pharmacodynamic C5 suppression.

Eculizumab, a humanized monoclonal antibody targeting C5, has proven transformative for patients with paroxysmal nocturnal hemoglobinuria (PNH), atypical hemolytic uremic syndrome (aHUS), refractory generalized myasthenia gravis (MG) and aquaporin-4 IgG positive neuromyelitis optica spectrum disorder. Ravulizumab is a second-generation anti-C5 antibody therapy that has recently received regulatory approvals for the treatment of patients with PNH and aHUS. Notably, ravulizumab binds the same epitope on C5 as eculizumab, but has an extended half-life compared with eculizumab. Ravulizumab also has a differentiated PK/PD and clinical profile to eculizumab.

We sought to better understand the potential for formation of multivalent complexes in recipients who receive other non-competitive anti-C5 antibodies, while maintaining detectable levels of eculizumab in circulation. As an initial step, size exclusion chromatography, in combination with Multi Angle Light Scattering, was used to assess the size distribution of ternary complexes formed *in vitro* between eculizumab, C5 and two other anti-C5 antibodies having the same sequences as antibodies currently undergoing clinical trials that bind to different epitopes on C5. We identified the formation of a range of complexes, some of which were of a size consistent with the incorporation of multiple anti-C5 antibodies.

Such multivalent complexes, resulting from the combination of C5 with two antibodies binding different epitopes, have unknown pharmacologic properties, and the safety of such complexes if formed *in vivo* is presently unknown. This highlights the importance of rigorous functional characterization of such complexes and incorporation of monitoring and mitigation processes to assess and prevent their potential formation in patients. Other methods of avoiding multivalent complexes would include switching among antibodies recognizing and binding the same epitope, such as switching from eculizumab to ravulizumab, or switching to a monovalent C5 antagonist which is unable to promote the formation of multivalent immune complexes.

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BSH2020-PO-155

Hydroxyurea therapy does not impact TCD velocity in the East London sickle cell newborn cohort study

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Trans Cranial Doppler (TCD) is used as a screening test to prevent acute ischaemic stroke in children with sickle cell disease (SCD). Hydroxyurea (HU) is a disease modifying agent which has been shown to reduce maximal TCD velocity in children with SCD. We wished to evaluate the longer-term effects of HU on TCD velocities and TCD stroke risk criteria in HbSS children in the East London Newborn Cohort Study

Inclusion criteria were diagnosis of homozygous sickle cell disease, birth in the London boroughs of Hackney and Tower Hamlets, diagnosis by newborn screening (local programme from 1983 to 2004, and national programme from 2004 onwards), referred and seen at least once in this specialist referral centre. Permission to study all patients in the cohort without individual consent was obtained from the UK National Confidentiality Advisory Group and National Research Ethics Committee. TCD screening programme was initiated in the early 1990's and became a universal screening programme from 2002. TCD was evaluated by non-imaging methodology up until 2016, when imaging was also introduced. Scans have been done by 4 practitioners over this time period (PT, BK, FK) and imaging by one practitioner (NN) TCD Max velocity values were compared between groups of patients not receiving any treatment and those on HU falling within these age brackets [0-5 yrs], [6-10 yrs] and [11-15 yrs]. Patients under 2 years of age were excluded. Patients on chronic transfusion programme for > 6 months, or undergoing haematopoietic stem cell transplant were excluded at the onset of this treatment. Subjects were followed from birth to age 15 or 31st December 2018, whichever occurred first.

During follow up of 266 patients 3 patients died during FU, 3 patients had BMT. 63 (23.7%) initiated regular transfusion, 53 (31.93 %) patients with HbSS have been treated with HU for > 6 months for a total of 224.49 patient yrs. The median age at start of HU was 10 yrs. (IQ range 6 - 13). 2 patients (3.39%) died while on HU. Overall, the average maximal TCD velocity for patients not on HU was 133.6 cm/sec and for those on HU was 135.6 cm/sec. The difference between these was not significant. The average maximal TCD velocity for male not on HU was 130.2 cm/sec and for those on HU was 134.5 cm/sec. Whilst for female, the average maximal TCD velocity not on HU was 137.0 cm/sec and 136.3 cm/sec for those on HU. When categorised according to age group: 0-5 yrs old no treatment 1 (141.5) & HU (144); 6-10 yrs old no treatment (135.3) & HU (136); 11-15 yrs old control (118.8) & HU (123.6). For follow-up not on HU 80 patients remained Standard Risk (SR), 2 (2.15%) progressed to conditional and 1 (1.08%) progressed to abnormal. For follow up on HU 25 remained SR, 3 (6.98%) progressed to conditional, 0 progressed to abnormal.

In this observational, real world study, a significant effect of HU on reducing TCD velocities has not been demonstrated, particularly at a young age. One possible confounding factor is that the indication for initiation of HU may have been influenced by higher TCD velocity. Many of these children were not on maximal tolerated dose of HU and therefore may not have achieved optimal benefit in TCD velocity reduction. Results will be analysed in more detail in a larger population of E London patients using multivariate analysis. Earlier

initiation of disease-modifying treatment and longer-term follow-up will be required to provide evidence of treatment effects.

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Haemoglobin distribution width is low in alcoholic liver disease

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Alcoholic liver disease (ALD), B₁₂ & folate deficiency and drug treatments are common causes of red cell macrocytosis. Traditional red cell indices such as red blood cell count (RBC), haemoglobin (Hb), haematocrit (Hct), mean cell volume (MCV), mean cell haemoglobin (MCH), mean cell haemoglobin concentration (MCHC) and red cell distribution width (RDW) are unable to reliably differentiate these in routine practice.

The haemoglobin distribution width (HDW) is a patient reportable parameter derived from the standard deviation of the red cell haemoglobin concentration histogram using ADVIA 2120i haematology systems (Siemens Healthcare Diagnostics, Tarrytown, NY, USA). Whereas the RDW is an automated measurement of anisocytosis, the HDW is an automated measurement of anisochromia (haemoglobin concentration variance). We evaluated the clinical utility of the RBC, Hb, Hct, MCV, MCH, MCHC, RDW and HDW parameters to differentiate alcoholic liver disease from other causes of red cell macrocytosis using an ADVIA 2120i.

Eighty samples sent for a routine full blood count (FBC) that presented with an MCV > 105 fL were used in this study. Three categories were investigated; alcoholic liver disease ($n = 26$), B₁₂ or folate deficiency ($n = 15$) and medication related ($n = 39$). Analyzer quality control and maintenance tasks were performed according to the manufacturer's instructions.

Traditional red cell indices such as RBC, Hb, Hct, MCV, MCH, MCHC and RDW were shown to be unable to differentiate ALD from B₁₂ & folate deficiency or drug treatments. Our study showed that only the HDW was significantly different across all three categories. In ALD the HDW measurement was found to be lower than most cases of B₁₂ & folate deficiency or treatment related causes of macrocytosis. The red cell haemoglobin concentration histogram from which the HDW is derived is correspondingly narrow in ALD and the 2-dimensional red cell volume vs. haemoglobin concentration plot also has a characteristic shape not typically seen in cases other than ALD. Stomatocytes and target cells are associated with a

low HDW and both are a frequent finding in ALD which would explain the observed results.

Our study showed that an MCV > 105 fL combined with a low HDW were suggestive of alcoholic liver disease. In cases presenting with a raised MCV and low HDW the possibility of alcohol misuse should be suspected and investigated accordingly.

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BSH2020-PO-157

Transcranial doppler long-term follow-up in children with sickle cell disease

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Children with Sickle Cell Disease (SCD) are known to be at increased risk of stroke. Screening for risk of stroke using Transcranial Doppler (TCD) is now standard care, and those identified with abnormally high TCD velocities (≥ 200 cm/s) receive regular Exchange Blood Transfusion (EBT). Following the report of the TCD with Transfusion changing to Hydroxyurea (TWITCH) study, Hydroxycarbamide (HU) has been offered as an alternative to transfusions in children who have received at least 1 year of transfusions with no evidence of severe cerebral vasculopathy. Long term follow-up (FU) data of cerebral velocities with and without EBT or HU therapy is limited. The objective of this retrospective analysis was to determine the progression of TCD velocities in children with SCD on a long-term FU irrespective of the intervention.

Children with SCD aged 2-16 years with at least 5 consequent TCD scans (time-point evaluations T0 to T4) recorded not less than 6 months apart between 2010 and 2018 were included. The highest Time Averaged Maximum Velocity (TAMV) measured in the Middle Cerebral Artery (MCA) and distal Internal Carotid artery (dICA) were recorded. Patients were divided into 3 groups: those who received No Treatment at any time (NoT), those who started HU and those who were initiated on EBT regimen for multiple etiologies. For those included in the treatment groups (both HU and EBT), the T0 TCD recorded was the one before starting any therapy. Statistical analysis was performed to compare clinical characteristics and TCD scans over time among the 3 groups.

During the study period, 135 children with SCD were screened with TCD at a tertiary hospital in London, UK. After excluding 73 patients due to insufficient time-point evaluations, 62 were retained in the final analysis (36 males, 58%, all HbSS). Mean age at the beginning (T0) and at the end (T4) of FU was 6.9 ± 3.5 and 10.8 ± 3.5 years respectively: 25/62 (40%) patients received NoT; HU was started in 28/62 (45%) children at a mean age of 8.2 ± 2.4 years, while 9/62 (15%) were included in EBT program at a mean age of 7.9 ± 4 . No significant differences were shown between groups in terms of demographic characteristics. At baseline TAMV was higher in the EBT group (153.7 ± 45.3 cm/s) than in the HU (121.6 ± 23.1 cm/s) and NoT groups (118.6 ± 19.9 cm/s; $P = 0.003$). Children in the NoT group showed no significant changes over time of TAMV measured. A significant decrease over time of TAMV was observed for patients in the HU group ($P = 0.034$), with no significant differences between single time-point evaluations. Children in the EBT group showed a significant change

over time either for MCA TAMV ($P = 0.004$) and dICA TAMV ($P = 0.039$), with significant decreases from T0 to T3 (mean age 9.1 ± 4.4 ; $P = 0.043$) and from T0 to T4 (mean age 10.1 ± 3.8 ; $P = 0.009$).

We provided additional evidence that HU therapy can effectively lower TCD values and that this effect is sustained over long time. EBT is effective in lowering TCD values as well, and this is effect is more homogenous over time.

Disclosure of Interest: None Declared

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Characteristics and outcomes of patients with sickle cell disease admitted to pediatric intensive care, a retrospective review

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Sickle cell disease (SCD) is an autosomal recessive disorder common in Saudi Arabia (KSA). This is the first study in KSA and middle-east revealing the characteristics of pediatric SCD patients admitted to pediatric intensive care (PICU).

We retrospectively reviewed the records of pediatric SCD patients < 14 years who were admitted to PICU over 6-years-period from Dec 2013-Dec 2019. Study was done in King Salman Armed Forces hospital located in Tabuk, KSA. Data were collected using patient's files, electronic records and PICU admission book.

Average number of pediatric SCD patients < 14 years following in our hospital/year is 240 patients, 58 SCD patients admitted 93 times to PICU, accounting for 4.5% of total PICU admission over 6-year-period, rating 6.5 admissions/100 active SCD patients/year. Sixteen patients were admitted more than once. Majority of admissions were from the station and emergency room (ER) (38, 33 admissions respectively). Admissions categorized into emergency and elective (59, 34 admissions respectively). Average length of stay 1.9 days.

Acute chest syndrome (ACS) ($n = 28$) represented 30% of total admissions and majority (47.5%) of emergency admissions, patients with ACS were admitted first to the station suffering vaso-occlusive crisis (VOC) or respiratory manifestations before developing ACS that necessitated transfer to PICU in 20 (71.4%) admissions. Furthermore, in a significant percentage of admissions, patients with ACS had high WBC count and high HBS level > 50% (89%, 93% respectively). Cerebrovascular accident (CVA) was the second most common cause of admission (18.2%), most from ER (64%). Five patients (5.37%) admitted for other neurological causes, 4 admissions for gastrointestinal complications, 2 were admitted in shock following splenic sequestration and 1 admission for each of sepsis, priapism and severe hemolysis. On the other hand, there were 25 (26.8%) elective admissions for stroke prevention, and 9 (9.6%) elective admissions for post-operative recovery.

A considerable number of exchange transfusion was performed in 65 (69.8%) admissions, no significant association between exchange transfusion and admission categories, $P = .56$. While patients were more likely to have simple blood transfusion in the emergency admissions ($n = 16$, 17.2%) than patients who were electively admitted, $P < .05$. Only 3 needed inotropes, 5 invasive ventilation and oxygen required in 43 (46.2%) admissions. Antibiotics prescribed in half of admissions. Central venous line in 81 (87%) admissions, 2 intubated, one had chest tube.

WBC counts were significantly higher in emergency group (mean, $18.37 \times 10^3/\mu\text{l}$) than in elective group (mean, $10.4 \times 10^3/\mu\text{l}$), $P < .05$. Whereas, hemoglobin was significantly lower in emergency group (mean, 7.8 g/dl) than in elective group (mean, 9.67 g/dl), $P < .05$. Contrarily, mean platelet counts were $383 \times 10^3/\mu\text{l}$ in emergency group, and $347 \times 10^3/\mu\text{l}$ in elective group, which was not statistically significant, $P = .36$. Out of 58 patients, 22.4% had previous stroke, 34.4% had previous ACS, whereas significant number ($n = 37$, 63.8%) had frequent VOC, 70% of patients were prescribed hydroxyurea and 27% were on regular transfusion. Mortality rate of SCD in PICU was low ($n = 2$, 3.4%), one died of sepsis and one from severe hemolysis.

PICU plays pivotal rule in management of SCD patients. Outcomes dramatically improved in presence of dedicated team, who are essential participants in comprehensive services for SCD patients.

Disclosure of Interest: None Declared

Table.

Total number of admissions: $n = 93$

Reason for admission	Number of admissions (% from total)	Invasive ventilator support	O2 support	Inotrope support	Exchange transfusion	Simple blood transfusion	Antibiotic usage
Emergency admissions $n = 59$							
ACS	28 (30.1)	2 (7.1)	28 (100)	0 (0)	16 (57.1)	11 (39.2)	28 (100)
Sepsis	1 (1.07)	1 (100)	1 (100)	1 (100)	0 (0)	1 (100)	1 (100)
CVA	17 (18.2)	0 (0)	0 (0)	0 (0)	17 (100)	0 (0)	4 (23.5)
Other neurologic (PRES, TIA, SAH, acute soft head)	5 (5.37)	1 (20)	2 (40)	1 (20)	4 (80)	0 (0)	3 (60)
GI emergency (pancreatitis, liver failure, hepatic sequestration)	4 (4.3)	0 (0)	3 (75%)	0 (0)	3 (75%)	1 (25%)	4 (100)
Splenic sequestration	2 (2.15)	0 (0)	1 (50)	0 (0)	0 (0)	2 (100)	1 (50)
Others (priapism, hemolysis)	2 (2.15)	1 (50)	2 (100)	1 (50)	0 (0)	1 (50)	1 (50)
Elective admissions $n = 34$							
Stroke prevention	25 (26.8)	0 (0)	0 (0)	0 (0)	25 (100)	0 (0)	0 (0)
Post-operative	9 (9.6)	0 (0)	6 (66.6)	0 (0)	0 (0)	0 (0)	3 (33.3)
Total (% of total)	93(100)	5(5.37)	43(46.2)	3(3.2)	65(69.8)	16(17.2)	45(48.3)

Specifications of emergency and elective PICU admissions for SCD patients. ACS, acute chest syndrome; CVA, cerebrovascular accident; PRES, posterior reversible encephalopathy syndrome; TIA, transient ischemic attack; SAH, subarachnoid hemorrhage

BSH2020-PO-159

A retrospective laboratory assessment of the pyruvate kinase enzyme assay and analysis of 10 novel variants

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Pyruvate kinase (PK) deficiency is an autosomal recessive disease caused by mutations in the *PKLR* gene, which reduce PK enzyme activity and result in decreased energy synthesis in erythrocytes. This causes the red blood cells to break down rapidly, resulting in haemolytic anaemia. The severity of the disease varies between patients from mild anaemia to hydrops fetalis. Diagnosis is primarily by an enzyme assay. The reference range for the enzyme assay varies between different diagnostic and academic groups, and sometimes individuals with borderline low PK levels can be diagnosed differently upon repeat testing. The aim of this study is to review the enzyme assay reference range of 11–19 IU/g Hb using genetic analysis. Demographic data was also audited on individuals registered at King's College Hospital to determine whether PK levels differ with variables such as gender, ethnic origin and age.

Nine novel variants, five of which are likely pathogenic, and a large deletion were found in the *PKLR* gene during this study (c.319 A>G, c.508-2 A>G, c.508-57 G>A, c.627 G>C, c.822 C>G, c.959T>C, c.965 + 2 T>C, c.1076G>C, c.1299 C>A and c.1437-131_c.1619-114). These were assessed using the PK activity and *in silico* prediction tools. No individuals with a PK activity > 11 IU/g Hb possessed definitive pathogenic variants. However, individuals with a PK level ≤ 11 IU/g Hb were found to have a mixture of zero, one and two or more pathogenic variants. Based on this data, it is proposed that an enzyme activity of 7.8–11 IU/g Hb be reported as "Possible PK deficiency", with confirmation by genetic analysis for results below 11 IU/g Hb. There was no significant difference found in PK activity between those of African and White European ethnic origin, showing a single reference range is suitable for these groups. A significant difference was found between males and females with regards to age, indicating PK activity decreases with age in females. Further studies on how the enzyme activity is affected by non-*PKLR* factors, and extended protein studies to further understand the effect of the novel variants found, would be useful.

Disclosure of Interest: None Declared

BSH2020-PO-160

Luspatercept significantly reduces red blood cell transfusion burden, regardless of gene mutation frequency, spectrum, and prognostic significance, among patients with lower-risk myelodysplastic syndromes enrolled in the MEDALIST trial

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The phase 3 MEDALIST trial (NCT02631070) evaluated luspatercept in patients with anaemia due to lower-risk myelodysplastic syndromes (LR-MDS) (Revised International Prognostic Scoring System [IPSS-R]-defined Very low-, Low-, and Intermediate-risk) with ring sideroblasts (RS), who were refractory, intolerant, or ineligible to receive erythropoiesis-stimulating agents (serum erythropoietin > 200 U/L) and required red blood cell (RBC) transfusions. This study explored the associations of mutations with response to luspatercept in the MEDALIST trial.

DNA was isolated from bone marrow mononuclear cells from 222 of 229 patients (148 luspatercept, 74 placebo) at screening and following treatment. Next-generation sequencing of 23 MDS-relevant genes was performed at screening and every 24 weeks. Bone marrow cell populations were analysed by cytomorphology. Response status (RBC transfusion independence for at least 8 weeks within the first 24 weeks of treatment) was used for correlative analyses.

Of evaluable patients, 91% had *SF3B1* mutations at baseline. A median of 2 of the 23 MDS-relevant genes analysed were mutated per patient. Mutation profiles were balanced between luspatercept and placebo arms. Luspatercept responders and non-responders had similar numbers of mutated genes at baseline, and comparable response rates were achieved regardless of the number of mutations. Response to luspatercept was independent of mutation in any of the genes analysed, whether individually or when grouped by functional categories. Similar mutation profiles were seen in responders and non-responders. Response rates were similar regardless of baseline *SF3B1* allelic burden.

No statistically significant differences were observed in gain or loss of mutations between treatment arms at Week 24, and no difference was seen in variant allele frequency for mutations in genes associated with adverse prognosis.

Patients enrolled in MEDALIST had mutations consistent with LR-MDS with RS, with a preponderance of *SF3B1* mutations. Responses to luspatercept were achieved regardless of *SF3B1* allelic burden, number of baseline mutations, and presence of individual mutations.

Luspatercept is a potential new therapy for the treatment of anaemia in patients with LR-MDS with RS, regardless of the number of associated mutations. Further study of the predictors of response is warranted.

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Thrombosis and Haemostasis

BSH2020-PO-161

Bone and joint health markers in persons with haemophilia A (PwHA) treated with emicizumab in HAVEN 3

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Emicizumab is a recombinant, humanised, bispecific monoclonal antibody that restores the function of missing activated factor VIII (FVIII) by bridging activated FIX and FX in persons with haemophilia A (PwHA). Prophylaxis with emicizumab once weekly or every two weeks resulted in significant reductions in joint bleeds, and a favourable safety profile in PwHA without FVIII inhibitors in the HAVEN 3 study (NCT02847637). To explore the potential effect of emicizumab prophylaxis on bone and joint health beyond bleed prevention, we measured joint health scores and bone and joint biomarkers in HAVEN 3.

Haemophilia joint health scores (HJHS; v2.1) were evaluated at baseline and Week 49 of emicizumab prophylaxis in 107 PwHA in HAVEN 3. Biomarkers of bone formation (osteocalcin [OC], N-terminal propeptide of type I procollagen [P1NP]), bone resorption (C-terminal telopeptide of type I collagen [CTX-I]), osteoblasts (osteoprotegerin), osteoclastogenesis (soluble receptor activator of nuclear factor- κ B Ligand), cartilage turnover (cartilage oligomeric matrix protein [COMP]), cartilage degradation (C-terminal telopeptide of type II collagen), cartilage synthesis/repair (a chondroitin sulfate epitope—CS-846), and inflammation (interleukin 1 beta [IL-1 β], IL-6, and tumour necrosis factor) were measured in 117 PwHA (Table 1) receiving emicizumab at baseline and after 3, 6, 12, and 18 months of treatment. In all, 78 PwHA were part of both the biomarker analysis and HJHS evaluation.

PwHA previously on FVIII prophylaxis and those with no target joints at study entry had lower (indicating healthier) HJHS at baseline. Mean improvements from baseline of -2.25 (95% confidence interval [CI]: -4.12 , -0.39) in total HJHS and -2.23 (95% CI: -4.07 , -0.38) in HJHS joint-specific domain (excluding gait) were observed after 48 weeks of emicizumab prophylaxis in PwHA with one or more target joints at study entry ($n = 71$). Improvement was consistent across HJHS for different joints. No significant differences in the measured biomarkers between PwHA previously on FVIII prophylaxis or on on-demand treatment, or in those with or without target joints, were seen at baseline. Mean baseline values of most bone and joint biomarkers were within normal ranges, or similar to published levels in healthy individuals, although large variability was observed between individuals. None of the measured biomarkers changed significantly during emicizumab prophylaxis. Higher OC, P1NP, and CTX-I levels were observed in adolescent vs adult PwHA at all time points, which is consistent with reported increases of these biomarkers during skeletal growth.

This analysis showed clinically relevant improvements in HJHS (defined as a ≥ 2 -point reduction in HJHS joints domain) after

48 weeks of emicizumab prophylaxis. The biomarkers measured in blood as surrogates of bone and joint health did not show significant changes during 18 months of emicizumab prophylaxis. This may reflect heterogeneity between individuals, and factors other than joint health. However, improvement in bone and joint biomarkers would have been unexpected as the observed means at baseline were already similar to levels reported in healthy individuals. We observed no indication of worsening of the measured bone and joint health markers that might have resulted from reduced exposure to FVIII. Additional data are needed to better understand the long-term effect of emicizumab prophylaxis on bone and joint health.

Table. Characteristics of PwHA receiving emicizumab* who had evaluable HJHS and biomarker measurements

Characteristics	PwHA in HAVEN 3 with evaluable HJHS, $n = 107$	PwHA in HAVEN 3 with biomarker data, $n = 117$
Mean age, years (min–max)	35.7 (13–77)	38.4 (13–77)
≥ 65 years, n (%)	2 (1.9)	5 (4.3)
< 18 years, n (%)	7 (6.5)	7 [†] (6.0)
Mean BMI (min–max)	26.0 (19.2–40.6)	26.0 (16.8–40.6)
Race, n (%)		
White	64 (59.8)	80 (68.4)
Asian	28 (26.2)	22 (18.8)
Black/African American	5 (4.7)	5 (4.3)
Other	1 (0.9)	1 (0.8)
Unknown	9 (8.4)	9 (7.7)
Prior FVIII prophylaxis, n (%)	47 (43.9)	50 (42.7)
Prior FVIII episodic treatment, n (%)	60 (56.1)	67 (57.3)
No target joints, n (%)	36 (33.6)	38 (32.5)
≥ 1 target joints, n (%)	71 (66.4)	79 (67.5)
History of HIV infection [‡] , n (%)	17 (15.0)	31 (26.5)
Any osteoporosis [‡] , n (%)	1 (0.9)	5 [†] (4.3)
Treated osteoporosis [‡] , n (%)	1 (0.9)	4 [†] (3.4)

BMI, body mass index; FVIII, factor VIII; HIV, human immunodeficiency virus; HJHS, haemophilia joint health scores; PwHA, persons with haemophilia A; QW, once weekly; Q2W, once every 2 weeks.

*Received loading doses of 3 mg/kg emicizumab QW for 4 weeks followed by either 1.5 mg/kg QW or 3 mg/kg Q2W maintenance doses, according to the HAVEN 3 protocol.

[†]All in Arm D of HAVEN 3 (participants who had received prior FVIII prophylaxis, and received loading doses of 3 mg/kg emicizumab QW for 4 weeks followed by 1.5 mg/kg QW maintenance).

[‡]HIV and osteoporosis numbers are based on CSR information applied on these subsets.

Data cut-off: 4 Oct 2018.

Disclosure of Interest: A. Kiialainen Conflict with: Employer and Equity Ownership interests - F. Hoffmann-La Roche Ltd., M. Niggli Conflict with: Employer - F. Hoffmann-La Roche Ltd., C. Kempton Conflict with: Research funding - Novo Nordisk, Octapharma, Pfizer, Genentech, Inc., Spark Therapeutics., G. Castaman Conflict with: Research funding - Pfizer, CSL Behring, Sobi, Speakers Bureau and Membership on an entity's Board of Directors or advisory committees - Bayer, Takeda (SHIRE), CSL Behring, Sobi, Novo Nordisk, Kedrion, Uniqure, F. Hoffmann-La Roche Ltd, Werfen, Sanofi., T. Chang Conflict with: Employer - Genentech, Inc., Equity Ownership interests - Genentech, Inc./F. Hoffmann-La Roche Ltd., I. Paz-Priel Conflict with: Employer - Genentech, Inc., Conflict with: Employer - Genentech, Inc., J. Adamkewicz Conflict with: Employer - Genentech, Inc., Equity Ownership interests - F. Hoffmann-La Roche Ltd., G. Levy Conflict with: Employer - Genentech, Inc., Equity Ownership interests - F. Hoffmann-La Roche Ltd.

BSH2020-PO-162

Plasma MicroRNAs dysregulation in postpartum pre-eclamptic women: implication for the development of cardiovascular disease events and deep vein thrombosis later in life?

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Pre-eclampsia (P-EC) is associated with long-term complications including two to four-fold increased risk of developing cardiovascular disease (CVD) and Deep Vein Thrombosis (DVT). MicroRNAs (miRNAs) are essential post-transcriptional regulators of gene expression and have been implicated in several disease conditions including CVD and DVT. The aims of this study are 1) To investigate miRNAs expression in postpartum women who had a history of P-EC and 2) To examine if alterations in plasma miRNA dysregulation are linked to subsequent development of CVD events and DVT. The study was performed in two phases, Discovery and Confirmation. *Discovery Phase*, blood specimens were obtained post-delivery from women who had P-EC ($n=30$) at yearly intervals including 2007, 2012, 2013, 2014, 2015 and 2016, i.e., five women in each year. In addition, blood samples ($n=5$) were collected from healthy women who had uncomplicated pregnancies in 2016 (controls). Blood specimens were collected in vacutainer tubes containing Sodium Citrate. Next-Generation Sequencing (NGS) was performed to identify miRNAs differential dysregulation according to the manufacturer's protocol (QIAGEN Genomic Services, Germany). A total of 417 miRNAs were established, 14 were identified as significantly and differentially downregulated and annotated, with log fold change (log FC). *Confirmation Phase*, in order to confirm and validate the miRNAs which were significantly and differentially downregulated in the Discovery Phase, the sample size was increased to 66 subjects i.e., 2007 ($n=7$), 2012 ($n=7$), 2013 ($n=18$), 2014 ($n=11$), 2015 ($n=10$), 2016 ($n=13$) and controls 2016 ($n=14$). The validation was performed using quantitative reverse transcription-polymerase chain reaction. We found 14 miRNA which were significantly and differentially downregulated at the Discovery Phase, these are (*hsa-miR-103a-3p*, *hsa-miR-190a-5p*, *hsa-let-7f-5p*, *hsa-miR-6734-5p*, *hsa-miR-454-3p*, *hsa-let-7b-5p*, *hsa-miR-301a-3p*, *hsa-miR-26a-5p*, *hsa-miR-4467*, *hsa-miR-144-5p*, *hsa-miR-374a-5p*, *hsa-miR-652-3p*, *hsa-miR-191-5p*, *hsa-miR-5010-3p*). Subsequently, in the Confirmation Phase, plasma *miR-103a-3p*, *has-let-7f-5p* and *has-miR-193a-5p* were significantly and differentially dysregulated in the P-EC group compared to controls, with log FC = -1.3, -1.6, +2.1; BH-adj, $P < 0.05$,

$P < 0.05$ and $P < 0.01$, respectively). In conclusion, this is the first report that directly links plasma miRNAs dysregulation with P-EC, mainly 1) *SCN8A* target gene; an ion channels associated with cardiac excitability and arrhythmia, 2) *miR-103a-3p*; contributes to Endothelial Progenitor Cell Dysfunction and implicated in DVT through targeting phosphatase and tensin homolog (PTEN), 3) *let-7f-5p*; implicated in post-translational modification of ion channels leading to fatal arrhythmias and heart failure, and 4) *miR-193a-5p*; involved in various disease processes, especially ischemic injury. Thus, our results provide novel findings and strong evidence linking gene dysregulation in P-EC and subsequent development of cardiovascular events and DVT.

Keywords: microRNA, Pre-eclampsia, Cardiovascular Disease, Deep Vein Thrombosis, Next Generation Sequencing, Quantitative Reverse transcription-polymerase Chain Reaction.

Disclosure of Interest: None Declared

BSH2020-PO-163

Prothrombin complex concentrate versus fresh frozen plasma in adults undergoing heart surgery (PROPHECY) - a pragmatic pilot randomized controlled trial

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Background: Fresh Frozen Plasma (FFP) is the standard treatment for clotting factor replacement in bleeding patients undergoing cardiac surgery in the UK. However, prothrombin complex concentrate (PCC) is being used off-license in this setting due to higher concentration of clotting factor levels, smaller volume, and more rapid administration as compared to FFP.

Method: PROPHECY is a pragmatic single-centre, pilot randomized, controlled trial (RCT) that assessed the feasibility of a large trial to compare FFP with PCC in cardiac surgery. The trial was registered on the EudraCT database (2018-003041-41) and Clinicaltrials.gov (NCT03715348). Primary outcome was to evaluate the proportion of subjects who are consented and receive the intervention. *Secondary objectives* were to assess the delivery of different components of the trial, protocol violation/compliance and ability to collect outcome data up to 90 days after surgery. *Inclusion criteria:* Adult patients (≥ 18 years, not on vitamin-K antagonists) undergoing cardiac surgery who develop active bleeding that requires FFP transfusion within 24 hours of surgery, were randomized by the transfusion laboratory to FFP (15 ml/kg) or PCC (500 IU if < 60 kg, 1000 IU if 61–90 kg, 1500 IU if > 90 kg) using block randomization. If bleeding continued after administration of the first study dose, standard care was administered (no subsequent PCC dose was given). Follow-up was 90 days after surgery.

Results: From January 2019 to November 2019, 180 patients were screened, of which 134 were consented, 54 (40% recruitment rate)

received an intervention (25 FFP and 29 PCC) and 50 were appropriately randomised (25 on each arm). Of those that were randomised, the median age was 66 years (SD 12), 64% were male and 76% (PCC arm) and 80% (FFP) had elective surgery. There were 18 trial protocol deviations relating to randomisation ($n = 10$), intervention ($n = 3$), documentation ($n = 2$) and research blood samples collection ($n = 3$). In randomised patients 113 adverse events (55 PCC vs. 58 FFP) and 19 serious adverse events (6 in 5 patients for PCC vs. 13 in 8 patients for FFP) were reported. There was no increase in thromboembolic event relating to PCC. Mean days alive and out of hospital within 90 days was 71.8 (SE = 1.8) for PCC and 73.8 (SE = 1.7) for FFP. No randomised patients were withdrawn from the study, 5 were lost to follow-up and 4 died (2 FFP and 2 PCC). Follow-up data up to 90 days were collected in 89% of randomised patients.

Conclusion: There have been no RCTs to date that have compared the efficacy/safety of FFP versus PCC in cardiac surgery patients who are bleeding and this pilot study demonstrates that it is feasible to perform a large trial in the future.

Disclosure of Interest: None Declared

BSH2020-PO-164

Platelet storage pool disease with excessive bleeding associated with a novel FLI1 variant

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Inherited platelet disorders (IPDs) are a heterogeneous group of disorders associated with normal or reduced platelet counts and bleeding diatheses of varying severities. The identification of the underlying cause of IPDs is clinically challenging due to the absence of a gold standard platelet test, and is often based on a clinical presentation and normal values in other haematology assays.

Dense granule disorders typically result in defective platelet aggregation of variable severity, specifically absent secondary aggregation response to exogenous stimuli and significant reduction in the content and ratio of ADP to ATP or a reduced ATP release.

Next-generation sequencing (NGS) technologies are allowing the rapid analysis of genes that have been previously implicated in IPDs or that are known to have a key role in platelet regulation, as well as novel genes that have not been previously implicated in platelet dysfunction.

A 30 year old woman with a family history of platelet storage pool disorder was referred for investigation. She had a history of easy bruising, recurrent epistaxis and suffered increased post-partum bleeding after the birth of her second child. Her ISTH BAT score was 6. Routine coagulation tests were normal, platelet count was 212, platelet aggregation showed primary disaggregation to ADP at all concentrations, and ATP:ADP ratio was elevated at 7.0 (0.8 – 2.2).

A variant in the FLI1 gene (transcription factor friend leukaemia virus integration 1) was identified through Thrombogenomics, c.1019G>A p.(Arg340His). However the significance of the variant was initially unknown. Further family testing was undertaken in 2

males and a female who all had a dense granule deficiency and a male who had normal platelets. The variant was only detected in the individuals with the dense granule deficiency and the mode of inheritance was autosomal dominant in this family. The variant was then reclassified as pathogenic and the family had bleeding disorder, platelet type 21.

FLI1 belongs to the ETS-domain transcription factor family, in which members bind a specific DNA consensus sequence to control the expression of genes that are essential in the regulation of cellular proliferation, differentiation and programmed cell death. To date a further 4 variants have been associated with platelet dense granule secretion defect and excessive bleeding, 2 variants have been associated with macrothrombocytopenia and 1 variant has been associated with thrombocytopathy.

Six of the 7 variants are missense mutations and 1 is a small deletion and they are all in the highly conserved C-terminal ETS DNA binding domain, in exon 9 between codons 324-345. These include another variant at codon 340, c.1019G>C p.Arg340Pro, which was found in an individual with macrothrombocytopenia and dense granule deficiency.

As more individuals with platelet dense granule secretion defects are screened by NGS and their genetic cause is found, this will allow us to have a better understanding of the numerous platelet disorders and the effect they have on these individuals.

Disclosure of Interest: None Declared

BSH2020-PO-165

Shear flow-based assay with HUVECs shows von Willebrand Factor (VWF) strings anchored to the endothelial surface and platelet aggregation in acute immune-mediated Thrombotic Thrombocytopenic Purpura

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Immune-mediated Thrombotic Thrombocytopenic Purpura (iTTP) is a rare, life-threatening disorder characterised by the abnormal presence of ultralarge VWF multimers as a consequence of acquired severe deficiency of the plasma metalloprotease ADAMTS13. Under physiological conditions, VWF is secreted from endothelial cells as ultralarge multimers, and ADAMTS13 progressively reduces the size of these VWF multimers as they circulate in the blood. In the absence of functional ADAMTS13, the accumulation of platelet-hyperadhesive ultralarge VWF multimers leads to the formation of platelet-rich microthrombi within small arterioles inducing thrombotic macroangiopathic anemia and thrombocytopenia. The aim of this study was to investigate acute thrombotic iTTP patients to assess disease severity by studying the impact of reduced/absence ADAMTS13 activity on VWF adsorption to collagen, platelet adhesion to VWF bound to collagen and platelet aggregation under flow.

A shear flow-based assay with HUVECs was developed. Endothelial cells were seeded on gelatin coated μ -slides VI (Ibidi, Germany) at a density resulting in confluent monolayers and were cultured for 48 hours before the flow assays. μ -slides were connected to a pump system (Harvard Apparatus, Holliston, MA, USA) under temperature control (37 °C). After histamine stimulation, plasma with the addition of platelets at a concentration of 0.5×10^8 /mL was perfused. A constant wall shear stress of 2.5 dynes/cm² (0.25 MPa) was maintained throughout the experiments. Finally, cells were fixed under

flow with 4% formaldehyde and immune-labelled for VWF and CD41. Stacks of confocal images were combined to generate tilescans to quantify platelet recruitment and VWF coverage.

Twenty acute iTTP presentation patients with severe deficiency (<10%) of ADAMTS13 activity were included. Median presenting anti-ADAMTS13 antibodies levels was 67% (range, 2% - 140%) and median presenting ADAMTS13 antigen levels was 3.0% (range, 0.7% - 47.6%). VWF antigen and VWF activity levels were also analysed, and their pre-treatment median levels were respectively 1.71 IU/mL (range, 0.72 IU/mL - 3.16 IU/mL) and 0.84 IU/mL (range, 0.15 IU/mL - 2.10 IU/mL). Pre first plasma exchange (PEX) citrated samples were analysed utilising the shear flow-based assay with HUVECs. When analysing with the shear flow-based assay with HUVECs, significant increase of platelet recruitment was observed in patients compared to pool normal plasma ($P < 0.0001$). VWF coverage was significantly higher in patients compared to pool normal plasma ($P = 0.0035$). In addition, significant increase in median string number was reported in patients ($n = 83$, range 46 - 184) compared to pool normal plasma ($n = 38$, range 20 - 66) ($P < 0.0001$).

In conclusion, these data provided a model of the *in vivo* pathophysiological mechanism by which thrombus formation is the consequence of ultralarge VWF string formation and elongation on endothelial cells, followed by platelet adhesion and aggregation under flow. This study may help to understand the complexity of the disease and shows that static assays are not the only parameter to be considered when evaluating disease severity.

Disclosure of Interest: None Declared

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Investigation of von Willebrand factor polymorphisms in congenital thrombotic thrombocytopenic purpura

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Von Willebrand factor (VWF) is a multimeric glycoprotein with a central role in primary haemostasis. Under high shear stress stimulation, VWF is released as ultra-large and highly adhesive multimers that form long strings to facilitate platelets adhesion to the endothelium and aggregates formation. This prothrombotic activity is significantly inhibited by ADAMTS13 cleavage in the A2 domain (Y1605-M1606 bond) of the VWF multimers. Many polymorphisms in VWF gene have been reported and investigated for their role and association with thrombogenicity and cardiovascular diseases. The aim of this study was to investigate the presence of polymorphic variants in the coding region of the functional domains of VWF in congenital Thrombotic Thrombocytopenic Purpura (cTTP).

60 cTTP patients from the UK TTP registry were analysed with Sanger Sequencing. Genomic DNA was extracted from peripheral EDTA whole blood samples (QIAamp DNA Blood Midi Kit, Qiagen). VWF exons from 28 to 32, and exon 45, were amplified by polymerase chain reaction (PCR) with corresponding introns primers. PCR products were sequenced by GENEWIZ and sequences data were analysed using Sequencher 5.4-6 (Gene Codes Corporation). Genomic database Ensembl was used to identify known VWF polymorphisms.

The cTTP patients were included in the study: 77% ($n = 46$) female and 23% ($n = 14$) male, median age 35 years (range: 2-74). The A1, A2, A3 and C4 domains, corresponding to the respective binding sites for GPIb, ADAMTS13, fibrillar collagen and GPIIb/IIIa on VWF, were investigated. In total, VWF variants were identified in

45% ($n = 27$) cTTP patients. DNA sequencing revealed two missense variants in exon 28, p.T1381A (rs216311) and p.D1472H (rs1800383), and one missense variant in exon 45, p.F2561Y (rs35335161). Both variants in exon 28 were located in the A1 domain. The first polymorphism (rs216311) was located in the middle of the A1 loop, while the second polymorphism (rs1800383) was just after the C-terminal disulphide bond on the A1 loop. Fifteen percent of patients ($n = 9$) were T/T1381 homozygotes, 45% ($n = 27$) A/T1381 heterozygotes and 40% ($n = 24$) A/A1381 homozygotes. Twenty-seven percent ($n = 16$) D/H1472 heterozygotes and 7% ($n = 4$) F/Y2561 heterozygotes were detected.

Although all the variants identified in this observational study have been defined to be "likely benign" by PolyPhen-2, they may have a prognostic meaning in our patients cohort. Homozygous T/T1381 variant has been demonstrated to have a higher affinity for the platelet receptor GPIIb/IIIa. Similarly, F/Y2561 has been investigated as a gain-of-function variant which increases the force sensitivity of VWF interaction with platelets. The aminoacidic substitution in D/H1472 variant seemed to require an increased concentration of ADAMTS13 for the cleavage of VWF. In conclusion, VWF variants were more common than anticipated in cTTP cases and may alter the prothrombotic risk and phenotypic diversity in cTTP patients.

Disclosure of Interest: None Declared

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Prospective cohort study of haemostatic activation markers and correlation with bleeding and thrombotic complications in patients receiving Venovenous Extra-Corporeal Membrane Oxygenation (VV-ECMO) -HAE study

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Bleeding and thrombosis are common complications during Venovenous Extra-Corporeal Membrane Oxygenation (VV-ECMO) but their etiology remains unclear. Coagulation and inflammatory systems are immediately activated when blood contacts the ECMO circuit, necessitating systemic anticoagulation. In this prospective cohort study, we aimed to determine the changes in haemostatic activation markers during initiation and continuation of VV-ECMO and to correlate these with the development of bleeding and thrombotic complications.

Blood samples were collected into citrate tubes with and without corn trypsin inhibitor (CTI) at the time of insertion VV-ECMO (prior to bolus dose unfractionated heparin[UHF]) (day 0) and after 24 hrs (D1) and 5 days of ECMO (D5) from 23 consecutive patients from September 2018 to Jan 2019. If the duration of ECMO was > 5 days, blood was collected at 5-day intervals until the completion of ECMO or termination of ECMO due to death of the patient. Samples were also collected 1 hr and 24 hrs after de-cannulation. Samples were double centrifuged and stored in -80°C freezer within 4 hrs of collection. All assays were performed after neutralising UFH with heparinase. We used time to event analysis (ECMO circuit change, development of thrombosis [TV] and major bleeding [MB]) to determine the relationship of the event with haemostatic markers. Risk of development of events was expressed as the hazard ratio (HR).

One patient died during study period due to multiorgan failure and was excluded from the analysis. Mean age (standard deviation) of study patients was 45.2 ± 11.4 yrs and 60% were male. Median (range) on VV-ECMO was 9.5 days (5–29). Factor II, V, VIII, IX, X, XI, XII and von Willebrand antigen (VWF:Ag) / VWF Ristocetin Cofactor (VWF: RCo) ratio fell and nadirs were seen on day 25 (D25) which was the longest VV-ECMO duration for which samples were collected. After de-cannulation and 24 hrs post VV-ECMO, these factors returned to the levels present at day 0 or above. Thrombin generation parameters followed a similar pattern with reduced endogenous thrombin potential [ETP] and peak thrombin and increased lag time and time to peak thrombin at D25. Activated partial thromboplastin time (APTT) and D dimer also achieved their highest values at D25. There was no specific pattern of change in observed in factor VII and prothrombin time. 27% (6/22), 54% (12/22) and 32% (7/22) patients respectively had circuit change, thrombosis and MB during VV-ECMO. Patients who had circuit changes and MB were on ECMO significantly longer (21.5 vs 9.6, $P = 0.0001$ and 18.4 vs 10.3, $P = 0.012$; days respectively) compared to those without these events. No difference was seen in patients with thrombosis (14.5 vs 10.9 days, $P = 0.27$). With time to event analysis, increased factor V and VIII were linked to reduce circuit changes whilst increased factor V, VIII, IX and ETP linked to increased thrombosis. Increased D dimer was predictive of circuit change. The only factor linked to bleeding was increased ETP and reduced risk of MB. Neither heparin anti-Xa levels or APTT were predictive of bleeding or thrombosis.

In conclusion, bleeding and thrombosis in VV-ECMO were as high as 32% and 54% respectively. Although intrinsic coagulation factors reduced gradually during VV-ECMO suggesting continuous activation of the contact pathway, increased factor V, VIII and ETP show linked to development of thrombosis whilst ETP was a predictor of both bleeding and thrombosis.

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Factor VIII levels and bleeding according to factor 8 mutation in pregnant carriers of haemophilia A: a multi-centre retrospective cohort study

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Heterozygous carriers of haemophilia A have lower levels of factor VIII than non-carriers and are at a higher risk of bleeding. Factor VIII levels are known to increase towards the third trimester of pregnancy, although carriers are still at a higher risk of primary postpartum haemorrhage (PPH). We aimed to investigate the effect of genotype on factor VIII levels during pregnancy for haemophilia A carriers and to examine the incidence of primary PPH.

We conducted a retrospective cohort study of 64 pregnancies in 53 heterozygous carriers of haemophilia A seen at the Oxford

Haemophilia and Thrombosis Centre between January 2011 and August 2019. Baseline, first trimester, third trimester, and postpartum (at least six weeks post-delivery) factor VIII levels were collected and patients were stratified according to the severity of the underlying mutations. The severity of the underlying mutations was also compared to the incidence of primary PPH (blood loss ≥ 500 mL within 24 hours after birth) and severe primary PPH (blood loss ≥ 1000 mL within 24 hours after birth).

The mean baseline factor VIII level was 0.73 ± 0.25 IU/mL, rising to a mean third trimester level of 1.40 ± 0.41 IU/mL. There were no significant differences in factor levels between the severity of mutations at either baseline ($P = 0.53$) or during the third trimester ($P = 0.90$). Four patients with four different mutations in the A1 domain were identified whose factor VIII levels did not respond well (third trimester levels less than 1.00 IU/mL). The overall incidence of primary PPH in carriers was 17/41 (41.5%), including three (7.3%) with a severe primary PPH (≥ 1000 mL), compared with a nationwide incidence of 18.8% in 2017–18. Thirteen of the 17 incidences of primary PPH that occurred were in severe carriers, with 54.2% of severe carriers having a primary PPH.

These results suggest that severity of genetic mutation is not a useful prognostic indicator of factor VIII response in haemophilia A carriers during pregnancy. Additionally, haemophilia A carriers, particularly those with a severe mutation, have an increased incidence of primary PPH.

Disclosure of Interest: None Declared

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A fast and simple UPLC-MS/MS method for the simultaneous quantification of apixaban, edoxaban and rivaroxaban in human plasma and breast milk: for investigating direct oral anticoagulants in emerging lactation studies

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Direct oral anticoagulants (DOACs) are increasingly prescribed in clinical practice. They provide the possibility of replacing low molecular weight heparin (LMWH) for the prophylaxis and treatment and of venous thromboembolism (VTE) during the postpartum period, but to date, information on the transfer of DOACs into human breast milk is scarce. To support emerging clinical studies on investigating DOAC's transfer into breast milk, a sensitive quantification assay must be established. Although several LC-MS methods have been developed to quantify DOACs in plasma, there are no published studies validating DOAC measurement in human breast milk. Breast milk is a complex matrix whose components vary with the time of day, the day of lactation, from mother to mother, and among species. Our aim was to develop and validate an ultra-performance liquid chromatography-tandem mass spectrometry (UPLC-MS/MS) method for evaluating apixaban, edoxaban and rivaroxaban in plasma and breast milk.

Calibrators and quality control (QC) samples were prepared by spiking blank plasma or human breast milk from sub-stock solutions, which contain all analytes in acetonitrile. For sample preparation, each calibrators or QC sample was mixed with internal standard working solution (100 ng/mL in acetonitrile) and pure methanol. The mixture was vortex-mixed and then centrifuged.

Table 1 Intra-assay and inter-assay accuracies and precisions of all analytes ($n = 6$).

	In plasma Apixaban				In breast milk Apixaban			
	LLOQ	QC 3	QC 2	QC1	LLOQ	QC 3	QC 2	QC1
Expected conc. (ng/mL)	5.36	16.07	214.20	428.40	5.12	15.35	81.85	204.62
Mean (ng/mL)	5.92	16.56	227.51	454.70	4.99	15.04	81.65	191.37
Accuracy (%)	110.52	103.08	106.21	106.14	97.52	97.99	99.76	93.52
Within-run imprecision (%)	4.63	3.73	2.34	2.34	10.34	7.42	5.38	4.34
Between-run imprecision (%)	1.93	1.11	0.31	1.61	8.96	4.01	3.19	1.30
	Edoxaban				Edoxaban			
	LLOQ	QC 3	QC 2	QC1	LLOQ	QC 3	QC 2	QC1
Expected conc. (ng/mL)	5.12	15.35	204.60	409.20	5.08	15.23	81.22	203.06
Mean (ng/mL)	5.47	16.10	219.10	440.29	4.92	15.09	81.26	198.07
Accuracy (%)	106.91	104.90	107.09	107.60	97.02	99.06	100.05	97.54
Within-run imprecision (%)	7.64	3.85	3.02	2.61	6.00	3.72	6.71	5.23
Between-run imprecision (%)	7.21	1.54	0.17	2.10	3.33	1.31	5.10	1.61
	Rivaroxaban				Rivaroxaban			
	LLOQ	QC 3	QC 2	QC1	LLOQ	QC 3	QC 2	QC1
Expected conc. (ng/mL)	4.78	14.35	191.30	382.60	4.78	14.35	76.52	191.30
Mean (ng/mL)	4.54	14.01	189.64	379.63	4.80	13.98	75.98	180.96
Accuracy (%)	95.00	97.63	99.13	99.22	100.31	97.42	99.30	94.59
Within-run imprecision (%)	3.78	3.78	1.98	3.71	6.63	6.19	5.55	3.77
Between-run imprecision (%)	1.72	2.54	1.32	2.42	5.36	4.12	3.48	2.12

Conc., concentration.

Following the centrifugation, 100 μ L of the supernatant was mixed with 100 μ L of deionised water, before being injected onto the UPLC-MS/MS system. The UPLC system was coupled to a triple quadrupole mass spectrometer, operating with an electrospray ion source set on positive mode and performed in MRM mode. Chromatography was achieved with a C18 column. Gradient elution was utilised with the mobile phase composing of 0.1 % aqueous formic acid (A) and acetonitrile (B). The method was validated in accordance with the European Guideline. To ensure the assay's reliability in quantifying analytes in samples originating from different subjects, at various time points after dosing, and from different population pools in practice, plasma (K₃EDTA, lithium heparin, citrated plasma, serum, hyperlipidemic plasma and hemolysed plasma) were used to evaluate matrix effects (ME) and matrix factors (MF). Additionally, preterm and term breast milk expressed by different mothers and on different days post-delivery were used, to evaluate ME and MF from the breast milk.

The separation of 3 analytes was obtained within 5 minutes. Calibration ranging between 5 and 500 ng/ml in plasma, and between 5 and 250 ng/mL in milk, resulted in good linearity with correlation coefficients (R^2) ≥ 0.99 for all analytes. The within-batch and between-batch variability remained $< 9\%$, and the accuracy was $\pm 15\%$ for all analytes (Table 1). Recoveries ranged from 106.13% to 109.05% in plasma and 93.40% to 107.91% in breast milk. MF's lot-to-lot variability was within $\pm 15\%$ for all analytes. All analytes were stable when stored for 24 hours at room temperature, 7 days at 2–8°C, and for at least 5 weeks at –20°C in both plasma and breast milk.

The assay developed was shown to be fast and reliable in all matrices tested. It will now be used in a small clinical trial evaluating the transfer of apixaban and rivaroxaban into human breast milk, and it is capable of analysing apixaban, edoxaban, and rivaroxaban in future clinical lactation studies and in the clinical setting for the plasma therapeutic drug monitoring of DOACs.

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EDTA-associated platelet degranulation and thrombocytopenia mimics grey platelet syndromeK. Burley^{1,*}, R. Aungraheeta², M. Hopkins³, A. Calvert³, T. Latham³, C. Knechtli⁴, A. Mumford²¹*School of Physiology, Pharmacology and Neuroscience,*²*School of Cellular and Molecular Medicine, University of Bristol,* ³*Filton Centre, NHS Blood and Transplant, Bristol,*⁴*Department of Haematology, Royal United Hospitals Bath NHS Trust, Bath, UK*

Grey platelet syndrome (GPS) is a rare autosomal recessive disorder in which pathogenic variants in *NBEAL2* result in enlarged platelets with a characteristic pale appearance on light microscopy. We report a case presenting with thrombocytopenia and platelet morphology highly suggestive of GPS but resulting from EDTA-dependent *ex vivo* platelet degranulation.

The study case was a 61-year old male with incidental detection of thrombocytopenia in primary care. There was no personal or family history of abnormal bleeding or autoimmune disorders. Analysis of blood anticoagulated with 1.8 mg/mL EDTA using a Sysmex XN-series analyser confirmed thrombocytopenia (platelet count $22 \times 10^9/L$) but no other haematological abnormalities. Wright-Giemsa stained blood films showed thrombocytopenia and large agranular 'ghost' platelets. DNA sequencing demonstrated no pathogenic variants in *NBEAL2* or other genes associated with bleeding or platelet disorders.

Platelet counts in blood samples collected into 3.2% sodium citrate or 17 IU/ml heparin were consistently higher ($41\text{--}59 \times 10^9/L$) than in EDTA samples and did not show platelet degranulation by

light microscopy. Platelets in EDTA-anticoagulated blood samples showed markedly increased surface exposure of the alpha granule marker P-selectin (PS) and dense granule/lysosome marker CD63 compared to samples collected into citrate, consistent with secretion of both granule subtypes. Following stimulation of platelets with TRAP-6 10 μ M, there was an increase in PS and CD63 exposure in case platelets collected into citrate, similar to control platelets in citrate or EDTA, indicating stimulated granule secretion. In case platelets collected into EDTA, there was no further increase in PS or CD63 exposure suggesting maximum degranulation had already occurred. Platelet degranulation increased rapidly following exposure to EDTA and appeared complete within 2 hours. Mixing studies in which case platelets collected into citrate were washed and resuspended in either case or control plasma, showed that EDTA-dependent platelet degranulation was mediated by a component of case plasma.

These data suggest that the abnormal appearance of platelets from the case resulted from EDTA-mediated platelet degranulation affecting both alpha and dense granules/lysosomes occurring rapidly *ex vivo*. These characteristics are similar to rare previous reports of an acquired disorder termed pseudo-grey platelet syndrome (p-GPS) in which loss of platelet alpha granules was attributed to an EDTA-dependent antibody (Cockbill *et al.* 1988). We were unable to corroborate this finding since the platelet immunofluorescence test (PIFT) and monoclonal antibody immobilization of platelet antigens (MAIPA; targeting GPIIb/IIIa, GPIa/IIa, GPIb/IX/V and CD109) did not show platelet-associated antibodies. Moreover, platelets from a patient with Glanzmann's thrombasthenia with absent surface expression of glycoprotein IIb/IIIa (the most common epitope for clinically significant anti-platelet antibodies) also degranulated in the presence of EDTA-anticoagulated case plasma.

This case extends the description of p-GPS by demonstrating EDTA-dependent loss of both platelet alpha granules and dense granules/lysosomes mediated by an uncharacterised plasma component. The disorder superficially resembles classical GPS associated with genetic variants in *NBEAL2* but can be distinguished by the absence of bleeding and by normal platelet morphology in citrate-anticoagulated blood.

Disclosure of Interest: None Declared

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A systematic review of efficacy and safety of chemical thromboprophylaxis in renal transplantation

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Individuals undergoing renal transplantation are at increased risks of thrombosis and bleeding. When these individuals undergo transplant surgery, it is unclear what the risk benefit ratio is of administering chemical thromboprophylaxis (TP) as there have been no previous systematic reviews to address this important question, which is reflected in the variations in national guidelines.

The primary objective of this systematic review was to assess the efficacy (symptomatic, asymptomatic venous thrombosis and renal

allograft thrombosis) and safety (major bleeding and mortality) of a form of chemical thromboprophylaxis intervention compared with another form, no intervention or placebo for up to 3 months post-transplant. Secondary objectives assessed similar outcomes beyond 3 months post-transplant.

Pubmed, MEDLINE, Embase, Cochrane, CINAHL, World Health Organisation (WHO) International Clinical Trials Registry Platform and ClinicalTrials.gov databases were searched from 1946 to present for randomised controlled trials (RCTs), controlled clinical trials, single and multiple intervention studies. Inclusion criteria included (1) participants undergoing renal transplantation only with no contra-indication to TP i.e. no underlying bleeding disorder (2) no history/clinical suspicion of acute organ rejection and (3) studies describing a form of chemical TP intervention compared with another form, no intervention or placebo.

8,682 citations were identified, with 13 studies included in the analysis ($n = 1600$ patients), of which five were randomised trials (RCTs), one a non-randomised controlled trial and 7 were cohort studies. There were no ongoing trials. Each study used a different form of chemical TP, different timing of onset and TP duration.

Symptomatic and asymptomatic VTE were very poorly reported, with only one retrospective study reporting on this outcome ($n = 320$ patients). Rates of renal allograft thrombosis were more likely to be reported and proportions were higher in the no intervention arm (upto 18.8%) compared to intervention arm (upto 6.3%).

Only six studies reported on major bleeding, the definition of which was poorly defined between studies.

Comparisons of different chemical TP showed huge variations in bleeding rates between studies with one RCT reporting no bleeding events with UFH ($n = 25$) and 4% with low molecular weight heparin (LMWH). The rates of major bleeding in cohort studies varied from 3% (prophylactic heparin) to 62.5% (treatment doses of heparin).

RCTs did not demonstrate a significant difference between chemical TP and no intervention. However, studies were small with very wide confidence intervals. A single meta-analysis for minor bleeding (for RCTs with follow-up to 3 months) was possible and showed no evidence of a difference between the unfractionated heparin (UFH) and no intervention arms (RR 1.65, 95% CI 0.89-3.07).

There is lack of good quality evidence to demonstrate the benefits of chemical TP during renal transplantation. There are signals that chemical TP (compared to no TP) may reduce the rate of renal allograft thrombosis but increase the risk of bleeding. However, large-scale randomised controlled trials are needed to determine the risk benefit ratio of TP in renal transplantation surgery.

Disclosure of Interest: None Declared

BSH2020-PO-172

The optimal anticoagulation during catheter ablation of atrial fibrillation: a systematic review and network meta-analysis

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The optimal periprocedural anticoagulation strategy for patients with atrial fibrillation (AF) undergoing catheter ablation (CA) remains

Table. Summary of network meta-analysis of major bleeding, minor bleeding and SCI

Intervention	Major bleedingOdds ratio (95% CrI)		Minor bleedingOdds ratio (95% CrI)		SCIOdds ratio (95% CrI)	
	Direct [†]	NMA	Direct	NMA	Direct	NMA
Compared to U-VKA INR 2.0-3.0						
U-Apixaban	0.67 (0.31 to 1.47)	0.64 (0.29 to 1.38)	0.84 (0.55 to 1.28)	0.84 (0.55 to 1.28)	1.09 (0.67 to 1.75)	1.07 (0.68 to 1.70)
I-Apixaban		0.84 (0.19 to 3.90)		0.79 (0.39 to 1.63)		1.30 (0.62 to 2.75)
U-Dabigatran	0.22 (0.08 to 0.58)	0.20 (0.07 to 0.51)	1.12 (0.74 to 1.68)	1.07 (0.72 to 1.62)		NA
I-Dabigatran	0.27 (0.08 to 0.90)	0.22 (0.05 to 0.74)	1.21 (0.37 to 4.01)	0.97 (0.39 to 2.39)		NA
U-Rivaroxaban	0.98 (0.17 to 5.76)	1.23 (0.21 to 7.58)	1.15 (0.71 to 1.87)	1.26 (0.78 to 2.04)	0.98 (0.38 to 2.55)	0.99 (0.45 to 2.16)
I-Rivaroxaban		5.13 (0.29 to 197)*		0.83 (0.33 to 2.01)		0.70 (0.25 to 1.92)
U-Edoxaban	0.96 (0.25 to 3.61)	1.04 (0.30 to 5.09)	1.51 (0.65 to 3.54)	1.55 (0.69 to 3.97)	1.02 (0.37 to 2.87)	1.56 (0.56 to 5.10)
I-Edoxaban		4.06 × 10 ⁻⁷ (4.27 × 10 ⁻²² to 1.25)*		2.64 (0.55 to 14.1)		2.43 (0.70 to 9.82)

INR: international normalised ratio; U-Apixaban/I-Apixaban: uninterrupted/minimally interrupted apixaban; U-Dabigatran/I-Dabigatran: uninterrupted/minimally interrupted dabigatran; U-Rivaroxaban/I-Rivaroxaban: uninterrupted/minimally interrupted rivaroxaban; U-Edoxaban/I-Edoxaban: uninterrupted/minimally interrupted edoxaban. (Minimally interruption applied in this study means withholding one or two doses of DOAC before ablation).

[†]Data presented here for direct estimates were generated using fixed effect model.

*Imprecisely estimated.

unclear. We aimed to assess the efficacy and safety of all commonly used oral anticoagulation (OAC) strategies in this setting.

A systematic review of Medline, Embase, Cochrane Library for clinical trials, ClinicalTrials.gov, and Clinicaltrialsregister.eu was done to find all randomised controlled trials (RCTs) in which direct oral anticoagulants (DOACs) were compared against each other or uninterrupted vitamin K antagonists (U-VKA) for CA of AF. The outcomes of interest were thromboembolic events (TE), silent cerebral ischemia (SCI), death, major bleeding and any bleeding. A fixed-effect model with Bayesian network meta-analysis (NMA) of was performed to calculate the odds ratio (OR; with 95% credible intervals [95%CrI]) for each DOAC strategy compared to U-VKA. The study was done in accordance with Cochrane and PRISMA guidelines.

A total of 14 RCTs ($n = 4375$) were included. TE and death were not included in NMA as endpoints due to the rare occurrences. Nine OAC interventions were included in the NMA (table). In comparison with U-VKA, uninterrupted dabigatran (U-Dabigatran, OR 0.20 [95% CrI 0.07-0.51]) and minimally interrupted dabigatran (I-Dabigatran, OR 0.22 [95% CrI 0.05-0.74]) were associated with lower rates of major bleeding. No noticeable differences were observed in major bleeding for other DOAC strategies compared to U-VKA. No noticeable differences were found in minor bleeding or SCI for all available DOAC strategies compared to U-VKA. No data was reported to compare dabigatran with other interventions regarding the risk of SCI.

The results indicate that DOACs are reasonable alternatives to U-VKA in patients with AF ablation. Minimally interrupted DOAC is comparable to uninterrupted DOAC in terms of major bleeding, minor bleeding, and SCI. Dabigatran (without interruption or with minimal interruption before ablation) appears to be the optimal strategy regarding its lowest rate of major bleeding. To aid physician's decision making, further well-powered RCTs are required to directly compare each available DOAC, and to compare dabigatran with other oral anticoagulants regarding the incidence of SCI.

Disclosure of Interest: None Declared

BSH2020-PO-173

UK NEQAS for blood coagulation D-dimer Point of care testing programme

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In May 2014 UK EQAS BC introduced a POCD-dimer programme. This programme is open to any user of either the Roche Cobas h232 or the Quidel Triage Meter pro devices. One EQA sample is provided in each survey and 4 surveys are provided per year. Current numbers enrolled in the programme are 106 (H232 = 67 and Triage = 39).

Samples are lyophilised citrated plasma together with a pre measured volume of diluent and a disposable pipette such that no laboratory equipment is needed to perform the test. A brief patient scenario is also provided and users are also asked to choose either "no further investigation required" or "further investigation" required based on their local test result. This post analytical aspect is scored if there is a consensus of 80% or more. Peer group performance assessment is undertaken so Cobas h232 results are compared only with Cobas h232 results (and Triage only with Triage). Median values are calculated and the furthest 10% from the median (both higher and lower) are considered "outwith consensus". If a centre has 3 outwith consensus surveys consecutively they will be classified as "persistently outwith consensus" and receive contact from the programme director.

23 surveys have been completed with a range of median values of 100-2310 ng/ml FEU for the Triage device and 100-2300 for the Cobas h232 device. The Triage cut off as stated by the manufacturer is 400 ng/ml FEU and the Cobas h232 manufacturers stated cut off is 500 ng/ml FEU. For the interpretation aspect the samples have been scored in 17/23 surveys for the Triage and 20/23 surveys for the Cobas h232. Although overall interpretations have been in line with the centres results we have noted a small number of centres that have given inappropriate interpretations for their results. For example one centre that had a result of 740 ng/ml FEU but stated "no

further investigation". For samples that were provided with a patient scenario of a Wells score of 3 or 4 (4/23 samples) there were a small number of centres on each occasion that stated no further investigation required and whilst these centres results were below the cut off they did not take into account the high Wells score.

The average coefficient of variance for these POC tests for D-dimer was 21.4% for Triage and 19.0% for Cobas h232. These values are comparable with D-dimer methods in our laboratory programme. On average 14.6% of Triage users were outwith consensus for the result and 3.8% outwith for the interpretation. For the Cobas h232 users on average 14.2% were outwith consensus for the result and 5.6% were outwith consensus for the interpretation.

We conclude that EQA is essential for D-dimer testing whether in the laboratory or in a POC setting. Whilst the interpretations of these results overall is good we are still seeing some centres that do not take into account the patient scenario provided which is a concern. We have also seen results which are clearly over the manufacturers stated cut off which have been deemed by the user to not require further investigation and this is also of concern.

Disclosure of Interest: None Declared

BSH2020-PO-174

Observational study of outcomes in patients with antiphospholipid syndrome anticoagulated with Rivaroxaban

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The management of antiphospholipid syndrome (APS) has changed recently as a result of the Medicines and Healthcare products Regulatory Agency (MHRA) interpretation of the TRAPS. The TRAPS study, 2016, suggested little difference in the surrogate end point of thrombin generation when rivaroxaban and warfarin were compared in the management of patients with APS with prior venous thromboembolism (VTE). Based on this clinicians started to use rivaroxaban in APS patients with VTE. Following review of TRAPS the MHRA stated that DOACs should not be used in patients with APS. This decision surprised some as the criteria for inclusion in the TRAPS study included triple positivity for the antiphospholipid antibodies; lupus anticoagulant (LA), anticardiolipin (ACA) and anti-beta 2 glycoprotein 1 (anti-beta-2GPI), whereas most patients with APS are diagnosed on the basis of persistent positivity of a single abnormal test. In response to the MHRA advice we performed a retrospective study of thrombotic and haemorrhagic outcomes in patients with APS and VTE who had been anticoagulated with rivaroxaban in our centre.

A single centre retrospective study was conducted in the department of Haematology, Aberdeen Royal Infirmary. Patients newly diagnosed with APS between November 2016 and August 2019 and treated with rivaroxaban were included. Patients with prior exposure to warfarin were excluded. The diagnosis of APS was made based on persistent positivity of LA and/or ACA Ig over a 12 week period. We do not routinely test for anti beta-2 GPI. We reviewed clinical records for evidence of thrombotic and haemorrhagic events whilst on rivaroxaban. The definitions of thrombotic events and major, clinically relevant non-major bleeding (CRNMB) and minor bleeding episodes were based on ISTH criteria. Patients were contacted by phone to ratify their history using a standard questionnaire. 31 patients were identified. All were persistently positive for either LA and/or anticardiolipin. 30 cases were single positive, 1 was double positive. Amongst the 30 single positives deep vein thrombosis (DVT) was the presenting feature in 53.3% ($n = 16$), 36.6 % ($n = 11$) had pulmonary embolism (PE), 6.6 % ($n = 2$) had both

DVT and PE and 3.3 % ($n = 1$) had ischemic stroke at presentation. No new venous or arterial thrombotic events were reported throughout the study period with a median treatment duration of 723 days per patient and a total follow up period of 21,691 patient days. No major bleeding occurred, 6.6 % ($n = 2$) patients reported CRNMB; gastrointestinal bleeding ($n = 1$), epistaxis ($n = 1$). The overall rate of CRNMB bleeding was approximately 1 in 10,000 days.

Our study shows that no new thrombotic events occurred in 21,691 days of treatment in our cohort of patients with newly diagnosed non-triple positive APS treated with rivaroxaban. The incidence of new/recurrent thromboembolism in the TRAPS cohort was 12% in 59 patients followed for median 569 days. The incidence of major bleeding and CRNMB was also lower than reported in the TRAPS study. Many clinicians have been surprised at the MHRA response to the TRAPS data and the extrapolation that this decision should pertain to all cases of APS. This small retrospective study suggests that the event rate in non-triple positive APS patients on rivaroxaban is less than that seen in TRAPS. Our results suggest further large prospective studies are required to address whether the TRAPS data is generalisable to non-triple positive APS.

Disclosure of Interest: None Declared

BSH2020-PO-175

External quality assurance (EQA) for the thromboelastometry devices: UK NEQAS BC programme update

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Thromboelastography and Thromboelastometry are Point of Care testing (POC) technologies for global haemostasis tests which measure the elasticity of a developing clot. Thromboelastometry is performed using Rotem® devices, the Rotem delta device requires reagents to be pipetted whereas the newer Sigma device uses cartridge based technology with no pipetting required. The TEG devices are used to perform Thromboelastography and can be either TEG5000 which requires reagents to be pipetted or TEG6 which is cartridge based.

Both the TEG6 and Rotem Sigma have recently been introduced to the market and UK NEQAS BC have introduced a formal programme for these devices. The majority of devices are in theatres and specialist wards and usually operated by staff without laboratory training and therefore pipette use training. It is suggested by the manufacturers that the cartridge based devices should show improved precision.

UKNEQAS BC has been providing EQA samples for the Rotem Delta and the TEG 5000 for 5 years and have recently introduced EQA for the cartridge based technology.

One lyophilised sample plus a pre measured diluent and a disposable pipette is distributed 3 times per year. Users are allowed 24 days to test and return results. Median values are calculated and the R time for TEG parameters and CT time for Rotem parameters are performance assessed. Peer group assessment is undertaken so that each device is only compared to other of that type. Samples ranged from obtained from a sample from a normal donor, donor plasma spiked with heparin, rivaroxaban, apixaban and a sample designed to mimic excessive blood loss.

Results for Rotem Delta and TEG 5000 (both pipette based devices) show CVs ranged from 7.2% to 73.2% for Rotem delta (mean CV = 21.4%) and 17.4% to 59.5% for TEG5000 (mean CV = 27.7%). For Rotem Sigma and TEG6 (both cartridge based technology)

results ranged from 3.9%>22.9% for Rotem Sigma (mean CV = 9.3%) and 13.6%>30.7% for TEG6 (2 results only).

We conclude that EQA is essential for this technology and that preliminary data on precision of cartridge based technology has shown an improvement compared to traditional testing

Disclosure of Interest: None Declared

BSH2020-PO-176

Thromboembolism in adult patients with acute lymphoblastic leukaemia / lymphoma: past and future practice in a tertiary centre

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Venous Thromboembolism (VTE) is a common and impactful complication for patients with acute lymphoblastic leukaemia / lymphoma (ALL/LBL) with up to half of events occurring as cerebral vein sinus thrombosis (CVST). Older children and adults receiving asparaginase-containing treatment regimens appear particularly at risk. Most VTE events occur within 2 months of treatment initiation.

Practice at the Bristol Haematology and Oncology Centre was reviewed for all adult patients diagnosed with ALL/LBL between 1st January 2015 and 31st September 2018. During the data collection period (3rd January 2019 until 27th March 2019), 32 patients had completed induction. The median age at diagnosis was 31 years (range 18-77), 17 were male and 15 female. The majority of patients had B-cell ALL/LBL (81%), and the median presenting white cell count was $10 \times 10^9/L$ (range 0.3-113 $\times 10^9/L$). Lymphadenopathy was present in 28%, of which one third had bulky mediastinal disease over 70 mm (all of whom had T-ALL/LBL). Patients were treated following either UKALL2011 (41%), UKALL14 (44%) or UKALL60 + (16%) protocols. Asparaginase was administered to 70%, and half of patients were treated with allogeneic haematopoietic stem cell transplantation. With a median follow up of 21 months, 78% of patients were alive.

Thirteen VTE events occurred in 12 patients (38% overall; 43% within the asparaginase-receiving cohort) with most events (75%) occurring within the first 2 months of treatment. VTE sites included line-associated (38%), CVST (23%), lower limb deep vein thrombosis (23%) and upper-limb deep vein thrombosis not line-associated (15%). All VTEs were grade 1-2 except for the three CVSTs (all grade 4). Eight thromboses (62%) occurred following the administration of asparaginase after a median of 12 days (range 1-20). Six patients were successfully re-exposed to asparaginase with ongoing anticoagulation ($n = 6$) and antithrombin monitoring/replacement ($n = 4$). Provisional laboratory data demonstrated a linear correlation between antithrombin and peak antiXa activity (clexane 1 mg/kg bd ($R^2=0.55$) and 1.5 mg/kg od ($R^2=0.84$)) supporting the known importance of antithrombin *in vivo* for heparin's mechanism of action. There were no thrombosis-related deaths. Bleeding events were documented in 5 patients (16%). Only one patient was receiving anticoagulation at the time of bleeding, and 2 patients bled in the context of severe thrombocytopenia. Bleeding sites included subdural haematoma ($n = 2$), upper gastrointestinal bleed ($n = 2$) and a bleeding haemorrhoid ($n = 1$).

Given the high thrombosis and relative low bleeding rate in our patient cohort, we have now implemented local guidance for thromboprophylaxis of adults with ALL/LBL, and this will be prospectively audited. Enoxaparin at a dose of 1 mg/kg daily is offered to all patients who are not bleeding, not at high risk of bleeding and have an unsupported platelet count greater than $30 \times 10^9/L$. This is

commenced at diagnosis, and continued throughout the initial asparaginase-containing treatment block until 3 weeks after the last dose. This applies to both inpatient and ambulatory care. Treatment of significant VTE is with therapeutic enoxaparin (minimum of 3 months), dose adjusted for thrombocytopenia (as per BSH guidance) and peak anti-Xa and antithrombin activity monitoring 3 times per week for 3 weeks after each asparaginase dose. Antithrombin replacement is administered if recent VTE, low antithrombin and/or low anti-Xa activity.

Disclosure of Interest: None Declared

BSH2020-PO-177

The differential surface expression of tissue factor between monocyte subsets

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Sepsis-associated coagulopathy has a high mortality rate, there is a need to better understand the underlying pathophysiology to improve patient outcomes. The surface expression of tissue factor (TF) on monocytes is associated with the development of sepsis-associated coagulopathy. Monocytes can be divided into three groups using the level of CD14 and CD16: classical (CD14 + +CD16), intermediate (CD14 + +CD16 +) and non-classical (CD14 + CD16 + +). The subsets vary in function but the differential expression of TF between monocyte subsets is poorly understood.

Total monocytes and individual monocyte subsets were isolated from the blood of healthy volunteers using fluorescence-activated cell sorting (FACS). Informed consent was given by all participants. Ethical approval was granted by a medical ethics committee. Monocytes were cultured for 24 hours with lipopolysaccharide (LPS) using doses between 1 and 1000 ng/mL ($n = 7$). Human primary microvascular endothelial cells (hPMVEC) were grown to 90% confluence and used between passage 2 and 8. FACS isolated monocytes were incubated together with hPMVEC for 24 hours ($n = 5$). TF activity was measured using a human TF activity colorimetric assay. The significance level was set at $P < 0.05$ and calculated using a repeated measures ANOVA.

Following LPS incubation, there was an increase in proportion of classical monocytes ($P < 0.001$) without a change in either cell viability or cell recovery. Although the CD14 expression was unchanged, there was a decrease in CD16 median fluorescence intensity ($P < 0.05$). Classical and intermediate subsets had a higher percentage of TF positive cells compared to the non-classical ($P < 0.05$). Following LPS incubation, only the classical and non-classical monocyte subsets increased TF surface expression ($P < 0.05$). This effect appeared to be dose-dependent. The TF-activity per TF-positive cell was the same between all subsets and the increase in TF surface expression following LPS incubation did not lead to an increase in TF activity (p value non-significant).

o-culture with hPMVEC (without the presence of LPS) increased the monocyte surface expression of TF in both classical and intermediate subsets ($P < 0.01$). Repeating the co-culture of isolated monocytes and hPMVEC with and without cellular contact did not affect the monocytic surface expression of TF (p value non-significant). The incubation of monocytes and hPMVEC also affected the

fibrinolytic response with an increase in the endothelial release of plasminogen activator inhibitor-1 ($P < 0.05$).

Monocytes as a whole increase TF surface expression following incubation with LPS or endothelial cells; however, individual monocyte subsets respond differently to these stimuli. Classical and non-classical monocytes increase TF surface expression following exposure to LPS, whilst the classical and intermediate subsets increase TF expression following co-culture with endothelial cells. These data demonstrate that monocyte subsets differ in TF surface expression, something that must be considered during future work investigating the pathophysiology of sepsis-associated coagulopathy.

Disclosure of Interest: None Declared

BSH2020-PO-178

A case series of anticoagulation in mechanical and valvular heart disease in pregnancy

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Anticoagulation of patients with metallic heart valves (MHV) in pregnancy remains a controversial subject lacking consensus guidance. Recent data from UKOSS and the European Registry (ROPAC) report a very high risk of complications in these women. These include valve thrombosis and the bleeding complications and poor fetal outcome associated with anticoagulation itself.

Warfarin is the most efficacious anticoagulant in preventing valve thrombosis (0-4%), but carries a high risk of fetal morbidity as it crosses the placenta. Heparin does not cross the placenta and therefore carries less fetal risk, however Low Molecular Weight Heparin (LMWH) has been shown to carry a higher thrombotic risk than warfarin and there is no consensus on its use in pregnancy. Due to increased renal clearance and circulating blood volume the bioavailability of LMWH changes during pregnancy. It therefore requires dose adjustment and monitoring of LMWH, through peak and trough Anti-Xa levels and continual dose adjustment is recommended to reduce the thrombosis risk.

Our retrospective study looks at 16 pregnancies in 10 women at Bart's Health NHS Trust with mechanical heart valves (5 in women with Aortic Valve Replacements (AVR), 7 with AVR and Mitral Valve Replacements (MVR) and 3 with MVRs) with an average age of 34 years old (range: 25-44). 14 pregnancies were managed with LMWH throughout and 2 pregnancies were managed with warfarin throughout. All LMWH was enoxaparin 1 mg/kg BD as a starting dose except 1 patient on tinzaparin OD, all but one patient also took low-dose aspirin. Monitoring of LMWH was with weekly peak anti-Xa measurements, aiming for levels of 1.0-1.2 iu/l 4-6 hours post dose, post 2014 we introduced measurements of trough levels (prior to giving LMWH), aiming > 0.6 iu/l. These two measurements improved anti-Xa monitoring.

Both pregnancies on warfarin resulted in pregnancy loss at 12 weeks and 24 weeks respectively. 2 of the 14 pregnancies managed with LMWH ended in the first trimester; 1 miscarriage and 1 woman opted for termination. There were no reported fetal embryopathies and 1 instance of intrauterine growth restriction. The average gestation at delivery was 36 weeks. 2 pregnancies were delivered by elective C section, 6 by emergency C section and 5 vaginally (2 required ventouse). 5 pregnancies were delivered under spinal anaesthesia, 3 under general anaesthetic and the rest with neither.

Bleeding complications included haematomas post delivery, epistaxis and bruising. 19% of the cohort required a blood transfusion.

10 units of red blood cells were transfused across the cohort in total (40% in one patient with placenta praevia), with the average blood loss being 551mls (100-2200mls) peri-delivery. In the LMWH group 1 patient had placenta praevia and experienced an antepartum and post partum haemorrhage.

Valve thrombosis occurred in 1 patient with AVR (6.25%) who had been changed from warfarin to daily tinzaparin 13 days prior with no anti-Xa monitoring. There were no thrombotic complications in women on monitored BD enoxaparin. There were no other cardiac complications reported.

In conclusion Whilst our numbers are small, pregnancy in women with MHV is rare and our study adds to the evidence to support tightly controlled LMWH dosing with trough and peak level monitoring on a weekly basis in pregnant women. We show that with this approach and dedicated multidisciplinary care pregnancy outcomes can be good.

Disclosure of Interest: None Declared

BSH2020-PO-179

Anticoagulation in palliative and end of life care - results of a supra-regional audit and literature review

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Although the evidence for the use of anticoagulants for the prevention and treatment of venous thromboembolism (VTE) is abundant, there remains limited evidence for the use of anticoagulation in palliative care, particularly in the last hours to days of life. The question of how we best manage VTE and anticoagulation in this cohort of patients therefore remains largely unanswered by clinical trials.

Pallaborative North West is a palliative care network in the North West of England, whose scope includes writing evidence-based regional guidelines to improve the quality of care for people with life-limiting illness. A guideline development group (GDG) of healthcare professionals in both palliative care and haematology met bi-monthly for 12 months, conducting both a comprehensive literature review and supra-regional audit. These contributed to the development of updated NICE accredited regional guidelines on the use of anticoagulation in palliative and end of life care.

The GDG conducted a literature search using MedLINE and EMBASE which, after initial review and exclusions, identified 32 papers for contribution to the guideline writing process. These papers were individually assessed and the level of evidence graded by at least 2 members of the GDG.

Data for the supra-regional audits was collected using online surveys; the first a case note review of current clinical practice, and the second a healthcare professional questionnaire to ascertain knowledge and confidence with the use of anticoagulation in different clinical scenarios, including cancer-related VTE and stopping anticoagulation in the last days or hours of life.

From our literature review, 833 papers were identified based on search criteria. Following review by the GDG, this was reduced to 32 papers for inclusion in our final review. 11 were RCTs, of which 10 were investigating treatment of cancer-associated thrombosis (CAT). Of the RCTs, all had exclusion criteria that would exclude the majority of patients receiving palliative care (life expectancy less than three or six months and ECOG performance status > 2). From a

palliative care perspective, there were two observational trials on quality of life in patients with VTE, four observational trials focused on patients with brain tumours (including metastases) and two trials related to patients receiving palliative care.

For the case note audit, data for 189 patients receiving anticoagulation (either prophylactic or therapeutic) was collected from 12 sites within the region. Of these, 70% of patients were in a hospice, 23% were hospital inpatients and 7% were in the community. 42 patients died during the audited episode of care. In this group, anticoagulation was continued until within 48 hours of death in 60%, with 43% of patients continuing anticoagulation until death. However, from the healthcare professional survey, 69% of clinicians responded as feeling mostly or fully confident with knowing when to withdraw anticoagulation in patients with a life-limiting illness.

Although the management of CAT is a rapidly developing field, the evidence for the management of both VTE and anticoagulation in patients at the end of life remains limited. Currently, patients with VTE and those on anticoagulation are managed on an individual basis when they reach the last days or hours of life. However, the results from our audit highlight that further research in this field would be helpful to support these often complex decisions.

Disclosure of Interest: None Declared

BSH2020-PO-180

The CAVEaT multicentre audit of venous thrombosis in patients with haematological malignancy and thrombocytopenia: preliminary results.

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Venous Thrombosis (VTE) in patients with malignancy and thrombocytopenia is uncommon but presents a complex management situation with high rates of thrombosis recurrence and risk of bleeding. There is highly variable use of platelet transfusion. The British Society for Haematology and International Society on Thrombosis and Haemostasis (ISTH) have issued guidance, whilst accepting very limited data to inform optimal management. In order to better document UK practice, the CAVEaT (Cancer Associated Venous thrombosis and Thrombocytopenia) multicentre audit has been established in collaboration with the HaemSTAR network of Haematology trainees and NHS Blood and Transplant. We present interim data from the first 9 months.

Consecutive patients with haematological malignancy, platelets $< 50 \times 10^9/L$ and VTE within 28 days are identified prospectively by the local haematology team at participating sites. Fully anonymised and routinely available data are collected at entry, 4 weeks and 12 weeks from VTE. Participating sites are required to register the audit locally.

Between May and December 2019, 27 patients have been included from 13 sites. Patient characteristics are: median age 66 years (range 15-90); 16 male, 11 female. Underlying diagnosis: 13 Acute Leukaemia (2 Acute Promyelocytic Leukaemia), 4 MDS, 5 Aggressive NHL, 3 Low grade NHL / CLL, 1 Myeloma. Current treatment: 10 inpatient chemotherapy, 4 bone marrow transplantation, 9 outpatient palliative chemotherapy, 4 supportive care. 3 patients received peg-asparaginase in the last 8 weeks. Thromboses by ISTH risk

Table. Venous thrombosis management and outcomes at 4 weeks from VTE by ISTH risk assignment.

High risk VTE ($n = 16$)

Management	Total patients entered	Outcome data at 4 weeks from VTE $n = 8$	Median Inpatient days in the last 4 weeks	VTE progression / recurrence	CRNMB	Major bleeding	Deaths	Median platelets transfused
LMWH with Platelet transfusion threshold $40-50 \times 10^9/L$	8	4	11	1	1	0	0	3.5
LMWH with Platelet transfusion threshold less than $40-50 \times 10^9/L$	4	2	12	0	1	0	0	6.5
Unfractionated heparin	0	N/A	N/A	N/A	N/A	N/A	N/A	N/A
DOAC	3	2	14	0	0	0	0	9.5
No anticoagulation	1	1	28	0	0	0	0	1
Low risk VTE ($n = 11$)								
(Upper limb and subsegmental PE)								
LMWH with Platelet transfusion threshold $40-50 \times 10^9/L$	3	1	0	0	0	0	0	Unknown
LMWH with Platelet transfusion threshold less than $40-50 \times 10^9/L$	4	4	15	1	1	1	2	8
Unfractionated heparin	1	0	N/A	N/A	N/A	N/A	N/A	N/A
DOAC	0	N/A	N/A	N/A	N/A	N/A	N/A	N/A
No anticoagulation	3	3	9	0	0	0	2	0

VTE venous thromboembolism, LMWH low molecular weight heparin; CRNMB Clinically relevant non-major bleeding; PE pulmonary embolus; DVT Deep vein thrombosis

classification: 16 high risk: 9 PE, 4 Proximal lower limb, 2 Internal jugular vein (both catheter associated), 1 other. 11 low risk: 9 upper limb (8 catheter associated), 2 subsegmental PE. All 11 low risk VTE and 8 of 16 high risk VTE patients were symptomatic. Median platelet count at entry was $35 \times 10^9/L$ (range 7-49).

10 of 16 (62.5%) patients with high risk thrombosis were managed with the standard ISTH guidance approach of low molecular weight heparin (LMWH) with platelet transfusion threshold $40-50 \times 10^9/L$; 4 of 11 (36.4%) of low risk patients were also managed with this approach. Amongst the remaining patients, 5 received LMWH with platelet transfusion thresholds less than $40 \times 10^9/L$, 1 received unfractionated heparin, 3 a DOAC, and 4 no anticoagulation.

17 patients have data from the 4 weeks from VTE timepoint as presented in table 1. There were 2 cases of VTE progression. 4

patients had clinically relevant non-major bleeding and 1 patient major bleeding. 1 patient had an IVC filter placed, which has not been removed. The anticoagulation strategy was changed for a further 2 patients because it was not practical to maintain the target platelet count by transfusion. There were 3 deaths, including the patient with major bleeding. In the first 28 days median number of platelet units transfused was 6. Median inpatient stay was 11 days.

These preliminary data from the CAVEaT audit project demonstrate substantial variation in practice, with significant morbidity and mortality in this group of patients with cancer associated thrombosis and thrombocytopenia. We plan to continue to a target of a total of 100 patients. Sites wishing to join the project as it continues should contact CAVEaT@nhsbt.nhs.uk.

Disclosure of Interest: None Declared

Transfusion

BSH2020-PO-181

Dynamic and time-to-event analyses demonstrate marked reduction in transfusion requirements for janus kinase inhibitor-naïve myelofibrosis patients treated with momelotinib compared head to head with ruxolitinib

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Momelotinib (MMB) is a potent, selective, orally-bioavailable, small-molecule inhibitor of Janus kinase (JAK) 1, JAK2 and ACVR1 being developed for the treatment of intermediate and high risk myelofibrosis (MF). Systemic inflammation integral to the pathogenesis of MF leads to increased ACVR1 activity which in turn increases secretion of hepcidin, resulting in perturbed iron homeostasis and an iron-restricted anemia (Physiol Rev. 2013;93:1721–41, Am J Hematol. 2014;89:470–9). MMB's inhibition of ACVR1, unique amongst the JAK inhibitor (JAKi) class, leads to a reduction of hepcidin, restoring iron homeostasis and red blood cell (RBC) production and alleviating anemia and transfusion dependency (TD). Chronic, progressive anemia is the key hallmark feature of MF; anemia and TD are strongly predictive of reduced survival (Am J Hematol. 2013;88:312–6). MMB is the only clinical stage JAKi to possess potent ACVR1 inhibitory activity, resulting in improvement of anemia in contrast to ruxolitinib (RUX) which results in worsening.

The SIMPLIFY-1 (S1) study, a double-blind, active-controlled Phase 3 study in which 432 patients received randomized treatment with MMB or RUX for 24 weeks was previously reported (JCO. 2017;35:3844–50). In addition to a significant reduction in splenomegaly and improvement in constitutional symptoms, the study demonstrated that patients in the MMB arm achieved nominal-statistical significance for all anemia endpoints tested, including a higher rate of transfusion independence ($P < 0.001$) and lower rates of TD ($P = 0.019$) at Week 24, compared to patients on RUX, consistent with MMB's pro-erythropoietic effect. Overall, a demonstrably decreased transfusion requirement was noted in patients who received MMB vs RUX.

Since transfusion burden is of significant concern to clinicians and patients, to better understand the dynamics of RBC transfusions we further examined the S1 data through statistical models utilizing

a variety of novel anemia benefit endpoints. Kaplan-Meier estimates of the proportion of patients who required zero units transfused during the randomized treatment period were 73% and 46% for MMB and RUX respectively ($P < 0.0001$), while the proportion of patients requiring 4 or fewer units were 83% and 62% ($P < 0.0001$). When examining units transfused as recurrent events, patients receiving MMB possessed a hazard ratio (HR) of approximately one-half that for patients on RUX (HR 0.522; $P < 0.0001$) both with and without patients' baseline characteristics as covariates. A mixture model, based on a zero-inflated negative binomial (ZINB) distribution fit to the transfusion data demonstrated that MMB increased the odds of having zero units transfused by a factor of 9.3 ($P < 0.0001$) vs RUX.

Taken together, the novel dynamic and time-to-event analysis methods described are relevant and informative additions to standard measures of transfusion burden in patients with MF. The results of these analyses allow more detailed description of MMB's differentiated anemia benefit as compared to RUX in a double-blind study of JAKi-naïve patients. These results when combined with additional data from the SIMPLIFY studies demonstrate that MMB is able to address the three hallmark features of MF, namely anemia, constitutional symptoms and splenomegaly, differentiating it from other JAK inhibitors. The benefit of MMB in reducing transfusion burden will be further evaluated in MOMENTUM, a Phase 3 study of MMB in MF patients.

Disclosure of Interest: D. McLornan Conflict with: Honoraria; Jazz Pharm. & Novartis, Conflict with: Honoraria; Jazz Pharm. & Novartis, R. Mesa Conflict with: Novartis/Sierra Onc/La Jolla/Pharma, Conflict with: Celgene/Incyte/Abbvie/Samus/Genotech/Promedior/CTI, J. Catalano Conflict with: Travel support: Celgene, F. Cervantes Conflict with: Advisory board / speaker's bureau; Novartis and Celgene, T. Devos: None Declared, M. Egyed: None Declared, J. Gotlib: None Declared, J.-J. Kiladjian Conflict with: Celgene, Conflict with: Novartis, AOP Orphan, Conflict with: Honoraria: Novartis, AOP Orphan, K. Shimoda Conflict with: Bristol's Myers Squibb, Novartis, Chugai Pharma, Kyowa Kirin, Pfizer, MSD, Conflict with: Lecture fee: Novartis, Shire, Celgene, Bristol's Myers Squibb, E. Coart Conflict with: Employment: IDDI, K. D'hollander Conflict with: Employment: IDDI, R. Donahue Conflict with: Employment: Sierra Oncology, M. Kowalski Conflict with: Employment: Sierra Oncology

BSH2020-PO-182

A national survey and simulation study of pre-hospital whole blood transfusion: what are the benefits?

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Background: A balanced 1:1 haemostatic resuscitation with red blood cells (RBC) and fresh-frozen-plasma (FFP) reduces mortality

in traumatic haemorrhage, and is standard of care in hospital. Achieving a balanced resuscitation whilst delivering different blood components pre-hospital is a logistical and clinical challenge. A product containing all components in one bag (i.e. leuco-depleted red cell and plasma [RCP] or whole blood [WB] (red cells, plasma and platelets in a single bag) could overcome these challenges.

Aim and methods: This study aimed to: 1) survey the pre-hospital community and major trauma centres (MTCs) on the optimal transfusion strategy for use pre-hospital, and 2) establish the logistical and practical benefits of pre-hospital RCP use (2 units) versus other transfusion strategies (2 RBC + 2 FFP units and 2 RBC + 2 Lyoplas units) by conducting a three-arm cross-over simulation. The simulation outcomes measured were: median flow time (overall scene time), median touch time (time pre-hospital teams are directly involved in blood transfusion and unable to perform other tasks), and the number of steps, checks, people, and equipment needed from decision to transfuse (DTT) to complete transfusion.

Results: 17 of 27 MTCs and 17 of 22 air ambulance trusts responded to the survey. 82% of services stated that WB would be their preferred pre-hospital component, followed by 65% with RCP as their second choice. Over 80% stated that a combined component would improve the time to a balanced haemostatic resuscitation and the ease of pre-hospital blood product administration. 100% stated that the development and trial of pre-hospital WB versus other transfusion strategies would be beneficial for both patients and clinicians.

The cross-over simulation showed that transfusion of 2 units RCP had a shorter flow time and touch time compared to alternative transfusion strategies. Transfusion of 2 units RCP had a median touch time of 2 minutes 31 seconds for LD-RCP, compared to 5 minutes 21 seconds and 15 minutes 3 seconds for RBC/FFP and RBC/Lyoplas, and a flow time of 6 minutes 31 secs against 12 minutes 20 seconds and 18 minutes 29 seconds. The LD-RCP arm required fewer steps, pieces of equipment and checks to produce a balanced pre-hospital haemostatic resuscitation than both other arms (see table).

Conclusion: There is national interest in developing pre-hospital WB component use. There is clinician demand for a future trial to demonstrate its efficacy and benefits. The benefits of a pre-hospital combined component are a decreased touch time, and reduced crew cognitive burden with less interruption of time critical processes. This liberates the pre-hospital team to perform other tasks on scene, and reduces potential for human error.

Combined component transfusion leads to earlier haemostatic resuscitation, decreased touch time, more effective resource use, and reduced overall scene time. The decrease in tasks, touch time, and clinician cognitive burden leads to increased scene efficiency, earlier effective treatment and more rapid hospital transfer for traumatically injured bleeding patients.

Disclosure of Interest: None Declared

BSH2020-PO-183

A feasibility study on the provision of extended matched red cells for transfusion support in chronic haematological diseases

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Myelodysplasia (MDS), Chronic Myelomonocytic Leukaemia (CMML), Myelofibrosis (MF) and Aplastic Anaemia (AA) are haematological disorders that result in bone marrow failure; therefore, a patient will become transfusion dependent to improve their quality of life. The risk of antibody production is thought to be raised with an increased exposure to different donor red blood cells (RBC).

Antibody production is a burden to society and causes provision of donor RBCs for transfusions to become challenging. This can be detrimental to the patient as they are at a greater risk of having a transfusion reaction and becoming non-transfusable.

24 patients were identified as having MDS, CMML, MF or AA and with no transfusions in the last three months. A Direct Antiglobulin Test (DAT) and serological phenotype (consisting of the Rhesus (Rh), Kell, Duffy, Kidd, MNS blood group systems) was performed. A molecular RBC genotype was performed by NHSBT. Discrepancies were identified between eight patient's results.

It was investigated if three fully phenotypically matched donor RBC could be provided from routine stock, three times monthly, for three months for each of these patients as a measure of feasibility.

Providing fully phenotype matched donor RBC for every transfusion from routine stock was only possible for five patients. However, donor RBC matched for Rh/Kell only could be provided for all 20 patients. Performing a serological phenotype and providing Rh/Kell matched RBC was determined to be the most cost-effective option.

200 patients with the same diagnoses were retrospectively identified who had been transfused with non-phenotype matched blood. The rate of alloimmunisation due to transfusion was determined.

Of the 200 patients, 23 became alloimmunised (12%). The non-alloimmunised group received a median of 23 units, versus 61 units in the alloimmunised group ($P = 0.0001$). Seven of the patients developed antibodies to Rh/Kell only and five patients to Rh/Kell plus an additional antibody. Anti-E was the most common antibody.

Rh/Kell matching of donor RBCs would have prevented 30% of alloimmunisation in this transfusion dependent group of patients. If fully phenotype matched RBC were to be given, 43% of alloimmunisation could be prevented but delays in provision could have been significant. We are currently discussing next steps with Clinical Haematology colleagues with a view to implementing Rh/Kell matching in the next few months.

Disclosure of Interest: None Declared

Table.

Blood Component(s)	Median Time taken from DTT to administration (Flow time)	Median Touch Time	No. of tasks	No of pieces of equipment needed	No of checks	No of people needed
2 RBC + 2 FFP	12 min 20 sec	5 min 21 sec	40	10	16	2
2 RBC + 2 Lyoplas	18 min 29 sec	15 min 03 sec	46	12	16	3
2 LD-RCP	6 min 31 sec	2 min 31 secs	28	4	8	2

BSH2020-PO-184

Rh antibodies in patients with sickle cell disease – an on-going challengeC. S. Booth^{1,2,*}, F. Barroso¹, S. Bennett¹, D. Kafuko¹, P. Telfer¹, B. Kaya¹¹Haematology, Barts Health NHS Trust, ²Haematology, NHS Blood and Transplant, London, UK

Patients with sickle cell disease are at increased risk of forming red cell alloantibodies after transfusion. This can result in haemolytic transfusion reactions, particularly when patients are transfused at other hospitals where their antibody history is not known. It can also create challenges in sourcing compatible blood and cause delays when patients require transfusion in an emergency. In 1997, UK blood banks adopted a policy for issuing extended Rh (CcEe) and K matched red cells to patients with sickle cell disease to reduce exposure to these highly immunogenic antigens. We examined the transfusion records of all patients transfused at our centre to assess the effectiveness of this intervention.

The National Haemoglobinopathy Register (NHR) was used to identify all patients with sickle cell disease registered through Royal London Hospital. Data regarding past transfusions and Rh phenotype were recorded from the Laboratory Information Management System (LIMS). Where available, information on red cell genotype was obtained from the International Blood Groups Reference Laboratory (IGBRL) via the Sp-ICE database.

727 patients were registered on the NHR via our centre. Of these, 405 had received a blood transfusion in our Trust and 97 (24%) of these had formed alloantibodies. 51 (53%) had a single alloantibody, 18 (19%) had two, 18 (19%) had three and 13 (13%) had four or more.

A total of 26 different antibodies were recorded, 14 of which would necessitate antigen-negative units for transfusion. The commonest specificities were anti-C (23 patients), anti-Kpa (23), anti-S (19), anti-Jkb (17) and anti-E (17).

48 (50%) of alloimmunised patients had formed antibodies to Rh antigens: 28 of the 48 (58%) patients born in 1997 or earlier and 20 of the 49 (40%) patients born after 1997. All transfused patients born after 1997 had a full Rh phenotype and appropriate flags for special requirements on our LIMS at the time of this study, but some may have been transfused in countries without access to Rh matched products, or at hospitals whose laboratories were not informed of their diagnosis of sickle cell disease.

9 of the 20 patients born after 1997 who had formed Rh antibodies had been phenotyped as positive for the corresponding antigens. 3 of these were shown on genotyping to have Rh variants (one D variant and two e variants), 2 (one having anti-c and one anti-e) had not been genotyped and 4 (three with anti-e and one anti-D) were confirmed positive for the antigen by genotyping, consistent with true autoantibodies.

Alloimmunisation remains a challenge in patients with sickle cell disease. Even when antigen-negative units are not required, a positive antibody screen will inevitably lead to delays in supplying red cells. Antibodies to Rh antigens are common in spite of extended Rh matching. Good communication between clinical teams and laboratories is essential to ensure all special requirements are met and patients should be empowered to remind clinicians they need 'special' blood, particularly when receiving a transfusion at a hospital for the first time. Red cell genotyping can help provide insight into unexpected antibodies where patients have variant Rh alleles and may influence decision-making around selecting most compatible products.

Disclosure of Interest: None Declared

BSH2020-PO-185

Group O D negative blood – promoting the appropriate use of an essential but limited resourceS. Allard^{1,*}, D. Foukaneli^{2,3}, M. Bend⁴, L. Estcourt⁵, L. Sherliker⁶¹Clinical, Medical, NHS Blood and Transplant, London,²Clinical, Medical, NHS Blood and Transplant, ³Clinical, Cambridge University Hospitals NHS Foundation Trust, Cambridge, ⁴Blood Stocks Management Scheme, NHS Blood and Transplant, Sheffield, ⁵Clinical, Medical, ⁶Patient Blood Management, NHS Blood and Transplant, Oxford, UK

While overall red cell usage has decreased significantly (at least by 25% in 15 years) with the implementation of Patient Blood Management (PBM), the relative demand for O D negative blood remains strong and has actually been rising for the past 12 months. This is an essential resource for prompt transfusion of red cells in an emergency where the patient group is unknown. However, over dependence may have a negative impact on blood stock management and the balance between supply and demand of O D negative red cells remains a major challenge for blood services in many countries.

Approximately 7% of the UK population is O D negative, but current average demand of O D negative blood cells by hospitals in England supplied by NHS Blood and Transplant (NHSBT) is > 13% causing significant challenges to the supply chain. Considerable efforts and resources are being focussed on improved donor marketing and recruitment to achieve a blood group mix to fulfil patient needs.

There are also ongoing challenges in meeting the demand for Ro blood. The Ro phenotype (D positive, cc and ee) is common in the black population. In patients with Sickle Cell Disease requiring Rh phenotype matched blood, the demand for Ro red cells has doubled in the last 3 years with an increased number of patients moving to automated red cell exchange programmes. NHSBT can currently fulfil ~50% of Ro demand with actual Ro units and therefore has to issue ABO rr as a substitution placing additional pressure on O D negative stocks.

There are ongoing significant collaborative efforts between NHSBT and Hospitals to promote optimum usage with minimum wastage of this finite resource. The National Blood Transfusion Committee in conjunction with NHSBT has updated guidance for hospitals in 2019 outlining best practice principles for the appropriate clinical use and stockholding of O D negative blood.

Repeated surveys have been undertaken of the clinical use of O D negative red cells (2008, 2010, 2015, 2018) with the latest key findings showing that overall 59.3% of O D negative red cells are transfused to O D negative patients with a significant change in practice from the 2010 O D negative audit where 70% of O D negative red cells were transfused to O D negative patients. 16% (794/4970) of the total units were used as "emergency units" whereas in 2010, "emergency units" were used in 280/5108 (5.5%) transfusion episodes. 12.6% O D negative red cells were transfused to avoid wastage due to time expiry and 5% were wasted. 6 % O D negative red cells were transfused in an emergency to males and females aged over 50 years. It is important to note that 31% of sites do not have a policy to provide O D positive red cells in an emergency to unknown males and females aged > 50 years. 10.8% of O D negative red cells were used as a substitution by hospital laboratories. Approximately half of those needs could have been met by suitable O D positive red cells. Availability of extensively phenotyped units of O D positive red cells as well as red cells meeting other special requirements (irradiated, CMV negative) could have reduced those substitutions. O D Negative stockholding greater than 12.5% has increased from 46% of sites in 2010 to 64% of sites in 2018.

We have also developed an O D negative toolkit as a web-based resource for clinical and laboratory teams and are also currently working in a bespoke manner with hospitals that are the highest users to review and influence practice.

Disclosure of Interest: None Declared

BSH2020-PO-186

Does early discharge affect administration of anti-D Ig prophylaxis

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Introduction: Haemolytic disease of the foetus and newborn (HDFN) can be prevented with routine antenatal anti-D prophylaxis (RAADP). The RAADP should be administered during the pregnancy, according to local guidelines, and within 72 hours of delivery.

This qualitative review was undertaken to determine if early discharge is negatively affecting the administration of Routine Antenatal Anti-D Prophylaxis (RAADP) within 72 hours of delivery.

Method: The SHOT data base was reviewed for the two years 2017 – 2018 and incidences where early discharge may have led to late administration or omission of RAADP were reviewed.

Results: Each year SHOT receives in excess of 3000 error reports. Errors relating to RAADP in the two years 2017 and 2018 received by SHOT totalled 892 (13.6% of all reports). There were 138 reports (15.4%) where RAADP was not administered within 72 hours of birth. Of those 138 reports 43 (31.1%) were solely attributed to the early discharge of the women. There were only 4 incidents noted to be out of hours. The time range for Out of Hours is from 20:00 hours to 08:00 hours. The remaining 95 (68.8%) involved early discharge but had other attributable causes which cannot solely be attributed to the early discharge.

On a review of the reports made, SHOT has noted that some areas have checklists in place which include the need for RAADP but in reports where the use of a checklist is noted this has not resulted in timely administration of RAADP. Similarly there are areas which have a sticker on the case notes to alert the midwife and again where this is mentioned in the report it was not a sufficiently robust system to prevent the omission. There does not appear to be a robust system in place to identify those who require RAADP and subsequently a robust system to administer it within 72 hours.

In all of the reported cases the women were identified and did receive RAADP > 72 hours after delivery. This was largely administered in the community. Error reports relating to RAADP have been increasing year on year, which suggests awareness is increasing but it is likely there are a number that remain unreported or unrecognised. The concern that failure to administer RAADP in a timely manner or at all will lead to an increase in Anti – D sensitisation continues but that information is not yet available for this group of women.

Recurring themes identified

Lack of communication between hospital and community midwifery teams – particularly in those patients discharged early after delivery

Lack of understanding amongst staff of when anti-D Ig is required Checklists ticked when RAADP not administered Conclusion

Maternity teams need to have robust systems to administer RAADP within 72 hours of birth. As more women are being discharged early bloods should be checked prior to discharge and women who require RAADP should be identified prior to discharge. Administration of RAADP could then be arranged in the community with a simple standard that each woman who requires RAADP

should receive this on day two after delivery day. This standard could form part of the ongoing patient and staff education. Setting such a standard could also improve communication between hospital and community teams.

Disclosure of Interest: None Declared

BSH2020-PO-187

Red cell alloimmunisation in patients receiving Veno-Venous Extracorporeal membrane oxygenation (VV-ECMO)

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Veno-venous Extracorporeal Membrane Oxygenation (VV-ECMO) is lifesaving for patients with severe respiratory failure. These patients require frequent repeated red cell transfusion which can lead to development of alloantibodies against one or more red cell (RC) antigens, complicating subsequent transfusions. The frequency of alloimmunisation in patients receiving EMCO is unknown. Although the optimal transfusion threshold for patients receiving VV-ECMO remains a source of controversy, our threshold for transfusion is a hemoglobin (Hb) of ≤ 70 g/L compared to > 120 g/L as per international guidelines on transfusion in ECMO. The aim of this retrospective single centre observational study was to assess the frequency of RC alloimmunisation in patients receiving VV-ECMO.

All patients who received ECMO from September 2017 to August 2019 were included in the study. Patients with red cell alloantibodies at the initiation of VV-ECMO were excluded. ABO and Rh blood groups and the RC antigens (D, C, c, Cw, E, e, K, k) were typed (using DiaClon Rh-subgroups +K, Bio-RAD, UK) in all patients on admission prior to receiving any RCs. ABO, Rh and Kell matched blood was given to all patients requiring RC transfusion. An antibody screen was performed three times a week in all patients on VV-ECMO. If the screen was positive, antibody identification was performed locally, and a sample was sent to the national blood service reference laboratory for confirmation.

A total of 179 of patients were studied with a median (range) age of 47(16-76) and 61% were male. Rh(D)-positive phenotype was observed in 88% (159/179) patients and 60 % were male. Twelve% (20/179) patients were Rh(D)-negative of which 70 % were male. The median (range) number of RC units transfused during VV-ECMO was 8 (2-59) during a median duration of 7 (range 4-45 days) on ECMO. Six patients developed alloantibodies (anti-K in 4 and anti-E in two) during this period hence the frequency of alloimmunisation in patients receiving VV-ECMO was 3.4% (6/179). Median age (range) of the patients who developed RC antibodies was 46.6 (43-54) years and 50% were male. The median days (range) of transfusion whilst on VV-ECMO and the number of RC units (median and range) transfused prior to development of RC alloantibodies were 7 days (4-21) and 6 units (5-29 units) respectively. There was no difference in the age, gender, median days of transfusion whilst on VV-ECMO and the number of red cell units transfused in patients who developed RC alloantibodies compared to the patients who did not developed RC antibodies.

In conclusion, despite the lower Hb threshold for transfusion and extended red cell phenotyping and matched red cells transfusion, the frequency of RC alloimmunisation in patients receiving VV-ECMO was 3.4% where they received only 6 RC units (5-29 units) over a median of 7 days (4-21) whilst on VV-ECMO prior to development of alloimmunisation. The frequency of RC alloimmunisation in

patients receiving VV-ECMO was similar to reported incidence of RC alloimmunisation in patients who received repeated RC transfusion for haemoglobinopathies (3%) where all alloimmunised patients received more than 10 RC units (range 18 to 152) in 6 months or more prior development of RC antibodies. This could be due to acute inflammatory process in patients receiving VV-ECMO which can predispose to development of alloantibodies to transfused red cells and may need frequent monitoring for development of alloantibodies.

Disclosure of Interest: None Declared

BSH2020-PO-188

Reduction of platelet thresholds to adhere with British Society of Haematology (BSH) guidance for tunnelled central line insertion does not increase bleeding complication in haematology patients; a single centre experience

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The use of platelet transfusion increased year on year from 2007, up to and including 2015. This, in part, was due to an increase in haematological malignancy diagnosis in the ageing population in addition to more intense treatment strategies for older patients including stem cell transplant. Alongside platelet support for disease and treatment-induced bone marrow failure, platelet transfusions are indicated to raise the platelet count above a defined threshold allowing the safe undertaking of invasive procedures in thrombocytopenic patients.

The 2016, British Society of Haematology (BSH) guideline 'Guidelines for the Use of Platelet Transfusions' recommended a platelet threshold of $> 20 \times 10^9/l$ for the insertion of tunnelled central venous access. Prior to this guidance, a platelet threshold of $> 50 \times 10^9/l$ was widely used. Despite the recommendation in 2016, national variation in practice continues with hesitance to lower the platelet threshold expressed from centres due to concerns of increased bleeding complication risk.

At Manchester Royal Infirmary, the interventional radiology and clinical haematology teams trialled the implementation of the recommended BSH procedural platelet transfusion threshold of $> 20 \times 10^9/l$ for the insertion of ultrasound guided, tunnelled central catheters by experienced radiology staff members. Previously, a platelet threshold of $> 50 \times 10^9/l$ was used at the Trust. This prospective, un-blinded observational study compared complication rates between the patient group with a pre-procedural platelet count between $20-50 \times 10^9/l$, to those with a pre-procedure platelet count $> 50 \times 10^9/l$.

Seventy-three patients with a diagnosis of a haematological malignancy who received a central tunnelled line insertion at Manchester Royal Infirmary between April 2018 and October 2019 were included in data analysis. Those patients under the age of 16, with a non-haematological malignancy diagnosis and those with deranged coagulation profile parameters prior to intervention were excluded. A pre-procedure haemoglobin concentration, platelet count and coagulation screen were recorded for each patient and complications during and following the procedure were recorded.

A total of 15/73 (21%) patients had a pre-procedure platelet count of less than $50 \times 10^9/l$; of which 3/15 had a pre-procedure platelet count between $40-49 \times 10^9/l$, 8/15 between $30-39 \times 10^9/l$ and 4/15 between $20-29 \times 10^9/l$.

The overall complication rate was 2/73 (3%). In both patients, the complication described was a small haematoma requiring

mechanical compression only. Of the 15/73 patients with a platelet count of $< 50 \times 10^9/l$, 1/15 had a minor complication as described (patient had a pre-procedure platelet count of $40 \times 10^9/l$). The other patient who developed a post-procedural haematoma had a pre-procedure platelet count of $357 \times 10^9/l$.

Although small patient numbers, the experience from our centre is that the national platelet threshold recommendations of $> 20 \times 10^9/l$ for central tunnelled line insertion is safe to implement with no observed increase in bleeding complications.

Adherence to national guidance also resulted in less delays and a reduction of cancellations of line insertions for those patients with a pre-procedure platelet count between $20-50 \times 10^9/l$. By reducing this threshold, 15/73 patients avoided an unnecessary pre-procedure transfusion of platelets, the risks of which include transfusion reactions, transfusion transmitted infection and alloimmunisation.

Disclosure of Interest: None Declared

BSH2020-PO-189

Alloimmunisation in sickle cell disease in the North Central London sickle cell cohort: providing a baseline for extended red cell antigen matching and targeted donor genotyping

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In the UK there are approximately 14,000 patients with sickle cell disease (SCD). For a significant number of these patients, their medical lives are defined by access to, need for and complications of blood transfusion. Alloimmunisation remains one well-recognised complication and sourcing blood as an emergency or as part of a long-term transfusion programme for those with certain types or combination of alloantibodies is well recognised to be complicated and laborious. In some cases, sufficient numbers of typed units may not be available rendering the patient 'un-transfusable'. The aim of this project was to document and characterise the numbers and patterns of red cell antigens and alloantibodies in a large current cohort of patients with sickle cell disease.

We gathered transfusion data on 2,026 haemoglobinopathy patients across the North Central London Sickle Cell and Thalassemia Network (NCLSTN), which links UCLH, Whittington Health and North Middlesex Hospital data. Patients were identified from local databases, the National Haemoglobinopathy Registry and Blood Transfusion Laboratory Information Management Systems (LIMS).

Their antibody and antigen status was identified from the Blood Transfusion LIMS as well as NHSBT SpICE. Data was pseudo anonymised and gathered into an excel spreadsheet and analysed. The researcher had an honorary contract at all three sites.

In this unselected population 205 of 1455 SCD patients (14%) had alloantibodies. Of the 1455 patients 918 had HbSS (63%), 376 were < 18 years old (26%) and 309 were on regular transfusion (21%). 74/309 of those on regular transfusion were alloimmunised, for those not on regular transfusion it was 131/1146. Rh and Kell were the most common antibodies, then: Anti S 44/1455 – 3%, Anti Kpa 33/1455 – 2.2%, Anti M 22/1455 – 1.5%, Anti Cw 20/1455 – 1.4%, Anti Le a 19/1455 – 1.3%, Anti Fya 18/1455 – 1.2%. Only 567/1455 had a genotype performed with variant analysis. In this group: Blood Group O 50%, no one was AB, 8% had C variants (49/567), 35/532 U neg 6.6%, 528/567 were Fya neg.

Our analysis provides a picture of the profile of alloimmunisation in a large population. A significant number of patients fail to develop any alloantibodies despite regular exposure to non-extended matched blood products. The findings support the need to better understand the national burden of alloimmunisation in haemoglobinopathies and the impact on patient care, but also to explore new strategies for optimising matching and timely delivery of blood. Further work is planned to add to this cohort so that we can ascertain risk of alloimmunisation and with donor genotyping develop a matching strategy for a precision transfusion model algorithm and a pilot study of better matched blood. One approach is the application of new technologies for cost-efficient genotyping, which might support a reconfiguration of the entire process of blood grouping, matching and selection of blood. Furthermore, by identifying the optimal antigen profiles for our patient cohort, the donor recruitment strategy can be refined and donors could be targeted for donations with precision.

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BSH2020-PO-190

Incorrect plasma components transfused: when it comes to ABO compatibility Do we need to stop only ever 'seeing red'?

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The Serious Hazards of Transfusion (SHOT) haemovigilance scheme collects and analyses anonymised information relating to serious adverse reactions and serious adverse events of blood transfusion reported in the UK (UK). SHOT then makes recommendations to improve patient and transfusion safety. Unintentional ABO-incompatible (ABOi) blood transfusions are regarded as Never Events by NHS England, NHS Wales and in Northern Ireland (serious incidents that are wholly preventable), or as Red Events by NHS Scotland (significant adverse events). ABOi transfusions demonstrate a preventable breakdown in transfusion protocols and standard operating procedures. They can have severe consequences, with significant morbidity and mortality. Although the risk of haemolysis and serious

harm is more likely with red cells, ABO incompatibility for plasma components still carries significant risk, particularly in infants.

The number of ABOi reports made to SHOT from 2012 to 2018 were analysed to determine what proportion of these errors were related to plasma components (standard and pathogen reduced fresh frozen plasma, and cryoprecipitate), and to identify the key steps where improvements could be made to reduce these errors.

Analysis of ABOi errors shows 22/70 (31.4%) involved plasma components. Plasma components accounted for 2,645,596/18,095,566 (14.6%) of all blood components issued in the UK from 2012-2018. Therefore, a disproportionately high percentage of plasma component ABOi events are reported.

The proportion of ABOi due to plasma components has also increased over time, from 3/15 (20.0%) in 2012 to 3/7 (42.9%) in 2018. Education surrounding red cell compatibility has improved and the number of ABOi red cell transfusions has decreased, this is not the case with plasma components.

A total of 20/22 (90.9%) plasma related ABOi errors originated in the laboratory: most errors 12/20 (60.0%) occurred at component selection, 5/20 (25.0%) during testing and 3/20 (15.0%) at sample receipt and registration. As the majority of initial errors occurred during laboratory component selection, the following learning recommendation was made in the 2018 Annual SHOT Report: "It is important to ensure stock that may possibly be issued in ABO-mismatched scenarios is of the correct specification and the standard operating procedure (SOP) is clear about replacement group issues".

Additionally, in all ABOi transfusion events, this initial error is not detected at the bedside administration check. Therefore, the Annual SHOT Report 2017 contained the recommendation "Training in ABO and D blood group principles is essential for all laboratory and clinical staff with any responsibility for the transfusion process. This should form part of the competency assessments".

Whilst there is heavy emphasis on the ABO compatibility of packed red blood cells (RBC) in transfusion education, education and awareness regarding compatibility of plasma components and platelets is often lacking, contributing to errors. It is essential that everyone involved in transfusions are aware about compatibility of plasma components as well to reduce the risk to patients and improve transfusion safety.

Disclosure of Interest: None Declared

BSH2020-PO-191

Transfusion 2024 – defining a 5-year national strategy for clinical and laboratory transfusion practice

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Around 2 million units of blood and components are used in England to support the care of adult and paediatric patients with a wide range of clinical conditions. It is essential to provide clear policies for the many healthcare teams involved in the care of these patients to ensure safe and appropriate transfusion practices.

The Transfusion 2024 symposium was organised in March 2019 by the National Blood Transfusion Committee (NBTC) and NHS Blood and Transplant (NHSBT) with the aim of defining a strategy

for clinical and laboratory blood transfusion for England for the next 5 years. The topics covered included patient blood management (PBM), transfusion laboratory safety, and research and development. With > 100 invited participants, the symposium provided an effective forum for NBTC and NHSBT to work collaboratively with hospital teams, Royal Colleges, professional bodies, regulators, healthcare providers, NHS England (NHSE) and above all patients to determine priorities for transfusion care in line with key NHS policies.

Multi-professional discussion informed by short presentations emphasised the importance of building on the successes of previous initiatives including the recommendations provided by the Better Blood Transfusion Health Service Circulars (1997, 2002 and 2007) and subsequent PBM Seminar (2012). Over the last 20 years, there has been considerable improvement in transfusion practice supported by evidence from clinical trials, guideline implementation and process improvements. These have resulted in improved patient safety, a 25-30% reduction in blood use and significant cost savings for the NHS. However, there remains considerable variability in transfusion practice within and between hospitals that may impact on patient outcomes, and this needs further action.

The outcome of Transfusion 2024 was the development of recommendations for improving clinical and laboratory transfusion practice in England. It was agreed that a core aspect of a future transfusion strategy must include ensuring a trained and competent laboratory workforce to support patient needs across all clinical disciplines where transfusion may be needed. Key actions to strengthen hospital transfusion laboratory safety were the development of advanced information technology and exploration of integrated models for hospitals in collaboration with NHSBT. Other key recommendations were the development of 'PBM' self-assessment tools for hospitals, improved support for Hospital Transfusion Teams especially the Transfusion Practitioner role, and research into novel blood components and genotype matching of blood for multi-transfused patients at risk of red cell alloimmunisation such as patients with sickle cell disease.

The final recommendations are currently awaiting approval by NHSE and are in line with 1) the NHS Long Term Plan with an emphasis on a skilled and trained workforce, better use of data and technology, and integrated models of working; 2) the Department of Health and Social Care in promoting inter-operable digital solutions; and 3) the NHS Patient Safety Strategy in promoting safer patient care.

Disclosure of Interest: None Declared

BSH2020-PO-192

Patient Identification errors in transfusion continue to be a problem

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Patient identification is pivotal to several steps of the transfusion process; it is integral to ensuring that the correct blood component is given to the correct patient. Patient misidentification at any step can have potentially fatal consequences for the recipient. It is imperative that patients are robustly and correctly identified at all stages of the transfusion process and that all spelling and number sequences on the patient ID band and all transfusion-associated documentation are correct and legible. Reports of incidents where a patient was transfused the correct blood component but had an identification error in the process are still being received by SHOT (Serious Hazards of Transfusion).

A retrospective analysis was performed of all Right Blood Right Patient (RBRP) cases reported to SHOT between January 2016 to December 2018 to identify the frequency of patient identification errors and areas for improvement. This category includes any incident where a patient was transfused correctly despite one or more serious identification or prescription errors which in other circumstances might have led to the transfusion of an incorrect blood component.

RBRP incidents accounted for 643/9647 (6.6%) of the total SHOT reports received during this period. Majority of these 392/643 (60.9%) were patient identification errors. Errors relating to patient name on the pre-transfusion blood sample was the most common error 164/392 (41.8%) and were mainly due to mis-spelling of the patient's name 105/164 (64.0%).

Most errors occurred in the clinical setting 283/392 (72.1%) especially with emergency/urgent transfusions 213/392 (54.3%). Of these 93/213 (43.6%) were administered outside normal working hours of 8am to 8 pm. There were 75/392 (19.1%) errors within the emergency department where the urgency of transfusion may have contributed to errors being made. Laboratory errors accounted for 109/392 (27.8%). Errors that originated in the laboratory included wrong demographic data entered manually or incorrect donation number noted.

Patient identification errors continue to occur and can potentially result in life threatening complications in patients due to the wrong component being transfused. The data shows that errors occur more frequently in emergency out of hours transfusions. The use of a bedside checklist may reduce the number of identification errors and it is imperative (where possible) that the patient is involved at the sampling and administration stage of the process. It is important that all staff involved in the transfusion process are aware of the correct patient identification process. There is a need for increased vigilance in both the clinical and laboratory settings. It is very important to review and learn from these incidents, including near misses to prevent future occurrences and potential patient harm.

Disclosure of Interest: None Declared

BSH2020-PO-193

Re-evaluation of therapeutic plasma exchange and compliance with policy at a Tertiary Referral Hospital

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Therapeutic plasma exchange (TPE) is used to treat a variety of pathologies. TPE involves removal of patients' plasma and replacement with a physiological fluid. Human albumin solution (HAS) is commonly used, plasma is used in thrombotic microangiopathies (TMA) or when dilutional coagulopathy due to TPE occurs. It is recommended that clotting tests are monitored regularly and alternate-day exchanges are done to reduce dilutional coagulopathy (ref besh 2015).

Following publication of British Committee for Standards in Haematology (BSCH) 2015 and American Society For Apheresis (AFSA) 2016 guidelines on use of TPE, an audit was undertaken in 2016 to assess use of TPE in a large tertiary referral London hospital. Following this, a local hospital policy was introduced. This study aims to provide an updated evaluation of TPE in the same hospital to assess compliance with the policy.

All requests for TPE within a 2 month period were matched with electronic and paper patient records to gather data on clinical indication, duration, replacement fluid type, volume requested and used, coagulation monitoring and adverse reactions.

27 patients underwent TPE of which 19 (70.4%) were elective. Total number of TPEs was 98, median of 3 per patient (range 3-7). 8 (29.6%) were the patient's first documented episodes of TPE.

Patients with TMA were not included. 21 (78%) were Neurology patients. 11 (40.74%) indications fell into category I/II which guidelines define TPE as recommended or established treatment. 10 (59.25%) were category III indications which guidelines state TPE may be considered.

24 (88.9%) patients had PT, APTT and fibrinogen monitored before the first exchange. 20 (74%) patients had this repeated prior to every subsequent exchange. 1 patient had no coagulation monitoring during TPE. 9 (34.6%) patients with documented fibrinogen had levels ≤ 1 g/dL during TPE. 8 of these (88.9%) received plasma-based products to compensate. Only 1 patient had TPE scheduled on alternate days to avoid dilutional coagulopathy.

22 (81.5%) patients received HAS 5% as their initial replacement fluid. However, 19 (70.3%) patients received plasma at some stage during TPE. 5 received Octaplas (solvent detergent FFP), the remaining 14 received FFP. There was no documentation why Octaplas was given over FFP.

7 patients had documented adverse reactions during TPE which were wide ranging and non specific. 1 patient had bleeding WHO grade 2 with concurrent fibrinogen drop from 1.8 to 1. Octaplas was given on day 2 and no further bleeding symptoms recorded.

In 7 cases where HAS was the planned replacement fluid, volumes requested did not match volume used totalling 6L difference.

In conclusion, following introduction of documented treatment plans and a hospital policy for TPE, all indications for TPE met guidelines during the data collection period in 2019, however the majority were class III indications. Monitoring of coagulation has improved and the use of plasma products has reduced. However there remains disproportionately more patients receiving plasma than have a recorded coagulopathy and documentation of indication is lacking. Poor documentation extended to all adverse reactions.

Based on these data there is scope to improve compliance by using order-sets, treatment plans and prompts via electronic health-care records in this Trust. Blood tests, TPE indications and adverse reactions would be amenable to electronic prompts and drop-down lists for electronic population.

Table. Coagulation monitoring and plasma product use in 2016 and 2019.

	2016 (% patients)	2019 (% patients)
PT, APTT and Fibrinogen checked prior to first exchange	75	88.9
PT, APTT and Fibrinogen checked prior to subsequent exchanges	71	74.0
Documented fibrinogen ≤ 1 g/dL		34.6
FFP or Octaplas given following documented fibrinogen ≤ 1 g/dL	n/a	88.9%
FFP or Octaplas given during TPE	84	70.3
Documented reaction	17	25.9

Disclosure of Interest: None Declared

BSH2020-PO-194

Red blood cell antibodies and adverse transfusion events in multi-transfused patients with sickle cell disease in Ghana

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In Ghana, the main indications for red blood cell (RBC) transfusions in patients with sickle cell disease (SCD) are hyperhaemolysis, recurrent acute pain crisis and severe malaria. Routine pre-transfusion investigations involve only ABO compatibility testing with no screening for RBC antibodies. Little is known on the prevalence of RBC antibodies and adverse transfusion reactions in previously transfused patients with SCD in sub-Saharan Africa.

This was a multi-centre cross-sectional study that investigated multi-transfused Ghanaian patients with SCD, for the presence of, and risk factors for, RBC antibodies and adverse transfusion reactions. Transfusion history and information on adverse transfusion reactions were retrieved from patients' hospital files supplemented by information from the patients or caretakers. RBC antibody testing was performed in the experimental immunohaematology lab, Sanquin, the Netherlands, with standard serologic techniques, using LISS as enhancer and enzyme treated RBC panel cells. Logistic regression was used to determine the association of patient characteristics with the presence of antibodies.

Between July and December 2018, 226 transfused patients with SCD were recruited. Female to male ratio was 1.3 and the median age was 17 years (range 1.7-66). A total of 25 patients (11.1%) had developed 28 RBC antibodies. The rate of antibody formation was 2.5 per 100 units. Antibodies were predominantly of Rhesus specificity. Receiving the first transfusion after the age of three years and a higher number of transfusions were both associated with antibodies (aOR 3.27 and 1.78, respectively). 68 patients (30%), had experienced adverse reactions to transfusions, of which 23 (10%) patients had signs and symptoms suggestive of haemolytic reactions. Adverse transfusion reactions were associated with the number of transfusions (aOR 2.06).

Our observed prevalence of alloimmunisation (11%) is within the range (2.9%-28.6%) reported in other cross-sectional studies in patients with SCD in Sub Saharan Africa. It is however, lower than that reported from high income countries prior to the start of RBC antigen matching (18%-76%), probably due to the differences in RBC exposures. The proportion of overall adverse transfusion reactions is similar to that reported in a previous study in Ghana and other parts of Africa, among SCD and other transfused populations. The proportion with haemolytic reaction is however higher than reported in other African countries. This may be because in our study population we documented adverse reactions during, at 24 hours and beyond the transfusion episode(s) but in other studies, patients were monitored up to 24 hours after transfusion.

In conclusion, red blood cell antibodies and adverse transfusion reactions are common in multi-transfused patients with SCD in Ghana and the prevalence increases with the number of transfusions. Routine pre-transfusion antibody testing, and component transfusion therapy (preferentially leucodepleted) are potential strategies to reduce adverse transfusion events in Ghana.

Table. Summary of the 68 transfusion reactions in 226 multi-transfused Ghanaian patients with sickle cell d

Characteristic	All patients	Females (<i>n</i> = 126)	Males (<i>n</i> = 100)	p-value*
SCD genotype: Hb SS, <i>n</i> (%)	200 (88)	104 (83)	96 (96)	0.005
Hb SC + Other, <i>n</i> (%)	24 + 2 (12)	21 + 1 (17)	3 + 1 (4)	
Age at enrolment (years)	17 (2-66)	21 (2-66)	15 (2-54)	<0.001
Age at first transfusion (years)†	5 (0.1-61)	7 (0.1-61)	3 (0.1-31)	<0.001
Number of transfused units	3 (2-40)	3 (2-40)	3 (2-30)	0.67
Number of transfusion episodes	3 (1-21)	3 (1-21)	3 (1-19)	0.90
Single unit transfusion episodes (%)	74	72	76	0.16
Patients with a single transfusion episode, <i>n</i> (%)	34 (15)	23 (18)	11 (11)	0.13
Years since first transfusion‡	8.6 (0.55-5)	7.8 (0.55-5)	9.7 (0.34-8)	0.13
Years since last transfusion‡	2.0 (0.55-5)	2.2 (0.55-5)	1.8 (0.31-8)	0.63
Years between first and last transfusion†	4 (0-34)	2.8 (0-24)	5.0 (0-34)	0.09
Multiple transfusion hospitals, <i>n</i> (%)§	58 (30)	23 (22)	35 (39)	0.01
RBC antibodies present, <i>n</i> (%)	25 (11.1)	11 (8.7)	14 (14.0)	0.21

Data presented as median (range), unless stated otherwise; *p-value comparing female and male patients; †Data available from 174 patients (84 females and 90 males); ‡Years between transfusion and current study; §Calculated for the 192 patients (103 females and 89 males) with multiple transfusion episodes.

Summary of the 68 transfusion reactions in 226 multi-transfused Ghanaian patients with sickle cell disease

Transfusion reaction	Frequency
Allergic reaction	16(23.5)
Febrile non-haemolytic reaction	13(19.1)
Possible haemolytic reaction*	23(33.8)
Unclassified	16(23.5)

In seven patients a transfusion was abruptly stopped because of problems with the blood unit (clotted, haemolyzed, etc), in six patients because of venous occlusion (i.e. not flowing well) and three patients did not recall the reason.

*Clinical signs suggestive of haemolytic reactions were dark urine (*n* = 12), abdominal/lower back pain (*n* = 9) and abdominal pain with dark urine (*n* = 2). In seven of these patients, fever/chills accompanied these reactions.

Disclosure of Interest: None Declared

BSH2020-PO-195

Transfusion in iron deficiency anaemia: retrospective audit of the frequency and nature of transfusion in the context of iron deficiency in Scottish hospital

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Using data from Account for Blood, the SNBTS Transfusion Team linked the ICD10 code for iron deficiency and identified that in 2017/18 in 36% of episodes of iron deficiency patients received transfusion of at least one unit of red cells. Patients coded as having IDA received 6509 units of blood, approximately 5% of the Scottish blood supply. Guidance from British Society for Gastroenterology, NICE and Patient Blood Management programmes recommend that patients with IDA should only be transfused if iron therapy has been ineffective, not tolerated or in acute bleeding with haemodynamic instability.

A representative random 20% sample (*n* = 371) of patients coded as IDA who were transfused during that episode was audited. 61% of cases coded were confirmed as absolute or functional IDA using a specifically designed algorithm and individual case review.

No cause of anaemia was assigned to 38%. The commonest cause was gastrointestinal blood loss; upper GI (23.3%) and lower GI (18.6%). 2.7% had dietary cause alone.

219 cases had endoscopy, or similar investigations, 14.2% had active bleeding, 38.8% had a previous potential source of blood loss

identified, 43.8% had no source of blood loss identified. There are particular areas for improvement in exclusion of GI malignancy in men and post menopausal women as per BSG guidelines. This audit showed that males are more likely to receive endoscopy as a diagnostic tool compared to females- only 28% of males did not undergo endoscopy compared to 44.1% of females.

Restrictive transfusion practice has been shown by many studies to be safer or equivalent to liberal transfusion practice. Aside from acute blood loss, decision to transfuse should be based on significant symptoms of anaemia such as chest pain or syncope first and Hb level second. In 16.4% cases transfusion would be deemed appropriate supported by guidance. 54 cases were transfused for active bleeding. In 14.8% of cases there was no specific indication leading to the conclusion that the decision was made on Hb level alone. However, in the majority of cases (68.8%) transfusion was justified based on symptoms or co-morbidities which represents a more comprehensive and holistic assessment of need.

Oral iron was given to 72.2% of cases and 19.7% also went on to have intravenous iron. Some had IV iron alone as this was given to 25.9% of cases. 17.3% (*n* = 64) were reported never to have received iron treatment so transfusion had been their only treatment at the point of audit. In 41.2% of this subset of patients, the documented indication was 'fatigue/pallor'.

Prior to transfusion the risks, benefits and alternatives should be discussed and consent documented on the transfusion record. In this audit where there was an alternative to transfusion it was expected that a high proportion would have documentation of a discussion, however, 57.5% did not. Similarly, 54.3% did not have valid consent documented.

Every effort should be made to characterise anaemia to guide treatment, especially before transfusion is considered in haemodynamically stable patients with no severe symptoms. The Haematology

and Transfusion Scotland Network will soon be producing anaemia guidelines for both primary and secondary care clinicians, including algorithms for the diagnosis of IDA and functional iron deficiency hoping to change practice to avoid transfusion when iron treatment is more appropriate. Clinical practice will need to be re-audited once these guidelines are implemented.

Disclosure of Interest: None Declared

BSH2020-PO-196

Avoidable Red cell transfusion due to Inadequate Investigation of Anaemia (ARIIA): A national study to determine if the patients that were transfused in an acute care setting were appropriately investigated for the causes of the anaemia

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Previous studies suggest that there is a population of patients receiving avoidable red blood cell transfusions for anaemia in the acute care setting. Patients may be receiving blood transfusions to treat unexplained anaemia and enable their discharge from hospital without undergoing adequate investigation for the cause of their anaemia. Because of this, important diagnoses may be missed, treatments for anaemia other than blood transfusion may not be considered, and patients may be exposed to the risks associated with blood transfusion unnecessarily, possibly even on several occasions.

Data from 336 patients were collected from 6 hospitals across the UK (Oxford University Hospitals, Queen Elizabeth Hospital Birmingham, Queen Elizabeth Gateshead, Sunderland Royal Hospital, Hastings Hospital and Wexham Park Hospital) between 1-31/1/2019. The population of interest were patients who received blood transfusion in an acute care setting. We investigated if their blood transfusion was appropriate and if they were adequately investigated for the cause of their anaemia according to NICE guidelines and recommendations (1-3). We also identified further blood transfusions for up to 6 months after that date which may have been avoided if a cause for anaemia had been found. The collection of the data was coordinated by the HaemStar network of Haematology Registrars.

70 patients out of 336 (18.6%) were transfused inappropriately. In 40/70 (57%) patients, their pre-transfusion Hb was 70-79 g/L, while the cause of the anaemia was unknown in 32/70 (45.7%) of these patients. No cause for the anaemia was found in 80/336 (24%) patients in total. Only 21/80 (26.5%) of those patients were adequately investigated. In 5 cases, a specific decision was made not to investigate further, while 54/80 (67.5%) cases were inadequately investigated. 23 out of these 54 (42.5%) patients required further transfusion in the next 6 months. This is in contrast to 5 out of 21 (23.8%) patients who were investigated adequately and required further transfusion; the number needed to prevent a blood transfusion by adequate investigation was 5.3 patients.

This national study shows that adequate investigation of anaemia of unknown cause in an acute care setting prevents further

transfusions. It may also prevent other health hazards due to delayed diagnosis of underlying causes for anaemia and quality of life but these were not investigated in this study. The study also found that a significant percentage of patients (18.6%) were transfused unnecessarily. There are limitations to this study in that the data collection was retrospective and missing data could have impacted the result. In summary, efforts should be made to prevent avoidable transfusion in patients with unexplained anaemia in the acute care setting through better adherence to the appropriate investigation for anaemia and to restrictive transfusion thresholds.

1 National Institute for Health and Care Excellence (NICE) (2015) *Blood Transfusion* URL <https://www.nice.org.uk/guidance/ng24>

2 NHS Clinical Knowledge Summaries Anaemia - iron deficiency, February 2013 cks.nice.org.uk/anaemia-iron-deficiency.

3 NHS Clinical Knowledge Summaries, Anaemia - vitamin B12 and folate deficiency, July 2015 cks.nice.org.uk/anaemia-b12-and-folate-deficiency

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BSH2020-PO-197

Impact of Fetal RHD screening upon a UK hospital's practice for Routine Antenatal Anti-D Prophylaxis (RAADP)

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UK 2002 NICE and 2014 BSH guidelines specify pregnant women who are D negative should be offered routine antenatal anti-D prophylactic immunoglobulin to prevent sensitisation to the potentially D positive fetus. However, a new NHSBT non-invasive pre-natal immunohaematology test using the woman's blood, allows prediction of fetal D status prior to delivery thereby negating prophylactic anti-D injections where fetus is predicted as D negative. In our hospital this has changed patient experience, patient safety, and care pathways for the better.

Go-live for Fetal RHD screening began March 2018. We had to offer and capture consent as prediction had an error rate of 0.1% for D negative predictions turning out to be D positive after birth. Sample taking had to be within the permitted time window of between 11 + 2 and 28 weeks. Community midwives were tasked to consent and plan when to take the sample e.g. at dating scan, drop-in to the phlebotomy room, or midwife appointment. Next, they sent notification of the plan as a referral email letter to the Anti-D midwife. The Anti-D midwife had a real-time database, listing all antenatal D negative D status unknown women, created to facilitate tracking. This safety net was used to find who was eligible but had not been offered the test, and who had a result showing prophylactic anti-D was indicated. Potentially sensitising events (PSE) were treated with anti-D injection unless the test had been done where prediction was D negative. To prevent omission of anti-D post-delivery due to incorrect D negative predictions, a cord blood samples were required.

Audit over one year (01-03-2018 to 28-02-2019) showed 511 patients were eligible out of which 54% were offered the test. Rejection rate was small with 1.7% ($n = 5/282$) declining. Completion of NHSBT forms and sample labelling was excellent with 99% ($n = 274/277$) fulfilling the criteria for processing. There was adequate notification from date of referral email to planned test date with a median of 14 days and mode of 20 days. The actual date of the test showed 56% ($n = 38/67$) were taken on the planned date.

Only one sample ($n = 1/277$) was taken too early i.e. before 11 + 2. The test was commonly taken 23 days after D group was established. Where prediction was D negative a cord was taken in 53% ($n = 51/96$) of cases, and where the cord had a result it matched prediction in 100% of cases ($n = 48/48$). Reasons for not taking a cord, or no result, were failure to meet sample labelling criteria (5%), delivered elsewhere (9%), and problems with obtaining or forgetting to take it (15%). Where prediction gave an inconclusive result (5% $n = 13/274$), available cord results showed 5/6 were D positive.

Anti-D product usage declined for both 1500iu (28 weeks and post-delivery) and 500iu (PSE events) doses from 56 to 43 vials monthly and 9 to 6 vials monthly respectively. Patients who appeared to have missed postnatal anti-D, 7% ($n = 8/105$), was less than in a local 2015 audit where it was 11%.

A patient survey ($n = 15$) showed motivation for consent related to safety (53%) and avoidance of injections (33%), and most would recommend it (85%).

In summary, improved early detection of D negative women, coupled with predictive fetus D blood group testing, enabled a reduction in patient population exposure to an unnecessary blood product, and was an empowering experience as the ease by which they could align the test with their existing appointments and other blood tests was up to them.

Disclosure of Interest: None Declared

BSH2020-PO-198

What do they know and what should we tell them- knowledge, attitude and education on blood transfusion therapy among antenatal mothers

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Introduction: Obstetric hemorrhage is one of the common indications for blood transfusion. It is recommended to obtain consent before blood transfusion but it may not be practical and possible in critical emergencies. Most of the antenatal management guidelines have not recommended the routine education on blood transfusion to the patients.

Objective: Aim was to analyze the knowledge and attitudes regarding blood transfusion among antenatal mothers and the need of routine education regarding blood transfusion among the antenatal population.

Methodology: A descriptive study was conducted among transfusion naive 320 antenatal women in their 3rd trimester attending antenatal clinic at De Soysa Maternity Hospital, Colombo, Sri Lanka.

Data was collected with a self-administered questionnaire and analyzed with SPSS software and Microsoft excel.

Results: Study population included 37.5% (120) primi and 62.5% (200) of multi gravida. Religion-wise, 49.7% (159), 28.1% (90), 16% (51), 9.3% (30) of them were Buddhists, Hindus, Christians and Muslims respectively. None of them were Jehovah's witnesses.

None of the mothers were counseled about blood transfusion during their antenatal visits by the health care workers. Out of the total population, 31.9% (102) were aware of at least one indication for the blood transfusion. 20.3% (65) and 14.7% (47) of mothers knew bleeding and anaemia were indications of transfusion respectively.

Only 45.3% (145) of mothers knew at least one undesired effect of blood transfusion. Transfusion transmitted infections and allergic reactions were the complications known by 31.2% (100) and 21.6% (69) of mothers respectively.

Among study population 94% (301) agreed that they would agree for blood transfusion irrespective of the indication if it was suggested by health care staff whereas 6% (19) participants mentioned they would agree for blood transfusion only in an acute life threatening situation. None of them mentioned that that they would refuse the transfusion even during a life threatening situation.

Among the mothers who'd agree only in a life-threatening situation, 2.8% (9) of them decided so since they considered blood transfusion to be dangerous, whereas 2.5% (8) due to fear of needles and 0.6 % (2) mothers were scared of the sight of blood. Religious reasons were not mentioned by any of them as a restricting factor for receiving blood transfusion.

Among study population 97.1% (311) of the mothers suggested that all pregnant mothers should be routinely be counseled about blood transfusion in the antenatal care.

Conclusion: Although the knowledge on blood transfusion was inadequate, most of the mothers accepted blood transfusion as an important clinical therapy. Majority of the study population favored the inclusion of education and counselling on blood transfusion during the antenatal period. It is recommendable to educate the mothers at antenatal clinics prior to their admission to wards for child birth.

Disclosure of Interest: None Declared

BSH2020-PO-199

Five years of collaboration towards achieving total voluntary non-remunerated blood donation in Trinidad and Tobago

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Trinidad and Tobago (TTO) is a high income developing Caribbean country with advanced medical services but a limited blood supply. Over 80% of blood donors are family replacement donors (FRD) at six Regional Health Authority (RHA) hospital-based blood donation centres. The remainder make donations on a mobile unit and retain the right to reclaim their donations, constituting remunerated donors. In contrast with Curacao, a country in the same region with 100% voluntary non-remunerated donors (VNRDs), TTO's blood service is plagued by low annual donation rates in all sociodemographic groups, high deferrals and high donor transfusion transmissible infections (TTIs). Donation is lowest among the 18-25 year age group and females constitute < 30% of prospective donors. Following an unsuccessful attempt to switch to total VNRD in 2010, a programme was initiated at The University of the West Indies, the Caribbean's regional university to recruit VNRDs.

Recognizing Pan American Health Organization/World Health Organization (PAHO/WHO) resolutions to transition from family replacement and remunerated to exclusively voluntary non-remunerated donors, research, sensitization and collaboration were used to establish a voluntary non-remunerated blood donor blood programme. Historical and sociological research was undertaken to examine the origins of the FRD-predominant blood donation system. Sensitization was initiated within the Faculty of Medical Sciences by introducing blood donation lectures, practicums, community blood donation knowledge, attitudes and practice (KAP) surveys along with conference presentations and training in appropriate blood use for clinical undergraduate and postgraduate students. The University of the West Indies Blood Donor Foundation (UWIBDF) was established in 2011 and collaborated with the Departments of History, Social Sciences, Surgical Sciences and the Institute of Gender Studies to produce 23 KAP surveys, seven international publications and four conference abstracts. Outreach activities were undertaken to include

secondary school students, places of worship and military organizations. A voluntary non-remunerated blood donor programme was initiated in collaboration with the North Central Health Authority (NCRHA). Data from this activity were collected on Microsoft Excel spreadsheets and analysed using SPSS for descriptive and inferential analysis and the chi square test to measure significance of proportions. $P < 0.05$ was used to assign statistical significance.

At fifteen quarterly blood drives over the first five years, 1,258 units of blood were collected, 57% from females, 49% from 18–24 year olds and 58% from repeat donors. Donor deferral and initially reactive TTI test rates were 5.2% and 1.3% compared to 43.1% and 3.04% respectively for the national blood donation system ($P < 0.05$ for both). The programme was accepted by the Ministry of Health and commended by PAHO/WHO for national application at all donation centres and for mobile unit donations.

A university-led programme has demonstrated the feasibility of voluntary non-remunerated blood donation in TTO and provided a model for transitioning to exclusively VNRD in accordance with PAHO/WHO recommendations.

Disclosure of Interest: None Declared

BSH2020-PO-200

Audit of red cell genotyping in sickle cell disease patients in two London trusts

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Red cell (RC) transfusion plays a crucial role in the management of patients with sickle cell disease (SCD). However, they are at increased risk of alloimmunisation from both higher numbers of units transfused, and differences in RC antigens expressed (particularly RH genetic diversity) between recipient and donor populations. Alloimmunisation rates are reduced by provision of extended RH and K matched units. Current BSH guidelines recommend that all SCD patients have extended RC antigen typing at baseline to allow, at least, provision of extended RH and K matched blood, but also to guide RC selection in those with haemolytic transfusion reactions or other complex transfusion requirements (e.g. multiple alloantibodies).

RC antigen typing can be done serologically if the patient had no transfusion within 3 months or by genotype if recently transfused. Genotyping can also identify RH variants, which can cause sensitisation even when extended RH matched blood phenotypically has been provided, so can guide better selection of units. From April 2015 to June 2016, NHSBT offered free genotyping to all haemoglobinopathy patients in England (Haemoglobinopathy Genotyping Project) with results available on Sp-ICE.

We reviewed all adult and paediatric patients with SCD under follow-up at two London trusts: Imperial College Healthcare NHS Trust (ICHNT) and London North West University Healthcare NHS Trust (LNWH). Primarily, we aimed to identify how many patients had RC genotyping, and for those who had not, to identify if they had extended phenotyping instead, albeit this does not identify RH variants. We also identified how many patients had RH variants and antibodies to RC antigens (auto and allo). Patient lists were provided by the clinical teams. All data was collected from Sp-ICE.

482 (298 adult, 184 paediatric) and 346 (226 adult, 120 paediatric) SCD patients are followed-up at ICHNT and LNWH respectively. At ICHNT, 311 (64%; 238 adult, 73 paediatric) patients had genotyping and of the remaining 171 patients, 61 (13%; 14 adult, 47

paediatric) had extended phenotyping. At LNWH, 181 (52%; 130 adult, 51 paediatric) patients had genotyping and of the remaining 165 patients, 30 (9%; 15 adult, 15 paediatric) had extended phenotyping. 68 (14%) patients at ICHNT and 31 (9%) patients at LNWH had RH variants. 62 (13%) patients at ICHNT and 32 (9%) patients at LNWH had antibodies to RC antigens.

Most SCD patients at ICHNT (77%, $n = 372$) and LNWH (61%, $n = 211$) had extended RC antigen typing in accordance with BSH guidance. However, a significant number (ICHNT 23% $n = 110$; LNWH 39% $n = 135$) had neither genotyping nor extended phenotyping. This may compromise selection of suitable RC units, so can increase the risks from undetected antibodies, particularly in those with complex transfusion requirements. We also showed, as expected, a significant number of patients with RH variants and antibodies to RC antigens, thus highlighting the complex transfusion requirements in SCD patients. Our data also shows that fewer paediatric than adult patients had extended RC antigen typing, which is significant as they may potentially receive more transfusions throughout their lifetime.

To address these issues, we recommended that all patients without extended RC antigen typing should have their notes flagged to prompt a sample for genotyping at the next clinical encounter, ideally within 12 months. We aim to re-audit in 12 months' time.

Disclosure of Interest: None Declared

BSH2020-PO-201

Re-evaluating the shelf-life of whole blood for use in trauma: impact of storage on platelet function

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Background: There is an immense interest in using Whole Blood (WB) transfusion for early resuscitation of trauma patients with major bleeding in the pre-hospital setting. The shelf-life of WB component (21–35 days depending on jurisdiction) was established decades ago, primarily based on the viability of red cells. This study aimed to determine the platelet function of leucocyte-depleted WB (LD-WB) over storage in order to re-establish the shelf life.

Methods: LD-WB was prepared using a platelet-sparing LD filter (Terumo WB-SP Imuflex). LD-WB was held at 2 – 6 °C alongside comparator arms consisting of: A) UK standard pooled buffy coat platelets stored at 20 – 24 °C (RT-PLTS) B) pooled platelets held at 2 – 6 °C (COLD PLTS) and C) platelet rich plasma (PRP) produced using the Terumo Imuflex WB-SP kit, for 21 days. A series of *in vitro* assays were assessed.

Results: There were fundamental changes in the platelet function of LD-WB over storage at 2 – 6 °C for 21 days. Platelet count was retained to a mean of 57 ± 14 % of starting number at day 21 in 80 % of LD-WB components compared with 97 ± 2 % at end of shelf life (day 7) in RT-PLTS. Over time the platelets in LD-WB become more activated, as seen in increased CD62P expression (Day 1 – 7 ± 3.7 % vs. Day 21 – 59 ± 17.1 %) and annexin V binding (Day 1 – 2 ± 0.2 % vs. Day 21 – 21 ± 15.1 %). For comparison, 18.6 ± 6 % of platelets in RT-PLTS demonstrated CD62P expression at day 7 whilst Annexin V binding in RT-PLTS at day 7 was 2.6 ± 0.5 %. LD-WB demonstrated decreased response to agonists over time (CD62P expression with ADP: (Day 1 – 73 ± 9 %, Day

21: $13 \pm 8 \%$), **TRAP-6:** (Day 1 - $80 \pm 6 \%$, Day 21 - $13 \pm 8 \%$), **CRP-XL:** (Day 1 - $64 \pm 15 \%$, Day 21 - $11 \pm 12 \%$). From day 12 onwards, LD-WB showed very little ability to aggregate in response to ADP and TRAP-6. Platelet-derived microparticles increased steadily throughout storage, which corresponded with increased potential to promote thrombin generation (Day 1 - 72 ± 5.7 s to clot vs. Day 21 - 38 ± 3.5 s to clot) in LD-WB. Metabolism was higher in RT-PLTS than cold-stored components.

Conclusion: During storage of LD-WB, there is a substantial loss of platelet count, coupled with progressive loss of aggregatory response,

and increased activation. These data, along with previously published work to understand red cell function and plasma quality in LD-WB *in vitro* and international experience, were used to recommend a shelf life of 14 days for LD-WB in the UK. This work is part of ongoing collaborative efforts to establish the clinical need and feasibility of WB in the treatment of major haemorrhage, particularly in the pre-hospital environment.

Disclosure of Interest: None Declared

Transplantation

BSH2020-PO-202

Prophylactic donor lymphocyte infusion reduces the risk of relapse and increases survival in patients with mixed T cell chimerism after reduced intensity allogeneic stem cell transplantation

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Reduced intensity conditioning (RIC) transplant regimens extend the potentially curative graft versus leukaemia (GVL) effect to older adults with high risk haematological malignancies. Nonetheless, relapse is a major cause of treatment failure and strategies with the potential to augment a GVL effect are required. Patients with persistence of host haematolymphopoiesis (measured by host-donor T cell chimerism) are reported to be at higher risk of relapse, and prophylactic donor lymphocyte infusions (pDLI) have been used to reduce the risk of disease relapse. DLI is however associated with a significant risk of graft-versus-host disease (GVHD) and data surrounding the toxicity and efficacy of pDLI are limited. We have therefore characterised the safety, toxicity and efficacy of pDLI in patients allografted for a high-risk haematological malignancy at a major UK centre.

Consecutive patients allografted using a RIC regimen who received pDLI as treatment for mixed T cell chimerism between August 2008 and January 2019 were studied. We constructed multivariate Cox-proportional hazards models to identify predictors of relapse and survival post DLI, using occurrence of GVHD, and achievement of full donor chimerism (FDC) as time dependent covariates in separate models. Primary outcomes were the incidence and severity of acute and chronic GVHD (as defined and graded by the revised Glucksberg criteria and NIH criteria respectively), the achievement of FDC (defined as $\geq 98\%$ donor T cells), relapse, and overall survival.

122 patients received pDLI for mixed host-donor T cell chimerism in the time period. 77 (63%) were male and median age at first DLI was 58 (range 20-73). 64 (52%) patients had acute myeloid leukaemia or myelodysplastic syndrome. Mean donor T cell chimerism prior to pDLI was 48%. Prior to pDLI, 19 (16%) patients had mild GVHD (acute $\leq$ grade II, mild chronic). 34 (28%) developed GVHD after pDLI which was severe (grade III-IV acute, or moderate to severe chronic) in 12 (10%). pDLI caused or contributed to death in 6 patients (5%). 85 (70%) converted to FDC following pDLI. The median time between conversion to FDC and the commencement of pDLI was 136 days (range 21-644). 28 (23%) converted to FDC after one pDLI infusion at a median of 70 days. The median number of infusions required to achieve FDC was 2 and the median cumulative DLI dose was $6 \times 10^6/\text{kg}$ (range $1 \times 10^6/\text{kg} - 2.2 \times 10^8/\text{kg}$). 5-year overall survival after pDLI was 71.1%. In multivariate analyses, achievement of FDC as a time-dependent covariate was predictive of survival (HR 0.24, 95% CI 0.11-0.54, $P = 0.00056$) and protective against disease relapse (HR 0.23, 95% CI 0.071-0.75, $P = 0.015$). The development of severe GVHD as a time-dependent covariate was a predictor of mortality (HR 5.69 95% CI 2.31-13.99, $P = 0.0002$).

To our knowledge, this is the largest study of pDLI for mixed T cell chimerism. Our data demonstrate the safety and efficacy in restoring FDC. pDLI was relatively safe with a 10% risk of severe GVHD and 5% mortality. Mild GVHD prior to pDLI did not increase the risk of severe GVHD post treatment. 70% converted to FDC after pDLI and this was associated with reduced risk of disease relapse and improved survival. These data support the use of pDLI for mixed chimerism and underline the importance of the ongoing proDLI randomised trial through the IMPACT transplant trial network.

RB and SN are joint first authors and contributed equally to the work.

Disclosure of Interest: None Declared

BSH2020-PO-203

Defibrotide treatment for veno-occlusive disease/sinusoidal obstruction syndrome after haematopoietic cell transplantation: outcomes by severity and multi-organ failure status in an observational registry study

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Severe hepatic veno-occlusive disease/sinusoidal obstruction syndrome (VOD/SOS) is a potentially life-threatening complication of haematopoietic cell transplantation (HCT); the most severe form is associated with multi-organ dysfunction/failure (MOD/MOF) and a mortality rate of $> 80\%$ if untreated. This multicentre, multinational, prospective, observational study (NCT03032016), performed by the EBMT, enrolled patients (pts) with VOD/SOS post-HCT, including severe VOD/SOS, and who received defibrotide (DF) from April 2015 to July 2018. Investigators diagnosed VOD/SOS using classical/standard criteria (including but not limited to hyperbilirubinaemia, hepatomegaly, ascites, and weight gain $> 5\%$) and graded VOD/SOS severity based on typical clinical practice. Pts with renal, pulmonary, or cerebral dysfunction, as judged by the investigator, were diagnosed with MOD/MOF. The primary endpoint was incidence of serious adverse events (SAEs) of interest up to 12 months post-HCT in pts with severe VOD/SOS. Secondary endpoints included Day 100 survival and overall rate of VOD/SOS (and MOD/MOF, if present) resolution (as defined by the investigator) up to 12 months post-

HCT. This exploratory subgroup analysis evaluated outcomes in pts with non-severe VOD/SOS and with severe VOD/SOS, including MOD/MOF. Overall, 104 DF-treated pts with VOD/SOS post-HCT were enrolled: 42 had non-severe VOD/SOS (median age: 13.1 years [range: 0–69]) and 62 had severe VOD/SOS, of which 26 had no MOD/MOF (median age: 11.7 years [range: 1–65]) and 36 had MOD/MOF (median age: 21.1 years [range: 0–68]). SAEs of interest occurred in 8/42 (19%) pts with non-severe VOD/SOS. Among pts with severe VOD/SOS, 7/26 (27%) with no MOD/MOF and 12/36 (33%) pts with MOD/MOF had an SAE of interest (Table). The Kaplan-Meier (KM)–estimated Day 100 survival rate was 91% (95% CI: 77%, 96%) in pts with non-severe VOD/SOS; among pts with severe VOD/SOS, Day 100 survival was 85% (95% CI: 64%, 94%) in pts with no MOD/MOF and 64% (95% CI: 46%, 77%) in pts with MOD/MOF. Median KM–estimated survival post-HCT was not reached in pts with non-severe VOD/SOS or severe VOD/SOS with no MOD/MOF and was 249.5 days (95% CI: 83, not reached) in pts with MOD/MOF. Cause of death was VOD/SOS in 0/11, 2/10, and 11/21 pts with non-severe VOD/SOS, severe VOD/SOS with no MOD/MOF, and pts with MOD/MOF, respectively. In pts with non-severe VOD/SOS, the Day 100 cumulative incidence of VOD/SOS resolution was 95%. The Day 100 cumulative incidence of VOD/SOS resolution in the severe VOD/SOS population was 88% in pts without MOD/MOF and 61% in pts with MOD/MOF. Resolution of MOD/MOF was achieved in 19/36 (53%) pts. Among pts with VOD/SOS post-HCT, the safety profile of DF was consistent with previous reports. Rates of Day 100 survival and VOD/SOS resolution were consistent with previous reports of DF in the post-approval setting, with better outcomes observed in pts with less severe disease.

Table.

	Non-severe VOD/SOS (<i>n</i> = 42)	Severe VOD/SOS	
		No MOD/ MOF (<i>n</i> = 26)	With MOD/MOF (<i>n</i> = 36)
Any SAE of interest ^a , <i>n</i> (%)	8 (19)	7 (27)	12 (33)
Most common (≥5% of pts)	2 (5) 1 (2) ^b	1 (4) 2 (8) ^c	3 (8) 2 (6)
SAEs of interest, <i>n</i> (%)			
Gastrointestinal			
bleeding			
Pneumonia			
Sepsis			
Urinary tract bleeding			

^aHaemorrhage, hypotension, coagulopathy, allergic/hypersensitivity reactions, injection-site reactions, infection/septicaemia, and thromboembolic events.

^bRecorded as “infection, other: lung”.

^cRecorded as “bleeding, other: urinary”.

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Pharmaceuticals, Conflict with: Has received research funding from Jazz Pharmaceuticals

BSH2020-PO-204

Outcomes of Allografts for Acute Myeloid Leukaemia and Myelodysplastic Syndromes in Patients Aged over 60 years and over 65 years are not inferior to those of younger patients: a single centre study

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Acute myeloid leukaemia (AML) and myelodysplastic syndrome (MDS) are haematological malignancies that have an increasing incidence in patients > 60 years. Allogeneic stem cell transplantation (HSCT) remains an important curative treatment for these conditions but, has previously been denied to older patients due to fears of high treatment related mortality in this population. However, the development of comorbidity scores (HCT-CI) which are more reflective of fitness than chronological age, together with reduced intensity conditioning (RIC) regimens is now allowing increasing numbers of older patients to benefit from HSCT.

We analysed the outcomes of a cohort of 152 consecutive patients undergoing sibling and matched unrelated donor allogeneic transplantation at Nottingham University Hospitals Trust between 2014–2018 and stratified outcomes by age (above and below 60 and 65 years), HCT-CI scores (above and below 3 and 4) and by HLA matching (9/10 and 10/10).

The median age of the patients was 62.2 years (range 18–74.2 years) and most were aged over 60 years (*n* = 92, 60.5%), with 38.2% of patients being aged > 65 years. Very few patients (*n* = 33, 22%) were aged < 50 years. 61% of patients were male and 75% of patients had AML with 25% having MDS, which was commoner in the > 60 years group (31.5%, *P* = 0.021). The vast majority of patients received RIC protocols (91%) reflecting their age, and MUD donors were commoner than siblings in the both older and younger cohorts (79.6%).

60.5% of patients had a HCT-CI score of < 3 and 39.5% of patients had scores ≥ 3. There was no difference in the number of patients aged > 60 yrs having a score of > 3. The median comorbidity score was 2 for both groups and all patients had a comorbidity score in the range 0–5 except 1 outlier in the older group who had a comorbidity score of 8. All patients were matched at a minimum of 9/10 HLA loci. Most patients were fully matched (86.8%) at the 10 loci while the rest (13.2%) were matched at 9/10 loci and were similar for both groups.

With a median follow up of 2.84 yrs (0.98–5.84 yrs) 76 patients were alive without relapse and 72 patients died (48.6% due to relapse and 54.8% due to TRM). There was no difference in OS, EFS, RR or TRM in the patients aged > 60 years compared to those aged < 60 years (3 year OS 44% and 53.5% in < 60 and > 60 respectively). Furthermore re-stratifying using the older age of 65 years also had no effect on outcomes.

Analysis of outcomes by HCT-CI score revealed no difference in OS, EFS, RR or TRM in patients with HCT-CI scores > or < 3 or > or < 4. The only variable affecting outcome was HLA matching with inferior OS and EFS in patients receiving 9/10 matched allografts compared to 10/10 matches (3 yr OS 33% vs 50%, *P* = 0.006) due to a higher TRM (50% vs 33%, *P* = 0.027).

We conclude that patients aged > 60 years now comprise the majority of patients undergoing allografting for AML / MDS but found that age > 60 and > 65 years did not impact outcomes. This

may be because older patients are highly selected since the majority of HCT-CI scores were low. In contrast to other studies HCT-CI scores of > 3 and > 4 did not influence outcomes and the only factor associated with inferior outcome was 9/10 HLA match. This suggests allo-HSCT can be successfully performed in patients > 60 yrs provided that they are selected carefully on fitness, receive RIC transplants from 10/10 matched donors and that nihilistic past attitudes can be replaced with cautious optimism

Table,

Variable		< 60 years		> 60 years		Total		P value
Sex	Male	33	55	60	65.2	93	61.2	0.118
	Female	27	45	32	34.8	59	38.8	
Diagnosis	AML	51	85	63	68.5	114	75	0.021
	MDS	9	15	29	31.5	38	25	
Conditioning	MA	9	15	0	0	9	5.9	0.001
	RIC	51	85	100	100	143	94.1	
Donor	MUD	48	80	73	79.3	121	79.6	0.92
	Sibling	12	20	19	20.7	31	20.4	
Matching	9/10	9	15	11	12	20	13.2	0.589
	10/10	51	85	81	88	132	86.8	
HCT-CI	> 3	33	55	59	64.1	92	60.5	0.26
	< 3	27	45	33	35.9	60	39.5	
Status	Alive	29	48.3	51	55.4	80	52.6	0.551
	Dead	31	51.7	41	44.6	72	47.4	
Cause of death	Relapse	14	45.2	21	51.2	35	48.6	0.611
	TRM	17	54.8	20	48.8	37	51.4	

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BSH2020-PO-205

The use of Autologous Haematopoietic Stem Cell Transplants in patients with Relapsing Remitting Multiple Sclerosis (RRMS): A Meta-Analysis

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Objective: To investigate the safety and efficacy of administering autologous haematopoietic stem cell transplants (AHSCT) in patients who have been diagnosed with relapsing remitting multiple sclerosis (RRMS) as an alternative to current disease modifying therapies (DMTs) such as Alemtuzumab and Interferons.

Design: Meta-analysis using published data from clinical trials including a newly published randomised control trial (RCT).

Data Sources: Medline, PubMed, Google Scholar, clinicaltrials.gov, and European Marrow and Bone Transplant-Autoimmune Diseases Working Party (EMBT-ADWP), searched between 1st January 2002 and 15th January 2019.

Main Outcome measures: Change in Expanded Disability Status Score (EDSS) in patients treated with AHSCTs compared to DMTs. Secondary outcomes include percentage Event Free Survival, Progression Free Survival, and Treatment Related Mortality (TRM).

Results: After exclusion criteria were applied, a total of 277 RRMS patients treated with AHSCT across 6 studies were evaluated. The

mean EDSS across all studies either remained unchanged or reduced to at least 3 years follow-up. Studies that had a lower mean EDSS documented the largest decrease in EDSS. When comparing the control arm of the RCT, which used the most clinically appropriate DMTs, the EDSS increased markedly.

Event free survival and progression free survival were considerably higher in patients that received AHSCT compared to DMTs. In the RCT, progression free survival in AHSCT was 90.3% at 5 years compared to 24.7% in DMT ($P < 0.0001$); Event free survival in AHSCT patients was 84.6% compared to 14.8% in the DMT group ($P < 0.0001$). Overall TRM was calculated as 0.722% (3sf) across the studies included. There appears to be a trend towards better progression and event free survival in studies post-2015 vs pre-2015 however given the data available, it is not possible to calculate P-values.

Conclusion: After relatively recent refinement to AHSCT that reduces mortality and morbidity, it is a reasonable treatment for patients with auto-immune conditions such as RRMS. Compared to DMTs, it offers greater efficacy and cost effectiveness. The benefits are seen particularly in younger patients, particularly those with a lower EDSS, a shorter disease duration and patients who have not yet progressed to secondary progressive MS. With this study along with additional trials, particularly with adverse effects and long-term outcomes beyond 5 years as primary outcomes, it is hoped that AHSCT will become a mainstream treatment for patients with early-stage RRMS.

Disclosure of Interest: None Declared

BSH2020-PO-206

Alemtuzumab containing renal transplant protocols are associated with late onset EBV+ PTLD, with poorer overall survival

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Post-transplant lymphoproliferative disorder (PTLD) is a rare aggressive disease, complicating 2.6% of renal transplants. The renal transplant unit at Imperial College NHS Trust, London undertakes ~150 transplants/year and uses alemtuzumab induction protocols which have been previously associated with higher rates of PTLD: Between 2005-2011 a fixed 30 mg alemtuzumab dose was used and after 2011, a 0.4 mg/Kg protocol was adopted. The aim of this retrospective study was to determine incidence of PTLD, risk factors, and quantify survival outcomes of this cohort, including the impact of two different alemtuzumab dosing regimens.

The local renal transplant database was established in 2005 and was interrogated to determine the PTLD cohort and to identify a matched control cohort of sequential patients matched for age and type of transplant (live/cadaveric, related/unrelated). Clinical data on these cases were retrospectively investigated using a combination of local clinical databases (CERNER, ICE, Aria, medical letters). Survival data and risk factors were analysed using Prism statistical software (version 8).

A total of 2236 patients underwent renal transplantation between 2005-2019 with 37 confirmed cases of PTLD (1.65% cohort) which arose at a median of 53 months (range 10 – 142 months) post-transplant. 64.9% of PTLD cases were associated with Epstein-Barr Virus (EBV, $n = 24$) and 35.1% were not EBV associated ($n = 13$). The median time to PTLD in the EBV-associated cohort was 53 months, but it was not reached in the EBV negative cohort. The median OS in the PTLD cohort was 118 months, whereas the median was not reached in the control population.

A low IPI score (0-2), achieving a complete response to frontline therapy and surprisingly, EBV negative PTLD were significantly

associated with improved survival. There were no major risk factors (such as high IPI and monomorphic histology) significantly associated with EBV status nor alemtuzumab dosing regimen. EBV DNA viraemia was significantly higher in the PTLD cohort compared to the control (1500 copies/ml, 95% Confidence Interval (CI) 3262 - 80774 versus < 500 copies/ml, 95% CI, 1424 to 12319), $P = .0087$.

PTLD remains to have poor prognosis following renal transplantation. Using alemtuzumab-containing induction regimens there was no increase in the incidence of PTLD, but EBV+ PTLD occurred significantly later than expected, and EBV+ PTLD was associated with inferior survival and outcomes. Whilst EBV viraemia is associated with PTLD it is not a good predictor for future PTLD and novel surveillance strategies are urgently required.

Keywords: Post-Transplant Lymphoproliferative disorder, Alemtuzumab, renal transplant, solid organ transplant, Epstein-Barr Virus, International Prognostic Index

Disclosure of Interest: None Declared

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First report of post-transplant autoimmune hyperlipidaemia and treatment with rituximab

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Autoimmunity after stem cell transplantation has been reported over decades mainly with autoantibody production, or more rarely, the appearance of autoimmune disease. Most commonly described is autoimmune cytopenia thought to be triggered by transplantation-induced lymphopenia leading to loss of self-tolerance and subsequent proliferation of autoreactive T-cells.

Glycosylphosphatidylinositol-anchored high-density lipoprotein binding protein 1 (GPIHBP1) is a protein which plays a critical role in the lipolytic processing of triglycerides. Here, we present the first reported case of positive autoantibody against GPIHBP1 in a patient who had undergone a T-replete reduced intensity allograft for acute myeloid leukaemia (AML), resulting in significant hypertriglyceridaemia (>50 mmol/L) and resolving only with rituximab therapy.

A 50-year old gentleman with relapsed AML in CR2 underwent an ABO compatible FluCy T-replete allogeneic stem cell transplant in January 2018. In April 2018, biopsy confirmed grade 1 acute gut GvHD; ANA + 1/640; T-cell receptor analysis showed a clonal population; whole blood donor chimerism was 92% (CD3 82%; CD15 99%). In July 2018, weaning of ciclosporin was started following resolution of symptoms. In August 2018, his haemoglobin fell to 77 g/L with positive DAT (IgG+++ C3d+++), raised lactate dehydrogenase and reticulocytosis. He was diagnosed with autoimmune haemolytic anaemia (AIHA) and commenced on high dose prednisolone. Concurrently it was noted that his blood and bone marrow were extremely lipaemic with triglycerides > 50 mmol/l (NR: 0-5), total cholesterol 8.7 mmol/l (NR: 0-5); HDL 0.4 mmol/l (NR: 0.9-1.5). He had no personal or family history of diabetes or hyperlipidaemia; BMI 29.7 kg/m² and consumed no alcohol. Atorvastatin and fenofibrate were commenced. After one month of high dose prednisolone his AIHA showed no significant response hence Rituximab (375 mg/m² weekly x4) was started and he was admitted for intravenous insulin/dextrose for persistently elevated triglycerides. His ciclosporin was stopped, in case it was a contributing factor.

Given two autoimmune phenomena post-transplant (AIHA and GvHD), it was postulated that his marked hypertriglyceridaemia was also autoimmune in nature. Samples were sent to a reference

laboratory for analysis by ELISA for autoantibodies to GPIHBP1 and lipoprotein lipase (LPL) quantification.

Autoantibodies were detected at a level of level of 2442u/ml with a low LPL level of 6.7 ng/ml (NR: 50-77 ng/ml). Following Rituximab therapy, his triglycerides normalised, his reticulocytosis disappeared and his haemoglobin improved. Repeat autoantibodies were 1975U/ml; LPL level had increased to 14.1 ng/ml.

GPIHBP1 binds LPL on the capillary endothelium and transports it to the site of action within the capillary lumen. LPL is important for intravascular hydrolysis of triglycerides. Autoantibodies against GPIHBP1 have been described previously as a cause of severe hypertriglyceridaemia in patients with known autoimmune conditions, notably lupus and Sjogren's syndrome.

Dyslipidaemia has been reported in patients with overwhelming haemolysis, presumed to be due to a high lipid burden from red cell membrane damage, as well as in the setting of steroid use, or ciclosporin and sirolimus toxicities. Our case is the first reported case of such an autoimmune phenomenon post-allograft. Further work is required to elucidate the frequency of such autoimmune phenomenon in the transplant population.

Disclosure of Interest: None Declared

BSH2020-PO-208

Reduced Intensity Conditioning with Thiotepe-Fludarabine-Busulphan (TBF) and Post-Transplant Cyclophosphamide (PTC) Results in Rapid Complete Donor T-cell Chimerism in Haploidentical Recipients with Acute Myeloid Leukaemia

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Allogeneic stem cell transplant (SCT) offers the best prospect of long-term survival for the majority of patients with intermediate and high-risk acute myeloid leukaemia (AML). However, huge diversity in the human leukocyte antigen (HLA) gene complex means identifying a suitably matched donor can be challenging. While up to 30% of patients may have a fully matched sibling, the probability of identifying a matched unrelated donor (MUD) for the remaining patients ranges from 80% in Caucasian patients to less than 20% in ethnic minority groups. Alternative donor sources are therefore an appealing option, and estimates suggest that 95% of patients have at least one haploidentical match. The optimal conditioning regimen for haploidentical transplant remains unclear, but the emerging tolerability and success of post-transplant cyclophosphamide-based regimens has placed them at the forefront of development. Robust T-cell engraftment is critical to the graft versus leukaemia effect, which is pivotal to the success of SCT in AML. The inherent bidirectional alloreactivity of a haploidentical transplant risks poor engraftment, therefore, we assessed whole blood and T-cell engraftment in haploidentical recipients with reduced-intensity TBF- PTC conditioning.

We retrospectively analysed chimerism results of all recipients of reduced-intensity conditioned haploidentical stem cell transplants for AML at our institution and compared these to contemporaneous sibling and matched unrelated donor transplant. Institution policy dictates that transplant recipients undergo whole blood and CD3 T-cell selected donor-recipient chimerism assessment routinely at day + 30, +60, +90 and + 120, and thereafter as clinically indicated. The Pro-mega PowerPlex[®] 16HS system was used for DNA typing by Small

Tandem Repeat (STR) analysis, and complete chimerism was defined by > 95% donor cells. The probabilities of achieving complete whole blood, and T-cell donor chimerism, and of relapse were estimated using the Kaplan Meyer method and compared by log-rank test.

73 patients underwent reduced intensity SCT for AML between 2005 and 2018 and had sufficient chimerism follow-up. The number of patients undergoing haploidentical, sibling and matched unrelated donor transplants were 10, 24 and 39 respectively. At 93%, the overall achievement of complete whole blood donor chimerism was high in the cohort, but was significantly higher and attained more rapidly in the haploidentical group (probability of complete chimerism 100%, 73.9% and 92.3% at day + 120 in the haploidentical, sibling and MUD recipients respectively ($P = 0.001$), occurring at a median of 36, 52 and 40 days ($P < 0.001$)). The rates of T-cell engraftment for the whole cohort was 82% and was significantly different between the three groups. At day + 120 the probability of complete T-cell chimerism was 90%, 58.3% and 45.7%, for haploidentical, sibling and MUD recipients respectively ($P < 0.001$). The median time to complete T-cell chimerism was 36, 77 and 93 days. Failure to achieve complete T-cell chimerism by day + 120 was associated with a significantly higher rate of relapse (3-year probability of relapse 39.5% vs 16.9%, $P = 0.015$).

The reduced intensity TBF-PCT-based haploidentical regimen results in rapid complete whole blood and T-cell donor chimerism. In this cohort, early complete T-cell chimerism is associated with lower rates of disease relapse.

Disclosure of Interest: None Declared

BSH2020-PO-209

Morbidity of very long term (>30 year) survivors of allogeneic haemopoietic cell transplantation 1979-1989: a single centre experience

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Background: The number of patients undergoing allogeneic haemopoietic cell transplant (HCT) is increasing. Although

potentially curative for a range of malignant and non-malignant disorders, the success of HCT is limited by the intensity of treatment which can result in organ toxicity and/or death. Ten-year survival of patients surviving disease free 2y post-HCT was 85% in a multi-centre study of > 10000 patients (median follow up 9y), but life expectancy remained below that of the general population¹. The cumulative incidence of developing a chronic health condition after HCT has been estimated as 59% at 10 y² but there is a paucity of data on morbidity in patients surviving > 15y after transplant. In this study we have looked at the long term (LT) outcome of patients transplanted 30-40y ago.

Methods: Data on survival and cause of death were obtained from local online databases, death certificates and the NHS portal. The latter records patient death as an outcome if 1. the patient is living in the UK when they die and 2. they have an NHS number. Data on morbidity were obtained retrospectively from correspondence and electronic records.

Results: 226 patients were transplanted in the 10y period 1979-1989 of whom 164 (72%) died (see table). Indications for HCT were as follows: CML $n = 202$, AML $n = 14$, ALL $n = 5$, MDS $n = 4$, MPN $n = 1$. 207/226 patients received TBI- conditioning in the range 10-12 Gy. Among 62 presumed survivors of HCT, 15 lived abroad and further information was unavailable. Of 47 patients presumed living in the UK for whom there was no record of death on the NHS portal, detailed data on morbidity was available for 26 (55%); 18 /47 were lost to follow up for > 4y and there was a lack of data for 3/47. Among these 26 patients (13 male), the median age at follow up was 63y (range 49-76), median follow up 33y (range 30-40) and median age at HCT 27y (range 14-42). Most (20/26) had been transplanted for CML. Morbidity was frequent in this group of survivors with a median number of chronic health conditions of 4 (range 1-9). The most frequent problems were acquisition of cardiovascular risk factors: dyslipidaemia (17/26), hypertension ($n = 16/26$) and diabetes ($n = 5/26$). Overt ischaemic cardiac disease was present in 5/26 and cerebrovascular disease in 2/26. Osteopenia or osteoporosis occurred in 13/26. Endocrinopathies included hypothyroidism (12/26) and low testosterone ($n = 3/13$). Second cancers had occurred in 10 survivors with 3 patients having two cancer types. These included tongue ($n = 2$), breast ($n = 1$), renal ($n = 1$), bladder ($n = 1$), testicular ($n = 1$) and 5 patients had non melanoma skin cancers. Additional causes of morbidity were cGVHD ($n = 5$), renal impairment ($n = 5$), asthma ($n = 4$), gout ($n = 3$), irritable bowel syndrome ($n = 2$).

Discussion and conclusion: Detailed follow up information relating to death or survivorship decreases with time after follow-up and this

Table.

MORTALITY	0-100 days	100 days-2 years	2-10 years	10-20 years	20-30 years	30-40 years
N=	55/226	45/ 171	28/126	20/98	11/78	5/67
Relapse	2	18	12	5		
Graft failure	10	3	-	-		
GVHD	12	5	3	3		
pneumonitis	14	10	2	-	2	
infection	10	6	6	2	1	1
VOD	2	-	-	-		
MOF	2	-	-	1	1	
Other	3 ^a	2 ^b	2 ^c	1 ^d	-	-
Second malignancy		1 ^e	1 ^e	4 ^e	3 ^e	1 ^e
Cardiac					1	1
Unknown			2	4	3	2

a. Cerebrovascular, cardiac failure, respiratory failure b. ARDS x 2 c. Suicide, brain haemorrhage d. Pulmonary embolism e. Second cancers were as follows: breast x 2, bladder x2, colorectal x 2, oesophagus, lymphoma, renal, stomach

Transplantation

remains a challenge for transplant physicians. Amongst patients surviving > 20y, relapse and GVHD were notably absent as recorded causes of death while second cancers became more prevalent in patients surviving > 10 y. In LT survivors, multi-morbidity was high with half of our survivors having four or more chronic health conditions. It is important that evolving strategies for following up this

group of patients, take into account the clinical complexity of patients > 30 y after HCT.

Refs: 1. Wingard *et al.* 2011, JCO 29:2230-2239. 2. Majhail and Rizzo 2013, BMT 48:1145-1151.

Disclosure of Interest: None Declared

ALL and AML

BSH2020-EP-001

Outcomes following treatment discontinuation in chronic myeloid leukaemia (CML): real-world experience from three regional UK centres

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Emerging evidence from clinical trials has demonstrated the feasibility of TKI discontinuation in CML. However, real-world data of treatment discontinuation has been limited, particularly where this is due to clinical considerations such as drug toxicities. To address the real-world outcomes of treatment discontinuation and factors influencing this, we retrospectively analysed all patients from three hospitals in Birmingham, UK who have discontinued CML treatment since 2010.

Altogether 28 patients (11 males and 17 females) discontinued TKI. Prior to discontinuation, these patients received TKI for chronic-phase CML for a median of 8.9 years (range 4.3–15.2 years). At the time of discontinuation, the majority ($n = 22$) were receiving frontline TKI therapy. Six patients had 1 or 2 prior treatments including interferon-alpha, allogeneic SCT and other TKIs. Twenty-two patients (78.6%) discontinued imatinib and three patients each (10.7%) discontinued nilotinib and dasatinib respectively. The median age at the time of discontinuation was 61 (range 18–83). The most common reasons for TKI discontinuation were patient preference ($n = 10$), enrolment in a TKI discontinuation trial (DESTINY; $n = 9$) and TKI drug toxicity ($n = 8$).

Seventeen patients (60.7%) achieved MR4.5, six patients (21.4%) achieved MR4 and five patients (17.9%) had remission status below MR4 at the time of TKI discontinuation. Sixteen patients (57.1%) had been in deep molecular remission (MR4 or MR4.5) for at least 5 years prior to TKI discontinuation, including 11 patients (39.3%) who had been in MR4.5 for ≥ 5 years. Only one patient experienced withdrawal syndrome following discontinuation. With a median follow-up of 2.5 years (range 0.2–5.4 years), the median treatment-free remission (TFR) was not reached. Ten patients (35.7%) experienced molecular relapse following TKI discontinuation, and in the majority (9 of 10) of these patients relapse occurred within 6 months of TKI discontinuation. All the 10 patients were restarted on TKI, with six of 10 patients returning to a major molecular remission. With the exception of two patients who died of a second malignancy unrelated to CML, all patients remained alive.

Remission status at the time of TKI discontinuation significantly predicted for the duration of TFR ($P = 0.018$). The relapse rate in the initial 6 months following TKI discontinuation was 18.5% (3 of 16 patients), 33.3% (2 of 6 patients) and 80% (4 of 5 patients) for patients in MR4.5, MR4 and <MR4 respectively at the time of discontinuation. There was also a trend for the TFR to be positively associated with the total duration of TKI therapy ($P = 0.057$) and the duration at MR4.5 ($P = 0.066$) prior to TKI discontinuation. There was no significant association between TFR and the type or line of TKI at discontinuation, the patient's age or their reason for discontinuation.

Of note, long-term TFR was also observed in a patient who remained in MR4 following the discontinuation of interferon-alpha 4 years ago; the patient had been in MR4 for 9.6 years.

Overall, our results are largely consistent with published data on TKI discontinuation in CML, demonstrating the feasibility of discontinuation in clinical practice. Importantly, outcomes following discontinuation significantly correlated with the depth and duration of remission at the time of TKI discontinuation. Given the paucity of real-world data on treatment discontinuation in CML, our study may give additional confidence to clinicians on TKI discontinuation. **Disclosure of Interest:** F. Wandroo: None Declared, M. Raghavan: None Declared, J. Ewing Conflict with: none, M. Kwok: None Declared, Y. Hasan: None Declared, S. Pancham: None Declared, A. Shenouda: None Declared.

BSH2020-EP-002

Could CD45 have a role in the development of myeloid malignancies?

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CD45, also known as protein tyrosine phosphatase receptor type C (PTPRC), is a transmembrane glycoprotein tyrosine phosphatase, expressed on almost all haematopoietic cells except for mature erythrocytes, and is essential regulator of T- and B-cell antigen receptor-mediated activation. Disruption of the equilibrium between protein tyrosine kinase and phosphatase activity (from CD45 and others) can result in immunodeficiency, autoimmunity or malignancy. Therefore, CD45 is vital for the normal blood functioning and alteration in this molecule results in serious consequences and may have potential as a drug target.

The objectives of this study were to investigate the effects caused by knockdown of CD45 in myeloid cell lines and to gain insight into pathways induced by CD45, which are disrupted in myeloid malignancies (AML, MDS and MPN) in an attempt to develop novel therapeutic strategies.

CD45 expression was examined in a panel of leukaemia cell lines including OCI-AML3, OCI-AML2, SKM, HL-60, THP1, REH, NALM-6, NB4 and UT-7 cells by flow cytometry, western blot and PCR. AML cells exhibit higher levels of CD45 expression, with ALL cell lines REH and NALM-6 and the MPN cells particularly UT-7 having the lowest expression. CD45 was knocked down, by siRNA, in the MDS/AML cell line SKM and in OCI-AML3 to study the effect of reducing CD45 expression on cellular survival and proliferation. CD45 knockdown, ~eightfold, was confirmed by PCR and western blotting.

To further explore the effects of lowered CD45 levels, cells were incubated with increasing concentrations of Cytarabine for 48 and 72 h, followed by cytotoxicity study by Cell Titre Glo (CTG) assays. The knockdown cells showed a gradual decrease in cell proliferation with an increased resistance to Cytarabine at 48 and 72 h relative to DMSO controls (100%). The IC50 for the control cells were 0.008 $\mu\text{mol/l}$ and was 10-fold higher at 0.097 $\mu\text{mol/l}$ for the CD45 knockdown cells.

Furthermore, we identified several genes whose expression positively or negatively correlated with that of CD45 including *ACTR2*,

THAP-3, *PBX-1* and *Serglycin*. The top 20 positive and negative genes were used imported as a signature into Quadratic, a powerful scalable gene expression connectivity mapping software. We identified that three drugs, Alendronate, Allopurinol and Balsalazide, may have the potential to act as CD45 inhibitors. This was explored by western blot and cytotoxic studies, which indicated that alendronate, had an effect on viability, allopurinol and Balsalazide were non-toxic. However, each drug resulted in a decrease in CD45 expression although surprisingly allopurinol and Balsalazide induced the expression of *ACTR2*. The combination of these drugs with Cytarabine has indicated some degree of synergy.

In conclusion, CD45 is highly expressed in AML cell lines and the inhibition of the gene, by siRNA or novel repurposed agents targeting CD45, have given an insight into the molecular pathways and potential therapeutic options associated with this gene.

Disclosure of Interest: None Declared.

BSH2020-EP-003

The impact of PCR thermocycling conditions, DNA template concentration and intraoperator variation on determining allelic ratio in the detection of FLT3-ITD mutations in AML

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Following the RATIFY trial of Midostaurin in AML (Stone *et al*: N Engl J Med. 2017 Aug 3;377(5):454–464) and the introduction of new 2017 European Leukaemia Net guidelines (Döhner H *et al*. Blood. 2017;129(4):424–447), accurate determination of allelic ratio (AR) for *FLT3* ITD and TKD mutations has become crucial in determining AML risk stratification and disease management. Typically *FLT3*-ITD mutation length and disease burden are determined by capillary electrophoresis fragment analysis following PCR amplification of patient DNA. However, this assay is not standardised between laboratories, raising concerns that differing protocols may cause variations in AR determination. This study aimed to determine whether PCR cycle number, patient DNA template concentration and intraoperator handling variations, impacted on calculated AR values.

To investigate the potential impact of PCR conditions and DNA concentration in determining AR, six bone marrow or peripheral blood samples previously obtained at diagnosis, from patients entering the NCRI AML trials, and containing *FLT3*-ITD mutations of varying length and disease burden, were identified. These were amplified by PCR over a range of PCR cycle numbers and at different DNA concentrations, and determined the AR. In all but one sample, the number of PCR cycles and concentration of the initial DNA template had only a minor impact [standard deviation (SD) ≤ 0.1] on determining the AR, which was consistent across all conditions tested. In the sample where AR fluctuation was observed, this did not appear attributable to ITD base pair length and may have been a consequence of a high AR (>1) and therefore disease burden in this patient. Further work is required to test this hypothesis.

To determine the intraoperator effect on AR variation, DNA (1 μ l) from the patient samples identified above was amplified multiple times under the same PCR conditions and the variation in AR value determined. In addition, one PCR product per patient sample was analysed multiple times by capillary electrophoresis and the variation in AR value also determined. As above, the largest variation in

AR (SD ± 0.15) between different PCR runs occurred in the sample with the high AR, while all other tested samples had a SD of ≤ 0.07 . The impact of operator handling between multiple capillary electrophoresis runs was negligible in all cases (SD range ± 0.01 – 0.03).

Overall, neither DNA template concentration nor number of PCR cycles appear to affect AR determination, and no sample tested crossed the 0.5 (high *versus* low) boundary set by ELN 2017 guidelines for AML risk stratification as a result of varying these conditions. Additionally, intraoperator variation when determining AR on a single sample was found to be negligible in all cases. Currently, the presence and length of *FLT3*-ITD mutations in patient samples are determined by PCR amplification followed by fragment analysis by capillary electrophoresis. Data from this study suggest that this methodology is also a robust method for determining AR values and hence AML risk stratification.

Disclosure of Interest: None Declared.

BSH2020-EP-004

Identification of a new target patient population for ibrutinib treatment

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Acute lymphoblastic leukaemia (ALL) is a haematological malignancy that is characterized by the uncontrolled proliferation of immature lymphocytes. About 80% of cases occur in children where ALL is well understood and treated. However it has devastating effects on adults, where multi-agent chemotherapy is the standard of care with allogeneic stem cell transplantation for those who are eligible. Increasing interest in the potential of immunotherapy offers adults with B-ALL an opportunity for improved survival. To try to identify new targets for treatment we performed sero-profiling to identify the antigens that may act as targets in this difficult to treat disease.

Thirteen sera samples from nine presentation adult B-ALL patients were directly compared to age- and sex-matched healthy volunteer (HV) sera. We identified 69 differentially sero-recognised antigens ($P \leq 0.02$; fold change >1.5) including 19 that were preferentially recognised by patients sera. We identified three antigens for further analysis, bone marrow tyrosine kinase on chromosome X (BMX), dCTP pyrophosphatase 1 (DCTPP1) and vestigial like 4 (VGLL4). Each antigen had decreased transcript levels in B-ALL patients compared with HVs while immunocytochemistry showed that each antigen was expressed in patient samples at moderate to high levels but at low levels in HV samples. BMX plays a crucial role in the Bruton's tyrosine kinase (Btk) pathway that can be targeted by ibrutinib. In clinical trials for relapsed/refractory B-cell disorders such as chronic lymphocytic leukaemia (CLL) and mantle cell lymphoma, ibrutinib shows no dose-limiting toxicity and high response rates, even as a single agent.

We have identified, using proto-array analysis, a number of antigens that are uniquely recognised by adult B-ALL sera at disease diagnosis. These antigens have provided novel insights into the antigens recognised by adult B-ALL patient sera and we have identified a

new target patient population for existing, and seemingly effective, small molecule therapy.

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The clinical significance of FLT3-ITD allele burden in AML – a long-term retrospective single centre study

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FMS-like tyrosine kinase 3 ligand (FLT3)-internal tandem duplication (ITD) is an important prognostic marker in patients with acute myeloid leukaemia (AML). One of the most common mutations in AML is it represents a particular clinical challenge, conferring a higher risk of treatment failure and a poor prognosis. There is increasing evidence to suggest a patient's FLT3-ITD allelic burden may be able to further refine this prognostic information and potentially influence treatment decisions especially with the introduction of FLT3 inhibitors. We studied all newly diagnosed FLT3-ITD mutated AML cases with the aim of correlating FLT3-ITD allelic burden with long-term clinical outcome and to assess the utility of incorporating allelic burden ratios into our integrated diagnostic reports.

FLT3-ITD mutated AML cases were identified over a 9-year time period (2007–2016) to ensure a minimum of 36 months follow-up data. FLT3-ITD, FLT3-D835 and NPM1 mutations were detected by multiplex PCR followed by capillary electrophoresis on peripheral blood and bone marrow samples. The size of the FLT3-ITD was recorded. Allelic ratios were retrospectively calculated (area FLT3-ITD/(area FLT3 WT + area FLT3-ITD) X 100); with a high allelic ratio defined as >50% mutant to wild type. Patient demographics, presentation details, cytogenetic and molecular data, management and clinical outcomes were collected.

Thirty-one patients were identified as FLT3-ITD mutated AML, 26 (83.9%) of whom were treated with intensive chemotherapy. The intensive treatment group comprised of 11 men and 15 women with an age range at diagnosis of 21–78 (median of 51). Fifteen patients were also NPM1 positive, other concurrent mutations included PML-RARA ($n = 4$) and DEK/NUP214 ($n = 1$). Cytogenetic results were available on 21 patients; 20 intermediate and one favourable risk. The majority of patients were treated out with a clinical trial (six patients enrolled in NCRI AML17 and one in AML15) with the most common induction regimen being DA ($n = 19$), followed by ATRA+Ida ($n = 4$), FLAG-Ida ($n = 2$) and ADE ($n = 1$). Eight patients underwent allogeneic stem cell transplant. None of the patients received a FLT3 inhibitor.

The median FLT3-ITD allelic ratio was 47% (range 4–99%). Nine patients (34.6%) had a high FLT3-ITD allelic ratio (range 50.1%–99.7%; median 63.9%). There was a significant age difference between groups: the high allelic ratio group (mean: 38.6, SD: 13.0) was younger than the low allelic ratio group (mean: 60.1, SD: 11.5); $P < 0.01$. Those with a high ratio were more likely to present with WBC $> 100 \times 10^9/l$ (high: 33%, low: 18%). However, there was minimal difference between the rates of complete response after their

first cycle of induction chemotherapy (high: 89%, low: 82%) and non-significant difference in 2-year relapse free survival (high: 44%, low: 53%, $P = 0.95$). Interestingly, patients with both FLT3-ITD and NPM1 mutations a high allelic ratio did not predict an inferior 2-year relapse-free survival.

Analysis of this retrospective single centre cohort was not able to demonstrate any meaningful differences between the patients with high and low FLT3-ITD allele burden AML. As a result, FLT3-ITD allele burden is not routinely reported. Given the multitude of methods and lack of national or internationalised standardisation for FLT3-ITD testing, this study underpins the importance of institutional validation before the integration of allele burden to routine diagnostic reporting and clinical risk assessment.

Disclosure of Interest: None Declared.

BSH2020-EP-006

Platelet recovery after the first cycle of DA chemotherapy for AML is an independent predictor of survival

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While overt refractoriness to the first cycle of AML induction therapy is an established predictor of poor outcome in AML, the remaining categories of complete remission (CR), complete remission with incomplete count recovery (CRi) and partial remission (PR) define a heterogeneous population with large variations in survival outcome. Remission status among these groups is conventionally used as a predictor of survival; however, haematological count recovery following intensive chemotherapy may more clearly define prognostic groups, having already shown to be an independent predictor of stem cell transplantation survival. The WHO criteria identify CRi as a distinct entity from CR, where the former describes <5% residual blast count with persistent thrombocytopenia ($<100 \times 10^9/l$) or neutropenia ($<1 \times 10^9/l$); however, do not currently specify a time point at which count recovery thresholds should be met. We investigated the value of a day-30 platelet count post-induction therapy as a predictive indicator of overall survival (OS) and identified which patient and disease factors were associated with count recovery delay.

All non-M3 patients undergoing DA induction therapy for AML between January 2006 and December 2018 with full data available for review were entered into the study. Count recovery was retrospectively analysed and designated as 'delayed' or 'rapid' based on the isolated day-30 platelet count after the first induction cycle, stratified relative to the median platelet count of the cohort. The primary outcome was overall survival at 3 years. Differences in the distribution of patient and disease characteristics between count recovery groups were also assessed. Day-30 platelet count was used as a surrogate marker for absolute bone marrow recovery.

Of 112 patients, 23 were refractory to cycle 1, and 89 achieved complete or partial remission. The 3-year OS was significantly worse in refractory patients, but no significant survival differences were seen between CR and PR groups (3-year OS: 55.6% (95% CI 42.3–68.9%), 48.3% (95% CI 26.0–70.6%) and 16.4% (95% CI 0.0–32.9%) for CR, PR and refractory groups respectively). Among non-refractory patients, the median day-30 platelet count was $261 \times 10^9/l$. Delayed platelet count recovery was significantly associated with poorer OS in non-refractory patients ($P = 0.024$) and maintained independent prognostic significance in multivariate analysis (HR 2.21, CI 1.17–4.17; $P = 0.015$). A 3-year OS was 58.8% (95% CI

41.9–75.7%) in the rapid count recovery group and 46.1% (95% CI 30.0–62.3%) in the delayed. The frequency of AML patients with myelodysplasia-related changes and secondary disease was significantly higher in the delayed count recovery group, as was those with positive minimal residual disease status. Delayed recovery was further associated with older age at diagnosis and a higher day-30 post-chemotherapy CRP.

Platelet count at day 30 offers a feasible surrogate for absolute marrow recovery, which has been identified as an independent predictor of overall survival. Platelet recovery to within the normal range may be a more suitable prognostic indicator than the existing $100 \times 10^9/l$ WHO threshold; however, a spectrum of response was evident, with those achieving the highest and lowest quartiles of platelet count recovery exhibiting the most extreme differences in survival. Platelet recovery offers an accessible index with potential utility in the prognostication of non-refractory patients.

Disclosure of Interest: None Declared.

BSH2020-EP-007

Antibody immunotherapy treatment for adults with B-acute lymphoblastic leukaemia—a real-world analysis

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Blinatumomab (Blina) is recommended for routine use in the NHS for patients with Philadelphia-chromosome-negative (Ph-neg) CD19-positive B-precursor acute lymphoblastic leukaemia (ALL) with minimal residual disease (MRD) of at least 0.1% whose disease is in first complete remission (CR1) and in relapsed/refractory (R/R) Ph-neg ALL. Inotuzumab ozogamicin (Ino) is recommended as an option for treating R/R CD22-positive B-cell precursor ALL irrespective of Ph status. We sought to investigate the safety and efficacy of these immunotherapy agents in the real-world setting.

This real-world study includes patients with B-ALL who received Blina or Ino at St Bartholomew's hospital between 1 Jan 2014 and 31 Dec 2019. Data were extracted from medical notes. Patients were followed from immunotherapy initiation until death or the end of the study period (1 Jan 2020), whichever occurred first.

Seventeen patients were eligible for the study, two were excluded as the relevant medical notes were not available providing a final study population of 15. Of these 13 of 15 (87%) received immunotherapy for the indication of R/R disease and two of 15 (13%) for MRD-positive disease. Adverse cytogenetics was documented in 40% (6/15); 67% Ph-pos, 33% MLL rearrangement. In 87% (13/15) of patients UKALL 14 (85%) or UKALL 60+ (15%) clinical trials were administered as primary therapy. The median age at relapse was 47.5 years (range 36, 76).

Of the 15 R/R patients seven (47%) received Blina and seven (47%) received Ino, one patient received both Blina and Ino sequentially. About 85% ($n = 11$) of those treated for R/R disease had prior allogeneic hematopoietic stem cell transplant (HSCT). The mean WCC at relapse was $9.45 \times 10^9/l$ (Range; 0.7, 23.9). Median duration of first remission was 23.5 months (Range; 3, 63). In 13 of 15 (87%) patients the immunotherapy was given as first line salvage treatment, the median number of prior salvage therapies was 1 (Range; 1, 2). The remission rates for those with R/R disease after cycle 1 of immunotherapy were 2/6 (33%) and 4/8 (50%) with Blina and Ino respectively. Two of eight (13%) patients received Blina for MRD-positive disease prior to HSCT. All (2/2 patients) remain in remission 32 and 40 months post-transplant.

In the R/R patients, toxicity in the form of cytokine release syndrome (CRS) was observed in 50% (3 of 6) of patients receiving Blina, and two thirds (67%) of these were grade 3/4; requiring Tocilizumab. One third (33%) of patients who experienced CRS ceased Blina during cycle 1 due to grade 4 liver injury. One of six (12.5%) patients developed grade 4 neurotoxicity 4 days after commencement of cycle 1, necessitating the cessation of therapy. In the Ino-treated patients there were no events of veno-occlusive disease to date inclusive of the single case proceeding to allo-sct. No early terminations of therapy occurred in Ino-treated patients. There were no toxic events in the two patients treated for MRD-positive disease prior to allo-SCT.

This study follows the outcomes of immunotherapy used primarily as first salvage therapy after CR1 in relapsed B-ALL. A significant proportion of patients achieve morphological remission after 1 cycle substantiating the efficacy of these therapies as previously reported.^{1,2} Real-world toxicity also appears comparable. Larger cohorts are needed to fully address the safety and efficacy of these agents in unselected patient cohorts.

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BSH2020-EP-008

Estimating the relative treatment effect and corresponding cost-effectiveness estimates of inotuzumab ozogamicin versus blinatumomab for adults with Philadelphia chromosome-negative relapsed/refractory B-cell acute lymphoblastic leukaemia in the United Kingdom

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Inotuzumab ozogamicin (InO) is approved in relapsed/refractory B-cell acute lymphoblastic leukaemia (R/R B-ALL) and is recommended for reimbursement by the Scottish Medicines Consortium (SMC). The Phase III INO-VATE trial compared InO to investigators choice; InO was associated with a remission rate of 81%. Blinatumomab (Blina) was also recommended by the SMC, prior to InO; the phase III TOWER trial compared Blina to investigators choice but only for Philadelphia chromosome-negative (Ph-) disease. As Blina was already reimbursed at the time of InO's submission to the SMC, a comparison of costs and effects was required as part of InO's appraisal. This comparison was done via indirect treatment comparisons (ITC) and a United Kingdom (UK)-based cost-effectiveness model (CEM) that explored the impact of treatment differences on mean life years and quality-adjusted life years (QALY).

Stem cell transplant (SCT) is paramount in increasing overall survival (OS) in an otherwise terminal disease. Achieving complete response/complete response with incomplete count recovery (CR/CRi) is a typical prerequisite to undergo SCT. InO and Blina precede SCT, thus their CR/CRi and SCT rates are key. CR/CRi rate, SCT rate, and OS were compared using patient-level data from INO-VATE and aggregated data from TOWER.

A CEM was built to submit to the SMC which included the health states 'No CR/CRi & no SCT', 'CR/CRi and no SCT' and 'SCT/Post SCT'; progression-free survival (PFS) was modelled within

each state and the probability of death was specific to each state. Data from INO-VATE allowed InO patients to be modelled into these states and matching-adjusted indirect comparisons (MAIC), Bucher and multinomial ITCs were explored to allocate Blina patients into states. Quality of life data, including time-varying utilities, were applied to the states, derived from INO-VATE and the literature. A UK perspective with 2017 sources was used to apply costs to the CEM, with results discounted at 3.5% annually.

The ITCs found that InO was associated with higher odds of CR/CRi [Odds ratio (OR) 2.81–3.38] and SCT (OR 3.24–4.11) *versus* Blina. Blina was consistently found to lead to lower chances of SCT (19.1–22.5%) and CR/CRi (25.2–33.3%) than InO, with 49.3% of InO patients reaching SCT in the model. ITCs comparing observed OS for InO *versus* Blina showed similar results; however, due to a proportion of patients receiving SCT for their disease (causing a plateau in the tails of the Kaplan-Meier's) the mean OS is important.

Using UK list prices, the CEM resulted in an incremental QALYs gain of 0.91–1.14 with InO over Blina, with an incremental cost-effectiveness ratio (ICER) of £3700 to £7010 per QALY. Scenario analyses indicated that InO remained a cost-effective option compared to Blina, falling consistently under a willingness to pay threshold of £20 000/QALY. The SMC recommended InO as a cost-effective use of resources citing an ICER of £6754 based on the MAIC and a mean survival increase of >29 months.

The methods explored and submitted to the SMC indicated InO provides patients with a greater chance of CR/CRi and SCT *versus* Blina; data are not currently published to substantiate long-term survival without follow-on therapies such as SCT. The mean OS gain cited by the SMC is thus intuitive given the superior odds of CR/CRi and SCT for InO. Further research is needed to determine the long-term PFS and OS in R/R B-ALL, both with and without SCT.

Disclosure of Interest: T. A. Russell-Smith Conflict with: Employment at Pfizer, M. Cooper Conflict with: BresMed Health Solutions Ltd., I. van Oostrum Conflict with: Pfizer, Merck, Janssen, AstraZeneca, Conflict with: Employment at Ingress Health, V. Welch Conflict with: Employment at Pfizer, S. Critchlow Conflict with: BresMed Health Solutions Ltd.

BSH2020-EP-009

Determination of therapeutic potential of luteolin for Philadelphia chromosome positive acute lymphoblastic leukaemia cells

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Acute lymphoblastic leukaemia (ALL) is a malign hematologic cancer characterized with increased number of immature lymphoblast in bone marrow and peripheral blood. Reciprocal translocations between BCR and ABL genes that results information of Philadelphia (Ph) chromosome with constitutive tyrosine kinase activity occur in ALL and named as Ph+ ALL. BCR/ABL translocation regulates the several pathways related to proliferation, anti-apoptotic and drug resistance through its aberrant tyrosine kinase activity. In addition to classical chemotherapy, tyrosine kinase inhibitors (imatinib and dasatinib) are used in the treatment of Ph+ ALL patients. However, current therapies do not provide complete cure for the disease. Therefore, novel strategies are needed for the treatment of Ph+ ALL. Luteolin which is a member of flavones family of flavonoids is found in vegetables and medical herbs. Luteolin was shown to have antioxidant, pro-oxidant, anti-inflammatory and anticancer activity in various cancers such as breast, lung, colon, gastric and acute myeloid leukaemia in both *in vitro* and *in vivo*.

In this study, cytotoxic effect of luteolin on Ph+ ALL cells was determined by MTT proliferation assay and Trypan blue dye exclusion method. Luteolin significantly decreased the cell proliferation and viability of SD-1 cells in dose and time-dependent manner. Besides, the effect of luteolin on cell growth and proliferation of two different healthy cell lines was shown. Luteolin not significantly decreased the cell proliferation and viability of healthy cell lines. Moreover, the apoptotic effects of luteolin on SD-1 cells were determined by Annexin-PI double staining and mitochondrial membrane potential assay. Luteolin significantly enhances the apoptosis in SD-1 cells in different doses. Furthermore luteolin induces cell cycle arrest at G2/M phase and S phase in SD-1 cells. Moreover, the involvement of bioactive sphingolipids genes, which regulate the several pathways including cell proliferation, apoptosis, drug resistance and senescence in cell, in luteolin-induced apoptosis in SD-1 cells was determined by qRT-PCR and western blotting. Expression levels of ceramide synthase (CERS) genes increased while the expression levels of sphingosine kinase-1 (SK-1) and -2 (SK-2) genes not significantly changed in SD-1 cells. Next, the protein levels of SK-1 were significantly decreased while protein levels of SK-2 and GCS not significantly changed in SD-1 cells. Moreover, therapeutic potential of combination of luteolin-imatinib for SD-1 cells was determined by MTT and trypan blue dye exclusion methods for the first time.

As a consequence, therapeutic potential of luteolin on Ph+ ALL cells and the roles of bioactive sphingolipids genes in luteolin-induced apoptosis in Ph+ ALL cells were determined for the first time.

Disclosure of Interest: None Declared.

BSH2020-EP-010

Incidence and management of PEGylated-asparaginase related thrombosis in adults with ALL

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Introduction: Asparaginase is a cornerstone of acute lymphoblastic leukaemia (ALL) therapy; however, its use is associated with a number of toxicities including increased risk of thrombosis. There is considerable uncertainty about the frequency of asparaginase-related thrombosis in adult ALL, with reported incidences ranging from 1% >36%. While there are guidelines for the paediatric population, there are no specific adult guidelines on the prevention and management of venous thromboembolism (VTE) prevention in adult ALL.

Methods: We retrospectively analysed 33 consecutive adult patients treated with PEGylated asparaginase (PEG-ASP) as part of the UKALL14 regime from 2014 to 2019 at St Bartholomew's Hospital. The protocol specifies one (age ≥40 years) or two doses (age <40 years) of 1000 IU/m² PEG-ASP during standard phase 1 induction. Clinical notes and imaging were used to determine the frequency of thrombotic events during Phase 1 and 2 induction therapy incorporating PEG-ASP, the choice of anticoagulation strategy and the use of antithrombin (AT) concentrate.

Results: Of the 33 patients (13F, 20M – age range 23–60, mean 39.8) who received PEG-ASP, there were 14 symptomatic thrombotic events in 10 patients (30%, 5F, 5M, mean age 40.7). Of these 14 events, three occurred prior to PEG-ASP being given. The majority were deep vein thromboses (DVT; 64%). There were two superficial vein thromboses (SVT; 14%), one internal jugular vein thrombosis, one pulmonary embolism (PE) and one episode of sagittal sinus venous thrombosis with cortical vein thrombosis. The average time from PEG-ASP dose to thrombosis was 18 days. All patients were

Table.

Age and gender	Thrombosis site	Days since last dose of PEG-ASP administered	Antithrombin level at time of thrombosis (normal range 81–119 IU/dl)
31M	L upper DVT	10	35
	R upper DVT	11	35
	R lower DVT	8	33*
	Bilateral upper DVT	24	65*
30F	Cerebral venous sinus thrombosis	13	27*
26M	Internal jugular vein thrombosis	Pre-asparaginase	40
	PE	3	Not re-tested
52F	L upper DVT	5	Not measured
31M	L upper DVT	Pre-asparaginase	Not measured
31M	L lower DVT	27	41†
47F	L lower DVT	17	Not measured
59F	R lower SVT	Pre-asparaginase	42
47F	R upper DVT	60	Not measured
53M	R upper SVT	21	Not measured

*Antithrombin concentrate used.

†Checked 3 months after thrombosis diagnosed.

treated with therapeutic low molecular weight heparin (LMWH) with five patients (50%) having antithrombin (AT) levels checked, all of which were below the normal range (see table 1). Two patients received AT concentrate and five had further asparaginase after the thrombotic event with two patients still awaiting further treatment. Three events were PICC line associated.

Discussion: Our single centre data demonstrate a high penetrance of venous thrombosis of varying severity, with a predominance of DVT site. Despite demonstration of low antithrombin levels, there was significant variation in replacement therapy. Our data show a similar mean age in those with and without thromboses which differs from previous data [Grace, 2011] – although age over 30 was identified as being high risk in previous studies. Unfortunately, data were not reliably available on other thrombotic risk factors such as BMI and smoking status. Further data on the impact of antithrombin concentrate and risks of re-exposure to asparaginase are necessary.

Reference: Grace RF, Dahlberg SE, Neuberger D. The frequency and management of asparaginase-related thrombosis in paediatric and adult patients with acute lymphoblastic leukaemia treated on Dana-Farber Cancer Institute consortium protocols. *Br J Haematol*. 2011;152:452–459.

Disclosure of Interest: None Declared.

BSH2020-EP-011

Spontaneous remission of acute myeloid leukaemia – how should this be handled by haematologists?

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Prior to 1955 approximately 100 case reports detailed spontaneous remission of acute myeloid leukaemia (AML). The number of cases has dwindled possibly due to the more prompt initiation of therapy masking the natural progression of disease. We present two cases from the last 12 months of spontaneous remission of AML in young

male patients – both with intercurrent infection at the time of diagnosis.

A 35 years old male with no significant past medical history was repatriated to the Royal Sussex County hospital, Brighton, from Azerbaijan in September 2019. He reported 1 week history of pharyngitis and general malaise. After subsequent presentation to a local hospital, he was found to have pancytopenia. On repatriation to the United Kingdom, he was diagnosed with proctitis attributed to his profound neutropenia. Initial bone marrow biopsy showed left shifted myelopoiesis with a blast population that was difficult to quantify due to the variability in morphology. He was informed of a diagnosis of acute myeloid leukaemia. With ongoing broad spectrum antibiotics his symptoms began to improve, as did his counts particularly his neutrophils, a second BMT, 2 weeks after the first, showed considerable improvement in the myeloid maturation and no convincing blasts by morphology or immunophenotype.

The second patient, a 28-year-old, also presented with sudden onset illness in March 2019 reporting 3 days of profound diarrhoea. He was thrombocytopenic and neutropenic, rapid clinical deterioration led to admission to intensive care at Kings College Hospital London. He developed a bleeding diathesis requiring blood products. Bone Marrow biopsy performed 4 days after admission showed a large population of blasts consistent with AML FAB M4/5. After a protracted and complex ITU stay, and a short course of high-dose steroid, repeat bone marrow biopsy displayed complete AML remission.

While small case series have followed such patients we know very little amount this mechanism underlying this presentation and currently have no way of identifying if spontaneous remission is likely to occur. A literature review of 32 cases from 1979 onwards reported that most cases were associated with infection and with diagnosis of FAB M4/M5 AML. The median age of patients was 58, the majority comparable to our two cases had normal cytogenetics. The length of the remission was variable from months to years, the two patients reported here have maintained normal blood counts and are regularly monitored. With longer waiting time prior to initiation of therapy to allow results of diagnostic tests which inform treatment choice, this phenomena may be increasingly observed. The added emotional burden of informing patients and relatives of an initial leukaemia diagnosis, then having to revise this in the context of

spontaneous remission and further explain its uncertain prognostic features, should not be dismissed. Further investigation of these patients may provide insight into the pathology of AML and perhaps identify a similarity with the transient abnormal myelopoiesis akin to infants with Down syndrome.

Disclosure of Interest: None Declared.

BSH2020-EP-012

Relapsed acute lymphoblastic leukaemia diagnosed using femoral head histology

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With modern therapies many patients now diagnosed with acute lymphoblastic leukaemia (ALL) achieve remission. Unfortunately a proportion of these will go on to relapse, with results from the international MRC UKALL XII/ECOG2993 trial^[1] showing that 50% of under 55 s and 60% of over 55 s relapse following first remission. Relapsed ALL presenting with extramedullary disease has been rarely reported.

Avascular necrosis is a recognised complication of ALL treatment and is frequently managed with joint replacement surgery. Recently at Nottingham University Hospitals, we have diagnosed three patients with relapsed ALL from femoral head histology following joint replacement for avascular necrosis of the femoral head.

The first case was a 21 years old lady with precursor B-ALL. She was treated on UKALL2011 protocol. 14 months after completing maintenance therapy the patient underwent a total hip replacement. Femoral head histology showed an interstitial infiltrate of lymphoblasts, with immunophenotyping consistent with B-ALL. She was treated with FLAG-IDA, followed by blinatumomab due to ongoing minimal residual disease (MRD) positivity, before undergoing an allogeneic transplant. Bloods at the time of hip replacement were normal.

In the second case a 51 years old man was treated for B-ALL on UKALL14 trial, on the high-risk arm. As part of this he had an allogeneic stem cell transplant. Nine months later, relapsed disease was diagnosed on routine post-transplant bone marrow. The patient received FLAG-IDA, inotuzumab and gemtuzumab, following which he was in remission. Three years after completing this treatment he underwent a total hip replacement. Femoral head histology showed replacement of the medullary cavity with necrotic tumour, with immunophenotyping consistent with B-ALL. Bone marrow biopsy at the time was negative for ALL, although multiple areas of active disease were evident on PET-CT. The symptomatic areas of disease have been treated with localised radiotherapy and the patient has had active monitoring with repeated bone marrow biopsies. Should a haematological relapse be found, he will be considered for CAR-T cell therapy.

In the final case, a 22 years old with T-lymphoblastic lymphoma and hepatitis C was treated off trial using UKALL 11 protocol. Eight months after completing maintenance the patient presented with spinal cord compression and was found to have relapsed. This was treated with nelarabine, cyclophosphamide and etoposide, followed by autologous stem cell transplant. About 20 months later the patient underwent a total hip replacement. Femoral head histology showed sheets of small lymphoid blast cells, with immunophenotyping consistent with T-ALL. Blood tests at the time showed only thrombocytopenia, which was longstanding post-transplant. The patient was re-treated with nelarabine, cyclophosphamide and etoposide with high-dose methotrexate. An allogeneic stem cell transplant is currently planned.

From literature there are only case reports of relapsed ALL presenting this way and the exact number of patients with evidence of disease in the femoral head is unknown. We would recommend as a result of our experience always sending bone specimens for histology in patients with a history of ALL undergoing joint replacement surgery.

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Disclosure of Interest: None Declared.

BSH2020-EP-013

19 courses of CPX-351 for the treatment of AML

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CPX-351 is a liposomal encapsulated formulation of daunorubicin and cytarabine in a fixed 1:5 molar ratio. In phase 3 trials has shown a survival benefit in high-risk or secondary AML [1, 2]. It has been approved by NICE for use within its marketing authorisation of 'the treatment of adults with newly diagnosed, therapy-related acute myeloid leukaemia (t-AML) or AML with myelodysplasia-related changes (AML-MRC)' [3].

It has, however, been more widely used in a clinical trial context and the experience of its use within Belfast City Hospital for patients enrolled on the AML19 trial have been collated and analysed. Each course of treatment with CPX that commenced between 01/01/2017 and 31/05/2019 in Belfast City Hospital has been reviewed: 19 courses of treatment involving 12 patients. For each of these courses data were recorded on the time to platelet/neutrophil nadir, last red cell/platelet transfusion (as proxies for red cell and platelet recovery), neutrophil recovery ($>0.5 \times 10^9/l$) and discharge were recorded. In addition total duration of antimicrobials was recorded as well as any adverse events that occurred until discharge. Furthermore, for each patient cytogenetics at diagnosis were recorded and a timeline of treatment and outcome established.

In this sample, from day 1 of treatment the median time to: platelet nadir was 12 days (1–17 days), neutrophil nadir was 15 days (1–20 days), last platelet transfusion was 27 days (16–74 days), last red cell transfusion was 26 days (10–77 days), neutrophil recovery was 28 days (20–56 days) and discharge was 34 days (21–54 days). The median duration of antimicrobial treatment was 23 days (0–58 days). Time to platelet and neutrophil nadir was usually shorter in the case of first courses of chemotherapy than for subsequent courses due to pre-existing pancytopenia, although there did not seem to be any significant difference in count recovery time.

Excluding febrile neutropenia, which was the most common adverse event, 32 other adverse events were noted across all courses of treatment. The most common was rash in 12 of 19 courses often similar to folliculitis in appearance. Other events included peripheral swelling (6 of 19 – 2 of which were DVTs), joint swelling (4 of 19), constipation (3 of 19) and non-cardiac chest pain (3 of 19).

Prolonged time to count recovery has already been noted in previous clinical trials with CPX-351 with median time to neutrophil recovery reported at 35 days compared with 29 days for DA 7 + 3 [1]. This difference was not as evident in our, much smaller sample, although the range was quite wide at 20–56 days.

Overall CPX-351 was well tolerated in the patient cohort reviewed. One aspect that was not frequently recorded in the medical notes but which was evident on discussion with patients is that hair

loss is less frequent with CPX-351 than with other intensive chemotherapy regimens.

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Disclosure of Interest: None Declared

BSH2020-EP-014

The role of nutrition in aggressive haematological malignancies – a single centre review of patients receiving intensive chemotherapy

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Intensive chemotherapy regimens are needed to cure aggressive haematological malignancies including acute myeloid leukaemia (AML) and high-grade non-Hodgkin Lymphoma (NHL). To achieve this, patients require good pre-morbid fitness and maintenance of fitness throughout treatment, for which adequate nutrition is key. Malnutrition is common in patients with haematological malignancies, reflecting a combination of patient-, disease- and treatment-related factors. Malnutrition is associated with markers of higher risk disease, poorer prognosis and poorer quality of life in these patients. Methods for assessing nutritional status, however, are unsatisfactory, with no consensus on how to identify and manage patients at risk.

We performed a service evaluation of patients newly diagnosed with AML and high-grade NHL undergoing intensive inpatient chemotherapy. Indicators of nutritional status, weight changes during admission, level of dietetic support and limited clinical details were collated from patient records.

We evaluated 21 consecutive patients admitted between June 2018 and May 2019 (15 cases of AML and 6 cases of NHL). Body mass index (BMI) at presentation categorised most patients as at least overweight, none as underweight. Published nutritional risk scores determined most patients to be at low risk of malnutrition. Nearly all patients (20/21, 95%) lost weight during their admission: mean weight loss 5.6 kg (range 0–12.1 kg), six patients lost over 10% of admission body weight. Average weight at discharge was 4.4 kg less than admission weight; only four patients recovered to their presentation weight.

Specialist dietician input was received by 15 of 21 (71%) of patients, typically within the first week of admission. Patients receiving dietetic care were more likely to have a record of nutritional intake. Enteral feeding (nasogastric) was commenced in five patients under dietetic supervision, two of whom continued it post-discharge.

Mean length of stay was 33 days. On average, patients had 2.1 separate episodes of febrile neutropenia, amounting to 15 days of fevers and 22 days (64% of admission length; range 27–100%) on intravenous antibiotics.

Exploratory analysis showed that weight loss was associated with admission C-reactive protein concentration, and strongly correlated with increased length of stay (Pearson's $r = 0.68$, $P = 0.001$). No correlation was found between weight loss and age, admission weight, albumin or nutritional risk scores.

There is currently no satisfactory measurement of nutritional status in patients undergoing intensive chemotherapy for aggressive haematological malignancy. Indicators used elsewhere are not directly transferable to this setting: there is often no history of acute weight loss at disease presentation; normal/high BMI at presentation does not equate to adequate nutritional status; nutrition biomarkers are confounded by intercurrent acute illness; weight changes are significantly affected by fluid accumulation.

Better metrics need to be identified to assess the nutritional status of these patients both at presentation and during treatment. Possible parameters may include mid-upper arm circumference, patient-generated subjective global assessment, bioelectrical impedance analysis and handgrip. Specialist dietetic input from the outset of treatment can improve early identification of patients at risk of malnutrition and, through appropriate nutritional intervention, can optimise patient's nutrition.

Disclosure of Interest: None Declared.

BSH2020-EP-015

Rapidly progressive vision loss following CAR T-cell therapy

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Immunotherapy with chimeric antigen receptor (CAR) T-cell therapy has emerged recently as a remarkable innovation and an alternative to cytotoxic chemotherapy in the treatment of relapsed/refractory acute lymphoblastic leukaemia (R/R ALL). Several trials have shown that CAR T-cell therapy achieves 70–90% complete remission (CR) in R/R ALL. Despite the encouraging response rates, 30–60% may still relapse post-CAR-T therapy. Bone marrow (BM) remains the most common site for relapse even post-CAR-T therapy with a few isolated CNS relapses being reported. It is unclear if relapses in other extramedullary sites were reported post-CAR-T therapy. Overall, intraocular relapse in ALL has been reported as a rare event. In this report, we describe a case of a relapsed ALL post-CAR-T therapy presenting as an isolated intraocular relapse in the form of rapidly progressive subretinal leukaemic infiltrates.

A 7-year-old boy was diagnosed with very high-risk (VHR) B-ALL based on his cytogenetics (hypodiploidy and Philadelphia (Ph)-like chromosome). He achieved CR-1 after a standard 4-drug induction chemotherapy followed by match unrelated donor hematopoietic stem cell transplant (HSCT). On day +100 post-transplant, surveillance bone marrow aspirate (BMA) confirmed a relapse with 50% blasts. The blasts were positive for CD19 and CD22. He had a traumatic lumbar puncture (LP) therefore he was stratified as CNS 2c. He was then treated with inotuzumab ozogamicin (IO) as a bridge to CAR-T therapy. He achieved CR-2 with three cycles of IO in addition to multiple intrathecal methotrexate. Following CR-2, the patient underwent CAR T-cell therapy successfully with no major adverse events. Six months into the CAR-T therapy, the patient presented with cough, runny nose and conjunctival erythema of his left eye. He was initially managed as a viral conjunctivitis. Ten days later, his symptoms progressed with more erythema and photophobia. He was diagnosed by ophthalmology as Herpes Simplex Virus uveitis and was treated with oral Acyclovir. His symptoms however continued to worsen and a B-scan ultrasonography (USG) showed

subretinal infiltrates that could represent an infectious *versus* a leukaemic process. Four days later, the vision in his left eye was started to be affected. A repeat B-scan USG showed a new hyperechoic subretinal mass. The biopsy of the mass performed by a retina specialist confirmed a relapsed pre-B ALL with leukaemic cells being positive for CD19 but negative for CD22 markers. Meanwhile, virology tests were negative. His LP and BMA were negative at the same time, confirming an isolated intraocular relapse. In spite of orbital radiation therapy (1800 cGy in 10 fractions), the patient experienced progressive vision loss and unfortunately lost vision in the affected eye.

In this case we have reported the challenges in diagnosing an intraocular ALL relapse as its manifestations are non-specific. Conjunctivitis/uveitis-like symptoms may resemble an early sign of leukaemia relapse. Therefore, post-CAR-T therapy patients presenting with such symptoms should be thoroughly monitored and investigated for the possibility of a leukaemic relapse. Subretinal leukaemic infiltration may aggressively progress, and prompt diagnosis and therapy may preserve vision.

Disclosure of Interest: None Declared.

BSH2020-EP-016

The utility of flow cytometry in the diagnosis of acute monocytic and monoblastic leukaemia

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Immunophenotyping remains an important tool for diagnosing and classifying acute leukaemia. However, stages of monocytic maturation can be difficult to delineate by both morphology and immunophenotyping. This is of particular relevance for detecting early monocytic blast equivalents (monoblasts and promonocytes) in the diagnosis of acute myelomonocytic, acute monocytic and acute monoblastic leukaemia. Currently, CD14 and CD64 are most often utilised within this setting. Monoblasts typically show a CD64⁺, CD14⁻ phenotype but in our experience the expression of CD14 by promonocytes is heterogeneous (CD64⁺, CD14^{+/−}).

This study aimed to identify a flow cytometry gating strategy for monocytic maturation, particularly identifying promonocytes in an acute monocytic and monoblastic leukaemia cohort. The secondary aim was to determine if CD14 conjugated to different fluorophores, targeting different CD14 epitopes (CD14 FITC Mo2 epitope; Beckman coulter CD14 FITC clone 116 and CD14 APC; Becton Dickinson CD14 APC-Cy7 clone MΦP9) and if this would improve the identification of promonocytes.

Eight patients with acute monocytic or monoblastic leukaemia were assessed. Mature monocyte populations were referenced from a control group for gating comparison. Monoblast and mature monocyte populations were gated and compared to a manual morphological differential of monocyte populations. The proposed promonocyte region between monoblast and mature monocyte gate was analysed and percentage difference between manual differential made.

In all patients, promonocyte populations were underquantified using either antibody compared to morphology. The mean percentage difference from morphological promonocyte differential between FITC Mo2 epitope was 58.4% (SD 21.22%, $P = 0.01$) and CD14 APC 58% (SD 23.12%, $P = 0.006$). No discernible difference between either epitope was noted; both antibodies demonstrate promonocyte CD14 expression similar to that seen in mature monocytes.

In summary, promonocyte populations cannot be easily gated and quantified using standard immunophenotyping methods to help ascertain blast equivalents, and secondly, the FITC Mo2 epitope for CD14 is no more accurate than CD14 APC for discriminating

different monocyte populations. This reinforces the importance of morphological assessment of monocyte populations compared to immunophenotyping alone in defining monocyte lineage acute leukaemia.

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Disclosure of Interest: None Declared

BSH2020-EP-017

CAR-T therapy as a curative option for childhood acute lymphoblastic leukaemia (ALL) in developing countries?

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Acute lymphoblastic leukaemia (ALL) is the commonest childhood cancer. Although largely curable, survival remains suboptimal in developing countries such as those in South East Asia (SEA). The high mortality is due to both relapse and treatment-related mortality (TRM). Inherent difficulty in achieving the optimal balance between under and over-treatment is compounded by challenges such as inadequate manpower and infrastructure, high therapy abandonment, limited access to risk stratification tools and expensive allogeneic stem cell transplant (allo-SCT) in developing countries.

We hypothesize the wider utilisation of chimeric antigen receptor T cell (CAR-T) could represent a solution for developing countries to achieve the high cure and low TRM rates of developed countries. CAR-T possesses several attributes making it a potential leapfrog health care technology:

- Easy to administer single-dose therapy, avoiding the need to adhere to prolonged treatment
- Relatively low-cost therapy, possibly even lower than chemotherapy. A regional laboratory which manufactures CAR-T in SEA has priced the therapy at only USD25,000.
- Highly effective with remission rates of 70–90%
- Safe with practically zero TRM. It is certainly safer than allo-SCT and likely safer than prolonged chemotherapy in this setting too.

We report here efficacy and safety data from the pilot phase of a program in SEA.

This is an observational cohort study evaluating the outcome of children with relapsed ALL who received re-induction chemotherapy followed by CAR-T as consolidation. Between October 2018 and December 2019, we treated seven children with relapsed ALL (six from Indonesia and one from Sri Lanka). Thirteen children referred to the program failed to make it to treatment due to subsequent death from relapse or TRM, or were unable to raise the financing.

The mean age of the seven children treated to date was 11 (range 6–19). All scored ECOG 1 or 2; one had prior allo-SCT; three had

persistent CNS leukaemia despite repeated triple therapy (including one with leptomeningeal disease (LMD)).

All patients had received a 6-week course of re-induction therapy, and all had failed to achieve remission or had minimal residual disease prior to CAR-T infusion. The mean dose of CAR-T infused was 2.4×10^6 cells/kg body weight. CAR-T was administered in the general ward in all except one (Transplant Unit- local preference). All children were discharged by day-3 post-infusion, except one who had cytokine release syndrome- easily managed with a single dose of Tocilizumab. None had neurotoxicity.

Three patients achieved complete remission, with the longest event-free survival being 14 months. Two patients with CNS leukaemia failed to achieve remission; the one with LMD subsequently died at 7 months post-CAR-T. The other had treatment failure possibly related to inadequate cells shipment, and is now undergoing second dose of CAR-T. One patient has yet to be evaluated.

Our observation, even with the limited data to date, attests to the administrative simplicity and safety of CAR-T. However, the efficacy of CAR-T, especially its ability to achieve long-term durable remission in this setting, has yet to be demonstrated. There remain several challenges including logistical and regulatory issues related to shipping cells across borders in the region, early preparation of patients for CAR-T soon after relapse before they succumb to relapse-related complications or TRM and the residual financial barrier to access this therapy.

Disclosure of Interest: None Declared.

BSH2020-EP-018

Evaluating Vyxeos use in routine clinical practice; data from the South East Scotland Cancer Network

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Acute myeloid leukaemia (AML) encompasses a group of aggressive haematological neoplasms. It is a clinically, morphologically and genetically heterogeneous disease involving one or all of the myeloid lineages and the most common malignant myeloid disorder in adults. Many factors, both of the disease and the patient, govern treatment decisions and crucially the patient's prognosis.

Vyxeos is a liposomal, synergistic, fixed combination of cytarabine and daunorubicin, preferentially taken up by leukaemic cells within the bone marrow for a prolonged period of time. Phase III data, showing Vyxeos to improve median overall survival compared to conventional intensive treatment in older patients (60–75 years) with poor risk AML ($P = 0.003$), led to NICE and SMC approval of Vyxeos as a treatment option for untreated therapy-related AML and AML with MDS-related changes (December 2018 and March 2019 respectively). We undertook a review of patients treated with Vyxeos following SMC approval within the Haematology South East Scotland Cancer Network (SCAN) to assess its safety and efficacy in routine clinical practice.

Nine patients were retrospectively identified from SCAN Multidisciplinary Meeting records. The median time at census post-diagnosis was 159 days (range 28–529 days). The data collected included demographics, diagnostics, treatment and outcomes. The median age at diagnosis was 61 years (range 17–70 years), with $n = 3$ female. Two patients were diagnosed with therapy-related AML (22.2%). Six patients (66.7%) had a diagnosis of AML with myelodysplasia-related changes; three had preceding MDS (two had received Azacitidine), one had preceding CMML. A complex karyotype was observed in two cases, with monosomy 7 found in a third case. Dysplasia was

seen in ≥ 2 lineages in all cases. The final patient had refractory *FLT3* positive AML. The mean white cell count prior to treatment was $27.3 \times 10^9/l$ (range $2.6 \times 10^9/l$ to $113 \times 10^9/l$) with a mean bone marrow blast percentage of 22.1% (range 15–94%). All had a good performance status (0–1) with no significant co-morbidities. Toxicities included neutropenic sepsis (of four patients, two required high dependency admission), bleeding (three patients including one intracranial bleed), a diffuse follicular rash (three patients) and mucositis (one patient). Five patients (55.6%) developed an isolated fever within 24 h of chemotherapy. All required blood product support (packed red cells: mean 12 units, range 9–16 units; platelets: mean 13 pools, range 10–19 pools). The mean time to neutrophil and platelet recovery was 28 days (range 19–39 days) and 34 days (25–59 days) respectively. Four (44.4%) patients achieved CR or Cri, two patients proceeded to allogeneic stem cell transplant. Three patients have died, two with refractory disease.

Our experience is consistent with a diverse cohort of intensively treated AML patients, and comparable to the 301 Study. Our use of Vyxeos did not raise any unique issues compared with other intensive chemotherapy regimens for AML, in terms of practical delivery or toxicity profile. Given its proven efficacy in this patient group Vyxeos will no doubt remain a useful weapon in our ever-expanding armoury of AML therapies.

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BSH2020-EP-019

High-risk cytogenetic abnormalities in acute myeloid leukaemia – single centre experience

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It is widely established verity that even with the advent of specialized molecular testing, conventional cytogenetic studies have a fundamental role in identifying novel translocations and risk stratification of acute myeloid leukaemia (AML) in under-resourced laboratories with limited molecular testing facilities. In the WHO classification, some of the recurring clonal cytogenetic abnormalities are considered diagnostic of acute leukaemia regardless of blast percentage. Identification of high-risk cytogenetic abnormalities at the time of diagnosis defines the course of disease, choice of management like standard chemotherapy or hemopoietic stem cell transplant and risk of relapse in AML. AML with high-risk cytogenetic abnormalities have poor response to standard chemotherapy and poor survival rate. In Pakistan, lack of technical expertise in the field of cytogenetics and resource limitation does not allow risk stratification of these patients and hence they do not receive treatment tailored to their cytogenetic abnormalities. As a result treatment outcome has been far inferior to that in the developed countries where this facility is available. So far, only two centres existed in the country that performs cytogenetic analysis of only a limited number of these patients as most patients pay out of their pocket. All the remaining patients receive empirical chemotherapy and majority succumb to complications. Ours is the third centre in Pakistan that started cytogenetic analysis in January 2016 to date and since then has performed successful cytogenetic analysis of 124 AML cases. Based on the findings of these analyses, patients were risk stratified and received treatment tailored to their cytogenetic profile including bone marrow transplant. Study was conducted at Cytogenetics Laboratory, Pathology department, Shifa International Hospital, Islamabad. Peripheral blood and bone

marrow aspirate samples were subjected to 24 and 48 hours culture without mitogen, followed by Giemsa-Trypsin banding. Twenty metaphases were analysed in all cases except few cases with low mitotic index. Unsuccessful cultures were not included in the study. Cytogenetic findings were described according to International System for Human Cytogenetics Nomenclature (ISCN 2013). Out of 124 cases of AML, 15 cases were stratified as good-risk group, 69 cases intermediate-risk group whereas 33 cases were categorized as poor prognosis. Seven cases were included intermediate/poor risk groups as few cases were reported according to review of literature. Age range was 11 months to 80 years. Male to female ratio is 2.4:1. High-risk abnormalities included monosomal karyotypes, complex karyotypes, rare novel translocations, trisomies and presence of marker chromosome alone or in addition to other cytogenetic abnormalities. Our observations illustrate significant role of conventional cytogenetics in routine diagnostic work-up of AML to provide framework for risk stratification and early detection of high-risk abnormalities. An ongoing challenge in our country is the integration of pretreatment parameters, including cytogenetics and available molecular markers, with early assessment of treatment response to develop robust algorithms that further refine risk stratification of AML to guide consolidation therapy.

Keywords: AML, high-risk cytogenetic abnormalities, risk stratification

Disclosure of Interest: None Declared.

BSH2020-EP-020

Single hospital retrospective review of patients in treatment-free remission (TFR) following therapy with tyrosine kinase inhibitors (TKIs) for chronic myeloid leukaemia (CML)

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Background: Chronic phase CML (CML-CP) patients who obtain a durable major molecular response (MMR) or better on first and second generation TKIs can achieve long-term survival. TKI discontinuation is now being considered in patients with optimal molecular response and is a potential treatment goal for such patients. We did a retrospective review of patients who discontinued TKIs, if they had sustained molecular response (MR) of 4 (MR4) for at least 3 years on second generation TKIs and 5 years on first generation TKIs.

Objectives: To retrospectively assess the duration of molecular response, relapse and withdrawal syndrome in evaluable TKI discontinued patients in a district general hospital (DGH) over a 2 years period.

Methods: Data were collected from patient notes and health board laboratory portal from patients with CML-CP. Variables assessed included age, sex, lines of treatment, molecular response at time of TKI discontinuation, duration of continuing molecular response following TKI discontinuation, time of relapse and withdrawal syndrome.

Results: There was a total of 15 CML-CP patients of whom 11 were female and four male. Age range was 44 to 85 years with average age of patients being 69 years. Out of 15 patients, two patients with advanced age and associated co-morbidities received modified once daily dose of Nilotinib 300 mg. Nine patients fulfilled the TFR eligibility and were included in our retrospective analysis. An additional patient who achieved MR4.5 followed by continuous MR of 24 months on first-line standard dose Nilotinib was also included in our review after discontinuing it for pancreatitis. Among these 10 patients, seven received first line, one patient received second line

and two patients received third line TKIs. All patients receiving second- and third-line TKIs were primarily due to intolerance and not resistance. Of these, seven had discontinued TKIs, while one patient on modified once daily Nilotinib refused to stop and two patients were considering to do so. One patient who had discontinued TKI subsequently died due to relapsed Hodgkin's lymphoma with her last response being MR4. Both elderly patients on once daily Nilotinib had deep MR though one has died, cause of which was unrelated to CML. Average duration of MR was 10 months (range 2–26) with no relapse. Four of seven patients developed symptoms of withdrawal syndrome with one patient requiring re-institution of TKIs.

Conclusion: Our patient numbers are small as we are a small DGH. All our patients maintained MMR or better off TKIs. A significant proportion of our patients developed withdrawal syndrome. Elderly patients with co-morbidities receiving modified reduced doses could achieve and maintain optimal molecular response including deep MR. In our hospital, practice of TKI discontinuation in patients with optimal MR was adopted late in view of a single patient report we published in 2016. We had reported (BSH abstract number 520–2016) a rare occurrence in a patient with continuing MMR of over 5 years on TKIs who had sudden transformation to Philadelphia positive acute lymphoblastic leukaemia (ALL). A recent correspondence to *British Journal of Haematology (BJH)* also reported of a single patient developing sudden blastic lymphoid transformation in TFR CML (BJH 2019, 187, 543–545). Although in CML, TFR is a potential treatment goal in majority who achieve optimal MR, rare instances of blastic transformation in such patients is a concern and the trigger for this event remains unknown.

Disclosure of Interest: None Declared.

BSH2020-EP-021

Avascular necrosis diagnosed in young adults patients who had treatment for acute lymphoblastic leukaemia

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Abstract: Avascular necrosis of the bones has been diagnosed in patients treated for acute lymphoblastic leukaemia and had a high dose of steroids as part of the chemotherapy protocol. Avascular necrosis can be a debilitating condition and if progressing to severe stages it will require surgical treatment (arthroplasty).

Methods: Retrospective observational study of a number of 16 young adult patients who have been treated with UKALL 2011 trial, 10 of them had standard dexamethasone (R1) and six had short dexamethasone course (R1). Patient's notes reviewed including results from MRI, CT and x-rays. Patient's demographics, co-morbidities, calcium, vitamin D3 level and the time since chemotherapy started and avascular necrosis diagnosed were also considered.

Results: From 16 patients investigated, six of them developed avascular necrosis with one of them requiring arthroplasty of the shoulder and hip.

Conclusion: Young adult patients diagnosed and treated for acute lymphoblastic leukaemia and developing steroid-induced avascular necrosis have a reduced quality of life due to pain and disability. There are no current guidelines or randomised controlled trials to advocate medical treatment with bisphosphonates, calcium and vitamin D3 replacement in early stages of avascular necrosis, but research articles published by orthopaedic specialists have positive results in treating avascular necrosis with bisphosphonates.

Disclosure of Interest: None Declared.

BSH2020-EP-022

Single centre experience in use of leukapheresis for management of leucostasis in high count leukaemia**N. Melamed^{1,*}, R. Valeza², M. Mayhew², J. Ortizo², A. Jumah², A. Reed², J. Chua², F. Willis²**¹Medical School, St George's, University of London,²Haematology, St. George's Hospital NHS Foundation Trust, London, UK

Leucostasis is an oncological emergency defined by a white blood cell (WBC) count of $\geq 50\text{--}100 \times 10^9/\text{l}$ in the presence of clinical signs and symptoms of decreased tissue perfusion. Rapid depletion of the WBC count via leukapheresis is often the treatment of choice used to prevent a fatal course of the disease in leukaemic patients with high counts. However, there is contrasting evidence in the current literature over its successes and pitfalls and little available that compares its efficacy between the leukaemia. Since 2009, St. George's Hospital in London has been using leukapheresis to treat select patients with symptoms of leucostasis. This retrospective study reviews the outcomes of patients treated with this therapeutic approach at this tertiary centre.

Data were collected from the apheresis database, electronic patient records and PowerChart. All adult patients treated with leukapheresis between 2009 and 2019 were selected for this study.

Twenty-four adult patients were included in the study, 12 with acute myeloid leukaemia (AML), eight with chronic myeloid leukaemia (CML) and four with acute lymphoblastic leukaemia (ALL). In the AML group, the median age was 54 years (range 18–75); the pre-apheresis average white cell count (WCC) was $248 \times 10^9/\text{l}$ (range 122 to $480 \times 10^9/\text{l}$). Of these patients, three had secondary AML and four had documented high-risk disease. The AML patients presented with the greatest number of complications including tumour lysis syndrome (3), type 1 respiratory failure (6), acute

kidney injury (4), neurological compromise (4) and 10 patients required intensive care (ITU) support. Survival was poorest in this AML group with six patients dying during induction treatment predominantly due to complications of their advanced disease. In the CML group, the median age was 39 years (range 27–76); the pre-apheresis average WCC was $411.1 \times 10^9/\text{l}$ (234.9 to $739.9 \times 10^9/\text{l}$). Two patients died, one within 24 h of fulminant disseminated intravascular coagulation (DIC) and multiorgan failure. The second died of relapse post-stem cell transplant 2 years later. Both of these patients presented in blast crisis. In the ALL group, the median age was 22 years (range 18–67); the pre-apheresis average WCC was $279.9 \times 10^9/\text{l}$ (range 110.9 to $407.7 \times 10^9/\text{l}$). All ALL patients achieved complete remission with induction chemotherapy; however two patients relapsed and died post-stem cell transplant. Both these patients had high-risk cytogenetics.

With leukapheresis, the best reduction in WCC was seen in the AML group (45.5%). This is partly due to the good separation of the buffy coat from the red cell and plasma layers in the apheresis procedure. The procedure was well tolerated in all patients, regardless of diagnosis, except one patient who presented with advanced multiorgan compromise and was unable to tolerate the procedure due to cardiac compromise. The study confirms that leukapheresis is potentially a life-saving procedure in patients with leucostasis, when used wisely and in conjunction with appropriate disease-targeted chemotherapy treatment. These results demonstrate that in patients with leucostasis due to high count leukaemia, the procedure is well tolerated. Reduction in WCC is most efficient in the AML group. Poor outcomes were associated with advanced disease and diseases associated with poor prognosis.

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Education

BSH2020-EP-023

The feasibility and impact of teaching haematology registrars via a web-based videoconferencing system

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Background: The Barts Haematology has 25 haematology Registrars who are split over six hospitals in central and east London. Trainees used to commute between base hospitals and teaching hospitals to attend teaching sessions. We wanted to improve access to teaching and reduce travelling by sharing sessions online via GoToMeeting. We upgraded the multi-header microscope with a new digital camera which was then connected to a PC via USB and the camera feed was shared online via GoToMeeting. Videos were also recorded and stored on the cloud for review later. Power Point slides could also be shared via GoToMeeting.

Methods: Attendance and student/teacher interaction were recorded for 26 teaching sessions and a survey was undertaken to get feedback from Registrars.

Results: Remote teaching is feasible and sessions were most often attended via Smart Phone or Web browser. On average 50% of attendance was from remote users with more users choosing remote attendance over time. The average number of interactions per session was 3.6 from local users and 1.4 from remote users.

About 47% of registrars reported that they attended teaching remotely weekly or more frequently and 59% reported they used recorded video sessions.

Conclusion: Delivering teaching via videoconferencing systems is feasible. The system is popular among trainees. Overall teaching attendance increased with the intervention, with a trend to more trainees choosing to attend remotely over time. However, remote teaching probably requires a different teaching style and raises new questions about how to best facilitate interaction at distance

Disclosure of Interest: T. Erbllich Conflict with: Clinical Research Fellow, S. Hibbs Conflict with: Registrar, S. Araf Conflict with: Registrar, T. Butler: None Declared, L. Bowles: None Declared.

BSH2020-EP-024

A toolkit for undertaking a volunteer OOPPE visit to a low resource country

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For my OOPPE (Out Of Programme Experience) I spent a year volunteering at Queen Elizabeth Central Hospital (QECH) in Malawi assisting with the development and provision of Haematological services at the hospital.

QECH is a 1000-bed tertiary referral hospital for Southern Malawi serving a population of more than 7 million people. Malawi is one of the poorest countries in the world with over 50% living below the poverty line.

My experience in Malawi has been an eye-opener to the enormous need and opportunities for professionals working within Haematology to support low-income countries in developing their clinical and laboratory haematology services.

Arranging such a visit, particularly for those without experience of visiting countries such as Malawi, can be very challenging and daunting.

This poster will include a checklist/toolkit of aspects which need to be considered in advance, and during, a visit to a low-income country. This resource aims to assist and support others within our society when arranging visits to low-income countries.

The checklist will include:

UK aspects: getting approval and support for an OOPPE/Visit

Finding a suitable country and hospital: how to find a centre, visa applications, medical council registration, travel permits, making links with specialists in the country/at the centre

Financial considerations: securing funds to support living costs and other expenses

Travel and accommodation

Health considerations: travel insurance, dealing with illness, vaccinations and infection prophylaxis

Developing attributes required to work in low-income country: tropical medicine experience, language skills, cultural experience.

The opportunity to learn from my experience will hopefully aid others considering similar projects and lead to our society using our collective expertise to benefit low-income countries/settings. The benefits for those undertaking the visits must also not be understated.

Disclosure of Interest: None Declared.

BSH2020-EP-025

A little learning is a dangerous thing – how little do we know about when to treat pulmonary embolism?

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It's subjunctive history. You know, the subjunctive? The mood used when something may or may not have happened. When it is imagined. (Alan Bennett, The History Boys)

The General Medical Council (GMC) guide good medical practice states, 'You must keep your professional knowledge and skills up to date'. Guidelines from the National Institute of Clinical Excellence (NICE) help doctors to provide care according to the best available evidence without sifting through it. However there is a risk that, by focusing only on guidelines and neglecting individual trials, doctors overestimate the strength of evidence on which their practice is based. This in turn prevents them asking important, unanswered questions with an open mind.

Considering pulmonary embolism (PE), NICE guidelines (2015) recommend 'offer... low molecular weight heparin (LMWH) or fondaparinux'. The advent of direct oral anticoagulant drugs (DOACs) has increased the anticoagulation on offer, but the recommendation to offer anticoagulant therapy to all patients with PE is still based on a single randomised trial (Barritt & Jordan, 1960). This recruited 35 patients before the control arm was closed, and used no imaging beyond chest radiographs. There was no blinding or placebo treatment. Recruitment based on clinical signs for pulmonary infarction likely included patients with infectious pneumonia as well as PE.

This trial was preceded by the haphazard introduction, in the 1930s, of anticoagulation with heparin to treat deep vein thrombosis (DVT), according to physicians' preferences. The observed mortality drop in patients with DVT from 18% to 0.4% was generally accepted as compelling evidence. However bedside diagnosis of PE is infamously unreliable, and some physicians withheld heparin due to bleeding risk prior to 1960.

The introduction of ventilation-perfusion (V/Q) scintigraphy in the 1960s made diagnosis of PE more reliable, but it is unreliable for detecting small emboli. CT pulmonary angiography (CTPA) was introduced in the early 1990s, and its widespread adoption was associated, in the USA, with an 81% increase in reported PE incidence, but only a very small decrease in mortality. A randomised trial comparing CTPA with V/Q scanning found an increased rate of PE with CTPA, but no corresponding decrease in recurrent venous thromboembolism during the 3 months follow-up period. (Anderson *et al.* 2007) These data suggest that the additional, smaller, emboli diagnosed by the more sensitive technology may not be clinically significant, and may not need treatment. Unfortunately, following the advent of CTPA, randomised trials re-testing anticoagulation against placebo are still awaited.

Anticoagulation therapy is associated with significant bleeding risk. In the 1960s trial, one of two deaths in the heparin arm related to a haemorrhagic peptic ulcer, and bleeding affected 27% of heparin patients in the urokinase pulmonary embolism trial (1970). The risk may well outweigh any benefit in patients with subsegmental and asymptomatic PE, but this is unknown. By understanding the history of a treatment and its evolving indications over time, shortcomings in the trial data can become visible. This may not alter clinical practice immediately, but it informs trial design and recruitment to address the critical unknowns in a field, rather than merely seeking to identify a market niche for the latest pharmaceutical variant.

Disclosure of Interest: None Declared.

BSH2020-EP-026

Infection control and prevention-beyond horizons – analysis of practices at newly established hospital in Pakistan

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Establishing hospital in a community is blessing for community members, but nothing great comes easy. One of the main downside of a hospital is health care-associated infections. Patients go to hospital to seek a cure and relief from their suffering but that same place can give them infection, hence may contribute to their misery. Health care-associated infections are threat to health care systems despite many advances. These infections not only impact death rate but also increase the economic burden. Practices for infection control and prevention are imminent to overcome these problems. This cross-sectional study was conducted over a period of 1 year and was segregated into two phases. First phase of the study comprised initial 6 months and second phase for later 6 months of hospital establishment. Out of 1151 cultures, 681 were collected in first phase of study and infection control team was assigned and sessions for staff and other health professionals were conducted to improve the infection control and prevention practices at the hospital. Then, in second phase of study remaining 470 cultures were collected and results were compared to determine the impact of the measures taken to improve the infection control. Our study revealed that the number of organisms isolated in initial phase were more than the later phase but in later phase organisms depicted significant resistance to antimicrobials.

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General Haematology

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Real-world data from the United Kingdom on the sequencing of second generation tyrosine kinase inhibitors for the treatment of chronic phase Philadelphia chromosome-positive chronic myeloid leukaemia

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For patients resistant or intolerant to tyrosine kinase inhibitors (TKIs), a practice of sequencing through different available drugs is

commonly used in the clinical management of chronic myeloid leukaemia (CML) patients. In the United Kingdom (UK), this practice has been affected by the reimbursement status of the drug. Currently, there are few clinical studies which discuss the real-world clinical benefits of TKI sequencing. Therefore, this study aimed to determine the outcomes of routine, real-world sequencing of second generation (2G) TKIs in patients who received at least three or more lines of TKIs, with particular reference to the number of patients in major molecular response (MMR) in lines 3 and 4.

A multicentre retrospective UK chart note review was executed between 1 November 2018 and 31 July 2019. Patients needed to be aged ≥ 18 , with chronic phase, Philadelphia chromosome-positive CML and have had at least 3 TKIs, with the third started between 1 June 2013 and 28 February 2018. Patients were followed up until study end. Exclusion criteria were a >3 months gap in treatment before progression or relapse, or receipt of a 2G TKI on an interventional clinical trial in third line. Molecular responses (BCR-ABL International Scale values, or unconverted BCR-ABL/ABL ratios, if the prior was unavailable), duration of therapy and reasons for stopping were recorded up to the last hospital follow-up (LHF) or death. Analyses were descriptive in nature.

In total 80 patients were included from 14 UK sites (Table). Eleven patients received 12 transplants in total as one patient received a second transplant in line 4. Four of 11 allogeneic transplant recipients died of disease/transplant-related causes.

When sequencing TKIs, the most common TKI in first line was imatinib (83.8% of prescribing), and in second line was nilotinib (63.8%). This reflected NICE guidance at the time. During the study period 29 unique TKI sequences were seen, the most common of which proceeded sequentially with imatinib, nilotinib then dasatinib (19 patients, 23.8%).

In total 55 patients (68.8%) were in MMR in line 3; of these, 25 patients (31.3%) had maintained MMR from line 2. Twenty-three patients (28.8%) achieved MMR for the first time during line 3 and 22 patients (27.5%; 78.6% of line 4) were in MMR in line 4; of these, 14 (17.5%; 50.0%) had maintained MMR from line 3 and five

Table. Study demographics and patient destinations.

Field	Overall (N = 80)	Line 2 (n = 80)	Line 3 (n = 80)	Line 4 (n = 28)	Line 5 (n = 2)
Median age, years (range)	51.1 (18–89)	–	–	–	–
Male/female, n (%)	42 (52.5%)/38 (47.5%)	–	–	–	–
High-risk sokal, n (%) of 80; % of 33, the number of scores recorded)	9 (11.3%; 27.3%)	–	–	–	–
Median time from line 1 to line 3 initiation, months (range)	33.7 (3.6–145.3)	–	–	–	–
Median duration from line 3 to death or LHF, months (range)	28.0 (9.3–63.7)	–	–	–	–
Stem cell/bone marrow transplants, n (%)	11 (13.8%)	2 (2.5%)	4 (5.0%)	5 (17.9%)	1 (50.0%)
Deaths/CML related, n (%)	10 (12.5%)/7 (8.8%)	–	6 (7.5%)/4 (5.0%)	4 (14.3%)/3 (10.7%)	0 (0.0%)
Ongoing therapy at study end, n (%)	56 (70.0%)	–	36 (45.0%)	19 (67.9%)	1 (50.0%)
Lost to follow-up at study end, n (%)	8 (10.0%)	–	7 (8.8%)	1 (3.6%)	0 (0.0%)

patients achieved their first MMR in line 4 (6.3%; 17.9%). Only 14 patients failed to achieve MMR in any line (17.5%).

The commonest reason for treatment switch in line 1 was lack of efficacy, and in lines 2 to 4 it was intolerance. The main reason for switch from imatinib and bosutinib was lack of efficacy, and from nilotinib, dasatinib and ponatinib was intolerance.

During this study, TKI availability changed from solely imatinib to later access to imatinib, nilotinib, dasatinib, bosutinib and ponatinib in first or later lines. Over time, with the reimbursement of each TKI, median duration of therapy (mDoT) from line 1 initiation to line 3 reduced. When only imatinib was available, mDoT was 110.1 months; when all TKIs were available mDoT reduced to 3.6 months.

Sequencing of 2G TKIs led to an increased number of patients who achieved MMR by line 3; however, some patients never achieved MMR. A number of patients died following line 3 and several patients received transplant with disease/transplant-related mortality. These patients may benefit from earlier identification for the optimal timing of treatment switch and sequence personalisation.

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Bosutinib real-world data research study: a single centre, retrospective, non-interventional, cohort study analysis of pre-treated chronic myeloid leukaemia patients

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Bosutinib is a second-generation tyrosine kinase inhibitor (TKI) indicated for adult patients with Philadelphia chromosome-positive (Ph+), chronic myelogenous leukaemia (CML). Recent second-line, and later line, clinical trials have shown consistent efficacy and safety profiles with bosutinib. The objective of this study was to provide real-world experience of bosutinib efficacy and safety, in previously treated patients with CML.

This single centre, retrospective, non-interventional, cohort study was undertaken at the Beatson West of Scotland Cancer Centre. Data were gathered from the medical records of patients prescribed bosutinib for the treatment of Ph+ CML. Study protocol was approved by the Greater Glasgow and Clyde Health Board.

Patients with diagnosed Ph+ CML > 18 years old receiving bosutinib, prescribed either during routine clinical practice or via a

compassionate use program, were included in this study. Patients prescribed bosutinib as part of an interventional clinical trial initiated on bosutinib less than 3-month prior to data collection and patients prescribed bosutinib as exclusively post-allograft therapy were excluded.

A total of 17 patients were included in this study. Median age at bosutinib initiation was 57.5 years (range 20.6–82.0). Median time from first TKI to bosutinib initiation was 1.4 years (range 0.2–10.5). Intolerance ($n = 8$), resistance ($n = 5$), and both intolerance and resistance ($n = 4$) to prior TKI therapies were the reasons for switching to bosutinib. Most patients received bosutinib in the second ($n = 8$) or third line ($n = 5$) settings. A large proportion of patients suffered concomitant co-morbidities at diagnosis of CML (8/17), with six of 17 patients suffering three or more co-morbidities simultaneously. The most common co-morbidities were: vascular (hypertension [$n = 5$], ischaemic heart disease [$n = 3$], peripheral vascular disease [$n = 2$], cerebrovascular disease [$n = 1$]), chronic pulmonary ($n = 4$), cardiac ($n = 2$) and endocrine ($n = 2$) disease.

The proportion of patients with best cumulative partial haematological response, complete haematological response, partial cytogenetic response, complete cytogenetic response, major molecular response, molecular response 4.0 and 4.5 were: 76%, 76%, 65%, 59%, 41%, 24% and 12% respectively. Patient follow-up was: all of 17 at year 1 (59% continued bosutinib [$n = 10$]), 14 of 17 at year 2 (43% continued bosutinib [$n = 6$]) and 12 of 17 at year 3 (42% continued bosutinib [$n = 5$]). Overall median duration of therapy during data collection was 19.0 months (continued bosutinib median = 44.6 months; discontinued bosutinib median = 3.9 months). One death occurred during this period which was not related to bosutinib. No observed patients progressed to accelerated phase or blast phase disease.

A total of 15 AEs were reported by 11 of 17 patients. Respiratory AEs were most common (3 of 11). Grade 3/4 AEs included: respiratory ($n = 4$), lipase increase and lymphatic disorders ($n = 2$), alanine transaminase increase ($n = 1$), hepatotoxicity ($n = 1$) and thrombocytopenia ($n = 1$). It was not possible to attribute the discontinuation of bosutinib directly to treatment-associated AEs from the data.

This real-world study revealed that bosutinib, implemented predominantly in second and third line therapy, yields cytogenetic and major molecular responses consistent with those seen in clinical trials. Bosutinib was generally well tolerated, with most patients not suffering a grade 3/4 AE. However, common AE rates were inconsistent with previous clinical trials.

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Cardiovascular risk factors and events in a district general haematology population undergoing cytotoxic chemotherapy

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The cardiovascular (CV) impact of cytotoxic chemotherapy is under closer scrutiny with the advent of new treatment modalities. The European Society of Cardiology (ESC) has defined baseline risk

factors for cardiotoxicity. Within our district general haematology population the burden of these risk factors is unknown.

We undertook a retrospective analysis of patients commenced on chemotherapy within the last year. At baseline we assessed an individual's CV risk factors and followed them for CV events.

A total of 107 consecutive cases were analysed, seven (6.5%) were excluded due to paucity of data, final sample $n = 100$. Median follow-up was 219 (31–339) days. Median age was 73 (61–81), 60% male and 85% Caucasian.

Myeloma was the most common malignancy (45%), 37% had lymphoma, 17% chronic leukaemia and 1% acute leukaemia. Total mortality was 15%; most common cause was infection (53%), with one CV death due to heart failure (HF) from cardiac amyloidosis.

The burden of CV risk factors was: 50% hypertension, 45% hypercholesterolaemia, 28% diabetes, 43% ever-smoker and 8% excess alcohol intake. Pre-existing CV disease was present in 35% with 16% having coronary artery disease, 12% atrial fibrillation, 1% ventricular tachycardia and 7% HF. On average patients had a cumulative risk factor burden of $2 (\pm 1.8)$. Age was significantly different ($P = 0.03$) between malignancy groups, with lymphoma patients being younger than myeloma and leukaemia patients. There were no significant differences between malignancy groups for the burden of CV risk factors or current CV disease.

During the follow-up period seven patients were noted to have CV complications. The majority $n = 5$ (71.4%) occurred in patients with myeloma, with one in lymphoma and one in chronic leukaemia. There were three cases of HF, one acute coronary syndrome, two new onset atrial fibrillation and two patients were investigated for new onset chest pain. CV complications required hospital admission in four (57.1%), interrupted chemotherapy in three (42.9%) and changed chemotherapy in three (42.9%) patients. Those with CV complications had a significantly higher ($P = 0.01$) cumulative risk factor burden of 3–9.

In summary, our DGH haematology caseload has a moderate burden of CV risk factors and current CV disease. There was no difference in the burden of these between different malignancy groups. CV complications during follow-up were uncommon but when occurring, had a significant impact: hospital admission was common and treatment was usually changed. Those with CV complications had a significantly higher burden of risk factors at baseline. The authors therefore recommend increased vigilance and early specialist referral in those at high baseline risk.

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Review of 5 years experience of intravenous iron in pregnancy

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Between September 2014 and October 2019 367, iron-deficient pregnant ladies were treated with intravenous iron isomaltoside 1000. Mean haemoglobin level was 90 g/l with a range from 62 to 120 g/l. ($Hb = >10$ –120 14%, $=80$ –<100 71%, <80 15%). 91% had Ferritin levels of <30 mcg/l, 7% 30–100 μ mcg/l. Iron isomaltoside 1000 was given at a dose of 20 mg/kg based on the early booking weight of the woman. Mean dose was 1290 mg with a range of 500–1800 mg. 85% of patients had >1000 mg. Follow-up was at variable times post-infusion. Post-infusion mean Hb was 110 g/l with a range of 83–143 g/l (>120 g/l 18%, 100 –<120 61%, <100 18%). Only one patient required a blood transfusion after 1800 ml obstetric haemorrhage. Iron isomaltoside 1000 was largely well tolerated. There were

four severe Fishbane reactions and two patients went on to complete the infusion at a reduced rate. There was one mild Fishbane reaction where the infusion was completed after a short pause. Intravenous iron is an effective and safe treatment for iron deficiency in pregnancy. Intolerance and non-compliance with oral iron is common. Our experience is that many women, when found to be anaemic, are commenced on high doses of oral iron which they do not tolerate, then go on to be referred after 34 weeks to our service. Earlier referral for intravenous iron would be beneficial as would oral replacement with doses of iron that are better tolerated.

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Updated audit of the use of pegylated interferon in patients with myeloproliferative neoplasms. Are patients maintained on less frequent dosing than once weekly?

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Pegylated interferon (peginterferon) is commonly used as a cytoreductive agent for myeloproliferative neoplasms (MPN); however, this indication is unlicensed. An audit was originally undertaken in November 2018 to establish:

- Indication for use (including diagnosis and mutational status) and whether use was in line with commissioning policy
- Starting dose
- Discontinuation rate and reason for stopping

Anecdotally at NUH, pegylated interferon has been administered less frequently than once weekly. A secondary aim was to establish the proportion of patients dosed less frequently than once weekly. The original dataset was expanded in December 2019.

A single-centre, retrospective analysis of case notes, for adult patients who received peginterferon was undertaken. Patients were identified through dispensing records via the pharmacy stock management system. The original dataset captured patients between 17 February and 18 November and a report was later repeated to include patients initiated on peginterferon between 18 November and 19 December. Patients were excluded if they did not have a diagnosis of MPN.

About 33 patients (age range 21–87 years) were identified as having received at least one prescription for peginterferon between 17 February and 19 December. Of these, 55% (18) of patients were diagnosed with essential thrombocythaemia (ET), 36% (12) polycythaemia vera (PV) with the remaining patients either MPN – unclassified, ET/PV overlap or post-ET myelofibrosis. 92% of PV patients carried the JAK-2 V617F mutation (1 not documented). 39% of ET patients were CAL-R mutated and 39% JAK-2 V617F mutated.

Peginterferon was initiated in 45% (15) of patients following intolerance to standard short-acting interferon. A further four patients (12%) were predicted to not tolerate standard interferon. Three patients (9%) were started on peginterferon due to the withdrawal of standard interferon. For the remaining 33%, reasons for initiation were intolerance/treatment failure associated with other therapy or was not documented. 36% (12) patients discontinued peginterferon. The most common reasons for discontinuation

were toxicity relating to thyroid function ($n = 3$) or diagnosis of a second malignancy requiring chemotherapy ($n = 3$).

Most patients (67%) were initiated on a starting dose of 90 µg weekly, with 18%, 6% and 9% started on 45, 135 and 180 µg once weekly respectively. Once stabilised, nearly half (45%) of patients required less frequent dosing, with a minimum interval of fortnightly injections (range 2–6 weeks).

Historically, peginterferon has been reserved for patients intolerant to standard interferon-alpha due to the lack of comparative data between peginterferon and standard therapy. However, recent phase 3 data have shown peginterferon to be equivalent to hydroxycarbamide. In addition, since the original analysis was started, Roche Products Limited has discontinued the manufacture of Roferon A. Nearly 50% of patients required dosing every 2 weeks or less. Based on NUH prices, compared to a standard interferon-alpha dose of 3 million units three times a week, peginterferon therapy at this reduced frequency is cheaper. Less frequent administration of peginterferon may mitigate the cost of introducing this agent first line, where hydroxycarbamide is not indicated.

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3. Disclosure of Interest
3. C. Loughran Conflict with: Novartis, F. Wadelin: None Declared, R. Fiskien: None Declared, A. Thompson: None Declared.

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Fluid replacement in therapeutic plasma exchange: survey report

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For some conditions such as thrombotic thrombocytopenia purpura there is clear guidance to use plasma (Octoplas) as the exclusive replacement fluid for plasma exchange (PEX). There is less guidance for other conditions with informal discussions suggesting significant variation in practice. Human albumin solution (HAS) is commonly used as the replacement fluid but some of the fluid may be exchanged by 0.9% sodium chloride and/or gelofusine and/or plasma (particularly to correct deranged clotting or manage/reduce the risk of haemorrhage).

There is an absence of clear guidance on the recommended replacement fluid used for many of the indications for therapeutic plasma exchange. This survey assesses current practice in the use of replacement fluid in therapeutic plasma exchange within the United Kingdom and Republic of Ireland. Variation in practice will provide the parameters to inform the basis for a clinical trial to determine best practice for replacement fluid use in therapeutic plasma exchange.

Since there is no definitive list of apheresis providers in the United Kingdom, the main apheresis providers in England were identified using intelligence from NHSBT (National Health Service Blood and Transplant) and apheresis machine manufacturers. The main renal providers were identified through the Renal Association database. The Scottish Blood service and The Irish Blood Transfusion Service were included in the survey.

A telephone/email-based survey was distributed to all identified providers and the haematology and renal units that perform plasma exchange treatments

The survey facilitated by the NHSBT audit department was composed of 15 questions to cover technology used, type of patients treated, and specialties covered and standard exchange fluid used.

Sixty contacts were made to renal and haematology/apheresis units with 27 units completing the survey (response rate 45%). 78% of responding units use centrifugal technology with 22% using filtration. 25.9% of units perform more than 500 treatments a year; 11.5% between 350 and 500; 14.8% between 200 and 300; 18.5% between 100 and 200; and 29.6% performing less than 100 plasma exchange treatments a year.

Representing the group of patients treated, 59% of the units treat adults and children and 41% treat adults only. None of the units treat only children. 44% of the units/departments treat patients from renal, neurology and haematology; 19% treat neurology and haematology patients; 15% treat haematology patients only; 11% treat renal and neurology patients; 7% treat renal patients; and 4% treat renal and haematology patients only.

Albumin 4.5% or 5% is used as standard fluid replacement for 65.5% of the units enquired. 17.2% use a combination of albumin and NaCl 0.9%. A combination of albumin and gelofusine is referred by 13.8% and 3.5% use a combination of albumin, NaCl 0.9% and gelofusine. 67% reported changes in approach under certain patient settings, that is, need for biopsy.

Albumin 4.5% or 5% is the most common standard fluid replacement used. Alternatives are used as standard and there are currently insufficient data to assess inferiority of these approaches. Deviation from standard replacement fluid depending on patient diagnosis or/and conditions is common. 76% of units were interested in taking part of a clinical trial of replacement fluid.

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BSH2020-EP-033

Environmental and occupational determinants of myelodysplastic syndromes: exposure, association and risk: a case-control study from Pakistan

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Myelodysplastic syndromes (MDS) are a heterogeneous group of haematopoietic stem cell disorders characterised by ineffective dysplastic haematopoiesis involving one or more cell lineages. Despite several large epidemiological studies and methodologically sound in vitro data, there remains little evidence for a significant contribution of environmental/occupational exposure to MDS risk. In Pakistan and the surrounding regions, patients do not have access to prompt diagnostic and treatment facilities. Moreover, the characteristics of many patients, such as a low literacy rate, exposure to hazardous chemicals, inadequate access to milk and water, and habitual narcotics use, make them more vulnerable to diseases like MDS. We conducted a case-control study that evaluated environmental and occupational determinants as risk factors for MDS patients. To the best of our knowledge, this is the first study conducted in Pakistan in this context. This was a case-control study conducted from January 2010 to August 2018. The participants were 153 de novo MDS cases and 447 age- and sex-matched controls without a history of cancer during the same time period as the cases. For each case of MDS three controls were selected. This study was approved by the institutional review board and informed consent was obtained from

all participants prior to the collection of epidemiological data. The Statistical Package for the Social Sciences Software was used for analysis. Qualitative variables were presented in terms of frequency and percentages. Chi-squared test was used to determine the association between categorical variables. Furthermore, binary logistic regression was used to determine odds ratios (OR) for significant variables. Univariate logistic regression analyses were used, and OR with 95% confidence intervals were reported to determine the association between exposures and disease occurrence (MDS). Among the study participants ($n = 600$), there was a significant association between cases and controls regarding marital status, type of job, educational status, water source and arsenic ($P < 0.0001$). There was also significant difference found in study group with language ($P = 0.037$) and benzene ($P = 0.021$). The association with residency in rural and urban areas and milk source was not found significant. We concluded that people in developing countries such as Pakistan could be at higher risk than those in industrialized countries of getting diseases like MDS owing to the unrecognised exposure to perilous environmental and occupational factors. This study will be helpful to understand the aetiology of MDS in the Pakistani population.

Disclosure of Interest: None Declared.

BSH2020-EP-034

Autoimmune myelofibrosis associated with Sjogren's syndrome successfully treated with steroids and improved myelofibrosis

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Myelofibrosis is a disease in which bone marrow fibrosis occurs extensively due to the proliferation of reticulum fibres, collagen fibres and myeloblast fibroblasts. It is roughly classified into primary myelofibrosis (PMF) of unknown cause and secondary myelofibrosis secondary to other diseases. Among them, myelofibrosis secondary to autoimmune abnormalities is called autoimmune myelofibrosis (AIMF). In particular, there have been only five reports of AIMF associated with Sjogren's syndrome, which is extremely rare.

We herein report a case of a 49-year-old man who complained of intraoral bleeding and petechiae of the extremities and subsequently was diagnosed with AIMF. His platelet count dropped significantly to 1000/ μ l. Bone marrow aspiration revealed a dry tap, and bone marrow biopsy confirmed infiltration of small lymphocytes and an increase in reticular fibers. JAK2 V617 gene, MPL gene and CALR gene mutations were all negative. Since he had a history of Raynaud phenomenon, collagen disease was suspected. We diagnosed him with Sjogren's syndrome because examination for anti-SS-A antibody, Schirmer's test and fluorescein staining test were all positive. In addition to the bone marrow findings, a diagnosis of autoimmune myelofibrosis was made. Treatment with steroids improved the platelet count immediately.

AIMF is a rare condition that can be diagnosed by bone marrow biopsy and is difficult to distinguish from primary myelofibrosis. Autoimmune myelofibrosis is a disease with a good prognosis for which steroids are highly effective, and it is necessary to distinguish it from primary myelofibrosis. Differentiating AIMF from PMF is important; however, the pathological features of PMF and AIMF overlap in the retrospective diagnosis of AIMF patients, and it is often impossible to meet all diagnostic criteria. In cases where only thrombocytopenia is remarkable, as in this case, it is difficult to differentiate from ITP. In fact, one case that was initially misdiagnosed as ITP has been reported. Since both steroids are the mainstay of

treatment for both ITP and AIMF, it is presumed that some patients have been misdiagnosed and unnoticed. Therefore, immunological search and bone marrow biopsy are very important for accurate diagnosis.

Disclosure of Interest: None Declared.

BSH2020-EP-035

A review of first trimester anaemia

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In total 1715 pregnancies were reviewed during the 5 months of November and December 2018, March, April and May 2019. We looked at the estimation of haemoglobin (Hb) for those booking prior to 13 weeks gestation. Our population has a very low carriage of haemoglobinopathy.

About 148 (8.6%) women had Hb concentrations <120 g/l with 25 (1.5%) <110 g/l. Median Hb was 132 g/l; minimum 90 g/l, maximum of 160 g/l. The 95% lower limit confidence level was 116 g/l.

Guidance suggests that Hb values >110 g/l are adequate in the first trimester. Our data show the lower limit of normal in our cohort was 116 g/l.

We further assessed the outcome in pregnancies in November, December and March in those pregnancies where the booking Hb at less than 13 weeks gestation was <120 g/l. Of 1001, 90 women during these months had Hb <120 g/l giving a 9% incidence of anaemia. Of 81 evaluable cases Hb fell from booking to 28 weeks gestation by a median of 8 g/l (range +39 to -27 g/l) with 33 (41%) dropping by 10 g/l or more. No data on iron supplementation were collected.

Of the women with Hb <120 g/l the average MCV and MCH were 87.5 fl and 29 pg respectively. Of these 17% had an MCV <80 fl and 27% had an MCH <28 pg. Most therefore had normal red cell indices.

In the 3 months assessed, only 16 women (18%) had their serum ferritin (SF) checked with a median value of 6.5 mcg/l (range 3–45). Thirteen of 16 women had SF <30 mcg/l. Although only a few cases, we saw an average fall in Hb of 3.5 g/l (median rise of 2 g/l, range -22 g/l to +15 g/l). One assumes that in those pregnancies found to have a low SF, iron supplementation was given. Of note the SF was only requested at booking if the MCH was found to be <27 pg (in all but one patient who had a normal MCH) as part of the National Sickle cell and Thalassaemia screening programme.

Only four pregnancies were associated with blood transfusions. While we cannot show that low Hb at booking predicts for a transfusion requirement we can show that the incidence of anaemia in our first trimester population is around 9%. We suggest that Hb values <120 g/l in the first trimester are not physiologically acceptable. For our cohort there is a fall in Hb between booking and 28 weeks of 8 g/l which would generally be the accepted norm. The fall is less in the small number of cases that were shown to have low SF and therefore likely treated.

Our study suggests that for all women at booking with Hb less than 120 g/l we should offer low dose iron supplementation in accordance with the BSH guideline. If the SF is <30 mcg/l, irrespective of the Hb, iron should also be offered. Using a low MCV or MCH to decide if SF testing is required seems to be wholly inappropriate since both MCV and MCH are in the majority normal in pregnant women at booking even with iron deficiency.

Implementation plan:

- 1 All women at booking require FBC and SF testing.

- All women who do not have thalassaemia with Hb <120 g/l should be offered iron replacement therapy (ferrous sulphate) at 100 mg/day.
- All women not anaemic (irrespective of Thalassaemia) with SF <30 mcg/l should be offered iron replacement therapy (ferrous sulphate) at 100 mg/day
- Low red cell indices should not be relied upon as surrogate indicative markers of iron status.

Once established we will reassess the impact of this change on pregnancy outcomes.

Disclosure of Interest: None Declared.

BSH2020-EP-036

An audit of complications arising in central venous access devices (CVADs) in adult haematology patients, Chesterfield Royal Hospital

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We undertook an audit of all central venous access devices (CVADs) inserted between 1-October-2016 and 30-September-2018 in adult haematology patients at CRH. Audit standards included (i) samples taken during investigation of possible infection; (ii) culture of line tip where removed for possible/ proven infection; (iii) CVAD removal for specific organisms or non-resolving colonisation; and (iv) compliance with local guidelines for use of systemic antibiotics and antibiotic line locks.

Central line-associated blood stream infections (CLABSI) were defined in accordance with the international consensus. Tunnel infection was diagnosed on clinical grounds. Exit site infection was diagnosed by a combination of clinical features and positive swab from the site. A pragmatic decision was to define positive blood culture from a CVAD (with an organism known to be associated with line colonisation) with concurrent negative peripheral blood culture as 'line colonisation'. Other CVAD complications were recorded.

A total of 28 CVADs were inserted during the audit period (17 Hickman lines and 11 PICC lines) in 17 patients. Patient diagnoses were nine AML and three MDS (all treated with curative approach), five lymphoma (two poor peripheral access and three salvage chemotherapy requiring CVAD). CVADs were in place for a total of 1778 days (range 10–237 days).

A total of 146 blood cultures were undertaken. Sixteen samples were not identifiable by site (11%). On six occasions all available sites were not sampled with no reason documented in the patient record.

There were 13 episodes of CLABSI (nine definite and four probable), giving a CRBSI rate of 7 per 1000 line days. Organisms were four *E. coli*, two vancomycin-resistant enterococcus (VRE), one *E. faecium*, one *E. cloacae*, one *Sphingomonas*, one *Capnophytophaga*, one *Corynebacterium*, one coagulase-negative staphylococcus (CNS) and one alpha haemolytic staphylococcus. Fifteen separate episodes of colonisation occurred in 10 CVADs (multiple colonising species were detected in two CVADs); 10 were CNS, one *Stenotrophomonas*, one VRE, one *E. coli*, one *Chryseobacterium* and one *Sphingomonas*.

Antibiotic line locks were used in seven episodes where CVAD colonisation with CNS occurred; where the CVAD was not removed, re-culture after the completion of antibiotics was only was

undertaken in two of six episodes. Tunnel site infection occurred in three and exit site infection in one CVAD.

Eleven CVADs were removed for infections (CLABSI, colonisation, tunnel or exit site infection). Other line complications resulting in removal were one PICC-related thrombosis and two damaged CVADs.

This audit brought to light local issues with poor specimen identification, missed samples and failure to re-culture where indicated.

Disclosure of Interest: None Declared.

BSH2020-EP-037

A case of refractory heparin-induced thrombocytopenia

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Type II heparin-induced thrombocytopenia (HIT) is caused by antibody production which binds complexes between heparin and platelet factor 4 leading to platelet consumption and thrombosis. Rarely the antibodies continue to activate platelets in the absence of heparin and can present with thrombosis over 10 days post-heparin exposure.

We present the case of a 71 years old man who attended Accident and Emergency on day 10 post-op following tissue aortic valve replacement for aortic stenosis with hypoxia, dyspnoea and pleuritic chest pain.

Blood results showed platelets of $20 \times 10^9/l$, Hb 106 g/l, WCC $31.46 \times 10^9/l$. A blood film showed left-shifted neutrophils. He had a D dimer of 60 938 mcg/l, PT 15s, APTT 28s and fibrinogen 4.5 g/l. A CT pulmonary angiogram confirmed a saddle PE extending into major branches of the pulmonary artery, with CT evidence of right ventricular strain. HIT was suspected with a HIT score of 7. HIT antibodies with Werfen Chemiluminescence AcuStar method (specific IgG PFH- H assay) was positive at >125 U/ml.

The patient received unfractionated heparin (UFH) while on cardiopulmonary bypass during his cardiac surgery and prophylactic low molecular weight heparin (LMWH) post-operatively for 5 days. The pre-operative platelet count was $156 \times 10^9/l$ which fell to $99 \times 10^9/l$ on day 4 post-op but rose to $153 \times 10^9/l$ on day 5 post-op when the patient was discharged. The patient had also received 3000 units of intra-arterial UFH during a diagnostic coronary angiogram performed 77 days prior to his operation. Platelet counts were not checked then and there were no obvious complications.

Argatroban was commenced with a target APTT ratio of 1.5–3.0 (APTT 42–84 s) but the patient developed haemoptysis and bleeding from his sternal wound requiring manual pressure for several days. On day 14 post-op, he also developed acute bilateral lower limb ischaemia with multiple areas of necrosis on both feet. The APTT ratio was consistently at the lower end of the therapeutic range. We elected to increase the target APTT to 50–60 s and subsequently to 60–70 s due to worsening ischaemia while attempting to avoid bleeding complications. However, the platelet count remained between $7 \times 10^9/l$ and $24 \times 10^9/l$ at day 20 post-op and a repeat HIT test remained at >125 U/ml.

Refractory HIT was therefore suspected although other causes of thrombocytopenia were considered including sepsis. The patient was on broad spectrum antibiotics for pneumonia and was clinically responding.

Given the concerns regarding refractory HIT, intravenous immunoglobulin (IVIg) 0.5 g/kg/day was given for 3 days without steroids due to the concurrent infection. The platelet count rapidly responded and normalised ($155 \times 10^9/l$) on day 22 post-operatively.

The patient improved clinically, with decreasing oxygen requirements and improving peripheral ischaemia avoiding the loss of any digits. Fondaparinux was commenced on day 23 post-op and warfarin started as per protocol. The platelet count remained normal throughout the remainder of the admission and the patient was discharged on day 44 post-op.

Refractory or delayed-onset HIT with features of prolonged severe thrombocytopenia and lack of response to conventional anti-coagulants is an important clinical scenario that clinicians need to be mindful of. Case reports have shown successful treatment with IVIg +/- steroids as in this case although caution is advised given the potential prothrombotic risk of IVIg in patients with HIT.

Disclosure of Interest: None Declared.

BSH2020-EP-038

Early pre-microscopic differentiation of acute promyelocytic leukaemia: the convergence of cell population data and artificial intelligence tools

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A targeted and timely offered treatment can be a benefitting tool for patients with haematological emergency like acute promyelocytic leukaemia (APML). Complete blood count (CBC) is a key test in diagnostic workup of haematological neoplasms, for which modern haematological analysers generate cellular measurements as cell population data (CPD) in background for reporting blood cell's counts. We hypothesized that pre-microscopic flagging of 'acute leukaemia (myeloid/ lymphoid) of analysed blood sample by the haematology analysers could be life saving. Such a flagging will alert the technologist to bring the slide of the flagged sample to the haematologist for early review. Using artificial intelligence (AI) algorithm and based on CPD parameters, we have built artificial neural network (ANN) model to early flag of (predict a) APML. We collected classical CBC items along with cell population data (CPD) from haematology analyser at diagnosis of 1067 study subjects with haematological neoplasms. For morphological assessment, peripheral blood films were examined. Statistical and machine learning (ML) tools including principal component analysis (PCA) helped in the evaluation of predictive capacity of routine and CPD items. Then selected classical CBC and CPD parameters-driven ANN predictive modelling was developed to smartly use the hidden trend by increasing the auguring accuracy of these parameters in differentiation of APML cases. We found a characteristic triad based on lower (53-73) platelet count (PLT) with decreased/normal (4-72) immature fraction of platelet (IPF) with addition of significantly higher value (65-5) of DNA/RNA content-related neutrophil (NE-SFL) parameter in patients with APML against other haematological neoplasm's groups. On PCA, APML showed exceptionally significant variance for PLT, IPF and NE-SFL. Through training of ANN predictive modelling, our selected set of attributes successfully classifies the APML group from non-APML groups at highly significant (0-894) AUC value with lower (2-3%) false prediction rate. Practical results of using our ANN model were found acceptable with value of 95-7% and 97-7% for training and testing datasets, respectively, which indicates that our set of parameters can represent a relevant 'fingerprint' of a APML.

This approach expands the application of AI tools/models in clinical practice and indicate the results of basic blood test like CBC contain more information than clinician generally recognize. This CBC data-driven AI modelling is a novel approach that substantially strengthen the predictive potential of CBC items, allowing the clinicians to be confident by the typical trend raised by these studied parameters and could initiate new prospects for clinical diagnosis.

Disclosure of Interest: None Declared.

BSH2020-EP-039

Investigating compliance of haematology patients with recommended pneumococcal and influenza vaccines

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Introduction: National guidance in the Greenbook of Immunisation states that those who are immunosuppressed falls in a clinical risk group who should be offered an annual influenza vaccine. In addition to this, they should be offered a pneumococcal vaccine which may need to be repeated every 5 years. The aim of this audit was to investigate a cohort of haematology patients at Maidstone hospital to see whether they were compliant with national guidance. If they were not, we aimed to find out why and how we could improve the compliance.

Methods: An audit collection tool consisting of 14 questions was used to collect data from 89 haematology patients who were receiving chemotherapy at Maidstone Hospital during February 2019. This was either on the haematology ward or outpatient chemotherapy suite. Our key performance indicators were as follows:

- Was advice given regarding need for influenza and pneumococcal vaccines, and if so, was it verbal or written?
- Has the patient received an influenza vaccine in the last year?
- Has the patient received a pneumococcal vaccine in the last 5 years?

Results: 23% of patients had received verbal advice and 0% had received written advice regarding vaccines. 76% of those surveyed had been given an influenza vaccine in the last year and 36% had received a pneumococcal vaccine in the last 5 years. There were multiple reasons for patients not having vaccines but the most frequent were that they did not think they needed one or they had concerns about the vaccine itself. Of our cohort of patients 30% were admitted to hospital with either flu or pneumonia. Of those admitted for flu, 73% had received a vaccination and 27% had not.

Conclusion: Advice needs to be given to patients who are having chemotherapy, either verbally or written. We propose, as a result of this audit, to provide patients with a leaflet outlining the recommendations as per national guidance. The study is limited by both time and size to make a significant comment between whether certain conditions or chemotherapy regimens had a differing proportion of admissions.

Disclosure of Interest: None Declared.

BSH2020-EP-040

Acute haemolysis in a glucose-6-phosphate dehydrogenase-deficient patient receiving high-dose vitamin C therapy for metastatic caecal cancer: a case reportR. Sen¹*, L. Stockton¹, D. Saleh¹¹General Medicine, North Middlesex University Hospital, London, UK

Introduction: The use of intravenous high-dose vitamin C in cancer remains an area of contention. It is occasionally used as an adjunct to chemotherapy by physicians and in naturopathic settings. Evidence for its use is inconclusive with some papers reporting it has the potential to slow cancer growth, improve survival and reduce chemotherapy toxicity. Despite a lack of definitive evidence, it has a favourable safety profile and potential anti-tumour effects, therefore continues to be used off-licence. We present a case of life-threatening haemolysis caused by high-dose vitamin C administration in the context of glucose-6-phosphate dehydrogenase (G6PD) deficiency that disputes the safety of this therapy for particular patients.

Case report: A 46-year-old male presented with a 2-day history of light-headedness and dyspnoea. He had a diagnosis of metastatic caecal carcinoma for which he was receiving high-dose vitamin C intravenously from a private clinic. He was severely hypoxic on admission requiring 15 l of oxygen to maintain his saturations. Initial investigations demonstrated acute anaemia with a haemoglobin level of 49 g/dl (from 118 g/dl 3 weeks prior), a raised bilirubin, and a high methaemoglobin and carboxyhaemoglobin on arterial blood gas. Reticulocytes were markedly elevated, haptoglobin levels were low (<0.1) and DAT screen was negative. A blood film reported polychromasia, spherocytes and occasional red blood cells in keeping with haemolysis. GP6D testing during this acute haemolytic episode was negative; however, records obtained from before administration documented a mild deficiency. The patient was diagnosed with haemolytic anaemia secondary to intravenous vitamin C therapy; he was transfused four units of blood and discharged 3 days following.

Discussion: There is a growing body of evidence that suggests erythrocytes from G6PD-deficient patients show increased oxidative stress in response to high-dose vitamin C therapy and are more susceptible to haemolysis. To our knowledge there are six reported cases of haemolysis secondary to high-dose vitamin C treatment in G6PD deficiency, this is the first case of haemolysis reported in vitamin C therapy for cancer. In many high-dose vitamin C trials for sepsis, G6PD deficiency is considered an exclusion criterion; however, it is not listed as a contraindication.

This case highlights the need for more comprehensive research into the association between haemolysis in GP6D deficiency and high-dose vitamin C therapy to ensure safe administration.

Disclosure of Interest: None Declared.

BSH2020-EP-041

Nucleic acid containing units (NACU); a new parameter in haematologyG. Gibbs¹*, G. Precht², V. Jones³¹Siemens Healthcare Ltd, Surrey, UK, ²Siemens Healthcare Diagnostics GmbH, Vienna, Austria, ³Siemens Healthcare Diagnostics Manufacturing Limited, Dublin, Ireland

We would like to introduce and briefly describe a new parameter, the nucleic acid-containing units (NACU) count, developed for ADVIA 2120 and ADVIA 2120i Haematology Systems (Siemens Healthcare Laboratory Diagnostics, Tarrytown, NY).

Neutrophil extracellular trap (NET) formation is a rapidly expanding area of clinical interest across a wide range of pathophysiological states. For example, NETs have been shown to have prognostic value in sepsis correlating both with the severity of sepsis and degree of organ dysfunction. With sepsis, cell-free DNA (cfDNA) is suspected to be released by activated neutrophils via the process of NETosis while cfDNA in trauma has been shown to originate mainly from injured or necrotic cells. cfDNA has also been shown to be a good predictor of patient outcome in ICU patients. For the first time, NETs may be rapidly detectable via a routine blood count.

While developing the reticulated platelet parameter as part of the v6-10-9 software update (released in 2019) it became apparent that the capability of this channel extended to the detection of other events that stain positive for nucleic acid.

ADVIA 2120 and 2120i haematology systems utilise oxazine 750 for staining cells according to their nucleic acid content in order to detect and enumerate reticulocytes and reticulated platelets. In addition, the NACU count identifies NACU by counting signals with high absorption values and low refractive index that are below the reticulated platelet area. Preliminary investigations suggest that these particles may be from NETs, bacteria, fungi, RNA from lysed red blood cells (reticulocytes), malarial parasites, DNA strands from destroyed white blood cells or other sources of cfDNA and cell-free RNA (cfRNA).

NACU counts are also suspected to have a role in detecting damaged endothelial cells from diabetes as well as atherosclerosis. Certainly, NET formation has been implicated in these conditions and others as well as revising our understanding of thrombus formation.

Our preliminary research regarding a normal range for NACU counts suggests a value <15. NACU values above the reference range indicate that abnormal levels of cfDNA or cfRNA may be present. Currently the NACU count is research use only (RUO) and no diagnostic claims can be made.

We encourage our clinical and laboratory colleagues to evaluate the NACU count in clinical settings where NACU may play a role. Clearly, as more research is carried out, we will gain more clinical understanding of the role of the NACU count as a disease marker. Further nucleic acid-containing structures in the peripheral blood may remain to be discovered, as well as further applications of the NACU count.

The ability to detect cfDNA and cfRNA offers a new approach to what haematology analysers can do in the future.

Disclosure of Interest: None Declared.

BSH2020-EP-042

The advent of the extremely elderly clinic: not if, but when? A District General Hospital experienceR. Haji¹*, C. Liu¹¹Haematology, Chelsea and Westminster NHS Trust, West Middlesex Hospital, London, UK

With the trend of increasing life expectancy with time, the use of secondary care by an aging population is likely to increase. Extremely elderly patients bring unique challenges to the assessment and management of haematological conditions. Currently there is a lack of a consistent or uniform approach in assessing a patient's fitness for treatment or their tolerance to treatment.

We performed a retrospective review of patients aged 90 and above attending the general haematology clinic over a 6-month period from June 2019. The age range was 90–99 (mean age 92.4 years). Sixteen patients were male and 18 were female. In total, there were 52 clinic attendances. Of these, 16 were new appointments and 36

were follow-ups. The average number of co-morbidities was 3–4 per patient. Most patients did not require active intervention, three patients were treated with chemotherapy, six were discharged back to the GP and three patients died. In cases where no treatment was required, repeat follow-up appointments were still made to actively monitor patients. This in itself could have a detrimental impact on the well-being of the extremely elderly, due to issues such as difficulties with transport, and long waiting times in clinic.

The management of the extremely elderly age group needs to consider a number of factors, including appropriateness of the referral, the health impact of the presenting problem, and the patient's view on their management. Special consideration needs to be given to the physiological age, number of co-morbidities and estimated life expectancy of the patient. Quality of life must remain a key point. Use of a geriatric assessment toolkit can improve the evaluation of the care required, resulting in more personalized medicine. Consideration should be given for joint haematology/elderly care clinics with robust primary care links to adopt a more holistic approach, which takes into account the physical, psychological and socio-economic circumstances of extremely elderly patients.

Disclosure of Interest: None Declared.

BSH2020-EP-043

Pica in women in rural South Africa

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Pica is defined by the International Statistical Classification of Diseases (ICD-11) as the consumption of non-nutritive substances, such as soil, chalk or paper. Derived from the Latin word for magpie, it alludes to the non-discriminatory eating habits exhibited by people with the condition. Pica is a persistent behaviour that is inappropriate for the person's developmental stage and severe enough to warrant clinical attention.

Evidence of geophagia, the eating of soil, can be found in ancient writings by Hippocrates and Celsus, which also discussed anaemia as a common cause of geophagia. While geophagia is generally more common in women, specifically pregnant women, low levels of educational attainment, emotional stressors and mental health conditions such as obsessive-compulsive disorder and schizophrenia have also been linked. Geographically, it is most common in Africa, where its prevalence is approximately 44–8% in pregnant and postpartum women. In America and Eurasia the prevalence is estimated at 23% and 18% respectively.

Tintswalo Hospital in the eastern South African province of Mpumalanga has approximately 400 beds and serves the surrounding rural population. Local markets sell packets of soil for people to consume. For the following project, a questionnaire was designed to determine who ate soil, the nature of the soil they ate, where they acquired it from and their reasons for doing it.

Hundred women attending a variety of outpatient clinics at Tintswalo Hospital in May 2019 completed the questionnaire. 39% answered that they currently or previously consumed soil. 74% bought packets from the local market and 26% scraped it off termite mounds. 46% of women were anaemic and 41% ate soil during their pregnancy. Interestingly, some women stopped eating soil once their anaemia was resolved, for example once their anti-retroviral treatment for HIV had begun. There is a debate in the literature about the causal relationship between geophagia and anaemia. This study found that most women consumed soil in addition to their regular meals and only ate a small amount once or less than once a day. This was presumed not to be enough to cause anaemia.

Most women in the present study responded that they ate soil simply because they craved the taste. There are several theories about

the cause for geophagia, including cultural practice or that specific cravings are triggered by various nutritional deficiencies, such as iron deficiency. Termite mounds have been shown to be much richer in certain minerals than the surrounding soil. A physiological theory claims that the particle clay content in soil consumed by people with pica relieves nausea and helps control diarrhoea. Some women who completed the questionnaire stated it helped their nausea and heartburn. This theory is supported by animal models, for example the mountain gorillas in the Virunga mountains that eat yellow volcanic rock. These gorillas most commonly mine the rock in the dry season, when they are forced to change their diets to plants like bamboo, which causes diarrhoea.

Despite varying theories regarding its origin, geophagia cannot be attributed to a single cause. Cultural practices, physiological issues and nutritional deficiencies as well as the patient's personal and psychiatric history all tie together as causative factors. This project presents findings from a subset of people with pica in eastern South Africa and explores the various aspects that surround it.

Disclosure of Interest: None Declared.

BSH2020-EP-044

Assessment of functional iron status in patients with chronic kidney disease attending University of Ilorin Teaching Hospital, Ilorin, Nigeria. Sanni E.O, Olawumi H.O, Durotoye I.A, Olarewaju T.O, Babatunde AS, Shittu A.O, Biliaminu SA

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Background: Functional iron deficiency has been found to be a common cause of poor response to erythropoietin-stimulating agents in anaemic patients with chronic kidney disease. However little is known about the functional iron status of patients with CKD in our environment. This study evaluates functional iron status in patients with CKD.

Methods: This was an hospital-based cross-sectional study of 113 patients with CKD who are attending Nephrology clinic at the University of Ilorin Teaching Hospital and 113 age- and sex-matched control. Informed consent and relevant information were obtained using a study proforma. Some haematological parameters (full blood count, reticulocyte count) and serum ferritin, soluble transferrin receptor (sTfR), total iron-binding capacity (TIBC), percentage transferrin saturation (TSAT), C-reactive protein (CRP) levels of both patients and controls were carried out. Full blood count (FBC) was done using automated haematology analyser; Sysmex Kx21, serum ferritin, TIBC and CRP were done using ELISA; serum iron and TIBC were done using the colorimetric method with use of POINTE SCIENTIFIC, INC iron/TIBC test kit. Data were analysed using descriptive and inferential statistics on an SPSS software version 22.0. The level of statistical significance was set at $P \leq 0.05$.

Results: The mean age of patient with CKD was 55.00 ± 15.37 years, while that of controls was 52.73 ± 13.59 years. The mean value of Hb and RBC in the patients was 9.69 ± 2.25 g/dl and $3.62 \pm 0.92 \times 10^{12}/L$, respectively, and were significantly lower than the controls participants; $P < 0.001$. The mean corrected reticulocyte count in the patients was $0.32 \pm 0.22\%$ and was significantly lower than that of control participants; $P < 0.001$. The platelet count and white blood cell count in the patients were statistically higher compared to the control participants $P < 0.001$. The prevalence of anaemia in patients with CKD was 88.5% and was significantly higher than control participants, 21.2%; $P < 0.001$

Table. Serum ferritin, serum iron, TIBC, TSAT, sTfR and CRP levels in patients with CKD and controls

Variable	Patients Mean $\pm$ SD	Control participants Mean $\pm$ SD	<i>T</i>	<i>P</i> value
Ferritin (ng/ml)	290.12 $\pm$ 161.52	99.46 $\pm$ 53.05	11.922	<0.001*
Serum iron (μ g/ml)	122.99 $\pm$ 77.29	104.72 $\pm$ 34.62	2.294	0.023*
sTfR (nmol/ml)	18.27 $\pm$ 11.38	17.61 $\pm$ 6.14	0.544	0.587
TIBC (μ g/ml)	405.56 $\pm$ 147.34	316.75 $\pm$ 72.71	5.745	<0.001*
TSAT (%)	30.72 $\pm$ 14.36	32.74 $\pm$ 6.83	-1.354	0.177
CRP (μ g/ml)Creatinine (μ mol/l)	3.02 $\pm$ 1.45434.92 $\pm$ 368.34	1.51 $\pm$ 0.9568.73 $\pm$ 10.01	9.31610.564	<0.001*<0.001*

The mean values of serum ferritin, serum iron, TIBC and CRP was significantly higher in patients compared with control participants ($P < 0.001$, 0.023, <0.001 and 0.001), respectively, although there was no significant difference in the mean TSAT and sTfR between patient and the control participants. Normal iron status was seen in 60.2% of the patients, functional iron deficiency in 19.5% and absolute iron deficiency was seen in 15.0%. There was no statistically significant relationship between CRP and FID; however, using the coefficient of binary logistic regression, the risk of having functional iron deficiency increases by 1.033 for every unit increase in CRP.

Keywords: Chronic kidney disease, Functional iron status, Anaemia, ESA. *T*, independent samples *T* test; sTfR, soluble transferrin receptor; TIBC, total iron-binding capacity; T SAT, percentage transferrin saturation; CRP, C-reactive protein.

* $P < 0.05$ (i.e. statistically significant).

Disclosure of Interest: None Declared.

BSH2020-EP-045

Rhapsody alerts – how effective are they? A quality improvement project

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Introduction: Imaging plays a tremendous role in identifying incidental and non-incidental malignancies. The Royal College of Radiologists state that there should be fail-safe alert procedures for significant unexpected findings. Sandwell and West Birmingham NHS Trusts (SWBH) have integrated a Rhapsody alert system platform into their Trust to overcome such issues. This quality improvement project aimed to establish whether the Rhapsody alert system improved the management of the haemato-oncology cancer care pathway.

Methods: A retrospective data collection was carried out in SWBH. The haemato-oncological Rhapsody alerts that were sent to the clinical nurse specialist and haematologist multidisciplinary cancer lead were analysed to find eligible patients from January 2019 to March 2019. Electronic databases 'Clinical Data Archive' and 'Somerset' were used to access further information: patients who were referred by; the modality of imaging that was flagged; whether haematological cancer was diagnosed; quality of flagging; and other relevant parameters. Duplicates of patients that were flagged throughout the Rhapsody alert were excluded.

Results: Seventy-six patients were flagged during the 3-month data collection period. The majority of referrals were flagged due to suspicious radiographs and CT scans, indicative of potential lymphoma. The majority transpired to be haematological malignancies. Results were compared to the Trust's recommended standard and the cancer care pathway recommendations, to assess whether the Rhapsody alert

system improved the cancer care pathway and whether variables could be implemented to improve patient care.

Discussion: The Rhapsody alert system is an efficient modality for identifying suspicious imaging findings and alerting the haemato-oncology department in a timely manner. However, there are a number of limitations that must be considered, such as: the data collection was done retrospectively, over a 3-month period and referrals from different radiographers may have introduced interobserver bias. Therefore, in the future, it would be useful to repeat this project and include a larger cohort (for example, a 1-year prospective data analysis). This would be more representative of the referrals the SWBH Haematology department encounters over a year and could be used to identify seasonal trends.

Disclosure of Interest: None Declared.

BSH2020-EP-046

Review of anti-fungal treatment prescriptions for haematology inpatients at Nottingham University Hospital

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The incidence of invasive fungal infections is increasing each year with improvements in medicine and availability of wider selection of immunosuppressive agents. Despite improved treatment strategies, there is potential for patient harm due to toxicities and development of resistance. According to the national anti-fungal stewardship program, anti-fungal therapy/treatments for inpatients should be reviewed 48–72 h after initiation and every 7 days thereafter by specialist stewardship team. A review at 24 h should ensure appropriate diagnostic investigations have been organized; with further reviews at 48–72 h and 7 days to evaluate available diagnostics and to assess the continued need for anti-fungals. By undertaking timely clinical review of the anti-fungal treatment prescriptions by senior members of the patient team, inappropriate use of the anti-fungal agents can be reduced; there will be improvement in patient outcomes and reduction in the development of resistance to anti-fungals.

This study aimed to measure the proportion of anti-fungal treatment prescriptions reviewed at 24, 48–72 h and 7 days; to look at whether appropriate investigations had been arranged within 24 h of decision to prescribe; and to assess at patients on empiric treatment who had proven or probable infection.

Inpatients prescribed anti-fungals medications on more than one occasion and in large quantities within the Haematology department of Nottingham University hospital during a month period (November 2019) were selected by pharmacy for this study. The anti-fungal stewardship CQUIN data collection proforma was used to collect data.

Out of the 15 patients who received anti-fungal treatments as inpatients, 100% of the prescriptions were reviewed at 7 days, 100% were reviewed at 48–72 h. At 24 h, only 20% of the anti-fungal prescriptions were documented as being reviewed by a consultant; the remaining 80% of prescriptions were reviewed by haematology specialty registrars. 83% of the patients had HRCT requested, 25% had HRCT before the treatment was started, 50% had the HRCT within 3 days of initiation of treatment and 25% had the HRCT after 3 days of initiation of treatment. 86% of the patients had fungal markers tested. 33% of the patients on empiric treatment had a proven or probable infection (EORTC criteria).

This study has shown that the current practice of inpatient anti-fungal treatment reviews at 48–72 h and 7 days have met the national anti-fungal stewardship requirements. However improvement is required to meet the requirement of a 24 h review by a consultant. The initiation of anti-fungal treatment for inpatients is usually a consultant-led decision. Regular review of prescriptions are undertaken during the daily verbal board rounds and discussed in the multidisciplinary team meetings twice a week. An improvement in documentation of these reviews is recommended to meet the national requirements.

Disclosure of Interest: None Declared.

BSH2020-EP-047

Abstract Withdrawn.

BSH2020-EP-048

A quantitative approach to differentiate haemolysis from dilutional changes in patients with Malaria

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Introduction: Anaemia occurs in one third of hospitalized patients who are initially not anaemic on admission. In patients with malaria, it is important to differentiate whether changes in haemoglobin during their hospital stay are occurring as a result of haemolysis due to malaria itself, or as a consequence of haemodilution due to administration of intravenous (IV) fluids. The aim of this study was to use a recently recommended quantitative approach to differentiate between haemolysis and haemodilution in patients with malaria.

Materials and Methods: This retrospective study was conducted at Shifa International Hospital, Pakistan. Adult patients who were hospitalized due to malaria from 2016 to 2019 and received IV fluids were included. We excluded patients who were dialysis dependent or receiving diuretics. Using the four-step method suggested by Hale et al in 2019⁽¹⁾, the patient's body weight, initial haemoglobin levels and volume of IV fluids were used to calculate the expected haemoglobin levels after IV fluid administration. The calculated expected haemoglobin levels were compared to the patient's actual haemoglobin levels measured after receiving IV fluids. A drop in haemoglobin of greater than 1 g/dl over a period of 24 h was considered significant.

Results: Seventy-two adult patients who met our inclusion criteria were admitted due to malaria during the study period. Seventeen (23.6%) of these patients developed a significant decrease in

haemoglobin while receiving IV fluids. The mean drop in haemoglobin was 2.02 ± 0.66 g/dl over a period of 24 h. Based on this quantitative method, it was identified that 11 patients were in a state of haemolysis (Group A) whereas haemodilution had occurred in six patients (Group B). The mean drop in haemoglobin was significantly higher in Group A as compared to Group B (2.31 ± 0.62 g/dl vs. 1.5 ± 0.33 g/dl, $P = 0.01$). There was no significant difference in the volume of IV fluids administered to Groups A and B (1.2 ± 0.5 l vs. 1.44 ± 0.49 l, $P = 0.349$). Two of the patients identified to be in haemolysis received blood transfusion.

Conclusion: We recommend that a quantitative approach should be used in order to reliably differentiate between haemolysis and dilutional changes. This approach can aid clinicians in making an informed decision regarding patient management and can potentially prevent any unnecessary blood transfusions along with their associated adverse effects.

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Disclosure of Interest: None Declared.

BSH2020-EP-049

T-cell imbalance or decreased Th:Tc ratio in patients with idiopathic thrombocytopenic purpura: is it clinically significant?

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Immune thrombocytopenic purpura (ITP) is an autoimmune disorder. Besides the platelet-specific auto-antibodies, T-cell imbalance might also be a cause of this disease. This study was planned to evaluate T-cells disparity in ITP patients and its association with disease persistence and severity. A case-control study (1:1) was conducted from January 2015 to July 2017 with 111 ITP patients and 111 age-matched controls. ITP patients were grouped as newly diagnosed/persistent ITP and chronic/refractory ITP patients. Peripheral blood was collected and complete blood profile parameters were recorded using XN 1000. The helper (Th) and cytotoxic (Tc) T-cell ratio was enumerated by flowcytometric analysis. The significant difference between newly diagnosed/persistent ITP or chronic ITP and healthy control groups was determined by *F*-test using SPSS ver. 19. Significantly low haemoglobin levels, platelet count with high immature platelet fraction were detected in newly diagnosed/persistent and chronic/refractory ITP patients ($P < 0.05$). T-cell imbalance was also confirmed by significantly low Th:Tc ratio ($P < 0.05$) in newly diagnosed/persistent ITP (0.88 ± 0.47) as compared to control group (1.73 ± 0.46). Within the chronic/refractory ITP group children ≤ 16 years had a mean Th:Tc ratio of 0.84 ± 0.34 . No significant difference ($P > 0.05$) was observed between children with non-severe chronic (0.84 ± 0.42), severe chronic (0.82 ± 0.49) and refractory ITP (0.85 ± 0.51). Significantly low Th count and Th:Tc ratio was observed in ITP patients compared to healthy controls as a sign of immune dysregulation. The age-dependent increased Tc production observed in children (≤ 16 years) with chronic/refractory ITP may be linked to chronicity but not with severity in this group.

Disclosure of Interest: None Declared.

BSH2020-EP-050

Investigating the relationship between iron deficiency with and without anaemia and patients' quality of life using the Fact-An questionnaire at a London teaching hospital

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Iron deficiency is the most common cause of anaemia globally. However, there are only a small number of studies on the quality of life in patients with iron deficiency with or without anaemia. Conlon et al (1) found that lower haemoglobin levels post-operatively were associated with a lower quality of life score in geriatric patients. Another small study which showed levels of fatigue improved in patients who were iron deficient but not anaemic but went on to receive intravenous iron (2).

We set out to investigate the relationship between the degree of iron deficiency with or without anaemia and patients' quality of life.

This study was performed on patients with iron deficiency with or without anaemia attending haematology services at Royal Free Hospital, London. Iron deficiency in our study is defined as ferritin <50 and/or transferrin saturation <20%. Anaemia is defined as haemoglobin (Hb) <120 g/l in women and <130 g/l in men.

Patients attending either the anaemia clinic or the day unit at Royal Free Hospital with the above criteria were given the Functional Assessment of Cancer Therapy – Anaemia (Fact-An) questionnaire to complete (3). Based on the responses, the Fact-General and Fact-Anaemia scores were calculated (4). Fact-G assesses the quality of life in four different domains – physical, social, emotional and functional. The FACT-An score assesses additional anaemia-related parameters.

The two-tailed *t* test is used to analyse the FACT-G and FACT-An scores in iron-deficient patients with anaemia and without anaemia. Linear regression is used to identify any correlation between haemoglobin, ferritin and transferrin saturations and the FACT-G and FACT-An scores.

In total, 21 patients were included in this study. Of these 15 were female and six were male. The median age was 47. The median Hb was 120. The median ferritin was 37 and the median transferrin saturation was 15%.

Of the patients with iron deficiency defined by our criteria, 11 were anaemic (52%). There was no significant difference between either the FACT-G score or FACT-An score between iron-deficient patients with or without anaemia ($P = 0.67$ and $P = 0.38$ respectively).

The linear regression, R^2 , value was calculated to correlate the haemoglobin, ferritin and transferrin with the quality of life measures, FACT-G and FACT-An scores. The results are in the table below.

Table. Correlating the degree of iron deficiency with patients' quality of life scores

Parameters	R^2	95% confidence interval
Haemoglobin and FACT-G score	0.0072	$-0.054 \leq R^2 \leq 0.068$
Haemoglobin and FACT-An score	0.00056	$-0.017 \leq R^2 \leq 0.018$
Ferritin and FACT-G score	0.016	$-0.074 \leq R^2 \leq 0.11$
Ferritin and FACT-An score	0.040	$-0.010 \leq R^2 \leq 0.18$
Transferrin saturation and FACT-G score	0.070	$-0.11 \leq R^2 \leq 0.25$
Transferrin saturation and FACT-An score	0.082	$-0.18 \leq R^2 \leq 0.34$

Our study demonstrates that in patients with iron deficiency, there is no significant difference in the quality of life scores between anaemic and non-anaemic patients. In addition, there is no direct correlation between the degree of patients' iron deficiency and their quality of life. Future studies could be performed with a larger sample size. In addition, it would also be useful to compare the quality of life between iron replete and iron-deficient patients.

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Disclosure of Interest: None Declared.

BSH2020-EP-051

Is MDS really treatable? An experience from Pakistan

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Myelodysplastic syndromes (MDS) represent a diverse group of haematopoietic stem cell disorders characterizing ineffective hematopoiesis, morphological dysplasia and leukaemic transformation. Although the international prognostic scoring system (IPSS) represents gold standard for risk assessment, it is still difficult to predict survival and median time to leukaemic transformation. MDS is being considered as relatively uncommon disorder in our region and lacks in national registry that may help in evaluation of disease burden as it is presumed to be an incurable disease with majority of patients conventionally being treated by only blood product support. Most of the patients are unable to avail standard treatment and those who are treated with recommended options are presumed to have hopeless outcomes. Hence, the study was planned to evaluate the fate of this disorder in our region with respect to various given treatment options and to establish local data. It was an observational cross-sectional study conducted at National Institute of Blood Diseases and Bone Marrow Transplantation, Karachi Pakistan from 2010 to 2018. Baseline investigations done were complete blood counts, bone marrow biopsy and cytogenetics. Clinical parameters of all patients were recorded. Patients were classified according to World Health Organization (WHO) 2008 classification and IPSS was calculated. Approval from the institutional ethics committee was obtained prior to the study. Variables were expressed as percentages. Differences in clinical measurements were evaluated using SPSS version 23. Overall survival was defined as the time elapsed between diagnosis and last follow-up. A total of 186 patients were included in the study. The median age was 55 (range 3–90) years. There were 139 males and 47 females. In our patients the most common presenting complaint was loss of appetite 110 (59.1%) followed by weakness and fever in 76 (40.9%). The mean haemoglobin was 7.94 ± 2.17 g/dl, total leucocyte count $7.51 \pm 13.3 \times 10^9/l$, platelet count $81 \pm 93.7 \times 10^9/l$. Cytogenetic data were available for 104 patients which revealed normal karyotype in 58 (31.1%) and abnormal karyotype in 46 (24.73%). The mean IPSS of the patients was 1. Out of total, supportive treatment was

offered to ($n = 80.43\%$), hematinics (vitamin B12 and folic acid; $n = 45.24\%$), recombinant erythropoietin ($n = 25.14\%$), lenalidomide ($n = 19.10\%$), thalidomide ($n = 11.06\%$), hypomethylating agents in ($n = 6.03\%$), steroids ($n = 6.03\%$) and growth factors (G-CSF) in ($n = 4.02\%$). Death of 124 patients was observed during the study period. Cause of death was ascertained in ($n = 40.32\%$). Septicaemia was the most common cause of death and found in ($n = 27.21.7\%$) followed by shortness of breath ($n = 26.21.4\%$), severe anaemia ($n = 21.16.9\%$) and intracranial bleeding in ($n = 10.08\%$). For survival analysis, median overall survival was 16 months and was analysed by Kaplan–Meier curve. Risk of progression to AML was found in six patients with median time of 15 months. Although MDS is considered as an uncommon disease yet the number of patients is increasing in Pakistan. There is a need to develop disease registry in our region to understand and approximate better measures which will help to treat patients with definitive treatment. Moreover, further studies with large sample size are needed to convince state's health department to finance treatment and research in the context so that better treatment options are made available.

Disclosure of Interest: None Declared.

BSH2020-EP-052

Abstract Withdrawn.

BSH2020-EP-053

The challenges associated with the treatment of haematological malignancy in patients with dementia

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With an aging population and diagnosis of malignancy, we are more often caring for patients who have permanent significant memory impairment. These patients need more complex treatment and interventions, the risks of which are often significant. Potential challenges are consenting and counselling about the potential impacts of treatment, 24-h supervision to monitor for adverse events, impact of treatment on Instrumental Activities of Daily Living (IADL), frequent hospital visits, ability to follow written instruction, potential effect of chemotherapy and effect of supportive medications on mental status.

In our centre we have noted a significant number of such cases presenting with a wide range of challenges. One example is a 69-year-old female lived alone with a diagnosis of dementia, living at home and managing with carers and family support. A diagnosis of chronic lymphocytic leukaemia was made. Upon best interest meeting, involving her family, consultant in charge, clinical nurse specialist and the dementia team, it was felt that treatment would be in her best interest because of rapidly progressing disease. Unfortunately, this patient was often not compliant with undergoing blood tests and would become extremely distressed if they were attempted. Another consideration was of the monitoring for serious complications of chemotherapy, such as neutropenic sepsis. Living alone there was a concern that this patient may not be able to recognise illness or fever or would not know how to seek urgent medical review. Furthermore, should she be assessed and we are unable to obtain blood tests then this would create a monitoring issue. In this case the

patient unfortunately would not have tolerated most treatment options and the required monitoring; therefore the risks of treatment outweighed the benefits.

The treatment of patients with systemic anti-cancer therapy without capacity as we have found is complex. Few patients have advanced directives or family members with power of attorney for health. This leaves the clinicians alongside the MDT and family members to make a best interest decisions. In this case the family of the patient was accepting and agreeable of the best interest outcome; however, this would surely be even more challenging should there be conflicting views of clinicians and relatives.

We often make decisions about patient's fitness for invasive investigations and other interventions based on their performance status and physical fitness. Cognitive and memory assessment is not routinely incorporated into the evaluation and management of older adults with malignancy. Several screening tools are available for use in non-cancer patients, for example Mini Mental State Examination and Montreal Cognitive Assessment. Studies are needed to develop feasible methods to incorporate cognitive screening into cancer care to allow identification of this potentially higher risk group of patients.

Literature and guidance is limited in evaluating therapeutic or supportive care interventions in older patients with pre-existing memory impairment receiving chemotherapy. Developing evidence-based, patient-tailored supportive care interventions to improve outcomes in this vulnerable population during cancer treatment is needed.

Disclosure of Interest: None Declared.

BSH2020-EP-054

Clinical association of increased antioxidant enzymes levels in immune thrombocytopenic purpura

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Immune thrombocytopenic purpura (ITP) is a disorder characterized by immune-mediated accelerated platelet destruction and suppressed platelet production. Increased Oxidative stress and reduced antioxidant levels may be a cause of this disease. Oxidative stress occurs as a result of increased activity of free radical-producing enzymes, decreased activity of free radical-removing enzymes and insufficient levels of antioxidants. Current study has been done to evaluate antioxidants levels in ITP patients and their possible association with occurrence of ITP. This 1:1 case-control study was conducted on 111 ITP patients selected at NIBD, Karachi from January 2016 to December 2017. CBC was performed by Sysmex XN-1000. Total antioxidant status (TAS), superoxide dismutase (SOD) and glutathione reductase (GR) enzymes levels were determined using Randox® kits on Selectra Chemistry analyzer. Fisher's exact test was used to evaluate difference on SPSS version 23. The patients' age ranged from 9 months to 83 years. All ITP patients were further divided into newly diagnosed/persistent ($n = 64$) and chronic ITP ($n = 47$) respectively. Mean platelets count was 32.9 ± 35.1 ($P \leq 0.05$), 39 ± 39.2 ($P \leq 0.05$) and 278 ± 58.9 for newly diagnosed/persistent ITP, chronic ITP and controls. There was no significant difference of TAS in patients and controls but significantly increased levels of GR and SOD were observed with mean of 90.1 ± 33.5 and 9818.5 ± 525 in newly diagnosed/persistent ITP, 100.5 ± 32.7 and 8486.9 ± 498 in chronic ITP as compared to controls which were 68.4 ± 11.3 and 757.4 ± 425.5 . Oxidative stress may be one of the cause of ITP. Higher production depicts hyper-activity of antioxidant enzymes thereby confirming

oxidative stressed status in ITP patients irrespective of their type of disease. Evaluation of oxidative stress and antioxidants levels can be helpful in understanding the pathogenesis of ITP.

Disclosure of Interest: None Declared.

BSH2020-EP-055

A case of Niemann-Pick disease type C diagnosed on bone marrow biopsy: a rare event

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Niemann-Pick type C disease is a lysosomal storage disorder characterized by an inability of the body to transport cholesterol and other fatty substances inside of cells. Lysosomes are organelles that contain digestive enzymes involved in the breakdown of proteins, nucleic acids, carbohydrates and lipids. Four basic types are identified categorized on the basis of genetics and symptomatology. Niemann-Pick type C disease is due to mutation in NPC 1 gene located on long arm of chromosome 18 (18q11-2) and NPC-2 gene located on long arm of chromosome 14(14q24-3). Diagnosis involves detailed history, thorough physical examination along with laboratory investigations such as blood-based testing for biomarkers (oxysterols, lysosphingolipids and bile acid metabolites) and genetic analysis. Psychiatric manifestations often mislead physicians towards dementia and schizophrenia. This case report depicts an occurrence of Niemann-Pick disease diagnosed on bone marrow biopsy. We report a case of 39 years old patient who presented with fever and abdominal pain. Abdominal examination revealed tenderness in right iliac fossa while ophthalmic exam revealed vertical gaze palsy. Patient also reported behavioural changes and depressive episodes. Complete blood count showed thrombocytopenia. Bone marrow biopsy was, therefore, suggested to assess the cause of thrombocytopenia. Aspirate showed a normocellular specimen with active granulopoiesis and erythropoiesis but highlighted few scattered large histiocyte-like cells having abundant cytoplasm with soap—bubble like appearance with morphology suggestive of storage cells. Bone marrow trephine biopsy further confirmed the diagnosis by highlighting histiocyte-like cells having abundant vacuolated cytoplasm with appearance suggestive of storage cells. These cells were negative for periodic acid Schiff stain. On immunohistochemistry, these cells were slightly positive for CD 68. Increased histiocytic activity was noted. These histiocytes were strongly positive for CD 68. Overall findings were, therefore, consistent with lipid storage disorder, most likely Niemann-Pick disease.

Disclosure of Interest: None Declared.

BSH2020-EP-056

Pondering possibilities: PRES versus PML

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Posterior reversible encephalopathy syndrome (PRES) is a rare neuro-radiological syndrome. It can manifest as variable neurological symptoms making it difficult to diagnose. Although a trigger is usually identifiable, the pathophysiology of PRES remains controversial with uncontrolled hypertension and endothelial dysfunction the main mechanisms implicated. We describe an unusual presentation of PRES and the associated diagnostic difficulties.

A 68-year old with stage IV classical Hodgkin lymphoma presented with vertigo, ataxia and diplopia. She was otherwise systemically well

and afebrile. Four days earlier she had commenced treatment for shingles affecting the C4 and C5 dermatomes. She had completed treatment (ABVD chemotherapy escalated to PVACEBOP due to a positive interim PET scan) 12 weeks before presentation. While treatment had been complicated by significant cytopenias and admissions with febrile neutropenia she had obtained clinical and radiological remission. Since she remained lymphopenic she continued with anti-viral prophylaxis (co-trimoxazole and acyclovir). Blood tests showed normal haematological indices, CRP 9.1 mg/l. A CT head was reported to show no abnormalities. CSF examination demonstrated a white cell count of 14 in 2 ml with predominance of lymphocytes. CSF protein was slightly elevated at 672 mg/l. She was empirically commenced on treatment for meningo-encephalitis with IV ceftriaxone and IV acyclovir. Initial virology for VZV, HSV1, HSV2 and enterovirus were negative. Despite antimicrobials her neurological symptoms persisted. An MRI was subsequently performed and revealed extensive bihemispheric, symmetrical leuco-encephalopathy without diffusion, restriction or enhancement. The differential diagnosis considered in this case included viral encephalitis, progressive multifocal leuco-encephalitis and posterior reversible encephalopathy syndrome. Upon further questioning the patient reported mild, persistent neurological symptoms since her first cycle of chemotherapy 6 months ago. The acute clinical picture was felt to be more in keeping with PCR-negative VZV encephalitis. She therefore received 14 days of IV acyclovir. She remained clinically stable, with subtle neurological improvement during her admission. Extended CSF analysis, prompted by her MRI findings, confirmed the absence of JC Virus, HHV6 and HHV7 infection. Interval imaging, 6 weeks later, showed stable extensive leuko-encephalopathy. Clinically the patient made a progressive recovery with complete resolution of all clinical signs by 3 months. In view of the relatively mild neurological symptoms along with the improvement on anti-viral therapy, a diagnosis of acute PCR viral encephalitis was made along with persistent radiological findings and clinically history of PRES, secondary to chemotherapy 6 months before.

This case highlights the importance of considering rare diagnoses in immunocompromised patients presenting with neurological symptoms. While it highlights the benefits of MRI it also shows the importance of clinical correlation and consideration. There are increasing reports linking PRES and PML with newer chemotherapy and immunotherapy regimens; thus clinicians involved in the use of such medicines must be mindful of the risk of significant adverse effects, and remain vigilant when neurological complications arise.

Disclosure of Interest: None Declared.

BSH2020-EP-057

The effectiveness of tyrosine kinase inhibitors and molecular response in newly diagnosed patients of chronic myeloid leukaemia

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Introduction: The Aim of oral tyrosine kinase inhibitor (TKI) treatment in chronic myeloid leukaemia (CML) is to get ideal haematological, cytogenetic, molecular responses at the critical time points. The depth of the response obtained with TKI and the time to achieve this response are both important in predicting the prognosis in patients with CML.

Objective/Rationale: This study observes the response of tyrosine kinase inhibitors in patient of chronic myeloid leukaemia in public sector hospital.

Study Design: Cross-sectional observational study.

Setting: This study was conducted for a time period of 5 years from 2013 to 2018 at Civil Hospital Karachi.

Subjects: 1233 CML patients were enrolled in the study.

Material & Methods: After gaining informed consent from the patients, a performa was filled which included details that is age, gender, presence and duration of the symptoms. Samples of the patient including the blood sample, bone marrow aspirate and trephine along with the cytogenetics sample was obtained. Diagnosis was done on the morphological basis and BCR-ABL mutation by PCR/FISH. All patients received TKIs including imatinib and nilotinib. 1123 patients received Tab imatinib 400 mg od and 289 patients received Tab nilotinib 300 mg bd.

Results: A total of 1233 patients were included in the study out of which 519 were females and 713 males. The mean age of the patients was calculated to be 38.5 years (range: 2–82 years). Most of the patients fall in age group of 31–40 years. These patients came from different districts of Sindh province. After the start of TKIs, most of the patients achieved complete haematological response. Major molecular response of 836 patients was monitored through PCR at 1 year of treatment. 469 patients achieve major molecular response at 1 year and 367 patients fall in suboptimal response and failure of response.

Conclusion: Among community-based CML patients who received first-line TKI therapy, patients treated with either imatinib or nilotinib achieved higher rates of CCyR and MMR. Our study showed the most of patients diagnose in younger age as compare to European data whose patients diagnose in older age.

Increased understanding of the dynamics of leukaemic response to TKI therapy has led to definitions of molecular responses that are optimal and those that are suboptimal. Suboptimal responses are associated with a significantly higher risk of mutations and loss of response. Regular molecular monitoring will allow loss of response to be recognized at an early stage in most cases and may also facilitate better compliance. The resistance profile of each TKI to specific mutations has been better defined so that clinical recommendations based on these findings are now being established.

Keywords: Chronic myeloid Leukaemia, Tyrosine kinase inhibitor, Major molecular response.

Disclosure of Interest: None Declared.

BSH2020-EP-058

Additional cytogenetic abnormalities in resource constraint countries; an additional burden on chronic myeloid leukaemia patients?

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Chronic myeloid leukaemia (CML) is a myeloproliferative neoplasm characterized by the presence of Philadelphia chromosome (Ph) resulting from translocation between chromosomes 9 and 22, that is t(9;22)(q34;q11). In addition to this, 5–10% of patients have variant translocations in which at least a third chromosome is involved in the rearrangement 8, 9 addressed as additional cytogenetic abnormalities (ACA) and are generally marked as high-risk attributes in CML. Molecular and cytogenetic response monitoring in CML is recommended and it is endorsed by current treatment guidelines. The option to omit cytogenetic assessment has been proposed in 2013 European LeukemiaNet (ELN) recommendation; however, on the other hand, it is also emphasized to perform cytogenetic analysis until the achievement of complete cytogenetic response (CCyR) and major molecular response (MMR). The rationale of the current study was to analyse the presence of ACA at baseline, review its characteristics and implicate

that performing cytogenetics at baseline is doing some extra benefit or putting more burden on patients in countries like Pakistan where health is beared solely by patients and not government unlike west. In this context we followed the patients identified with ACA at baseline, assessed their clinical outcome, response to treatment and observed if there was any difference in molecular response as compared to the cohort not identified with ACAs at baseline. A case-control study carried out at our institute from 2010 to 2018. Participants included 18 CML cases with additional cytogenetic abnormalities. For each case, two controls were taken. Controls were diagnosed CML patients without additional cytogenetic abnormalities. Response to treatment was assessed according to ELN recommendations 2013. Descriptive and inferential statistics including chi-squared and independent *t* tests were applied to observe the association by using SPSS. Survival outcomes were also computed. Post-treatment response was assessed in cases and controls which revealed that both the groups had same response to treatment with respect to increment in haemoglobin levels and normalization of total leucocyte count and platelet counts. Attainment of molecular response at 6 months was same in cases and controls while at 12 months, higher in controls than cases. Survival outcomes were also found same in cases and controls. Patients with additional cytogenetic abnormalities in our cohort showed similar haematological and molecular response as controls. In developing countries with limited resources, omission of cytogenetic analysis at baseline may be considered when molecular level tests are available for decisions to be taken in the management of CML patients.

Disclosure of Interest: None Declared.

BSH2020-EP-059

Erythropoietin in low-risk MDS: difficult to convince for safety due to affordability; an experience from Pakistan

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Myelodysplastic syndromes (MDS) represent a diverse group of haematopoietic stem cell disorders characterized by ineffective hematopoiesis, morphological dysplasia and tendency for leukemic transformation. MDS being considered a relatively uncommon disorder in our region is presumed to be an incurable disease with majority of patients conventionally being treated by blood product support only. Most of the patients are unable to receive standard treatment due to financial reservations and presumed to have worst outcomes. To achieve better outcome, erythropoietin administration is one of the better option to raise haemoglobin levels and avoid transfusion. Therefore the study was planned to evaluate the outcomes of MDS patients treated with erythropoietin versus pack cell transfusion. The aim of the study was to assess outcomes in MDS patients given erythropoietin versus those who received pack cell transfusion. It was an observational study conducted at National Institute of Blood Diseases and Bone Marrow Transplantation, Karachi from June 2010 to December 2018. Approval from Institutional Ethics Committee was obtained prior to the study. Patients were risk stratified according to IPSS scoring and included low and intermediate-1 risk MDS patients requiring red cell transfusion. Division in two groups was made that is those who received erythropoietin and those who received pack cell transfusion. Baseline investigations were done including complete blood counts, bone marrow biopsy and cytogenetics. Analysis

of variables was done by using SPSS version 23.0. Out of 177 MDS patients enrolled at our hospital, 108 (61%) patients were offered erythropoietin. Only 26 (24%) chose to get treated by erythropoietin and rest 82 (76%) opted for pack red cell transfusion. The average erythropoietin dose administered per week was 30 000 IU and mean blood transfusion was 2 units pack red cells per month. The mean haemoglobin level at baseline of erythropoietin and transfusion group was 4.95 ± 1.48 mg/dl and 5.1 ± 1.27 mg/dl respectively. Outcome was evaluated, 16 (61.5%) patients in erythropoietin group and 36 (46.3%) in transfusion group were alive and 10 (38.4%) and 46 (57%) died respectively. Leukaemic transformation was observed in six patients in transfusion group while in only one patient in erythropoietin group. The mean rise of haemoglobin in erythropoietin group was

4.95 ± 1.48 mg/dl to 9.2 ± 2.1 mg/dl while in transfusion group average 1 gm/dl increment in haemoglobin level was observed after transfusion. Our findings are suggestive of the fact that erythropoietin has an influential impact on haemoglobin levels. However, limited number of patients chose erythropoietin as their treatment considering it expensive as compared to transfusion. Evidence by our statistics suggested that in our region cost of erythropoietin is same what comparable as that of transfusion which includes replacement donors, transfusion and day care expenditures. Moreover, hazards and potential risks associated with transfusion can also be minimized. Further studies with large sample size are needed to evaluate the outcomes of treatment options in MDS.

Disclosure of Interest: None Declared.

Lymphoma/CLL

BSH2020-EP-060

Abstract Withdrawn.

BSH2020-EP-061

Contemporary real-world treatment patterns of chronic lymphocytic leukaemia (CLL) in England

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Background: CLL is the most common type of leukaemia in adults with men twice as likely to develop the disease. The incidence of CLL increases with age and the average age of onset is 72 years. Although incurable at present, there have been many recent treatment advances that have resulted in improved outcomes for patients

suffering from CLL. In this study, we investigated the patient characteristics and treatment patterns of CLL in England using Public Health England's (PHE) national cancer registration and analysis service (NCRAS).

Methods: Patients in England with a diagnosis of CLL and first recorded systemic treatment between 1 July 2014 and 31 March 2018 were included from NCRAS. Combination therapies were identified by overlapping cycle dates of individual drugs. The study feasibility and methodology were supported by The Simulacrum, a synthetic version of NCRAS which was then applied by PHE to NCRAS data. Only fully anonymous aggregate results were released.

Results: In total, 4095 patients were included in this study. The median time from CLL diagnosis to first recorded systemic treatment was 2.5 years (interquartile range (IQR): 0.6–5.4), which is likely to be reflective of the initial 'watch and wait' period in this group of patients. Fludarabine/cyclophosphamide/rituximab (FCR) and bendamustine/rituximab (BR) were the two most frequently used treatment modalities for CLL patients with a mean age at the start of treatment of 62 and 71 years respectively. Twelve per cent of patients

Table. First recorded systemic treatments and characteristics of CLL patients in England ($n \geq 50$) between 2014 and 2018.

First recorded treatment*	Number (%)	Mean age at first treatment, years (SD)	Number of males (%)	Performance status (PS) at first treatment, number (%)	Median time from diagnosis to first treatment, years (IQR)
All	4095 (100)	71.0 (10.4)	2752 (67)	PS 0–1: 2532 (61)PS $\geq$ 2: 301 (8)Missing: 1262 (31)	2.5 (0.6; 5.4)
Fludarabine/cyclophosphamide/rituximab (FCR)	740 (18)	62.3 (8.5)	522 (71)	PS 0–1: 503 (68)PS $\geq$ 2: 11 (2)Missing: 226 (31)	1.6 (0.3; 4.1)
Bendamustine/rituximab	725 (18)	71.2 (7.1)	508 (70)	PS 0–1: 467 (64)PS $\geq$ 2: 40 (6)Missing: 218 (30)	2.3 (0.5; 5.3)
Trial	516 (13)	65.9 (10.1)	373 (72)	PS 0–1: 381 (73)PS $\geq$ 2: 7 (1)Missing: 128 (25)	2.0 (0.5; 4.6)
Ibrutinib	506 (12)	71.4 (9.4)	327 (65)	PS 0–1: 310 (61)PS $\geq$ 2: 48 (9)Missing: 148 (29)	5.2 (2.4; 8.0)
Chlorambucil/obinutuzumab	329 (8)	76.5 (6.5)	211 (64)	PS 0–1: 222 (68)PS $\geq$ 2: 36 (11)Missing: 71 (22)	2.3 (0.6; 4.4)
Chlorambucil/rituximab	308 (8)	79.7 (6.5)	179 (58)	PS 0–1: 166 (54)PS $\geq$ 2: 34 (11)Missing: 108 (35)	2.5 (0.5; 6.4)
Chlorambucil	270 (7)	82.4 (6.6)	154 (57)	PS 0–1: 101 (37)PS $\geq$ 2: 70 (26)Missing: 99 (37)	2.6 (0.7; 5.5)
Other†	230 (6)	71.1 (9.8)	156 (68)	PS 0–1: 122 (53)PS $\geq$ 2: 16 (7)Missing: 92 (40)	2.2 (0.3; 5.2)
Rituximab	198 (5)	72.8 (12.0)	139 (70)	PS 0–1: 102 (52)PS $\geq$ 2: 17 (9)Missing: 79 (40)	2.6 (0.7; 5.3)
Chlorambucil/ofatumumab	84 (2)	77.5 (7.0)	49 (58)	PS 0–1: 47 (56)PS $\geq$ 2: 10 (12)Missing: 27 (32)	2.2 (0.5; 4.3)
Idelalisib/rituximab	83 (2)	73.7 (9.1)	57 (69)	PS 0–1: 55 (67)PS $\geq$ 2: 7 (8)Missing: 21 (25)	4.2 (1.2; 6.8)

SD, standard deviation; IQR, interquartile range.

*Following treatments had less than 50 patients: Alemtuzumab monotherapy, Bendamustine monotherapy, Idelalisib monotherapy, Obinutuzumab monotherapy, Ofatumumab monotherapy, Venetoclax monotherapy use not supported by the Cancer Drugs Fund (CDF), Fludarabine + Cyclophosphamide, Ibrutinib + Rituximab. Compounds supported by the CDF were excluded.

†Includes rare treatments.

received Ibrutinib with a median time from diagnosis to treatment of 5.2 years (IQR: 2.4–8.0). This may, in part, be influenced by the restricted reimbursement conditions of Ibrutinib in England. Chlorambucil-based combinations were found to be preferentially used in older patient groups with worse performance scores. A summary of our findings is shown in Table below.

Conclusion: In this contemporary study of CLL patients with a first recorded systemic treatment, chemo-immunotherapy (FCR / BR) continues to be the mainstay treatment modality of choice. The relative low usage of Ibrutinib in England likely reflects the restricted reimbursement conditions of this agent to patients with high-risk cytogenomic features in the first-line setting.

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Utility of FDG-PET CT-directed splenic biopsy in lymphoma diagnosis

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Lymphoma is the fifth most common type of cancer in United Kingdom. Therapy requires an accurate tissue diagnosis in all cases. Variegated clinical presentation can cause diagnostic difficulties, especially in those people without accessible lymph nodes. We report on the utility and safety of FDG-PET CT-planned splenic core biopsy in people suspected of having lymphoma.

We retrospectively identified people who underwent splenic core biopsy between 2009 and 2019. We also collected data on FDG PET-CT scan patterns observed prior to splenic biopsy. We determined the diagnostic yield by reviewing histology reports and the safety of the procedure by looking at the complications reported on an electronic record.

Twenty-one patients were identified. All had imaging performed which failed to show any accessible tissue for diagnosis, except the spleen. Prior investigation included bone marrow (14 patients) and skin biopsy (1). Fifteen (diagnostic yield 71.5%) achieved a diagnosis following splenic biopsy. Two of the six without a diagnosis following splenic puncture underwent splenectomy. Liver and bone marrow biopsy were performed on one person subsequent to splenic biopsy.

Ten of 15 people achieving a diagnosis were found to have lymphoma (see table: six with diffuse large B-cell lymphoma, two small lymphocytic lymphoma, one with follicular and one marginal zone lymphoma); three had fungal infection, one had sarcoidosis and one infection with mycobacterium tuberculosis. Subsequent follow-up of those who were undiagnosed after splenic biopsy demonstrated lymphoma (three patients) and polycythemia vera (one patient).

FDG PET-CT scans were performed on 16 patients who had splenic biopsies. All demonstrated abnormal findings in the spleen.

Fourteen had abnormal homogeneous uptake while two had an abnormal heterogeneous pattern. The spleen was enlarged in 11 (69%) but of normal size in 5. FDG uptake was variable with SUVmax ranging from 1.5 to 24.8 for spleen with median of 8.5, while spleen: liver SUVmax was ranging from 1.2 to 20.3 with a median of

Table.

Histopathology of splenic biopsy	Final diagnosis	No. of cases
Lymphoma	Lymphoma	10
Granulomatous inflammation	Fungal infection	3
	Mycobacterium tuberculosis	1
	Sarcoidosis	1
Non-diagnostic	Lymphoma	3
	Polycythaemia vera	1
	No diagnosis	2
		21

3.2. None of the patients had a normal FDG-PET-scan prior to splenic biopsy.

Among the 16 with positive PET-CT findings we established a diagnosis in 14 (87.5%) after further splenic biopsy; 2 people with an abnormal PET (12.5 %) were undiagnosed after a positive PET and splenic procedure.

Regarding the safety of splenic biopsy, there were no procedure-related deaths; two (13.3%) developed perisplenic haematomas which were managed conservatively.

Splenic biopsy is safe and has a high diagnostic yield. Although FDG PET-CT scanning is a useful modality to guide targeting of the spleen and to ensure that no other, safer target is available, there is no level of avidity which can be said to represent lymphoma. It can also help prompt splenic biopsy in those with normal spleen size. We recommend FDG-PET CT and consideration of splenic biopsy in those people with suspected lymphoma, an abnormality on FDG-PET CT, no accessible lymph node and a normal bone marrow.

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The GELOX-radiotherapy sandwich regime is a highly effective outpatient regimen for limited stage extranodal NK/T-cell lymphoma nasal type with acceptable toxicity

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Extranodal natural killer/T-cell lymphoma, nasal type (ENKTL) is a rare subtype of non-Hodgkin lymphoma. The upper aerodigestive tract is the most commonly involved site, primarily the nasal cavity. Optimal treatment strategies for patients with ENKTL have not been fully established as there have been very few randomised trials and existing data is usually retrospective, involving small patient numbers. Here we present the outcome of patients with limited stage ENKTL treated with the GELOX-radiotherapy sandwich regimen at our Centre. (Wang L et al. Cancer. 2013;119: 348-355).

We retrospectively evaluated the efficacy and safety profiles of this combined modality regimen in patients with early-stage ENKTL. Two cycles of GELOX chemotherapy (gemcitabine 1250 mg/m², oxaliplatin 85 mg/m² and PEGasparaginase 2500 units/m² all given on day 1 of a 15 day cycle) were followed by radiation 55 Gy in 30 fractions and then 2 more cycles of GELOX in patients with early-stage ENKTL.

Efficacy endpoints were overall response rate (ORR), complete response rate (CMR), progression-free survival (PFS) and overall

survival (OS). Toxicity data were collected from patients electronic records as well as their radiotherapy and chemotherapy records.

Between 2013 and 2019, nine patients aged between (34–76 years) were newly diagnosed with ENKTL. Six patients were stage 1E and three were stage 2E. Eight of nine patients completed treatment. One patient died of complications of progressive haemophagocytic lymphohistiocytosis after one cycle of treatment before we could assess his response. All of the eight patients who managed to complete treatment attained a complete metabolic response. By intention to treat the overall response rate and CMR rate was 89%.

Median follow-up was 11.5 months (range 2–55). All eight patients who completed treatment are alive and free of lymphoma. One year overall and progression-free survival were both 89%.

GELOX treatment was well tolerated with few grade 3 and no grade 4 toxicity identified. There were no treatment-related deaths. Grade 3 neutropenia was observed in three patients (37.5%) although full blood count was not consistently performed. To our knowledge none of the patients required blood or platelet transfusions. One patient required admission for febrile neutropenia (grade 3) following chemotherapy. Most common non-haematological toxicities were nausea and vomiting (87.5%; grade 1:71.4%, grade 2:28.6%), mucositis (25%; grade 2) and deranged ALT (12.5%; grade 2). Two patients required admission for 5 and 7 days respectively for grade 2 nausea and vomiting. Two patients (25%) had reactivation of previous infections, one had pulmonary Tuberculosis who did not receive any prophylaxis during treatment while other patient developed genital herpes simplex virus infection despite aciclovir prophylaxis. No deterioration of renal functions was observed in any patient.

The median time from day 15 of cycle 2 of GELOX to initiation of radiation was 3 days (range 1–7 days) and median time from last day of radiation to day 1 cycle 3 GELOX was 3 weeks (range 2–8 weeks). Over all treatment time from day 1 cycle 1 GELOX to day 15 cycle 4 was 18 weeks (range 17–22 weeks).

Our results indicated that GELOX radiotherapy sandwich regimen is a highly effective treatment strategy for limited ENKTL which is well tolerated and can be administered as an outpatient therapy.

Disclosure of Interest: None Declared.

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Improved outcomes in patients with chronic lymphocytic leukaemia/small lymphocytic lymphoma (CLL/SLL) treated with ibrutinib during earlier lines of therapy

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Ibrutinib (ibr) is the only once-daily BTK inhibitor that has demonstrated significant overall survival (OS) benefit in three randomized,

phase 3 studies in patients (pts) with CLL/SLL (RESONATE, RESONATE-2, and ECOG 1912). This analysis evaluated long-term follow-up (FU) data from RESONATE and RESONATE-2 to compare efficacy and safety of ibr by number of prior lines of therapy.

This integrated analysis included pts treated with single-agent ibr in first-line (1L; RESONATE-2, NCT01722487) and relapsed/refractory (R/R; RESONATE, NCT01578707) settings. R/R pts were grouped by 1–2 vs. ≥ 3 prior lines of therapy. Pts with del(17p) were excluded, consistent with their exclusion from RESONATE-2. High-risk prognostic features included TP53 mutation, del(11q) and/or unmutated IGHV. Outcomes included investigator-assessed progression-free survival (PFS) and objective response rate (ORR), OS, and safety.

This analysis included 271 pts (1L, $n = 136$; 1–2 prior, $n = 68$; ≥ 3 prior, $n = 67$). 1L pts were older (median age [range], 73 years [65–89]) than those with 1–2 prior (65 years [30–86]) or ≥ 3 prior lines (67 years [44–83]). The proportion of pts with high-risk prognostic features was lowest in 1L pts (1L, 54%; 1–2 prior, 81%; ≥ 3 prior, 76%). Median FU times were 59.8, 66.2 and 65.1 months for 1L, 1–2 prior and ≥ 3 prior lines respectively. Median PFS was not reached (NR) for 1L or 1–2 prior lines and was 40.1 months for ≥ 3 prior lines. The proportion of pts remaining progression-free or alive at 60 mo was greatest in pts who received ibr in earlier lines (1L: 70%; 1–2 prior: 60%; ≥ 3 prior: 33%). 1L treatment (tx) resulted in a 34% reduction in risk of disease progression or death (HR: 0.66 [95% CI, 0.40–1.09]) compared with 1–2 prior lines. PFS was significantly prolonged for 1L vs. ≥ 3 prior lines (HR: 0.32 [95% CI, 0.21–0.49]) and 1–2 prior vs. ≥ 3 prior lines (0.48 [0.30–0.77]). In pts with high-risk prognostic features, median PFS was NR for 1L and 1–2 prior lines and was 42.5 months for ≥ 3 prior lines; prolonged PFS occurred in these pts treated with ibr in earlier lines (1L vs. 1–2 prior, HR: 0.64 [95% CI, 0.35–1.18]; 1L vs. ≥ 3 prior, 0.33 [0.19–0.57]; 1–2 vs. ≥ 3 prior, 0.51 [0.30–0.87]). OS medians were NR (range, 0.10+ to 66.04+) for 1L, NR (5.98 to 71.56+) for 1–2 prior, and 67.4 mo (1.15 to 69.78+) for ≥ 3 prior lines. ORRs were 91%, 94% and 82% for 1L, 1–2 prior and ≥ 3 prior lines respectively. Complete response (CR) rate (CR+CR with incomplete marrow recovery) was highest for 1L pts (1L: 30%; 1–2 prior: 12%; ≥ 3 prior: 10%). At the time of analysis, 58% of pts remain on ibr in the 1L group. Prior to study closure, 38% of pts with 1–2 prior and 18% of pts with ≥ 3 prior lines remained on ibr. Overall, eight pts (6%) in the 1L group discontinued due to progressive disease (PD). PD was the most common reason for discontinuation in the 1–2 ($n = 15$, 22%) and ≥ 3 prior lines ($n = 25$, 37%) R/R subgroups. Across all pts, 52 pts (19%) discontinued due to adverse events (AEs) (1L: $n = 29$, 21%; 1–2 prior: $n = 13$, 19%; ≥ 3 prior: $n = 10$, 15%). AEs leading to dose reduction occurred in 20% of 1L, 13% of 1–2 prior and 22% of ≥ 3 prior lines.

Overall, this integrated analysis of CLL pts with up to 6 y of long-term FU demonstrates that using single-agent ibr in earlier lines of tx improves PFS, OS and ORR with sustained efficacy, including in pts with high-risk prognostic features. Of pts treated in the 1L setting, only 6% discontinued due to PD. Ibr was well tolerated; only 19% of all pts discontinued due to AEs.

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A systematic evaluation of physiologically relevant conditions for the maintenance of chronic lymphocytic leukaemia cells *ex vivo*: towards an optimal system for assessing the impact of therapeutic compounds

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Chronic lymphocytic leukaemia (CLL) is the most common adult leukaemia in the Western world. While recent advances in treatment strategies have dramatically improved outcomes for patients, CLL is still considered an incurable disease. Hence the quest for new treatment modalities continues apace. Alongside an upsurge in the development of drugs targeting CLL, comparable efforts are being directed towards modelling *in vitro* the pathophysiological niches that support CLL clonal growth and survival *in vivo*. Here we have compared three of what are currently considered exemplar *ex vivo* culture conditions in supporting CLL cells – both *ab initio* and with the components of each system incrementally combined. Components of such 'proliferation niches' modelled here included: the cytokines IL-4, IL-15 and IL-21; the TLR-9 agonist CpG-ODN; CD154 (CD40 ligand)-transfected fibroblasts; and supplements considered (at least partial) substitutes for stromal cells. Together with monitoring the capacity of the culture systems to maintain CLL cell viability and support clonal growth, we compared the utility of each in disclosing the impact and efficacy of three classes of selective therapeutic inhibitors: (i) carfilzomib (an irreversible proteasome inhibitor); (ii) duvelisib (a PI3-kinase inhibitor); (iii) zanubrutinib (a Bruton's tyrosine kinase inhibitor). CLL, despite a high level of phenotypic uniformity is a highly heterogeneous disease genetically and clinically and, unsurprisingly, there was a degree

of variability in the response profile of individual patient's clones with respect to both basal cultures and drug impact. Notwithstanding, some general trends emerged. The most consistently effective culture systems with regards supporting clonal growth were those that included human fibroblasts that express cell surface recombinant CD154. In stroma-/CD154-free cultures, optimal or near-optimal growth support could be achieved with CpG-ODN plus IL-15, but only where (and consistent with an earlier report) the 'stroma-replacing' additives insulin, transferrin, selenium and 2-mercaptoethanol were also included. Other combinations of signals either further enhanced modestly one or other of the aforementioned conditions or had a 'null' effect. Of the therapeutic compounds tested, carfilzomib had the most consistent (negative) impact on 'niche'-supported CLL cells in terms of both viability and proliferation. The detailed outcomes from all three classes of compounds under each culture condition studied will be presented and discussed for individual cases of CLL and the sample cohort in its entirety. In summary, this study demonstrates the utility of testing different classes of drugs in a variety of *in vitro* models reflecting CLL niches believed to be operative *in vivo*.

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Risk stratifying patients presenting with lymphadenopathy – an audit of 2 weeks wait referrals for suspected haematological malignancy compared to NICE guidance

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Lymphoma is the 6th commonest cancer in the United Kingdom and given that the 5-year survival rates in the United Kingdom are lower than in the rest of Europe, there is need for improvement in the diagnosis and management of patients with lymphoma. One aspect to consider is the effectiveness of the 2-week wait (2ww) referral system for suspected lymphoma and how it compares with NICE guidance. The current NICE guidance suggests that a 2ww referral should be made 'in adults presenting with unexplained lymphadenopathy or splenomegaly...[taking into account] associated symptoms, particularly fever, night sweats, shortness of breath, pruritus and weight loss'.

The population for this study was 100 patients seen in 2017 at Good Hope Hospital via 2ww referral for suspected haematological malignancy. This audit focuses on the subset who presented with lymphadenopathy.

1. Lymphadenopathy was the second most frequent symptom, after fatigue, which generated a 2ww referral in these 100 patients. However, only 12% (5 of 41) of these patients were found to have lymphoma while 78% were found to have reactive lymphadenopathy or no pathology was identified. Therefore, lymphadenopathy had a low specificity for lymphoma and the majority of patients referred with this finding had no malignancy. This suggests a need for improvement in the ability of primary care clinicians to use clinical

history and examination of lymphadenopathy to help differentiate between benign and malignant causes.

2. Of the 41 patients referred with lymphadenopathy, only 68% had associated symptoms listed in the NICE referral criteria. This suggests that 2ww referrals for suspected lymphoma had a relatively low concordance with NICE guidance.

3. Out of the 34 patients who presented with lymphadenopathy but did not have a haematological malignancy, 50% had other NICE defined symptoms that prompted a suspected lymphoma referral. Of the five who presented with lymphadenopathy who were diagnosed with lymphoma, only one had these other symptoms. This suggests that in this population, the associated symptoms in the NICE guidelines were not specific or sensitive for lymphoma. This emphasises the variable presentation of lymphoma and the difficulty of using associated symptoms to risk stratify patients with lymphadenopathy.

4. Only one patient out of those presenting with lymphadenopathy who did not have haematological cancer had an elevated LDH level, while of the five who had lymphoma, 80% had an elevated LDH level. Although not used as a diagnostic tool for lymphoma, this suggests the potential value of testing for raised serum LDH in primary care before making a 2ww referral to help risk stratify patients with lymphadenopathy.

5. Of the 34 patients found not to have haematological malignancy, 41% were referred from the haematology clinic for imaging (mainly ultrasound scanning (USS)) which ruled out malignancy. Given that 78% of those presenting with lymphadenopathy had no malignancy and that USS has a high sensitivity and specificity in differentiating lymph node pathology, this suggests the potential value of an urgent USS of any suspicious lymphadenopathy where a 2ww referral is likely to be made.

In conclusion, requesting an LDH level and USS scanning from primary care for patients with suspected lymphoma could potentially reduce the number of unnecessary referrals.

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Vaccination recommendation in newly diagnosed chronic lymphocytic leukaemia: a retrospective audit

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Chronic lymphocytic leukaemia (CLL) is a chronic lymphoproliferative disorder characterised by a progressive accumulation of functionally incompetent lymphocytes of monoclonal origin. After initial diagnosis there is low risk of progression over the following 5 years and treatment often involves a 'watch & wait' approach. CLL is characterised by a gradual reduction in humoral immunity and hypogammaglobulinaemia. Studies have indicated there is a high incidence of infection in CLL patients even in the early stages of disease. This has been attributed to suboptimal antibody response in patients with normal IgG concentration and a reduction in overall IgG concentration in others, that is, hypogammaglobulinaemia. Strategies to reduce the risk of infection include vaccination and antibiotic prophylaxis. The British Society of Haematology guidelines recommend that all patients with CLL should be offered seasonal influenza vaccination and pneumococcal vaccination (Prenvar 13-conjugate vaccine and then Pneumovax 23-polysaccharide vaccine).

The aim of this audit was to determine if newly diagnosed CLL patients attending 1st Haematology Clinic were routinely being advised to get seasonal influenza and pneumococcal vaccinations.

Data were collected retrospectively from 1 April 2017 to 30 September 2019. Patients were identified from laboratory data which

contained a phenotype report of 'CLL'. One hundred and forty patients were initially identified with a final cohort of 115. Patient information was obtained from the electronic record system and their clinical Haematology letters reviewed for vaccination recommendations.

Our study revealed of the 115 patients reviewed only 31 had documentation that advised the above vaccinations. This means that 73% of newly diagnosed CLL patients are not being routinely advised regarding vaccinations at their first haematology clinic review. This may impact the number of CLL patients being vaccinated which in turn leads to an increased risk of infection in this patient group. This is an important finding as recurrent infections lead to increased number of hospital admissions. It is also important when one considers that the median age of diagnosis of CLL is 65 meaning many of these patients will have a number of co-morbidities which can be exacerbated by infection. This can lead to higher mortality and morbidity in this population.

This audit highlights that vaccinations are not being routinely advised at first clinic visit. Vaccinations can offer some prophylactic protection against common viruses. It can also be used to identify patients with a suboptimal vaccination response. These patients may later benefit from immunoglobulin replacement therapy.

Our recommendations include:

- 1 Education: Poster & presentation of results to Haematology Department to increase awareness which was completed in December
- 2 Proforma for clinic letters/checklist for newly diagnosed CLL patients to ensure recommendations regarding vaccination are carried out. A standard paragraph advising on vaccinations for all new patients with CLL. This will be added to clinic letters before sign off.
- 3 Re-audit once awareness raised and proforma implemented in 6 months time

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Implementing venetoclax treatment into day case setting for patients at high risk of tumour lysis syndrome; University Hospital Southampton NHS Foundation Trust experience

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Patients requiring Venetoclax therapy are rapidly increasing. While complications during and after treatment are generally not severe, in the firsts weeks of therapy there is a variable risk of tumour lysis syndrome (TLS), which may be fatal if not managed with adequate prophylaxis through the dose-escalation phase. The initial recommendation for patients at high risk of TLS is hospitalisation. However, this imposes a high turnover of admissions/discharges, highly trained staff to manage drug-related complications, while being more costly. Administration of Venetoclax in a day case setting with specialised haematology personnel (i) can improve patient experience (ii) can reduce use of inpatient beds (iii) can improve time spent on co-ordination and administration of Venetoclax. We investigated the number of patients actually required admission to inpatient wards for TLS management, in order to address the following questions: (i) how to improve patient's experience (ii) how to minimise hospital stay (iii) how to reduce the administrative workload (iv) how to set up a safe and effective service (v) whether the day case setting is a safe way to deliver Venetoclax. We reviewed all the patients having

commenced Venetoclax treatment in our Haematology Day Unit between May 2018 and Sep 2019, excluding patients receiving Venetoclax on clinical trials. Primary endpoint was to quantify the number of patients were admitted as inpatients for TLS management during dose escalation. Bishop's scoring system was used to distinguish TLS high-risk (including medium-risk) and low-risk patients. Management protocols included monitoring of potassium, uric acid, calcium, phosphate, creatinine in the serum; hydration, diuretics, ion-chelating agents or anti-hyperuricemia drugs according summary of product characteristics (SmPC) by the European Medicines Agency (EMA). Financial cost of treatment was calculated using the SUHT hospital charges (Day-case stay/patient/day=£107; In-patient stay/patient/night=£339). Total of 13 patients ($n = 4$ low-risk and $n = 9$ high-risk) were identified. Only one of the nine high-risk patients required admission to hospital. Although grade 2 hyperphosphataemia was documented, this specific patient was admitted because of severe lower respiratory tract infection. The other patients all experienced a laboratory grade 1–2 TLS which resolved using our internal TLS protocol without extension of hours of chair occupation, opening time of Day Unit or involvement of additional administrative, nursing or medical staff. By analysing the cost of a high-risk patient with potential indication for admission to in-patient wards (Hospital stay/patient = 2nights/weekX5 weeks = 10nightsX£339 = £3390/patient) versus a patient treated in the Day Unit (Day case treatment/patient = 3visits/weekX5 weeks = 15visitsX£107 = £1605/patient), we calculated a potential save of £14 280 by our 8 high-risk patients, that is, £178 500 every 100 patients receiving 5 weeks of dose-escalated Venetoclax therapy. We find that high-risk patients rarely require admission to hospital. Treatment of high-risk patients in the day case setting is feasible and significantly less expensive than admission to hospital. In light of our results, we have modified our internal policies and taken the following actions: (i) revision of treatment protocols (ii) writing nursing guidelines (iii) regular teaching to Day Unit personnel, including Haematology Consultants, doctors in training and nurses.

Disclosure of Interest: None Declared.

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18Fluoromethylcholine –positron emission tomography computed tomography is not inferior to gadolinium-enhanced magnetic resonance imaging in central nervous system lymphoma imaging

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Primary central nervous system lymphoma (PCNSL) accounts for approximately 4% of newly diagnosed central nervous system (CNS) tumours and 1–2% of all lymphomas. Secondary CNS involvement occurs in 2–3 to 10% of patients with diffuse large B-cell Lymphoma (DLBCL) but can also occur in other lymphoma subtypes.

The gold standard for central nervous system (CNS) imaging in primary and secondary CNS lymphoma is with gadolinium-enhanced magnetic resonance imaging (MRI). End of treatment (EoT) imaging is used to predict if patients have achieved a remission or whether further treatment is warranted.

There have been recent safety concerns over exposure to gadolinium-based contrast agents (GBCAs) leading to the European Medicines Agency (EMA) restricting or suspending the use of linear products in 2017.

¹⁸Fluoromethylcholine (FCH)- positron emission tomography (PET) computed tomography (CT) uses an analogue of choline, which was first developed as a PET tracer in 2000. The expression of choline kinase is upregulated in tumour cells and allows increased uptake of FCH. FCH-PET is used in the detection of prostate cancer, breast cancer, parathyroid adenomas and hepatocellular carcinoma. As there is minimal background grey matter uptake of FCH, FCH-PET can be used to detect tumours with a high lesion-to-CNS background ratio and has been used in diagnostic and follow-up imaging in brain tumours, particularly high-grade gliomas. Based on the glioma studies our centre has been using FCH PET alongside MRI in CNS lymphoma since 2011.

We identified 41 patients with a diagnosis of primary or secondary CNS lymphoma between 1 November 2011 and 10 October 2019, who had completed chemotherapy and had end of treatment imaging with both gadolinium-enhanced MRI and FCH-PET. Twenty-seven of the 41 patients (66%) had concordant results. Of the discordant results, 9 of 14 had partial response (PR) on MRI but showed a complete metabolic response (CMR) on FCH-PET. For the whole cohort the overall response rates (ORR – complete response (CR) or PR) were similar for both modalities, with an ORR of 88% for MRI and 93% with FCH-PET. For the whole cohort 100-day PFS was 78% and 2-year PFS was 51.2%. Thirty-eight patients achieved ORR on both MRI and FCH-PET. Of these, 17 patients (47%) relapsed. PFS was 86.1% at 100 days and 55.6% at 2 years. Whether patients achieved CR or PR on EoT imaging did not predict early relapse (at 100-days) or relapse by 2 years. Patients were further categorised into three response levels. Response level 1 achieved CR in both modalities ($n = 15$), response level 2 achieved PR on MRI and CR on FCH-PET ($n = 11$) or CR on MRI and PR on FCH-PET ($n = 1$), and response level 3 achieved PR on both modalities. Again, there was no difference in early relapse at 100 days or by 2 years.

FCH-PET is a reasonable alternative or additional mode of imaging to gadolinium-enhanced MRI brain for assessment of CNSL, but further studies are warranted to assess whether it is superior in predicting relapse and for scanning post-treatment on ad hoc basis if concerned for relapse.

Disclosure of Interest: None Declared.

BSH2020-EP-070

Local experience of the use of IVE methotrexate chemotherapy for the treatment of systemic T-cell lymphoma

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T-cell lymphoma is a rare subset of lymphoma often associated with poor outcomes. The data for optimal management can be conflicting due to the small numbers of patients and lack of randomised data. There is a paucity of clear standards of care for treatment.

Many centres use CHOP chemotherapy, but the outcomes are poor. The addition of etoposide only benefits patients with ALK-positive anaplastic large cell lymphoma (ALCL) under 60 years of age and these patients receive CHOEP chemotherapy as a result. At the Freeman Hospital, Newcastle-upon-Tyne we have been using a regime of IVE-methotrexate (Ifosfomide, vincristine, etoposide and methotrexate) chemotherapy followed by autologous stem cell transplantation (ASCT) for patients with a good performance status, following previous data suggested that this is a viable and effective regime for enteropathy associated T-cell lymphoma.

Consecutive patients were retrospectively reviewed, 14 patients with systemic T-cell lymphoma were treated between 2013 and 2018: 6 Angioimmunoblastic T-cell lymphoma, 4 ALK negative ALCL, 2

peripheral T-cell lymphoma with systemic involvement and 2 enteropathy type intestinal T-cell lymphoma.

Patient demographics show an age range of 33–65 with a median of 53. All patients had a performance status 0–1. Ten patients had stage 3–4 disease and three had bulky disease of greater than 5 cm.

The median progression-free survival (PFS) was 29.5 months with six patients remaining in remission following IVE MTx and ASCT alone. During treatment three patients had progressive disease; one patient subsequently had CHOEP chemotherapy and is now in remission after ASCT, one patient had DHAP but progressed and died and one patient died with no further treatment. CR rate at the end of treatment (before proceeding to ASCT) was 77%. Four patients had a subsequent relapse ranging from between 12 and 49 months, one had DHAP and three had salvage chemotherapy followed by allogeneic transplantation.

Median overall survival (OS) has not been reached, as 9 of the 14 patients remain alive. This is better than would be expected, in the Vose et al international T cell project 2-year survival would be around 50–60% for patients in our cohort. Of the five who have died, two were due to progressive disease, one relapsed disease who had DHAP refractory disease and the 4th patient died of relapse disease post allogeneic stem cell transplant. One patient died of treatment related toxicity.

Several patients have not reached either PFS or OS consequently making averages potentially inaccurate. Nevertheless, of the patients who have achieved ongoing remission the mean PFS is 28 months and OS 34 months.

Haematological toxicity was seen, 7 patients developed neutropenic sepsis with all 14 having grade 3 or 4 neutropenia, 7 had grade 3–4 anaemia and 9 grade 3–4 thrombocytopenia. One patient died as a consequence of treatment related toxicity. One patient was diagnosed with lung adenocarcinoma after four cycles of chemotherapy and did not proceed to have an autologous stem cell transplant but remains in remission and alive. One patient developed aplasia post allogeneic stem cell transplant again not thought to be due to the IVE methotrexate initial treatment.

In summary, with the limitations of retrospective data and small numbers of patients, these outcome data are superior to previously documented CHOP responses and supports local practice to offer IVE methotrexate then autologous stem cell transplant to patients with systemic T-cell lymphoma.

Disclosure of Interest: None Declared.

BSH2020-EP-071

Indirect treatment comparison analysis of venetoclax + obinutuzumab with standard front-line regimens for chronic lymphocytic leukaemia

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The novel regimen of 12-month fixed duration treatment of venetoclax + obinutuzumab (GA-101 [G]; VenG) has recently demonstrated

improved efficacy *versus* G + chlorambucil (G-Clb) as first-line (1L) treatment in chronic lymphocytic leukaemia (CLL). However, there is an absence of head-to-head clinical evidence on the comparative efficacy of VenG *versus* standard 1L chemoimmunotherapy regimens like fludarabine + cyclophosphamide + rituximab (FCR) and bendamustine + R (BR). In addition, ibrutinib (I) + R have recently demonstrated improved efficacy *versus* FCR and BR. This study aimed to indirectly compare relative efficacy, including progression-free survival (PFS), overall response rate, (ORR) and complete response (CR) rate of VenG *versus* FCR, BR and IR.

Indirect treatment comparison (ITC) was conducted using two methodologies, the Bucher (frequentist) method (Bucher et al, *J Clin Epidemiol* 1997) and a Bayesian method (network meta-analysis [NMA]; Hoaglin et al, *Value Health* 2011). The Bucher method compared VenG to FCR and BR by linking through common treatment arms across five phase 3 randomized clinical trials in 1L CLL with available data (Table 1) using investigator-assessed endpoints. The Bayesian approach applied fixed-effect modelling to a network of 10 randomized clinical trials and allowed for probabilistic interpretation of uncertainty around the PFS estimates. Approximately 60 000 Monte Carlo iterations were used in the simulations to estimate the 95% confidence interval (CI).

ITC analyses suggested improved PFS for VenG *versus* FCR [hazard ratio (HR) 0.429, 95% CI = 0.221–0.831] and BR (HR 0.261, 95% CI = 0.142–0.479). The likelihood of achieving ORR and CR was also higher with VenG compared to FCR (ORR odds ratio 5.029, 95% CI = 1.593–15.879; CR odds ratio 2.450, 95% CI = 0.858–6.998) and BR (ORR odds ratio 4.677, 95% CI = 2.053–10.653; CR odds ratio 3.623, 95% CI = 1.347–9.748). When compared to IR, PFS for VenG was similar (HR 1.218, 95% CI = 0.540–2.749). Overall survival data were not mature enough to make any meaningful comparison at this time. Inherent limitations of this modelling are acknowledged, including trial differences such as treatment durations and patient demographics; however, consistency of results between the two methods support the overall findings.

These ITC analyses suggest that VenG may lead to prolonged PFS and higher ORR and CR rates compared to FCR and BR. Moreover, the PFS achieved with a 12-month fixed duration VenG regimen appears similar to that achieved with a continuous ibrutinib-based regimen. Although there are inherent limitations of ITCs, our study provides useful estimates of the relative efficacy of these regimens while we await the results of definitive, prospective, comparative trials such as GAIA/CLL13 (NCT0295005) and FLAIR (2013-001944-76).

Table 1. Cross-trial linking based on treatment combinations.

Trial	Treatment combinations					
	VenG	FCR	BR	G-Clb	R-Clb	IR
CLL14 (NCT02242942)	X			X		
CLL11 (NCT01010061)				X	X	
MaBLLe (NCT01056510)			X		X	
CLL10 (NCT00768522)		X	X			
ECOG-E1912 (NCT02048813)		X				X

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research funding from TG Therapeutics, DTRM, Celgene, Debiopharm, C. Moreno Conflict with: Honoraria as advisor/speaker from Janssen, AbbVie, Pharmacyclis, Gilead, Roche; research funding from Gilead, Roche, Janssen, Conflict with: Honoraria as advisor/speaker from Janssen, AbbVie, Pharmacyclis, Gilead, Roche; research funding from Gilead, Roche, Janssen, G. Follows Conflict with: Consulting/paid lecturing for Roche, AbbVie, Janssen, Bayer, A. Cuneo Conflict with: Advisory board and/or speakers' bureau for Janssen, AbbVie, Amgen, Gilead, Novartis, Roche, Conflict with: Advisory board and/or speakers' bureau for Janssen, AbbVie, Amgen, Gilead, Novartis, Roche, K. Humphrey Conflict with: Employee of Roche and may own stock or stock options, W. Scharly Conflict with: Employees of AbbVie and may own stock or stock options, K. Sail Conflict with: Employees of AbbVie and may own stock or stock options, Y. Song Conflict with: Employees of AbbVie and may own stock or stock options, Z. Gao Conflict with: Employees of AbbVie and may own stock or stock options, Z. Tang Conflict with: Employees of AbbVie and may own stock or stock options, G. Pena Conflict with: Employees of AbbVie and may own stock or stock options, R. Foà Conflict with: Advisory board and/or speakers' bureau for Janssen, AbbVie, Amgen, Incyte, Novartis, Roche, P. Hillmen Conflict with: Honoraria for speaker's fees/consultancy from Janssen, AbbVie, Roche, Celgene; institutional research funding from Janssen, Pharmacyclis, AbbVie, Roche, Novartis, Gilead.

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Impact of lisocabtagene maraleucel (liso-cel) treatment on health-related quality of life and health utility in patients with relapsed/refractory aggressive B-cell non-Hodgkin lymphoma: TRANSCEND NHL 001 (NCT02631044)

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TRANSCEND NHL 001 is a phase 1 study in patients (pts) with relapsed/refractory (R/R) B-cell non-Hodgkin lymphoma (NHL) receiving liso-cel, an investigational, CD19-directed, defined composition, 4-1BB, chimeric antigen receptor (CAR) T-cell product administered at target doses of CD8⁺ and CD4⁺ CAR⁺ T cells. Here, we report patient-reported outcomes (PROs) from the large B-cell lymphoma (LBCL) cohort.

Eligible pts (≥18 years of age with ≥2 lines of prior therapy) with R/R diffuse LBCL not otherwise specified (including transformed from any indolent lymphoma), high-grade B-cell lymphoma with *MYC* and *BCL2* or *BCL6* rearrangements, primary mediastinal B-cell lymphoma, or follicular lymphoma grade 3B received lymphodepletion followed by liso-cel infusion. PROs in the LBCL cohort were evaluated using the European Organisation for Research and Treatment of Cancer Quality of Life Questionnaire Core 30 (EORTC QLQ-C30) and the EuroQol 5-Dimension 5-Level questionnaire (EQ-5D-5L; utility index comprising the EQ-5D based on US tariffs and EQ-visual analogue scale [VAS]).

The EORTC QLQ-C30 (score, 1–100; higher scores indicate improved health-related quality of life [HRQoL] or higher symptom

Table. Summary of mean (SD) EORTC QLQ-C30 and EQ-5D-5L scores

	EORTC QLQ-C30 domains of interest				EQ-5D-5L	
	Global health status	Physical functioning	Fatigue	Pain	Health index score	EQ-VAS
General US population norm data, ^{*,†} mean (SD)	63.9 (22.9)	80.8 (25.2)	31.9 (27.8)	27.5 (30.2)	NA	NA
Baseline, <i>n</i>	181	181	181	181	186	186
Mean (SD)	62.3 (20.3)	77.8 (19.2)	38.2 (21.8)	25.6 (25.8)	0.817 (0.120)	68.3 (19.5)
Month 1, <i>n</i>	160	160	160	160	165	164
Mean (SD)	66.4 (21.9)	72.7 (24.8)	37.7 (26.0)	20.0 (24.8)	0.794 (0.157)	70.3 (20.2)
Month 2, <i>n</i>	146	147	147	147	150	149
Mean (SD)	74.5 (18.9)	80.8 (19.7)	27.1 (21.1)	17.7 (21.7)	0.837 (0.136)	77.1 (17.2)
Month 3, <i>n</i>	135	134	134	135	138	138
Mean (SD)	72.7 (19.7)	79.6 (19.5)	30.2 (22.8)	22.7 (24.2)	0.822 (0.145)	75.4 (18.9)
Month 6, <i>n</i>	81	81	81	81	85	84
Mean (SD)	74.3 (18.6)	80.2 (18.9)	28.5 (20.9)	17.9 (23.2)	0.833 (0.122)	78.0 (18.0)
Month 9, <i>n</i>	65	65	65	65	65	65
Mean (SD)	75.8 (16.5)	81.1 (18.1)	27.5 (19.7)	24.4 (24.7)	0.833 (0.122)	80.1 (15.6)
Month 12, <i>n</i>	38	38	38	38	38	38
Mean (SD)	79.8 (20.1)	83.2 (20.3)	23.4 (23.5)	18.4 (25.9)	0.848 (0.157)	82.1 (17.8)

NA, not available; norm, normal; SD, standard deviation.

*EORTC QLQ-C30 norm data for general US population (*N* = 1009) (Nolte S, et al. *Eur J Cancer*. 2019;107:153–163).

†PRO assessments were implemented into the study after a protocol change; therefore, only pts enrolled after this change were eligible for the assessments. The number of assessments declined over 12 months due to loss to follow-up, deaths, length of time on study and other reasons.

burden) and EQ-5D-5L (EQ-5D score, 0–1; EQ-VAS score, 0–100; higher scores indicate improved health) were completed at baseline (BL) before liso-cel infusion and postinfusion at months 1 (day 29), 2, 3, 6, 9 and 12. HRQoL was evaluated using prespecified EORTC QLQ-C30 domains: global health status (GHS), physical functioning (PF), fatigue and pain. A ≥ 10 -point score change (increase for functional/health status; decrease for symptoms) from BL was considered a clinically meaningful improvement. Proportions of pts with a clinically meaningful change (i.e. ≥ 10 -point improvement or worsening) in prespecified EORTC QLQ-C30 domains were calculated based on change scores from BL to months 6 and 12. Mean change from BL was calculated for the utility index (EQ-5D) and VAS scores.

At data cut-off, 181 and 186 of 268 pts (mean age, 60 years) were evaluable for the EORTC QLQ-C30 and EQ-5D-5L respectively. Mean EORTC QLQ-C30 GHS scores were improved from BL at months 1–12, while mean PF scores declined at month 1 and then improved from BL at months 2–12. Pt-reported fatigue symptom burden remained similar at month 1, followed by improvement from BL at months 2–12 postinfusion (Table). Mean pain symptoms were consistently lower from BL at months 1 to 12.

The mean change from BL score ranged from 4.3 to 16.9 for GHS, –4.8 to 6.5 for PF, –15.5 to 0.0 for fatigue and –8.8 to –1.8 for pain scores through month 12. A notable proportion of pts demonstrated clinically meaningful improvement at months 6 and 12 in GHS (52% and 61%), PF (30% and 37%), fatigue (53% and 61%) and pain (35% and 47%) respectively. Mean EQ-5D-5L US-based health utility index scores decreased 1 month after liso-cel infusion, fluctuated between months 2 and 3, and then improved from months 6 to 12 compared with BL. Mean EQ-VAS scores were improved across all timepoints from BL at months 1 to 12.

Treatment with liso-cel in the LBCL cohort of TRANSCEND NHL 001 improved patient's HRQoL and health utility, as well as reduced fatigue and pain symptom burden, through month 12 after liso-cel infusion.

Disclosure of Interest: D. Patrick Conflict with: Honoraria: BMS, A. Powers Conflict with: Salary and stock ownership interest from BMS, M. Parisi Jun Conflict with: salary and stock ownership interest from BMS, Y. Kim Conflict with: salary and stock ownership interest from BMS, J. Garcia Conflict with: salary and stock ownership interest from BMS, C. Dehner Conflict with: salary and stock ownership interest from BMS, D. Maloney Conflict with: Juno, a Celgene/BMS company, Celgene, a BMS company, and Kite Pharma, a Gilead company, Conflict with: Honoraria: Juno, a Celgene/BMS company, Celgene, a BMS company, and Kite Pharma, a Gilead company, BioLineRX, Gilead, Novartis, Pharmacyclis, A2 Biotherapeutics; Patents: Juno, a Celgene/BMS company; Royalties: Juno, a Celgene/BMS company; Stock: A2 Biotherapeutics

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Real-world treatment patterns, adverse events and clinical outcomes in patients with chronic lymphocytic leukaemia treated with ibrutinib in the United Kingdom

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Bruton tyrosine kinase (BTK) is a key component of the B-cell receptor pathway and is a validated target for the treatment of chronic lymphocytic leukaemia (CLL). Ibrutinib is a covalent, small

molecule BTK inhibitor approved for treatment of CLL. We analysed treatment patterns, outcomes and adverse events (AEs) in patients with CLL treated with ibrutinib in a real-world setting.

A retrospective chart review was conducted in patients with CLL who were treated with ibrutinib in UK oncology centres. Centre selection was non-random, but all geographic regions were represented. Patients were eligible if they initiated ibrutinib after diagnosis of CLL, between January 2017 and June 2018, with at least 12 months of follow-up data available. Patients who died <12 months after ibrutinib initiation were also eligible. Haematology/oncology physicians reviewed selected medical records for eligibility and completed web-based data collection forms. Baseline medical history and data on treatment characteristics and AEs were collected. All analyses were descriptive.

Thirty-four physicians contributed data for 259 ibrutinib-treated patients. In these patients, median follow-up was 16.7 months from ibrutinib initiation (index date) and 62.0 months from initial CLL diagnosis; median age was 71 years at ibrutinib initiation, 56% were male, 22.0% had del(17p) deletions, 21.6% had TP53 mutations, and hypertension was the most prevalent comorbidity (43.2%). First-line ibrutinib was initiated in 52 patients (20.1%) while 151 (58.3%) and 48 patients (18.5%) initiated ibrutinib as second- or third-line therapy respectively. Median time to ibrutinib initiation was 43.7 months from CLL diagnosis and 84.6% of patients were still using ibrutinib at end of follow-up. Other therapies received included the combination of fludarabine, cyclophosphamide and rituximab (first-line, 24.3%), bendamustine plus rituximab (first-line, 20.9%; second-line, 13.5%) and chlorambucil plus rituximab (first-line, 8.5%). Overall, the median progression-free survival was 28.7 months and median overall survival was not reached. The most common AEs observed during ibrutinib therapy were bruising (19.3%), cytopenias (17.0%), diarrhoea (13.9%) and arthralgia (11.6%). Forty patients (15.4%) discontinued ibrutinib, mainly due to progressive disease (42.5%) or toxicity (22.5%); the median crude time to discontinuation was 10.1 months. The overall median crude time to dose reduction was 4.2 months (0.8–22.2 months) in 37 (14%) patients and was similar regardless of when ibrutinib was initiated. Toxicity was the most common reason (81.1%) for dose reduction. Among patients who initiated further therapy following discontinuation of first- or second-line ibrutinib ($n = 13$), venetoclax was the most common therapy (53.8%).

This analysis describes patient characteristics, outcomes and treatment patterns in ibrutinib-treated patients in the UK. The majority of ibrutinib use was in the second-line or later. AEs such as bruising and cytopenias were commonly reported in patients treated with ibrutinib.

Disclosure of Interest: J. Xie Conflict with: AstraZeneca LP – Employment, A. S. M. Yong Conflict with: AstraZeneca – Employment, C. Waweru Conflict with: AstraZeneca LP – employment, T.-A. Sorof Conflict with: Acerta Pharma (a member of the AstraZeneca group) – Employment, R. K. Goyal Conflict with: RTI Health Solutions (full time employee), which received funding from AstraZeneca to conduct this research. RTI Health Solutions is a business unit of Research Triangle Institute, which is an independent, nonprofit, research organization that does work for government agencies and private companies. K. L. Davis Conflict with: RTI Health Solutions (full time employee), which received funding from AstraZeneca to conduct this research. RTI Health Solutions is a business unit of Research Triangle Institute, which is an independent, nonprofit, research organization that does work for government agencies and private companies. P. Hillmen Conflict with: AbbVie, Janssen, Pharmacyclis LLC, an AbbVie Company, Conflict with: Roche, Gilead, Apellis, Pharmacyclis LLC, an AbbVie Company.

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Is the all-oral DEVEC metronomic schedule a better alternative to infusional chemotherapy protocols in elderly with peripheral T-cell lymphoma?

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Background: From registry data, median overall survival (OS) and progression-free survival (PFS) of peripheral T-cell Lymphoma patients (PTCLs) is 20 and 10 months respectively. Instead, in relapsed/refractory PTCLs (rr-PTCL) OS and PFS is 5.8 and 3.7 months respectively (for transplant ineligible patients). Prognosis is even worst in the elderly, for whom there is not a standard of care. Therefore, we designed a new metronomic regimen termed DEVEC (Prednisolone, Etoposide, Vinorelbine, Cyclophosphamide) to be used as a palliative, all-oral chemotherapy schedule in elderly/frail PTCLs.

Methods: DEVEC foresees an induction and a deescalated maintenance phase both consisting in six cycles. Consecutive patients with PTCL, considered unsuitable for standard chemotherapies because elderly, frail or rr were enrolled in five Italian Institutions. HIV+ve or subjects with central nervous system involvement, were excluded. Pretreatment comprehensive geriatric assessment (CGA) was suggested for elderly patients. Restaging was scheduled after the 2nd induction cycle, at the end of induction phase and every 6 months thereafter. Patients were eligible for analysis if they had completed at least two cycles of DEVEC. Data were prospectively collected and retrieved as of 30 November 2019 (EC approval n° 4640). In vitro cell proliferation assays (144 h) were also performed on PTCL cell line HH with metronomic vinorelbine.

Results: From September 2013 to January 2019 17 PTCLs started treatment with DEVEC. **Patients features:** treatment-naïve = 5, rr = 12 [Refractory = 10; Relapsed = 2; median number of previous line in rr = 2 (1–4)]; age range = 56–87 (median 72 years); 8/17 (47%) were FRAIL (CGA); histologic subtype PTCL-NOS = 12; AITL = 2, ATLL = 1, NK/T nasal, Transformed-MF = 1; IPI 3–5 = 78%, PIT 3–5 = 76.5% (evaluated at the time of first diagnosis); relevant haematologic toxicities = 24% (i.e. ≥G4 for >6 days); extra haematologic toxicity of G ≥ 3 = 24% (one bacterial meningitis, one pneumonia, two neutropenic sepsis). No treatment-related deaths were recorded. Outcome: the ORR in naïve and rr was 100% and 58%, respectively; complete remission rate (CRR) was 40% in naïve and 25% in rr respectively. After a median follow-up time of 41 months (range = 14–61) the median PFS in naïve was 20 months (95% CI = 0.00–45 months) and in 11 months (95% CI = 4.2–17.7 months) respectively. Metronomic vinorelbine was highly effective on HH cells, with an IC50 of 43.1 ± 0.9 pmol/L.

Conclusion: Devised with palliative intent, DEVEC proved promisingly effective in a proportion of both naïve and rr PTCLs, with a possible direct activity on lymphoma cells. This schedule seems

particularly suited for elderly and frail subjects who need to reduce individual toxicity with tailored treatments as well as limit hospital admittance. In fact, most elderly patients reported as critical the chance of having an oral schedule.

Disclosure of Interest: None Declared.

BSH2020-EP-075

Exercise training is a feasible and active complementary therapy in adult and elderly patients receiving anti-lymphoma treatments

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Background: Lymphoma and its repeated treatment can produce adverse effects on the short and long term including cardiovascular diseases, body composition changes, physical deconditioning, fatigue and depression, finally leading to a reduced health-related quality of life (HRQoL). This issue is particularly relevant in elderly patients, who may experience an accelerated geriatric decay.

Methods: We conducted a prospective controlled pilot study in a real-life setting to establish the feasibility, safety and the activity of a supervised and combined Exercise Training (ET) program in adult lymphoma patients undergoing cancer treatments. This was part of a National Project of the Italian Ministry of Health (CCM2014) aimed at improving QoL in long-term survivors.

Results: 16/30(53%) patients were enrolled in the Intervention group (IG) and 14/30(47%) in the Control group (CG). The median age was 65.5 years (range 29.0–80.0). The recruitment and adherence rates to ET were 86% (95% CI: 0.706–0.937) and 50% (95% CI: 0.364–0.636) respectively. At the 6 months follow-up the IG

Table. Percentage changes of the means of the EORTC QLQ-C30 domains from T0 to T2 (6 months)

Outcome Δ (T0–T2)/T0	Intervention group (IG)	Control group (CG)	P-value
Mean % change			
Global health status/ quality of life	+100	+33.2	0.249
Physical functioning	+33.8	–82.3	0.000*
Role functioning	+88.0	–63.9	0.062*
Emotional functioning	+57.0	–69.5	0.013*
Cognitive functioning	+26.4	–85.6	0.003*
Social functioning	+42.6	–77.8	0.001*
Fatigue	–58.4	–4.0	0.470
Nausea and vomiting	–75.0	–70.0	0.624
Pain	–100	–66	0.025*
Dyspnoea	–71.4	–50.0	0.695
Insomnia	–75.0	0.0	0.073
Appetite loss	–88.9	0.0	0.035*
Constipation	–78.6	–50.0	0.456
Diarrhoea	–37.5	–100.0	0.194
Financial difficulties	–50.0	–66.7	0.721

EORTC QLQ-C30, item European Organisation Research and Treatment of Cancer-Quality of Life-C30 questionnaire.

(median-age = 65.5 years) had a significant improvement ($P < 0.05$) compared to the CG (median-age = 63.0 years) in muscle strength, cardiorespiratory fitness, physical mobility, balance and QoL.

Conclusions: This study confirms supervised ET is an effective supportive measure during anti-lymphoma treatments to counteract most short term toxicities. Importantly, it showed ET is feasible and beneficial also in patients >65 years. Therefore, larger studies are needed in elderly subjects to definitely assess, if during anti-cancer therapies, ET intervention can be considered a standard complementary measure.

Disclosure of Interest: None Declared.

BSH2020-EP-076

The effectiveness of different modes of anti-viral therapy in patients with hepatitis C-associated (HCV) lymphoma marginal zone (LMZ)

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Currently the association between HCV and LMZ is supported by data from epidemiological and therapeutic studies in the last 20 years. A large number of studies have demonstrated that different variants of anti-viral therapy (AVT) are an effective treatment for patients with HCV-associated MZL. The use of AVT, consisting of interferon (IFN) and ribavirin (RBV) and the new anti-viral drugs direct action (DAA), at any time during the life of these patients (pts) can cause a high level so anti-viral anti-tumour responses.

The aim of our study is to evaluate the clinical features and results of treatment of patients with HCV + MZL depending on the variant of the anti-viral treatment.

Methods: Our study included 46 patients with MZL and infected with HCV, which determined HCV-RNA. Patients received treated from 2005 to 2019 in NMRC oncology by N. N. Blochina and RNRMU by N.I. Pirogov. The median age was 47 years (31–75 years), 27(59%) were men, 40 (86%) pts had III+IV stage, b-symptoms was detected in 10 (21%). The distribution of subtypes of MZL was as follows: node 10 (21%), extranodal MALT 11 (24%), splenic, 23 (50%), disseminated MZL 22 (48%) pts. The viral load ranged from 2×10^4 to 1×10^8 copies/ml. Depending on the treatment the patients were divided into two groups (G): 28(61%) patients received treatment with anti-viral regime on the basis of IFN (G1), 18 (39%) patients (G2) receive direct anti-viral drugs (a regimen based on sofosbuvir).

Results: 1. Frequency of anti-viral response G1 was 21(72%), in G2 – 16(89%) $P = 0.003$. The time to achieve virological remission in G1 was 3.5 months and in G2 – 1 month ($P = 0.005$).

2 ORR was in G1 24(85%) pts and in G2 13(75%) pts respectively, $P = 0.02$. However, the time to reach remission in G1 was 3.5 months and in G2 – 6.5 months, $P = 0.005$. After remission of lymphoma was achieved, patients receiving DAA therapy continued for another 2 months to consolidate. The median time of DAA

therapy was 8.5 months. Two-year survival without disease progression (PFS) was in G1 85 % and in G2 82%, $P = \text{ns}$. The median follow-up was 40 months.

3. In G1 to 2 pts, therapy was cancelled due to the development of depression,

In G2 any patient therapy was not modified.

Conclusion: Anti-viral therapy is the preferred first-line option in patients with HCV-associated MZL. Due to the safety, speed and effectiveness of DAA in obtaining a virological response, as well as its good tolerance, DAA therapy is probably preferable to anti-viral treatment with IFN. Very important is the time to achieve remission on the DAA scheme – 6.5 months. Thus, the duration of DAA therapy should be at least 9 months. Perhaps the combination of these two regimens will increase the effectiveness of treatment for hepatitis C-associated LMZ

Disclosure of Interest: None Declared.

BSH2020-EP-077

Anthracycline prescribing in lymphoma: traffic light guidance using NT-proBNP and echocardiogram

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Anthracyclines have been utilized in certain malignancies for decades with cardiotoxicity related to anthracycline use widely recognised among clinicians. While estimated data of worldwide prevalence of anthracycline-induced cardiotoxicity have been lacking, the mechanism of cardiotoxicity is thought to be due to inhibition of the topoisomerase 2 β enzyme resulting in impairment of subsequent mitochondrial biogenesis in non-proliferating cardiomyocytes.

To pre-emptively risk assess patients and minimize irreversible cardiotoxicity we have developed traffic light-based guidance for our lymphoma cohort, implemented in March 2019.

In any patients with: age >60, history of hypertension, diabetes or previous cardiac event, smoking history, or the presence of ECG changes consistent with left ventricular (LV) hypertrophy, baseline investigations including NT-proBNP and transthoracic echocardiogram are recommended. Results then guide anthracycline dosing from cycle 1 (or before second cycle in urgent cases).

A green light 'go ahead' was given to anthracycline administration in patients with LV ejection fraction (EF) $\geq 55\%$, no ventricular dilatation and NT-proBNP <300.

Amber caution was advised for patients with preserved LVEF $\geq 55\%$ but NT-proBNP >300 in whom anthracycline was administered at cycle 1 but a cardiac MRI and cardio-oncological opinion expedited pre cycle 2.

Anthracyclines were not administered if baseline LVEF was found to be 40–55% or patients had LV dilatation – until the echocardiogram had been reviewed or repeated and a cardio-oncological opinion sought.

Anthracyclines were contraindicated in those with LVEF <40%. Cardio-oncological optimization was sought and alternatives to anthracyclines considered.

Retrospective data collection of patients treated at our institution between March and October 2019 highlights a total of 58 patients who had both been diagnosed with lymphoma and where anthracycline containing regimens were considered.

Patients were aged 22–89 years; 35 were diagnosed with DLBCL, 11 with Hodgkin's, 3 with T-NHL and the remaining 9 with other high-grade B-NHL subtypes. Nine patients were excluded from further analysis as they were not deemed suitable for chemotherapy in

view of performance status and comorbidities and one was excluded as they were treated outside of our institution. Of the 48 patients who were considered for treatment with an anthracycline, 10 did not meet the criteria for use of the traffic light guidance. Twenty-four of the 38 patients who met traffic guidance criteria use had both NT-proBNP and an echocardiogram performed. Only one patient had an EF precluding anthracycline use, and they had received historical anthracycline chemotherapy for prior breast cancer and were now aged >60. Fifteen patients had a straight forward green go ahead for anthracycline use. Eight had amber warnings, managed with dose adjustments and cardiological advice. Only one patient went on to have a cardiac MRI, highlighting potential difficulties in obtaining this test.

The importance of risk assessing patients for cardiotoxicity will be covered in departmental teaching sessions with a more robust roll out of the guidance. The traffic light guidance will be reassessed in future with larger numbers to assess its ease and practicalities of use and impact on cardio protection, ensuring no deleterious effect on patient outcomes.

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BSH2020-EP-078

Post discharge care needs for patients with treated Primary CNS Lymphoma – a baseline single centre retrospective study

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Primary central nervous system lymphoma (PCNSL) is a rare lymphoma localised to the brain, leptomeninges or eyes. It is a subtype of diffuse large B-cell lymphoma accounting for 1% of non-Hodgkin lymphoma. It presents with a range of non-specific symptoms such as seizure and hemianopia which can be confused with other diseases of the older population hence it takes longer to reach a diagnosis.

Modern multiagent induction chemotherapy, MATRix (high-dose methotrexate, cytarabine with thiotepa and rituximab) followed by whole-brain radiotherapy or stem cell transplantation consolidation has significantly improved survival rates and created, in parallel, a survivor population with unmet and poorly defined treatment and disease-related care needs. Since there is a lack of research and experience of issues and support needs of these patients so we performed a retrospective analysis to describe the supportive care needs, available services and unmet needs of patients with PCNSL following treatment at a single tertiary care centre.

Newly diagnosed patients with PCNSL consecutively treated at The Christie NHS Foundation Trust between January 2012 and December 2018 were identified from hospital records. Patients receiving initial treatment with at least one cycle of multiagent chemotherapy or whole-brain radiotherapy and survived for a minimum of 3 months from the start of treatment were included as this study focused on later manifestations of the condition. Clinical records for 10 randomly selected patients were reviewed to identify the most common care needs during and after completion of planned treatment. Worse grade symptom severity was graded using

the common terminology criteria for adverse events grading system version 4.03. Referrals for supportive care in the community following completion of planned treatment and unmet care needs were identified and recorded.

Sixty-two patients were identified from the initial search, and 42 met the inclusion criteria. The patient population was made up of 20 males and 22 females with the age ranging from 36 to 91 and peaks at 56 and 66 for males and females respectively. Thirty-six patients received chemotherapy with just under 50% completing the four cycles. In terms of survival, 26 patients were alive as of 1 April 2019 when this study was completed. Poor mobility was the most common support need – present in 80% of patients – followed by fatigue in 60%, poor nutrition in 48%, inability to perform activities of daily living in 43% and cognitive problems in 24% of patients. Over 73% of the study population were referred to community support services – physiotherapy (31 patients), occupational therapy (29 patients), dietician (23 patients) and specialist allied health professionals were directly involved in the care of 15 patients. We also found that in most patients, their needs were met as an inpatient, however, following discharge due to reasons such as weak documentation of symptomatic needs, lack of referral or shortage of services in their area needs were missed.

A high proportion of treated PCNSL patients have ongoing care needs following discharge from the hospital, but this information is incompletely and inconsistently recorded in the patient record. We are developing a patient-reported outcomes measures tool to identify common care needs for referral and follow-up. This tool will be prospectively audited and gaps in service provision across Greater Manchester identified for future service development.

Disclosure of Interest: None Declared.

BSH2020-EP-079

A drug repositioning approach identifies a combination of compounds as a potential regimen for chronic lymphocytic leukaemia treatment

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Drug repositioning, identifying a new indication for an existing drug, is a promising and powerful innovative strategy in the field of drug discovery. In this study, we screened a compound library containing 800 Food and Drug Administration approved drugs for their anti-leukaemic effect. All screening activities made use of human peripheral blood mononuclear cells (PBMCs), isolated from healthy or leukaemic (CLL, CML, ALL, AML) donors. In the first round of screening (at 10 µM compound concentration), compounds which yielded a higher cytotoxic effect in leukaemic cells compared to normal PBMCs were identified and subjected to a second round of screening (dose-response studies to determine compound potencies). Compounds with confirmed cytotoxicity were selected and classified in three groups: (i) anti-neoplastic compounds which are drugs used in leukaemia treatment, (ii) compounds known to have an anti-cancer effect and (iii) compounds demonstrating an anti-leukaemic potential for the first time. The latter group was the most interesting from a drug repositioning perspective and yielded a single compound, namely Isoprenaline which is a non-selective β-adrenergic agonist. Analysis of the cytotoxic effect of this drug indicated that it induces sustainable intracellular ATP depletion leading, over time, to necrotic cell death. We exploited the Isoprenaline-induced intracellular ATP depletion to sensitize primary leukaemic cells to fludarabine

(purine analogue) treatment. *In-vitro* treatment of primary leukaemic cells with a combination of Isoprenaline and fludarabine showed a very high synergistic effect. The combination of Isoprenaline with fludarabine could constitute a new efficient regimen for CLL treatment following successful evaluation in animal models and clinical trials.

Disclosure of Interest: None Declared.

BSH2020-EP-080

Palliative care needs during inpatient stay for commercial chimeric antigen receptor T-cell infusion at the Newcastle-Upon-Tyne Hospitals NHS Foundation Trust

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Retrospective case note analysis for 10 adult patients who received chimeric antigen receptor T-cell (CAR-T) infusion for the treatment of refractory or resistant large B-cell lymphoma and primary mediastinal B-cell lymphoma at the Newcastle-upon-Tyne Hospitals NHS Foundation Trust between March 2019 and August 2019.

Aims were to assess patients' symptomatic burden and palliative care needs during the admission in which the patients underwent cell transfer. Outcomes used included symptom burden (defined in this instance as evidence of the patient experiencing pain, nausea, dyspnoea or anxiety), referral to specialist palliative care, clinical minutes spent by the specialist palliative care team with referred patients, and preferred and achieved place of death where relevant. The geographical distances between the patients' homes and the hospital were calculated and the ease of facilitating discharge was assessed.

Patients experienced a high symptom burden at the time of their cell infusion. One hundred per cent of patients experienced at least one of the symptoms assessed, and 80% of patients experienced more than one. All patients experienced nausea and vomiting.

Four patients (40%) were referred to specialist palliative care during their admission for CAR-T infusion. Among these referrals, during the admission in which the patient underwent cell transfer, the specialist palliative care team spent a total of 4054 clinical minutes, with a median time of 1041 min per patient seen. All those referred to specialist palliative care were experiencing multiple symptoms.

Eight patients (80%) lived in a place outside of our locality with a mean distance of 153 miles travelled to receive treatment.

At the time of writing, four of the patients (40%) had died, all of whom had input from the specialist palliative care team. In all cases, preferred place of death was established and achieved. One patient required rapid discharge on a palliative basis, which caused logistical difficulty due to the distance between their preferred place of death and the treating hospital (more than 100 miles).

Limitations of this work included retrospective case note analysis being unable to identify anything not documented in the case notes; reliance on proxy measures; and lack of data on whether referral to the specialist palliative care team improved the patient experience.

A potential area of service delivery improvement would be the proactive and early involvement of specialist palliative care clinicians in patients being referred for CAR-T therapy. The potential role for a system for rapid discharge on a palliative basis which crosses geographical boundaries could be an important area of research, as the current model for delivering commissioned CAR-T therapies can mean a patient's place of origin is distant from the hospital delivering the infusion. We appreciate that this may change as services develop; however, there are many forthcoming trials in this

developing therapeutic area, indicating that centralised modes of delivery for some patients are likely to continue for some time yet.

Disclosure of Interest: None Declared.

BSH2020-EP-081

Epidemiology of haematological malignancies in National Hospital Abuja: a 10-year retrospective study

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Haematological malignancies are a collection of heterogeneous malignant conditions originating from cells of the bone marrow and the lymphatic system and are derived from a single cell in the marrow or peripheral lymphoid tissue which has undergone genetic alteration. They are a significant cause of morbidity and mortality in Nigeria. There is limited information about the pattern of distribution of these malignancies in our environment.

Therefore, the aim was to determine the epidemiology of haematological malignancies and the prevalence of the different haematological malignancies.

This is a hospital-based retrospective study that was carried out over a 10-year period (2010–2019). Data from the clinic attendance register, the admission/ discharge register, patient's case note and cancer registry were used for the study. The socio demographics and disease demographics of the 475 patients with haematological malignancies out of a total of 2514 patients with various malignancies were analysed using the statistical package for social science (SPSS) version 15.

The outcome showed that haematological malignancies accounted for 18.9% of all cancers cases seen within the study period. The male to female ratio is 1.4:1. The median age of the study population is 44.6 years with the oldest and youngest being 90 years and 17 months respectively.

Non-Hodgkin lymphoma (NHL) has the highest prevalence of 24.6% while myelofibrosis (MF) has the least prevalence of (1.3%). The prevalence of other disease entities includes multiple myeloma 16.6%, Chronic myeloid leukaemia (CML) 15.4%, acute myeloid leukaemia (AML) 9.9%, Hodgkin lymphoma/disease (HL) 9.7%, Chronic lymphocytic leukaemia (CLL) 8.2%, acute lymphoblastic leukaemia (ALL) 7.4%, polycythaemia rubra vera (PRV) 2.5%, essential thrombocythaemia 2.3% and myelodysplastic syndrome (MDS) 2.1%.

Haematological malignancies are common in our environment and results were similar to few studies carried out in other tertiary facilities in Nigeria. It was observed that multiple myeloma was significantly higher in our study and this may be due to the advanced diagnostic tools present in our facility for proper diagnosis of this disease.

Disclosure of Interest: None Declared.

BSH2020-EP-082

Experiences of providing commercial Chimeric Antigen Receptor (CAR-T) cell therapy at the Northern Centre for Cancer Care

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To date, 17 patients have received commercial Chimeric Antigen Receptor (CAR-T) cell infusion at the Northern Centre for Cancer Care (NCCC), Newcastle upon Tyne Hospitals NHS Foundation Trust.

Data collection will continue until the time of presentation.

Between January 2019 and the end of November 2019, the NCCC received 25 referrals for commercial CAR-T therapy.

Seven patients (28% of all referrals) did not receive CAR-T cells and have died. Of these, four did not undergo leukapheresis because of disease progression. Three underwent leukapheresis but died due to disease progression before cells could be infused.

CAR-T cells were manufactured within specification for 90% of patients who underwent apheresis. Axicabtagene ciloleucel (Yescarta) was successfully manufactured in all cases. Tisagenlecleucel (Kymriah) was successfully manufactured in all but two cases. In one case the patient died due to disease progression after production failure on two occasions. In the second, the patient is awaiting a second attempt to manufacture cells.

Of the patients who have received CAR-T cells ($n = 17$), 12 (71%) had diffuse large B-cell lymphoma, 3 (18%) had transformed follicular lymphoma, and 2 (12%) had primary mediastinal B-cell lymphoma. Five (29%) had two lines of prior treatment, 10 (59%) had three lines and two (12%) had four or more lines. Eight (47%) received bridging therapy: six received DECC (lomustine, etoposide, chlorambucil and dexamethasone), one received IVE (ifosfamide, epirubicin and etoposide) and one received pulsed dexamethasone.

Twelve patients received axicabtagene ciloleucel. Five received tisagenlecleucel. There have been no treatment-related deaths. Median time from leukapheresis to infusion of cells was 37 days (range 33–76). All were conditioned with fludarabine and cyclophosphamide.

Fifteen patients have received CAR-T infusion and completed their inpatient admission for purposes of infusion. Median duration of inpatient stay after CAR-T infusion was 16 days (range 13–38). Six (40%) required admission to the intensive care unit during their admission: median duration of stay there was 2.5 days (range 2–10).

During inpatient admission, 10 patients (67%) experienced cytokine release syndrome (CRS) of any grade. Four (27%) experienced CRS of grade 3 and above. Three (20%) experienced immune effector cell associated neurotoxicity syndrome (ICANS) of any grade. Two (13%) experienced ICANS of grade 3 and above. Ten (67%) received tocilizumab. Six (40%) received steroids (in all cases this was IV dexamethasone) for issues related directly to CAR-T therapy.

Follow-up results are available for 14 patients. Overall response rate was 50% (seven patients), of which four achieved complete response as best overall response, and three achieved partial response. Three (21%) had stable disease as best overall response, and four (29%) had progressive disease. Overall survival rates were 100% at 28 days, 80% at 3 months and 50% at 6 months after CAR-T infusion. Four patients (29% of cohort available to follow-up) have died after CAR-T infusion. Progression-free survival rates as calculated at standardised follow-up points were 71% at 28 days, 31% at 3 months, and 27% at 6 months after CAR-T infusion.

One patient who had initial disease progression later had a subsequent complete response after receiving another line of treatment.

Disclosure of Interest: None Declared.

BSH2020-EP-083

Btk inhibitors ibrutinib and acalabrutinib impair FcγRIIa-mediated responses of healthy donor platelets and chronic lymphocytic leukaemia platelets to bacteria

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Chronic lymphocytic leukaemia (CLL) is the most common form of leukaemia in the Western World. Numerous therapeutic options can be employed for the treatment of CLL, however, ibrutinib is a preferable first-line treatment. Ibrutinib disrupts B-cell receptor signalling by inhibiting Bruton's tyrosine kinase (Btk), preventing proliferation and survival signals. Ibrutinib is associated with increased risk of bleeding, due to the role of Btk in platelet aggregation. This treatment is also associated with infections in CLL patients, but the potential effects of ibrutinib, or second-generation Btk inhibitors like acalabrutinib, on platelet immune function have not been evaluated. This is of relevance as platelets can be activated in response to IgG-opsonised bacteria via the platelet low-affinity IgG immune receptor, FcγRIIa, of which Btk is downstream. Upon platelet activation, changes in the platelet surface and released immune mediators, including antimicrobial cytokine Platelet Factor 4 (PF4), have an impact on platelet interaction with other immune cells and pathogens. We hypothesised that Btk plays an important role in platelet FcγRIIa signalling in response to bacterial agonists, and that ibrutinib and acalabrutinib inhibit such responses, contributing towards the increased risk of infection seen in CLL.

Herein we show via light transmission aggregometry that ibrutinib and acalabrutinib inhibit healthy donor FcγRIIa-mediated platelet aggregation in response to *Staphylococcus aureus* Newman and *Escherichia coli* RS218, as well as upon direct FcγRIIa crosslinking with monoclonal antibody IV.3 (anti-FcγRIIa). The effect of Btk inhibitors on platelet secretion was also investigated by measuring dense granule release via an ATP luciferin-luciferase assay, and α-granule release by quantifying PF4 secretion by ELISA. Both ATP and PF4 release were inhibited by ibrutinib and acalabrutinib in healthy platelets stimulated with bacteria or crosslinked IV.3. Conversely, these drugs did not affect aggregation nor secretion upon platelet stimulation with thrombin receptor activator peptide 6 (TRAP-6). When activated, Btk is phosphorylated on tyrosine (Y) 551, leading to the autophosphorylation of Y223. We demonstrate that Btk has increased phosphorylation at Y223 upon platelet activation by *S. aureus* Newman, *E. coli* RS218, and IV.3 crosslinking; and this phosphorylation is completely abolished upon the use of both ibrutinib and acalabrutinib. These results indicate that ibrutinib and acalabrutinib inhibit FcγRIIa-mediated platelet activation that might modulate platelet bacterial responses.

We next investigated CLL platelet responses to bacteria, and how ibrutinib alter these responses. We show that both treatment-naïve and ibrutinib-treated CLL patients have similar platelet aggregation and PF4 release responses to TRAP-6 and ADP compared to healthy controls. Treatment-naïve platelets can also respond to bacteria in the presence of autologous plasma. However, ibrutinib-treated CLL patients have significantly diminished aggregation and PF4 release to *S. aureus* Newman, *E. coli* RS218, and IV.3 crosslinking.

Additionally, platelet Btk is phosphorylated at Y223 in treatment-naïve, but not ibrutinib-treated CLL patients, for all FcγRIIa agonists.

In conclusion, we propose that Btk inhibitors impair the FcγRIIa pathway in platelets, leading to reduced platelet responses to bacteria, possibly contributing to the increased risk of infections seen in CLL.

Disclosure of Interest: None Declared.

BSH2020-EP-084

ACOPP chemotherapy for front-line treatment of older patients with Hodgkin lymphoma – a pilot study

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Approximately 20% of patients with Hodgkin lymphoma (HL) are aged 60 years or older. There is no standard of care for such patients, who have a markedly inferior prognosis compared with younger counterparts and are underrepresented in clinical trials. BEACOPP chemotherapy has been demonstrated to be highly effective in front-line HL management, but when used in older patients (even at non-escalated doses) there is an unacceptable rate of toxicity-related death. We proposed that a modification of the BEACOPP regime by removing bleomycin and etoposide and dose reduction of cyclophosphamide ('ACOPP') would potentially be a well tolerated and effective regimen for older HL patients.

This pilot study investigates the feasibility of ACOPP chemotherapy for front-line management of older patients with HL who were deemed unfit for more intensive management approaches. Patients received intravenous (IV) Doxorubicin (35 mg/m²) and Cyclophosphamide (650 mg/m²) on day 1 along with oral Procarbazine (100 mg/m²) for 7 days and Prednisolone (40 mg/m²) for 14 days. IV Vincristine (1.4 mg/m², max 2 mg) was given on day 8 and granulocyte colony stimulating factor on days 9–13. Treatment was delivered as an outpatient every 21 days to a maximum of 6 cycles.

Ten patients with a median age of 78 years (range 58–92) were treated between March 2018 and August 2019, eight at the Beatson Cancer Centre, Glasgow and two within Newcastle MDT. All patients had histologically confirmed classical Hodgkin lymphoma. Nine of 10 had advanced stage disease, with one patient having unfavourable stage 2 disease. Median Charlson Comorbidity index was 6. Seven (70%) patients completed six cycles of therapy, with the other three patients receiving two, four and five cycles respectively. Reasons for stopping treatment early were frailty in two patients and postural hypotension in one.

Of the eight patients who had interim PET (post cycle 2), seven achieved a complete metabolic response and one had an area of Deauville 5 PET positivity at the site of a recent wound infection. To date, nine patients have had end of treatment CT assessment and all were in complete remission. With a median follow-up of 26 months, one patient has relapsed (15 months after completion of ACOPP) and is currently undergoing Brentuximab monotherapy. One patient died 5 months after ACOPP completion from unrelated causes.

Grade 3/4 neutropenia was seen in 6/10 patients but only two patients had febrile neutropenia. Thrombocytopenia (two patients, both grade 3, none requiring platelet transfusion) and anaemia (six patients, all grade 3 and required red cell transfusion) were also seen. Grade 2 peripheral neuropathy occurred in two patients. Seven of 10 patients required admission to hospital during therapy – reasons for admission were infection ($n = 3$), constipation ($n = 2$), anaemia ($n = 1$) and autonomic neuropathy ($n = 1$).

We have demonstrated the feasibility of ACOPP as a regimen which can be delivered in an outpatient setting to older patients with HL. We would highlight the median age of 78 years and significant co-morbidity burden of the patients in this pilot study. Toxicity was mainly haematological but was manageable with only two episodes of febrile neutropenia and no platelet transfusion required. Definitive conclusions regarding efficacy and safety cannot be drawn from this small number of patients but the high rates of early metabolic response are encouraging and we feel the regimen merits further assessment.

Disclosure of Interest: M. Wilson: None Declared, P. McKay Conflict with: Epizyme: Consultancy, Honoraria; Janssen: Honoraria, support to attend conferences; Roche: Honoraria; Gilead – Honoraria, support to attend conferences; Celgene – Honoraria, K. Parsons: None Declared, C. Ogilvie: None Declared, J. Travers Conflict with: Celgene: Speakers Bureau; Takeda: Membership on an entity's Board of Directors or advisory committees, M. Martin: None Declared, J. Allan: None Declared, W. Osborne Conflict with: Roche, Takeda, Pfizer, Servier, Kite Gilead, MSD, Novartis, M. Leach Conflict with: Abbvie: Honoraria; Janssen: Consultancy, Honoraria; Roche: Honoraria.

BSH2020-EP-085

Novel synthetic thienopyridine derivatives induce tubulin disruption and cell cycle arrest in malignant B cells with level of activity related to their chemical structure

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Disclosure of Interest: None Declared. Thienopyridines are well established for the treatment of atherothrombosis, however, a series of novel synthetic compounds within the thienopyridine family have recently been identified as potential anti-cancer agents. The mechanism of action of these thienopyridines is still being established, with B-cell receptor signalling cascade proteins PLCγ and Lck suggested as direct targets, and mitotic spindle inhibition via microtubule de-stabilisation demonstrated *in vitro*. Targeted treatments for mature B-cell malignancies have made significant advances in overall and progression-free survival, however, even with newer agents, treatment resistance and relapse is still prevalent. Microtubule-targeting agents (MTAs) are effective in the treatment of lymphoma and leukaemia but multidrug resistance pathways against MTAs mean there is a requirement for continued discovery of newer, more effective agents.

This study aimed to assess the effects of three novel thienopyridine derivatives (DJ0109, DJ0171 and DJ0199) each synthesised with a different chemical structure, on B-cell apoptosis, cell cycle arrest and mitotic disruption with the intention of identifying a new family of compounds with potential MTA-activity for use in the treatment of mature B-cell neoplasms.

DAUDI Burkitt Lymphoma cells were incubated in the presence of novel thienopyridines for 48 h at a range of concentrations (100 μmol/l to 1 nmol/l). Synthesis of the compounds, and description of aliphatic and phenyl ring substitutions has been previously published. MTS assay was used to assess cell biochemical activity. Flow cytometry using dual staining with Annexin V/Propidium Iodide (AV/PI) was used to observe apoptosis and necrosis while

single staining with Propidium Iodide was used to determine cell cycle effects. Alpha-tubulin expression was visualised by confocal microscopy following staining with α -Tubulin (DM1A) Mouse mAb (Alexa Fluor® 488 Conjugate) antibody and DAPI nucleic acid stain. The rate of tubulin polymerisation was measured spectrophotometrically by Tubulin Polymerisation Assay. Paclitaxel was used as a positive control for tubulin stabilisation.

Across all assays, compound DJ0199 showed greater activity compared to the other tested compounds, inducing a significant reduction in cell viability, and an increase in early and late apoptosis at 10 and 100 μ mol/l concentrations. Necrosis was not induced by these compounds. DJ0171 showed no significant effect on cell viability by MTS assay, but was found to exhibit a similar but weaker activity to DJ0199 when all other assays were considered. Increased G2/M arrest was also seen following treatment with these compounds at 10 μ mol/l, with DJ0199 showing greater potency at this concentration. These compounds showed a significant inhibitory effect on tubulin polymerisation at 10 μ mol/l dosage and signs of microtubule disruption via confocal microscopy. Compound DJ0109 did not show any activity across any of the assays, presumably due to the para-substitution on the phenyl ring in its chemical structure, which is a confirmation of previously published data.

The study demonstrates the anti-mitotic activity of two novel thienopyridine derivatives in malignant B cells and the potential for use in the treatment of mature B-cell malignancies. It also confirms previously published data on the impact of the structure of the thienopyridine compounds on their *in vitro* activity.

BSH2020-EP-086

A retrospective study to look at outcomes of those over 80 treated with RCHOP chemotherapy

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It is generally acknowledged that with increasing age, patients with diffuse large B-cell lymphoma (DLBCL) have poorer outcomes. A significant decrease in relative survival occurs from the 5th decade and this worsens progressively with age (Gustaf Hedström et al 2015). Many patients in the 9th decade are deemed not fit for treatment and those who are have varied outcomes.

We carried out a retrospective analysis of the outcomes of our older patients (≥ 80 years) who were deemed suitable for chemotherapy with R-CHOP (Rituximab cyclophosphamide, vincristine, doxorubicin and Prednisolone) or R-miniCHOP between February 2009 and February 2018 at the Norfolk and Norwich University Hospital. We identified 42 patients in this category. Of these patients, 16 (38%) were treated with R-CHOP (median age 82.6 years) and 26 (62%) were treated with R-mini CHOP (median age 83 years).

Patients suffering major adverse effects (defined as incidents resulting in cessation of treatment or death) were comparable in both groups (2/16 (12.5%) in R-CHOP vs. 3/26 (11.5%) in R mini CHOP). Of those remaining, 8/16 (50%) of the R-CHOP group suffered serious adverse effects (defined as effects that resulted in change of dose or delay of one or more treatment cycles), compared with 12/26 (46.2%) in the R mini CHOP group.

Complete remission was achieved in 14/16 (87.5%) patients receiving R-CHOP compared with 14/26 (53.9%) patients in the R mini CHOP group. Of those who received R-CHOP, 3 (18.75%) relapsed with a median time to relapse of 10 months. In the R mini CHOP group, there was one (3.9%) relapse after 3 years.

This left nine patients (21.4%) who did not achieve complete remission but did not have major adverse effects preventing them from completing treatment. Of these nine only one had a partial response. The remaining eight had progressive disease. Four of these went on to have further chemotherapy and none of these achieved a complete remission following this.

In conclusion, 28/42 (66.6%) patients over the age of 80 deemed suitable for RCHOP or RminiCHOP achieved a complete remission. However, high rates of adverse events were seen in both groups. We acknowledge the dataset is small and there would be a selection bias based on suitability and fitness for treatment, but it does indicate that elderly patients who are suitable for treatment with RCHOP / RminiCHOP can achieve durable remission. Better treatment strategies are needed in this group of patients to reduce toxicity and maximise effectiveness of treatment.

Disclosure of Interest: D. Gordon-Walker: None Declared, N. Shah Conflict with: Abbvie, Roche, Janssen, D. Sparksman Conflict with: Abbvie.

BSH2020-EP-087

Outcomes of patients with refractory peripheral T-cell lymphoma, angioimmunoblastic and other nodal lymphomas of T follicular helper-cell origin (OPerA)

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Introduction: Peripheral T-cell lymphoma (PTCL) – not otherwise specified(NOS) and angioimmunoblastic T-cell lymphoma (AITL) are two most common subtypes accounting for 45% of diagnoses. WHO created an umbrella category of nodal T-cell lymphomas with T-follicular helper phenotype (TFHL) which includes AITL and PTCL with TFH phenotype. 25–30% of patients (pts) are refractory to initial therapy and even among responders relapse is common. Survival after relapse or progression (R/P) is poor. Outcomes have not changed significantly even with introduction of several new agents. There is no standard 2nd line therapy or guidance on sequencing of therapies. The aim of study was to determine outcomes in large, well-defined pts with either primary refractory PTCL-NOS or TFHL.

Methods: We performed a multicentre retrospective study to determine outcomes to 2nd line therapy for adult pts diagnosed b/w 1.1.09–6.30.18 with PTCL-NOS or TFHL, who were primary refractory defined by either progression during initial therapy, st CR prior to relapse (almost all autologous haematopoietic cell transplant [AHCT]). At R/P, 48% received single agent therapy [mostly romidepsin (15/39)], while 36% received multiagent salvage therapy [mostly ICE (9/29)]. Thirteen of 80 pts were placed on hospice or only received local therapy after R/P. Median OS from diagnosis and following R/P was 19 m/12.9 m respectively. Median PFS2 (defined

as time from 2nd line therapy to 2nd progression) for pts receiving systemic therapy was 73 days (d) (range 41–175). There was no significant difference in survival by histologic subtype (74 days for PTCL and 73 days for TFHL; $P = 0.29$), irrespective of single agent *versus* combination salvage therapy (97 vs. 51 days, $P = 0.09$), platelet count $<150\ 000$ (75 vs. 59 days $P = 0.37$), presence of B symptoms at R/P (78 vs. 63 days, $P = 0.16$), ECOG PS ≥ 2 (75 days vs. XX, $P = 0.62$), elevated LDH (63 vs. 75 days, $P = 0.31$) and EN disease at R/P (59 vs. 75 days, $P = 0.72$). There was no statistically significant difference in survival if pts had relapsed disease or induction failure (75 vs. 50 days, $P = 0.97$). OS was not improved if pts HCT as consolidation after 2nd line therapy (alloHCT 29.1 m, AHCT 48.5 m vs. 16.1 m for no HCT; $P = 0.61$). Median number of therapies after salvage was 1 (range 1–6). OS was not improved if pts received alloHCT after 2nd line therapy (29.1 m vs. 16.1 m) or AHCT (48.5 m; $P = 0.06$).

Conclusions: Outcomes in this population of primary refractory PTCL-NOS and TFHL were poor, but better compared to most other retrospective series. EN disease and advanced stage were common in this cohort of primary refractory pts. Relapse after HCT in CR1 had a particularly poor prognosis. Our findings suggest that single agent therapy following R/P in primary refractory pts and transplant may be associated with better survival, but statistical power was limited.

Table.

Baseline characteristics $n = 80$	At progression/ relapse $n = 80$
Sex	57% male 43% female
Histologic subtype	59% PTCL-NOS 41% TFH lymphoma
Age at diagnosis in years	62 (21–92)
Time from diagnosis/ progression to treatment	22 days (1–126)
LDH elevated	75%
Platelets $<150\ K$	26%
Stage	I 4% II 2% III 25% IV 69%
ECOG > 2	22%
BM involvement	43%
Non-BM extranodal disease	56%
B symptoms	59%
Treatment	CHOP: 53% CHOEP: 30% HCVAD: 2% Other: 15%
Response to treatment	CR: 41% PR: 11% SD: 6% PD: 42%
Consolidative HCT	Autologous: 13% Allogeneic: 1%
	Romidepsin: 18% ICE: 10% ESHAP: 4% Belinostat: 4% Other: 64%
	CR: 29% PR: 10% SD: 4% PD: 58%
	Autologous: 5% Allogeneic: 6%

BM, bone marrow; HCT, haematopoietic stem cell transplant. Disclosure of Interest: None Declared.

BSH2020-EP-088

Incidence of cardiotoxicity in anthracycline-based chemotherapy in lymphoma patients: a single centre experience

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Background: Anthracyclines are an important component of lymphoma treatment but they can induce cardiac dysfunction, ranging from an asymptomatic decline in left ventricular ejection fraction to the development of severe heart failure. Although several risk factors have been identified, predication, prevention and management remain a challenge. Currently there is no consensus on managing cardiac dysfunction in patients receiving anthracycline-based treatments.

Aim: To identify the incidence of cardiac toxicity in lymphoma patients following treatment with anthracycline and to identify possible risk factors and monitoring strategies.

Methods: A prospectively maintained database of adult lymphoma patients treated at University Hospital of Wales was reviewed to identify patients who received CHOP+/- R (Cyclophosphamide, Doxorubicin, Vincristine, Prednisolone and Rituximab) ABVD+/-R (Doxorubicin, Bleomycin, Vinblastine, Dacarbazine and Rituximab) between January 2016 and March 2019. Patients who received a minimum of four cycles were included. Information about diagnosis, risk factors, pre- and post-treatment echocardiogram and management of patients diagnosed with cardiac dysfunction was analysed.

Results: A total of 93 patients were identified. Mean age at diagnosis was 58 years old (range 27–78). The most common subtype was DLBCL (48.4%) with stage IV disease (49%). Seventy-one per cent of patients received R-CHOP. Thirty-three per cent had hypertension, 11% had diabetes mellitus and pre-existing cardiac disease was found in 10%. Pretreatment echocardiogram results were available for 76.3% patients, all of which were normal (ejection fraction $>50\%$). Post-treatment echocardiogram results were available for 47.3%, range 3–40 months post-treatment, abnormal results were detected in six patients (6.4%): Two had asymptomatic reduction in left ventricular function, three had severe left ventricular failure and one had severe right ventricular failure (see table one). All patients with abnormal results were referred to cardiology. Medications commenced by cardiology included angiotensin-converting enzyme inhibitors, angiotensin-receptor blockers, beta-blockers and diuretics. Of the 93 patients, 11 (11.8%) died, main cause of death was progressive disease (45%), one patient had severe right side heart failure which prevented further treatment delivery.

Summary: In this retrospective analysis we identified a significant number of patients who developed cardiotoxicity following anthracycline treatment. No common risk factor was identified, all patients were above the age of 60. However, the number of patients is too small to draw definitive conclusions. There is an increasing need for a national guideline on the selection of patients appropriate for anthracycline-based treatment, and the monitoring and management of anthracycline-induced cardiotoxicity. Until such guidelines are available in our centre, we propose to carry out a pre- and post-treatment echocardiogram on all patients having anthracycline, a thorough review of cardiac risk factors, with prompt referral to cardiology when dysfunction is recognized and to continue to collect information on patients who develop anthracycline-related

Table.

	Age	Diagnosis	Treatment	Risk factors	Pretreatment echocardiogram	Post-treatment echocardiogram	Outcome
Patient 1	71	DLBCL	R-CHOPX6	Nil	Not available	Left ventricular ejection fraction (LVEF) 45%	Referred to cardiology
Patient 2	62	DLBCL	R-CHOPX6	Nil	LVEF 55%	LVEF 45%	Referred to cardiology
Patient 3	73	DLBCL	R-CHOPX6	Hypertension and mild tricuspid regurgitation	Mild tricuspid regurgitation and normal bi-ventricular ejection fraction	Severe right ventricular failure	Started on cardiac medications. Died of progressive disease
Patient 4	70	DLBCL	R-CHOPX6	Nil	Normal	LVEF 10%	Started on cardiac medications repeat LVEF 40–45%
Patient 5	77	Hodgkin lymphoma	CHOPX6	Nil	Normal	LVEF 27%	Started on cardiac treatment awaiting repeat echocardiogram
Patient 6	64	DLBCL	R-CHOPX6	Hypertension	Normal	LVEF 10%	Started on cardiac medications. LVEF initially increased to 30–40% but dropped to 19% on most recent echocardiogram, remains under cardiology care

cardiotoxicity. The CARDIAC Care study, which is currently investigating the use of cardiac markers in identifying high-risk patients and the utilization of angiotensin receptor blockers and B-blockers to prevent heart muscle injury related to chemotherapy, will be vital to future guidelines.

Disclosure of Interest: None Declared.

BSH2020-EP-089

A single centre experience of using ciclosporin and mycophenolate mofetil immunosuppression in relapsed/refractory angio-immunoblastic T-cell lymphoma

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Angio-immunoblastic T-cell lymphoma (AITL) has a poor prognosis with overall survival of 5.5 months in transplant-ineligible relapsed–refractory disease. AITL is characterised by immune dysregulation and immunosuppressants are thought to have an immunomodulatory effect. Suggested mechanisms for ciclosporin in AITL are by inhibiting T-cell activation and T follicular helper cell differentiation. Mycophenolate mofetil suppresses B and T lymphocyte proliferation. We describe three cases of relapsed/refractory AITL with long remissions following treatment with ciclosporin and mycophenolate mofetil.

A 45-year-old with Stage 3 disease presenting with bilateral neck swelling and weight loss completed six cycles of CHOP, achieving a Deauville 3 response on end-of-treatment PET-CT. He progressed at 29 months and failed to respond to three lines of salvage therapy (alternating IVE and methotrexate, single agent lenalidomide followed by cisplatin with gemcitabine). He was commenced on ciclosporin 250 mg twice daily, achieving a complete response, facilitating allogeneic stem cell transplantation. Ciclosporin was discontinued

5 months post-transplant and he continues in remission 46 months later.

A 65-year-old with Stage 3A disease presenting with a blistering polymorphic rash and cervical lymphadenopathy. She was treated with six cycles CHOP on clinical trial. End-of-treatment PET-CT confirmed refractory disease and she soon developed autoimmune haemolytic anaemia. She was salvaged with IVE and intermediate dose methotrexate, followed by consolidation with autologous stem cell transplant. Despite achieving complete response on PET-CT 3 months post consolidation, she progressed 9 months later with FDG avid disease. Relapse was complicated by haemophagocytosis which responded initially to dexamethasone. She was ineligible for further intensive salvage chemotherapy due to poor performance status (3). She was commenced on ciclosporin, currently 100 mg twice daily (dose reduced due to renal impairment). The patient remains in ongoing remission 12 months later.

A 71-year-old with Stage 4B disease presented with rash, lethargy and supraclavicular lymphadenopathy. Staging confirmed bone marrow infiltration and bowel involvement. She was treated with CHOP and prophylactic intrathecal methotrexate as first-line treatment. Despite metabolic remission on end-of-treatment PET-CT, she re-presented with a rash and florid lymphadenopathy 5 weeks later. She was unable to proceed with intensive salvage therapy due to poor performance status and was commenced on ciclosporin. She subsequently developed ciclosporin induced haemolytic anaemia and was switched to mycophenolate mofetil 750 mg twice daily, initially combined with four cycles of rituximab and IVIG. PET-CT at 8 and 13 months showed complete metabolic remission. Response is currently sustained at 17 months.

We described above favourable outcomes with ciclosporin and mycophenolate mofetil in relapsed/refractory AITL in variable settings; as a bridge to consolidation with transplant, post multiple lines of salvage therapy and as palliative treatment. Toxicity appears manageable. Other case reports suggest renal toxicity and EBV reactivation as most frequent complications. We suggest future trials incorporating oral immunosuppressive agents in this rare disease.

Disclosure of Interest: None Declared.

BSH2020-EP-090

Single centre experience of efficacy and tolerability of attenuated immunochemotherapy (R-miniCHOP) in elderly patients with high-grade non-Hodgkin lymphoma

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Background: The increasing incidence of high-grade Non-Hodgkin Lymphoma in ageing populations places a significant burden on healthcare systems. Co-morbidity, frailty, and reduced organ and physiological reserve contribute to treatment-related complications. R-mini-CHOP is emerging as the standard of care for these patient with good efficacy and tolerability in clinical trial settings. We reviewed our single centre experience of delivering R-mini-CHOP therapy to elderly patients with high-grade lymphoma over a 5 years period in a rural British hospital.

Aims: We sought to assess the clinical efficacy and tolerability of attenuated immunochemotherapy (R-mini-CHOP) in elderly patients with high-grade lymphoma in a 'real-world' setting with medium term follow-up of up to 7 years.

Methods: We conducted a retrospective case review of patients who presented to our haematology service with aggressive non-Hodgkin lymphoma and received R-mini-CHOP attenuated immunochemotherapy as outpatient therapy between 2012 and 2017. Patients received rituximab (375 mg/m²), cyclophosphamide (400 mg/m²), doxorubicin (25 mg/m²) and vincristine (1 mg/m²) on day 1 of each cycle, and prednisolone (40 mg/m²) on days 1–5. We examined baseline patient characteristics, ECOG performance status, Ann Arbor stage, serum lactate dehydrogenase (LDH) concentration and International Prognostic Index (IPI) at diagnosis. Furthermore, we undertook a review of interim or post chemo disease response by PET/CT imaging and renal, cardiac, hepatic and bone marrow

toxicities through to the eventual cause of death in all patients with follow-up until January 2019.

Results: We identified 23 patients treated with R-mini-CHOP between 2012 and 2017. Median age at diagnosis was 82 years (range 74–88) and 52% were female. Diffuse large B-cell lymphoma (DLBL) was the most common diagnosis (74%). Mean serum [LDH] at diagnosis was 320 U/l (range 162–1202 U/l). Ann Arbor staging and IPI are detailed in Table 1 but patients of all IPI stage were represented in our cohort. The mean number of immunochemotherapy cycles administered was 4 (range 1–8) and only three patients required dose/regimen modifications. Fatigue (52%), peripheral neuropathy (26%) and alopecia (22%) were the most commonly identified toxicities. Two patients (9%) experienced significant cardiotoxicity with echocardiographic evidence of reduced left ventricular ejection fraction. There were no intensive care admissions or deaths related to neutropenic sepsis. Overall, 30% of patients experienced a partial treatment response and a further 39% were in clinical remission at the time of reporting. Ten patients (43%) died over the follow-up period, with five deaths attributable to disease progression, one died of GI bleeding which may have been associated with chemotherapy and the other four died of alternative unrelated causes.

Summary/Conclusion: In our single centre experience at a small rural general hospital, attenuated immunochemotherapy proves an efficacious and generally well-tolerated treatment regimen for elderly patients with high-grade non-Hodgkin lymphoma and can provide lasting remissions for a substantial number of patients in this age group. These data are broadly in line with a previous phase 2 study of R-mini-CHOP in patients aged over 80 years with DLBL and support the continued use of this regimen in elderly patients with high grade NHL receiving treatment in a variety of settings. (Peyrade et al., Lancet Oncology 2013).

Disclosure of Interest: None Declared.

BSH2020-EP-091

Treatment decisions in Waldenstrom's Macroglobulinaemia; outcomes from the South East Scotland Cancer Network

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Waldenstrom macroglobulinaemia (WM) is an indolent non-Hodgkin lymphoma characterised by an IgM paraprotein and MYD88 mutation. At present there is no standard of care for WM, treatment tends to combine Rituximab with a range of chemotherapy. Ibrutinib targets cell death via the MYD88 mutation and is effective in WM. NICE approved the use of Ibrutinib in WM in adults who have had at least one prior therapy or as first-line treatment when chemoimmunotherapy is unsuitable. We undertook a review of patients diagnosed with WM between 2014 and 2019 within the Haematology South East Scotland Cancer Network (SCAN) to assess efficacy of current management and review access to Ibrutinib as a potential unmet need.

Seventy patients were retrospectively identified from SCAN Multidisciplinary Meeting records. All had a minimum of 12 months follow-up. The data collected included demographics, symptoms, diagnostics, treatment and outcomes. The median age at diagnosis was 74.5 years (range 35–90 years), with *n* = 15 female. Twenty-three patients had symptomatic disease at presentation. Sixty-eight patients had an IgM paraprotein with a mean concentration of 16 g/l (range 2–55 g/l). All had either bone marrow (*n* = 68) or lymph

Table.

Patients (*n* = 23)

Female	12 (52%)
Median age	82
ECOG performance status	
0	7 (30%)
1	8 (35%)
2	6 (26%)
3	2 (9%)
Serum [LDH] > 246 U/l	8 (35%)
Ann Arbor staging	
I	3 (13%)
II	5 (22%)
III	3 (13%)
IV	12 (52%)
IPI	
0–1	3 (13%)
2	2 (9%)
3	6 (26%)
4–5	12 (52%)
Deaths due to disease progression	5 (22%)

node biopsy ($n = 2$) to confirm the diagnosis. *MYD88* mutations were detected in 58 of the 68 patients tested. The majority ($n = 64$) had a CT scan at diagnosis, abnormalities included lymphadenopathy ($n = 17$) and splenomegaly ($n = 4$).

Twenty-eight of the 70 patients received treatment; 20 started within 60 days of diagnosis. Anaemia was the main reason for initiation of therapy ($n = 17$). Overall the mean haemoglobin concentration at the start of treatment was 97 g/l (range 58–147 g/l). Other indications for treatment included B-symptoms ($n = 3$), hyperviscosity ($n = 2$), Bing-Neel syndrome ($n = 2$), thrombocytopenia, renal heavy chain disease, neuropathy and lymphadenopathy.

Rituximab, Cyclophosphamide, Dexamethasone (RCD) was the most common first-line treatment ($n = 18$). Frail patients ($n = 4$) received either Rituximab and Chlorambucil or single agent Rituximab; 50% did not respond to treatment in this cohort. Alternative frontline regimens included R-CVP ($n = 4$), R-CHOP ($n = 1$) and R-Bendamustine ($n = 1$).

Of those with symptomatic disease, nine required more than one line of therapy, two required three or more lines. Median duration of response to first-line therapy was 33 months. Three patients received Ibrutinib. Two patients had Bing-Neel Syndrome and the third was frail and relapsed following first line Rituximab and Chlorambucil. All three continue on treatment, mean duration of ibrutinib therapy so far is 10 months. Ibrutinib has been well tolerated with Grade 3 atrial fibrillation in one patient the only significant toxicity.

Twelve patients died during the follow-up period. Four died from progressive disease prior to the routine availability of ibrutinib. One of these patients had received WM treatment, one patient declined treatment, two patients were unfit for chemoimmunotherapy.

The majority of patients with WM have not required treatment as yet. Prior to the introduction of Ibrutinib, we found that patients unfit for RCD were experiencing high rates of treatment failure. Indeed, some unfit patients declined chemoimmunotherapy or were offered symptomatic management only. It is clear that many of these patients may have benefited from Ibrutinib and there will no doubt be an increase in Ibrutinib use going forward in this patient group.

Disclosure of Interest: None Declared.

BSH2020-EP-092

Safety, efficacy and tolerability of the NORDIC regimen in the front-line treatment of transplant-eligible patients with mantle cell lymphoma

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Mantle cell lymphoma is a mature B-cell neoplasm comprising 3–10% of all non-Hodgkin lymphomas. It is incurable with standard therapies, with a historic median survival of 4–5 years. The use of high-dose therapy and autologous stem cell transplantation (ASCT) has significantly improved outcomes in younger (<65), fitter and transplant-eligible patients. There is growing evidence that cytarabine-based induction regimens offer a survival advantage in this group of patients, and hence the current standard of treatment for these patients combines upfront anthracycline and cytarabine-based chemoimmunotherapy followed by ASCT. There is no gold standard induction regimen for these patients, and centres vary in their choices depending on local experience. The NORDIC regimen incorporates rituximab and augmented CHOP (R-maxi CHOP)

alternating with high dose cytarabine followed by a BEAM autograft; 15-year follow-up of this regimen has shown median survival of up to 12.6 years. However, this improvement in survival comes at a cost of increased toxicity, including early death and secondary malignancies.

At the Royal Derby Hospital (RDH), patients with mantle cell lymphoma who are deemed suitable for high-intensity chemotherapy are treated with the NORDIC regimen. The aim of this project was to look at the outcomes of all patients treated using this protocol between 2012 and 2018, to assess real life safety, efficacy and tolerability of this regimen.

Overall, 22 patients were treated under the Nordic Protocol (mean age = 59.3 years, M:F 13:9). All patients had stage 3 or 4 disease at diagnosis, and a Mantle Cell Lymphoma International Prognostic Index (MIPI) score was documented for 16 patients (mean = 5.6). The median follow-up period post-treatment was 3.4 years.

Two patients died (9%) during treatment under the Nordic Protocol (including one ischaemic stroke as a cause of death). Four patients relapsed (18%) and underwent treatment escalation to ibrutinib or RBAC (rituximab, bendamustine and cytarabine), of whom one died. Eighteen patients (82%) completed all six cycles of the Nordic Protocol, as one patient de-escalated to R-CHOP plus maintenance rituximab due to recurrent sepsis. Of these, 16 also received an autograft. During treatment with the Nordic Protocol, patients experienced a mean of: 0.7 delayed chemotherapy cycles due to cytopenias, side-effects or intercurrent infection; 3.0 admissions, of which 1.8 were with neutropenic sepsis; and received a mean of seven units of red cells and two pools of platelets. Four patients received dose reductions or alterations for at least one chemotherapy cycle due to toxicity or comorbidities, whilst a further four patients saw their maintenance rituximab discontinued prematurely (out of the 14 who received it). At the time of analysis, two patients had been lost to follow-up, whilst all of the 17 remaining alive patients were in a complete remission (including those whose treatment was escalated due to relapse).

Overall, these data suggest that although this regimen is reasonably well-tolerated and efficacious in the majority, there is considerable associated morbidity and toxicity. More long-term follow-up of a larger patient cohort is required to look at overall survival and late toxicity. Furthermore, similar data from comparable regimens such as RDHAP (rituximab, dexamethasone, cytarabine and cisplatin) would enable clinicians to make a more informed choice of front-line therapy in these younger patients.

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Combination of Bruton's Tyrosine Kinase (BTK) inhibitor, ibrutinib, with rituximab versus placebo with rituximab in previously untreated patients with follicular lymphoma (FL): a randomized, phase 3 study (Perspective)

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Follicular lymphoma (FL) is a slow-growing subtype of non-Hodgkin lymphoma originating from germinal-centre B cells. Very few chemotherapy-free treatment options are available for previously untreated patients with FL, particularly those who are older or who have comorbidities. Rituximab, administered as a single agent, offers a chemotherapy-free treatment option for the first-line treatment of patients who are old, frail or have low tumour burden. Ibrutinib, a once-daily Bruton's tyrosine kinase inhibitor, is approved in the US and EU for the treatment of various B-cell malignancies. A combination of ibrutinib and rituximab was evaluated in a phase 2 trial for the treatment of FL in the first-line setting. Ibrutinib was administered orally, once daily, in combination with once-weekly doses of intravenous rituximab for 4 weeks. Ibrutinib treatment was continued until disease progression or unacceptable toxicity. At a median follow-up of 34 months, an objective response rate of 85% and a complete response rate of 40% were observed. At 30 months, the progression-free survival (PFS) rate was 67% and the overall survival rate was 97% (Fowler, BJH. In press). To further evaluate the clinical activity of ibrutinib-rituximab combination therapy in first-line FL, the phase 3 PERSPECTIVE study (PCYC-1141-CA, NCT02947347) was designed with two objectives: (1) to compare PFS with ibrutinib in combination with rituximab versus placebo-rituximab combination, and (2) to compare PFS with continuous ibrutinib treatment versus fixed-duration ibrutinib treatment. This multicentre, randomized, double-blinded, placebo-controlled phase 3 study will be recruiting approximately 440 previously untreated patients with FL who meet at least one of the Groupe d'Etude des Lymphomes Folliculaires (GELF) criteria (Solal-Célgny, NEJM 1993). Additional inclusion criteria are patient age of either ≥ 70 years or 60–69 years along with one or more of the following comorbidities: creatinine clearance 30–59 mL/min or ECOG performance status of 2. Patients will be excluded if they have received any prior treatment for FL or demonstrated any evidence of transformation or central nervous system involvement. In Part 1 of the study, eligible patients will be randomized into two treatment arms. Arm A will include patients receiving once-daily oral ibrutinib plus rituximab and Arm B will include patients treated with oral daily placebo plus rituximab. In both treatment arms, rituximab will be administered as 4 weekly doses, followed by maintenance rituximab therapy. Part 2 of the study will evaluate patients randomized to Arm A (ibrutinib plus rituximab) who have continued ibrutinib treatment for at least 2 years. These patients will be re-randomized to continue receiving once-daily oral ibrutinib or switch to once-daily oral placebo. The primary end-point of both components of the study will be PFS. Enrolment in

PERSPECTIVE is ongoing at sites in multiple regions, including the US, Europe and Asia Pacific.

Disclosure of Interest: S. Rule Conflict with: Janssen, Conflict with: Janssen and Pharmacyclis LLC, an AbbVie Company, Conflict with: Janssen, Conflict with: speakers bureau for Janssen, I. Flinn Conflict with: AbbVie, Seattle Genetics, TG Therapeutics, Verastem, Conflict with: Institution received from AbbVie, Acerta Pharma, Agios, ArQule, Astra Zeneca, BeiGene, Calithera, Celgene, Constellation, Curis, Forma, Forty Seven, Genentech, Gilead, Incyte, Infinity, Janssen, Juno, Karyopharm Therapeutics, KITE, Merck, Morphosys, AG, Novartis, Pfizer, Pharmacyclis LLC, an AbbVie Company, Portola, Roche, Takeda, Teva, TG Therapeutics, Trillium, Unum, Verastem, R. Chen: None Declared, L. Kwei Conflict with: employment with Pharmacyclis LLC, an AbbVie Company, and stock ownership in AbbVie, A. D. Chu Conflict with: employment with and stock ownership in AbbVie, L. I. Gordon Conflict with: Bristol-Myers Squibb-Celgene-June, Gilead-Kite, and Bayer, Conflict with: speakers bureau for Janssen, Conflict with: patents, royalties, or other intellectual property with Gold nanoparticles in cancer, N. Fowler Conflict with: AbbVie, Celgene, Janssen, Roche, and TG Therapeutics, Conflict with: AbbVie, Celgene, Janssen, Roche, and TG Therapeutics, Conflict with: speakers' bureau for Janssen; travel expenses from Celgene, Janssen, Roche, and TG Therapeutics.

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Real world outcomes for treatment of diffuse large B cell Lymphoma in patients aged 80 and over

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Diffuse large B-cell lymphoma (DLBCL) is an aggressive lymphoma. Treatment of older patients is challenging due to an increased risk of treatment-related adverse events. Studies looking at the oldest patients with this diagnosis are limited and have tended not to include patients with higher levels of comorbidity and poor performance status. We present real-world outcomes for patients aged 80 and over diagnosed with DLBCL in a single NHS Trust.

Electronic records were analysed for all patients in Northumbria Healthcare NHS Trust coded as new DLBCL between 01/01/2013 and 31/12/2017. Demographics, staging, comorbidity and treatment received were collected and survival measured up to a data cut-off of 18/03/2019. Cause of death was collected to allow assessment of disease specific survival (DSS). Differences in overall and disease specific survival were analysed using Kaplan-Meier curves with log rank test.

One hundred and thirteen patients were diagnosed with DLBCL within this 5-year period of which 22 (19.5%) were aged 80 or over; 13 (59%) male and nine (41%) female. Eleven of the patients aged over 80 were treated with curative intent and 11 with palliative intent. The patients treated with curative intent had a better ECOG performance status (ECOG 2+ in 2/11 vs. 10/11) and less comorbidity (HCT-CI 3 +2/11 vs. 4/11). Of those treated with curative intent initial treatment was R-CHOP in 10 cases, given with attenuated doses in 9/10 cases. One patient was treated with R-GCVP. Of those treated palliatively eight received steroids, one R-DECC (Rituximab and oral lomustine, etoposide, chlorambucil and dexamethasone), one diagnosed with testicular and ocular lymphoma had radiotherapy to the orbits following orchidectomy and then R-DECC. One patient deteriorated and died following biopsy before a diagnosis was reached. Twenty of the 22 patients died within the study period (9/11 in the curatively treated group and 11/11 in the palliative group).

Median OS and DSS for all patients was 3 months with 41% 1-year survival. Five deaths (two due to dementia, and one each for cardiac failure, duodenal perforation and lung cancer) were thought to be unrelated to DLBCL, four in the curatively treated group and one in the palliative group. The OS and DSS were significantly better for those treated with curative intent (33 months and not reached vs. 2 and 2 months). Significantly more patients treated curatively were alive at 1 and 3 years (72.7 and 43.6% vs. 9.1 and 0%). In the group treated curatively the five deaths from lymphoma occurred within the first 25 months whilst the four deaths from other causes occurred later (33–49 months).

The decision surrounding treatment in patients over 80 is difficult, as many of these patients are frail and have multiple other comorbidities that influence treatment decision. The median OS of 33 months for patients treated with curative intent is similar to 29 months reported previously by the GELA group for R-miniCHOP in patients aged over 80. However only 50% of this age group were felt to be fit enough to receive this treatment in our cohort and the survival for those treated palliatively was short indicating an unmet need for effective well-tolerated treatment in this group.

Disclosure of Interest: None Declared.

BSH2020-EP-095

Immunochemotherapy in patients with hepatitis C associated diffuse in large cell lymphoma and direct-acting antiviral agents (DAA)

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Hepatitis C virus (HCV) infection is a major public health issue worldwide. HCV is both hepatotropic and lymphotropic. It is a proven fact etiopatogenetic the relationship between HCV and marginal zone lymphomas and DLBCL. A large number of studies have shown high antitumor efficacy of antiviral therapy interferon and ribavirin in patients with hepatitis C-associated indolent lymphomas. There are not many studies showing anti-tumour efficacy in patients with hepatitis C-associated indolent lymphomas in direct-acting anti-viral drug therapy (DAA). Data on the efficacy of new direct-acting antiviral drugs (DAA) in the treatment of patients with DLBCL + HCV are limited.

The aim of our study is to evaluate the clinical features and treatment results of patients DBCL + HCV depending on the treatment option: immunochemotherapy +/- DAA.

The study included 49 patients with HCV-associated DLBCL who received treatment between 2016 and 2019 at the National medical research center of Oncology. N. N. and the Blokhin Russian National Research Medical University N and Pies. The median age of the patients was 48 years (32–75). The ratio of men and women was 31/18. Forty-five (91%) patients had stage III-IV. B-symptoms were 25 (52%) patients. Histological variants: in 28 (57%) patients were DLBCL GCB type, in 21 (43%) patients not GCB DLBCL type. The

defeat of the spleen was in 18 (37%) patients, bone marrow in 15 (31%) patients, liver damage – in 10 (20%) patients. When diagnosing lymphoma, increase in LDH levels ≥ 450 IU/l was found in 40 (81%) patients, ALT ≥ 40 IU/l in 46 (95%) patients, AST ≥ 40 IU/l in 34 (70%) patients, GGTP ≥ 35 IU/l in 15 (44%) patients and albumin ≤ 35 g/l in 11 (37%) patients. HCV RNA was detected in the blood of all patients. The median RNA level was 5×10^5 IU/l. 1-St virus genotype was detected in 31 (63%) patients.

Depending on the treatment the patients were divided into two groups: group 1 G1 -18 patients who received immunochemotherapy (R-CHOP) together with DAA therapy in accordance with the virus genotype, group 2 G2 -31 patients who received only immunochemotherapy.

In G1 after 30 days R-CHOP + DAA-therapy virological remission was achieved in 94% (16) patients. The ALT level of ≤ 40 IU/l is normal in 14 (77%) patients, AST ≤ 40 IU/l-16 (90%) patients, GGTP ≥ 35 IU/l was 16 (90%).

In G2 after 30 days on immunochemotherapy R-CHOP viral load increased in all patients.

The overall response (ORR) to therapy in G1 and G2 was the same and amounted to 56 and 57%. The median follow-up was 24 months.

2-year disease progression-free survival (PFS) in G1 was 76%, in G2 – 52%. In G1, 15 (83%) patients managed to complete the full treatment programme for DBCL (without increasing the time between courses of treatment and without reducing the doses of R-CHOP), in G2 only 18 (56%) patients completed the full treatment programme.

Conclusion: DAA therapy allows completing a therapy programme without complications for all patients with DLBCL + HCV. The safety, speed and effectiveness of DAA in obtaining a virologic response, as well as the good tolerance profile of DAA therapy can be used in combination with immunochemotherapy

Disclosure of Interest: None Declared.

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Hepatitis C (HCV) associated diffuse B-cell large lymphoma (DBCL) and maintenance antiviral therapy (AVT) with interferon

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In the last two decades on the basis of epidemiological, molecular and therapeutic studies the aetiopathogenetic link between HCV and b-cell non-Hodgkin lymphoma has been proven. It is now shown that various modes of anti-viral therapy (AVT) can induce remission in indolent lymphomas. The role of AVT in the treatment of DBCL is not so obvious. The role of maintenance therapy in the treatment of DBCL is also a cause for great debate.

The aim of our study was to evaluate the role of maintenance therapy with interferon in patients with HCV-associated DBCL lymphoma.

Our study included 80 patients(pts) with HCV associated DBCL lymphoma, which from 2005 to 2019 in Cancer Research Center named N.N. Blokhin and Russian National Research Medical University named N.I. Pirogov received therapy about HCV associated DBCL lymphoma.

The age of patients ranged from 21 to 76 years, median 47 years, ratio of male to female was 1:1.3. The median follow-up was 49 months.

In the group of pts with HCV associated DBCL lymphoma I + II stage was in 9 (11%), III+ IV stage was 71 (89%) pts. Histological variants were: 44 (55%) pts have GCB variant DBCL and non-GCB — 36 (45%) pts. Extranodal lesions were present in 58 (72%) pts. The involvement of the spleen was in 34 (43%) pts, bone marrow — in 25 (31%) pts, the lesions of the liver was in 16 (20%) pts and in these group 3 patients had isolated lesions of the liver. At the time of diagnosis increase in LDH ≥ 450 IU/l was observed in 68 (81%) pts, ALT ≥ 40 IU/l in 72 (90%) pts, AST ≥ 40 IU/l in 52 (65%) pts, GGTP ≥ 35 IU/l in 30 (38%) pts and albumin ≤ 35 g/l in 15 (18%) pts. All pts were determined the viral RNA. Viral load ranged from 3×10^3 to 1×10^8 copies/ml, the median amounted to 1×10^5 copies/ml. The first genotype of HCV was found in 58 (73%) pts.

All pts received chemotherapy according to the scheme CHOP/R-CHOP. Complete remission was achieved in 48 (60%) pts. Among this group of pts 17 (21%) were primarily resistant to the immunochemotherapy.

Pts were divided into two groups: the first group (G1) included 49 patients who after chemotherapy received AVT with the inclusion of interferon; the second group (G2) included 31 patients without AVT. AVT was conducted from 12 to 30 months after the end of chemotherapy. The median duration of AVT was 21 months.

Median overall survival (OS) in G1 was 67 months and in G2 — 28 months ($P = 0.0003$). The median of progression-free survival (PFS) in G1 was 52 months, and in G2 — 30 months ($P = 0.0005$).

OS in pts with GCB variant of DBCL in G1 was 71 months, and in G2 — 32 months ($P = 0.0005$). The median of PFS in pts with GCB variant of DBCL in G1 was 61 months and in G2 — 22 months ($P = 0.001$).

The OS in pts with non-GCB variant DBCL in G1 was 33 months, and in G2 — 18 months ($P = 0.0001$). The median of PFS in pts with non-GCB variant of DBCL in G1 was 25 months and in G2 — 13 months ($P = 0.0002$).

Conclusions: HCV-associated DBCL is a separate group of lymphomas with characteristic clinical and morphological features: young age of patients, advanced stages of the disease and frequent extranodal lesions. The long-term results of PFS and OS therapy were significantly better than patients with HCV associated DBCL with AVT including alpha interferon therapy.

Disclosure of Interest: None Declared.

We present a rare and possibly unique case of synchronous dual haematological malignancy (SDHM) of Lymphoplasmacytic lymphoma/Waldenstrom macroglobulinaemia with concomitant Hairy cell leukaemia at our haematological malignancy diagnostic centre at Kings College Hospital, London, UK.

A 77-year-old gentleman was referred with symptoms of weight loss, lethargy and night sweats, and blood investigations showed anaemia (Hb 7.7 g/dl) and IgM kappa paraproteinaemia (IgM 5 g/l). Significant past medical history includes Type 2 diabetes mellitus and transitional cell carcinoma of the bladder 2010 (G2 pT1) treated with BCG. There was also evidence of hydronephrosis, hypothyroidism and folate deficiency and a CT abdomen and pelvis performed in 2017 did not show splenomegaly.

Bone marrow samples for immunophenotyping revealed two distinct B-cell clones. The larger clone exhibited an immunophenotype diagnostic of Lymphoplasmacytic lymphoma (LPL) whereas the smaller clone showed an immunophenotype typical of Hairy cell leukaemia (HCL). The LPL clone demonstrated kappa light chain restriction whereas the HCL clone displayed lambda light chain restriction. A small population of clonal kappa light chain restricted plasma cells was also detected. The bone marrow histopathology and immunohistochemistry (IHC) was reviewed; 70–80% cellularity, prominent lymphoid and plasmacytoid cell infiltrate and some plasma cells with Dutcher bodies were observed. A subpopulation of lymphoid cells showed IHC staining with DBA44, CD11c, CD25 and CD123. Also, CD38 and CD138 IHC stains highlighted 20% >30% of plasma cells which showed kappa light chain restriction. Findings were consistent with the involvement of bone marrow by LPL with HCL. The predominant population involving BM was LPL (70%) while HCL showed 30% involvement of the bone marrow. The aspirate immunophenotype and bone marrow IHC studies correlated for this unusual case. Subsequent molecular studies confirmed both MYD88 L265P and BRAF V600E variants by Real-Time PCR confirming the diagnosis of co-existent LPL and HCL.

The patient received six courses of rituximab, cladribine and dexamethasone, and a repeat bone marrow assessment showed reduction in the both clones but both were still detectable (LPL ~7% and HCL ~2%) respectively. The patient was subsequently lost to follow-up.

Multiparametric flow cytometry played a vital role in the complete diagnosis of our case clearly showing two phenotypically different clones with differing light chain isotypes. The incidence of this Lymphoid+Lymphoid group of SDHMs is extremely low (0.66% in a study by Rouslan Kotchetkov et al, 2018). Co-existent LPL and HCL have not been reported in the literature so far to the best of our knowledge.

Keywords: Synchronous dual haematological malignancy (SDHM), bone marrow (BM), lymphoplasmacytic lymphoma (LPL), Hairy cell leukaemia (HCL), polymerase chain reaction (PCR)

Disclosure of Interest: None Declared.

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Synchronous dual haematological malignancies: lymphoplasmacytic lymphoma and hairy cell leukaemia

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Dual diagnoses pose a diagnostic challenge and are often under-diagnosed. Literature is scarce about synchronous dual haematological malignancy (SDHM) though we have experience of several cases with precursor myeloid and peripheral lymphoid neoplasms.

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Asian variant intravascular lymphoma – the itinerant impressionistA. Sridhar^{1*}, S. Kassam², N. Cooper³, J. Salisbury⁴, U. Mahadeva⁵, S. Delpino⁶, S. Gupta⁷¹Haematology, Guys and St Thomas' Hospitals,²Haematology, Kings College Hospital, ³Haematology, Hammersmith Hospital, ⁴Pathology, Kings College Hospital, ⁵Pathology, Guys and St Thomas' Hospital, ⁶Pathology, Hammersmith Hospital, ⁷Haematology, Lewisham and Greenwich NHS Trust, London, UK

Background: Intravascular Lymphoma (IVL) is rare and aggressive, common B-cell type, observed as a myriad of symptoms with rapid and fatal progression. IVL cells are restricted to the vascular intima, and deposits are poorly visualised on imaging, posing a diagnostic challenge. The pathophysiology is elusive, though impaired lymphocyte trafficking may be involved. IVL is generally divided into 'Western' and 'Asian' variants, but haemophagocytic lymphohistiocytosis (HLH) features may be better defining than ethnicity. 'Western' IVL has skin and neurological involvement. 'Asian' IVL demonstrates HLH, hepatosplenic involvement and cutaneous sparing. Initially reported in Asian patients, it has since been described in Afro-Caribbean, Latin and Caucasian patients.

Aims: We describe three patients with IVL to further illustrate this rare entity.

Methods: We have reviewed patient notes to report clinicopathological features.

Results: Patient 1 was a 73-year-old Vietnamese man with a 2-month history of dry cough and weight loss. He suffered a stroke and was treated for pneumonia. HLH was diagnosed; persistent pyrexia, splenomegaly (143 mm), cytopaenias (Hb 74 g/L, plt 60 × 10⁹), hyperferritinaemia (8770 µg/L), high triglycerides (3.4 mmol/L), and successive marrow aspirates showed increasing macrophage activity, and eventually haemophagocytosis. He had intravenous immunoglobulins (IVIg), steroids and etoposide therapy, alongside broad-spectrum anti-microbials. IVL was diagnosed on autopsy, affecting the adrenal vasculature, with increased hepatosplenic macrophage activity.

Patient 2, a 63-year-old man of Caucasian origin with a background of Sarcoidosis, presented with a 4-month history of weight loss and leg weakness and acute sensorineural deafness. MRI Brain and Spine showed cerebral white matter lesions across vascular territories and degenerative spinal changes. EMG showed polyneuropathy. PET CT did not support neurosarcoidosis, but he had a brief response to steroids. Further investigations showed HLH: pyrexia, splenomegaly (162 mm), cytopaenias (Hb 75 g/L, Plt 41 × 10⁹), hyperferritinaemia (6564 µg/L), high soluble CD25 (>20 000 µg/L) and marrow HLH. He had IVIg therapy but deteriorated due to multi-organ failure. Autopsy showed diffuse cerebral intravascular lymphoma.

Patient 3 was a 69-year-old Caucasian man with pyrexia, weight loss and night sweats, and treated for sepsis, with no malignant findings on initial investigations. He developed left leg weakness and urinary retention, with no acute changes on MRI spine. A marrow aspirate appeared reactive. Repeat CT scan demonstrated splenic lesions. Biopsy of a thigh skin rash revealed IVL. He did not respond to R-CHOP therapy. Whilst disease locus favours Western IVL, he demonstrated HLH; splenomegaly (150 mm), recurrent pyrexia, high

ferritin (15210 µg/L), hypertriglyceridaemia (3.4 mmol/L) and cytopaenias (Hb 81 g/L, Plt 84 × 10⁹). Repeated marrow aspirates showed IVL infiltration and increasing macrophage activity, but not frank HLH. He presented an interesting picture of cutaneous IVL with HLH; a potential overlap between IVL variants.

Conclusion: IVL presentation is highly heterogeneous with insidious onset and rapid progression. The 'Asian' Variant IVL also occurs in non-Asians, with potential overlap between the variants. In conclusion, IVL is a disease posing diagnostic and therapeutic difficulty, but should be an important differential in patients with HLH in the absence of precipitating factors.

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Real world experience of Ibrutinib at the University Hospital of Wales, Cardiff

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Background: The Bruton's Tyrosine Kinase inhibitor (BTKi) Ibrutinib is licensed for the treatment of Chronic Lymphocytic leukaemia (CLL), Waldenström's Macroglobulinaemia (WM) and Mantle Cell lymphoma. Since entering the market in 2014 its use has increased significantly, but there remains a lack of real-world data. There is also increasing interest in the long-term effects of Ibrutinib use.

Aim: To review data on patients treated with Ibrutinib outside clinical trials at University Hospital of Wales, concentrating mainly on side effects experienced and dosing of Ibrutinib.

Method: A total of 32 patients were treated between 1st April 2014 and 30th of June 2019. Three (9%) accessed the drug on compassionate grounds, one (3%) via health insurance and the remainder 28 (87.5%) via the NHS. Twenty-six patients treated in clinical trials were excluded.

Results: The majority of patients had CLL or Small Lymphocytic Lymphoma (SLL) (21/32). Five out of thirty-two patients had Mantle Cell Lymphoma and 4/32 had WM. Interestingly, one patient had dual pathology (WM and CLL) plus another developed Chronic Myeloid Leukaemia after their diagnosis of CLL. Patients were mostly male (88%) with an average age of 68 years old. All patients received at least one prior line of treatment, median 2 (1–9).

Of the 23 patients treated for progressive CLL, 14/23 were Stage C. Eight out of twenty-three patients had either deletion of 17p or mutated TP53 on cytogenetic testing. The median length of treatment was 18.7 months (1–69).

Eighty-one percent of all the patients on Ibrutinib reported side effects. The frequency of side effects is reported in Table 1 below:

When a patient experienced an adverse event, Ibrutinib was either withheld or dose reduced. When able, the medication was reintroduced, and attempts were made to increase to the maximum dose tolerate to aid disease control. Fourteen out of thirty patients had their Ibrutinib stopped but then restarted, however only five of these managed to reach full dose Ibrutinib again. Five remain on a lower dose and are well. Two patients suffered disease progression while taking a lower dose, whilst two patients discontinued the medication entirely after failing dose titration.

As of January 2020, 20/32 patients continue on Ibrutinib. Of the 12/32 that have stopped treatment, 5/32 (16%) stopped due to side

Table.

Side effect	Rash	Diarrhoea	Arthralgia	Transaminitis	Cardiac	Bleeding	Infection	Cytopenia
Frequency	7 (22%)	3 (9%)	6 (19%)	1 (3%)	6 (19%)	3 (9%)	7 (22%)	7 (22%)

effects and (5/32) due to disease progression, with another one patient undergoing Richter transformation and another diagnosed with a lung malignancy.

In total eight of the patients died, seven from progressive disease and the final patient died from an unrelated malignancy. Two of the patients who died from progressive disease had been unable to tolerate Ibrutinib due to significant side effects- (namely rash and diarrhoea) and had discontinued the medication prior to progression.

Conclusions: In our cohort rash, cardiac arrhythmia and diarrhoea were the most clinically significant side effects. It is important to manage these side effects appropriately and use dose modifications to allow patients to gain maximum disease control with Ibrutinib.

Disclosure of Interest: None Declared.

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An analysis of patients treated with weekly salvage chemotherapy regimens for high grade lymphoma

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Diffuse large B-cell lymphoma (DLBCL) is potentially curable, however 13% of patients have refractory disease and a further 30–40% relapse after front-line therapy. Outcomes are poor among these patients with response rates of approximately 50% in the second-line setting. In transplant eligible patients, the aim of salvage chemotherapy is to reduce disease sufficiently to proceed to stem cell autologous transplant (ASCT). There is no clear evidence that one salvage regimen is more efficacious than another, often in the UK, R-GDP or R-GemP is preferred due to it being an outpatient regimen, associated with decreased toxicity and equivalent efficacy to other salvage regimens.

Prior to CAR-T being available in the UK, or in patients unsuitable for CAR-T, in our centre, weekly salvage chemotherapy regimens (WSC) were considered for patients with R/R DLBCL who have failed ≥ 1 salvage regimen. The aim of the chemotherapy regimen is to allow patients to proceed to ASCT in those achieving a sufficient response; as holding or bridging treatment prior to CAR-T or to prolong survival in those who are unsuitable for consolidation or CAR-T therapy.

We present the outcomes of patients treated with weekly salvage chemotherapy (WSC) regimens between 2002 and 2019 for R/R DLBCL. Patients were censored at last clinic date.

Specific WSC regimens were based on frailty, co-morbidities and prior therapy history, and consisted of:

- (+/-R)-PCE2BO(+/-M);
- (+/-R) PCEGEMBO(+/-M);
- (+/-R) PMITCEBO(+/-M)

Abbreviation (dose):

R: Rituximab (375 mg/m²); P: Prednisolone; E:Etoposide (300 mg/m²); C: Cyclophosphamide (300 mg/m²); B: Bleomycin (10 000 IU/m²); O: Vincristine (1.4 mg/m²); Mit: Mitoxantrone (7 mg/m²); M: Methotrexate (100 mg/m²).

Thirty patients (*de novo* DLBCL $n = 18$; transformed DLBCL $n = 12$) with a median age of 60 years (range 20–84) received WSC (median four cycles, range 2–6). Median prior lines of therapy were three (range 1–5).

Overall response rate was 20/30 (66%) (*de novo* DLBCL $n = 13$; transformed DLBCL $n = 7$).

Seven of the responding patients proceeded to ASCT and one to allograft. These eight patients had a median of 6 months of follow-up (range 2–16), all are still alive and only one has relapsed. Two patients were referred elsewhere for CAR-T therapy. Ten patients who achieved a response were unsuitable for transplant due to frailty or co-morbidities, with a median follow-up of 27 months of these patients, the median PFS for this cohort is 41 months.

With a median follow-up of 15 months for the whole cohort the median PFS is 41 months (CI 95%, 6–76) and 2-year OS is 77%.

The response rates, duration of response and the number of patients able to proceed to transplant in this heavily pretreated patient population are encouraging. We hypothesise that the reason for these results is the heavy steroid use that forms part of this regimen and the weekly 'metronomic' nature of this chemotherapy that does not have prolonged breaks in treatment between cycles. WSC was well tolerated, including in those with poorer performance status and co-morbidities.

Formal studies comparing this salvage regimen to alternatives should be performed for confirmation of the results. WSC could be considered for patients with R/R DLBCL in those patients who are potentially candidates for ASCT, in those patients who may need bridging for CAR-T, or in those who are ineligible for either as it may provide a reasonable duration of response.

Disclosure of Interest: None Declared.

Myeloma

BSH2020-EP-101

Real-world experience of KPD use in relapsed or refractory myeloma

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In multiple myeloma (MM), treatment options for patients' refractory to lenalidomide and bortezomib are very limited. In our centre heavily pre-treated MM patients who had received lenalidomide and bortezomib and were refractory or unable to tolerate these treatments were able to receive carfilzomib 20 mg/27 mg/m² day 1,2,7,8,14 and 15, dexamethasone 40 mg weekly and pomalidomide 4 mg 21 of 28 days (the KPD regimen). From 2016 to 2019, 14 patients received this combination of treatment. The male:female ratio was 7:1. The median age at MM diagnosis was 61.2 (46.8–78.5). This was a heavily pre-treated group with the median time to commencement of KPD from MM diagnosis of 3.8 years (0.5–15.5). The median number of prior regimens was 4 (1–11). All patients had received bortezomib and lenalidomide prior to starting this regimen. Eight patients (57.1%) had an autologous stem cell transplant and two patients (14.2%) had a second transplant after a prolonged period of remission. The median number of cycles of KPD in all patients including those still on the regimen was 5 (0.5–46).

Responses were evaluated in 82% (10/12) of patients eligible for assessment. The overall response rate was 50% (6/12). Complete remission (CR) was achieved in one patient (7.1%), very good partial response (VGPR) in three patients (21.4%) and partial remission (PR) in two patients (14.2%). Two patients had primary refractory disease having received bortezomib, lenalidomide and dexamethasone as first-line therapy but still responded to KPD.

Toxicity was common but manageable. Eight patients (57.1%) experienced grade 3 adverse event (AE) and no grade 4 AEs were recorded. Two patients (14.2%) developed deep venous thrombosis (DVT) despite thromboprophylaxis with low molecular weight heparin (LMWH) and required treatment with a direct oral anticoagulant (DOAC). Six (42.9%) patients experienced haematological toxicities necessitating dose reduction, transfusion or growth factor support. Five patients (35.7%) had grade 3 infectious complications (three episodes of sepsis and two of pneumonia). Other AEs of note included avascular necrosis of the hip (grade 2) and exacerbation of cardiac failure (grade 2). Two patients (14.2%) developed deep venous thrombosis (DVT) despite thromboprophylaxis with LMWH.

Thrombosis is a significant cause of comorbidity in MM patients treated with combination chemotherapy and can occur despite appropriate use of prophylactic anticoagulation. Recent studies have showed encouraging results with the use of the DOAC apixaban as thromboprophylaxis in this setting and this may be considered in future therapeutic strategies.¹

In conclusion, KPD was used successfully in our institution in a group of 14 heavily pre-treated MM patients with an ORR of 50% of 12 evaluable responses. Haematologic AEs predominated and thrombosis remains a significant concern. KPD remains an attractive option for salvage of relapsed and refractory MM in this cohort.

Reference: 1. Storrar NPF, Mathur A, Johnson PRE et al. Safety and efficacy of apixaban for routine thromboprophylaxis in myeloma

patients treated with thalidomide and lenalidomide-containing regimens. *Br J Haematol.* 2019;185(1):142–144.

Disclosure of Interest: None Declared.

BSH2020-EP-102

Mature osteoblasts induce myeloma cell death through inhibiting the Sp1-TAK1-Pim2-prosurvival pathway

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Background: Myeloma (MM) growth in bone has been demonstrated to be inversely correlated with bone formation in MM animal models treated with bone-anabolic non-cytotoxic agents, such as an anti-DKK1 antibody. We have reported that Pim-2 is constitutively over-expressed as a critical pro-survival mediator (Leukemia 2011, 2015) and further upregulated through TAK1 activation in MM cells upon co-culturing with bone marrow stromal cells (BMSCs) or osteoclasts. In this study, we explored the mechanisms of MM cell growth inhibition by mature osteoblasts (OBs) differentiated from BMSCs.

Methods: MC3T3-E1 preosteoblastic cells or BMSCs were cultured in osteogenic media with BMP-2 to differentiate into mature OBs with mineralized nodule formation. The expression of critical survival mediators in MM cells as well as their viability were analysed after co-culturing with mature OBs or the preosteoblastic cells/BMSCs.

Results: When co-culturing with mature OBs, MM cells underwent apoptosis with the activation of caspase-8, while promptly reducing TAK1 and Pim-2 levels followed by the phosphorylation of AMPK, a key energy sensor in the MM cells. Although the transcription factor Sp1 was overexpressed in MM cells, co-culturing with mature OBs reduced their Sp1 expression at protein but not at mRNA levels. Intriguingly, addition of the caspase-8 inhibitor z-IETD-FMK abolished the Sp1 reduction in the MM cells upon co-culturing with mature OBs, suggesting caspase-8-mediated Sp1 protein degradation. Importantly, the Sp1 inhibitor, terameprocol, induced MM cell death along with reduction of TAK1, Pim-2, IRF4 and cMyc. Interestingly,

mature OBs substantially reduced the expression of PGC-1 α , an activator of glycolysis and mitochondrial biogenesis, along with subsequent ATP reduction and phosphorylation of AMPK in MM cells. The Pim inhibitor, SMI16a, reduced PGC-1 α levels and thereby AMPK phosphorylation in MM cells, confirming the causative role of reduction of Pim-2 in PGC-1 α suppression and AMPK activation. Furthermore, the AMPK activator, AICAR, triggered MM cell death; the AMPK inhibitor, dorsomorphin, partially mitigated the OB-induced MM cell death, suggesting further impairment of MM cells by AMPK activation.

Conclusions: In sharp contrast to MM growth enhancement with activation of the TAK1-Pim-2 pathway by BMSCs and osteoclasts, OBs with mineralized nodule formation are able to induce apoptosis along with inhibition of the Sp1-TAK1-Pim-2-mediated prosurvival pathways and thereby impairing PGC-1 α -driven energy production and metabolism.

Disclosure of Interest: None Declared.

BSH2020-EP-103

Lenalidomide with low-dose dexamethasone in newly diagnosed transplant-ineligible myeloma. Real-world experience: Data from a Single Centre in North Wales, UK

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Background: In Wales, lenalidomide in combination with low-dose dexamethasone is indicated for the treatment of adult patients with previously untreated multiple myeloma who are not eligible for transplant and unsuitable for thalidomide-containing regimens.

Aims: To review the efficacy and side effects of lenalidomide with low-dose dexamethasone in newly diagnosed transplant-ineligible multiple myeloma in a real-world setting and to compare our results to outcomes from trial data as Wales was the first country in the United Kingdom to approve above regimen.

Method: A retrospective analysis of myeloma patients in our haematology centre who had received lenalidomide with low-dose dexamethasone as first-line therapy between November 2017 and January 2019 (15 months) was carried out. The data were collected from Welsh Clinical Portal (a national electronic patient record), electronic pharmacy records and clinic notes.

Results: Twelve patients received lenalidomide with low-dose dexamethasone between November 2017 and January 2019, with a median age at commencement of treatment of 76.5 years (63–86). 58% of patients were female.

Four patients (33%) showed a complete response (CR) defined by International Myeloma Working Group response criteria. One patient elected to have a break after 19 cycles of chemotherapy and remained in CR.

Another four patients (33%) were shown to have very good partial response (VGPR). Out of these four patients, one decided to stop treatment due to multiple pre-existing medical co-morbidities, and a second patient progressed later.

Three (25%) patients achieved partial response (PR).

Response rates in the trial: CR 15%, VGPR 28%, PR 32%.

Haematological side effects were more commonly encountered than non-haematological side effects: grade 3 anaemia in 33%, grade 3 neutropenia in 25%, grade 4 neutropenia in 17% and grade 3 thrombocytopenia in 8%. Seven (57%) patients developed infections with one case of neutropenic sepsis. Two patients (17%) developed sub-segmental pulmonary emboli (1 while on prophylactic low molecular weight heparin). Two patients (17%) developed grade 1 neuropathy.

The majority (83%) of patients tolerated this regimen well (10 out of 12 continued treatment).

In the trial, grade 3/4 haematological adverse effects were less compared to our data due to high numbers of patients. Grade 3/4 adverse events were: neutropenia 28%, anaemia 18%, thrombocytopenia 8% and venous thromboembolism occurred in 8%.

Conclusion: Our results indicates that lenalidomide with low-dose dexamethasone is well tolerated, with an excellent response by elderly transplant-ineligible patients.

Disclosure of Interest: C. Arachchilage: None Declared., E. Heartin Grant/Research from: Dr Earnest Heartin - Conflict with Celgene, Jansen, Takeda, Novartis. C. Smith: None Declared.

BSH2020-EP-104

Single centre retrospective cohort study of the occurrence of venous thromboembolism in patients receiving immunomodulator-based therapy for the treatment of multiple myeloma

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The outcomes of patients with multiple myeloma have improved drastically over the last few decades, and this success is partly attributable to therapies such as immunomodulators (IMiDs) encompassing thalidomide, lenalidomide and pomalidomide. Despite the nationwide use of IMiDs, and its perceived prothrombotic properties, there remains ambiguity regarding whether the choice of thromboprophylaxis, aspirin or low molecular weight heparin (LMWH), influences the risk of developing a venous thromboembolism (VTE). There is also uncertainty whether prophylaxis should be reserved for 'high-risk' patients and the appropriate risk stratification of these patients remains unclear.

Real-world data were collected retrospectively from all patients with plasma cell disorders who received IMiDs containing regimens between 1 July 2018 and 30 June 2019 at Guy's and St Thomas' NHS Trust, London; totalling 91 patients with 42 (46%), 38 (42%) and 11 (12%) receiving lenalidomide, thalidomide and pomalidomide respectively. All information was collected using the following online programs: Electronic Patient Records for pathology results and clinic letters, E-noting for comprehensive patient records, PACS SPECTRA for imaging reports and MOSAIQ for all prescribed chemotherapy.

There was a total of eight (9%) VTEs all occurring in patients who have IgG kappa myeloma of varying International Staging System severity (ISS). 4 (50%), 3 (37.5%) and 1 (12.5%) of the VTEs occurred in patients receiving lenalidomide, thalidomide and pomalidomide respectively. All IMiDs were taken in conjunction with high-dose dexamethasone which has known prothrombotic properties. All eight patients had received thromboprophylaxis in a 50:50 distribution between aspirin and LMWH. On average the VTEs occurred at cycle 4-25.

There were three reports of pulmonary embolism (PE) and one retinal vein occlusion in patients receiving lenalidomide (5–25 mg), three of which had received LMWH and one had received aspirin prophylaxis. Three of which had an ISS score of 3 and one patient with an ISS score of 1, with 75% of these patients having received previous IMiD therapy. There were three reports of PE in patients receiving thalidomide, two of which occurred in the same patient once Thalidomide was reintroduced. One DVT was reported in a patient with an ISS score of 2, who was receiving Pomalidomide (4 mg) and LMWH prophylaxis.

Only three (6%) of these patients had been deemed 'high risk' for having at least one additional thrombotic risk factor; previous VTE, malignancy, recent operation, reduced mobility or obesity. Each VTE occurred in patients who were fully compliant with thromboprophylaxis. There was no perceived difference in receiving either aspirin or LMWH prophylaxis. Routine anti-Xa levels were not performed.

Thromboprophylaxis was given in 92% of cases, and where it was not offered this was due to bleeding risk. The results show no difference between aspirin and LMWH in VTE risk reduction from this real-world population of patients. The question remains how best to reduce the risk of VTE in patients with myeloma receiving IMiD therapy as all of the VTEs reported occurred in patients reporting good compliance with thromboprophylaxis. DOACs or coumarins have not been historically employed in our centre.

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BSH2020-EP-105

Use of imaging in the management of newly diagnosed patients with myeloma at St. George's Hospital

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Cross-sectional imaging techniques have increased the sensitivity of detection of myeloma bone lesions compared to skeletal plain radiographs. However there is huge variation in practice as to the choice of modality due to differences in access to the techniques and the subtleties of the information gained from each type of imaging. At present whole-body (WB) diffusion-weighted (DW) MRI is not yet available in our hospital.

We aim to review compliance of our current practice with the 2017 British Society for Haematology (BSH) guideline 'Use of imaging in the management of patients with myeloma'. To review the number of imaging modalities performed in newly diagnosed myeloma patients.

Audit standard criteria range: 100%

(i) Patient with suspected myeloma underwent whole body MRI (WB-MRI) or CT. (ii) Patient without established myeloma-defining event but with bone marrow plasma cells 10–60% and/or M-protein > 30 g/l underwent WB-MRI, WB-CT or PET/CT. (iii) WB/DW-MRI or PET/CT performed in the assessment of oligosecretory myeloma. (iv) Whole spine MRI performed and reported within 24 h of suspected cord compression in myeloma. (v) PET/CT or WB/DW-MRI performed in the diagnostic assessment of possible solitary plasmacytoma.

Data collection period was between 1 July 2017 and 30 June 2018. We included all adult patients (age at least 18 years) who were newly diagnosed with myeloma or plasmacytoma in our Trust during the data collection period, and had their diagnostic imaging in our Trust. Data were retrospectively collected from patients' records.

Forty patients were included in the audit, which included 31 myeloma patients, eight smouldering myeloma patients and one plasmacytoma patient. There were no non-secretory or oligosecretory myeloma cases and no newly diagnosed myeloma patients presenting with suspected cord compression in our Trust during the data collection period.

All patients (100%) with newly diagnosed myeloma underwent cross-sectional imaging at diagnosis (Table 1). No skeletal surveys were performed. All smouldering myeloma patients underwent LDWB-CT. All solitary plasmacytoma patients had WB-MRI.

Twenty out of 40 (50%) patients had more than one imaging modality. For the majority of these patients this comprised a CT and MRI whole spine/pelvis (40%) and 10% had a CT and PET-CT. Two patients had three modes of imaging. The main reasons for requesting MRI spine/pelvis were back pain or a CT showing lytic lesion, equivocal lytic lesion, mass or abnormal marrow signal in spine. The main reasons for requesting PET/CT were equivocal lytic lesion on CT, or suspected cancer.

Our Trust has excellent compliance with the BSH guideline as all new myeloma patients underwent cross-sectional imaging with a CT. However, for half of these patients CT alone was inadequate to fully assess their bone disease, requiring an additional modality.

Conclusions:

- Patients with suspected myeloma are referred to the haematology department through different pathways, most commonly suspected myeloma and back pain (73%), and suspected cancer (27%).
- The route of referral often determines first imaging modality.
- For many patients more than one modality is required to accurately assess their bone disease, for example presence of focal lesions, active lesions, spinal cord compromise.

These factors may delay diagnosis and increase cost. Access to WB/DW-MRI may address some of these issues.

Disclosure of Interest: None Declared.

BSH2020-EP-106

Dynamic changes in serum complements during chemotherapy in patients with multiple myeloma and their correlations with disease markers

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Introduction: Multiple myeloma is associated with severe perturbations in immunity predisposing to infections. Reductions in normal

Table 1. Diagnostic imaging performed for patients with newly diagnosed myeloma.

Diagnosis	N	Cross-sectional diagnostic imaging requested				
		LDWBCT (N)	CT-CAP (N)	WB-MRI (N)	PET/CT (N)	MRI spine +/- pelvis (N)
Myeloma	31	20	11		3	13
Smouldering myeloma	8	8			1	3
Plasmacytoma	1	1		1		
Total number (%)	40	29 (73%)	11 (27%)	1 (2%)	4 (10%)	16 (40%)

immunoglobulins, abnormal T-cell functions and abnormalities of complement system have also been reported (Cheson et al., 1980; Kraut & Sagone, 1981; Lugassy et al., 1999; Zurlo et al., 1989). It is not known whether the clinical improvements seen with new anti-myeloma therapy is reflected in corresponding improvements in the immunological abnormalities involving complement system. We undertook a study to investigate the quantitative abnormalities of complement proteins (C3 and C4) during chemotherapy. In addition, the functional status of the complement system was studied by CH50 assay.

Methods: This was a prospective non-randomised laboratory-based study using surplus diagnostic material (serum). Institutional approval was obtained from the Trust Research & Development Committee. Patients with established multiple myeloma requiring chemotherapy and those with relapsed disease were identified. Serum concentrations of complement proteins C3 and C4 were studied by immunoturbidimetry using Roche Cobas analyser. CH50 assay used liposome-encapsulated G6PD to mimic invading microorganism. Antibody added along with complement from patient serum lyses the lysosome, releasing the G6PD which was measured (Optilite®CH50 Reagent from The Binding Site Group Ltd). Serum samples for CH50 assay were collected and frozen at -80°C within an hour of collection. The CH50 assay was performed in batches following the manufacturer's instructions. Correlations were made between the disease markers and the complement levels.

Results: Fifty-one complement assays were carried out on 31 patients between January and December 2019. Low C3 Levels (<0.75 g/l) with a median of 0.65 g/l was seen in three (6%) patients prior to starting chemotherapy. Low C4 levels (<0.14 g/l) with a median of 0.05 g/l was seen in 17 (38%) patients of which 10 (58%) were pre-chemotherapy and seven (41%) were during chemotherapy. The clinical characteristics of the patients with low C4 levels are summarised in Table 1. There were no discernible changes in complement levels during chemotherapy.

Batched CH50 testing was performed on 24 cases. The median CH50 was 87.75 U/ml (41.68 – 95.06). Only one (4%) patient had a low CH50 (38.44 U/ml). It is noteworthy that this patient had a normal C3 and C4 levels (1.13 g/l and 0.43 g/l respectively).

Conclusions: This study demonstrated that reductions in serum C3 level are uncommon in multiple myeloma. However, significantly reduced C4 levels are seen in more than a third of patients with

active disease. But there were no correlations between the disease parameters and the C4 levels. In addition, there were no dynamic changes in serum complement levels during chemotherapy. Significantly, the residual complement activity, as measured by CH50, remained within normal range even in patients with reduced C4 levels.

The results of this study raises questions regarding the performance and suitability of the CH50 assay technique employed to study functional complement activity in patients with myeloma. Incorporating additional tests to investigate the activities of alternate and lectin pathways may provide valuable insight into the functional state of complement system in multiple myeloma.

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Disclosure of Interest: S. K. Nagumantry Conflict with: Alexion, Shire, M. Smith: None Declared., M. Sivakumaran: None Declared.

BSH2020-EP-107

HaemBase Cymru (HBC): an electronic data collection platform for patients treated with haematological malignancies in Wales

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Background: Innovative partnership of NHS Wales, the Wales Cancer Network, Welsh Government and Janssen to drive improved data capture and patient-centred care for haematological malignancies. This five-year joint working agreement began in January 2018 between the four parties listed above, where this solution has been designed and developed by NWIS (NHS Wales Informatics Service) as part of the national data architecture confirming complete alignment with existing and new national systems.

Aims and Objectives: Welsh Government (WG) and Public Health Wales (PHW) are committed to ensuring the provision of sustainable services through NHS Wales' National Planned Care

Table 1. Disease parameters of patients with reduced serum C4 levels.

Paraprotein	Stage of disease	Chemotherapy	Albumen	C3 g/l (0.75–1.65)	C4 g/l (0.14–0.54)	CH50
IgG λ 47	Pre-treatment	Post-VCD	42	1.32	0.03	46.95
κ Light chain 8451	Pre-treatment	New diagnosis	40	1.64	0.03	
IgA κ 56	Pre-treatment	New Diagnosis	37	1.78	0.03	
IgG λ 43	Pre-treatment	New Diagnosis	36	1.82	0.03	
IgG λ 42	Pre-treatment	New Diagnosis	36	0.63	0.05	44.66
IgG κ 45	Pre-treatment	New Diagnosis	42	0.85	0.05	
IgG κ 10	During treatment	Daratumumab	22	1.1	0.05	
IgA λ 18	During treatment	CTD	38	1.59	0.05	
IgG κ 25	Relapse	Post-VCD	40	1.70	0.05	50.23
IgG κ 67	Pre-treatment	New Diagnosis	38	1.32	0.06	
IgA κ 1	During treatment	VTD	45	1.29	0.06	
IgA λ 25	Pre-treatment	New Diagnosis	41	1.32	0.06	
IgG λ 42	During treatment	VCD	37	1.33	0.06	66.12
IgG λ L 40	During treatment	VCD	36	1.47	0.06	
IgA λ 48	Relapse	Post-VCD	34	0.95	0.11	
IgG λ 27	Pre-treatment	New Diagnosis	40	1.17	0.13	
IgG κ 67	Pre-treatment	New Diagnosis	28	1.35	0.13	

Programme. The aim of this joint working agreement includes putting Wales at the forefront of measuring and tracking patient outcomes, using new technology in the form of a purpose-built haematological malignancy data solution, with the aim of making innovative new treatments available in line with the aspirations in the cancer delivery plan.

Vision: Patients with haematological malignancies have a clear and concise pathway, with clinicians able to digitally access records from any clinical setting in real time. Clinicians will be able to make informed treatment decisions based on accurate data capture. PROMS and PREMS will be captured on the national platform and integrated into HaemBase, to provide a comprehensive patient-level dataset that can inform patient/clinician interactions and support assessment of value in the pathway.

Mission: To develop a data solution for haematological malignancies that accurately gathers data during a patients' cancer treatment pathway. The solution will put the needs of the patients first and measure the impact of treatment on patient outcomes to enhance their quality of life.

Conclusions: This data solution goes live in a phased approach across all health boards in Wales, starting at University Hospital of Wales teaching hospital in December 2019. It is the first system in the United Kingdom, purpose built for myeloma patients initially, leading to all patients with haematological malignancies, with the aim of creating a single electronic point of data entry via a national NHS portal. This will facilitate the collection of robust, comprehensive real-world data as part of routine clinical work. The aims are to collate information from clinical consultations, MDT, prescribing, laboratory, radiology and patient facing apps and feed into the National Data Repository for Wales. This will provide a 360-degree view of the individual patient's journey at a single point and will also generate the quality of life and clinical outcome data for the entire Welsh population. As a result, NHS Wales will have meaningful analysis of the impact of treatment and support prudent use of healthcare resources.

Table. Desired outcomes of the project.

Outcomes	Project description
1	Support patients in their cancer care plan via the development of an electronic health record for haematology.
2	Support clinicians in their day to day decision-making via the development of an electronic health record for haematology.
3	Provide the data in a format to enable analysis of trends in demographics, diagnostic, staging and survival of patients with haematological malignancies.
4	Monitor and track patient pathway and outcomes.
5	Monitor response to treatment and quality of life.
6	Generate real-world evidence.
7	Operationalise complex Patient Access Schemes by providing the necessary data.
8	Support translational research studies with clinical data to combine with biobanked samples.
9	Support efficient multidisciplinary care of patients with haematological cancers.
10	Allow patients access to their records through mobile applications which give them greater control of their care.
11	Strengthen the haematology cancer site group and regional MDTs via sharing of good practice.

Disclosure of Interest: C. Bygrave Conflict with: Celgene, Janssen, Conflict with: Celgene, Janssen, Takeda, Amgen, Sanofi, BMS, Z. Khalaf: None Declared., L. Edwards: None Declared.

BSH2020-EP-108

Quality improvement project for the treatment of myeloma-related bone disease in Nottingham University Hospitals Trust

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International Myeloma Working Group (IMWG) guidelines recommend the use of intravenous bisphosphonates in all patients receiving first-line therapy for myeloma. Nottingham University Hospital NHS Trust guidelines recommend zoledronic acid as the bisphosphonate of choice due to data suggesting it confers an overall survival advantage compared to sodium clodronate. Contraindications to any bisphosphonate include dental issues which increase the risk of jaw osteonecrosis while zoledronic acid can only be used if the creatinine clearance is > 30 ml/min. Pamidronate and sodium clodronate are alternatives for those patients with renal impairment. The objective of our quality improvement project was to assess the adherence of the Haematology Department in Nottingham University Hospitals (NUH) to Trust guidelines regarding the treatment of myeloma-related bone disease.

We collected data for patients diagnosed with multiple myeloma in the years 2017 and 2018 with the assistance of the myeloma specialist nurses. We excluded patients with asymptomatic myeloma. The NUH guidelines were studied regarding first-line treatment in myeloma-related bone disease and alternative options/dose adjustments in case of contraindications. All the online resources of the Trust were thoroughly searched for administered treatments including electronic notes, drug charts, clinic letters and the Chemocare online prescribing programme.

Ninety-nine patients with active myeloma were included in the project. Zoledronic acid (first-line treatment according to NUH guidelines) was recommended for 90 patients. It was administered in 80 patients (89%). Ten patients (11%) did not receive zoledronic acid it being indicated by the guidelines. However, of the 10 patients not receiving zoledronic acid: one patient was unwell and died soon after the chemotherapy commenced, one declined treatment, one had severe side effects with the first dose of zoledronic acid, one preferred to continue sodium clodronate (had been started on another Trust) and one had relapsed myeloma. In five cases the reason was not documented. In nine of the 99 patients zoledronic acid was not recommended due to severe renal impairment. In these patients alternative options (pamidronate or sodium clodronate) were indicated by the NUH guidelines but not used.

Our findings showed that zoledronic acid was successfully given in the majority of cases where it was indicated. The main area to address is the use of bisphosphonates in patients with renal impairment where alternatives to zoledronic acid were not prescribed. The results were presented in the NUH myeloma multidisciplinary meeting. Haematology staff members have been educated on the trust guidelines and an outpatient clinic proforma has been developed to document planned treatments for bone disease. The project will be repeated in 12 months time.

Disclosure of Interest: None Declared.

BSH2020-EP-109

“Lenalodamide Induced ITP in myeloma patients with response to steroid and TPO agents - A rare but important event to consider”- case series from single institute from south India

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Introduction: Immune thrombocytopenia (ITP) is frequently encountered in patients with lymphoproliferative disorders. However this is only rarely reported in patients with multiple myeloma. The most likely explanation for this could be that monoclonal immunoglobulin (Ig), a hallmark of MM, does not exert activity as an antibody and often accompanies depressed normal Igs and consequently renders patients immuno-incompetent. (1) Although drug-induced thrombocytopenia is frequently observed, it is further rare phenomenon. We report a series of myeloma cases with ITP post-therapy.

Methods: We took all the myeloma cases presented with associated ITP defined as ‘sudden, abnormal lowering of the platelet count after ruling out all possible other causes like post-infection, with peripheral smear characteristics like giant platelets with normal megakaryocytes in marrow and presence of antibodies against platelets.(PA-IgG – though it have its limitations)’ between 2012 and 2019 at single institute. The clinical characters, treatment received and the response were evaluated

Results: In total four cases were presented during the period with following characters (Table 1). The one common finding is induction with lenalidomide + dexamethasone alone in all the cases.

We could observe all these cases met the ‘Naranjo adverse drug reaction probability scale-as high probability’ and hence we postulate that all these cases are probably lenalidomide-induced ITP.

Discussion & Conclusion: Although lenalidomide can induce immune dysregulations and cause autoimmunity (like myocarditis, Basedow’s disease, autoimmune haemolytic anaemia, cold agglutinin disease and acquired haemophilia), ITP is extremely rare phenomenon with first reported case in 2018. (2) However the appropriate treatment was not defined by the authors and was left as ‘a difficult to treat condition’. The two proposed mechanisms of this rare condition are

- 1 Immunoglobulin (specific against platelets) production from the myeloma cells, which were selected by the antitumor effect of lenalidomide
- 2 Antibodies from normal plasma cells, which were activated by immune dysregulation effect of lenalidomide

In our observation attempts like steroids or rituximab did not yield optimal results. This opens the possibility of megakaryocytic function suppression as possible mechanism and we suggest that a combination of steroids with stimulating agents like eltrombopag/romiplostim are the best choice as we could achieve best response with these agents in all the cases.

Disclosure of Interest: None Declared.

Table.

Characters	Case 1	Case 2	Case 3	Case 4
Age/Gender	65/M	42/F	54/M	69/F
Duration of symptoms	2 months	1 month	1 month	2 months
M band – levels	3.8 gm/dl	4.2 gm/dl	2.9 gm/dl (IgA)	6.3 gm/dl
Karyotyping	Normal	Normal	Normal	Normal
Beta 2 Microglobulin	26.92 mg/l	43.12 mg/l	16.83 mg/l	31.74 mg/l
ISS-stage	Stage II	Stage III	Stage II	Stage III
Calcium	8.6 mg/dl	8.0 mg/dl	9.1 mg/dl	7.8 mg/dl
Hb	10 g	9.8 g	9.5 g	9 g
Creatinine	1.0 mg/dl	0.8 mg/dl	1.2 mg/dl	1.1 mg/dl
% Marrow plasma cells	10%	12%	10–12%	10–12%
Treatment offered	4 cycles Vd f/b remission then on maintenance lenalidomide	4 cycles Vd then on maintenance lenalidomide f/b HSCT	4 cycles of Vd then on lenalidomide, on progression started on thalidomide then underwent HSCT	4 cycles of lenalidomide and dexamethasone with denosumab
Interval between drug and ITP	2.1 months	6 months	2 months	3 months
Recovery of ITP – Line 1 and best response	Poor response to steroids	Received 4 weeks of rituximab	Received steroids	Recovery after 2 weeks after receiving eltrombopag
Recovery of ITP – Line 2 and best response	Recovery within 2 weeks of eltrombopag but thrombocytopenia persisted	IV Ig – short-lived response for 3 months	Recovery within 4 weeks of romiplostim	
Recovery of ITP – Line 3 and best response	Best response with romiplostim	Recovery within 2 weeks with weekly romiplostim and VCR		

BSH2020-EP-110

A retrospective review of antimicrobial use and risk stratification of patients with multiple myeloma admitted due to suspected infectionS. Raffles^{1,*}, F. Jaffar², A. Bajric², T. Sugai³¹Haematology trainee, ²Foundation trainee, North West London Hospitals, ³Clinical Haematology Consultant, Hillingdon Hospital, London, UK

Infections remain a major cause of morbidity and mortality in patients with multiple myeloma. As patients survive longer and treatment regimens increasingly incorporate novel agents, rates of infection and early death related to infection is growing and thus is the need for risk stratification of first-line antibiotic use. Unlike in other haemato-oncology and general oncology patients, treatment in myeloma often does not render the individual neutropenic and therefore accepted neutropenic sepsis protocols are not formally applicable and the use of prophylactic antibiotics is not routine. Rather, immunodeficiency in myeloma is a result of B-cell immunoparesis, such as hypogammaglobulinaemia, as well as T-cell and NK cell dysfunction. Early identification of high-risk individuals and relevant microorganisms need to be incorporated into first-line antibiotic treatment regimens alongside local antibiotic guidance. Our audit evaluates antibiotic use in adult patients with multiple myeloma admitted due to suspected infection, as compared with local guidelines. We aimed to discern common positive organisms and antimicrobial resistance, alongside identification of any patient or disease characteristics that are associated with adverse outcomes that may be used to risk stratify patients and support tailored first-line antibiotic approach. Adult patients with multiple myeloma diagnosed between January 2015 and October 2019 admitted with suspected infection requiring antibiotics were retrospectively reviewed. Potential adverse outcome factors to include age, time from diagnosis, paraprotein and light chain level, percentage bone marrow infiltration and albumin were analysed alongside identified organisms and antibiotics administered. Of 97 patients, 43 (56%) required hospital admission for antibiotics due to suspected infection (median age 74.5, range 47–95 years) with median length of stay 7 days (range 1–70, mean 9.5). Of these, 28% patients required multiple admissions (median age 76.5) increasing to 44% where patients had an index admission within 3 months of diagnosis. There were 79 total episodes of inpatient admission resulting in four deaths. The most likely source of infection was chest (51%) with identification of likely causative microorganism in only 17% cases. Tazocin was administered first line in 41% of cases, despite only two presentations with neutrophil count on admission < 1, and less than 20% of admissions received first-line antibiotics in accordance with trust policy. None of the four positive blood cultures grew multi-drug resistant organisms. There was one recorded episode of c-diff. In conclusion, admission due to suspected infection in multiple myeloma results in significant length of stay and exposure to broad spectrum antibiotics despite infrequent identification of organism. There was no significant association between paraprotein, beta-2-globulin or lymphocyte count and recurrent admission; however, index admission within 3 months following diagnosis increased the likelihood of recurrent admission without association of increased mortality. Strikingly, there is poor compliance with local antibiotic guidelines and frequent use of neutropenic sepsis antimicrobial protocols, reflecting the recognition by clinicians of high-risk morbidity and mortality in patients with multiple myeloma but without individualised protocols available or reliable risk stratifications indices.

Disclosure of Interest: None Declared.

BSH2020-EP-111

Myeloma in Northland 2016–2017: incidence, risk factors and outcomeL. Chen^{1,*}, K. Breen², D. Hammer³, S. Krishnamurthy³, S. Poplar³¹NHS Thames Valley, Oxford, ²NHS Greater Glasgow and Clyde, Glasgow, UK, ³Jim Carney Cancer Centre, Northland Blood and Cancer, Whangarei, New Zealand

Background: The incidence of myeloma has shown a steady increase across the world in recent decades; however, there continues to be global variation in incidence. Australasia has the highest reported incidence of myeloma¹ and specifically, in New Zealand, the age-standardised rate is 5.28 per 100 000 person-years.² The possible factors that explain the geographical variation in myeloma incidence and relative high rates in Australasia are poorly understood.

Here, we present the analysis of new myeloma cases in Northland; the northernmost region in New Zealand. Northland has significant heterogeneity in its population and thus provided scope to investigate the possible impact of several epidemiological factors on myeloma incidence and outcomes.

Methods: We performed a retrospective analysis of clinical notes and laboratory results of all new myeloma cases presenting to Northland Haematology service between January 2016 and December 2017, collecting data on all patient and disease characteristics. Retrospective follow-up concluded in February 2019.

Results: Within the 2-year study period, 51 new cases of multiple myeloma were observed with a calculated crude incidence of 14.9 per 100 000 person-years and age-standardised incidence of 13.6 per 100 000 person-years. The median age at diagnosis was 67 years (range 42–86); 56.1% of whom were male. BMI, ethnicity and social-economic decile in our cohort were not statistically different from the census population for the region. There was no difference in incidence between rural and urban areas and no geographical hot-spots were identified.

R-ISS and cytogenetics at presentation were comparable to that described in the international literature^{3,4}. Fourteen patients (32.5%) presented with emergency admission to hospital, with those from a lower socio-economic decile (6–10) more likely to present as an emergency compared to those from higher decile (1–5) with an odds ratio of 8.8 ($P = 0.022$, CI 0.90–86.6).

Median follow-up was 621 days (21–1122 days), with 10 patients dying during the study period. The median progression-free survival (PFS) was 23 months; however, over half of the patients followed up ($n = 21$) had yet to progress. Eighteen patients had chemotherapy-related toxicity that lead to either a dose reduction or abandonment of treatment. There was no statistical correlation between these outcomes and ethnicity or socio-economic decile.

Discussion: Appreciating the relatively small scale of this study, the incidence of myeloma in Northland 2016–2017 appears significantly higher than the corresponding New Zealand average. Despite looking into several possible factors; this increased incidence remains unexplained. Interestingly, despite previous studies showing a higher incidence of myeloma in Maori populations⁵, this was not replicated in our study.

We found no statistical relationship between ethnicity or social deprivation on PFS or treatment tolerability. Those of lower socio-economic decile in our cohort were more likely to present to hospital as an emergency; however, this did not appear to be reflected in poorer long-term outcomes.

Disclosure of Interest: None Declared.

BSH2020-EP-112

The use of urine Bence-Jones protein testing in the investigation of a new paraprotein – a DGH perspective

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Current BCSH guidelines recommend that following the detection of a new paraprotein, an early morning urine sample is tested for the presence of a Bence-Jones protein (BJP). If this result is positive or testing is not possible, then it is recommended that serum-free light chains (SFLC) be tested.

In our trust, when a new paraprotein is found, a flag or alert is placed on the result to the patient's GP or ward team requesting early morning urine be sent for testing to the laboratory. It was noted that many of these urine tests were never sent. It was felt that this guidance was difficult to enact and follow at our hospital trust, a large BCSH Level 3 DGH, and a project to assess this was devised.

We retrospectively audited 51 new paraprotein results, detected over a period of 2 weeks from 24 June 2019 to 11 July 2019. Of these new paraproteins, 35 were on blood tests sent from general practice, 15 from the hospital wards and two from haematology. In all but two cases, a urine BJP test was suggested with the results.

Of the 49 samples requested, 21 urine BJP tests were sent to the laboratory. Of these 21, none were positive for a paraprotein or light chains. Six patients had a SFLC sent, of which five were on haematology advice. Two patients of the 51 patients that we know of have a diagnosis of multiple myeloma.

We felt it would be useful to know what proportion of urine BJP samples received were positive across a larger sample size. Data from the laboratory have shown that, on average, 300 urine BJP tests are processed per month. Of these, around 55 will lead to the process of immunofixation. This will be followed up in more detail.

There are several reasons why the urine tests may not be sent. It involves hassle, particularly in the community, for the GP and the patient. In addition, some urine samples may be sent in the wrong container or for the wrong test and were therefore discarded.

The best way of rectifying this is difficult, particularly in a DGH setting. It is not possible for laboratory staff to chase urine tests with the GPs or the wards, or indeed to specifically request SFLC on samples that are not followed up with the recommended urine BJP test.

A more consistent approach would be to send every sample with a new paraprotein for a SFLC automatically. However, in our trust, SFLC is not done on site, and is £10 more expensive than a urine BJP test. In addition to the monetary cost, there is labour involved for the laboratory staff in packaging and sending away the SFLC to the regional centre and processing the results when they come back.

The cost of sending every sample for SFLC rather than doing urine BJP tests will be more expensive, but reduces the risk of missing patients with a diagnosis of light chain myeloma. This is a currently a significant risk, given that 15% of patients with myeloma fall in to this group.

In summary, we have followed the BSH guidelines for the investigation of a new paraprotein by requesting the appropriate tests. The testing of early morning urine for a Bence-Jones protein has a very low uptake. This may result in missing patients with light-chain myeloma. However, the alternatives to urine BJP testing are more expensive and labour-intensive. From our perspective, this is a challenge. We would be interested to know what other DGH's recommend in terms of deviations from the BSH guideline for the investigation of a new paraprotein.

Disclosure of Interest: None Declared.

BSH2020-EP-113

Myeloma UK: outline of the Clinical Service Excellence Programme (CSEP)S. Renwick^{1,*}, A. Capper², I. Frost¹, H. Parkin¹, I. Laketic-Ljubojevic¹¹Healthcare Advocacy Services, ²Patient Advocacy, Myeloma UK, Edinburgh, UK

Myeloma UK is the only organisation in the United Kingdom dealing exclusively with myeloma and related conditions. We provide educational resources/programs for health care professionals involved in caring for myeloma patients. Our aim is to support the roles of haematologists, nurses, GPs and allied health care professionals in providing patients with the best possible treatment and care. Designed in conjunction with myeloma experts, our accredited and easily accessible programs include information about current developments and best clinical practice in myeloma. The resources can be found on the Myeloma UK website and include the Myeloma Nurse Learning Programme, Nurse Best Practice Guides, Clinical Trial Finder, Drug Tracker, medical animations, key clinical guidelines, a GP online training module and supporting materials for Myeloma Grand Round events. In addition, we produce Myeloma Spotlight, a bimonthly e-newsletter tailored for doctors and nurses.

The Clinical Service Excellence Programme (CSEP) is a Myeloma UK accredited programme, developed in partnership with patients, family members and health care professionals.

Hospitals apply to attain the CSEP award to demonstrate their commitment to providing quality and patient-focused care: to celebrate success and to identify areas for improvement.

CSEP has been developed to:

- Cover all aspects of clinical care
- Ensure that the patient perspective plays a pivotal role in decisions that affect their care
- Recognise and acknowledge areas of excellence in the hospital team
- Identify and support further developments in patient care
- Gather feedback from patients on their hospital experience

The first stage of the CSEP accreditation process involves the submission of a suitability survey. Upon successful completion, each centre completes a self-assessment against the eight key themes of the Myeloma UK Best Practice Standards: patient experience; multi-disciplinary teamworking; diagnosis and follow-up; treatment and supportive care; research and clinical trials; end of life care; awareness, education and advice; information and support.

We ask for a minimum of 20 completed Patient Experience Surveys, to corroborate the findings of the self-assessment. A visit from Myeloma UK, to the participating hospital, enables us to meet the multidisciplinary team and observe the service provided. Evidence gathered from the visit is used to support the results of the surveys and assessments and allows us to produce a detailed report. This report provides clarity regarding the areas of excellence demonstrated by the team and aspects of the service that could be improved. Participating centres need to be at least compliant in all eight areas and have an overall score rated as 'Excellent' to attain the CSEP accreditation.

In the past 4 years, over 30 hospitals have successfully completed the CSEP programme, with 10 hospitals registered for 2020. We are currently planning re-accreditations for centres in our first cohort and look forward to evaluating how they have progressed in the intervening time.

CSEP accreditation is rapidly becoming recognised as a respected, pivotal acknowledgement of myeloma good practice.

Disclosure of Interest: None Declared.

BSH2020-EP-114

Osteonecrosis of the jaw secondary to bisphosphonate treatment in multiple myeloma – real-world data

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Bisphosphonates are potent inhibitors of osteoclast-mediated bone resorption and form part of current treatment guidelines for active myeloma whether lytic bone disease is present. There is widespread variance in local protocol regarding optimal duration of this therapy as there is no clear evidence to support a certain strategy. It is widely agreed appropriate to treat during active disease but management during asymptomatic remission is unclear. Osteonecrosis of the jaw (ONJ) is a condition in which bones of the maxillofacial skeleton, in particular the tooth-bearing areas, become necrotic and exposed to the oral cavity. It can have long-term and potential catastrophic consequences. Intravenous bisphosphonates have a well-known causal relationship with this condition and several studies have shown incidences of up to 11% in patients treated with bisphosphonate for myeloma.

We reviewed a cohort of 77 myeloma patients, with a median age of 74, treated with bisphosphonates for treatment durations of between 1 and 98 months. All patients had appropriate renal dose-adjusted therapy. 86% received treatment with zoledronic acid as per UK guidelines. The frequency of treatment was highly variable despite clear locally agreed guidelines. In the majority bisphosphonate treatment was continuous from initiation but with frequency reduction when in remission from myeloma to either 2 or 3 monthly infusions. 44% of patients had a documented dental review prior to treatment initiation, the remainder having local dental assessment with retrospectively untraceable outcome. Of those with a documented assessment a third required dental intervention prior to commencing bisphosphonate treatment due to increased ONJ risk. A total of four patients (5%) were diagnosed with ONJ during bisphosphonate treatment. A subgroup analysis showed that none of these patients had a retrospectively accessible dental record prior to bisphosphonate. All were treated with zoledronate for a duration of greater than 18 months prior to development of the complication. Only one of the four patients had renal impairment necessitating dose modification.

From this local real-world data we have implemented and mandated a pathway for hospital-based dental assessment prior to bisphosphonate therapy. Our evidence supports limiting bisphosphonate administration to 4 weekly while on myeloma treatment and then 2 monthly thereafter to a maximum of 2 years in the absence of active disease.

Disclosure of Interest: None Declared.

BSH2020-EP-115

Evaluating the role of the Myeloma UK, Myeloma Infoline in the provision of personalised information and care for myeloma patients and their families

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Personalised medicine is a key driver in the effective delivery of treatment for cancer patients. This tailored approach enhances overall success by focusing on the individual rather than using a 'one-size-fits all' approach. To successfully deliver personalised care,

patient information would also require tailoring to individual needs. While health care professionals are skilled at providing medical information and engaging in shared treatment decisions during consultations, tailoring information to each patient in a busy clinic, with limited resource, can be challenging. This is more evident in myeloma, as this is a complex, rare and individual cancer, with an ever-changing treatment landscape. Myeloma UK carried out a mixed methods study to investigate how the Myeloma Infoline Helpline service assisted in delivering personalised care to the specific circumstances of myeloma patients and their families.

Myeloma UK is the only organisation in the UK that deals exclusively with myeloma and associated conditions and provides a range of information and services for everyone affected by myeloma. The Myeloma UK, Myeloma Infoline and Ask the Nurse Services (a freephone helpline and email service respectively) provide bespoke information and emotional support to service users. These services are delivered by highly trained Myeloma Information Specialists (MIS) with extensive experience of discussing the complexities of this cancer.

The team tailor information to the individual needs of service users, supporting myeloma patients to prepare for shared treatment decisions (by developing questions for their consultations) and reduce emotional distress by discussing their concerns. The MIS team record data and use this for internal evaluation and to improve service delivery. However, evaluation of the contribution of this service to the provision of personalised information and shared treatment decisions has not been undertaken.

To better understand the role of the Myeloma Infoline in the delivery of personalised care and support, we conducted an extensive service evaluation. We employed a mixed methods design and summarised the anonymised data collected from over 8 years of the service and an evaluation of the service to examine the provision of tailored information and support for shared treatment decisions. The results demonstrate service users better understand their individual diagnosis, the disease and treatment options and are better prepared to engage in treatment decisions with their consultants. This study highlights that tailored care and support has a central role in effective delivery of personalised medicine and care to myeloma patients.

Disclosure of Interest: None Declared.

BSH2020-EP-116

A retrospective single centre cohort study of the efficacy of weekly subcutaneous bortezomib containing treatment regimes in the management of plasma cell disordersS. Bassiony^{1,*}, L. Peck¹, M. Dekmak¹, S. Alimam¹, M. Kazmi^{2,3}, K. Bailey¹, M. Gleeson¹, G. Milner¹, R. Bone¹, M. Streetly¹¹*Haematology, 2Haematology, Guy's and St Thomas' NHS Foundation Trust, 3Haematology, King's College Hospital, London, UK*

In 2004, NICE approved the use of intravenous bortezomib, a proteasome inhibitor, as a third-line therapy for multiple myeloma. In 2012 subcutaneous administration was favoured as it caused fewer adverse events (AE). Subsequently bortezomib has been given weekly instead of biweekly to improve tolerability, yet further data were needed regarding efficacy of this reduce frequency.

Data were collected retrospectively on 56 patients with a plasma cell disorders (multiple myeloma and plasmacytomas) receiving bortezomib between 01 January 2018 and 30 September 2019 at Guy's Hospital, London. Data were collected from: Electronic Patient Records for pathology results and clinic letters; E-noting for

comprehensive patient notes; SECTRA PACS for imaging reports; and MOSAIQ for chemotherapy prescriptions. Three patients were excluded as they were repatriated, one patient for incomplete data and three patients for receiving biweekly bortezomib.

The median age of 68 at the start of therapy, with 26 (53%) 53% receiving bortezomib as part of first-line therapy and 30 (61%) of this population being female.

Sixteen (32%) patients received VMP (bortezomib, melphalan, prednisolone), with majority (75%) receiving it as a first-line therapy. Averaged treatment cycles received was 8.31. It was generally well tolerated with two episodes of G3–4 neutropenia, two episodes of G3 nephrotoxicity, one episode of G3 hypercalcaemia. One patient discontinued treatment due to neurotoxicity and another due to gastrointestinal side effects. Median progression-free survival (MPFS) was 19.25 months and median overall survival (MOS) was 21.46 months.

Fifteen (31%) patients received VTD (bortezomib, thalidomide, dexamethasone) with eight (53%) receiving six cycles. It was well tolerated with: two episode of G3 anaemia, two cases of G3–4 neutropenia, one episode G4 nephrotoxicity and one episode of neurotoxicity leading to discontinuation of treatment. Thirteen (87%) patients achieved at least partial response (PR), on average at 7.13 months. Three (20%) achieved complete remission (CR) occurring between a range between 7 and 25.26 months. Two patients were refractory to treatment. Median OS was 12.13 months.

Nine (18%) patients received CVD (cyclophosphamide, bortezomib, dexamethasone) with 56% receiving it as second-line therapy. There was one episode of G4 neutropenia and one episode of G4 hyperbilirubinaemia. Treatment was discontinued for four patients due to suspected allergic reaction, one progression on treatment and two patients refractory to treatment. MPFS was 11.94 months MOS was 14.19 months.

Four (8%) patients received PVD (panobinostat, bortezomib and dexamethasone) which averaged as seventh-line therapy. There were one episode of G3 neutropenia and two episodes of G3 thrombocytopenia. One patient received 16 cycles and achieved CR. Two patients progressed on treatment. MPFS was 14.90 months MOS was 16.23 months.

Five (10%) patients received DVD (daratumumab, bortezomib and dexamethasone) with 100% of patients receiving treatment as at least second-line therapy. All had received at least eight cycles of treatment and are still alive. DVD was well tolerated. One patient exhibits G3 thrombocytopenia. Three achieved a PR over an average of 5.3 months and two patients showed NR to treatment. MPFS was 7.66 months and MOS was 7.05 months.

90% of patients achieving CR had IgG Kappa myeloma and 50% received VMP (all as first-line treatment). The quickest median time to PR was seen in PVD (3.19 months) and DVD (4.64 months). With the best overall survival achieved with VMP (20.46 months).

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IgD myeloma: a unique subtype with unclear prognostic significance

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Aim: IgD myeloma is a rare plasma cell dyscrasia which is traditionally reported to have a poor outcome compared to patients with

other types of M protein. This is thought to be due to higher rates of renal failure, chemotherapy resistance and a more aggressive clinical course. Recent studies have reported conflicting evidence into the survival outcomes of patients with IgD myeloma (Wang GR, Sun WJ et al. 2016, Chen L, Fan F et al. 2019). We therefore aimed to analyse the presentation and prognosis of IgD myeloma in a local cohort of patients in the era of novel therapies.

Method: Eight patients with an IgD M protein were identified from Monash Medical Centre, Victoria, based on serum electrophoresis records in patients between 2013 and 2018. Retrospective data were collected in regards to M-protein type and size, biochemical and bone marrow analysis at presentation, renal function, co-morbidities and treatment received.

Results: Median age at diagnosis was 75.5 years of age (range 51–83 years old). 50% presented with anaemia and 75% presented with renal failure with one patient requiring dialysis. The majority had high serum-free light chains at diagnosis with 87.5% having affected light chain above 1000 mg/l. All patients had ISS stage 3, 62.5% had R-ISS stage 3 disease at diagnosis. A single patient had smouldering myeloma without a myeloma defining event after 24 months of observation. Of the remaining patients, 85% received bortezomib-based induction, and 15% received thalidomide-based induction. Only one patient proceeded to autologous transplant. Overall survival in IgD myeloma patients was not statistically different from non-IgD myeloma patients (median 40 vs. 53 months; $P = 0.332$).

Conclusion: IgD myeloma remains a rare plasma cell dyscrasia and presents commonly with high serum-free light chains, renal disease and anaemia. In our cohort, survival did not appear worse than non-IgD myeloma patients.

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Disclosure of Interest: None Declared.

BSH2020-EP-118

Incorrect specimen collection for investigation of cryoproteins can give mis-leading results

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Cryoproteins are those that precipitate reversibly at temperatures below 37°C. Cryoproteins are classified as type I, II and III cryoprecipitating immunoglobulins and cryofibrinogen. Type I cryoproteins are monoclonal proteins associated with B-cell malignancies including Waldenstrom macroglobulinaemia, myeloma and lymphoma. Type II and III cryoproteins are associated with chronic infection or inflammation for example hepatitis C infection, Sjogren's syndrome. Cryoproteins can be associated with clinically significant symptoms, including hyperviscosity, purpura, neuropathy, renal impairment, Raynaud's phenomenon and even ulceration and necrosis. Cryoprotein symptoms can precede symptoms associated with any underlying malignancy.

Cryoproteins each have their own thermal profile; some precipitate within minutes of cooling below 37°C while others may only precipitate at room temperature or below, or precipitate slowly over days. It is vital that when investigating cryoproteins, samples are collected at 37°C and maintained at 37°C during clotting and sample

separation, otherwise incorrect results may be generated, particularly the quantification of the paraprotein.

Paraprotein concentration is important for assessing tumour burden at diagnosis, the need for urgent treatment for example plasmapheresis, and also as a measure of response to treatment. Underestimating the paraprotein concentration can have significant clinical consequences. Here we highlight the importance of cryoprotein sample collection with three cases where inappropriate sample collection resulted in significant underestimation of the paraprotein concentration.

Patient 1 had an IgM kappa paraprotein that quantified as 7 g/l in samples collected at room temperature, but 42 g/l when collected at 37°C. Patient 2 had an IgG kappa paraprotein that quantified as 8 g/L in room temperature samples but 51 g/l when samples were collected at 37°C. Patient 3 had an IgG lambda paraprotein that quantified as 44 g/l in samples collected at room temperature, but 74 g/l when the samples were collected at 37°C. The patients' paraprotein concentrations were underestimated by 90.5%, 84.3% and 40.5% respectively.

These three cases highlight the importance of appropriate sample collection when investigating suspected cryoproteinaemia. In the worst case, <10% of the true paraprotein concentration was detected; in two of these cases, the incorrect paraprotein concentration of < 10 g/l could have been interpreted erroneously if the patient was on treatment as evidence of response, or in a new patient at diagnosis, it could have been considered an unremarkable finding possibly associated with a monoclonal gammopathy of undetermined significance.

Clinicians and laboratory staff should be aware of the potential clinical consequences if samples are not collected correctly for cryoprotein investigations, in a way that maintains sample temperature at 37°C during sample transport, clotting and separation. It can be helpful if clinicians and laboratory staff discuss cases where the presence of a cryoprotein is suspected for example when laboratory staff notice an obvious precipitate in a sample that disappears when the sample is warmed to 37°C, or when significant variation in a patient's paraprotein concentration does not correspond to treatment, to consider whether cryoprotein studies are indicated.

Disclosure of Interest: None Declared.

Nursing

BSH2020-EP-119

Implementing a remote monitoring clinic for patients with stage A CLL

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Situation: Chronic lymphocytic leukaemia (CLL) is the most common leukaemia in the western world; however most patients are diagnosed at an early stage (stage A) and probably will not require intervention for couple of years but will need continuous monitoring to identify early signs of disease progression and assess and deliver treatment in a reasonable time.

Background: Our aim is to set up a remote monitoring stage A Chronic lymphocytic leukaemia (CLL) clinic. Our aim is to:

- 1 Deliver care to patients closer to their homes
- 2 To save patients time and waiting time in our clinics
- 3 Maintain patient contact with the specialist centre

- 4 To develop a database for patients with CLL stage A which will help identify risks and rate of progression
- 5 To introduce a key worker to the patients
- 6 Patient triggered follow-up which will give them more control of their disease

Assessment: To implement this we are:

- 1 Working closely with the continuing service improvement team (CSI),
- 2 Process mapping of the CLL service,
- 3 Monitoring clinic size and waiting times,
- 4 Patient feedback using patient forums,

Recommendations: This is an ongoing process with continuous work by the CLL team and CSI team to introduce this new service. We are going to have ongoing evaluation with the patients, CLL team and GP's of this service. This model maybe applicable to other chronic disease entities within haematology in the future or other specialities.

Disclosure of Interest: None Declared.

Paediatrics

BSH2020-EP-120

Audit of invasive fungal disease (IFD) in paediatric haematopoietic stem cell transplant (HSCT) recipients and compliance with a national guideline for the diagnosis, prevention and treatment of fungal infections during paediatric HSCT; the Royal Marsden Hospital (RMH) experience 2018–2019

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Invasive fungal disease (IFD) in paediatric patients undergoing allogeneic stem cell transplant remains a recognised cause of morbidity and mortality. The national UK paediatric bone marrow transplant (BMT) group agreed consensus guidelines for individual centre use in 2017 and a national multicentre prospective audit is in progress with the aim to harmonise practice and to collect data on IFD incidence and effectiveness of current prophylaxis, diagnosis and treatment strategies. We report single centre data for the Royal Marsden Hospital (RMH) over a one year period 2018–2019.

A prospective multicentre data collection tool was used (REDCap) to capture antifungal use and IFD incidence. RMH data were collected for 27 paediatric HSCT patients (2–18 years, median 7 years) on a weekly basis with the median follow-up of 27 weeks (1–66). The local aims of the study were to assess IFD incidence, assess heterogeneity of practice in reference to the national guidelines, and identify areas for local quality improvement.

The antifungal primary prophylaxis used in the first month following HSCT was liposomal amphotericin (12), itraconazole (10) and caspofungin (2). Three patients had a probable/proven IFD before transplant and were given antifungal treatment or secondary prophylaxis throughout the transplant including one patient who had active probable pulmonary IFD at time of transplant.

At last follow-up, 22 of 27 patients were alive, while four children died of transplant-related complications (no IFD) and one child died of leukaemia relapse. There were 28 episodes required empirical or pre-emptive antifungal treatment in 16 patients. Eight of 28 cases were suggestive of IFD (six pulmonary, one hepatic and one central nervous system) in the period from day + 7 until day + 369 after HSCT and one case of proven *Candida parapsilosis*-associated sepsis was diagnosed on day + 58 after HSCT.

Antifungal treatment occurred in line with the national guideline in all cases with liposomal amphotericin, caspofungin, voriconazole, posaconazole and anidulafungin used as agents. However, a moderate rate of none compliance with investigations in the cases of suspected IFD was identified in eight of 28 cases (28%) missing either diagnostic radiology or microbiology investigations. Out of eight cases suggestive of IFD, there were three cases of potential probable IFD, for which the necessary diagnostic investigations had not been completed. Out of six pulmonary IFD cases, three patients underwent bronchoscopy but only one of the three had 18s PCR testing. Therapeutic drug monitoring (TDM) occurred in all cases of azole prophylaxis but did not always occur with recommended frequency

(weekly while inpatient and every outpatient appointment). Compliant azoles TDM rate was 68.3% in inpatient and 83.8% in the outpatient settings. Appropriate dose adjustment following the level occurred in 40% of inpatient and 39% outpatient cases. There were 12 cases of liposomal amphotericin renal toxicity (five on the prophylactic dose and seven on treatment dose) and five cases of azoles hepatotoxicity.

This single centre data confirms expected incidence of fungal infection in a paediatric allogeneic population and when multicentre data are available will provide a new national benchmark for current morbidity and mortality related to fungal infections in UK allogeneic transplant recipients. Assessment of compliance with the national guideline highlighted areas for quality improvement and re-audit in relation to diagnostic pathways and TDM.

Disclosure of Interest: None Declared.

BSH2020-EP-121

Predicting treatment response in paediatric chronic ITP patients

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Background: Immune thrombocytopenia purpura (ITP) is a heterogeneous disorder characterised by increased destruction and suppressed production of platelets. There are no biomarkers to help direct treatment in children, who experience unpredictable response rates and a large disease burden. To help direct treatment pathways, we assessed whether there were any biological features that could predict responses to treatment.

Methods: This was a retrospective case review of 28 children with chronic ITP (>12 months) who attended St. Mary's Hospital (SMH), Imperial College NHS Trust Paediatric ITP clinic. This cohort was divided into two groups. The 'non-treatment' group were managed with observation ($n = 11$). The 'treatment group' required treatment due to bleeding symptoms ($n = 17$). In the 'treatment group', 11 are responding to a thrombopoietin receptor agonist (TPORA): seven to eltrombopag and four to romiplostim. Six patients failed to respond to romiplostim and/or eltrombopag and are on a combination treatment including immunosuppression.

Before treatment was initiated, immunoglobulin (Ig) levels, platelets, lymphocyte subsets, bone marrow biopsies and full blood counts were collected. Comparison of clinical data and treatment responses were completed using GraphPad Prism.

Results: Initial platelet count was significantly higher in the non-treatment group *versus* the treatment group ($21 \times 10^9/l$ vs. $6 \times 10^9/l$, $P = 0.0053$). Patients needing treatment had lower CD56 + numbers (285 vs. 327 cells/ μl , $P = 0.36$) and higher IgG (10.65 vs. 7.95 g/l, $P = 0.06$). Bone marrow biopsies of the treatment group had more megakaryocyte clustering (50% vs. 14%, $P = 0.13$).

Patients who failed to respond to TPORAs had the lowest CD56 + (Natural Killer, NK), CD4 + and CD8 + counts, although this did not achieve statistical significance in this small cohort. Both responders and non-responders had increased numbers of megakaryocytes (83% and 100%), which were small and hypolobated (83% and 77%).

Patients who responded to eltrombopag compared to romiplostim had a trend towards lower CD56 + (344 vs. 369 cells/ μ l, $P = 0.07$), higher CD8 + (1137 vs. 578 cells/ μ l, $P = 0.23$) and higher B-cell count (859 vs. 578 cells/ μ l, $P = 0.16$).

Discussion: Children with ITP are treated based on bleeding symptoms, consistent with our findings that patients with lower platelet counts are more likely to be given treatment. Increased clustering of bone marrow megakaryocytes in the treatment group may reflect more drive for the bone marrow to produce more platelets, or could reflect a different disease type.

Making decisions in those who need treatment can be difficult. At SMH, patients who need treatment are offered eltrombopag or romiplostim first line. Non-responders are either switched to the other TPORA or started on immunosuppressive treatment. For many patients, this results in rotation through treatments, which is frustrating and expensive.

Although there were few significant differences between groups, there was a trend towards lower NK cells, CD8 + cells and B cells in non-responders. Lower NK cell numbers and higher CD8 + cells were seen in patients who responded to eltrombopag, which is consistent with findings in our adult study.

Conclusion: Lower platelet counts are a predictor of needing further treatment and lower lymphocyte subsets can predict a failure to respond to TPORA. Further analysis in a prospective study may help to better define disease types and stratify patients to specific treatments.

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BSH2020-EP-122

Growth velocity in children with sickle cell disease is increased or maintained while taking hydroxycarbamide

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Growth velocity has been shown in several studies to be reduced in children with sickle cell disease. Poor growth velocity has also been correlated with lower haemoglobin.¹ Some studies have not shown a difference in height in peripubertal patients on hydroxycarbamide compared with no therapy.² There are very little data on the impact of hydroxycarbamide and growth. Due to the effect of the increase in haemoglobin it is possible that hydroxycarbamide can increase growth velocity.

We present the data of 32 children with sickle cell disease on hydroxycarbamide.

Methods: Height (cm), weight (kg) and centile were recorded in patients pre- and post-hydroxycarbamide, at 6 months, 1 year and 2 years and last recorded, including haemoglobin levels and haemoglobin F % at these time points.

Results: Thirty-two patients, median age 9 years (4–16 years) had been on hydroxycarbamide for a median time of 2 years (6 months to 4 years), with doses ranging from 20 to 35 mg/kg. Haemoglobin had increased significantly from baseline in all patients at 1 year post-hydroxycarbamide ($P = 0.008$) as well as haemoglobin F ($P = 0.006$). Height, weight and centile were maintained in all patients before and after taking hydroxycarbamide and overall there was no significant change ($P = 0.4$ and 0.5 respectively). Puberty occurred in one patient as expected without delay. However of note,

those on low weight centile (below 9th, $n = 4$) all had a significant increase in weight (11–25th centile; $P < 0.0001$).

Conclusion: Studies showing effect of growth in children on hydroxycarbamide are limited. Our study shows that height and weight are maintained in the majority of children, and pubertal development is normal. Of importance is the finding that low weight children (< 9th centile) can increase weight significantly to healthier levels while taking hydroxycarbamide. Parents do report better appetite in children while taking hydroxycarbamide. Weight increase may be due to a combination of increased appetite due to feeling less symptomatic, as well as a lower metabolic rate from reduced haemolysis. Larger studies are needed but parents should be counselled that underweight children may gain weight in beneficial ways while taking hydroxycarbamide.

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2. Nagalapuram et al. Effect of sickle cell anemia therapies on the natural history of growth and puberty patterns. *J Pediatr Hematol Oncol*. 2019;41(8):606–611.

Disclosure of Interest: None Declared.

BSH2020-EP-123

Retrospective audit of the use of cidofovir for the treatment of adenovirus in paediatric haematopoietic stem cell transplant (HSCT) recipients: a single centre experience of variable dosing schedules over a 7-year period at the Royal Marsden Hospital

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Adenovirus (AdV) remains a major cause of mortality and morbidity in paediatric haematopoietic stem cell transplant (HSCT) recipients. Standard treatment is the antiviral drug cidofovir (CDV) which is used for controlling adenoviraemia and preventing disseminated viral disease. However, this can be associated with renal toxicity and practice varies between centres regarding optimal dosing schedule.

Our aim was to do a retrospective audit of the dosing schedules and to examine patient outcomes, drug efficacy, need for rescue therapy and renal toxicity in a single centre over a 7-year period at the Royal Marsden Hospital. Twenty-eight episodes of adenovirus meeting the criteria for treatment in 25 patients between the ages of 2–17 years were observed. Data were collected from patient case records in order to assess dosing efficacy, response to treatment and potential incidence of drug toxicity.

Data were collected on 25 patients (one autograft and 24 allografts – 22 myeloablative conditioning and two reduced intensity conditioning) with 28 episodes of adenoviraemia. The outcomes included eight deaths with four children having mortality significantly attributed to AdV. All 28 episodes had positive blood AdV polymerase chain reaction (PCR) with AdV also detected in the stool (64%), nasopharyngeal aspirate (35.7%) and urine. Six patients (21.4%) were also noted to have positive urinary BK virus although this was not the primary reason for CDV treatment.

Twelve episodes (12/28, 42.8%) were treated using 1 mg/kg (thrice weekly) dosing regimen and 16 episodes (16/28, 57.1%) were

treated with the 5 mg/kg dosing regimen (5 mg/kg once a week for 2 weeks followed by once every alternate week). Nine children had evidence of renal toxicity (9/28, 32.1%) with one needing discontinuation of CDV after a single dose. One episode of neutropenia post-CDV was observed. Six needed rescue therapy for AdV not responding to initial treatment – Four received brincidofovir and the remainder received an alternate CDV regimen. Time to AdV detection ranged from D5 to D 867 (1/28) post-stem cell infusion and 75 % (21/28) had infection within the first 60 days.

The cumulative dose required to achieve AdV clearance in those receiving 1 mg/kg three times a week regimen ranged from 5 to 12 mg/kg with a mean dose of 8.57 mg/kg. In contrast, the cumulative dose in the 5 mg/kg regimen was between 5 and 25 mg/kg with a mean dose of 14.1 mg/kg to achieve AdV clearance. 41% (5/12) of those on 1 mg/kg dosing required rescue therapy as compared to 25% (4/16) on the alternate regimen. The time period needed for the clearance of AdV on the 5 mg/kg dosing schedule ranged from 4 to 60 days with a mean duration of 22.1 days and the clearance on 1 mg/kg regimen being 6 to 56 days with a mean duration of 26.4 days.

CDV was well tolerated in the majority of cases for the treatment of AdV. In terms of dosing schedule, 1 mg/kg three times per week appeared to be somewhat less nephrotoxic when compared to 5 mg/kg regimen (reversible renal dysfunction). However, with this (1 mg/kg) dosing schedule we noted a trend towards increased time to clearance and an increased need for alternate rescue therapy. There was also increased outpatient attendance as a result of the thrice weekly schedule. Novel therapies are needed for HSCT recipients to reduce the burden and mortality of AdV disease and its associated toxicities.

Disclosure of Interest: None Declared.

BSH2020-EP-124

Hydroxycarbamide: clinically effective, but parents still delay starting

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Introduction: Sickle cell anaemia (SCA) is an autosomal recessive haemoglobinopathy, identified by dysmorphic red blood cells (RBC), that in low oxygen states 'sickle' causing blood vessel obstruction. Clinical manifestations include unpredictable vaso-occlusive pain crises, limiting patient quality of life. Paediatric complications include acute chest syndrome (ACS), pain and stroke. SCA affects 15 000 UK patients. Management combines lifestyle measures, prophylactic medication and use of disease modifying therapies. The only cure for paediatric patients is bone marrow transplant.

Hydroxycarbamide (HU) has become gold standard for SCA treatment. Several studies, the baby HUG Trial (2011), INTO and Lobo (2013), demonstrate that HU usage correlates with reduced pain crises, hospital admissions, ACS and blood transfusions in child and adult patients. HU works by increasing percentage foetal haemoglobin (HbF), reducing percentage HbS. It also lowers white cell count; beneficial as inflammation can precipitate sickling. Despite evidence highlighting its efficacy, uptake of HU in children with SCA is low.

Aims: We aimed to assess the effect of HU on SCA, for up to 24 months post-treatment initiation. We also measured the number of discussions in children and carers prior to agreeing to HU treatment initiation.

Methods: Data from ($n = 172$) patients (0–18 years) with Hb SS or SC SCA were obtained from a South London Hospital Trust. Using electronic patients records (EPR) we obtained data retrospectively, on outcomes 24 months prior to and during HU use. We measured: no. of pre-treatment discussions, no. of pain crises hospital attendance, bed days, no. of ACSs and emergency transfusions. EPR were used to note side effects. Haematological parameters, haemoglobin (Hb), HbF and MVC and reticulocyte count, were measured at baseline, 12 and 24 months post-HU introduction.

Results: Among 59 patients started on HU, we found reduced hospital attendance for pain, ACS and emergency blood transfusions over 24 months. Hb and HbF percentage had upward trends over the 24 months. Only 10 patients discontinued HU treatment due to inefficacy or intolerance. Just 20% of children were started on HU on first discussion. The average discussion to treatment time was 5.2 months.

Discussion: The results corroborate the positive benefit of HU in SCA seen previously. Although annually HU uptake increased, just 46 of 172 patients were using HU regularly as of May 2019. Parental uncertainty commonly delayed treatment initiation and the main indications were worsening of symptoms rather than prophylaxis. Surprisingly, 20% of families required at least three discussions prior to HU commencement. The main limitations were the small study size. Patient care was spread over multiple sites restricting the sample size for assessing blood parameters as data were collected in the primary trust. Pain assessment was from EPR, meaning an under-reporting of pain managed at home. Finally, patients starting HU from 2018 onwards do not yet have the full 24 months of follow-up data.

Conclusions: The results suggest HU is an efficacious tool for managing SCA in children, reducing short-term adverse outcomes. However, exploring measures to address parental hesitancy to start HU treatment is necessary to encourage widespread and early paediatric uptake.

Disclosure of Interest: O. Oremakinde: None Declared., A. Thomas Conflict with: Consultant Paediatric Haematologist, Children's Directorate, St George's Hospital.

BSH2020-EP-125

Help me Grow! – a quality improvement project on neonatal blood transfusion

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Sixty percent of preterm infants receive blood transfusions during their neonatal admissions mainly due to repeated iatrogenic phlebotomy losses and ventilator requirements. There is a temporal association between red cell transfusion and necrotising enterocolitis, a disease that has caused much anxiety around neonatologists due to its high morbidity and mortality rate. Although the British Committee for Standards in Haematology published standards in 2016, there remains a large variation to blood transfusion practices between neonatal units. There are currently no large randomised controlled trials for withholding feeds and transfusion. Although there were local guidelines on blood transfusion, the team was aware this was not evidence based. Babies were kept nil by mouth (NBM) for prolonged periods surrounding transfusion, subsequently affecting babies' nutrition which impacts long-term growth and development. There were also concerns that babies were not being transfused in a timely manner despite meeting transfusion thresholds.

Aims of the project included adherence to blood transfusion cut-offs; 100% compliance with blood transfusion consent and care plan

documentation; and to review and develop an evidence-based local trust blood transfusion guideline.

Relevant stakeholders meetings were convened. Retrospective data were collected from case notes between April and October 2018. Using iterative QI methodology, we did multiple runs of Plan Do Study Act cycles, made small tests of change and used the adapt, adopt, discard methodology. Our initial findings were that the average time taken between decision to transfuse and start of transfusion was 5 h and the average time taken from starting feeds to achieving full feeds was 26.5 h. All patients had been consented for transfusion and had a care plan in place. In September 2018, we made an initial draft of neonatal blood transfusion guidelines and obtained feedback from frontline patient facing staff. In November 2018, we put up posters in the neonatal intensive care unit showing blood transfusion thresholds. Agreed changes were implemented that babies were only to be kept NBM during the period of transfusion (previously 4 h pre, during transfusion and 4 h post) and for babies to reach full feeds after two good feeds, reducing the duration of time babies were kept NBM. Data were recollected between November 2018 and February 2019, which found that the average time taken between decision to transfuse and start of transfusion was 2 h 50 min and the average time taken to reach full feeds was 8 h 15 min.

This QI project led to an evidence-based change in blood transfusion practice in our neonatal unit. A local blood transfusion guideline for neonates has also been created and awaiting to be published. Challenges of this project include the limited literature surrounding feeding and neonatal transfusion. There have also been set practices that have been ongoing for years in our neonatal unit where this change required a change in working culture. The next phase will be sustainability and monitoring transfusions given appropriately to set thresholds.

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Disclosure of Interest: None Declared.

BSH2020-EP-126

Impact of changing from co-trimoxazole to alternative pneumocystis pneumonia prophylaxis on the intensity of maintenance chemotherapy in the treatment of children with acute lymphoblastic leukaemia(ALL)

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Background: Patients treated for acute lymphoblastic leukaemia (ALL) receive co-trimoxazole prophylaxis against pneumocystis jiroveci pneumonia (PJP) as standard. There is some evidence that co-trimoxazole can impact on the oral chemotherapy dose intensity that patients can tolerate due to myelosuppression. Greater treatment intensity has been correlated with improved event-free survival in the treatment of ALL.

Aim: To investigate the impact of changing prophylaxis from co-trimoxazole to an alternative agent on the dose intensity of oral 6-mercaptopurine and methotrexate received by patients in the maintenance phase of ALL therapy.

Methods: We retrospectively studied patients treated in our centre on a UK ALL protocol over a 5-year period from May 2014–May 2019 who were switched to alternative PJP prophylaxis. We assessed the treatment intensity tolerated for two cycles of maintenance chemotherapy before and after changing and tolerance of second-line agents.

Results: Twelve patients were identified who had changed from co-trimoxazole because they did not tolerate full-dose oral chemotherapy due to myelosuppression. After the change, for all patients there was a reduction in the time spent off chemotherapy and for the six patients there was an increase in the average percentage of full-dose chemotherapy received. However, patients received fewer weeks of 100% dose. Half of the patients did not tolerate dapsone as a second-line agent and one patient developed PJP infection on dapsone.

Conclusion: In patients experiencing dose-limiting myelosuppression in the maintenance phase of ALL treatment, changing from co-trimoxazole to alternative PJP prophylaxis may improve the intensity of oral chemotherapy treatment that can be delivered.

Disclosure of Interest: None Declared.

BSH2020-EP-127

Management of life-threatening haemolytic anaemia in an adolescent Jehovah's Witness

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Management of life-threatening haemolytic anaemia in an adolescent Jehovah Witness

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Introduction: We describe a case of acute, life-threatening anaemia in a young patient who had recently signed his first Advance Directive refusing blood products.

Case Report: A 16-year Jehovah Witness with primary immunodeficiency presented as an emergency with abdominal pain, pallor and jaundice. He was hypotensive and tachycardic with Hb 50 g/dl. We confirmed acute exacerbation of previous compensated autoimmune haemolytic anaemia (AIHA). An Advance Directive, supported by his parents, indicating no primary blood products should be used in any circumstance (excluding human immunoglobulin, IVIG). He immediately received methylprednisolone and IVIG. We transferred to PICU to facilitate rapid escalation of care. We followed trust guidelines for blood-sparing management with erythropoietin-stimulating agent (ESA), intravenous iron, intramuscular cobalamin and folic acid. As he deteriorated with nadir Hb 40 g/l we considered all options, including Hb-based oxygen carriers. These are unlicensed, have a limited evidence base, safety profile with no previous use in Scotland. The company were willing to release product on compassionate use. Other conventional options including rituximab and splenic artery embolization were investigated. Elective intubation was offered in PICU to decrease oxygen demand though carried the risk of death on a ventilator with associated distress to family and staff. After the initial period of cardiovascular instability, his Hb began to rise and his haemolysis stabilised.

Our patient has subsequently diagnosed with a STAT3 gain of function mutation as his primary immunodeficiency, which is rare but there have been cases reported around the world. He was stabilised on sirolimus (on which he has remained) and has had no further episodes of haemolysis.

Discussion: Our case highlights the possibility of successfully managing life-threatening AIHA without allogenic blood products. It

also raised ethical dilemmas requiring input from medical and legal teams regarding capacity, consent and obligation to provide the best possible care to preserve life, in accordance to an Advance Directive. Should this include use of unproven products? It is argued that for this patient a precedent has been set as to the level of care expected which should continue following his transition to the adult sector.

Acknowledgement: All staff involved in this case and the patient who is keen to have his story heard.

Disclosure of Interest: None Declared.

BSH2020-EP-128

Building an effective and comprehensive transition service for paediatric and adolescent patients with inherited and acquired bleeding disorders

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Background: The historic transition process within Manchester Foundation Trust haematology service was limited and disjointed. Attendance rates at transition clinics had dropped below 50%.

Previously teenage transition clinics were based within the paediatric haematology department; with up to 15 patients on a teenage clinic. Patients being transferred rarely had the opportunity to visit the adult haematology department.

We outline the development of an effective and comprehensive transition service for patients with a severe bleeding disorder, aged 13–23 years.

Methods: Our transition team consist of adult and paediatric haematology consultants, specialist nurses, physiotherapists and social workers. Opinions were sought from patients and carers.

We piloted monthly clinics for three consecutive months located within the adult haematology department with consultants, nurse specialists and physiotherapists from both paediatric and adult teams. A maximum number of six patients aged 13–17 years were seen on the clinic and were encouraged to have the consultation on their own or with a carer present. All patients had a tour of the adult haematology unit.

Results: The revised transition clinics resulted in an increased attendance rate with the last two clinics receiving 100% attendance. Patients reported feeling very relaxed, found the staff very friendly, felt they had plenty of time to discuss their care and recognised how the clinic would help them transition to adult services. The tour of the adult unit was very much appreciated by the patient and parents.

Conclusion: What have we learnt?

- We have a large cohort of patients across bleeding disorders and haemoglobinopathy that will benefit from an improved transition service
- Ongoing work needs to explore how we best support patients aged 17–23 who have completed transfer

Disclosure of Interest: None Declared.

Red Cell Disorders

BSH2020-EP-129

Anaemia and its correlates in HAART-Naïve patients in Port Harcourt, Nigeria

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Anaemia is one of the complications of HIV-infection, it is multifactorial, a marker of many co-morbidities in this immunodeficiency-state such as renal disease, bone marrow dysfunction, progressing disease and poor prognosis as well. We did a retrospective assessment of the severity and correlates of anaemia in HAART naïve patients in our hospital between January and December 2018.

Their case notes and electronic registers were assessed to extract relevant data on age, sex, packed cell volume (PCV) and CD4 at first presentation to the clinic after diagnosis. The data were analysed with SPSS version 21.

The records of three hundred and twenty-one (321) new registrants in the ARV Clinic in RSUTH were evaluated. Their mean age was 40.43 ± 10.51 years (range 19–76 years). Most of the participants fell between the ages of 28–47 years – 225(70%). The females made up more than two-thirds – 208 (64.8%) of the study participants. Most of the males were in the 38–47 age group 50 (44.2%), whereas most of the females were in the 28–37 age group 85 (40.9%).

From the study, almost half of the participants presented in Stage 2 by the World Health Organisation staging of HIV-infection – 151 (48.09%) and 123(39.17%) were in stage 3 at presentation to the ARV clinic; using the WHO staging.

More than 70% of the participants – 259 (very low 38.7% and low 43.5%) had low or very low CD4 counts, participants with high CD4 counts were 56 (17.8%). The median CD4 count in these patients was 245 cells/μl, whereas 50% of the patients had CD4 counts of 289 cells/μl and below.

Anaemia was prevalent in 92 patients(28.7%) ; of this number, 22 patients (23.9%) had severe anaemia. A higher proportion of females had anaemia (23.1%), compared to males (5.9%) There is an association between the presence of anaemia and sex ($\chi^2 = 12.526$; $P < 0.001$). The proportion of participants with anaemia increased with increasing WHO stage from 18.2% in stage 1 to 71.4% in stage 4 ($\chi^2 = 24.39$, $P < 0.001$). The Spearman's correlation depicts a moderately positive association (0.256). There is a gradual decrease in the proportion of participants with anaemia from 15.9% in very low CD4 category to 4.4% in participants in the high CD4 category ($\chi^2 = 17.94$, $P < 0.001$). The Spearman's correlation depicts a weak positive association, 0.17($P = 0.002$).

Table.

Variable	Crude OR (95% CI)	P-value	Adjusted OR (95% CI)	P-value
Age	0.980 (0.95–1.004)	0.100	0.987 (0.962–1.013)	0.319
SEX	0.366 (0.207–0.646)	0.001	0.351 (0.191–0.642)	0.001
CD4 count	0.999 (0.997–1.001)	0.001	0.999 (0.998–1.001)	0.641
WHO stage	2.398 (1.612–3.569)	<0.001	2.402 (1.543–3.739)	<0.001

The WHO stage of participants was found to be the strongest determinant of the occurrence of anaemia among the study participants. The regression analysis (both crude and adjusted) showed that for every increase in the WHO stage of study participants, there was a 2.4 times likelihood of having anaemia and this likelihood was statistically significant ($P < 0.001$). Sex of a study participant was also found to be a predictor of anaemia such that the females were more (0.3 times) likely to have anaemia than the males, and this likelihood was found to be significant ($P = 0.001$).

The findings from this study shows that a significant proportion of people infected with HIV present with low immunologic status and low haemoglobin and there is need to strengthen the pathways for early diagnosis, prompt treatment by trained physicians and supportive care by haematologists so as to reduce the severity and impact of anaemia in this group of patients.

Disclosure of Interest: None Declared.

BSH2020-EP-130

Sickle cell disease pain management in the emergency department: a single-centre experience

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Background: Sickle cell anaemia (SCA) is an autosomal recessive disorder caused by point mutation of the β -globin gene, resulting in abnormal forms of haemoglobin that cause increased red blood cell rigidity and haemolysis. It is one of the most common hereditary blood conditions, affecting over 14 000 adults in the UK.

One of the manifestations of SCA is vaso-occlusive crises. These typically cause severe pain that may require emergency department (ED) attendance for pain management, typically with opioids. Pain relief should be given quickly and response to this assessed on a regular basis. Those patients in whom pain relief is not well-controlled are at the risk of further complications including acute chest syndrome.

The UK National Institute of Clinical Excellence (NICE) published a quality standard in 2014 stating that patients presenting to hospital with an acute painful sickle cell episode should have a pain assessment, a clinical assessment and appropriate analgesia within 30 min of presentation.

This study was performed to assess the Whittington Hospital's compliance to national recommendations and to establish which aspects of care in ED contributed to delays in management.

Methods: If a Whittington SCA patient attends ED, an automated email is generated that notifies the haematology team of the attendance. This system was used to identify acute sickle cell presentations to ED. Criteria for inclusion in the study were SCA patients presenting to ED with painful sickle cell crises between August 2017 and January 2018. Patients who received analgesia in the ambulance and patients with no documentation available were excluded. The time of presentation, analgesia prescription and administration for each attendance were noted from ED documentation.

A total of 104 ED SCA attendances were included. About 41% of patients presenting with an acute painful sickle cell crisis received analgesia during their first 30 min in ED. The average wait for analgesia was 47 min, with 75% of SCA patients receiving analgesia within 1 h of arrival.

The time taken to triage SCA patients was on average 8 min (range 0–29 min). Time from arrival to prescription of pain relief was much more variable with an average wait 40 min (range 10 min to 2 h 22 min). Time from prescription to administration also varied, with 56% administered within 10 min of prescription and 87% within 30 min.

Patients who frequently attended Whittington ED (defined as 7 or more attendances within the 6-month period studied) had a shorter average wait for analgesia. Analgesia was given 36 min after arrival on average for frequent attenders, whereas patients presenting six or fewer times had an average wait of 53 min.

We are not currently meeting our audit standard for provision of analgesia in the emergency department, and performance has worsened since our earliest available date from 2012. Most of the delay appears to be due to the time taken for medication to be prescribed, although the time taken to triage the patient and administer the medication was also not insignificant and often amounted to greater than 30 min. It remains unclear what is contributing to this delay, although it appears that performance is improved when the patient is a repeat attender and therefore known to the department. Education sessions with the ED department and availability of a SCA specialist nurse may improve management of SCA painful crises in ED. **Disclosure of Interest:** None Declared.

BSH2020-EP-131

An Investigation into the presence of complement factors on erythrocyte vesicles in sickle cell disease

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A single amino acid substitution in the globin gene segment is the underlying genetic defect in sickle cell disease (SCD) creating a haemoglobin molecule which polymerises on deoxygenation. Recent insights into the pathophysiology of the disease indicate that inflammation and hypercoagulability play a role in disease manifestation and severity. This may be caused, in part, by increased phosphatidylserine exposure and extracellular vesiculation by sickle erythrocytes.

Recent works have identified some of these vesicles as those released from reticulocytes in the process of maturation to erythrocytes. These vesicles are termed 'inside-out' autophagic vesicles as they are the result of fusion of endocytosed plasma membrane with autophagic endosomes and stain positively for intracellular glycoporphin A (GPA) epitope BRIC163. Further work has shown that some of these vesicles are also positive for an extracellular GPA epitope (BRIC256) raising the possibility that the vesicle membrane may have become permeabilised. One possible mechanism of

permeabilisation is the membrane attack complex of the complement system.

The aim of this project was thus to investigate whether complement proteins are present on erythrocyte derived vesicles during extrusion from the cell and while circulating freely in plasma.

Vesicles were isolated from SCD plasma and healthy control plasma using membrane-based isolation technique and fluorescence-activated cell sorting (FACS) before being studied by flow cytometry using antibodies against complement proteins C3, C4, C3c and C3d and intracellular and extracellular epitopes of GPA. Cultured reticulocyte, sickle erythrocytes and healthy control erythrocytes were analysed by confocal microscopy.

Complement proteins were found to be present on vesicles in both SCD plasma and healthy control in levels as high as 100% for BRIC163 vesicles in healthy control. Confocal analysis revealed that complement deposition on vesicles occurs prior to release and is associated with PS exposure.

This may have implications for the occurrence of chronic complement activation and hence depletion in SCD leading to reduced opsonisation of pathogens.

Disclosure of Interest: C. Obieze Conflict with: Shared Scholarship from the University of Bristol and Commonwealth Scholarship Commission for Masters degree.

BSH2020-EP-132

Association of the c.4517C>T mutation in exon 28 of the von Willebrand factor gene with sickle cell disease: another genetic modifier?

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Sickle cell disease (SCD) is a classic single-gene disorder, but has a wide spectrum of disease-severity due to multigenic interactions. The rate of haemolysis in SCD correlates with the quantity and activity of von Willebrand factor (VWF) in the plasma. The aim of this study was to investigate the possible association between genetic variations of VWF gene, plasma levels and functional activity of VWF and markers of haemolysis with the frequency of occurrence of vaso-occlusive crises reflecting disease severity in SCD.

A total of 51 consecutive adult Saudi SCD patients and 51 healthy age- and ABO blood group-matched control subjects formed the study population. All recruited patients were in the stable asymptomatic phase. Subjects with a history of smoking, hypertension, cardiovascular disease, diabetes, pregnancy or chronic disease were excluded. Plasma haemoglobin, lactate dehydrogenase, VWF antigen (VWF:Ag) assay and functional activity (VWF:RCO) and ADAMTS13 activity were determined. VWF multimer analysis was carried out in 30 patients. VWF exon 28, where majority of mutations in the gene occur, was selectively amplified and sequenced.

Significantly reduced VWF:RCO/VWF:Ag ratio was observed in patients, but VWF:Ag, VWF:RCO, ADAMTS13 activity and ADAMTS13 activity/VWF:Ag ratio were not significantly different between patient and control groups.

DNA sequence analysis of the control population identified 16 polymorphic sequence variants. Comparative analysis of the genotypes and allelic frequencies showed that there was no significant difference between the control group and the sickle cell disease group

with respect to these polymorphisms. These variants were then screened out to identify true mutations in sickle cell disease patients.

Remarkably, mutations within VWF were found to be limited to the SCD group only. In total, three non-synonymous mutations (c.4453G>C; c.4517C>T; c.4706G>A); and one synonymous mutation (c.5049A>C) were identified in 12 patients. Only the c.4517C>T mutation was predicted to have an impact on VWF protein functionality by all five *in silico* prediction tools used. This variant, located in the A2 domain of the VWF protein, has been described in cases of type 2A von Willebrand disease (VWD) and classified as likely pathogenic on the ClinVar database. Multimer analysis showed reduced HMW multimers in two out of eight patients harbouring the c.4517C>T mutation. This was not, however, significantly different from five out of 22 patients lacking this mutation – suggesting that other (genetic/acquired) factors besides ADAMTS13 level and mutations affecting the VWF A2 domain influence cleavage of HMW multimers in these patients.

Selective propagation of the potentially deleterious c.4517C>T mutation in the patient group strongly suggests that it provides survival advantage in this disorder. However, genotype–phenotype correlation analysis showed that of the genetic and laboratory parameters analysed in this study, only the plasma haemoglobin level strongly correlated with number of crises-events, hospital-days and transfusion episodes with potential as a useful prognostic marker.

Disclosure of Interest: None Declared.

BSH2020-EP-133

Characteristics of menstruation in sickle cell disease patients attending a tertiary healthcare centre in Northern Nigeria

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Characteristics of menstruation in sickle cell disease patients attending a tertiary health care centre in Northern Nigeria

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Background: Sickle cell anaemia a monogenic inherited disorder, is the most common haemoglobinopathy affecting millions of people predominantly in Africa, the Middle East and Mediterranean.

Objectives: There has been a significant reduction in mortality observed in patients over the years due to wide exploration of the pathophysiology, improved diagnostic and therapeutic approaches. Amongst which are neonatal screening, vaccination for microbial prophylaxis, multidisciplinary treatment approach and use of hydroxyurea. The aim of the study was to determine the menstrual characteristic in sickle cell anaemia and to compare the age of attaining menarche in them with normal controls.

Methods: A cross-sectional study of 100 patients with homozygous (HBSS) sickle cell anaemia recruited from routine outpatient clinic visit at the haematology unit of the University of Maiduguri Teaching Hospital. Thirty (30) apparently normal individuals (controls) amongst consenting medical students matched for age were also recruited. The study was carried out over a period of six (6) months. A questionnaire on menstrual characteristics and demographic variables was administered. Anthropometric parameters were also

recorded. The responses were entered into Microsoft excel sheet (version 2013) and subsequently analysed using SPSS version 20.

Results: A total of 100 HBSS with mean age of 24.86 yrs (range 16–40) and 30 HBAA with mean age of 24.6 years (range 16–39) were studied (Table 1). The mean age at menarche in is 15.00 yrs and this was observed in 91% (91) of the subjects leaving 9% (9) premenarcheal, whereas that of the control was 13.57 years observed in 100% (30) of the subjects ($P = 0.000$). The mean duration of flow was shorter in the study group than that of the control at 4.23 days vs. 5.43 days ($P = 0.008$), whereas the difference in the cycle length showed no statistical significance 25.84 days vs. 27.43 ($P = 0.224$) (Table 1). Using anthropometric measurements, comparison between menarcheal and premenarcheal subjects showed significant difference in weight ($P = 0.002$), body mass index (BMI) ($P = 0.004$) and body surface area (BSA) ($P = 0.005$) respectively.

Conclusion: Our study revealed a delay in the predicted time of onset of menarche in patients afflicted by sickle cell disease in accordance with previous reports. We also observed shorter menstrual cycles with a scanty blood flow thought to serve as check and balance system for the chronic anaemia in the patients compared to the normal controls. This study gives an insight into the characteristics of menstruation in sickle cell patients and its knowledge will provide avenue for further counselling and management of these group of patients.

Table 1. Age at menarche and Characteristics of menarche among Sickle cell anaemia subjects

Parameters	Mean	SD	P
Age at menarche (years)			
HBSS	15.00	5.581	0.000*
HBAA	13.57	1.524	
AGE (year)			
Attained menarche	25.65	6.78	0.000*
No menarche	16.89	2.42	
WEIGHT (kg)			
Attained menarche	49.52	9.60	0.002*
No menarche	38.63	6.63	
BMI (kg/m ²)			
Attained menarche	19.71	3.09	0.004*
No menarche	15.87	2.23	
BSA (m ²)			
Attained menarche	1.49	0.15	0.005*
No menarche	1.30	0.16	

Disclosure of Interest: None Declared.

BSH2020-EP-134

Thalassaemia awareness among Iraqi people in 2018

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Background: Thalassaemia is an autosomal recessive disease which is common in Iraq with prevalence of 35.7 per 100000. It is the most common type of hereditary anaemia registered in Iraq in 16 thalassaemia centres in 2015. It is a life-threatening condition with many complications which if not managed could cause death in early age.

This study aimed to assess the awareness of Iraqi people about thalassaemia regarding disease transmission and prevention and to find their source of information about the disease, as developing

good awareness is the first and the most advantageous road to establish a successful prevention programme.

Methods: This study involved 418 participants who were from medical and non-medical field, those in medical field were considered as control group for comparison. It was conducted for one month duration as an online survey using a self-structured questionnaire which was tested for validity, unidimensionality and reliability in a pilot study of 40 participants. Each participant who had heard about the disease was given a score (0–5) based on their knowledge.

All participants were informed about the goal of the survey and they gave their consent.

The data were analysed using Social Sciences program (SPSS) version 24.

Results: About 69.1% had heard about the disease previously, those had a mean score of 3.47 out of 5.

About 87.6 % knew that consanguineous marriage increase the risk of the disease and 89.4% claimed that it is a non-communicable disease. The lowest rate of knowledge was about the preventability of the disease with only 46.2% confirmed that the disease can be prevented.

The family history of thalassaemia was more common among non-medical field, but this predominance was non-significant.

No significant correlation was found between the score with the age, the family history and gender.

The difference in Knowledge among medical and non-medical was statistically significant.

Conclusion: People awareness about thalassaemia was relatively good, the highest awareness was for the contiguity of the disease and the lowest awareness was for the preventability. A Control strategy should be directed to elevate the awareness level about Thalassaemia in the community.

Key words: Thalassaemia, Awareness and prevention.

Disclosure of Interest: None Declared.

BSH2020-EP-135

Frequency of anaemia and causes in postliver transplant patients from tertiary care hospital in Pakistan: a retrospective audit

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Liver transplant is a surgical procedure which replaces a dysfunctional liver with a healthy liver from a deceased donor or a portion of a healthy liver from a living donor. Owing to multifactorial aetiologies it can be associated with anaemia. The aim of our study was to analyse the frequency of anaemia and its causes after liver transplant with several risk factors precipitating this complication. We attempted to find out the frequency and predictors of anaemia postliver transplant, in order to help define subgroups at risk for the development of anaemia.

A retrospective audit was undertaken for patients who underwent liver transplant at Shifa International Hospital Islamabad. Data from 303 patients were reviewed from January 2018 to October 2019. Haemoglobin count was recorded pre transplant, and at intervals of one, two and three months post-transplant. Anaemic patients (Hb = <10 g/dl) were identified and included in this study and evaluated for different causes of anaemia. Relevant clinical information was collected telephonically by interviewing the patients, their relatives or physicians.

Among the 303 patients who underwent liver transplant, 81 were identified as anaemic post liver transplant. There were 54 (66.6%)

males and 26 (32.3%) females with an age range of 6–65 years (median age: 45.5 ± 10 years). Sixty two patients were included in this study. The prevalence of post-transplant anaemia (PTA) at 3 months was 26.7%. Out of all these patients, five tested positive for parvovirus B19, 30 were CMV positive, five renal insufficiency, five due to immunosuppressants, three streptococcal infection, seven enterococcus, two Methicillin-resistant staphylococcus aureus (MRSA), three haemolytic anaemia, one aplastic anaemia, one graft-versus-host disease GVHD and four were due to unknown causes. Anaemia in recipients on treatment for recurrent hepatitis C (HCV) is common. These patients were treated accordingly. A few patients with life-threatening or severe anaemia were transfused with red cell concentrate (RCC).

Post-transplant anaemia is significantly associated with increased mortality and decreased graft survival when compared with non-anaemic liver transplant recipients.

Disclosure of Interest: None Declared.

BSH2020-EP-136

Evaluation of high-performance liquid chromatography pattern and prevalence of thalassaemia trait in Port Harcourt, Nigeria

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Sickle cell disease (SCD) is the commonest haemoglobinopathy in Nigeria which is among the nations that has the highest burden of SCD. About 150 000 new births a year in Nigeria carry the HbS gene, other forms of haemoglobinopathies such as thalassaemia were hitherto thought to be rare in Nigeria and probably restricted to Mediterranean regions. However, with increasing availability and the use of more sensitive haemoglobin screening methods such as high-performance liquid chromatography (HPLC) machine, there is increasing evidence of the prevalence of thalassaemia disorders in Nigeria. The HPLC machine quantifies the regular haemoglobin variants and estimates haemoglobin A₂ and F levels. The HbF and HbA₂ levels combined with evaluation of red blood cell indices serves as a suitable laboratory screening tool for the suspicion, diagnosis and categorization of patients with thalassaemia and those with compound sickle cell disease such as co-inheritance of S and thalassaemia genes.

Thalassaemia disorders is a spectrum of haemoglobin disorders characterized by the reduced or absence of one or more of the globin chains of an adult haemoglobin or the presence of a dysfunctional form of globin chain. Classical laboratory finding in thalassaemia include low haematocrit in severe disease, low red cell indices (mean corpuscular haemoglobin (MCH) and mean corpuscular volume (MCV)). On haemoglobin electrophoresis, a high A₂ levels >3.5% is suggestive of a beta thalassaemia. HbF levels is usually elevated (>1%). Combining these parameters, assessment of alpha and beta thalassaemia can be entertained. Globin chain and genetic studies can be used to make definitive diagnosis.

There is a growing number of researches on the burden of thalassaemia in Nigeria. The findings from these studies suggest the presence of thalassaemia disorder in Nigeria, but there is a dearth of such studies in Port Harcourt City, Nigeria.

This was a retrospective review of a 3 year records of adults and children who were investigated for haemoglobinopathies at our haematology clinic. Their haemoglobin analysis and blood counts were estimated using the HPLC system by BioRad D10 and Sysmex XP300 respectively.

One hundred and twenty-nine patients were evaluated during the period, 89 of them had complete details of their haemoglobin patterns including HbA₂ and HbF, red cell indices (MCV, MCH) were included in the analysis. Data were analysed with SPSS version 23 and summarized using descriptive statistics.

The median (IQR) of the subjects was 12 years (6–21). They included 54 (60.7%) males and 35 (39.3%) females. The haemoglobin phenotypes for the patients were SS in 54 (60.7%), AS in 10 (11.2%), S-βThal in 10 (11.2%), SC in 2 (2.2%) AA in 6 (6.7%), CC in 1 (1.1%) and 6 (6.7%) were thought to have a thalassaemia syndrome.

Majority of the patients, 77 (86.5%) had anaemia with PCV <30%, MCH was <27 g/pl in 54 (60.7%) and MCV was <76 fl in 29 (32.6%). Eighty-four (94.4%) of them had HbF >1%. A high HbA₂ >3.5% was found in 36 (40.4%) of the patients.

Using MCV <76 fl and HbF >1% in those with a high HbA₂, 13 (14.6%) had possible β-thal trait, whereas 15 (16.9%) had silent α-thalassaemia trait.

Our findings show that 60.7% of patients evaluated for haemoglobinopathy had haemoglobin SS, 3.3% were haemoglobin SC and for the first time we demonstrated that 6.7% of the patients had Sβ-thal in Port Harcourt. There is need for further family studies and molecular evaluation to confirm these findings.

Table.

Thalassaemia status	Criteria	n (%)
Possible β-thal	Low MCH, High A ₂	23 (25.8)
Silent α-thal trait	Low MCH, Normal A ₂	31 (34.8)
Silent β-thal trait	Normal MCH, High A ₂	13 (14.6)
Normal	Normal MCH, Normal A ₂	22 (24.7)
Thalassaemia status	n (%) (MCV <76, HbF >1%)	n (%) (MCV <76, HbF <1%)
Possible β-thal	13 (14.6)	0 (0.0)
Silent α-thal trait	15 (16.9)	1 (1.1)

Disclosure of Interest: None Declared.

BSH2020-EP-137

A case report of X-linked macrocytic dyserythropoietic anaemia

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X-linked macrocytic dyserythropoietic anaemia is an inherited red cell disorder caused by mutation of ALAS2 gene which encodes 5'-aminolevulinic synthase 2, a vital enzyme required for initiating synthesis of haeme. Haemizygous mutation in males and homozygous mutation in females are incompatible with life. Heterozygous females are affected but the phenotype is variable depending on the skewing of the X-linked inactivation. This rare condition is associated with dyserythropoiesis, macrocytic anaemia and iron overload. The prevalence remains unknown as it is a relatively newly recognised red cell disorder. There is a potential for misdiagnosis especially when patient has macrocytic anaemia and bone marrow suggestive of apparent myelodysplastic syndrome (MDS).

We report the case of a 60-year-old lady who presented to her general practitioner 12 years ago in her late 40s with increasing fatigue. Blood test showed significant macrocytic anaemia with HB 87 g/l and MCV 112 with normal WCC differential and platelets counts, normal B12 and

folate levels, raised ferritin and normal thyroid function which lead to a referral to the local haematologist. A presumptive diagnosis of low-risk myelodysplastic syndrome was made after extensive investigations. She was reviewed in a tertiary hospital where she declined transplantation. She was symptomatic and erythropoietin (EPO) was commenced 9 years ago with dose titration to maximal dose 60 000 units subcutaneously weekly with no response. Two years ago she was diagnosed with atrial fibrillation and symptomatic anaemia, hence weekly EPO injection was stopped and a transfusion programme commenced. At this stage we made a second referral to a tertiary hospital where she underwent a new bone marrow biopsy which again showed hypercellularity and features of MDS but comprehensive myeloid gene panel study failed to detect any mutations of pathogenic significance. A whole-genome sequence and inherited red cell disorder genetic screening demonstrated mutation in ALAS2 gene which confirmed the diagnosis of X-linked macrocytic dyserythropoietic anaemia. Her daughter and only sister had genetic testing and her sister was found to have heterozygous ALAS2 gene mutation as well but normal blood parameters.

This case highlights the importance of low threshold to search for inherited red cell disorders in a young patient who especially present with isolated anaemia. Bone marrow test in hereditary haemolytic anaemia can share many features of apparent MDS microscopically hence next-generation exome sequencing for red cell membrane studies and extended myeloid gene panel studies for MDS is essential to avoid the potential misdiagnosis and serious consequences.

Disclosure of Interest: None Declared.

BSH2020-EP-138

The successful use of eculizumab to switch off haemolysis in aggressive warm IgM autoimmune haemolytic anaemia: case report and literature review

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Eculizumab is a recombinant humanized anti-complement factor 5 monoclonal antibody which inhibits the activity of late complements' pathway. It is licensed for the treatment of paroxysmal nocturnal haemoglobinuria (PNH), atypical haemolytic uraemic syndrome (aHUS) in addition to some rheumatological indications. We report the successful use of eculizumab to switch off catastrophic haemolysis in warm IgM-mediated autoimmune haemolytic anaemia (wAIHA) unresponsive to standard immunosuppressive therapy. The patient has required transfusion of 30 units of red blood cells over 8 days. Eculizumab therapy resulted in immediate cessation of transfusion support requirements and normalization of haemolysis markers. The use of Eculizumab in immune haemolytic anaemia is investigational. We identified few case reports of Eculizumab use in immune haemolysis including cold agglutinin disease, ABO incompatible blood transfusion, paroxysmal cold haemoglobinuria and warm immune haemolytic anaemia. Few in vitro studies and clinical trials are investigating the use of Eculizumab and novel complement inhibitors in cold agglutinin disease. The pathogenetic role of complements in cold antibody haemolysis is well recognized, but its role in warm antibody haemolysis is not well understood. The successful use of Eculizumab to abort haemolysis in our patient should stimulate further evaluation of its role in refractory warm antibody autoimmune haemolytic anaemia.

Disclosure of Interest: M. Garg Conflict with: Takeda Janssen, Conflict with: Travel support: Takeda Janssen Ad board: Takeda Sanofi Amgen Novartis Janssen, S. Agarwal: None Declared., M. Altohami: None Declared.

Thrombosis and Haemostasis

BSH2020-EP-139

Discrepant interpretation of HIT screening results on the same sample – data from a UK NEQAS for blood coagulation exercise

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Heparin-induced thrombocytopenia (HIT) is an immune complication of treatment with heparin, in some cases resulting in life-threatening thrombosis. An accurate diagnosis is required to ensure appropriate clinical decisions are made. We have previously reported discrepant results obtained by different laboratories on a sample sent for external quality assessment (EQA).

We describe here a further EQA exercise in which laboratories employing different HIT assays were asked to test and interpret results for a sample from a different donor thought to have HIT based on clinical and laboratory findings.

Plasma was obtained with informed consent from a patient identified as having HIT antibodies by the Werfen Acustar assay. Plasma was buffered and lyophilised in aliquots and sent to participants in the UK NEQAS for Blood Coagulation (UK NEQAS BC) HIT programme.

Median, coefficient of variation (CV) and range of results were determined for each method and reagent group. Participants were also asked to make an interpretation (positive, equivocal or negative) for the sample.

Results were received from 66 centres. About 62% of centres reported a positive screen on the test sample, with 35% reporting a negative screen, and 3% equivocal results. In total, 15 out of 16 centres using ELISA-based assays reported a positive HIT screen, 12 of which used the Lifecodes/Immucor PF4 IgG kit (median OD 0.72). Thirteen centres using the Werfen Acustar assay reported a positive HIT screen (median 4.09 u/ml), compared to just five out of 12 using the HemosIL HIT-ab PF4-H kit. Eight out of 14 centres using the Biorad Diamed (ID-PaGIA heparin/PF4 antibody) test and 12 out of 15 centres using the Stago STic Expert kit reported a negative screen. One centre reported a positive functional HIPA screen on the sample and two centres reported heparin neutralisation or inhibition of their assay.

This exercise demonstrated further variability in interpretations that can arise for a sample with HIT antibodies, confirming discrepancies observed in a previous exercise and the need for laboratories and clinicians to understand the clinical implications of reported HIT assay results.

Disclosure of Interest: None Declared.

BSH2020-EP-140

Preanalytical variables in haemostasis: results of the UK National External Quality Assessment Scheme for Blood Coagulation haemolysis supplementary exercise

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Haemolysis is considered one of the major contributors of non-conformities and sample rejection in coagulation testing. UK National External Quality Assessment Scheme for Blood Coagulation (UK NEQAS BC) conducted a multicentre exercise in order to investigate

effects of haemolysis on methodology and results, and gather information about laboratory approaches to dealing with haemolysed samples.

In November 2018 UK NEQAS BC distributed two different lyophilised plasma samples to 800 participants comprising of centres registered for prothrombin time (PT), activated partial thromboplastin time (APTT) and either Clauss fibrinogen or thrombin time (TT) in the UK NEQAS BC programme. The same pooled normal plasma was used to prepare two samples, S18:19 and S18:20. A haemolysate consisting of freeze/thawed red blood cells was added to sample S18:20 in order to create haemolysis at 3 g/l concentration, which was confirmed by testing haemolysed plasma pool on a Sysmex XN 10 haematology analyser. Participants were also asked to provide answers to a questionnaire about their laboratory approach to dealing with haemolysed samples including strategies used to deal with different levels of haemolysis.

The variability in the number of received results for samples S18:19 and S18:20 was a subject to the test registered by participants.

Five hundred and one results were received for PT test. The range of PT ratios for sample S18:19 was 0.86–1.7 with a median PT ratio 1.05 and for sample S18:20 the range of PT ratios was 0.85–9.23 with median PT ratio 1.1. The observed difference between the means for two samples was 4.8%.

For APTT, 503 results were received. The range of APTT ratios for sample S18:19 was 0.76–1.47 with a median APTT ratio 0.95 and for sample S18:20 the range of APTT ratios was 0.86–1.60 with median APTT ratio 0.95. No difference was observed between the means for both samples.

Participants returned 504 results for Clauss fibrinogen testing. The range of results for Clauss fibrinogen for sample S18:19 were 1.67–8.3 g/l with a median 2.46 g/l and for sample S18:20 ranges of results were 1.62–8.0 g/l with the median 2.56 g/l. The difference between the means for two samples was 4.1%.

About 338 results were returned for thrombin time testing. The range of TT ratios for sample S18:19 was 0.67–1.37 with a median TT ratio 1.05 and for sample S18:20 the range of TT ratios was prolonged 0.71–1.42 with median TT ratio 1.01. Between the means of two samples the observed difference was (–3.8%).

Variability in practice for processing haemolysed samples was observed. For example out of 501 responses 226 centres indicated they reject haemolysed samples solely on visual checks, with 163 using initial visual checks with further sample rejection evaluation by analyser flags. About 333 centres indicated that the level of haemolysis affects sample rejection decisions while 132 stated it does not.

In this exercise, results for performed tests did not show great variability between two samples. This may be explained by artificial construction of the haemolysed sample in this exercise. Variability of responses for dealing with haemolysed samples reflects a lack of clear guidelines in the preanalytical area of sample processing.

Disclosure of Interest: None Declared.

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Effects of rivaroxaban on thrombophilia screening tests – results from a UK NEQAS (Blood Coagulation) Exercise

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Current recommendations suggest investigation of heritable thrombophilia should not be carried out on patients receiving anticoagulation. However, the information about the effects of direct oral anticoagulants (DOACs) on thrombophilia screening tests is useful, as clinical information is not always available when samples are received in the laboratory for testing. We report here a UK NEQAS (Blood Coagulation) proficiency testing exercise, in which a sample from a patient with protein S deficiency and taking rivaroxaban (level 93 ng/ml) was sent to laboratories for testing without relevant clinical information.

Participants were asked to perform thrombophilia testing [Protein C (PC), Protein S (PS), anti-thrombin (AT) and activated protein C resistance (APCR)] on the sample and interpret their results. About 347 centres returned results in this exercise. Median results were calculated for all methods employed for each test. Clot-based PC activity assay results were significantly higher than chromogenic-based assay results (122.7 u/dl vs. 101.7 u/dl, $P < 0.001$). Thrombin-based anti-thrombin assays gave lower results than Xa-based methods (96.0 u/dl vs. 107.0 u/dl, $P < 0.001$). PS activity assay results were higher than free PS antigen results (24.4 u/dl vs. 11 u/dl, $P < 0.001$). These differences were seen across the range of methods employed by participants, confirming the consistent effect of a direct Xa inhibitor on these assays. Despite this, all centres performing a full thrombophilia screen obtained the correct interpretation of PS deficiency, although three additionally reported APC resistance.

Anticoagulation with DOACs is not reported to affect chromogenic PC assays or free PS assays, so false-negative results are unlikely if these tests are employed in a thrombophilia 'screen', as they are by the large majority of UK NEQAS participants. However, AT assays may be affected by either thrombin- or Xa-based inhibitors dependent on the substrate used, and false-positive results may arise in patients with mild AT deficiency – our data show both these methods are widely used.

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BSH2020-EP-142

Management of pregnancy-associated acquired haemophilia A: a case-series from the Oxford Haemophilia and Thrombosis Centre, UK

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Pregnancy-associated acquired haemophilia A (AHA) is a rare bleeding disorder seen in about one in 350 000 births. Given the paucity of cases, management of this disorder is varied with no clear consensus guidelines. We describe here three patients with pregnancy-associated AHA, treated between February and October 2019.

The first patient was a 24-year old who had a prolonged activated partial thromboplastin time (APTT) two months after the birth of her second child (74 s, normal 20–30 s). She described mild bruising. She then fell pregnant with her third child and at 9 weeks gestation she was referred to a haemophilia centre where investigation of the prolonged APTT found a chromogenic factor (F) VIII <0.01 iu/ml, anti-human FVIII inhibitor titre 2.2 bethesda units (BU) and anti-porcine FVIII

inhibitor 0.5 BU. She was jointly managed with obstetric medicine, and, following discussion with obstetrics, was commenced on intermediate dose prednisolone 40 mg daily (0.5 mg/kg). Intermediate dose prednisolone was advised due to the potential risk of cleft lip/palate in the fetus with the use of prednisolone doses higher than 40 mg in the first trimester. A week after starting steroids her FVIII rose to 0.16 iu/ml; after 2 weeks to 0.27 iu/ml and at 4 weeks to 0.68 iu/ml. She had no major bleeding issues and is currently 16 weeks pregnant with FVIII level 1.12 iu/ml.

The second patient was a 36-year old who gave birth to her 1st child by spontaneous vaginal delivery after an uncomplicated pregnancy. She developed extensive bruising 4 weeks postpartum and her APTT was prolonged (47.1 s). Her FVIII level was <0.01 iu/ml with anti-human FVIII inhibitor 17.8 BU and anti-porcine FVIII inhibitor 3.4 BU. She was started on prednisolone 1 mg/kg and, after 5 weeks, given 4 doses of 375 mg/m² rituximab due to a poor response to steroids. Her FVIII levels increased to 0.54 iu/ml 1 week after completing rituximab and have remained at this level.

The third patient was a 35-year-old primigravida who presented with bruising 2 months postpartum. Blood tests showed a prolonged APTT (75 s), FVIII <0.01 iu/ml, anti-human FVIII inhibitor titre 7.1 BU and anti-porcine FVIII inhibitor 0.5 BU. She commenced prednisolone 1 mg/kg, but her FVIII level remained low with symptomatic bruising and so, after 7 weeks, she was started on 4 weeks of 375 mg/m² rituximab therapy and steroids were rapidly weaned. She continued to have low FVIII levels (0.02 iu/ml), and developed haematuria, and so was treated with NovoSeven 90mcg/kg. She was then started on mycophenolate mofetil (MMF) (initially 500 mg twice daily, increased to 1 g twice daily). Two weeks after starting MMF her FVIII level improved to 0.23 iu/ml and is now 1.00 iu/ml, 7 months postpartum.

These cases highlight the varied course pregnancy-associated AHA can take. There is very limited evidence on the management of AHA during pregnancy, with few reported cases in the literature. Careful patient counselling is essential at all times, as immunosuppressants can cross the placenta and pass into breastmilk, and so close liaison with the obstetric and pharmacy teams is vital in the management of these patients.

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BSH2020-EP-143

Haemostatic markers in cardiopulmonary bypass surgery (MIC-B Study)

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It has been shown in vitro that unfractionated heparin (UFH) can inhibit the von Willebrand Factor (VWF)-GPIIb/IIIa interaction, an anti-thrombotic effect distinct from its anticoagulant role via anti-thrombin. The physiological and clinical relevance of this effect is not known. This additional activity of heparin is likely to be of importance during cardiopulmonary bypass surgery (CPB), when high doses of UFH are needed to maintain the extracorporeal circuit and have to be reversed with protamine sulphate (PS) to restore haemostasis at the end of the procedure. This may in part explain why bleeding complications that occur during heparin therapy do

not always correlate with the heparin anti-Xa level or activated clotting time (ACT). To characterise the changes in VWF antigen (VWF:Ag) activity, function and platelet interaction, that occur pre and post UFH in patients undergoing CPB, we conducted a prospective, single-centre cohort study in a major cardiac surgery centre in London. Thirty patients aged >18, undergoing first time surgery for the correction of atrial septal defects or tissue mitral valve replacement were recruited over 6 months from January to July 2019. Patients undergoing aortic valve surgery or already on heparin were excluded. Blood samples were taken at six time points [baseline (Pre Heparin), 30 min after administration of intravenous UFH (Post Heparin), before reversal of UFH (Pre Protamine), 30 min after reversal of UFH with PS (Post Protamine) and 12 and 24 h postoperatively] into sodium citrate tubes, double centrifuged and analysed immediately or stored at -80°C until analysed. The mean age of the study group was 58.9 years ($\text{SD} \pm 16.1$) and 56.6% were male. There was no significant fall in the VWF:Ag intraoperatively with a significant rise by the end of surgery, $P < 0.0001$. The fall in collagen-binding activity (VWF:CBA) Post Heparin reached significance ($P = 0.018$), but was not significant when the VWF:CBA/VWF:Ag ratio was calculated ($P = 0.0916$). VWF:CBA rose significantly by the end of surgery ($P = 0.0133$) up to 24 post-op ($P < 0.0001$), but the VWF:CBA/VWF:Ag fell significantly ($P = 0.0015$) suggesting loss of VWF high molecular weight multimers, presumed due to shear induced proteolysis by the bypass circuit. The absolute VWF Ristocetin Cofactor (VWF:RCo) (an assessment of the VWF-platelet GPIIb/IIIa interaction), was not significantly impaired following UFH ($P = 0.0588$), but showed a trend towards significant fall. Although the absolute VWF:RCo recovered after UFH reversal ($P = 0.0003$), there was no significant recovery of the VWF:Ag/VWF:RCo which remained significantly reduced ($P = 0.0143$). This unexpected finding could be explained by published literature describing an inhibitory effect of protamine on the VWF-GPIIb/IIIa interaction. The platelet count fell throughout surgery and up to 24 h postoperatively. ACT and heparin anti-Xa level correlated poorly ($r = 0.039$) and neither predicted clinically significant bleeding. In conclusion, both heparin and PS affect many components of haemostasis including platelet function and coagulation. The complex interplay between components of haemostasis, well characterised consumptive and dilutional coagulopathy during CPB as well as these off-target effects of both heparin and PS, make it difficult to predict bleeding reliably in this group of patients. The poor correlation between the ACT and anti-Xa levels indicates extra caution is required when interpreting these results in this setting.

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Evaluation of the Abbott CP3000 coagulometer for factor assay and inhibitor detection

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Automated coagulometers are continually evolving to provide greater accuracy and precision as well as performing new haemostasis assays. Abbott Diagnostics (Santa Clara, CA) distributes the CP3000 Automated Coagulation System (Sekisui Diagnostics, Burlington, MA), which is stated to be analytically comparable to current validated automated coagulometers. The CP3000 also has a novel automated

immediate and incubated mixing study feature that is said to be able to distinguish between factor deficiencies and factor inhibitors.

The aim of the study was to compare the CP3000 against the Sysmex CS5100 (Sysmex, Milton Keynes) at the Haemophilia and Thrombosis Centre, Oxford. The suitability of the CP3000s novel mixing study feature was analysed for its ability to accurately distinguish between factor inhibitors and factor deficiencies on Haemophilia A and B patients. In addition, the sensitivity of the Coagpia PT-N and the Coagpia APTT-FS reagents to detect factor deficiencies was also determined. The study was carried out using de-identified leftover plasma that had been collected for diagnostic purposes and processed on the CS5100. Factor sensitivity curves were generated for the PT-N (II, V and X) and APTT-FS (VIII, IX, XI and XII) based assays by testing pooled normal plasma serially diluted with appropriate factor-deficient plasma in duplicate and plotting either PT or APTT clotting time against the factor activity. Data points were fitted with a power curve and the goodness of fit was evaluated based on the R^2 value. The factor activity corresponding to the PT and APTT normal thresholds was mathematically determined. The detection threshold of the PT-N-based assays (FII, FV and FX) were 37%, 41% and 32%, respectively, APTT-FS-based assay sensitivity were (FVIII, FIX, FXI and FXII) 53%, 45%, 61% and 58% respectively.

Twenty patients with factor FVIII and factor IX deficiency and 20 patients with inhibitors to factor VIII and factor IX had automated mixing studies performed on the CP3000 and Rosner Indexes (RI) calculated. All deficiencies had RIs of <12. Four of the 20 inhibitors had RIs of <12, the strongest being a commercial factor IX inhibitor plasma with 1.3 BU.

The study concluded that the CP3000s novel mixing study feature can distinguish between stronger factor inhibitors and factor deficiencies.

Due to limited numbers of samples for some factor deficiencies further work would be required to fully analyse the CP3000 and its ability to accurately measure factor levels at low levels.

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BSH2020-EP-145

A case of acquired Glanzmann thrombaesthesia post multivisceral transplant

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Acquired Glanzmann thrombaesthesia is a rare condition, with just 33 cases reported worldwide to date (1). It is characterised by acquired platelet dysfunction without a fall in platelet count, due to inhibiting antibodies blocking the adhesive receptor function of platelet glycoprotein IIb/IIIa thereby preventing platelet adhesion to fibrinogen. Acquired platelet dysfunction leads to a mucocutaneous bleeding phenotype similar to that seen in congenital Glanzmann thrombaesthesia. Due to the rarity of the condition, there is limited evidence to inform knowledge of disease characteristics, guide therapy or suggest prognosis.

We report a case of a 45-year-old gentleman of Polish descent, who underwent a small bowel, pancreas and colon transplant for complications relating to congenital portal vein thrombosis. 18 months post multivisceral transplant, he developed diffuse mucocutaneous bleeding symptoms. He had no prior bleeding history.

The following results were within normal limits: full blood count; factors II, V, VII, VIII, IX, X, XI, XIII and von Willebrand factor

antigen and ristocetin cofactor. Light transmission platelet aggregometry (LTA) revealed absent platelet aggregation to all agonists except ristocetin. Flow cytometric analysis demonstrated absent expression of CD41 on patient platelets.

Mixing studies demonstrated that patient platelet poor plasma (PPP), when mixed with donor platelet rich plasma (PRP), had an inhibitory effect on LTA and flow cytometric analysis. Despite this, a Monoclonal Antibody-specific Immobilisation of Platelet Antigen (MAIPA) assay did not prove the existence of an anti-GP IIb/IIIa antibody.

Multiple attempts at control of haemostasis were implemented sequentially, with limited success: desmopressin, tranexamic acid, platelet transfusions and plasma exchange. Immune regulation attempts were initiated alongside haemostatic measures, using IV methylprednisolone and intravenous immunoglobulins. Meanwhile, investigation for an underlying cause revealed a post-transplant lymphoproliferative disorder (PTLD).

The bleeding symptoms and laboratory abnormalities did not improve with the above measures. Second-line therapy was commenced with rituximab and vincristine chemotherapy, aiming to treat the underlying cause. Following this, the bleeding settled and flow cytometric analysis demonstrated re-expression of CD41. The patient remains in remission from the PTLD, has had no further bleeding sequelae and continues to demonstrate normal platelet aggregometric and flow cytometric results.

We conclude that on the basis of clinical and functional laboratory assay features, this represents a case of the rare disorder: acquired Glanzmann thrombocythaemia, most likely in association with underlying PTLD post multivisceral organ transplant. In particular, this case demonstrates the diagnostic challenges of this disorder, and the importance of identifying underlying causes. Of the existing other published cases of acquired Glanzmann thrombasthenia, eight occurred in association with a lymphoproliferative disorder and four occurred in the context of solid organ transplant recipients (1).

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BSH2020-EP-146

Management of traumatic intracranial haemorrhage in elderly patients on anticoagulation: a trauma centre experience

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Anticoagulation (AC) use is rising due to an ageing population, increased prevalence of atrial fibrillation (AF) and better guideline implementation. AC can cause major bleeding and management strategies need to be optimized to minimise adverse consequences. AC resumption after major bleeding is challenging with scarcity of high-quality evidence leading to variable practice. European Society of Cardiology AF guidelines (Kirchhof et al in 2016) suggest resumption of AC be considered >4 weeks after intracranial haemorrhage (ICH) after optimising risk factors.

The aim of this study was to review our experience of AC reversal and resumption in the management of traumatic ICH. We analysed available data from a trauma registry at a single urban Major Trauma Centre on patients that were admitted during a full 12-month period in 2018. Inclusion criteria were patients >65 years(y) with traumatic ICH. Data are presented as median (IQR).

Thirty-two patients were available for analysis: subdural 61.3% and intra-parenchymal ICH 38.7%. Age was 83 y (77–88). Indications for AC: AF 90% and venous thromboembolism 10%. Seventeen patients (53.1%) took warfarin with admission INR1.8 (1.2–3.0). Reversal agents were administered in 15 patients for warfarin therapy. About 58.8% had IV vitamin K. Time from admission to vitamin K was 2h58 m (1h58 m–4h29 m). About 64.7% had Prothrombin Complex (PCC). Time from admission to PCC was 3 h34 m (2h28 m–5h11 m).

Fourteen patients (43.8%) took a Xa-inhibitor: apixaban (34.3%), rivaroxaban (3.1%), edoxaban (6.3%). 64.3% had admission anti-Xa assay with level: apixaban 79 ng/ml (38–96), rivaroxaban 333 ng/ml and edoxaban 42 ng/ml. About 35.7% had reversal with PCC. Time from admission to PCC was 1 h35 m (1 h5 m–3 h45 m). One patient on low molecular weight heparin (LMWH) had no protamine reversal.

Tranexamic acid was given in 31.2% (<3 h in 20%). 35% had neurosurgery with age 77 years (72–82) vs. 86 years (78–89) without surgery.

Nine patients died (28.1%) due to ICH. Twenty-three patients were alive at discharge with GCS 15 (2–15) and ECOG 3 (2–4). At discharge 12 patients went home, two patients to rehabilitation unit and nine patients awaited nursing home. Length of hospital stay was 20 days (4–24).

Thirteen patients (56.5%) resumed AC prior to discharge. Time from admission to AC resumption was 13 days (8–24). Decisions regarding resumption of AC were made by neurosurgery 61.5% and geriatric medicine 23.1%. 76.9% decisions were consultant-led. 76.9% had the same AC at resumption, 15.3% had dose reduction and 23.1% changed AC (apixaban to treatment-dose LMWH, apixaban to prophylactic dose LMWH and apixaban to aspirin). 76.9% had >1 bleeding risk factor at resumption. Length of follow-up was 269d from the time of ICH (97–397). No recurrence of ICH occurred in this time.

Our trauma centre experience revealed delays in warfarin reversal despite availability of point-of-care INR testing and PCC in the Emergency Department. PCC administration was used less frequently, but was less delayed in Xa-inhibitor-associated ICH, which might be relevant in future to administration of the antidote andexanet, although numbers were small. The case-fatality is consistent with previous reports. Only 56.5% surviving patients resumed AC post-ICH but risk factors for bleeding were frequent. Time to resumption of AC was shorter than recommended and varied widely. Multidisciplinary review of complex cases to consider timing of AC resumption and bleeding risk factor optimisation would help standardise and individualise the best approach.

Disclosure of Interest: None Declared.

BSH2020-EP-147

Improving renal function monitoring for patients on direct oral anticoagulants after discharge from the anticoagulation clinic: a multifaceted quality improvement project in primary care

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There has been a significant increase in the prescribing of direct oral anticoagulants (DOACs) in recent years. While these agents offer major benefits over conventional oral anticoagulants, those with impaired renal function are at an increased risk of adverse events. International guidelines state that all patients require renal function

measurement at least yearly and in those with chronic kidney disease (CKD), more frequent measurement is recommended. Likewise, guidelines advocate the calculation of creatinine clearance (CrCl) in preference to estimated glomerular filtration rate (eGFR) to guide decisions. We aimed to assess and improve the compliance of a large general practice with these guidelines.

The cohort consisted of all patients taking DOACs in a single general practice who had been discharged from the local anticoagulation clinic. Patients were analysed retrospectively at three discrete time points, each one year apart (in 2017, 2018 and 2019). Assessment included whether patients were undergoing the recommended frequency of renal function measurement (for their level of renal impairment) and whether they were having CrCl calculated.

After the first audit cycle in 2017, an initial intervention involving education sessions for prescribers was designed and implemented. Then, after the second audit cycle in 2018, further interventions were introduced, including a new alert system that was integrated into the electronic records to remind prescribers if individual patients were not currently meeting recommendations as well as posters detailing these guidelines in consultations rooms.

The baseline audit in 2017 revealed that 111 out of 139 (80%) patients were undergoing appropriate renal function measurement but only 8/139 (6%) were having CrCl calculated. In 2018, these figures had dropped to 74% (132/170) and 2% (3/179), respectively, prompting the second intervention and assessment cycle. After reassessment in 2019, 90% (182/203) were having the appropriate frequency of measurement and 64% (130/203) had CrCl calculation. Likewise, the mean frequency of monitoring had improved from 8 monthly in 2017 to 4 monthly in 2019.

The above results suggest that after discharge from the anticoagulation clinic, renal function monitoring in patients taking DOACs is likely to fall short of international recommendations. Monitoring can be improved with a multifaceted intervention involving regular prompting, rather than single education sessions for prescribers. This is likely to be important in the prevention of adverse outcomes in patients taking DOACs in the long term.

Disclosure of Interest: None Declared.

BSH2020-EP-148

A systematic review and meta-analysis of randomized controlled trials on primary ambulatory thromboprophylaxis (PATP) in patients with ovarian cancer receiving chemotherapy

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Venous thromboembolism (VTE) is one of the most common complications in cancer patients and is associated with poor prognosis. Ovarian cancer (OC) is a highly thrombogenic cancer according to the Khorana score (KS) (score 1) in outpatient setting. OC cells promote thrombin generation and induce fibrin degradation which play a major role in thrombogenesis and cancer metastasis. Chemotherapy damages vascular endothelium and augments the procoagulant effect of cancer cells by multiple mechanisms, and therefore, imposes an additional risk factor for VTE. Cancer-associated thrombosis is the second leading cause of death in cancer patients. Thus, in recent years, PATP becomes an area of interest and has been considered as a potential therapy in ambulatory cancer patients undergoing chemotherapy with an ultimate aim of attempting to reduce the

incidence of VTE and potentially improve survival. We undertook a systematic review and meta-analysis of randomized controlled trials (RCTs) to determine the benefit of PATP in patients with OC who are on chemotherapy.

We systematically conducted a comprehensive literature search using MEDLINE and EMBASE databases through November 30, 2019 using the following keywords; 'low-molecular weight heparin' OR 'direct oral anticoagulant' AND 'ovarian cancer'. The references of all potential studies were also reviewed for any additional relevant studies. The RCTs with reduction in VTE as a primary or secondary end-point were incorporated in the analysis. The primary meta-analytic approach was a random effects model using the Mantel-Haenszel (MH) method. It was used to calculate the estimated pooled risk ratio (RR), and risk difference (RD) with 95% confidence interval (CI). Heterogeneity was assessed with I^2 and Cochran's Q- statistic. A 'P value' of <0.05 was considered significant and $I^2 > 50\%$ was considered substantially heterogeneous.

A total of 433 patients with OC from subsets of two RCTs (SAVE-ONCO and CASSINI) were included in our meta-analysis. SAVE-ONCO trial enrolled unselected cancer patients while CASSINI trial enrolled cancer patients with KS ≥ 2 . The prophylactic doses of anticoagulants were used; semuloparin in SAVE-ONCO and rivaroxaban in CASSINI trials. The PATP duration ranged from 3.5 to 6 months. The randomization ratio was 1 to 1 in both studies. The I^2 statistic showed moderate heterogeneity among RCTs, and random effects model was applied. The incidence of VTE was 2 (0.9%) in PATP group and 4 (1.8%) in control group. The pooled RR was not statistically significant at 0.69 (95% CI: 0.08–5.67, $P = 0.73$). The absolute RD was -0.03 (95% CI: -0.17 to 0.11 , $P = 0.66$).

Our meta-analysis demonstrated that the VTE incidence in OC patients is low in both PATP and control groups. There was no statistically significant reduction in VTE by providing PATP to ambulatory patients with OC who are on chemotherapy. Hence, routine PATP should not be recommended in ambulatory patients with OC. Further randomized trials are needed in the future to determine the high-risk subset of OC who may benefit from PATP.

Disclosure of Interest: None Declared.

BSH2020-EP-149

Outpatient Management of pulmonary embolism over ten years in a large teaching hospital: safe and evolving practice

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We report on a retrospective cohort study of all patients with pulmonary embolism (PE) diagnosed from targeted investigations, over ten years at the University Hospitals Plymouth NHS Trust. We review how many were treated as outpatients (OP), mortality and readmission rates. A modified Pulmonary Embolism Severity Index (PESI) was used to decide if OP management is suitable. We also examined the variance in management of hospital acquired (HAT) compared to community acquired PE (CAT) over the same period. Radiology data were used for detecting cases of PE from CT pulmonary angiograms and SPECT-CT scans. Outpatient management of PE is defined as being any positive PE event where the patient was discharged within 24 h of the scan. All-cause mortality, along with readmissions up to 28 days from PE diagnosis for both, were reviewed. Clinical notes were also reviewed to decide whether the readmission was associated with the PE diagnosis.

Table.

Year	Total PE total VTE	IP (%)	OP (%)	All mortality <28 days (IP)	All mortality <28 days (OP)	All OP readmitted	Total new PE on admission	IP (%)	OP (%)
2010	355 (55)	239 (67)	116 (33)	29	2	4	249	145 (58)	104 (42)
2011	465 (59)	339 (73)	126 (27)	52	0	3	321	195 (61)	126 (39)
2012	485 (61)	304 (63)	181 (37)	42	3	1	336	161 (48)	175 (52)
2013	399 (55)	245 (61)	154 (39)	53	2	3	294	147 (50)	147 (50)
2014	440 (58)	267 (61)	173 (39)	13	6	1	292	137 (47)	155 (53)
2015	518 (63)	296 (57)	222 (43)	27	4	2	360	162 (45)	198 (55)
2016	477 (55)	270 (57)	207 (43)	31	0	4	246	80 (33)	166 (67)
2017	509 (61)	269 (53)	240 (47)	28	4	4	278	93 (33)	185 (67)
2018	483 (59)	268 (55)	215 (45)	26	1	7	297	129 (43)	168 (57)
2019	452 (59)	290 (64)	162 (36)	38	3	6	271	126 (46)	145 (54)
Total	4583 (59)	2787 (61)	1796 (39)	339	25	35	2944	1375 (47)	1569 (53)

Over 10 years, there has been an increase in total PE diagnosis from 355 in 2010 to 452 in 2019, an increase of 27%. All PE diagnoses were reviewed from notes, including how they were managed. In 2010 116/355 (33%) were managed as OP compared with 162/452 (36%) in 2019, a modest increase of 9%. However, these figures include new PE diagnoses in current inpatients such as postsurgery and therefore smaller numbers. Therefore, data on patients who actually presented to the hospital and scanned within 24 h were collected separately. In 2010, 104/249 (42%) were managed as OP, increasing to 145/271 (54%) by 2019 a rise of 29%. All-cause mortality was significantly higher for inpatient PE events at 339, compared to 25 for OP management. None of the OP deaths were from recurrent PE events. This is not surprising, acknowledging that the inpatient group are likely to be more unwell. Over the ten years, there were 35 OP patients readmitted, approximately 2% of all patients managed this way. None of these readmissions were associated with them being managed as outpatients.

Patients with HAT events were significantly more likely to be treated as inpatients 1031/1279 (81%), compared to CAT events at 1756/3304 (53%), acknowledging they are likely to have more comorbidities as a more unwell group.

Over 10 years we demonstrate an increase in both total PE diagnoses and OP managed PE which could be linked to scanners picking up smaller emboli. The safety of the latter approach has been demonstrated when appropriate criteria are used. The increased use of direct oral anticoagulants and the introduction of ambulatory care units have contributed to the increase in OP management. This is a safe methodology (using the modified PESI score), with financial benefits (saving over 5000 bed days), over the study period.

Disclosure of Interest: None Declared.

BSH2020-EP-150

Influence of DOAC remove on coagulation assays in samples from patients on anticoagulation

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Background: Direct oral anticoagulants (DOAC) interfere with many haemostasis assays. DOAC-Remove (5-Diagnostics AG, Basel, Switzerland) claims to neutralise heparin and argatroban in addition to rivaroxaban, apixaban, edoxaban and dabigatran. We investigated

the DOAC-Remove effect on assays on plasmas collected from patients on anticoagulation.

Methods: Twenty plasmas from patients in each of the following groups were tested: lupus anticoagulant (LA) negative (no anticoagulant); dilute Russell's Viper Venom Time (DRVVT) positive; anticoagulated patients (20 each on warfarin, rivaroxaban, apixaban, edoxaban, dabigatran, heparin or argatroban).

DOAC-Remove was added to plasma, incubated for 5 min before centrifugation (2000 g, 5 min). Treated and untreated samples were tested for: prothrombin time [PT] (Siemens Innovin); activated partial thromboplastin time [APTT] (Siemens Actin FS); thrombin time [TT] (Siemens Thromboclotin); calibrated anti-Xa assay (Hyphen Biomed Heparin LRT) or anti-IIa assay (Hemoclot DTI); anti-thrombin [AT] (IIa substrate, Hyphen Biomed and Xa substrate, Hyphen Biomed LRT); DRVVT (Siemens LA1 and LA2).

Results: DOAC-Remove reduced DOAC activity to <20 ng/ml in 83% of rivaroxaban, 50% of apixaban, 100% of edoxaban and 100% of dabigatran samples. One sample with apixaban 555 ng/ml had residual activity equivalent to 109 ng/ml after adsorption. Argatroban levels were reduced but only 56% of samples had levels <150 ng/ml post-treatment. Heparin levels were unaffected.

PT and APTT were unchanged in LA negative, heparin or warfarin samples, but were reduced in direct Xa (DFXaI) or direct thrombin inhibitor (DTI) samples. TT increased slightly in LA-negative samples. There was no significant difference between these and the TT in adsorbed dabigatran samples, but adsorbed argatroban samples had significantly prolonged TT if the argatroban level remained >150 ng/ml.

DOAC-Remove reduced the rate of false positives in rivaroxaban or edoxaban samples whether calculated by percentage correction or by screen/confirm ratio. DRVVT was also shortened by DOAC-Remove in apixaban samples. 21% of non-anticoagulated samples previously shown to be DRVVT-positive were DRVVT-negative after treatment with DOAC-Remove.

Factor Xa-based anti-thrombin assays will be erroneously increased in the presence of DFXaI, and IIa-based assays will be erroneously increased in the presence of DTI. We observed an unexpected further increase in AT using Xa-based reagents in all sample types except heparin, and an unexpected further increase in AT using IIa-based reagents in warfarin, heparin, rivaroxaban and edoxaban samples. There were no significant differences in AT using IIa-based reagents in patients on dabigatran or argatroban.

Conclusions: DOAC-Remove removed anti-Xa activity from DFXaI samples, removed anti-IIa activity from dabigatran samples, partially removed anti-IIa activity from argatroban samples, and had no effect

on anti-Xa activity in heparin samples. DOAC-Remove removed some DOAC effects on screening and LA assays, but also normalised some LA assays in known LA-positive samples.

Anti-thrombin assays were affected by DOAC-Remove but not as expected, requiring further work.

Care needs to be taken in the interpretation of assay results in anticoagulated patients, with or without adsorption with DOAC-Remove. The preferred course of action when performing investigations should be for the patient to be off anticoagulation if possible.

Disclosure of Interest: None Declared.

BSH2020-EP-151

Clinical characteristics and outcomes in patients treated with ecilizumab for suspected atypical haemolytic uraemic syndrome (aHUS) – a single tertiary centre experience

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aHUS is a rare, complement-mediated thrombotic microangiopathy (TMA), commonly presenting with renal failure. Prompt diagnosis and treatment with the anti C5 monoclonal antibody ecilizumab may improve patient outcomes.

This was a retrospective study reviewing clinical characteristics and outcomes of patients treated with ecilizumab for suspected aHUS, in a centre providing a haematology-led specialist service for patients referred with TMAs. Data were collected from the centre's TMA patient database.

Eleven patients were treated with ecilizumab, 2013–2019. All the 11 patients were external referrals from a wide geographical distribution, median distance from referring hospital 40 miles (range 1–85 miles). Eight out of eleven patients were referred with suspected thrombotic thrombocytopenic purpura (TTP), three out of eleven with suspected aHUS. Eight out of eleven patients received a final diagnosis of aHUS, of which three out of eight demonstrated a genetic complement mutation; 2/8 complement factor H, 1/8 CD46. Three out of eleven remaining patients received a final diagnosis of Shiga toxin HUS, amphetamine-induced malignant hypertension and hypertensive nephropathy. Four out of eleven patients presented within 1 week postpartum. Ten out of eleven patients presented with hypertension; median systolic and diastolic blood pressure 178 and 100 mmHg respectively (range 139–260 mmHg systolic, 77–170 mmHg diastolic). Four out of eleven patients had extra-renal manifestations (3 neurological, 1 cardiorespiratory). Median presenting creatinine was 561 µmol/l (range 135–1621 µmol/l) with 7/11 patients requiring haemodialysis (HD) acutely. 0/8 patients with a final diagnosis of aHUS remain on HD. Median presenting haemoglobin was 81 g/l (range 50–135 g/l) and median presenting platelet count was $56 \times 10^9/l$ (range $12\text{--}164 \times 10^9/l$). Eight out of eleven patients presented with a platelet count $>30 \times 10^9/l$ and renal impairment, as classically seen in aHUS. Eight out of eleven patients underwent renal biopsy, 8/8 demonstrated TMA histology.

Nine out of eleven patients were admitted to the intensive care unit, 2/11 to the renal high dependency unit. Two out of eleven patients required intubation and ventilation, both of which had a final diagnosis of aHUS. Ten out of eleven patients received plasma exchange (PEX) prior to treatment with ecilizumab with a median duration of 3 days (range 1–16). The median time from consideration of diagnosis to ecilizumab treatment was 3 days (range 0–13).

Currently, five out of 11 patients remain on ecilizumab, all delivered by a healthcare provider at home. Reasons for ecilizumab discontinuation were; three out of 11 alternative diagnoses, two out of 11 sustained renal recovery, one out of 11 preconception. Two out of six patients experienced a relapse of aHUS <6 months following discontinuation and successfully recommenced ecilizumab.

The majority of patients referred with aHUS were initially suspected to have TTP and received PEX prior to ecilizumab. A proportion of patients presented critically unwell, with more than half requiring HD acutely. In most cases, ecilizumab was delivered promptly with all final diagnosis aHUS patients becoming HD independent. In more than half, ecilizumab was discontinued; however, a proportion of patients required reintroduction due to disease relapse.

This study demonstrates the diversity of clinical characteristics in patients suspected to have aHUS, emphasising the need for timely, pro-active, management ensuring better outcomes. As with other rare diseases, there is an important argument for patients with aHUS to be referred to regional specialist centres to facilitate comprehensive care for this complex patient group.

Disclosure of Interest: None Declared.

BSH2020-EP-152

Do patients taking Direct Oral Anticoagulants (DOAC) require regular systematic review? Evidence from a Comprehensive

Anticoagulation and Thrombosis Centre

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DOAC use has increased over past few years and these drugs are preferred anticoagulants for non-valvular AF and VTE treatment. Main advantages include fixed dose, lack of regular coagulation monitoring and low bleeding risk. Oral anticoagulants are classed among drugs associated with highest risk of medication related harm. Bleeding and thrombotic risk changes with age, co-morbidities, concurrent medication and liver-renal impairment. ESC guidelines 2018, therefore recommend regular follow-up review of patients receiving DOAC for Stroke prevention in AF.

Staffordshire Thrombosis and Anticoagulation Centre (STAC) oversee oral anticoagulation of over 8000 patients including patients on DOAC. Patients undergo regular review by a team of nurses via telephone consultation. This comprehensive review includes blood tests, review of co-morbidities, medication, compliance, bleeding, thrombotic problems and relevant personal and social information. All reviews are documented on the STAC electronic systems

Between 1st Jan 2018 and 31st Dec 2018, 7814 comprehensive reviews were performed of 5739 patients on oral anticoagulants. Of these, 2220 reviews were performed of 1280 patients on DOAC. Bleeding, thrombotic events and parameters likely to affect these risks are listed in Table 1.

The study confirms that DOAC are safe and effective in most patients with low risk of bleeding and recurrent thrombotic events. About 27% of our patient cohort had moderate or severe renal impairment and in over 20% patients significant medication interactions were identified. The review process resulted in change in treatment in 5.2% patients.

Although DOAC do not require routine coagulation testing, regular comprehensive review by specialized clinicians helps in identifying risks and taking appropriate clinical intervention.

Table.

Clinical parameter	Number of reviews of patients on DOACn = 2220 (100%)
Bleeding events (all)	97 (4.3%)
Major bleeding	3 (0.1%)
Clinically significant non-major bleeding	6 (0.3%)
Minor bleeding	88 (3.9%)
Recurrent thrombosis (all)	9 (0.4%)
Recurrent venous thrombosis	5
Stroke and systemic embolism	4
Thrombocytopenia	107
Severe (plat count < 50 × 10 ⁹ /l)	0
Moderate (plat count 50–80 × 10 ⁹ /l)	3
Renal impairment	607 (27.3%)
Severe (Cr Cl <30 ml/min)	76 (3.4%)
Moderate (Cr Cl 30–50 ml/min)	531 (23.9%)
Significant medication interaction	480 (21%)
Patient non-compliance/ dissatisfied with Rx	20
Outcome altered	117 (5.2%)
Anticoagulation stopped	9
Anticoagulant switched	36 (1.6%)
Dose altered	64 (2.9%)
Incorrect indication identified	0
Antiplatelet stopped	8
Others (pt declined review, advise given to GP, etc)	13

Disclosure of Interest: None Declared.

BSH2020-EP-153**Venous thromboembolism in a large teaching hospital over the last 10 years: changing patterns of disease****H. Rowswell*, T. Nokes***Haematology, University Hospitals Plymouth NHS Trust, Plymouth, UK*

University hospitals Plymouth NHS Trust (UHPNT) is a tertiary referral hospital of 900 beds, designated a Venous Thromboembolism (VTE) exemplar site in 2009. All VTE events diagnosed from targeted radiological data, between 2010 and 2019 inclusive have been included in metrics within UHPNT. These investigations include CT pulmonary angiograms, SPECT-CT scans and Doppler ultrasounds of the lower and upper limbs. Positive events, such as deep vein thrombosis (DVT) or pulmonary embolism (PE), are reviewed to decide whether they meet the criteria for hospital acquired thrombosis (HAT).

In 2010 there were 649 total VTE events with 329 DVTs and 320 PEs. Out of these 201 (31%) were HAT events, comprising 85 DVT and 116 PE; a significantly higher number of PEs ($P = 0.005$) compared to DVT. The total number of VTE events has increased by 120–769 (18%) over the 10-year study period. Over the same time, the total DVTs diagnosed has remained relatively constant, with 315 in 2019, a 4% decrease. In 2019 454 PEs were diagnosed, giving an increase of 42% (from 320) over the 10 years, representing a steady annual increase over that time. Compared to 2010 where the number of PE diagnoses was almost the same as DVT, by 2019 the split is 59% PE, compared to 41% in 2010 to 28% in 2019, although the

total number of HAT PE events has remained relatively similar except in the most recent data with a 14% increase in HAT PE.

We believe that the overall reduction in HAT events is linked to an improved awareness of this life-threatening condition. Accordingly, increased resources are being utilised for the prevention of VTE in UHP patients. The increase in total PE events may be explained by the implementation of more sensitive CT scanners. These are able to detect smaller, subsegmental PEs than the previous equipment. Generally, the number of HAT VTE events has plateaued at a lower number. We have previously shown that this reduction in HAT events is significant. The challenge now is to further reduce HAT events within UHPNT.

Table.

Year	Total VTE	All DVT (%)	All PE (%)	HAT DVT (%)	HAT PE (%)
2010	649	329 (51)	320 (49)	85 (42)	116 (58)
2011	782	325 (42)	457 (58)	73 (31)	165 (69)
2012	793	311 (39)	482 (61)	71 (33)	146 (67)
2013	805	342 (42)	463 (58)	66 (38)	106 (62)
2014	775	326 (42)	449 (58)	55 (33)	114 (67)
2015	835	340 (41)	495 (59)	60 (31)	131 (69)
2016	852	384 (45)	468 (55)	64 (38)	105 (62)
2017	836	329 (39)	507 (61)	56 (32)	119 (68)
2018	825	338 (41)	487 (59)	65 (35)	121 (65)
2019	769	315 (41)	454 (59)	51 (28)	132 (72)
Total	7921	3339 (42)	4582 (58)	646 (34)	1255 (66)

Disclosure of Interest: None Declared.

BSH2020-EP-154**Utility of Pro C Global assay and Factor VIII activity as anti-coagulant and pro-coagulant factors in Hepatitis C cirrhotic patients****A. Almeshawy^{1,*}, H. Azzam¹, R. Saad², H. Ghoneim¹**¹Clinical Pathology, ²Internal medicine, Mansoura University, El Mansoura, Egypt

For a long time patients with hepatitis C virus (HCV), liver cirrhosis are considered suffering from hypo-coagulability. It was thought that coagulation factors are decreased and therefore patients would be complicated by different types of haemorrhagic complications. Egypt is a country with the biggest number of HCV cirrhotic patients. So, it is important to assess the exact haemostatic state of the patients and whether it is affected by the stage of cirrhosis.

The aim of our study was to determine the utility of Pro C Global assay and factor VIII activity as anticoagulant and procoagulant factors in Hepatitis C cirrhotic patients. A case-control study was conducted for 45 adult patients with HCV liver cirrhosis (30 men, 15 women) with a median age of 51 years (range, 28–84) subdivided into three groups each composed of 15 patient according to Child-Pugh classification in addition to 20 healthy control subjects comparable for age and sex. Patients suffering from intrahepatic, extrahepatic malignancies or inflammatory conditions were excluded. Patients on drugs affecting blood coagulation were also excluded.

The haemostatic profile of our study groups showed that international normalized ratio (INR), activated partial thromboplastin time (aPTT) and thrombin time (TT) were significantly increased in all of liver cirrhosis patients compared to controls. The comparison between different Child Pugh classes and even with control group showed decreased levels of pro-coagulants with cirrhosis progression. Fibrinogen concentration was significantly lower in patients

Table. Correlation of PCAT-NR with other studied parameters in all studied groups.

	Control group N = 20		HCV liver Cirrhosis group N = 45	
	r	P	r	P
Age (years)	-0.280	0.232	-0.049	0.748
INR	0.128	0.591	-0.142	0.352
APTT (s)	0.122	0.607	-0.061	0.688
TT (s)	-0.185	0.435	0.098	0.520
PC antigen (%)	-0.273	0.244	0.721	<0.001
Free PS antigen (%)	-0.319	0.170	0.744	<0.001
F VIII activity (%)	-0.382	0.097	0.078	0.609

compared to controls. Regarding protein C pathway components, Protein C global test was used as screening test and expressed in PCAT – normalized ratio (PCAT-NR) by specific formula. It showed significant decrease in the three classes of Child Pugh compared to controls. Protein C and free protein S results were also decreased in patient groups compared to control group and this effect is increased with cirrhosis progression. FVIII activity results showed no significant difference in patient group in comparison with control group. ROC analysis showed an area under the curve of 0.912 ($P < 0.001$), and the best cut-off value for protein C global was 0.8, associated with 71.1% sensitivity, 100% specificity, 100% positive predictive value, 60% negative predictive value and accuracy of 80% for prediction of liver cirrhosis.

This study showed that it is no longer suitable to depend on PT only to assess the bleeding risks in hepatic patients. Protein C global assay and its modified assay can be used as a global test to assess protein C pathway components. Many complications of HCV liver cirrhosis as variceal bleeding are not directly caused by haemostatic changes, but may be affected by other factors as portal hypertension. The balance between anticoagulation and pro-coagulation may be preserved in many cirrhotic patients. The treatment of every patient should be adjusted according to the laboratory results and clinical status.

This table shows significant positive correlations between PCAT-NR and (PC and free PS) in HCV liver cirrhosis cases. Otherwise, there was no significant correlation of PCAT-NR with other studied parameters in HCV liver cirrhosis and control groups.

Disclosure of Interest: None Declared.

BSH2020-EP-155

Patients with acute thrombotic thrombocytopenic purpura (TTP) should be managed in a critical care setting – A Single TTP Referral Centre Experience

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TTP is a thrombotic microangiopathy caused by a deficiency in the von-Willebrand cleaving protease ADAMTS13. It is a haematological emergency, with a mortality of 90% if left untreated. Microvascular thrombosis leads to critical organ ischaemia resulting in sudden deterioration and patients often require intensive care support for multi-organ failure. Despite a lack of controlled clinical trials, there is a

growing consensus that patients presenting with TTP should be admitted and cared for within a critical care environment.

This was a retrospective review of patients admitted with acute TTP to a regional specialist centre between June 2016 and December 2019. The centre's local policy states that patients with suspected acute TTP should be admitted directly or referred into a Critical Care bed. Data were collected from electronic patient records.

A total of 24 patients had an acute episode of TTP, including congenital and immune, over a 42-month period. All patients were confirmed to have TTP with an ADAMTS13 <10% at presentation. The mean age of patients was 51 years and 67% of patients were female. Twenty-one out of 24 patients were admitted directly to intensive care (ICU); three patients did not have an ICU admission due to a low ADAMTS13 activity with a normal blood count and an absence of clinical symptoms.

In 4/24 episodes, patients were transferred to the regional centre with an anaesthetic escort. In 6/24 episodes, patients required intubation and ventilation, with the indication in all cases being reduced conscious level/seizures. Of those patients requiring intubation, three were intubated prior to transfer to the specialist centre, one prior to plasma exchange at the regional centre and two within 24 h of admission. Zero patients required re-intubation following extubation. The mean time to intubation from admission to the regional centre was 7.8 h.

In 19/24 episodes, a central line was inserted for plasma exchange therapy; 7/19 femoral, 12/19 internal jugular lines. One line was placed in interventional radiology, the remainder by Critical Care physicians. The median platelet count at the time of line insertion was $15 \times 10^9/l$ (range 8–59 $\times 10^9/l$); with one report of slight oozing. No other bleeding complications were reported.

There were 15/24 episodes of raised troponin, 12/24 acute kidney injury and 14/24 neurological symptoms on admission. Twelve out of 24 episodes demonstrated ≥ 2 organ failure.

The median duration of the ICU stay was 6.5 days (range 2–25 days), compared to a median duration of total hospital stay of 26 days, (range 7–51 days). Post discharge from ICU, none of the patients requires re-admission for ICU support. One patient died due to cardiac arrest.

Patients with acute TTP are at high risk of sudden deterioration, especially within the first 24–48 h. Our data highlight that at least ¼ require intubation either prior to or shortly after acute transfer. Central venous line placement by skilled critical care physicians in ICU, despite profound thrombocytopenia, appeared to be safe. Despite the number requiring organ support at presentation, >95% patients were successfully stepped down from ICU and discharged from hospital.

These findings support recommendations that patients with acute TTP should be admitted directly to the ICU, and provide a platform for local and national policy frameworks supporting improved outcomes for this patient group.

Disclosure of Interest: None Declared.

BSH2020-EP-156

Why do we use warfarin today? A retrospective study from a large comprehensive anticoagulation centre

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Introduction of direct oral anticoagulants (DOACs) is a major advance in anticoagulation over last 10 years. It is now defined as the preferred anticoagulant for patients with non-valvular AF and VTE. The use of warfarin and consequent workload of the

Table 1. Clinical reason for starting warfarin in new patients starting oral anticoagulation between Jan 2017–May 2019.

Year ⁿ = pt starting warfarin	Severe renal impairment	DOAC intolerance	Drug interaction with DOAC	Obesity	Physician preference	Patient preference	Rec VTE on anticoagulation	Mech heart valve	Unusual site thrombosis / APLS
Jan–Dec, 2017 <i>n</i> = 541	41 (7.6%)	4 (0.7%)	6 (1.1%)	4 (0.7%)	371 (68.6%)	13 (2.4%)	18 (3.3%)	64 (11.8%)	20 (3.7%)
Jan–Dec, 2018 <i>n</i> = 442	55 (12.4%)	15 (3.4%)	16 (3.6%)	0	216 (48.9%)	10 (2.3%)	24 (5.4%)	84 (19%)	22 (5.0%)
Jan–May, 2019 <i>n</i> = 149	12 (8.0%)	6 (4.0%)	2 (1.3%)	2 (1.3%)	78 (52.3%)	2 (1.3%)	12 (8.0%)	28 (18.8%)	7 (4.7%)

anticoagulation monitoring services is declining. Warfarin, however, is still the only anticoagulant option in patients with mechanical heart valves, anti-phospholipid syndrome, severe renal/hepatic impairment and some other select situations. Based at the Royal Stoke University Hospital (RSUH), Staffordshire Thrombosis and Anticoagulation centre (STAC) manages over 8000 patients serving a population of over 700 000. Over 800 new patients are referred to this service annually to start anticoagulation.

Since Jan 2018 STAC offers DOACs as the preferred option for patients with non-valvular AF or VTE unless clinical complexities preclude their use. In some cases, warfarin has already been started prior to the referral to STAC. We have assumed this is physician preference.

We performed a retrospective review of the choice of anticoagulant for all new patients started on oral anticoagulation from Jan 2017 until May 2019. The objective was to determine the reason for continuing use of warfarin. It was carried out to help future design of anticoagulation services. We reviewed the anticoagulant choice of 1431, 1301 and 176 patients with AF, VTE and MHV, respectively, who started oral anticoagulation during this period. Over the 29 months period, 1617 patients started DOACs and 1132 were started on warfarin. The use of DOACs increased from 38% to 68% and the use of warfarin decreased from 57% to 26% over this period.

Table 1 summarises the clinical reasons for starting warfarin in new patients starting oral anticoagulation between Jan 2017 and May 2019.

In Conclusion, warfarin use has declined by over half during the study period, 2017–2019. The declining trend is seen across all indications where DOACs are licensed. There is a small selection of patients who will continue to require warfarin for specific clinical indications and will require some service for INR monitoring and dosing. In most cases, patients were started on warfarin due to physician preference. About 15% of patients required warfarin due to mechanical heart valve where DOACs are not licensed. STAC has introduced a service for periodic follow-up reviews for all patients on DOAC to ensure its safe and effective use.

Disclosure of Interest: None Declared.

BSH2020-EP-157

Evaluation of nine APTT reagents for the detection of lupus anticoagulants

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Background: International Society for Haematology [ISTH] and British Society of Haematology [BSH] guidelines for lupus anticoagulant [LA] assays both state that Dilute Russell's Viper Venom Time [DRVVT] and Activated Partial Thromboplastin Time [APTT] should be used to screen for LA. We evaluated nine APTT reagents from five manufacturers on a Sysmex CS-5100 for their sensitivity to

LA: Siemens Actin FS [AFS], Actin FSL [AFSL] and Pathromtin SL [PSL]; Werfen SCT Screen [SCT-Sc] and SCT Confirm [SCT-Cf]; Hyphen Biomed Cephen and Cephen LS [CLS]; Diagnostica Stago PTT-LA; and PTT-LA mixed with Lupus Anticoagulant Confirm [LAC] platelet extract reagent from Alpha Laboratories.

Methods: Reference intervals [RI] were calculated from 100 samples from DRVVT-negative asymptomatic patients. Sixty-two samples DRVVT-negative samples from symptomatic patients and 73 DRVVT-positive samples were also tested. No samples were from anticoagulated patients.

Results: Normalised ratios [NR] for RI samples for all reagents showed no statistical differences although SCT-Sc and SCT-Cf RI were very wide. There was no statistical difference between the NR for RI and DRVVT-negative samples for all reagents, and all reagents showed a statistically significant difference between NR for RI and DRVVT-positive samples except SCT-Cf. LAC, AFS and Cephen all showed the least increase in NR in the DRVVT-positive samples, confirming their LA insensitivity (along with SCT-Cf) compared to the other reagents. SCT-Sc gave a significant number of prolonged NR in the DRVVT-negative population.

When calculating a percentage correction the most sensitive combinations of reagents for detecting DRVVT positivity were as follows: PTT-LA/LAC (sensitivity 74%); PTT-LA/Cephen (70%); PTT-LA/AFS (52%); SCT-Sc/SCT-Cf (50%); CLS/Cephen or CLS/AFS (32%); AFSL/AFS or PSL/AFS (13%). When combining DRVVT and APTT positivity by any reagent combination, the most sensitive reagents are as follows: PTT-LA/LAC (89%); PTT-LA/AFS (63%); PTT-LA/Cephen (55%); CLS/AFS (27%); SCT-Sc/SCT-Cf (17%); AFSL/AFS or PSL/AFS (15%); CLS/Cephen (13%).

Conclusions: The most LA-sensitive reagent tested was PTT-LA in combination with confirmation with the same reagent mixed LAC, AFS or Cephen. SCT-Sc and SCT-Cf showed unexpectedly poor results; this may have been to low numbers of samples tested due to sampling errors, and needs further investigation. Neither AFSL nor PSL were sensitive enough to LA to be considered for LA detection. CLS was moderately sensitive when used with AFS as confirmatory reagent, but not when used with Cephen.

Disclosure of Interest: None Declared.

BSH2020-EP-158

Clinical outcomes from lupus anticoagulant testing

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Introduction: Antiphospholipid (aPL) antibodies are a heterogenous group of antibodies directed against phospholipid-binding proteins. They are commonly tested when screening for anti-phospholipid

syndrome (APLS), investigating a prolonged APTT or as part of an autoimmune screen.

APLS is a systemic acquired autoimmune condition characterised by the presence of persistent aPL antibodies and the clinical manifestation of thrombosis and/or pregnancy complications. The 2006 revised Sapporo criteria for the diagnosis of APLS require at least one clinical and one laboratory criteria to be met.¹ Laboratory criteria are positive lupus anticoagulant (LA), anti-cardiolipin (aCL) and beta-2-glycoprotein-1 (β 2GP1) present on two or more occasions at least 12 weeks apart.

As no single test has shown 100% sensitivity for detecting a LA, current guidelines advise that at least two assays be used to ensure detection.² With multiple assays available, LA testing varies between centres and some centres rely on a single assay. Our laboratory uses the combination of dilute Russell Viper Venom time (DRVVT) and silica clotting time (SCT) with the patient regarded as having a positive LA if either test is prolonged and confirmed to be phospholipid dependant. However, it remains unclear whether positivity in one test alone is indicative of a clinically significant aPL antibody.

Methods: We performed a retrospective review of all patients with a new positive aPL antibody over a 12-month period in our laboratory. Requests from external sources were excluded. We looked at (i) reasons for testing (ii) compliance with re-testing guidance and (iii) final diagnosis. We specifically looked at the initial DRVVT and SCT to assess whether dual testing adds diagnostic value over a single test.

Results: A total of 1131 patients had LA tests performed in the 12-month period. One hundred and thirty-seven (12%) were positive. Of these 77 patients were being tested for the first time and further analysis focused on this group. Indications for testing were as follows: autoimmune screen $n = 35$ (45%), investigation of cause of thrombosis $n = 27$ (35%) and prolonged APTT $n = 13$ (17%). Two patients had no relevant clinical details provided.

Of the 77 patients that were testing positive for the first time, 18 (23%) were LA positive both on DRVVT and SCT, 38 (49%) just on SCT and 21 (27%) just on DRVVT. Forty-four (57%) of first time positive patients were re-tested within the next 18 months. Time periods for re-testing varied significantly from 4 weeks to over a year.

Of the 27 patients that were being investigated for thrombosis aetiology, five (19%) went on to be diagnosed with APLS. Two were positive on SCT alone on initial testing and three by both DRVVT and SCT.

Conclusions: Just over half of the LA-positive cases were re-tested after at least 12 weeks as per guidelines. Reasons for not re-testing include patients on long-term anticoagulation which would interfere with LA tests, normalisation of APTT and autoimmune screens where other tests have not supported a diagnosis of autoimmune disease. Our data suggest that there are cases where positivity on just the SCT, with negative DRVVT can lead to a diagnosis of APLS supporting the recommendation for using both assays. For centres where only one initial screening assay is used, there is the risk that cases of APLS could be missed.

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Disclosure of Interest: None Declared.

BSH2020-EP-159

ISTH SSC plasma standard lot #5 – results in a UK National External Quality Assessment Scheme for Blood Coagulation exercise

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The ISTH SSC plasma standard is a secondary coagulation standard developed to aid manufacturers in the labelling of coagulation calibrators. Values are assigned for 24 haemostasis parameters through assessment against relevant WHO International Standards in multi-centre international collaborative studies. However, it is useful to compare results obtained on this plasma with a wide range of reagents and methods in routine laboratory use, to confirm that commercial calibrants are accurately labelled.

UK NEQAS (Blood Coagulation) has regularly distributed the SSC plasma standard as a blinded sample for assay, and reported findings in a proficiency testing exercise compared to the assigned values. We report here the first comparison of proficiency testing results to the assigned values for the lot #5 of the SSC plasma standard.

The plasma was sent to 870 centres, with a request to perform fibrinogen assays, and 385 centres for FVIII, FIX and VWF assays. Results, along with method details for each assay were returned. Median results were determined, and compared to the SSC plasma standard assigned values.

Table 1 shows the median results for all methods for fibrinogen, FVIII, FIX and VWF:Ag assays. Method-specific VWF activity assay results are also shown. Reagent-related differences were noted, for example for FVIII:c assays, the median results for users of one widely used reference plasma was 81.3 u/dl ($n = 187$), compared to 91 u/dl for users of another reference plasma ($n = 52$, $P < 0.001$).

Conclusion – it is useful to compare values assigned to commercial calibrants against the SSC plasma standard in an exercise with results obtained by a large variety of methods in a proficiency testing exercise. We have demonstrated good agreement with the assigned values for fibrinogen, FVIII, FIX and VWF assays, including the method-specific target values for VWF activity, introduced for the first time by SSC with this lot of plasma. However, some method-related discrepancies require further investigation.

Disclosure of Interest: None Declared.

Table.

Test	Assigned value	NEQAS BC no.of users	NEQAS BC median value
Clauss fibrinogen	3.19 g/l	836	3.10 g/l
Factor VIII:C	82.0 IU/dl	361	81.0 IU/dl
Factor IX	109.0 IU/dl	319	110 IU/dl
VWF antigen	114 IU/dl	235	117.0 IU/dl
VWF activity (method)			
Aggregometry (VWF:RCO)	82.0 IU/dl	37	82.0 IU/dl
Chemiluminescence (VWF:GPIbR)	95.0 IU/dl	18	97.3 IU/dl
IL VWF:RCO (VWF: GPIbR)	95.0 IU/dl	71	101.0 IU/dl
Siemens (VWF: GPIbM)	80.0 IU/dl	42	83.6 IU/dl
VWF:Collagen binding	102.0 IU/dl	42	102.6 IU/dl

BSH2020-EP-160

The impact of education and audit to reduce the number of prophylactic low-molecular-weight heparin (LMWH) dose omissionsS. Rivera*, L. Roberts, E. Gee, J. Barker, R. Yorke, R. Arya
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Hospital-associated venous thromboembolism (HA-VTE) remains a key patient safety issue associated with significant morbidity and mortality. Root cause analysis of HA-VTE is an important tool for identifying areas for improvement. Investigation of a fatal PE in June 2017 revealed inadequate thromboprophylaxis due to 20 missed doses of LMWH prophylaxis associated with patient refusal related to fluctuating mental capacity. A survey of all prophylactic LMWH prescriptions between July 2016 and June 2017 revealed a mean of 1778 dose omissions per month. Thus a series of interventions were planned to reduce the number of unnecessary omitted doses with an audit comparing the number of prophylactic LMWH dose omissions pre and postintervention. The interventions implemented were a combination of teaching sessions to medical and nursing staff and information technology strategies to support staff in reducing LMWH prophylaxis omissions.

Total number of prophylactic LMWH doses omitted decreased from 13% in Oct to Dec 2016 to 8% in Jan–Mar 2018. The average proportion of doses omitted among the wards decreased from 15% to 10%. The proportion of dose omissions due to patient refusal reduced from 42% to 36%. The number of potentially preventable HA-VTE associated with inappropriate LMWH dose omissions reduced from 17% (18 out of 106 episodes from Oct 16 to Mar 17) to 3% (4 out of 118 episodes from Jan to June 18). The timeframe to collect this second set of data was extended from 3 to 6 months to capture the HA-VTE occurring within 90 days post admission.

The monthly proportion of prophylactic LMWH dose omissions has been subsequently maintained at 7–8% per month. The most common reason for dose omission was patient refusal, followed by omission due to clinical reasons or instruction by prescriber.

We recommend further audit to identify the appropriateness of omitted prophylactic LMWH doses documented on the drug chart as 'omission on instruction of prescriber' or 'due to clinical reasons' to explore themes and identify further knowledge gaps (if any) so that specific interventions can be implemented to potentially further reduce prophylactic LMWH dose omissions.

Disclosure of Interest: None Declared.

BSH2020-EP-161

Fixed dose prothrombin complex concentrate for direct oral anticoagulant and warfarin reversal: a rapid and effective solutionJ. Davies^{1,*}, V. Sansom¹, J. Piper¹, L. Ngu², P. Kerr²¹Blood Sciences, ²Haematology, Royal Devon and Exeter NHS Foundation Trust, Exeter, UK

In life-threatening bleeding or emergency surgery rapid reversal of anticoagulation is required. Pro-thrombin complex concentrate (PCC) is licensed for reversal of warfarin, with dosing based on body weight (kg) and INR value (standard dosing). Evidence is available supporting the use of fixed lower dose of PCC for warfarin reversal in these situations. PCC is also used for reversal of direct oral anticoagulants (DOAC) at a maximum dose of 50 iu/kg, but with sparse evidence. A protocol for fixed dose PCC (FD-PCC) at 1000 iu was implemented in our hospital for all indications and anticoagulants, except dabigatran where a specific antagonist is stocked. The Haemostatics BloodTrack system and laboratory computer system (Integrated Pathology System, DXC) were configured to support

'Emergency PCC' enabling removal of 1000 iu PCC from the blood fridge by clinicians without laboratory staff involvement. The BloodTrack system, used to administer blood products ensures electronic traceability for emergency PCC. Further doses of 500 iu PCC can be requested from laboratory if indicated. Data were collected from BloodTrack, pathology computer system and clinical notes for PCC use pre and postfixed dose regime for comparison. One hundred and thirty-six patients received FD-PCC for warfarin reversal; 46 bleeding, 52 intracranial haemorrhage (ICH) and 38 prior to surgical procedures. For standard dose PCC (SD-PCC) 21 patient records were available, six bleeding, five ICH and 10 prior to procedure. No significant difference ($P = 0.5656$) was found in pre-PCC INR values (FD-PCC mean 3.7, SD-PCC mean 3.4). No significant difference ($P = 0.0591$) was noted in INR post-PCC administration (FD-PCC mean 1.4 and SD-PCC mean 1.3). Eight patients (5.9%) given FD-PCC required additional PCC. Time from request to PCC administration (mins) was significantly reduced ($P = 0.0001$) in FD-PCC (mean 48) compared to SD-PCC (mean 128). Emergency FD-PCC significantly reduced time to administration ($P = 0.0001$) compared to laboratory issue. Dose of PCC per patient fell from average 2714 iu to 1040 iu. For bleeding patients, there was no significant difference in red cell ($P = 0.7650$), or fresh frozen plasma ($P = 1.000$) requirements between the two groups. No significant difference was noted in survival at 24 h ($P = 0.2649$) or 30 day ($P = 0.5138$) between the groups. Forty-seven patients were given PCC for DOAC reversal; 37 FD-PCC and 10 SD-PCC. 13 required FD-PCC for bleeding, 13 for ICH and 11 prior to procedure. Three patients were given SD-PCC for bleeding, six for ICH and one prior to procedure. FD-PCC led to a significant reduction ($P = 0.0001$) in time (mins) from request to administration (FD-PCC mean 32, SD-PCC mean 82). Time to administration (mins) was significantly faster ($P = 0.0007$) using emergency PCC (mean 23) compared to laboratory issue FD-PCC (mean 42). For bleeding patients, there was no significant difference in red cell ($P = 0.8926$) or fresh frozen plasma ($P = 0.9403$) requirement between the two groups. No significant difference was noted in 24 h survival ($P = 1.000$) or 30-day survival ($P = 0.4884$). This small review suggests that FD-PCC (1000iu) is as effective as SD-PCC for reversal of warfarin and DOAC. FD-PCC is simpler to use, reduces time to administration and leads to cost savings. Use of emergency rather than laboratory issued FD-PCC offers further advantage in time to administration. No inappropriate use of Emergency PCC has been found.

Disclosure of Interest: None Declared.

BSH2020-EP-162

Long-term outcomes in anti-phospholipid syndrome patients treated with vitamin K inhibitors. A single-centre retrospective study of 102 patients

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Background: Antiphospholipid syndrome (APLS) is an uncommon autoimmune disease in which venous and arterial thrombosis frequently occur with devastating and sometimes fatal effects. Vitamin K antagonists (VKAs) remain the mainstay for secondary prevention of thrombosis in APLS patients. Higher intensity warfarin therapy (target INR: 3–4) though used in high-risk APLS patients (pts), does not seem to further reduce the risk of recurrent thrombosis in randomized, controlled trials.

Aim: In this audit, we sought to distinguish APLS pts who received VKA anticoagulation treatment for more than 4 months and record their outcomes.

Patients: We identified 102 (female:72/male:30) pts with clinically confirmed APLS in the anticoagulant service database of Heartlands, Good Hope and Solihull Hospitals. Their median age at VKA initiation was 47.5 (18–89) years. The index events to offer anticoagulation were arterial (50/102), venous (31/102) and mixed arterial/venous (21/102). Nineteen pts had laboratory evidence for triple positivity, whereas 32 had a history of recurrent thrombosis; 20 pts had a co-existing autoimmune disease with a diagnosis of systemic lupus in 10 of them. The majority (64%) were put on VKA with a target INR of 2.5. Dual anticoagulation with VKA+low dose aspirin/clopidogrel was used in 13/102 pts. The median follow-up time on anticoagulation was 46 (4–259) months, whereas the median time where INR was within therapeutic range (TTR) was 57%.

Results: Recurrent thrombosis occurred in 13/102 (12.7%) pts with a median time of 38.5 (34.25–68.5) months following VKA initiation. The majority of breakthrough events were of arterial origin (stroke: 6, myocardial infarct: 2). At the time, 12/13 patients were on intensive anticoagulation with an INR target ≥ 3.0 ; 8/13 pts had a TTR $<65\%$. Postrecurrence, increasing the INR target was the most common therapeutic decision (5/13 pts). Still 3/13 pts suffered a further event. Catastrophic APLS was recorded in two cases. Bleeding occurred in 15/102 (14.8%) pts with major bleeds recorded in 11 pts (10.7%), Median time to major bleeding was 52 (27.5–73.75) months, whereas 6/11 had an INR target ≥ 3.0 . Overall, the 5-year probability for recurrent thrombosis was recorded at 7.8%, whereas major bleeding risk was 4.9% respectively.

Conclusions: Our study records the long-term outcomes of VKA use for secondary prevention of thromboembolism in a large real-life APLS patient cohort. Our results confirm that in the APLS setting, quality of anticoagulation (TTR) is probably more important than quantity in terms of setting a higher INR target. APLS pts represent a clinically heterogeneous group, with the risk of recurrent thrombosis and the intensity of anticoagulation being influenced by their clinical phenotype and risk profile. In this case, unfortunately we are still some way from identifying the optimal anticoagulation strategy.

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BSH2020-EP-163

A case of acquired von Willebrand syndrome becoming IVIg responsive following bortezomib treatment despite minimal paraprotein responseE. Blacker^{1,*}, V. MacDonald¹, D. P. Hart¹, L. Bowles¹, J. D. Cavenagh², K. Forsyth¹, A. Morris¹, C. Foley¹, C. Francklin¹, J. Pasi¹¹Department of Haematology, Royal London Hospital,²Haemato-Oncology, St Bartholomew's Hospital, London, UK

Acquired von Willebrand syndrome (aVWS) is a rare bleeding disorder caused by functional inhibition or increased clearance of circulating von Willebrand factor (vWF). Numerous conditions have been causally associated with aVWS through various mechanisms from antibody-mediated autoimmunity to shear stress across stenosed heart valves. Plasma cell dyscrasias including Monoclonal Gammopathy of Uncertain Significance (MGUS) are a frequent association of aVWS with a direct interaction between the M-protein and vWF being the presumed disease mechanism.

We present the case of a 44-year-old man who presented with a 6-month history of oral and rectal bleeding. Investigation prompted by a prolonged APTT revealed a vW antigen of 3.7iu/dL and Gp1bM activity of <2.2 iu/dL. A diagnosis of aVWS rather than congenital von Willebrand Disease (vWD) was made based on a lack of personal or family history of a bleeding disorder; the absence of pathogenic variants on targeted sequencing of the vWF gene and by the finding of an IgG monoclonal band of 4 g/L. Biochemistry and MRI whole spine and pelvis showed no evidence of symptomatic Myeloma. Light chain ratio was normal. Bone marrow examination was not undertaken due to bleeding concerns.

There was a poor increment with 40 units/kg vWF concentrate to a maximum Gp1bM activity of 17 iu/dL. There was also a poor response to IVIg (Octagam) 1 g/kg – maximum Gp1bM activity of 19.8 iu/dL at 30 min postinfusion with rapidly falling levels at 12 and 36 h. A 2nd dose of IVIg was given with the same poor result.

In light of the poor response to IVIg and vWF concentrate he was treated with bortezomib (Velcade) and dexamethasone, aiming to reduce the paraprotein and induce remission of the aVWS. After five cycles the paraprotein fell to 2 g/dL, however, treatment was stopped due to poor tolerance.

A re-trial with IVIg 1 g/kg was given ahead of an elective dental procedure. Following the infusion his vW Gp1bM activity rose from undetectable to 50 iu/dL. He was subsequently given vWF concentrate 30 units/kg with a postdose level of 119 iu/dL. 21 h later the level had risen to 127 iu/dL consistent with increased survival of endogenous vWF. This response is sufficient to facilitate invasive procedures and to manage acute bleeding episodes.

Given the fairly modest response in terms of paraprotein relative to the clinical response we speculate that the bortezomib-dexamethasone may have been most effective against a separate, non-clonal lymphoid population causing paraneoplastic autoimmune aVWS secondary to the MGUS. Alternatively, the M-band may have been a 'bystander' paraprotein as is seen in many autoimmune conditions as a result of immune dysregulation. This case also suggest bortezomib may be effective even in aVWS cases without a measurable paraprotein.

Disclosure of Interest: None Declared.

BSH2020-EP-164

Patients with below knee DVT are at risk of both under and over-treatment: results from a retrospective review of cases identified at the University Hospitals BristolS. Westbury^{1,2,*}, A. Webb³, A. Clark¹, T. Robinson³¹Department of Haematology, University Hospitals Bristol NHS Foundation Trust, ²School of Cellular and Molecular Medicine, University of Bristol, ³Department of Vascular Science, University Hospitals Bristol NHS Foundation Trust, Bristol, UK

NICE guidance for the diagnosis of lower limb deep vein thrombosis (DVT) specifically recommends proximal leg ultrasound scans for patients meeting the criteria for diagnostic imaging, including repeat proximal scans at 6–8 days for the highest risk patients with no proximal thrombosis identified (NICE guideline CG144). Below knee (BK) DVTs will therefore not be routinely diagnosed. There are no specific NICE recommendations for the management of BK DVT.

The Department of Vascular Science at the University Hospitals Bristol NHS Foundation Trust (UHBristol) performs a whole leg scan as standard, resulting in diagnosis of approximately three BK DVTs per week. Patients are referred via the nurse-led thrombosis clinic (predominantly from general practice or the emergency department) or through their hospital inpatient or outpatient team. Prior to January 2018, the majority of patients with BK DVT received therapeutic anticoagulation. Since January 2018, the thrombosis clinic has not routinely anticoagulated patients with BK DVT, with a repeat scan at 1 week to assess for proximal extension. In the absence of specific NICE guidance, we anticipated that the management of BK DVTs outside the thrombosis clinic would be inconsistent, with the risk of over or under treatment of patients.

We aimed to describe the management of BK DVTs at UHBristol before and after January 2018. We collected data retrospectively from the digital and paper notes of adult patients with BK DVTs identified between 1st September and 30th November in 2017 and 2018 and performed summary statistical analysis for each cohort of patients. Cases were categorised according to the healthcare professional requesting the scan and acting on the result. Patients were excluded from analysis if they were already on anticoagulation for any indication or if the thrombosis was chronic.

A total of 35 cases in 2017 and 24 cases in 2018 were identified for analysis. Patient demographics were similar across the years (overall, 56% male, median age 58 years). 69% (2017) and 79% (2018) of patients were managed through the thrombosis clinic. In 2017, 91% of non-thrombosis clinic patients and 73% of thrombosis clinic patients were anticoagulated. By 2018, this had reduced to 60% and 38% respectively. Of the patients not anticoagulated, all thrombosis clinic patients were given an appointment for a repeat scan within 7 days. Of the remaining three patients not anticoagulated, only one had a repeat scan organised.

These results confirm that local management of BK DVT is variable, particularly outside the thrombosis clinic, and that patients are at risk of over and under treatment. The rationale behind the management decisions was difficult to identify from retrospective analysis of the medical notes in all but two non-thrombosis clinic cases. A significant minority of thrombosis clinic patients continued to be offered anticoagulation for a BK DVT after January 2018, but the reasons were clearly documented in all cases.

This work raises questions as to the underlying cause of the variable management of non-thrombosis clinic patients while indicating the success of a dedicated thrombosis clinic. Further prospective work is required to understand the reasons for the observations reported here and to investigate whether a similar pattern is observed

in other Trusts. This will ensure appropriately directed training to increase the consistency of BK DVT management and potentially improve patient safety.

Disclosure of Interest: S. Westbury Conflict with: Delivered an educational symposium for Abbvie, A. Webb: None Declared., A. Clark: None Declared., T. Robinson: None Declared.

BSH2020-EP-165

A retrospective review of suspected thrombotic thrombocytopenic purpura cases from a tertiary referral centreA. Knott, C. Reilly-Stitt, S. Suntharalingham*, A. Clark
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Thrombotic thrombocytopenic purpura (TTP) is a rare and life-threatening diagnosis with a high rate of mortality without prompt treatment. Definitive diagnosis relies on demonstrating severe deficiency of the metalloprotease ADAMTS13. This assay is not available in many centres, therefore history, examination and standard laboratory investigations guide the decision to commence immediate treatment for suspected TTP. Diagnosis is challenging; the 'classic pentad' is present in only 35% of cases, and there can be overlap with the clinical presentation of a number of other conditions.

The University Hospitals Bristol NHS Foundation Trust (UHB) has provision for 24-h apheresis and over the last 2 years we have seen an increase in referrals for patients with suspected TTP from a wide geographical area across the South West of England. In order to estimate the resources required for the liaison and laboratory elements of a service in the South West, we conducted a retrospective review of patients who had ADAMTS13 activity assays at our trust over 2 years. The aim was to gain a better understanding of the actual diagnoses and outcomes of patients referred with suspected TTP.

From November 2017 to November 2019, 378 ADAMTS13 assays were performed on 99 patients (89% adults, 63% females) from 11 acute NHS trusts; the median number of tests per patient was 1, and the mean was 3.82. The number of assays performed increased by 269% from the first to the fourth quarter.

The final diagnosis was established in 86/99 (87%) patients. Thirty-eight out of 86 (44%) had TTP (36 immune and 2 congenital); of these 12/38 (32%) were de novo presentations, 7/38 (18%) had impending or confirmed relapse (ADAMTS13 <10%) and 18/38 (47%) were in remission. In patients not previously known to the unit the following diagnoses were established: 12/60 (20%) acute immune TTP, 18/60 (30%) other TMA (STEC HUS, aHUS, PET/HELLP, transplant-associated, drug-associated, malignant hypertension or pancreatitis), 8/60 (13%) primary haematological disorder (ITP, megaloblastic anaemia, congenital thrombocytopenia, MDS or inherited membranopathy), 5/60 (8%) autoimmune disorder (vasculitis or APLS), 3/60 (5%) advanced malignancy, 3/60 (5%) MAHA (malignancy or infection) and 3/60 and (5%) postoperative sepsis.

Thirty-three out of 60 (55%) of tests were carried for patients who presented acutely to other trusts; 14/33 (42%) were transferred to UHB to commence plasma exchange, 2/14 (14%) were subsequently found to have normal ADAMTS13 activity. Twenty-four out of 52 (46%) of patients required admission to an intensive care unit during their acute illness. Six out of 88 (7%) patients died within 6 months of the ADAMTS13 request.

These results reflect the complexities around the diagnosis of acute TTP and the plethora of differential diagnoses that need to be considered by specialist treating centres when consulted about these challenging patients. As the model of care for TTP moves towards

centralisation, clear pathways to initiate dialogue, access diagnostic investigations and link to other specialty disciplines, such as nephrology and rheumatology, will be essential. The team at UHB will respond to this study by educating all clinicians involved in the diagnosis and management of TTP on the results, strengthening local networks by sharing results and reaffirming referral pathways and supporting resources to provide a regional liaison service.

Disclosure of Interest: None Declared.

BSH2020-EP-166

Approach to manage thrombotic events while on anticoagulation

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Objective: The Audit Objectives are;

- 1 Assess break through events on DOACS and
- 2 Management of the events

Scope & Methods: We audited retrospectively on our cohort of patients who were started on direct oral anticoagulants (DOACS) for the various indications, either for stroke prevention due to atrial fibrillation (AF) or for treatment for venous thromboembolism (VTE) and those of them who have failed anticoagulation treatment in the last 5 years.

Data were gathered from 7200 patients who were started on DOACS and only 34 patients were reported to have failed anticoagulation treatment (accounting for only 0.5%) over 5-year period.

We looked into the duration when the treatment failed, after starting on DOACS and the findings were: **Triggering factors:** Out of 33 patients, five had very subtle triggering factors such as the long haul flights. One had bladder cancer hence changed to LMWH, and one had retroviral disease at the time of presenting with further VTE. Three were not compliant with medications.

Failure Pattern: We've tried to establish a failure pattern if any and noted the below

- Patient on anticoagulation for AF to present with VTE were 5
- Patients with AF who presented with stroke were 3 and
- Patients with VTE presenting with further VTE were 21, and
- Only 1 developed PE having been on anticoagulation previously for portal vein thrombosis.

Managing on Progression: Data also present how these patients were managed.

The data show that 20 patients (56%) were changed to Warfarin, but two patients were reverted back to same DOACS 6 months later due to patient preference, two patients continued on LMWH, and 11 were changed to different anti-XA Agent.

Thrombophilia Screening: Seven out of 33 patients had full thrombophilia screen. Two were detected to be factor V Leiden heterozygous, and one had protein C deficiency. The remaining patients had lupus anticoagulant screening checked only and were negative.

Every patient dose was weight appropriate except one patient who weighed 152 kg at presentation and the dose was adjusted according to the creatinine clearance (according to Cockcroft Gault calculation).

Measuring Assay: Six out of 33 had DOACS assay at Breaking Event. Five had normal levels and one patient had levels <11 ng/L.

Conclusion/Discussion: There is no universal agreement or It is an open question as to how to manage the patients who fail their anticoagulation treatment while on DOACS.

Do you change to different DOACS or change to warfarin?

Certainly, in the case of patients with anti-phospholipid syndrome, the suggestion would be to change to warfarin, and potentially with a higher INR target, as evidence from recent research is showing that DOACS are not offering adequate anticoagulation. There is no research in other groups of patients. We are aware from the trials conducted, that none of the anticoagulation treatments is offering a complete prevention from stroke in the cases of AF patients, or a complete prevention from VTE in patients who have presented with previous thrombosis.

No evidence is yet available to support a management plan, but from our 33 patients, 14 of them (42%) who continued on DOACS remain well now after being on DOACS for an average of 15 months.

Table.

POSTEVENT MANAGEMENT	
WARFARIN	18
LMWH	2
DOACS	12

Disclosure of Interest: None Declared.

BSH2020-EP-167

Heparin-induced thrombocytopenia (HIT): an audit of the diagnostic pathway at Barts Health NHS Trust

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Heparin-induced thrombocytopenia (HIT) is an uncommon, but serious complication of heparin therapy, affecting approximately 1 in 5000 hospitalised patients, which can cause significant morbidity and mortality. It is characterised by the development of antibodies against the platelet factor 4/heparin complex, resulting in thrombocytopenia and a prothrombotic state.

The diagnosis of HIT is difficult and requires integration of clinical and laboratory data. Diagnosis of HIT at the Royal London Hospital is based on British Society for Haematology guidelines starting with a pretest probability assessment using the 4T score.

In those with a moderate or high 4T score, or where this is difficult to estimate because of missing data, an initial screening test using the Diamed rapid particle gel immunoassay is performed, followed by an ELISA test if this is positive. A send-away heparin-induced platelet activation assay (HIPA) is performed in selected cases.

This audit examined all suspected cases of HIT at the Barts Health NHS trust in 2015–2018. Demographic information was collected, as well as reasons for admission, the 4T score from the request (if available) and subsequent laboratory testing. Documentation in the patients' notes pertaining to each diagnosis (if available) was also examined.

A total of 164 samples from patients with suspected HIT were received. The screening gel test was run on 85/164 samples – theoretically, these cases had a higher pretest probability or were deemed clinically necessary by the haematologist, but unfortunately data were not always available to support the rationale for each test. Out of these 85 gel tests, 32/85 (37.6%) were positive. None of the cases with negative results were sent for further testing. ELISA testing was performed on all screening-positive cases, plus three additional cases for which

Diamed screening cards were temporarily unavailable; out of the 35 ELISA tests run, 16 (45.7%) were positive. Three HIPA tests were sent away to the reference laboratory, of which 2 (66.6%) were positive. Taking all information into account, including the clinical information and 4T score, 12 positive diagnoses of HIT were made.

Out of the 164 requests, only 57 (34.8%) included the 4T score on the request form, despite this being a compulsory field. All but one of the positive diagnoses included a 4T score in the request, enabling this to be correlated with the other results: three out of 12 positive diagnoses had a score of 3 (low risk), 3/12 scored 5 (intermediate) and 5/12 scored 6 (high). Due to incomplete documentation, it is not clear why the three cases scoring 3 were tested, although it may be assumed that this was on clinical grounds.

Additional data were available for four of the positive diagnoses, in the form of the optical density (OD) of the ELISA (representing the degree of positivity, and unfortunately not available for all cases). Two of the cases with a high OD were correlated with a high 4T score. However, one case with a high 4T score had a lower OD; and, conversely, one case with a low 4T score returned a high OD. Documentation in these cases was not detailed enough to ascertain the rationale behind the diagnosis.

Overall, the results helped to illustrate the difficulties in diagnosing HIT. The low proportion of requests including the 4T score (an invaluable clinical detail given that diagnosis relies on clinical as well as laboratory data) is a concern and indicates a need for further education of clinicians in the diagnostic problems of HIT.

Disclosure of Interest: None Declared.

BSH2020-EP-168

Direct Oral Anticoagulants – the real story of their side effect profile

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Dabigatran was the first direct oral anticoagulant (DOAC) licensed for the use in 2008, followed by apixaban, rivaroxaban and edoxaban. The trials carried out highlighted the side effect profiles of the drugs, but with increasing use in clinical practice new data is emerging with regards to their safety profile.

In the Whipps Cross Hospital, we have been collecting data on our DOAC patients since June 2016. We felt it was important to analyse this data to ensure that we are counselling patients appropriately with regards to their side effects as well as improving our understanding of these drugs. We have a large anticoagulation and thrombosis service and a significant percentage of our patients are elderly, with multiple co-morbidities.

We used our database and warfarin dosing tool, DAWN 7.9, to access data collected on all patients on DOACs between June 2016 and May 2019. Using the events documentation tool and noting patients who have had their DOAC switched, we were able to collect data on all patients who had reported side effects on a DOAC.

The total number of patients was 3786 of which 64% were on a DOAC for AF and 36% for VTE. Out of these, 49% were on apixaban, 35% on rivaroxaban, 15% edoxaban and 1% on dabigatran.

There were 160 patient events reported. Out of these, rivaroxaban had 64 events, 40% of the total events, but 5% of the rivaroxaban patients. For patients on apixaban there were 56 events, 35% of the total events, but 3% of the apixaban patients. For patients on edoxaban there were 33 events, 21% of the total events, but 6% of the edoxaban patients and finally for dabigatran there were seven events, 4% of the total events, but 18% of dabigatran patients had an event.

The majority of the events were related to bleeding, and we found 35 events due to bleeding complications and 123 due to all other side effects. For two of the events there was no documentation.

RocketAF reported bleeding events of 20.7% (major and non-major clinically relevant bleeding) and other side effects of 81%. Aristotle reported bleeding rates of 4.3% for major haemorrhage and any bleeding of around 26% and other side effects of 81%. Engage-AF reported bleeding at around 30% and other side effects of 84.3%. RE-LY reported bleeding at 33% and other side effects at 93%.

We report an overall 4.2% event rate in our patient group most of which happened with rivaroxaban. The most common side effects were bleeding, skin, musculoskeletal, dizziness, fatigue, headache and gastrointestinal. We noted side effects that have not been reported in the BNF including hair loss and weight gain.

In conclusion, we have reported much lower events than within the trials. This could be related to over reporting within the study population of which many side effects may not have been directly related to the DOAC. The studies were also done in a significantly larger population. We have, however, found that in general, these agents are well tolerated and easier to take by most patients when compared with VKA antagonists.

Moving forward we need to ensure we are counselling patients appropriately and we have updated our counselling checklist to reflect this. We have reported side effects not previously reported in the BNF and will continue to collect data as more patients are being started on or switched to DOACs.

Disclosure of Interest: None Declared.

BSH2020-EP-169

Monitoring direct oral anticoagulant plasma concentrations – experience from the Princess Royal University Hospital

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The prescribing of direct oral anticoagulants, DOACs has increased exponentially in the past decade. Fixed dosing, combined with the more predictable pharmacokinetic profiles compared to traditional oral anticoagulants, has led to the general consensus that the routine therapeutic monitoring of DOAC therapy is unnecessary. However, there are certain situations, for example the very obese, where assaying the plasma concentration is considered desirable. This study aimed to explore the reasons behind DOAC assay requests at the Princess Royal University Hospital (PRUH) and establish whether there was any relationship between DOAC plasma concentration and patient outcomes, in those where an assay was requested.

Between 1 June 2017 and 30 June 2019, all DOAC assays requested by clinicians at the PRUH were retrospectively collated from DAWN® and Electronic Prescribing Records and reviewed. These assays were typically requested in patients where there was concern that exposure to the DOAC could significantly be altered, potentially impacting on outcomes. Information, including patient demographics, comorbidities and indication for assay, was recorded onto Excel and analysed. Patient outcomes (bleeding as defined by ISTH or thromboembolism) in the 90 days following the assay draw was also recorded.

Over the 2-year study period a total of 588 assays were requested, 500 of which had sufficient information to be included for review. These 500 assays were from 446 patients (median age 77 [IQR 64–86 years], 50% male). The assay requests were comprised of 218

(43.6%) apixaban, 186 (37.2%) rivaroxaban, 80 (16.0%) edoxaban and 16 (3.2%) dabigatran. These assays were most commonly requested for patients with renal impairment and patients at the extremes of body weight. In total 67 bleeding events were reported, two of which could be considered major bleeds; one fatal intracerebral haemorrhage on edoxaban and one episode of melena on rivaroxaban. Both major bleeds had a previously measured DOAC plasma concentration result that was considered *in range* (as defined by Gosselin et al 2018). There were also two thromboembolic events: an ischaemic stroke on apixaban and a DVT on rivaroxaban. The assay result for the patient who suffered a stroke showed an apixaban concentration that was *in range* and the rivaroxaban assay was below range. Overall, there appeared to be no link between the DOAC plasma concentration and the patient outcome in the frequency of bleeding observed in patients that had DOAC concentrations within the expected range and those who had a concentration above or below the range.

Our findings suggest there is no need for routine plasma concentration monitoring of direct oral anticoagulants. However, in situations of uncertainty, for example morbidly obese patients, they do provide reassurance to clinicians.

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BSH2020-EP-170

Overview of clinical characteristics of patients with TTP (thrombotic thrombocytopenic purpura) in a low socioeconomic population in New York: a 6-year center experience

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Background: Thrombotic thrombocytopenic purpura (TTP) is a thrombotic microangiopathic disease that occurs due to a deficiency of the von Willebrand factor (VWF) cleaving protease, ADAMTS13 (a disintegrin and metalloprotease with thrombospondin type 1 repeats, member 13).^{1,2} It may present either as a hereditary form or a more common acquired (autoimmune) type. There is consumption of platelets from spontaneous aggregation, leading to microangiopathic haemolytic anaemia (MAHA) resulting in multiorgan failure.

Methods: Electronic medical records using the EPIC system were used to conduct a search for all the patients admitted to a tertiary hospital serving a low socioeconomic population, between 2014 and 2019 with the diagnosis of TTP. A total of 29 patients was identified. Out of 29 patients, only 11 had confirmed TTP. One of the 11 patients had hereditary TTP and was excluded from the study. We herein present the baseline patient characteristics, clinical presentation and response to treatment in the remaining 10 TTP patients.

Results: Our patient population had an average age of 47 years with a slight female predominance (64%). 90% of our patients were obese

(BMI >30 kg/m²) of which 40% were morbidly obese (BMI >40 kg/m²). 80% of patients had a positive drug screen: cannabinoids (3), opiates (2), benzodiazepines (1), PCP (1), methadone (1). The classic pentad of TTP was observed in only 10% of patients. 100% of the patients had microangiopathic haemolytic anaemia and thrombocytopenia. About 91% of our patients had acute kidney injury on presentation whereas neurological complication was seen in only 36%. And about 70% of patients had gastrointestinal symptoms and 10% had a fever on the initial exam.

All of our patients received steroids and supportive plasma exchange. On average, patients received 15 sessions of plasma exchange with an average length of stay in hospital of 15 days. Complete remission was observed in 80% of our patients (days to remission: 10.5, SD 3.93). Unfortunately, around 70% had an exacerbation of TTP. Two out of 10 patients had relapses (time of relapse: 3 years and 14 years). 70% of patients received rituximab treatment with exacerbation. (4.5 doses on average). Around 40% of our patients also received other treatments including cyclophosphamide (2), vincristine (1) and caplacizumab (1).

Conclusion: We found that our patient population was mostly obese and drug abusers. A higher rate of TTP exacerbation was observed in our patient subgroup compared to the general population. Whether the high incidence of TTP exacerbation is related to the female gender, obesity or drug abuse needs to be further evaluated. An interesting finding of our study is the observation of a higher incidence of acute kidney injury compared to neurological complications. One explanation could be the renal pigment deposition secondary to the degree of haemolysis in conjunction with the renal insult caused by the illicit drugs.

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Disclosure of Interest: None Declared.

BSH2020-EP-171

Comparing patient experience and adherence with rivaroxaban or warfarin (PACT); 'Well, you know it (warfarin) kills rats, what's it doing to me?'

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Warfarin has been the oral anticoagulant drug of choice for around 50 years (Poulsen et al., 2012), however, it has been described as a cumbersome drug because of its interactions with food and other drugs, and the need for frequent laboratory monitoring (Connolly et al., 2009). More recently, in acknowledgement of the aforementioned restrictions associated with warfarin, direct oral anticoagulants (DOACs) have been approved for use. DOACs such as rivaroxaban, offer fixed dosing, more predictable pharmacokinetics and fewer interactions with drugs and food (Carter et al., 2008). Consequently, DOACs present strong intuitive appeal for patient populations who require anticoagulant medication (Hale, et al. 2017).

Despite a growing body of evidence, there remains very little data which helps us to understand the actual lived experiences of patients receiving anticoagulation treatment and currently none that enables

us to compare the experiences of warfarin and rivaroxaban treatment. As such, the PACT study seeks to provide an in-depth qualitative exploration of the lived experience of both warfarin and rivaroxaban patient populations.

In this study, 20 participants (all currently diagnosed with atrial fibrillation) were recruited (warfarin $n = 10$; rivaroxaban $n = 10$) via the employment of a stratified sampling methodology. Once consent was obtained, and preliminary questionnaire was completed (containing questions relating to medication adherence and health related beliefs) each participant took part in a semi-structured interview which explored topics relating to: initial experiences and adjustment to anticoagulant medication; current issues relating to medication; management and adherence; motivations for adherence and pros and cons of medication. Once completed, interviews were transcribed, coded and analysed following Braun and Clarke's (2006) thematic analysis guidelines.

Analysis of the interview transcripts revealed a number of salient themes which will be explored in detail at the BSH 2020 conference. These themes include the importance of trust in clinicians rather than information in guiding anticoagulant choice, the power of personal experience of stroke in guiding commitment to treatment, fixed personal beliefs about the practical benefits of their anticoagulant (whichever it is), the absence of adverse symptomatology from anticoagulation in general, and personal use of varied approaches to support adherence. In addition, we provide various recommendations from patients including: 'Do what you are told and just live your life and crack on with it' and (when you have a cut or injection) 'tell the doctor or nurse, so that they can press on the area to stop it bleeding'.

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BSH2020-EP-172

Investigating the response to desmopressin (DDAVP) in various subtypes of severe von Willebrand's disease

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Background: •von Willebrand disease (VWD) is the most common inherited bleeding disorder due to quantitative or qualitative deficiency of von Willebrand factor (VWF). •VWF, a large multimeric glycoprotein, is synthesised in megakaryocytic and endothelial cells and stored within specific storage granules, the α -granules of platelets and Weibel–Palade bodies within endothelial cells. •Desmopressin (DDAVP), a synthetic ADH analogue that stimulates VWF release from endothelial Weibel–Palade bodies, is generally used as a first line of therapy in most VWD patients. •Ristocetin cofactor assay (VWF:RCo) is a pharmacologic assay that measures the ability of VWF to bind platelet GPIb. VWF levels lower than 15 IU/dl demarcate the severe type of VWD. •High-molecular-weight (HMW) multimers of VWF-circulate in the blood-stabilise and serve as a carrier protein for plasma coagulation cascade, FVIII-provide the initial link between the platelet and the ruptured blood vessel.

Aims: •To understand the role of von Willebrand factor in haemostasis. •To compare and correlate patient's VWD types with vWF-ristocetin cofactor activity (vWF:RiCof) after the administration of a synthetic peptide desmopressin (DDAVP).

Table. Desmopressin use in various types of von Willebrand disease.

Type of VWD	Characteristic features	Desmopressin use
Type 1	Mild to moderate decrease in plasma VWF with a normal multimer pattern	Trial needed before use
Type 2A	Absence of multimers with high and intermediate molecular weight	Trial needed before use
Type 2B	Increased affinity for platelet receptor, ultimately leading to excessive platelet activation and thrombocytopenia	Contraindicated
	Mutation in FVIII-binding region of the VWF, disrupting FVIII-carrying capacity	Type 2N Trial needed before use
Type 2M	Mutation in the region of the VWF, disrupting FVIII-carrying function	Trial needed before use
Type 3	Undetectable levels of the VWF with severe bleeding diathesis	Trial needed before use

Methods: •Desmopressin trial data were collected retrospectively from 20 VWD patients being treated at the Royal Free Hospital in the years 1966–2019. •The VWD severity was used as an independent variable, using a pretreatment measurement of ≤ 15 IU/dl as the cut-off for 'severe' VWD; the postdesmopressin trial VWF:RiCof measurement was used to study response. •Data were collated into a spreadsheet and then analysed using excel. In order to maintain patients' confidentiality, we used the anonymous hospital identification numbers.

Results: •Comparison of RiCof response after administration of Desmopressin for severe and non-severe VWD patients. Desmopressin caused a 156% increase in RiCof in severe patients and a 275% increase in non-severe patients.

Discussion: •Our hypothesis assumes that desmopressin is a more useful therapeutic agent in the non-severe subtype of VWD (VWF:RCo above 15 IU/dl), where there is residual VWF in the endothelial storage granules. •Subtyping is essential in the management of VWD. Having higher baseline levels of VWF should cause desmopressin to be more effective as the mechanisms of synthesis and release are more intact compared to severe VWD patients. •Our results, however, showed no statistically significant difference between the two subtypes as confirmed by a paired Student's t-test. This indicates that there could be other factors that determine desmopressin response, potentially including genetics, and more research is required into the field.

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Disclosure of Interest

None Declared.

Transfusion

BSH2020-EP-173

A retrospective audit of perioperative management of anaemia in patients undergoing elective primary total hip replacement in a district general hospital

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Perioperative optimisation of haemoglobin is a key aspect of Patient Blood Management. The International Consensus Statement for perioperative management of anaemia and iron deficiency proposed a 9-point strategy to improve patient care and minimise transfusion associated adverse events. We conducted a retrospective audit comparing current practice at the Mount Vernon Hospital against best-practice recommendations made by the Consensus Statement. Adverse events up to 6 months following surgery were also recorded.

Data of elective primary total hip replacement (THR) cases occurring at the Mount Vernon Hospital between 2017 and 2019 were obtained from: SunQuest ICE, EPRO, Hillingdon Care Record and iReporter. Anaemia was defined as Haemoglobin (Hb) of <130 g/l; iron deficiency was defined as serum ferritin <30 µg/l. Adverse events up to 6 months following surgery were measured; this consisted of: vascular events (venous thromboembolism, cerebrovascular events and acute coronary syndrome), acute kidney injury (AKI), infection and falls.

A total of 82 patients with a median age of 78 years were reviewed. Preoperatively, 38 (46.3%) were anaemic, two (2.4%) had iron deficiency without anaemia and 42 (51.2%) patients did not have anaemia or iron deficiency. Of the 9-point best practice guidance highlighted by the Consensus Statement, only 1 point was followed: 100% of patients were investigated for the presence of anaemia with full blood count (FBC) and ferritin testing. Median time between initial preoperative assessment and surgery was 62 days. No patients with anaemia or iron deficiency received corrective treatment with oral or intravenous iron before surgery. Preoperative anaemic and iron deficient patients had a median length of stay of 5 days compared to 4 days in those without anaemia or iron deficiency. A total of 11 units of red blood cells (RBC) were transfused postoperatively among six out of 38 (15.8%) patients with preoperative anaemia compared to two 2 units in two out of 42 patients (4.7%) without preoperative anaemia or low ferritin. At 6 months following surgery, seven (18.4%) preoperative anaemic patients reported a total of 10 adverse events consisting of: six infections, two falls, one AKI and one vascular event. Ten (23.8%) patients without preoperative anaemia and iron deficiency reported a total of 10 adverse events which included: five infections, two falls, one AKI and two vascular events.

Our data highlighted a lack of compliance with eight of nine key strategies outlined in the Consensus Statement. All patients had preoperative FBC and ferritin testing but those with anaemia or iron deficiency had no corrective treatment prior to surgery despite a median of 62 days between initial preoperative assessment and surgery. Prolonged length of hospital stay and more than a threefold increase in postoperative transfusions were observed in this cohort. Although the rate of adverse events is comparable between patients with perioperative anaemia to those without, the audit highlights the need for incorporating the consensus recommendations to the trust

perioperative pathway. Regional or national collaboration is needed to increase the sample size to obtain more meaningful data on post-operative adverse outcomes for future audit cycles.

Disclosure of Interest: None Declared.

BSH2020-EP-174

Blood donation deferral pattern in a low-income blood banking service

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Blood transfusion has formed an integral component of essential healthcare delivery system globally of which blood donors play a central role. Blood supply for blood transfusions can be met if 2–3% of the population donates voluntarily. Shortage of blood and blood products for transfusion is a challenge in the developing countries because less than this percentage is willing to donate blood voluntarily and among these few, some are still deferred temporarily or permanently on account of transfusion transmissible infections (TTIs), low haemoglobin level other medical ailments. Literature on blood donation deferral pattern is scarce despite increasing need for blood and blood products in medical practice. Knowledge of these deferral patterns will be an important tool for blood transfusion safety. The aim of this study therefore was to determine the pattern of blood donation deferral in a low-income setting.

This was a retrospective study that was conducted at the blood bank of University of Ilorin Teaching Hospital, Ilorin, Nigeria. Donor demographics and reasons for temporary (low haemoglobin concentration) and permanent deferral (positivity to HIV, Hepatitis B and C) were extracted from prospective blood donor register over a 5-year study period (2015–2019). Data were analysed using SPSS 21.0.

A total of 12 359 (11 743 males and 616 females) prospective blood donors were attended to in the study period, with mean age 30.64 ± 8.1 years and male to female ratio of 19.1:1. The mean age of male donors was 30.73 ± 8.05 years, whereas that of females donors was 29.06 ± 8.9 years, $P > 0.005$. Majority of the donors were family replacement blood donors (97.6%) with voluntary blood donors constituting 2.4%. A total of 1895 donors were deferred bringing our deferral rate to 15.3%. Blood donation deferral due to HBsAg positivity was the commonest (7.6%), followed by low haemoglobin concentration (4.5%), anti-HCV positivity (2.4%) and HIV positivity (0.8%). There is, however, no statistically significant relationship between the sex of the prospective blood donors and their viral markers. HBsAg, HCV and HIV ($P = 0.578, 0.938$ and 0.454 respectively).

There is a statistically significant difference between type of donor and low haemoglobin concentration and types of donors and sex ($P = 0.000$ respectively). More family replacement donors were deferred based on positive viral markers (11.8% vs. 3.4%) and

Table.

	Reasons for blood donor deferral			
	Low haemoglobin	HBsAg positive	Anti-HCV positive	HIV positive
Sex				
Male				
Total	11 306	11 002	11 002	11 002
Deferred	434/78.3%	909/7.3%	285/2.3%	92/0.8%
Female				
Total	1053	432	432	432
Deferred	120/21.7%	36/0.3	10/0.3%	6/1.8%
P-value	0.000	0.578	0.893	0.454
Blood donor types				
FRD				
Total	12 072	11 348	11 348	11 348
Deferred	497/4.2%	938/7.3%	297/2.4%	98/0.8%
VBD				
Total	287	241	241	241
Deferred	57/19.9%	7/0.3%	0/0	1/0.1%
P-value	0.000	0.003	0.012	0.467

deferral based on low haemoglobin was more among voluntary blood donors (19.9% vs. 4.2%).

The deferral rate in this study period was found to be 15.3%. Positivity to HBsAg was the commonest reason for permanent blood donation deferral and low haemoglobin was the main reason for temporary deferral among the prospective blood donors. Family replacement donors constituted more of those that were permanently deferred, whereas voluntary blood donors constituted more of those that were temporarily deferred. The deferral patterns identified in this study will provide vital areas to focus on in the formulation of blood transfusion policy in this environment.

Disclosure of Interest: None Declared.

BSH2020-EP-175

Awareness and attitudes towards voluntary non-remunerated blood donation in two generational groups in Trinidad and Tobago

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Trinidad and Tobago is a multiethnic, multireligious, twin-island republic whose National Blood Transfusion Service relies on family replacement and remunerated donors with deleterious consequences for adequacy, equity and safety. Lack of information and inconvenience are principal contributors to poor donation practice. In October 2018, its Ministry of Health committed to total voluntary non-remunerated blood on a national scale and a blood donor education programme to facilitate transitioning. This study was conducted 6 months later to assess current knowledge, attitudes, practice and acceptability of voluntary non-remunerated blood donation in two generational donor groups, namely 18–30 and 45–60 year olds.

A questionnaire-based, cross-sectional, descriptive survey was administered to two conveniently selected samples from each group. Participants were informed of the definition of voluntary non-remunerated blood donation, the Ministry of Health's plan and

classified by age, gender, race, education, employment and donation history. They were assessed for motivators and demotivators to past donation, awareness of the current and proposed blood donor systems and likelihood to donate blood in the future. SPSS was used for descriptive and inferential data analysis, chi square analysis to measure significance and logistic regression to measure strength of associations. Ethical approval for the study was obtained from The University of the West Indies Ethics Committee.

The sample of 600 participants consisted of mainly students (34%), Christians (47%) and Indo-Trinidadians (57%) with a male: female ratio of 1.3: 1. Comparison of the 18–30 and 45–60 age groups showed differences in donor percentage (30% vs. 47%, $P < 0.001$) and the most common reasons for not donating blood in the past: never having thought about it (31% vs. 43%, $P < 0.001$), not meeting the requirements (27% vs. 23%, $P = 0.258$), fear of needles (21% vs. 23%, $P = 0.554$), not having the time (21% vs. 8%, $P < 0.001$) and being unaware of the blood donation centre location (17% vs. 7%, $P < 0.001$). Median knowledge score was unsatisfactory (2/8, satisfactory = 5/8) in both groups and there was no difference in attitude scores ($53.3 \pm \text{SD } 6.2$ vs. $51.4 \pm \text{SD } 7.2$, $P = 0.448$, positive = 53%) between them. More than half of participants in both groups did not know what type of blood donation system existed at the time of survey (51.3% and 52.0%, $P = 0.667$). Only 8.7% of participants had heard about the Ministry of Health's plan for total voluntary non-remunerated blood donation but the majority (50.7%) thought it would be successful. In fact, 76% of 18–30 year olds and 53.3% of 45–60 year olds ($P < 0.001$) intended to donate in the future. The main motivators to encourage future donation by non-donors were information about the blood stock (66.2%), closer blood drives (63%), information about how blood is used (62%) and shorter donation times (51%).

The survey revealed severe blood donation knowledge deficits but acceptability of voluntary non-remunerated blood donation in both age groups. The national information campaign must sensitize the public to the Ministry of Health's voluntary non-remunerated blood donation initiative, provide blood donation information, explain the demerits of the current system and highlight the benefits of truly altruistic blood donation. Access to blood donation and efficiency must be improved to recruit new donors.

Disclosure of Interest: None Declared.

BSH2020-EP-176

Blood loss, cell salvage, tranexamic acid and transfusion: an audit of major surgery at a tertiary hospital

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It is recognised that blood transfusion is an independent risk factor for morbidity and mortality across a range of major surgery.¹ Patient blood management (PBM) is a multimodal approach that seeks to promote the appropriate use of blood, and strategies to reduce or avoid the need for blood transfusion.² A key pillar of PBM is to minimise intraoperative bleeding. The NICE guideline for Blood Transfusion 2015³ recommends offering tranexamic acid to adults expected to have moderate blood loss (>500 mls) and consideration of intraoperative cell salvage with tranexamic acid for patients who are expected to lose a very high volume of blood.

We conducted a retrospective snapshot audit of major surgery performed at our hospital during April 2019. Of 1709 cases performed, 59 cases were included where blood loss 'may be expected

to be >500 ml'. This included procedures across a range of specialties. Data collected included: surgical procedure, pre and postoperative haemoglobin, documented blood loss during surgery, utilisation of cell salvage and tranexamic acid, administration of perioperative RBC transfusion, length of stay and 30-day mortality.

Operative blood loss was not documented in 13% of cases deemed to be at high risk of bleeding >500 ml. Of cases which had their blood loss documented, 59% had blood loss exceeding 500 ml. Preoperative Hb <130 g/l was associated with greater risk of transfusion. Intraoperative blood loss >500 ml was associated with increased risk of transfusion and increased LOS. Tranexamic acid was only utilised in 15% of cases. Tranexamic acid was only used in 18% of patients with blood loss >500 ml. Cell salvage was only utilised in 12% of cases. Perioperative transfusion occurred in 20% of patients. The majority of patients who received a blood transfusion did not receive tranexamic acid or cell salvage.

Intraoperative blood loss is difficult to predict and depends on numerous factors. Our audit shows blood loss was sometimes poorly documented, and we are not meeting the standards for utilisation of tranexamic acid and cell salvage as set out in the NICE guideline for blood transfusion.³ There is potential for greater utilisation of both tranexamic acid and cell salvage in major surgery undertaken at our hospital. Barriers to the implementation of both do exist, in particular concerns over thrombotic complications following utilisation of tranexamic acid and concerns regarding use of cell salvage in cancer surgery.

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Disclosure of Interest: None Declared.

BSH2020-EP-177

The impact of anticoagulation and perioperative blood transfusion in fragility hip fracture patients

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Transfusions of packed red cells (PRCs) are common in fragility hip fracture (FHF) patients, in the preoperative, operative and postoperative settings. FHF are the most common serious injury in older people, with significant morbidity and mortality (30% 1-year mortality). Many FHF patients are anticoagulated at presentation due to various comorbidities. The emergence of direct oral anticoagulant (DOAC) agents has presented new challenges in the perioperative optimisation of such FHF patients. Early operative intervention has been shown to improve outcomes in FHF patients but must be balanced with bleeding and subsequent risks of PRC transfusions.

This retrospective study examined the use, efficacy and appropriateness of PRC transfusions in 326 FHF patients who presented to a district general hospital in 1 year (1 October 2017 to 31 September 2018) against national and trust guidelines. Data were collected from the National Hip Fracture Database (NHFD), local transfusion laboratory, Electronic Patient Records (EPR), Patient Archived Computer

System (PACS) and discharge summaries. The rate of transfusion between FHF patients taking an anticoagulant agent at presentation was compared to non-anticoagulated FHF patients. The FHF patients most likely to receive a PRC transfusion were identified, grouped by fracture pattern and operation performed, with the goal of changing practice to identify them earlier. The relationship between PRC transfusion and 1-year mortality was assessed.

Almost a quarter (23%) of all FHF patients received a PRC transfusion. 86.3% had a pretransfusion haemoglobin (Hb) of <80 g/l, in keeping with national guidance. Most commonly two units were transfused per patient, followed by one unit per patient.

Forty-five (13%) of FHF patients were taking an anticoagulant at presentation. Almost half of these patients were taking a DOAC (22). Anticoagulated FHF patients had a 24.4% chance of receiving a transfusion, those who were not had a 22.8% chance. When comparing different anticoagulation agents, 27.2% of patients taking a DOAC and 21.7% taking Warfarin were transfused.

Intertrochanteric fracture patterns were identified as the group of FHF patients who had the highest association with PRC transfusion (34.4%). PRC transfusions were most likely in long intramedullary nailing operations (47%).

The 1-year mortality for FHF patients who received a PRC transfusion was 52%, compared to 30.5% for those who did not. Both groups had similar American Society of Anaesthesiologists (ASA) grading.

All PRC transfusions complied with trust policy of transfusion, of rechecking Hb after one unit before further transfusion. Extracapsular fracture patterns and their surgical fixations were more likely to result in a PRC transfusion than intracapsular fractures. FHF patients taking a DOAC at presentation were more likely to receive a transfusion than those taking Warfarin. Receiving a PRC transfusion was associated with higher 1-year mortality, warranting further investigation.

Disclosure of Interest: None Declared.

BSH2020-EP-178

An audit of cell free fetal DNA (cffDNA) screen to avoid administration of anti-D immunoglobulin in RHD negative pregnant women with RHD negative fetus

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Anti-D immunoglobulin prophylaxis has been the standard treatment in reducing the incidence of haemolytic disease of the fetus and newborn (HDFN). A significant proportion (40%) of RHD negative pregnant women carry RHD negative fetuses and therefore receive anti-D immunoglobulin unnecessarily. The National Institute for Health and Care Excellence (NICE) recommends fetal RHD genotyping as a cost-effective option to guide antenatal prophylaxis with anti-D immunoglobulin. International Blood Group Reference Laboratory (IBGRL) provides the diagnostic service to NHS Blood and Transplant (NHSBT). The Hillingdon Hospitals NHS Foundation Trust was one of the first NHS trusts in London to introduce the cell free fetal DNA (cffDNA) screen for RHD status in March 2017. The aim of the audit was to compare local clinical practice against the local and NICE guidelines in the context of cffDNA screening in RHD negative pregnancies. The proportion of pregnancies with RHD positive and negative fetuses, and false negative and positive cases were also reviewed.

The transfusion laboratory provided the list of RHD negative pregnant women who had blood test sent for cffDNA screening from

March 2018 to May 2019. Data were collected from Sunquest ICE and Winpath.

A total of 434 RHD negative pregnant women were identified. Of these pregnant women, 264 (60.8%) were identified to be carrying RHD positive fetuses, 147 (33.8%) were identified to be carrying RHD negative fetuses and 23 (5.2%) had inconclusive RHD status by cffDNA screen. There was no false positive or false negative result during the period of the data collection.

Cord blood at delivery for RHD typing was done in 39.4% ($n = 58$) of pregnant women with a RHD negative fetus according to cffDNA screen. A significant number of pregnant women in this group, 60.5% ($n = 89$) did not have a record of cord blood for RHD typing at delivery. Of the 23 pregnant women with inconclusive RHD status by cffDNA screen, cord blood at delivery for RHD typing identified 56.5% ($n = 13$) were RHD positive, 21.7% ($n = 5$) were RHD negative and 21.7% ($n = 5$) did not have a record of cord blood for RhD typing at delivery. 15.9% ($n = 42$) of pregnant women with RHD positive fetus by cffDNA screen did not have a record of cord blood for RHD typing at delivery. There were no sensitising events during the period of data collection.

We demonstrated that the cffDNA screen for RHD status not only provides an economic benefit by reducing use of anti-D immunoglobulin, but also demonstrates the added benefit of preventing exposure to unnecessary blood product. A significant proportion of RHD negative pregnant women did not have cord blood tested at delivery for RHD typing. The false negative result has clinical implications hence it is important to check RHD status on cord blood at delivery.

We have recommended the obstetric team to ensure cord blood is sent for RHD typing at delivery. This will help to identify any false negative results and initiate appropriate action. We aim to re-audit in 12 months time.

Disclosure of Interest: None Declared.

BSH2020-EP-179

The consultant clinical scientist in transfusion – overview and reflections on the journey to FRC Path

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The Higher Specialist Scientific Training (HSST) programme was established to enable a career pathway to Consultant Clinical Scientist for healthcare scientists working in the NHS. HSST is a bespoke 5-year doctoral level workplace-based programme which aims to train healthcare scientists to provide expert consultant level advice as part of a multidisciplinary approach to patient management, to meet the challenges of delivering healthcare at the limits of science.

The HSST programme is open to Clinical Scientists and Biomedical Scientists who have undertaken clinical scientist equivalence and comprises a Doctorate in Clinical Science (DClinSci) as well as Fellowship of the Royal College of Pathologists (FRCPath) qualification, and it endeavours to achieve a balance between academic learning and work-based training.

This session will provide insight into the different career paths available to healthcare scientists who have a passion for transfusion science. It will discuss the roles of scientists in the blood bank and in the blood services, and how they have evolved from more traditional scientific posts into a more clinical role in places with the introduction of transfusion practitioners and consultant clinical scientist posts. It will briefly describe current pressures in the NHS and how these have led to the creation and evolution of the Consultant

Clinical Scientist role. It will describe what impact pathology consolidation networks may have on scientists working in transfusion and will discuss some areas for suggested further development and discussion.

There will be a description of the HSST programme and the RCPATH Transfusion Science curriculum, in addition to a description of some of the roles that are currently being undertaken by Consultant Clinical Scientists within NHSBT. The talk will also give an overview of roles and responsibilities that potentially be undertaken within hospital trusts and in posts with shared roles between blood services and hospital trusts.

Tom Bullock is a Consultant Clinical Scientist Trainee, currently based at NHS Blood & Transplant in Filton, Bristol. He is one of the first cohort of trainees to undertake the HSST programme and is one of the first people to undertake the FRCPath Transfusion Science curriculum. He will give a unique perspective and a personal summary of his experience of the training programme, including a description of some of the challenges and obstacles involved, with suggestions for further development of both the process and eventual role of a Consultant Clinical Scientist in Transfusion

Disclosure of Interest: None Declared.

BSH2020-EP-180

Blood management in Volta Regional Hospital (VRH), Ho, Ghana: a quality improvement project

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Introduction: The VRH is a regional referral centre situated in South West Ghana with a potential catchment area of >2.4 million people. There is high demand for blood products due to presentations of severe anaemia secondary to malaria, polytrauma or postpartum haemorrhage. There is a limited pool of regular voluntary donors. This often leads to very low blood stocks in emergency situations. Due to local limitations in blood screening there is a significant risk of transfusion-associated infection and potential morbidity. Dhariwal et al.'s 2017 study (DS) revealed blood was being used inappropriately – 48% of products issued were not in keeping with National Transfusion Guidelines.

To reduce use of inappropriate blood and to optimise blood stocks, a new blood request form was introduced after training (laboratory/clinical staff). For blood to be released outside of the emergency situation a pretransfusion Hb and a recognised indication had to be stipulated on the form.

To further reduce demand for blood products, sickle cell clinic was restarted with the aim of offering hydroxycarbamide (HC) therapy to patients with severe phenotype Hb SS disease.

Methods: Blood use data: obtained from the blood issue book, blood request forms, patient notes and the electronic results service between 24 October 2018 and 20 November 2018. Sickle cell clinic recruitment: Patients with severe phenotype Hb SS were identified from inpatient admission records and in general medical and paediatric clinics. A protocol for establishment, and monitoring on, HC was formulated from Prof Rees' 2011 paper 'the rationale for using HC therapy' and a protocol from the Ridge Hospital, Accra. Patient data were recorded prospectively in the weekly clinics.

Results: One hundred and sixty-one products were issued to 98 patients. Fifty-nine per cent of patients were female and 41% male with a median age of 41 years. The mean pretransfusion Hb was 75.6 g/l. Of the blood products used, 80.1% were whole Blood, 7% were packed red cells and 2.2% were fresh frozen plasma. There was a significant reduction rate of blood issued compared to DS – products per week reduced from 63.8/week *versus* 40.3/week post intervention. Comparing June 2018 to June 2019 blood usage rate – pre and post intervention – there was a significant reduction from 72.3/week to 48.5/week.

In products issued there was a 68.1% concordance with Ghanaian Guidelines, 27.4% non-concordance and insufficient evidence to prove concordance in 4.4% cases – a significant improvement in comparison to DS ($P = 0.001$).

Between September and December 2018 47 patients were assessed in sickle clinic and 15 patients were commenced on HC with no recorded significant adverse events.

Conclusion: There was a significant increase in concordance with National transfusion guidelines and a subsequent reduction in Blood usage. Encouragingly, this reduction appears to have been maintained after the study period. Although only relatively small numbers of patients were commenced on HC in clinic – the protocol for establishing patients on this crucial treatment is now well founded and this service should grow with time.

Disclosure of Interest: None Declared.

BSH2020-EP-181

Detection of red cell antibodies in patient plasma after 3 months storage above the recommended -30°C

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The detection of clinically significant antibodies in patient plasma is an essential step in preventing transfusion reactions. Antibodies are detected using the indirect antiglobulin test (IAT) at 37°C and identification is achieved using a panel of prepared group O red cells with known phenotypes.

Samples are then stored for 3 months to ensure retesting of patient plasma is available. Longer term storage also allows for the investigation of delayed transfusion reactions.

BCSH^[1] guidelines state that plasma samples should be stored below -30°C ; however, the specimen freezers at St James' Blood Bank in Leeds (SJUH) run at a much higher temperature. A previous study² tested 118 frozen samples with specific antibodies using BioRad ScanGel, a similar method to the DG card technology used at SJUH, with 95.7% of the samples testing positive for antibodies after storage. The purpose of this experiment was to ensure that the storage of specimens at a higher temperature than recommended will not affect a repeat antibody screen and the results will not be changed. A risk assessment was produced after this experiment to determine whether the samples can continue to be stored at a higher than recommended temperature.

Ninety samples that had previously been tested positive for antibodies were retrieved from storage with an average age of 91 days (range 84–106 days). The two freezers that the samples were stored in had an average temperature of -17.7°C .

The plasma of separated specimens was tested over a period of 3 months on the Grifols Erytras at St James' Blood Bank for an antibody screen (IAT). The new results were recorded manually and compared to the previous results which were gathered from the laboratory information management system (LIMS). A grading difference of 1 was acceptable for a pass.

Of the 90 samples, 89 (98.9%) passed by obtaining a grading difference ≤ 1 ; however, 4 (4.4%) tested negative after 90 days storage. Forty-two per cent obtained the same grading as the previous result and 21% obtained a higher grading than the previous result. This may be due to the use of different batches of screening cells.

The sample that failed had a reaction grading difference of 1.5 and was identified as anti-M. This was still identified at day 97; however, anti-M is not normally known to cause a transfusion reaction. Three of the four antibodies that were no longer detectable (one anti-E and two anti-Jkb) will have antigen negative blood selected from the first result. The fourth antibody that was no longer detectable was anti-M which is not usually identified as a clinically significant antibody; only when tested at 37°C which was not confirmed at the time.

These results confirm that red cell antibodies can still be detected at higher than the recommended storage temperature of -30°C , with some exceptions. 38/90 (42.2%) were tested after 90 days which also verifies that samples are capable of being stored past the recommended 90 days.

References:

1. BCSH (2012) Guidelines for pre-transfusion compatibility procedures in blood transfusion laboratories. *Transfusion Medicine*;23 (1):3–35.

2. Bouix, O. et al. (2008) Erythrocyte-magnetized technology: an original and innovative method for blood group serology. *Transfusion*;48:1878–1885.

Disclosure of Interest: None Declared.

BSH2020-EP-182

Determination of appropriate deferral time for repeat apheresis platelets concentrate donation in Abuja, Nigeria

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Plateletpheresis is the process by which a therapeutic adult dose of platelet concentrate is produced from a single donor using automated cell separator equipment. Recently, the demand for apheresis platelets concentrate transfusions has been increasing in our hospitals. This has been attributed to increased cases of malignancy associated bone marrow suppression and various other causes of thrombocytopenia and thrombopathies. The increasing use of myeloablative chemotherapy in treatment of haematological and other malignancies also results in transient severe fall in platelet count. The demand generally outstrips the supply necessitating the need to search for additional donor sources to increase the supply of platelets for transfusion. Is it therefore possible to reduce the time interval between plateletpheresis while maintaining donor safety?

The aim of the study was to determine the platelets recovery time after platelets concentrate donation with a view to determining the appropriate deferral time for repeat plateletpheresis

The study was conducted at the National Hospital Abuja. Fifty-seven healthy plateletpheresis donors were recruited over a period of 7 months and their haematological parameters were analysed on days 0, 2 and 7 postdonation. We excluded donors whose predonation platelet count was less than $200 \times 10^9/\text{l}$.

Analysis was done in only forty-eight (48) donors as nine (9) participants withdrew before the end of the study. The results show there was a significant decrease in the platelets counts ($P < 0.001$), with subsequent good platelets recovery postdonation (Pre:

259.83 $\pm$ 44.61 $\times 10^9$ /l; Post: Day 0: 205.29 $\pm$ 36.77 $\times 10^9$ /l, day 2: 216.88 $\pm$ 32.71 $\times 10^9$ /l and day 7: 240.3 $\pm$ 41.83 $\times 10^9$ /l). The apheresis donors recovered 64.29% of the lost platelets by day 2 of donation and subsequently attain 92.49% of predonation count within 7 days of donation. There was a statistically significant transient increase in the immediate postdonation haemoglobin concentration (14.72 $\pm$ 1.32) g/dl but this gradually reduced on day 2 (13.75 $\pm$ 1.04) g/dl and 13.83 $\pm$ 1.04 g/dl on day 7 showing no significant different from the predonation Hb of 13.96 $\pm$ 1.02 g/dl.

In conclusion, this study has shown statistically significant platelets recovery 7 days after plateletpheresis making a 7 day deferral for repeat plateletpheresis feasible in our environment. The study also shows feasibility of a 2 days deferral as recommended by AABB. Strict monitoring and predonation evaluation of donors should be done to prevent any adverse events

Table 1. Platelet recovery after Apheresis-PC donation

Parameters	Apheresis-PC donors Mean $\pm$ SD/percentage recovery
Predonation	
Platelet count ($\times 10^9$ /l)	259 $\pm$ 44.41
Day 2 postdonation	
Platelet count ($\times 10^9$ /l)	216 $\pm$ 32.71
Platelet recovery (%)	83.47
Day 7 postdonation	
Platelet count ($\times 10^9$ /l)	240.31 $\pm$ 41.38
Platelets recovery (%)	92.49

Disclosure of Interest: None Declared.

BSH2020-EP-183

Intraoperative blood utilization and wastage in a low-resource tertiary Hospital in Northern Nigeria

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It is a common practice to group and cross-match blood for surgical patients scheduled for elective or emergency surgery. This is particularly so when the operation is expected to be long or associated with high blood loss. However, not all blood requested and collected from the blood bank is transfused to the patient. This causes wastage of the scarce and expensive blood.

This audit was conducted to assess the extent of blood utilization and wastage in our low-resource healthcare centre so that we will improve on the services rendered to our patients.

Over a 30-day period (24 October 2019–22 November 2019) relevant information was obtained from all consenting surgical patients undergoing elective and emergency surgical procedures in a low-resource tertiary hospital in Zaria. Data on age, sex, nature of surgery, preoperative packed cell volume (PCV), units of blood transfused and time taken to return unused blood to the blood bank were collected and analysed.

A total of 147 patients were operated upon during the period of audit. Females constituted 80 (54.4%) of cases. There were 80 (54.4%) elective operations performed while emergencies constituted 67 (45.6%) of surgical procedures. There was no blood transfusion in 110 (74.8%) of patients. One, two, three and four pints of blood were transfused to 26 (17.7%), 9 (6.1%), 1 (0.7%) and 1 (0.7%) patients respectively. Unused blood was not returned to the blood bank in 128 (87.1%) instances while in 2 (1.4%), 5 (3.4%), 8 (5.4%)

and 4 (2.7%) cases blood not transfused were returned to the blood bank in less than 30 min, within 60 min, within 2 h and after 2 h respectively.

In conclusion, most of the patients who had surgery during the period under review were not transfused. In a large number of cases unused blood was not returned to the blood bank and in the few situations where blood was returned it was done after 30 min.

Disclosure of Interest: None Declared.

BSH2020-EP-184

Cross-trust mandatory training in transfusion

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Around the country, significant Trust time and resources are invested in developing, updating and delivering the mandatory transfusion training (MTT) to hospital staff involved in the transfusion process.

At King's Health Partners (KHP) we are integrating this work by moving all of the MTT online and sharing resources across four hospitals in two Trusts (King's College Hospital, and Guy's and St Thomas', NHS Foundation Trusts) – this will realise significant benefits:

- Online training resources increase access for staff who work part time, shifts, weekends or who have difficulty attending face-to-face training.
- Release trainers from time spent developing and delivering training, maintaining training and updating training records.
- Standardise training resources across hospitals, utilising a wider group of expertise.
- Enabling portable training and compliance for staff transferring between sites.
- Allow learners to progress at their own pace to encourage more effective learning.

All current training resources, stated aims and learning outcomes were collected, assessed and collated. A pool of 12 modules was agreed from which specific role-based MTT programmes can be assembled based on training needs analyses.

Note: The MTT programmes developed will not replace the required competency observational assessment.

Training aims and outcomes were written and agreed for all modules. The modules were developed online using Articulate Rise® by the KHP Transfusion Team and an education expert and reviewed by peers and end-users.

Unnecessary or site-specific detail was removed where possible; where necessary, the modules took a branched approach (required in one module).

Care was taken to map existing compliance data on 'current' mandatory training to the new modules to ensure the continuity of existing compliance.

Ideally these modules would all be housed on a single learning management system (LMS) and all learners would access them at that location. This will become possible with joint or aligned IT.

Twelve training modules have been developed. These are:

- 1 Introduction
- 2 Safer Blood Transfusion
- 3 ABO & D

- 4 Appropriate Use and Prescribing
- 5 Informed Consent
- 6 Pretransfusion Sampling
- 7 Storage and Handling
- 8 Collection and Receipt
- 9 Administration
- 10 Emergency Transfusion and Major Haemorrhage
- 11 Monitoring and Acute Transfusion Reaction
- 12 Transfusion and Pregnancy (including cfDNA testing)

Modules are all stand-alone and take between 5 and 20 min to complete and training can be spread over a period of time to fit in with the learners' day-to-day responsibilities.

Organisations with multiple sites may benefit from a similar approach.

Disclosure of Interest: None Declared.

BSH2020-EP-185

Abstract Withdrawn.

BSH2020-EP-186

How do they measure bleeding? Analysing the triggers applied by the obstetric specialists for initiating blood transfusion for postpartum haemorrhage

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Introduction: Postpartum haemorrhage (PPH) is identified as the leading cause for the maternal deaths worldwide. Guidelines from the recognized institutions state that blood transfusion should be initiated based on clinical judgement but there are no well-defined fixed criteria to initiate the blood transfusion.

Aims: To analyse the clinical and biochemical parameters and triggers applied by the obstetric specialists to commence blood transfusion in clinical practice.

Methods: A descriptive study was conducted among 52 Consultant Obstetricians practicing in Sri Lanka. Data were collected with self-administered questionnaire and analysed with Microsoft excel and SPSS software.

Results: The mean years of experience of participants practicing as clinical specialists were 8 years. Among them, 88.4% (46) participants agreed that they considered initiating blood transfusion depending on the rate of bleeding irrespective of estimated total loss, when there was an evidence of hypovolaemia. All of them used 'increase in pulse rate above 100 or rise of pulse rate by 10 from base line' as an indicator. Only 11.1% (6) agreed that they considered blood transfusion on the rate of bleeding irrespective of the estimated total blood loss and clinical parameters. Regarding total blood loss in a patient with ongoing bleeding, all of them considered transfusion when total blood loss was above 1000 ml irrespective of clinical parameters and they considered transfusion with increasing pulse rate or dropping of blood pressure even if total loss is less than 1000 ml.

All participants agreed that visual assessment of total blood loss was an inaccurate method to define PPH.

All of them agreed that combination of clinical parameters was more reliable and effective method in assessing the need for transfusion than single parameters.

Thirty-five (67.3%) participants agreed that pulse rate was the most reliable parameter followed by blood pressure and respiratory rate and 32.7% (17) of them suggested change in the respiratory rate followed by pulse rate and blood pressure would be more reliable.

None of them suggested that they'd consider blood transfusion on the basis of clinical pallor or patients' consciousness status alone unless there was a defined blood loss or other evidence of hypovolaemia and 86.5% agreed that they'd wait for the haemoglobin assessment in a patient who was pale without considering the ongoing blood loss and normal clinical parameters.

All participants agreed that the involvement of a Haematologist or a Transfusion physician would be beneficial in making the decision of blood component therapy.

If a haematologist was not available, 59.6% (31) and 40.4% (21) agreed they'd have had initiated components transfusion with 4th unit and after 4th unit of red cells (before coagulation profile was available in low-risk patients).

Only 13.5% had the facility for point of care testing and they agreed that it could be beneficial, but it was not necessary for them to manage obstetric haemorrhage, whereas 86.5% never had the facility to manage a patient with such test.

Conclusion: All the specialists agreed clinical judgement was the important determinant and there were some specific parameters used by each of them in initiating the blood transfusion.

Disclosure of Interest: None Declared.

BSH2020-EP-187

Evaluation of preoperative haemoglobin level and blood transfusion practices among patients attending to a tertiary care unit in Sri Lanka – Colombo North Teaching Hospital

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Introduction and Justification: Preoperative anaemia is a very common problem in routine surgical patients worldwide. Causes for preoperative anaemia can be variable. Preoperative anaemia is associated with the increased risk of perioperative allogenic blood transfusion. Detection and management of preoperative anaemia will reduce the risk of allogenic blood transfusions. So it is justifiable to do a study on prevalence of anaemia and causes for anaemia in surgical patients to decide on corrective actions.

Objectives: To identify the haemoglobin levels of surgical patient attending to the Colombo North Teaching Hospital for routine surgical procedures and to find out whether there is a relationship between the preoperative haemoglobin and transfusion practices.

Methods: Study conducted as descriptive study. Data collected from all surgical units at Colombo north teaching hospital admitted for routine surgeries. Sample size was 384. We obtained data on haemoglobine level, surgery type and demographic data of the patient. Data analysed using SPSS software.

Results: Among 384 patients 156 (40.6%) were anaemic and 51% of females, 33% of males were anaemic. Severity of anaemia among the study population was as follows, 62% – mild anaemia, 37.8% moderate anaemia and 1.2% had severe anaemia. Only nine patients were investigated for anaemia. Thirteen patients had been transfused during the hospital stay but only one patient was non-anaemic prior to the surgery.

Conclusion and Recommendations: Preoperative anaemia is a significant finding among surgical patients in Sri Lankan setting with a female preponderance. But trends for investigations for cause for anaemia is lacking. Risk of blood transfusions was associated with

the haemoglobin level and the surgical risk category. It is recommended to investigate to find the presences of anaemia and cause for anaemia well before the surgery and to correct them at least in high to intermediate risk category surgeries.

Key words: Preoperative haemoglobin, Surgical patients, Blood transfusion, Investigation of anaemia

Disclosure of Interest: None Declared.

BSH2020-EP-188

Blood transfusion refusers: a clinical audit about referrals and their outcome

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Blood components and blood products are life-saving therapeutics with inherent, life-threatening risks. However, there is a cohort of patients who do not accept allogeneic blood components (including for religious reasons). St. George's is a teaching hospital with a large local population of Jehovah's Witnesses; therefore each year there are many patients who refuse blood transfusion. We aimed to evaluate the management of these patients between 2017 and 18.

All blood refusers for whom the transfusion team received notification between 2017 and 18 were included in this audit. Scanned referral forms and Electronic Patient Records were used to gather data from patients. The following standards were audited

- 1 Any patients who refuse blood components and/or blood products must be referred for a multidisciplinary team (MDT) meeting to determine a care plan.
- 2 This MDT should take place at least 3 weeks before a procedure and must involve the surgeon, anaesthetist, haematologist or transfusion practitioner and the patient.

Between 2017 and 2018, 120 notifications were received; 105 patients (88%) both attended an MDT and underwent an operation. Five patients underwent an operation despite not attending an MDT. Of the 105 surgical patients who had an MDT, 57% (60 referrals) occurred less than 3 weeks before the subsequent procedure.

Operations on patients who refuse blood transfusions occur weekly at St George's. In the majority of cases the local protocol is adhered to. The most common deviation was due to a short lead time between notification and surgical date. This shortening of lead time is due to the timing of preoperative assessment. The 3-week rule will need to be reviewed with the surgical teams in the light of this. A small number of procedures went ahead with no documented evidence of MDT. Awareness of the need for MDT will be raised within the surgical departments.

Disclosure of Interest: None Declared.

BSH2020-EP-189

Does measuring ferritin in the first or third trimester, optimise diagnosis of obstetric iron deficiency anaemia, thereby leading to prompt oral iron therapy and a reduction in blood transfusions?

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Obstetric iron deficiency anaemia (IDA) is still proving hard to diagnose despite the advancement in various haematological and biochemical markers related to iron storage; this results in lack of

targeted oral iron therapy (OIT) and in unnecessary use of transfusions. Currently haemoglobin (Hb) is in routine use for obstetric IDA diagnosis, however, ferritin has always been considered more sensitive. In this study, two groups were compared where one group ($n = 100$ after some patients were ruled out due to various reasons such as miscarriage) had ferritin measured in the first and third trimester and OIT administered as appropriate. The second group ($n = 100$) did not have ferritin measured or any OIT. The patients were followed to term to see if there was a significant difference in overall transfusions. There was a stabilising effect observed within the test group, where the labour Hb had dropped around 1 g/l from the third trimester to labour whereas within the control group the drop was around 7 g/l. Using Mann-Whitney U, OIT lead to a statistically significant difference between the labour Hb values of the two groups where $P < 0.05$. Four more people were transfused within the control group than in test group; the costs of blood transfusions and extra tests requested lead to a surplus of £94.84 for the control group. When analysing if OIT was more effective in first, third or both trimesters no significant difference was found, however, that is most likely due to the relatively small sample sizes ($n < 4$). To truly prove if ferritin really is sensitive, this study could be done again with the same conditions except a bigger sample size and a stricter OIT protocol.

Due to this study, the protocol for dealing with obstetric IDA is changing at East Surrey Hospital where measuring of ferritin is to be implemented as part of the routine booking bloods.

Disclosure of Interest: None Declared.

BSH2020-EP-190

An insight into the knowledge, attitude and practice towards blood donation among the students, health workers and general population of Hyderabad Sindh

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Background: Adequate and safe blood supply has remained a challenge in developing countries having a huge burden of bleeding disorders. The evidences indicate that there is a major shortage of safe blood and blood products and there is a high dependency on family replacement and remunerated blood donors. Thus recruitment of safe and low-risk donors is necessary. The growing body of peer-reviewed KAP surveys in developing settings is an important contribution.

Objectives: The study aimed at assessing knowledge, attitude and practice towards blood donation in population in order to identify motivating factors, the myths and fears that might deter blood donation to convert positive attitudes towards blood donation into the practice of volunteering to donate.

Study design: Descriptive cross sectional study.

Setting: Diagnostic & Research Laboratory Hyderabad
METHODOLOGY: A pretested, semi-structured self-administered questionnaire was used to interview total 350 respondents in which 160 healthcare workers (non-physicians, 90 students and 100 general people. The target population was residents of Hyderabad, Sindh. Descriptive statistics were used to analyse the data.

Results: The median age of respondents was 31 years (18–56) with males 57% and females 43%. Among 90 students, 74% had a positive attitude but only 22% had donated. Among healthcare workers 83% had knowledge and 34% had donated. General public lacking significant knowledge but had good practice towards donation.

Conclusions: The KAP surveys provide important information about blood donation that can be used to design rationale, targeted strategies, regularly scheduled awareness program and voluntary blood donation campaigns organized at the community level to utilize potential blood donors.

Keywords: Blood donation, Knowledge, Attitude, Practice.

Table.

	Healthcare workers	Students	General population
Knowledge	56%	77%	19%
Attitude	64%	56%	48%
Practice	41%	22%	78%

Disclosure of Interest: None Declared.

BSH2020-EP-191

An audit of out-of-hours transfusions across two hospital sites

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Blood transfusion outside of core working hours is associated with an increased risk of adverse events and should be avoided unless clinically essential. There are fewer available trained staff both in the laboratory and in clinical areas, often with multiple competing demands on their time and attention. This increases the risk of error at all stages in the transfusion process. With fewer doctors there may also be delays in obtaining a medical review if there is an adverse reaction. We audited the appropriateness of out-of-hours transfusions over a 1 month period across two hospital sites within our Trust.

Data relating to transfusions commencing between 20:00 and 08:00 in April 2019 were collected from electronic patient records, the laboratory information management system and the Blood-Track® transfusion management system. Patients on high dependency or intensive care units were excluded, as were patients on the Haematology ward, where supportive transfusions are often commenced before 08:00 for pragmatic reasons.

Sixty-seven components were transfused overnight to 56 patients – 11 patients received more than 1 component in the same overnight period. These comprised 61 units of red cells, 3 units of FFP and 3 pools of platelets. The indications for red cell transfusion were: Bleeding 24 (45%), Chronic anaemia 19 (36%), Anaemia of acute illness (e.g. sepsis) 7 (13%), Unknown 3 (6%). The median pretransfusion Hb was 68 g/l, range 46–86 g/l. 11 (21%) patients showed evidence of haemodynamic instability (defined as systolic BP <90 mmHg or HR >100 or a clear drop in BP from normal baseline). Out-of-hours transfusion was assessed to be clinically necessary in 22 (41.5%) patients. In a further 14 (26%), early morning transfusions were given for pragmatic reasons. The remaining 17 (32%) had neither a clinical nor a pragmatic reason for transfusion out-of-

hours. In eight patients (15%), transfusion was completely inappropriate as these had chronic iron deficiency anaemia with no evidence of haemodynamic instability.

In our local cohort, the proportion of overnight transfusions given with no clear reason (32%) was identical to the national average assessed in the 2008 National Comparative audit. Doctors requesting and prescribing blood products should be clear whether transfusion is required urgently and should proceed overnight or can be given electively the next day. This must be communicated to laboratory and nursing staff. Nurses are the final gatekeepers in the transfusion process and should be empowered to question decisions to transfuse overnight. During the day, laboratory staff will challenge transfusion requests, particularly for multiple units or in cases of iron deficiency. At night when staffing is short and there is less ready access to Transfusion Practitioners or Haematologists for advice, there may be less willingness to question clinicians. A cross-professional educational campaign targeting individuals at various steps in the process has potential to improve out-of-hours transfusion practice.

Disclosure of Interest: None Declared.

BSH2020-EP-192

High prevalence of haemolytic antibodies: a challenge for safe blood transfusion in Africa

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Allogenic blood has never been more in demand than it is today particularly in developing countries. One of the biggest challenges to blood safety particularly in developing countries is accessing safe and adequate quantities of blood and blood products. One strategy geared towards the optimum utilization of scarce donor units is the use of blood and blood products against ABO blood group barrier. Haemolysins are immune antibodies that can cause haemolysis during blood transfusion. Two hundred apparently healthy blood donors (183 males and 17 females), were screened for the presence of haemolysin using the standard tube method. Those that were positive were titrated for haemolytic antibodies. Study population included 121 blood group O donors (60.5%), 44 blood group A (22.0%) and 35 blood group B (17.5%) donors. The result showed that, out of the 200, a total of 37 (18.5%) had haemolysins in their serum. The prevalence of alpha, beta and alpha plus beta haemolysin, among group O donors were 6.5% for alpha haemolysin, 7.5% for beta haemolysin and 4.5% for alpha plus beta haemolysin and a total of 14.9% prevalence of haemolysin for blood group 'O'. While blood group A and B had 25.0% and 22.8% respectively. Haemolytic titre of 4 and above was seen in about 83.4% of the samples that had haemolysin. In conclusion, this study has shown that alpha and beta haemolysin exist in significance frequency among blood donors attending University of Calabar Teaching Hospital donor Clinic with a number of these donors having significant titre of 4 and above. We recommend that all group O, A and B blood donors whose blood is intended for transfusion against ABO blood group barrier be

Table 1. Distribution alpha and beta haemolysin among blood donors and titre of haemolysin.

Alpha haemolysin (%)	Beta haemolysin (%)	Alpha +Beta haemolysin (%)	Total number of donors positive for haemolysin	Titre >4 (%)	Titre <4 (%)
13 (6.5)	15 (7.5)	9 (4.5)	37 (18.5)	83.4	16.6

Total number of donors = 200; P = 0.067

screened routinely for alpha and beta haemolysins and that all blood group O red cell units intended for use as universal donor units must be screened and found negative for high titre alpha and beta haemolysins. Measures should also be put in place to ensure that blood component therapy become routine practice in our environment.

Disclosure of Interest: None Declared.

BSH2020-EP-193

Transfusion-related adverse event reporting and documentation compliance among healthcare professionals in developing country – Aiming towards improving practices

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Hemovigilance is a surveillance system that covers the whole transfusion process. In Pakistan, around 1.5 million bags are annually transfused. There are number of risks associated with transfusion and it is reported that the greatest risk to blood transfusion is human error. Standard guidelines regarding the screening of blood for infectious disease, genuine need of transfusion and ABO compatibility are followed and monitored drastically. However, the audit of bedside practices has not been conducted religiously and is under reported. We are a newly established hospital and are working towards the best possible management of patients. In this regard to minimize the transfusion errors and to highlight if any lacking being practiced during transfusion, we conducted this study to observe transfusion practices at bedside and promote its existence by calculating the compliance rate of documentation of transfusion form by the healthcare staff. Secondary objective of this was to observe the compliance of line of action taken in case of occurrence of transfusion reactions. This was an observational study conducted at NIBD and BMT from 2018 to 2019. Ethical approval was obtained prior to the study. Transfusion form for each transfusion was filled. The form provided information on documentation of blood product receiver name, employee identity number, date and time of receiving blood product, patient name, medical record number on units and on transfusion form. ABO compatibility on the unit and on form, medical record number from wrist band, name and employee identity number of two healthcare staff started transfusion, transfusion start and completion time. Time, temperature, blood pressure, pulse and initials of staff at the time of order, onset, after 15 min and at the completion of transfusion were also included. Transfusion reaction form was also filled by the healthcare staff. Data were analysed by using SPSS version 23.0. A total of 500 transfusions forms were analysed. Over all compliance rate was 18%. Out of 500, 115 (23%) forms were available in source notes and of 115, 88 (76%) were partially and completely filled. Higher compliance was seen in the initial months of hospital establishment than later months (P -value = 0.000). Highest non-compliance was seen in documentation of initials of duty doctors on transfusion form at the completion of transfusion (3%) and highest compliance was seen in documentation of name by healthcare nursing staff at the start of transfusion (89%). A total of 02 (0.4%) adverse events were reported from red blood cells and platelets. Mean time of start of symptoms was 2 h and 30 min for red blood cells and for platelets it was 1 h and 13 min. Transfusion was instantly stopped as the symptoms appeared with no delay of time

and actions were taken to resolve the reactions. Time of appearance of symptoms and time of start of medication were documented and error free. All blood bags were returned to the blood bank and discarded after 6 h as per the policy of hospital. Transfusion practices of healthcare staff during transfusion was found unsatisfactory. It was seen that higher compliance in documentation was observed in initial months than later months of establishment. Such errors may not be detrimental but they may cause serious fatal events. Limited data are available in Pakistan in this context and further studies are the need of the time for better transfusion services.

Disclosure of Interest: None Declared.

BSH2020-EP-194

Stop frightening start donating – An experience from resource constraint country

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Blood donation is an imperative procedure of healthcare saving millions of lives worldwide. The trend of blood transfusion is increasing globally, despite of this a massive number of patients requiring transfusion suffer delayed access to the safe blood. Among the different types of blood donors, voluntary donors are considered as a safest donors group because of their better health seeking gestures than replacement and remunerated donors. In Pakistan, majority of the donors are male and replacement donors while females and voluntary donations reserve a diminutive percentage. One of the observed fears that is keeping people to donate blood is anaemia and/or iron deficiency. However, it has been reported that risk of developing anaemia and iron deficiency increases only with repeated donations. By keeping the high proportion of male blood donors in mind, we conducted this study to investigate the prevalence of anaemia post-donation in first time male donors and compared it with non-donors group. This was a comparative study conducted at NIBD and BMT, PECHS campus, Karachi, Pakistan from February 2018 to December 2018. Ethical approval was obtained prior to the study. There were two study groups, group 01 included male donors who donated blood at our centre. Donors were selected and deferred as per the standard operating procedures based on the national blood donor policy developed by safe blood transfusion programme. Group 02 comprised of gender matched healthy non-donors. Laboratory investigation included complete blood counts. Anaemia was defined as haemoglobin level <14 mg/dl in males. SPSS version 23.0 was used for descriptive and inferential statistics. A total of 928 donations were accepted at our hospital including 09 (1%) females and 919 (99%) males and all were replacement donors. Out of 919, 318 (35%) were repeated donors and 601 (65%) were first time donors. One hundred and seventy-five (29%) first time donors were recruited for analysis after taking consent. Non-donors group comprised of 100 participants. Age range was 18–55 years in both groups. Mean haemoglobin level of donors and non-donors group was 14.6 ± 1.06 mg/dl and 14.1 ± 2.2 mg/dl respectively. The prevalence of anaemia in first time donors was 46 (26%) while in non-donors it was 28 (28%). Chi-squared test was applied and it was found that anaemia was equally prevalent in donors and non-donors (P -value 0.713). Blood donation is a life-saving procedure for patients and a healthy activity for donors. Few misconceptions have been associated with blood donations in which development of anaemia is included. However, our study results revealed the equal prevalence of anaemia in donors and non donors group. Some other realistic facts were revealed during the study that should be

considered including scarcity of voluntary donations and females participation in blood donation for that community should be counselled especially young population. Longitudinal prospective studies are needed in this regard.

Disclosure of Interest: None Declared.

BSH2020-EP-195

Platelet refractoriness during BONE marrow transplantation, comparison in aplastic anaemia and beta-thalassaemia major patients. An Experience of Public Sector BMT Unit in Pakistan

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Bone marrow transplantation (BMT) is curative treatment modality for many haematological and oncological diseases. This treatment is commonly available in developed countries but lower middle income countries (LMICs) like Pakistan this facility is not available easily. We had established first Public sector BMT centre at the Children Hospital & Institute of Child Health, Lahore, Pakistan in collaboration with NIBD and BMT, Karachi, Pakistan. This BMT centre is providing free of cost HSCT facility to the children of poor families. This unit is financially supported by the Government of Punjab, Pakistan. So we as BMT Physicians have the ethical and moral responsibility to save the expenses and decrease the undue expenditure.

In this regard we had noticed that a handsome amount of money is being consumed for platelet transfusions during BMT. Platelet transfusions reduce the major haemorrhagic complications during BMT but refractoriness to infused platelets is a major clinical problem as well as huge financial burden.

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middle income countries (LMICs) like Pakistan this facility is not available easily. We had established first Public sector BMT centre at the Children Hospital & Institute of Child Health, Lahore, Pakistan in collaboration with NIBD and BMT, Karachi, Pakistan. This BMT centre is providing free of cost HSCT facility to the children of poor families. This unit is financially supported by the Government of Punjab, Pakistan. So we as BMT Physicians have the ethical and moral responsibility to save the expenses and decrease the undue expenditure. In this regard we had noticed that a handsome amount of money is being consumed for platelet transfusions during BMT. Platelet transfusions reduce the major Haemorrhagic complications during BMT but refractoriness to infused platelets is a major clinical problem as well as huge financial burden.

To measure the frequency of platelet refractoriness and comparison in aplastic anaemia (AA) and beta-thalassaemia major (BTM) during BMT.

This study done at BMT unit, Children Hospital & Institute of Child Health, Lahore, Pakistan after approval from ethical committee. Platelet refractoriness is confirmed by the 'lack of platelet count rise ($>10\,000 \times 10^9/l$) after 1 h of a single Mega unit Platelet transfusion. No HPA antigen or antibodies or HLA antibodies identified because of financial constraints.

Total 19 patients included in study, AA patients were 11 and 08 were of BTM. In our experience, platelet refractoriness is more common in the patients of beta-thalassaemia than that of aplastic anaemia. The only 2/11 aplastic anaemia patients showed platelet refractoriness while 6/08 of beta-thalassaemia had platelet refractoriness.

Platelet refractoriness is more common in BTM patient in comparison with AA patients during bone marrow transplantation. Our study is also indirect evidence that the PRBC transfusions can also produce antibodies responsible for platelet refractoriness. We recommend to Physicians who are involved in the management of BTM patients that the Red cell Extended Phenotyping should be done before putting the patients on chronic PRBCs transfusion therapy. This strategy ultimately decreases the complications of alloimmunisation both for PRBCs and Platelet.

Hence decrease the financial burden on the family as well as on Hospital in LMICs.

Disclosure of Interest: None Declared.

Transplantation

BSH2020-EP-196

Autologous stem cell transplantation for selected myeloma patients aged >70 years is safe and effective and similar to that of younger patients: a single centre experience

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High-dose chemotherapy followed by autologous stem cell transplantation (ASCT) is known to be a safe and effective treatment for myeloma patients. Several studies have also shown benefits for high-dose therapy for older patients aged up to 70 years undergoing autologous transplantation, although due to the perceived greater risk of toxicity- and transplant-related mortality in older patients, the dose of melphalan has often been reduced from 200 to 140 mg/m² or 100 mg/m².

The safety and efficacy of ASCT in elderly patients aged >70 years was retrospectively analysed at our centre from January 2008 to December 2018. Of the myeloma patients receiving ASCT during this time period 37 (5.9%) were aged >70 years when they had the first ASCT. The median age at transplant was 71.3 (range 70.0–73.9). 91.9% of patients received the ASCT in first response with the remaining 8.1% when in second remission. All patients having a second ASCT were excluded. Most patients, 43.3%, achieved a partial response prior to transplantation, followed by 37.8% who achieved a very good partial response, 16.2% complete response and for 2.7% of patients, the disease status was unknown. The predominant stage (Salmon and Durie) was IIIA as this comprised 32.4% of the patients. The haematopoietic cell transplantation-specific comorbidity index (HCT-CI) was calculated for all patients, excluding one patient for whom these data were unavailable. 78.4% of patients had an HCT-CI score of 0, 13.5% of 1, 5.4% of 3, while 2.7% had a score of 4. The most common mobilisation regimen was cyclophosphamide and granulocyte-colony stimulating factor (G-CSF) 45.9%, 24.4% of patients received G-CSF alone and 18.9% of patients received G-CSF and Plerixafor. No data were available for the remaining 10.8%.

78.4% of patients received the full dose of melphalan (200 mg/m²) for conditioning while 21.6% received it at a reduced dose of 140 mg/m². The mean time to neutrophil and platelet engraftment was 11.6 days (range 10–15) and 12.0 days (range 10–15) respectively. The mean inpatient stay was 18.4 days (range 13–34). 24.3% of patients were diagnosed with neutropenic sepsis during their admission but only one patient succumbed to neutropenic sepsis within 30 days of transplant. There were seven patients who underwent a second transplant at >70 years of age, five of which received melphalan conditioning at 200 mg/m² the others at 140 mg/m². The HCT-CI score was 0 for all patients except one, for whom it was 2, and all patients were alive at 100 days from transplant. The transplant-related mortality rate for the patients >70 years undergoing a melphalan autograft was 1/44 (2.3%).

Our analysis shows that, despite most patients receiving full dose melphalan, the toxicity was very low. This can probably be explained by the fact that these elderly patients were highly selected since 93.2% of patients had low HCT-CI scores <3. This strategy appears

to be highly effective although we do not know how many other myeloma patients aged >70 years were not considered fit enough for high-dose therapy and who received conventional therapies.

Disclosure of Interest: None Declared.

BSH2020-EP-197

Lung function screening following allogeneic haematopoietic stem cell transplantation

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Chronic lung graft-versus-host disease is a significant cause of morbidity and mortality post-allogeneic haematopoietic stem cell transplantation (HSCT). Early lung cGVHD can be asymptomatic until significant lung function is lost. Deterioration in lung function tests can be detected prior to development of symptoms. Early intervention is important to reverse or stabilise further deterioration. There is international and interinstitutional variation in lung function follow-up and at the time of the service evaluation our centre did not have a formal policy on lung function testing post-allogeneic HSCT.

A service evaluation was performed to retrospectively assess whether allogeneic HSCT patients were receiving adequate lung function GVHD screening. The group under study were all patients who underwent allogeneic HSCT in 2015 in our centre ($n = 58$). Dates of lung function tests post-transplant recorded (in all cases simultaneous spirometry and gas exchange). The cohort was followed up until March 2018. A subgroup of those with suspected chronic GVHD of any type were examined in greater detail.

Data were collected for the patient group to assess if lung function was checked at least once post-transplant, if so it was noted whether this was done within the first 100 days post-transplant, or later than a year post-transplant.

Of the 32/58 patients who did not have lung function checked post-allograft there were 14 with relatively mitigating circumstances for not performing lung function post-transplant – either they did not attend their appointment, they had died within the first year post-transplant or lung function had been attempted but they had been too unwell to perform successfully.

Removing patients with these mitigating factors for not performing lung function left 44 patients in whom lung function could have been checked, 26 who had lung function checked post-transplant, 4 in the first 100 days, 11 within 12 months, and 15 later than 12 months post-HSCT. The rationale for performing lung function in several patients was not related to the original HSCT – for example as an unrelated investigation by another department – or was performed as workup prior to second HSCT.

Eight patients in the cohort were recorded as developing features of chronic GVHD. Of these 6/8 had lung function checked following the suspicion of chronic GVHD, however, this was only within 100 days of diagnosis of cGVHD in two patients.

One patient in whom lung function was planned at 50 days following suspicion of GVHD deteriorated, was admitted to hospital and died so lung function was not performed. Cause of death was listed on the medical certificate of death as 1a Pneumonia, 1b Chronic GVHD.

The service evaluation demonstrated that the majority of patients in our centre are not having lung function checked post allogeneic-HSCT, and that patients with features of chronic GVHD are not all having lung function checked following the diagnosis.

The local GVHD policy was updated as a result of this service evaluation with formal recommendation for lung function screening. Following attempts to address the logistical challenges of increased lung function monitoring the service evaluation will be repeated.

Screening lung function testing is now recommended within our service at day 100 post-HSCT, at 1 year post-HSCT, at diagnosis of cGVHD and 6 monthly for 2 years after the initial diagnosis of cGVHD.

Disclosure of Interest: None Declared.

BSH2020-EP-198

Abatacept acute graft-versus-host disease prophylaxis is feasible and safe when combined with post-transplant cyclophosphamide in the mobilised peripheral blood haploidentical donor setting

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Haploidentical haematopoietic stem cell transplantation (haplo-HSCT) is emerging as a preferred alternative donor source globally, utilising with post-transplant cyclophosphamide (PTCy). PTCy is typically combined with a calcineurin inhibitor and mycophenolate mofetil and successfully reduces severe chronic graft-versus-host disease (GVHD). However, breakthrough acute GVHD persists as a significant cause of non-relapse mortality (NRM) post-transplant.

CD28-CD80/CD86 axis T-cell co-stimulation blockade with CTLA4-Ig using abatacept (AB) is an emerging acute GVHD prophylaxis strategy, with US Food and Drug Administration (FDA) breakthrough designation. Our centre has successfully combined three doses of abatacept with PTCy in the matched related and unrelated donor myeloablative conditioning (MAC) settings to eliminate severe acute GVHD (*Blood* 2019;134(Suppl_1):4540; <https://doi.org/10.1182/blood-2019-130184>). In the unrelated donor mismatched donor setting, four doses of abatacept combined with conventional methotrexate and a calcineurin inhibitor is being investigated (NCT01743131), mostly in children. The feasibility and safety of combining three doses of abatacept with PTCy in the related mismatched (haploidentical) MAC transplant setting in unknown and presented here.

We report our single centre observational experience of seven patients with significant co-morbidities who received a haploidentical donor allogeneic stem cell transplant, with MAC: IV fludarabine 150 mg/m², IV treosulfan 42 mg/m² and 2 Gy total body irradiation (TBI). Post mobilised peripheral blood stem cell infusion, IV PTCy 50 mg/kg was administered on d+3 and +4, followed by IV tacrolimus 0.03 mg/kg/day infusion from d+5, and IV mycophenolate mofetil 15 mg/kg tds. Four patients received additional aGVHD prophylaxis utilising three doses of IV abatacept 10 mg/kg on d+5, +14 and +28.

The median patient age was 44 years, with median Karnofsky score of 80% and median HCT-CI of 6. Transplant indications included blood cancers (71%) and non-malignant (29%) bone marrow failure indications. Median time to engraftment of white cells was 17 days, neutrophils 18 days, platelets 25 days and 26 days for a target of $20 \times 10^{12}/l$ and $50 \times 10^{12}/l$ respectively. Median time to 100% donor chimerism (CD3+, whole blood or myeloid cells) was

d+36 post-HSCT, except for one patient who experienced primary graft failure and required stem cell top-up d+56 post-HSCT without conditioning but with repeated PTCy GVHD prophylaxis. This patient did not receive abatacept prophylaxis and did not experience aGVHD. No patients required donor lymphocyte infusions (DLIs) for mixed chimerism or relapse. Seventy-one per cent of patients experienced steroid responsive aGVHD with overall grades of 2–3, with median time to aGVHD onset of 94 days. Of these patients, those receiving abatacept had only gut GVHD whilst patients not receiving abatacept had both gut and skin involvement. No patients relapsed by 180 days and no patients died, with a median follow-up of 399 days.

Three doses of abatacept appears to be feasible and safe in the myeloablative mobilised peripheral blood haploidentical donor setting when combined with PTCy after MAC. A prospective clinical trial is planned.

Disclosure of Interest: None Declared.

BSH2020-EP-199

The process evaluation of the development of a complex physiotherapy intervention for teenagers and young adults undergoing allogeneic haematopoietic stem cell transplant

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Despite improvement in survival following allogeneic haematopoietic stem cell transplantation (Allo-HSCT), patients experience multiple physical and psychological issues related to prolonged hospitalisation, immobility, toxicities and complications of the treatment. Randomised controlled trials (RCTs) are regarded as gold standard for establishing effectiveness of an intervention, however, effect size does not always provide policymakers with information on how an intervention might be replicated in their specific context or whether trial outcomes will be reproduced. Evidence suggests exercise for patients undergoing Allo-HSCT had positive impacts on physical and emotional recovery. However, limited evidence exists to determine optimal exercise intervention within the context of teenagers and young adults (TYA). Based on data collected from a service evaluation (SE) we aimed to describe the development of a structured stratified physiotherapy intervention and its incorporation into the care of TYA patients 14–24 years undergoing Allo-HSCT over a 9-month period, April 2019 to current.

The UK Medical Research Council framework for Developing and Evaluating Complex Intervention was used as a guide for intervention development. Nine TYA patients have been enrolled into the physiotherapy programme. Assessment was made of the ability to assess exercise dose, compliance with the programme and the potential reach of a TYA specific physiotherapy programme. The intervention was designed to deliver the exercise component of prehabilitation before Day (0), supervised exercise during transplant, defined as Day (+1) until neutrophils were $1 \times 10^9/L$; and finally, for eight sessions following transplant.

Dose: The exercise dose advised was developed based on the outcome of a 10 m incremental shuttle walk test (ISWT). Patients were stratified to complete a level of intensity based on what they

achieved, measured against their predicted outcome. Patients were stratified into light intensity if they achieved less than 30% of their predicted, moderate intensity if they achieved 30% > 50% of their predicted and vigorous intensity if they achieved >50% of their predicted. Intervention intensity was then progressed on an individual basis.

Fidelity: When this SE commenced, patients were at different stages of their transplant pathway. Intervention delivery at prehabilitation was therefore not delivered entirely as per protocol. Eight of the nine patients were provided with information about the structured programme and exercise advice at least 1 week prior to Day (0). During transplant, each patient had an average of seven physiotherapy contacts with an average of four sessions delivered as per protocol. Following transplant, four patients completed eight sessions, one completed five, one opted for self-management and three are ongoing.

Reach: Funding of a physiotherapist within the TYA transplant clinic, multidisciplinary working and family support ensured that all patients engaged in some aspect of the intervention, albeit not in its entirety.

This intervention development provides a systematic approach to implementing exercise as integral to the transplant pathway. Prior to developing robust RCT in this complex patient population exercise interventions need to be clearly defined and replicable within clinical practice. The SE provides valuable insight into preliminary developments of a complex intervention but further work is required to implement fully on a larger scale and to determine acceptability.

Disclosure of Interest: None Declared.

BSH2020-EP-200

Feasibility, safety and efficacy of Abatacept for steroid refractory acute graft-versus-host disease

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Acute graft-versus-host disease (aGVHD) remains a major limitation to allogeneic haematopoietic stem cell transplantation (HSCT), associated with high patient morbidity and mortality. Steroid refractory acute GVHD (SR-aGVHD) has the highest mortality (around 80%) despite treatment intensification with available immunosuppressive therapy. This is complicated by poor response rates alongside increased toxicity and infectious complications from profound immunosuppression and prolonged uncontrolled GVHD. Novel immunotherapeutic strategies are therefore urgently needed for treatment of SR-aGVHD.

Immunological manipulation of T-cell activation is a novel therapeutic treatment approach for aGVHD. Abatacept, a CTLA4-Ig exerting CD28-CD80/CD86 axis T-cell co-stimulation blockade, has been shown in the phase 1 clinical trial setting to be safe, well tolerated without dose limiting toxicities (DLTs), and efficacious in the treatment of heavily pretreated chronic extensive GVHD (cGVHD). Marked improvement in National Institutes of Health cGVHD scores alongside significantly reduced steroid dose requirement in this cohort have led to a phase 2 trial (NCT01954979) and National Comprehensive Cancer Network (NCCN) recommendation. T-cell activation is a key component of aGVHD but it is currently unknown if Abatacept represents a potential therapy for SR-aGVHD.

We report the feasibility, safety and efficacy of Abatacept added to the management of SR-aGVHD in a cohort of four heavily pretreated patients in a single centre. Institutional compassionate access

was granted after exhaustion of all available therapies. Three of the four patients developed aGVHD after donor lymphocyte infusion (DLI), in the absence of pre-DLI conditioning or GVHD prophylaxis. Median overall MAGIC aGVHD score was 4, with two patients having predominant stage 4 gut aGVHD and two patients with predominant stage 4 skin aGVHD. The median prior lines of treatment were 5, including steroid therapy (Prednisolone or Methylprednisolone 2 mg/kg/day), calcineurin or mTOR inhibitor, mycophenolic acid and extracorporeal photophoresis (ECP) with 8-methoxypsoralen (8-MOP). Patients with gut aGVHD were also treated with vedolizumab and/or infliximab.

Abatacept 10 mg/kg was administered IV in addition to concurrent treatments for SR-aGVHD at d0, d+14, d+28, d+56, d+72 and d+90. If a clinical response was observed after 6 doses, Abatacept was continued every 4–6 weeks up to 12 doses. Two patients achieved a clinical partial response and one patient a clinical complete response at d+100 after initiation of Abatacept. Importantly all patients achieved a median dose reduction in steroids of 98% by d+100. Three patients had viral reactivation diagnosed prior to Abatacept initiation (median of d–155 before Abatacept). Importantly, no new viral reactivations were detected in any patients after Abatacept initiation. No DLTs were observed; bacterial infections were not considered a DLT if they had already occurred with prior therapies, being attributed to pre-existing severe immunosuppression and SR-aGVHD. We observed that patients with less than CR by d+100 developed chronic GVHD of the involved organ. Mortality was observed only in the two patients with gut involvement, as a result of chronic malnutrition and recurrent immunocompromised infection. Our experience suggests that Abatacept for SR-aGVHD is feasible, appears safe and may have efficacy in some patients. A prospective clinical trial is planned.

Disclosure of Interest: None Declared.

BSH2020-EP-201

The use of artificial nutritional support within a national allogeneic transplant unit

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Aim: To identify whether patients receiving nasogastric feeding have lower rates of weight loss and length of stay than those who receive parenteral nutrition.

Background: Patients undergoing allogeneic transplant are at risk of malnutrition which is associated with poorer outcomes and prolonged hospital stay. Enteral feeding would be first-line artificial nutritional support in our unit in line with ESPEN recommendations. Parenteral nutrition (PN) is not standard therapy if enteral feeding is refused and each patient will be assessed individually with risks of this method considered.

Methods: A retrospective review between 2017 and 2019 identified allogeneic transplant patients deemed high nutritional risk and requiring some form of artificial nutritional support (ANS). Patients ($n = 43$) were routinely assessed by a Specialist Dietitian pre transplant through individualised assessment. These patients were recommended to consider nasogastric (NG) feeding based on treatment plan, weight loss/BMI, current nutritional status and history. PN was considered as appropriate. Data on average length of stay (LOS), average weight loss, conditioning therapy and Karnofsky score were collected.

Results: Thirty patients agreed to placement of an NG tube (NGT). Only 57% tolerated insertion ($n = 17$). Overall 13 patients declined an NGT and received PN along with five patients who did not

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tolerate tube insertion ($n = 18$, 42%). As expected, myeloablative conditioning was the largest factor in those who received PN. Those who were NG fed had lower weight loss and length of stay compared to those on PN. Despite successful NGT insertion one patient required PN due to considerable gastrointestinal toxicity and was excluded from results together with one patient who died.

Conclusion: Data suggest nasogastric feeding can be used successfully to reduce weight loss and hospital stay when patients accept and tolerate tube insertion. Regardless of acceptance, some patients did not tolerate NGT insertion (43%). Despite the small sample size, further prospective studies could be warranted. Those commenced on PN had highest weight loss and hospital stay. Ideally, improved acceptance and insertion of NGTs could reduce need for PN. Future strategies to improve acceptance of feeding could include use of anthropometric measures to highlight lean muscle function and loss to enhance understanding of body composition when considering ANS. Furthermore, consideration of radiologically inserted feeding tubes if failed NGT at ward level could support first-line enteral feeding.

Table.

	PN (17)	NGT (16)
Average weight loss (%)	7	4
Myeloablative conditioning (%)	95	39
Average LOS (days)	40	30
Average Karnofsky score	80	90

Disclosure of Interest: None Declared.

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Comparison of apheresis peripheral stem cell collections undertaken using the Spectra Optia mononuclear and continuous mononuclear cell collection programs

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In 2018/19 NHS Blood and Transplant Therapeutic Apheresis Services (TAS) evaluated peripheral blood stem cell (PBSC) collections on the Spectra Optia, using both the Mononuclear Cell Collection

(MNC) and Continuous Mononuclear Cell (CMNC) programs. TAS has used the MNC program since 2013, to collect over 1000 PBSC products per year.

The CMNC program was developed several years after the MNC program. It collects cells continuously throughout the procedure, whereas the MNC program collects the stem cells in cycles, filling an elutriation chamber (cone) and emptying the cells into the collection bag when the cone is full. CMNC is used for collecting from small children, as pump speeds can go down to 5 ml/min compared to only 10 ml/min on the MNC program. This is a distinct reason for using CMNC over MNC, however, in evaluating CMNC against MNC, the aim was to compare collection results across the two programs for a wider patient population. The focus was to assess cell yields using each program, rather than the mechanical function of the machine.

A prospective evaluation commenced in two TAS units of approx. Seventy collections from each program. The machines had identical collection settings for Inlet:Anticoagulant (AC) ratio, AC infusion and blood volume processing end-points.

Measures recorded for comparison were:

- Initial peripheral pre procedure patient blood results
 - Total white cell count ($10^9/l$)
 - Peripheral CD34 count ($10^3/ml$)
 - Platelet count ($10^9/l$)
- Collected product content results:
 - Volume (ml)
 - CD34 cell content ($10^6/kg$)
 - Red blood cells (ml)
- Procedural measures
 - Volume of blood processed (ml)
 - Collection Efficiency (CE%)
 - Procedure time (min)
 - Platelet loss (%)
 - Citrate symptoms

The results indicate that the CMNC program gives higher CE% than the MNC and the MNC program results in lower red blood cell volumes in collections. Any statistical significance of this will be presented alongside the results of further statistical analysis, which is being undertaken to establish any relationship between initial peripheral white cell and CD34⁺ count, alongside the cell collection efficiency, using each program. The aim of this evaluation is to identify whether there is benefit to using one program over another, based

Table.

No of collections		CMNC		82	MNC		69		
Mean CE (%)		Median CE (%)		CD34 yield/Pre-CD34 Count (%)		Actual CD34/WCC (%)			
CMNC	MNC	CMNC	MNC		CMNC	MNC		CMNC	MNC
54.95	44.10	48.16	43.37	Mean	7%	16%	Mean	26%	17%
				Median	7%	7%	Median	9%	7%
				Range	3%>13%	3%>11%			
Mean RBC (ml)		Mean Plt loss (%)		Mean Vol proc (ml)		Mean Col Vol (ml)		Citrate (%)	
CMNC	MNC	CMNC	MNC	CMNC	MNC	CMNC	NMC	CMNC	NMC
6.34	4.12	25.2	24.5	12617	12480	211	216	65	62
Median RBC (ml)		Median Plt loss (%)		Median Vol proc (ml)		Median Col Vol (ml)			
CMNC	MNC	CMNC	MNC	CMNC	MNC		CMNC	MNC	
5.72	2.78	30.1	27.4	12553	12210		204	198	
				Range			Range		
				CMNC	MNC		CNMC	NMC	
				7423–17 298	8052–16 831		118–323	108–414	

on the patients' blood counts prior to the procedure and these conclusions will be presented alongside the data.

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The treatment of late-onset hepatic veno-occlusive disease following haematopoietic stem cell transplantation using defibrotide: a case study

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Defibrotide is approved for the treatment of severe hepatic veno-occlusive disease (VOD) posthaematopoietic stem cell transplantation (HSCT) in patients >1 month of age. Hepatic VOD, a rare complication of HSCT, occurs in approximately 13.7% of HSCT patients. Defibrotide was approved to treat VOD based on data demonstrating its clinical benefit in patients, showing consistently positive results where no other effective treatment has been identified. We present a case report describing successful treatment of VOD using defibrotide in a female who received chemotherapy for during pregnancy and subsequently relapsed, undergoing salvage treatment including myeloablative HSCT.

A 27-year-old woman was diagnosed with acute myeloid leukaemia at 24 weeks' gestation (mutated NPM1, normal cytogenetics, FLT3 wild type). She received a cycle of induction chemotherapy with daunorubicin and cytarabine (DA 3+7), achieving complete remission. After delivering a healthy baby, she declined consolidation chemotherapy. Seven months later and declining follow-up, the patient relapsed with a high white cell count $>150 \times 10^9/l$. After re-induction chemotherapy with FLAG-IDA, she achieved a second complete as well as molecular remission.

Following consolidation chemotherapy she underwent a myeloablative HSCT from a fully matched sister using cyclophosphamide and busulphur conditioning; engraftment was timely and she was discharged 24 days post-transplant. Eight days later, she presented with nausea and acute abdominal pain. Elevated C-reactive protein (82 mg/l; reference range: 0–5.0 mg/l) was detected, and she was admitted with suspected acute cholecystitis; no clinical findings indicated VOD.

Abnormal liver function tests were recorded; alkaline phosphatase levels were 198 IU/l (reference range: 30–130 IU/l) on admission, and alanine transaminase peaked at 248 IU/l (reference range: 1–35 IU/l) 4 days later. The patient did not comply with fluid balance and several attempts to measure urine output were unsuccessful; she subsequently displayed a distended abdomen with right upper quadrant pain on palpation, and weight increased from 50.5 to 59.5 kg (17.8%) over 10 days. Bilirubin was 22 $\mu\text{mol/l}$ (reference range: 0–21 $\mu\text{mol/l}$) 4 days after admission, rising rapidly to 37 $\mu\text{mol/l}$ 5 days later. These factors, along with the fluid retention and portal vein disruption noted on ultrasound scan, supported a diagnosis of late-onset severe VOD using the 2016 European Society for Blood and Marrow Transplantation grading criteria, although an element of graft-versus-host disease could not be excluded without biopsy.

Treatment with defibrotide 25 mg/kg in four divided doses was administered intravenously alongside careful nursing management, including analgesia and nutrition optimisation, strict fluid restriction and maintenance of platelets $>30 \times 10^3 \mu\text{l}$. Over the 21-day treatment course her liver function tests normalised and her weight reduced to baseline. Having improved significantly with normalisation of liver function tests and reduction of fluid load/weight, she was discharged home after completing 21 days of defibrotide.

In conclusion, this case highlights defibrotide as an effective treatment in severe hepatic VOD. In cases of suspected VOD, early diagnosis, swift initiation of treatment and close monitoring of supportive care are crucial to most effectively resolve this condition.

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Vyxeos in high-risk MDS as induction therapy prior to stem cell transplantation

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Myelodysplastic syndromes (MDS) are a heterogeneous group of clonal stem cell disorders characterised by cytopenias. There are 10 distinct subtypes, categorised into three broad risk groups on the basis of predicted survival time and risk of evolution to acute myeloid leukaemia (AML). Management ranges from supportive care to drug therapy and stem cell transplantation (SCT). Therapy decisions depend on risk stratification and patient factors, including performance status and age.

Vyxeos is a liposomal, synergistic combination of cytarabine and daunorubicin, preferentially taken up by leukaemic cells. Phase III data, showing Vyxeos to improve median overall survival compared to conventional intensive treatment in older patients (60–75 years) with poor risk AML ($P = 0.003$), led to NICE and SMC approval of Vyxeos as a treatment option for untreated therapy-related AML and AML with MDS-related changes.

Here we present a patient with high-risk MDS who received Vyxeos, through compassionate access, as induction therapy prior to stem cell transplant.

A 59-year-old female, with no notable past medical history, presented with a 3-week history of spontaneous bruising. Her full blood count demonstrated a mildly elevated total white cell count ($23 \times 10^9/l$; neutrophils $17 \times 10^9/l$, blasts 10.5%), anaemia (haemoglobin 108 g/l) and thrombocytopenia (platelets $78 \times 10^9/l$). The blood film revealed occasional tear drop poikilocytes, dysplastic neutrophils and circulating blasts at 7% ANC. The diagnostic bone marrow (BM) aspirate was aparticle; the trephine showed tri-lineage dysplasia with 7% blasts and Grade II fibrosis. A normal karyotype was observed (46, XX) with no mutation within *FLT3* or *NPM1*. Her R-IPSS (risk score) was intermediate. She was actively monitored. Four months later her full blood count deteriorated (Hb 110 g/l, WCC $17.4 \times 10^9/l$, neutrophils $9.13 \times 10^9/l$, blasts 12%, platelets $47 \times 10^9/l$). A repeat BM confirmed a rising blast percentage (15%). Next-generation sequencing reported the presence of *CSF3R*, *ASXL1* and *SRSF2*. Her revised R-IPSS was >4.8 (high) with a predicted median overall survival of 1.6 years. She proceeded to induction therapy with Vyxeos through compassionate access. She suffered bleeding (Grade 2) requiring platelet transfusions and tranexamic acid and febrile neutropenia (Grade 2) which resolved with first-line antibiotics. Neutrophil recovery was on day 48 and platelet recovery on day 41 (with recovery at 35 days and 26.5 days respectively within trial). Following count recovery a BM demonstrated response with a reduction in blasts ($<5\%$). She proceeded to a reduced-intensity conditioning allogeneic PBSCT with no treatment-related morbidity. Post-SCT she required treatment for RSV (Day 39) and CMV reactivation (Day 41) despite good engraftment although she remained systemically well throughout. She is currently 14 months

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post-SCT with a normal full blood count, continued good engraftment (100% donor DNA in the myeloid lineage) though with a mixed lymphoid chimerism and no treatment-related toxicities. She is currently being worked up for DLI to try to restore full lymphoid chimerism.

Through this case we have shown the efficacy and safety of Vyxeos as induction therapy prior to SCT in patients with high risk

MDS. We suggest Vyxeos should be considered alongside current therapies in patients with high-risk MDS as a bridge to SCT though the variation in count recovery may represent a planning issue for busy transplant centres.

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